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Aluminium adjuvants versus placebo or no intervention in vaccine development randomised clinical trials. A systematic review with meta-analysis and Trial Sequential Analysis

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Aluminium adjuvants versus placebo or no intervention in vaccine development randomised clinical trials. A systematic review with meta-analysis and Trial Sequential Analysis

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Abstract

Objectives: To assess the benefits and harms of aluminium adjuvants versus placebo or no intervention in randomised clinical trials in relation to human vaccine development.

Data sources: We searched CENTRAL, MEDLINE, Embase, LILACS, BIOSIS, Science Citation Index Expanded, and Conference Proceedings Citation Index – Science until June 29, 2021 and Chinese databases until September 2021.

Main outcome measures: Primary outcomes were serious adverse events (SAE), all-cause mortality, and proportion of participants with the disease being vaccinated against. Secondary outcomes were quality of life, non-serious adverse events (AE), and serological response.

Results: We included 102 randomised clinical trials (26 457 participants). Aluminium adjuvants versus placebo or no intervention may have no effect on SAE (RR 1.18, 95% CI 0.97 to 1.43; very low certainty). Trial Sequential Analysis (TSA) showed that the required information size was not reached. Aluminium adjuvants versus placebo or no intervention may have no effect on all-cause mortality (RR 1.02, 95% CI 0.74 to 1.41; very low certainty). TSA showed that the required information size was not reached. Only two trials reported on the proportion of participants with relevant disease being vaccinated against. No trial reported on quality of life. Aluminium adjuvants versus placebo or no intervention may increase AE (RR 1.13, 95% CI 1.07 to 1.20; very low certainty). TSA showed that there was sufficient information to confirm a harmful effect of aluminium adjuvants. We found no or little evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing geometric mean titres or concentrations. Aluminium adjuvants versus placebo or no intervention may have no effect on participants' seroprotection (RR 0.95, 95% CI 0.77 to 1.18; very low certainty).

Conclusions: Based on evidence at very low certainty, we were unable to clearly identify benefits or harms of aluminium adjuvants.

Protocol: PROSPERO CRD42017083013.

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Strengths and limitations of this study

- We seem to be the first to assess the benefits and harms of aluminium adjuvants by conducting a systematic review comparing aluminium adjuvants versus placebo or no intervention in combination with all types of vaccines.
- We included 102 randomised clinical trials from a comprehensive search with no language limitations or restrictions on outcomes reported in the trials, type of aluminium adjuvant or type of vaccine.
- The certainty of evidence is very low and this makes it difficult to draw firm conclusions.

review only

Introduction

Vaccination is one of the major triumphs of modern medicine.[1 2] Vaccination prevents infectious diseases, and the worldwide eradication of the deadly smallpox and the restriction of diseases such as poliovirus, measles, and tetanus can largely be ascribed to the numerous successful mass vaccination programs launched since the 1960s.[1 2] Similar successes are presently unfolding during the COVID-19 pandemic. [3]

Adjuvants are added to vaccines to enhance the ability to provoke an immune response to weak antigens and to improve the overall potency of vaccines.[3-5] Adjuvants may also pose other benefits, such as reducing the frequency of vaccination and the dose of antigen per vaccine, and some may provide 'cross-clade' immunity (i.e. immunity against different clades of viruses or bacteria) or improve the stability of the final vaccine formulation.[6 7] Several adjuvants with completely different mechanism of action are approved for use in vaccine production or in

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experimental use in humans. Different insoluble aluminium salts have been used as vaccine adjuvants since 1926.[8] Aluminium is also an adjuvant candidate for some of the COVID-19 vaccines.[9]

Our objectives were to assess the benefits and harms of aluminium adjuvants versus placebo or no intervention in randomised clinical trials in relation to human vaccine development.

Methods

Detailed description of the methodology is in our prepublished protocol,[10] PROSPERO protocol (CRD42017083013) and our Supplementary material. In short, we searched for randomised clinical trials irrespective of type, status, date, and language of publication. We included vaccine development trials comparing any type of aluminium adjuvant versus placebo or no intervention. We accepted any co-interventions of vaccines if planned to be delivered equally to the intervention groups. We used the trial results reported at maximum follow-up.

Primary outcomes were serious adverse events,[11] all-cause mortality, and proportion of participants with the disease being vaccinated against. Secondary outcomes were health-related quality of life, non-serious adverse events, and serological response.

We searched the Cochrane Central Register of Controlled Trials (2021, Issue 7) in The Cochrane Library, MEDLINE Ovid (1946 to July 2021), Embase Ovid (1974 to July 2021), LILACS (Bireme; 1982 to July 2021), BIOSIS (Web of Science; 1969 to July 2021), Science Citation Index Expanded (Web of Science; 1900 to July 2021), and Conference Proceedings Citation Index – Science (Web of Science; 1990 to July 2021). In addition, we searched (September 2021) the Chinese Biomedical Literature Database (CBM), China Network Knowledge Information (CNKI), Chinese Science Journal Database (VIP), and Wanfang Database (**Supplementary Table 1**). We also searched Google Scholar, The Turning Research into Practice (TRIP) Database, ClinicalTrials.gov (www.clinicaltrials.gov/), European Medicines Agency (EMA; www.ema.europa.eu/ema/), WHO International Clinical Trial Registry Platform

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(www.who.int/ictrp), The Food and Drug Administration (FDA; www.fda.gov), and pharmaceutical company sources for ongoing or unpublished trials (until March 2021). We applied EMA, FDA, and several national medicines agencies (Australia, China, India, Japan, United Kingdom, Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden) for clinical study reports on trials fulfilling our inclusion criteria.

Three review authors (SRK, SLK and MB) independently and in pairs screened titles and abstracts for inclusion of potentially eligible trials using Covidence (www.covidence.org). Following any disagreements, we asked a third author to arbitrate (JCJ or CG). The review author pair collected full-text trial reports/publications, and independently screened the full-texts and identified trials for inclusion. SRK extracted all data on all trials. SLK and MB each independently extracted half of the data. Extractions were compared and validated by SRK, SLK, and MB and in case of disagreement, the same review authors consulted JCJ or CG.

The review author pair (SRK, SLK and MB) independently assessed the risk of bias of each included trial according to the recommendations in the Cochrane Handbook for Systematic Reviews of Interventions.[12]

We visually investigated forest plots to assess the risk of statistical heterogeneity. We also assessed the presence of statistical heterogeneity using the Chi^2 test (threshold $P < 0.1$) and measured the quantities of heterogeneity using the I^2 statistic.[13 14] We assessed reporting bias using funnel plots where ten or more trials per comparison were included.

When the total proportion of participants experiencing any serious and non-serious adverse event was not reported, we extracted data from the highest proportion of participants experiencing an individual adverse event.

We performed subgroup analyses on A: outcomes at low risk of bias compared to outcomes at high risk of bias or unclear risk of bias (collectively termed high risk of bias); B: trials at low risk of vested interests compared to trials

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at high risks of vested interests;[15] C: according to aluminium adjuvants type; D: according to different vaccines; E: according to age groups; F: according to different maximal follow-up periods; G: according to participants' healt.

We assessed the potential impact of missing data with the 'best-worst' and 'worst-best case' scenario.

Intervention effects were assessed with both random-effects model[16] and fixed-effect model[17] meta-analyses. The more conservative point estimate of the two (the analysis with the highest P value) was reported. For analysis of three primary outcomes, a P value < 0.025 was considered statistically significant[18] because this would secure a family-wise error rate (FWER) below 0.05.

We analysed all primary and secondary outcomes using Trial Sequential Analysis to control random errors.[10]

We included aluminium concentration (as described by trialist or manufacturer) as a covariate in meta-regression to assess whether the concentration influences the effect of aluminium adjuvant administration on outcomes.

We used the GRADE system to assess the certainty of the evidence.[19]

Patient and public involvement

A patient and a representative of the public were involved in formulating the research question and the outcomes at the protocol stage. They were both involved in the interpretation and writing up of results. There are plans to disseminate the results of the research to the public and the relevant patient communities.

Results

Supplementary Figure 1 shows the flow of records obtained through the electronic searches. We identified 15 734 records through database searching. We obtained 396 full-text reports that were assess for eligibility. We excluded 280 records. We identified 6 ongoing trials and 8 trials awaiting classification (**Appendix 1**).

We identified 102 randomised clinical trials including a total of 26 457 participants that fulfilled our inclusion criteria. Characteristics of included and excluded studies are given in **Appendix 1**. We were unable to identify any

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clinical study report that was eligible for inclusion in this review. We approached all corresponding authors to request missing information or explanations on unclear information and received some additional information from seven authors. The 102 included trials were published between 1969 and 2021 and were performed in different countries: 35 in the USA;[20-54] 15 in more than one country;[55-69] six in Canada;[70-75] four in China;[76-79] four in Belgium;[80-83] four in Africa;[84-87] three each in the UK,[88-90] Taiwan,[91-93] and Australia;[64 94 95] two each in Thailand,[96 97] Poland,[98 99] Norway,[100 101] Italy,[102 103] Germany,[67 104] Cuba,[105 106] and Austria; [107 108] one each in Switzerland,[109] Sweden,[110] Singapore,[111] Mali,[112] Israel,[113] India,[114] France,[115] Colombia,[116] Chile,[117] and Bangladesh.[118] Three trials did not report a country.[119-121]

Trial participants

Type of participants

The trials randomised different types of participants. Nine trials randomised participants with a disease diagnosis;[24 31 34 45 64-66 103 108] 90 trials randomised healthy participants; and three trials did not describe the inclusion criteria of the participants.[29 73 74]

Age of participants

In regards of age, the trials randomised: infants (six trials[60 88 99 106 118 120]); children (11 trials[59 61 69 76 78 91 106 112 121-123]); adolescents (two trials[40 89]); elderly (nine trials[24 34 36 57 64 72 75 86 113]); and mixed populations (eight trials[31 52 66 77 79 83 87 97]). Two trials did not specify the population type.[29 108] The remaining 65 trials randomised adult participants.

Interventions and comparisons

Types of aluminium adjuvants

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The included trials assessed different types of aluminium adjuvants: aluminium hydroxide (37 trials[23 24 31 33 41-44 46 48 55-57 62 63 65 67 68 71 76-78 80 81 85 87 89 93 96-98 100 107 108 114-116 119]); aluminium phosphate (27 trials[20 21 27 34 36 37 40 49 54 60 61 72 83 84 86 90-92 95 105 106 110 113 120-122 124]); alhydrogel (21 trials[22 25 26 28-30 32 47 50-53 73-75 82 88 94 99 101 111]); amorphous aluminium hydroxyphosphate sulfate (two trials[38 45]); aluminium fluoride (one trial[69]); phosphate-treated aluminium hydroxide (one trial[118]); alhydrogel pretreated with phosphate buffer (one trial[109]); Adju-Phos® (aluminium phosphate gel) (one trial[70]); aluminium potassium sulphate (one trial[39]); aluminium chloride (one trial[35]); and aluminium oxide (one trial[117]). Eight trials did not describe the type of aluminium adjuvant used. [58 59 66 79 102-104 112]

Vaccines against different virus, bacteria, toxins, or diseases

The included trials assessed the effects of vaccines against different virus, bacteria, toxins, or diseases: influenza (25 trials[23 24 28 30 41 42 58 62 73-75 77 79 90 92 93 95 100 104 107 111 113-115 117]); Streptococcus pneumoniae (11 trials[48 59-61 69 83 86 109 118 120 121]); respiratory syncytial virus (11 trials[34 36 37 53 54 56 64 71 72 80 119]); human immunodeficiency virus (six trials[21 33 49 85 102 103]); Neisseria meningitidis (six trials[27 35 40 84 101 112]); Clostridium difficile (four trials[31 43 52 57]); dengue fever virus (four trials[29 32 94 125]); enterovirus (three trials[76 78 91]); Bacillus anthracis (three trials[22 25 26]); diphtheria and tetanus (two trials[88 110]); human papillomavirus (two trials[39 46]); Lyme borreliosis (two trials[63 68]); Haemophilus influenzae type B (two trials[105 106]); groups B Streptococcus (two trials[50 82]); Staphylococcus aureus (two trials[38 45]); poliovirus (two trials[98 99]); Pseudomonas aeruginosa (two trials[65 67]); Alzheimer's disease (two trials[66 108]); cytomegalovirus (two trials[20 70]); tetanus (one trial[89]); non-typeable Haemophilus influenzae (one trial[82]); Ross River virus (one trial[55]); hepatitis B (one trial[87]); malaria (one trial[116]); rabies and tetanus (one trial[96]); rabies (one trial[97]); Shigella flexneri (one trial[51]), and Staphylococcus aureus and Candida albicans (one trial[47]).

Vaccine doses

The included trials administered different numbers of vaccine doses: 24% of the trials administered 1 dose;[27 29 32 34 38 40 44 47 50 56 61 71 72 75 82 86 89 98 105 107 110 113 114 117] 40% of the trials administered two doses;[23-26 28 30 35-37 41 42 49 51 53 54 58 62 64 65 67 73 74 76-80 83 84 90-93 95 100 101 104 109 111 112 115 121] 21% of the trials administered three doses;[20 21 31 33 39 45 46 48 52 55 59 63 70 81 85 87 88 94 97 99 106 116 118] 10% of the trials administered four doses;[43 57 60 68 69 108 120] two trials administered five

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doses;[102] [103] and one trial administered seven doses.[66] Two trials did not specify the number of doses administered.[96] [119]

Aluminium concentrations

The included trials used different aluminium concentrations ranging from 125 mcg/dose to 6000 mcg/dose.

Control groups

Two comparisons (from two trials) extracted in this review did not involve a vaccine (i.e. comparison between saline placebo with or without aluminium).[44] [114] All the other control groups contained the same vaccine as the intervention group but without aluminium adjuvant.

Risk of bias within individual trials

Based on the information collected from the published reports and from authors, only 3/102 trials were at overall low risk of bias (all outcomes reported at low risk of bias). The remaining trials were at overall high risk of bias (**Supplementary Figure 2**).

Effects of aluminium adjuvants

Serious adverse events

170/7627 (2.2%) participants who received aluminium adjuvants with or without vaccines suffered a serious adverse event versus 149/13 936 (1.1%) participants receiving no aluminium adjuvants with or without vaccines (RR 1.18, 95% CI 0.97 to 1.43; 21 563 participants; 62 trials; I^2 0%; Bayes factor 146.8; very low certainty of evidence. **Figure 1, Table 1**). Visual inspection of the forest plot and I^2 reveal no statistical heterogeneity. Trial Sequential Analysis showed that the cumulative Z-curve (blue full line with quadratic indicating of each trial)

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touches the traditional boundary for harm. However, none of the trial sequential monitoring boundaries (etched curves above and below the traditional horizontal lines for statistical significance) were surpassed. The result is inconclusive, and the required information size has not been achieved. The TSA-adjusted confidence interval is 0.53 to 2.69 (Pc (proportion with an outcome in the control group) 1.0%, RR reduction or increase (RRR) 20%, alpha 2.5%, beta 10%, diversity 0%; diversity-adjusted required information size (DARIS) 110 696 participants) (**Supplementary Figure 3**).

Subgroup analyses

Test for subgroup differences showed no difference when comparing the effects of aluminium adjuvants in trials at high risk of bias to trials at low risk of bias; in trials at risk of vested interest to trials at low risk of vested interest; trials according to different aluminium types; trials with different vaccines; trials with different participants' ages; trials with different follow up durations; and trials with participants with different diagnoses compared to healthy participants (**Supplementary Figures 4-10**).

Sensitivity analysis

21/7648 (0.3%) participants in the intervention group versus 18/13 954 (0.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'worst-best' case scenario (**Supplementary Figure 11**). The 'best-worst' case analysis showed that incomplete outcome data did not have the potential to influence the result (**Supplementary Figure 12**).

Meta-regression showed that the proportion of participants with serious adverse events was not affected by the aluminium concentration used in the vaccine (P = 0.28).

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Due to several trials with zero events, we performed meta-analysis also with OR, however, the results showed no difference between aluminium adjuvant compared with placebo or no intervention (**Supplementary Figure 13).**

Individual serious adverse events analyses

Meta-analyses showed no evidence of a difference between aluminium adjuvants versus control when assessing individual serious adverse events (**Supplementary Analysis 1).** Individual serious adverse events reported only in one trial that were not possible to meta-analyze are shown in **Supplementary Table 2.**

All-cause mortality

61/7782 (0.8%) aluminium participants died compared with 57/14 104 (0.4%) control participants (RR 1.02, 95% CI 0.74 to 1.41; 21 886 participants; 62 trials; I^2 0%; Bayes factor 2.35; very low certainty evidence. **Figure 2, Table 1**). Visual inspection of the forest plot and I^2 indicated no statistical heterogeneity. Funnel plot showed no publication bias. Trial Sequential Analysis showed that the accrued information for all-cause mortality was below 5% of the DARIS (Pc 0.4%, RRR 20%, alpha 2.5%, beta 10%, diversity 0%; DARIS 278 247).

Subgroup analyses

Test for subgroup differences showed no difference when comparing trials at high risk of bias to trials at low risk of bias; trials at risk of vested interest to trials at low risk of vested interest; trials with no vaccine cointervention to trials with vaccine cointervention; trials with different aluminium types; trials with different vaccines; trials with different participants' ages; and trials with different participants' diagnoses (**Supplementary Figures 14-19**). Due

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to lack of relevant data, it was not possible to conduct the subgroup analyses on trials with different follow up durations.

Sensitivity analysis

28/7909 (0.35%) participants in the aluminium group versus 20/14 173 (0.14%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'worst-best' case scenario showing a harmful effect of aluminium adjuvants compared with placebo (**Supplementary Figure 20**). The 'best-worst' case analysis showed that incomplete outcome data did not have the potential to influence the result (**Supplementary Figure 21**).

Meta-regression showed that the proportion of participants with all-cause mortality was not affected by the aluminium concentration used in the vaccine (P = 0.88).

Due to several trials with zero events, we performed meta-analysis also with OR, however, the results showed no difference between aluminium adjuvant compared with placebo or no intervention (**Supplementary Figure 22**).

Participants with disease

Only two trials (one event) reported on the proportion of participants that developed the disease they were vaccinated against (**Supplementary Figure 23**).

Non-serious adverse events

Out of the 67 trials reporting non-serious adverse events, 34 trials reported the overall proportion of participants with one or more non-serious adverse event. From the remaining 33/67 trials reporting on non-serious adverse events, we extracted data from the highest proportion of participants experiencing an individual adverse event.

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3760/7098 (52.9%) aluminium participants experienced one or more non-serious adverse event compared with 4537/13 429 (33.8%) control participants. Meta-analysis of these trials showed that aluminium adjuvants compared with placebo or no intervention may increase the proportion of participants with one or more adverse events considered non-serious (RR 1.13, 95% CI 1.07 to 1.20; participants = 20 527; trials = 67; I^2 = 85%; Bayes factor 2.64*1084; the evidence was very uncertain. **Figure 3, Table 1**). Trial Sequential Analysis of non-serious adverse events shows that the cumulative Z curve crosses the boundary for harm, indicating that there was enough information to confirm that aluminium adjuvants compared with placebo or no intervention increases the risk of one or more non-serious adverse events (TSA Pc 33.5%, RRR 20%, alpha 2.5%, beta 10%, diversity 78%; DARIS 28 384. TSA adjusted CI 1.06 to 1.23. **Supplementary Figure 24**).

Visual inspection of the funnel plot and regression-based Harbord test showed no publication bias or small-study effects (beta = 0.99).

Subgroup analyses

Test for subgroup differences was statistically significant in the subgroup analysis according to age (P = 0.007. **Supplementary Figure 25)** and according to vaccine type (P < 0.00001. **Supplementary Figure 26).**

Test for subgroup differences showed no difference when comparing the effect of aluminium adjuvants in trials at low risk of bias to trials at high risk of bias; in trials at low risk of vested interest to trials at risk of vested interest in trials with different aluminium salts; in trials with different follow-up durations; and in trials with participants with different health status **(Supplementary Figures 27-31**).

Sensitivity analysis

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195/7392 (2.6%) participants in the aluminium group versus 186/13 341 (1.4%) participants in the control group were lost to follow up. Incomplete outcome data did not have the potential to influence the result.

We included aluminium concentration (as described by trialists) as a covariate in meta-regression to assess whether aluminium concentration has an impact on the effect sizes of the proportion of participants with adverse events considered non-serious. Meta-regression showed that the proportion of participants with adverse events considered non-serious was not affected by the aluminium concentration used in the vaccine (P = 0.68)

Individual non-serious adverse events

We performed meta-analysis on each of the 145 reported individual adverse event considered non-serious. Mainly local injection site reactions were increased in the aluminium group (**Supplementary Analysis 2 and Supplementary Table 3)**.

Serological response

Geometric mean titers (GMT) and geometric mean concentrations (GMC)

Serological response was assessed by different analytical assays and was reported as either geometric mean titre (GMT, 31 trials) or geometric mean concentration (GMC, 11 trials).

Meta-analysis showed no or little evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing geometric mean titres or geometric mean concentrations (**Figure 4** and **Figure 5**).

Subgroup and sensitivity analysis for serology

Subgroup and sensitivity analysis for the serological response is reported in **Supplementary figure 32-35**).

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Seroprotection

Meta-analysis showed that there was no evidence of a difference between aluminium adjuvants compared with placebo or no intervention when assessing seroprotection (RR 0.95, 95% CI 0.77 to 1.18; trials = 27; I^2 = 78%. Bayes factor 36.1; low certainty of evidence **Figure 6**).

Visual inspection of the forest plot and I^2-statistics indicated high heterogeneity (I^2 = 78%).

Subgroup and sensitivity analysis for seroprotection

Subgroup and sensitivity analysis for seroprotection is reported in **Supplementary text and Figure 36 and 37**.

Discussion

This review included 102 randomised clinical trials including a total of 26 457 participants. Aluminium adjuvants versus placebo or no adjuvants may have no effect on the proportion of participants with one or more serious adverse events and on all-cause mortality, but the evidence was very uncertain. Two trials reported on the proportion of participants with the disease they were vaccinated against. However, only one event was reported. None of the trials reported on quality of life. Aluminium adjuvants versus placebo or no adjuvants may increase the proportion of participants with one or more adverse events considered non-serious, but the evidence was very uncertain. We found no or little evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing geometric mean titres or concentrations. Aluminium adjuvants versus placebo or no intervention may have no effect on participants without seroprotection, but the evidence was very uncertain.

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Strengths and weaknesses

We seem to be the first to conduct a systematic review comparing aluminium adjuvants versus placebo or no intervention in combination with vaccines. We followed our peer-reviewed protocol which was published before the literature search began,[10] and we conducted the review using the methods recommended by Cochrane.[12 126] We reported our review according to the PRISMA statement[127] (**Supplementary Table 4**).

Our systematic review has several limitations. Despite our inclusion criteria being very broad, we could only find phase I/II trials that met our inclusion criteria. This limitation is because phase 3/4 trials of marketed vaccines are mainly designed with an active comparator (another vaccine or alleged 'placebo' with aluminium), and therefore these trial designs do not match the inclusion criteria of this review.

Another limitation of the applicability of our results is that we chose maximum follow-up as our time point of primary interest. This approach does not allow us to make conclusions on the effect of aluminium adjuvants on safety and immunogenicity after each vaccine dose (for those trials having multiple vaccine injections).

Another limitation is that we identified high clinical heterogeneity, especially within the immunogenicity outcome. Included trials did not use the same assays to assess the serological response and most of the trials had multiple assays performed. We chose to analyse only the geometric mean titre or concentration reflecting the data from the first assay presented in the publication, however, we are aware that this might limit the strength of our findings.

We chose to merge multiple groups in those trials that used vaccines with different antigen concentration. In so doing, we were unable to conclude whether the effect of aluminium adjuvants is correlated to the effect of different antigen concentrations.

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Only two trials (one event in total) reported on the proportion of participants with disease they were vaccinated against and none of the authors provided us with such data when contacted by email. Therefore, our conclusion regarding the effect of aluminium adjuvants on the immunogenicity is based on the surrogate outcome of the serological response to vaccine measured by different assays and on the seroprotection values as defined by the trialists.

Of the 62 trials that reported on serious adverse events, 8 trials reported only vaccine-related serious adverse events.[23 25 55 75 79 82 83 90] Of the 62 trials that reported on serious adverse events, 14 trials reported that serious adverse events occurred but these were not assigned per intervention group because they were considered unrelated to the vaccine (a total of 107 serious adverse events reported to having occurred but not described per intervention group.[24 40-42 51 57 58 60 66 72 93 95 108 121] Only 7/102 authors contacted provided us with all or some of the data requested (see Characteristics of included studies in the **Appendix 1**).

Out of the 67 trials that reported on adverse events considered non-serious, 34 trials reported the overall proportion of participants with one or more adverse event considered non-serious.[28 30 32 38 40 43 44 46 51-53 56 57 59 61 65 67 70 72 77 78 80-82 85 88 92 98 100 107 111 113 120 124 128] From the remaining 33/67 trials reporting on adverse events considered non-serious, we extracted data from the highest proportion of participants experiencing an individual adverse event.[20 23-25 27 36 50 54 55 58 60 66 68 75 76 84 86 89 90 93-95 98 99 102 103 109 110 114-118] A substantial number of trials reported only solicited adverse events instead of solicited and unsolicited adverse events as a combined outcome. This limitation may have resulted in an underestimation of the unsolicited adverse events that might have occurred but were not reported.

Agreements and disagreements with other studies or reviews

In 2004, Jefferson and colleagues reviewed evidence of adverse events after exposure to aluminium-containing vaccines against diphtheria, tetanus, and pertussis, alone or in combination, compared with identical vaccines,

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either without aluminium or containing aluminium in different concentrations.[129] The authors included three randomised trials, four semi-randomised trials, and one cohort study. They found that in young children, vaccines with aluminium hydroxide caused significantly more erythema and induration than plain vaccines and significantly fewer reactions of all types. In older children there was an association with local pain lasting up to 14 days. Despite a lack of good-quality evidence, the authors recommend against any further research on this topic.

In 2018, Lin and colleagues conducted the first meta-analysis on the efficacy of aluminium salts as an adjuvant for pre-pandemic influenza vaccines.[130] They included a total of 9 randomised clinical trials (published during 2006 to 2013), including 22 comparisons in 2467 participants that compared aluminium-adjuvanted H5N1 vaccines versus non-adjuvanted counterparts.[130] Their results showed an inferior seroprotection after aluminium-adjuvanted H5N1 vaccines compared with that conferred by non-adjuvanted counterparts. Furthermore, H5N1 vaccines with aluminium adjuvants were associated with a significantly higher risk of pain/tenderness at the injection site during the seven days after the first vaccination and after the second dose versus the non-adjuvanted counterparts.

In 2020, Jørgensen and colleagues set out to assess the benefits and harms of the human papillomavirus (HPV) vaccines in clinical study reports obtained from the European Medicines Agency and GlaxoSmithKline from 2014 to 2017.[131] They included 24 randomised trials comparing an aluminium-adjuvanted HPV vaccine with a placebo or active comparator in healthy participants of all ages. They found that at 4 years follow-up, the HPV vaccines decreased HPV-related precursors to cervical cancer and treatment procedures but increased serious nervous system disorders (exploratory analysis) and general harms.131 As the trials included in their review were primarily designed to assess benefits and not adequately designed to assess harms, the extent to which the benefits outweigh the harms was unclear.

In agreement with Lin,[130] our systematic review does not find an increased serological response of aluminium-adjuvanted vaccines compared with that conferred by non-adjuvanted counterparts. Also, in agreement with both Jefferson[129] and Lin[130] we find an increase in local injection site reactions after administration of aluminium-adjuvanted vaccines.

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Implications for practice and research

Considering the lack of good quality evidence to assess beneficial and harmful effects of adding aluminium to vaccines as presented here, relevance of this adjuvant should be investigated in future studies. Questions on aluminium form, concentration, and size remain unanswered due to scarcity or lack of data. Questions on the effects of aluminium adjuvants on vaccine effectiveness also remain unanswered.

Future randomised clinical trials in humans should be conducted according to the ethical principles that have their origin in the Declaration of Helsinki, and that are consistent with the ICH Good Clinical Practice and the applicable regulatory requirement(s).[132 133] Such trials should be designed in accordance with guidelines for clinical trials (SPIRIT)[134] and reported in accordance with the Consolidated Standards of Reporting Trials (CONSORT).[135]

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Declarations of interest

SRK: no known conflict of interest

MB: no known conflict of interest

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NL: no known conflict of interest

DZK: no known conflict of interest

JCJ: no known conflict of interest

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Figures

Table 1. Summary of findings table

Figure 1. Meta-analysis of the effect of aluminium adjuvant compared with placebo or no intervention on the proportion of participants with one or more serious adverse events.

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Figure 2. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on all-cause mortality.

Figure 3. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the proportion of participants with one or more non-serious adverse events.

Figure 4. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean titres (GMT) grouped by analytical assay.

Figure 5. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean concentrations (GMC) grouped by analytical assay.

Figure 6. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on seroprotection.

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Summary of findings table

Aluminium adjuvants compared with placebo or no intervention in vaccines

Patient or population: any population

Setting: any settings

Intervention: aluminium adjuvants

Comparison: placebo or no intervention

Outcomes	**Anticipated absolute effects*** (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo or no intervention	**Risk with Aluminium adjuvants**				
Proportion of participants with one or more serious adverse events	11 per 1,000	**12 per 1,000** (9 to 15)	RR 1.10 (0.88 to 1.38)	21563 (58 RCTs)	⨁◯◯◯ VERY LOW [a,b,c]	Aluminium adjuvants versus placebo or no adjuvants may have no effect on the proportion of participants with one or more serious adverse events, but the evidence was very uncertain. Imprecision and indirectness were considered 'very serious' and therefore downgraded twice
All-cause mortality	4 per 1,000	**4 per 1,000** (3 to 6)	RR 1.02 (0.74 to 1.42)	21886 (60 RCTs)	⨁◯◯◯ VERY LOW [d,e]	Aluminium adjuvants versus placebo or no adjuvants may have no effect on all-cause mortality, but the evidence was very uncertain Imprecision was considered 'very serious' and therefore downgraded twice

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Proportion of participants with one or more adverse events considered non-serious	338 per 1,000	**385 per 1,000** (362 to 409)	RR 1.14 (1.07 to 1.21)	20527 (64 RCTs)	⊕◯◯◯ VERY LOW [f,g,h]	Aluminium adjuvants may increase the proportion of participants with one or more adverse events considered non-serious but the evidence is very uncertain. Indirectness was considered 'very serious' and therefore downgraded twice.
Participants without seroprotection	118 per 1,000	**112 per 1,000** (91 to 139)	RR 0.95 (0.77 to 1.18)	7845 (14 RCTs)	⊕◯◯◯ VERY LOW [i,j,k]	Aluminium adjuvants versus placebo or no intervention may have no effect on participants without seroprotection, but the evidence was very uncertain. Publication bias was considered 'strongly suspected' and therefore downgraded twice.
Health-related quality of life	0 per 1,000	**0 per 1,000** (0 to 0)	not estimable	(studies)	-	no data
Participants with disease being vaccinated against	0 per 1,000	**0 per 1,000** (0 to 0)	not estimable	221 (2 studies)	-	too few data

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio

GRADE Working Group grades of evidence
High certainty: We are very confident that the true effect lies close to that of the estimate of the effect
Moderate certainty: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low certainty: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect
Very low certainty: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

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Explanations

a. Downgraded for risk of bias. Overall: 89% of the trials were at high risk of bias for this outcome and 11% of the trials were at low risk of bias for this outcome. Specifically, 17% of the trials reporting on serious adverse events were at high risk of bias in blinding of outcome assessors and 54% of the trials reporting on serious adverse events had unclear risk of bias in blinding of outcome assessors. 17% of the trials reporting on serious adverse events were at high risk of bias in missing outcome data and 27% of the trials reporting on serious adverse events were at unclear risk of bias in missing outcome data. 4% of the trials reporting on serious adverse events were at high risk of bias in selective outcome reporting and 27% of the trials reporting on serious adverse events were at unclear risk of bias in selective outcome reporting.

b. Differences in outcomes measures: of the 61 trials that reported on serious adverse events, 8 trials reported only vaccine-related serious adverse events without giving information on the serious adverse events that potentially occurred but were not related to the vaccine. Other 14 trials reported that serious adverse events occurred but these were not described per intervention group because they were considered unrelated to the vaccine (a total of 107 serious adverse events reported to having occurred but not described per intervention group).

c. Downgraded for imprecision. Optimal information size (n = 110,696) not reached.

d. Downgraded for risk of bias. Overall: 79% of the trials were at high risk of bias for this outcome and 21% of the trials were at low risk of bias for this outcome. Specifically, 16% of the trials reporting on mortality had high risk of bias in blinding of outcome assessors and 43% of the trials reporting on mortality had unclear risk of bias in blinding of outcome assessors. 9% of the trials reporting on mortality had high risk of bias in missing outcome data and 22% of the trials reporting on mortality had unclear risk of bias in missing outcome data. 1% of the trials reporting on mortality had high risk of bias in selective outcome reporting and 21% of the trials reporting on mortality had unclear risk of bias in selective outcome reporting. However, the overall limitations were unlikely to influence this outcome.

e. Downgraded for imprecision. Optimal information size (n = 278,247) not reached.

f. Downgraded for risk of bias. Overall: 75% of the trials were at high risk of bias for this outcome and 25% of the trials were at low risk of bias for this outcome. Specifically, 14% of the trials reporting on non-serious adverse events were at high risk of bias in blinding of outcome assessors and 49% of the trials reporting on serious adverse events were at unclear risk of

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bias in blinding of outcome assessors. 13% of the trials reporting on non-serious adverse events were at high risk of bias in missing outcome data and 9% of the trials reporting on non-serious adverse events were at unclear risk of bias in missing outcome data. 1% of the trials reporting on non-serious adverse events were at high risk of bias in selective outcome reporting and 26% of the trials reporting on serious adverse events were at unclear risk of bias in selective outcome reporting.

g. Downgraded for inconsistency. I^2 = 85%. Visual inspection of funnel plot may suggest potential publication bias for smaller trials reporting a harmful effect of aluminium adjuvants in the placebo group. Regression-based Harbord test showed no small-study effects (beta = 0.99)

h. Downgraded for indirectness. Differences in outcomes measures: a number of studies report only number of solicited or unsolicited adverse events, rather than total number of participants experiencing any adverse events. Out of the 66 trials reporting non-serious adverse events, only 34 trials reported the overall proportion of participants with one or more non-serious adverse event. From the remaining 32 out of the 66 trials reporting on non-serious adverse events we extracted data from the highest proportion of participants experiencing an individual adverse event.

i. Overall: 100% of the trials were at high risk of bias for this outcome. Specifically, 0% of the trials reporting on seroprotection were at high risk of bias in blinding of outcome assessors and 68% of the trials reporting on serious adverse events were at unclear risk of bias in blinding of outcome assessors. 74% of the trials reporting on seroprotection were at high risk of bias in missing outcome data and 0% of the trials reporting on serious adverse events were at unclear risk of bias in missing outcome data. 0% of the trials reporting on seroprotection were at high risk of bias in selective outcome reporting and 23% of the trials reporting on serious adverse events were at unclear risk of bias in selective outcome reporting.

j. Downgraded for inconsistency. I^2 = 78%.

k. Visual inspection of funnel plot may suggest potential publication bias for smaller trials reporting an harmful effect of aluminium adjuvants in the intervention group. Regression-based Harbord test showed no small-study effects (beta = 0.99).

Table 1. Summary of findings table.

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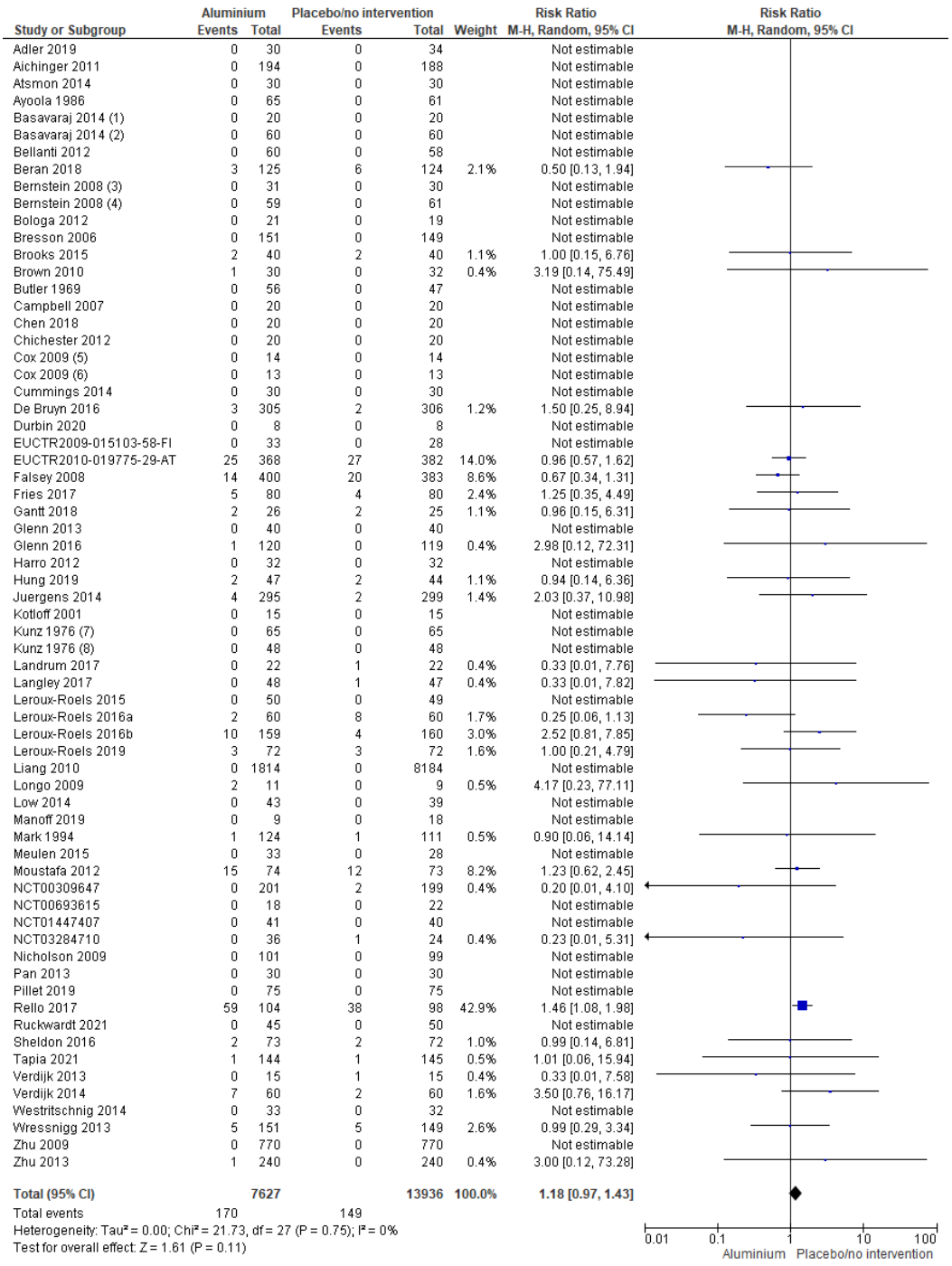
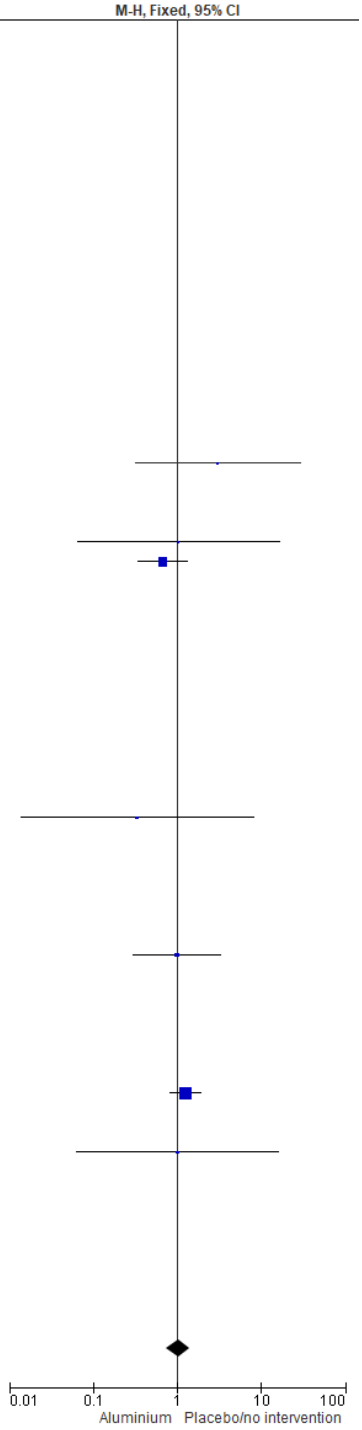


Figure 1. Meta-analysis of the effect of aluminium adjuvant compared with placebo or no intervention on the proportion of participants with one or more serious adverse events.

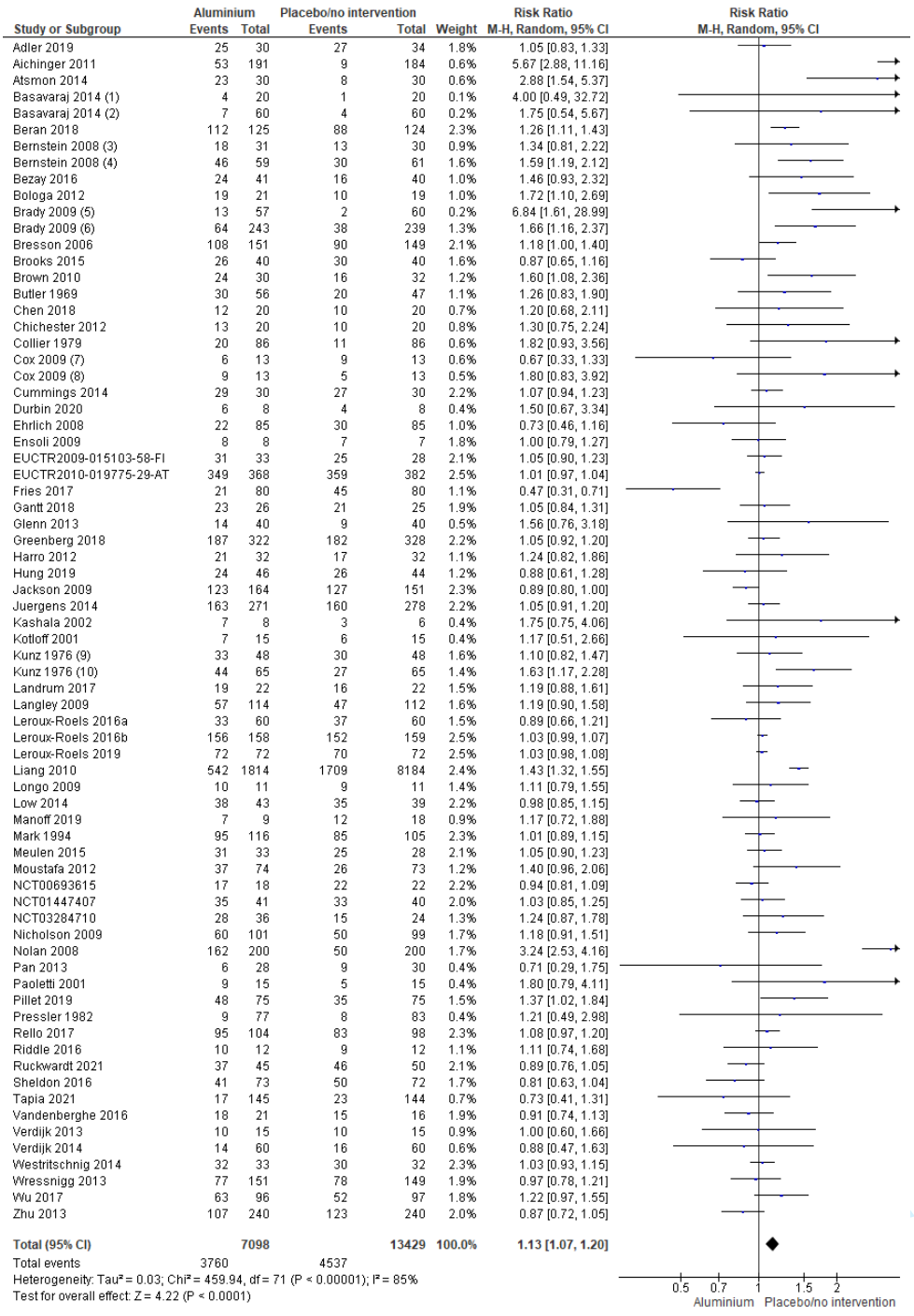
Study or Subgroup	Aluminium Events	Total	Placebo/no intervention Events	Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Adler 2019	0	30	0	34		Not estimable
Aichinger 2011	0	194	0	188		Not estimable
Atsmon 2014	0	30	0	30		Not estimable
Ayoola 1986	0	65	0	61		Not estimable
Basavaraj 2014 (1)	0	20	0	20		Not estimable
Basavaraj 2014 (2)	0	60	0	60		Not estimable
Bellanti 2012	0	60	0	58		Not estimable
Beran 2018	0	125	0	124		Not estimable
Bernstein 2008 (3)	0	31	0	30		Not estimable
Bernstein 2008 (4)	0	59	0	61		Not estimable
Bezay 2016	0	41	0	40		Not estimable
Bologa 2012	0	21	0	19		Not estimable
Bresson 2006	0	151	0	149		Not estimable
Brooks 2015	0	40	0	40		Not estimable
Brown 2010	0	30	0	32		Not estimable
Butler 1969	0	56	0	47		Not estimable
Campbell 2007	0	20	0	20		Not estimable
Chen 2018	0	20	0	20		Not estimable
Chichester 2012	0	20	0	20		Not estimable
Cox 2009 (5)	0	14	0	14		Not estimable
Cox 2009 (6)	0	13	0	13		Not estimable
Cummings 2014	0	28	0	30		Not estimable
De Bruyn 2016	3	305	1	306	1.7%	3.01 [0.31, 28.77]
Durbin 2020	0	8	0	8		Not estimable
Ehrlich 2008	0	91	0	90		Not estimable
EUCTR2009-015103-58-FI	0	33	0	28		Not estimable
EUCTR2010-019775-29-AT	1	368	1	382	1.7%	1.04 [0.07, 16.53]
Falsey 2008	14	400	20	383	34.8%	0.67 [0.34, 1.31]
Fries 2017	0	80	0	80		Not estimable
Gantt 2018	0	26	0	25		Not estimable
Glenn 2013	0	40	0	40		Not estimable
Glenn 2016	0	120	0	119		Not estimable
Harro 2012	0	32	0	32		Not estimable
Hung 2019	0	47	0	44		Not estimable
Juergens 2014	0	295	0	299		Not estimable
Keitel 2008	0	61	0	59		Not estimable
Keitel 2008	0	302	0	295		Not estimable
Kotloff 2001	0	15	0	15		Not estimable
Langley 2017	0	48	0	47		Not estimable
Leroux-Roels 2015	0	50	0	49		Not estimable
Leroux-Roels 2016a	0	60	1	60	2.6%	0.33 [0.01, 8.02]
Leroux-Roels 2016b	0	159	0	160		Not estimable
Leroux-Roels 2019	0	72	0	72		Not estimable
Liang 2010	0	1814	0	8184		Not estimable
Low 2014	0	43	0	39		Not estimable
Manoff 2019	0	9	0	18		Not estimable
Meulen 2015	0	33	0	28		Not estimable
Moustafa 2012	5	74	5	73	8.6%	0.99 [0.30, 3.26]
NCT00693615	0	18	0	22		Not estimable
NCT01447407	0	41	0	40		Not estimable
NCT03284710	0	36	0	24		Not estimable
Nicholson 2009	0	101	0	99		Not estimable
Pan 2013	0	28	0	30		Not estimable
Pillet 2019	0	75	0	75		Not estimable
Rello 2017	37	104	28	98	49.1%	1.25 [0.83, 1.87]
Ruckwardt 2021	0	45	0	50		Not estimable
Sheldon 2016	0	73	0	72		Not estimable
Tapia 2021	1	144	1	145	1.7%	1.01 [0.06, 15.94]
Verdijk 2013	0	15	0	15		Not estimable
Verdijk 2014	0	60	0	60		Not estimable
Vernacchio 2002	0	38	0	40		Not estimable
Westritschnig 2014	0	30	0	30		Not estimable
Wressnigg 2013	0	151	0	149		Not estimable
Wu 2017	0	100	0	100		Not estimable
Zhu 2009	0	770	0	770		Not estimable
Zhu 2013	0	240	0	240		Not estimable
Total (95% CI)		**7782**		**14104**	**100.0%**	**1.02 [0.74, 1.41]**
Total events	61		57			

Heterogeneity: Chi² = 3.80, df = 6 (P = 0.70); I² = 0%
Test for overall effect: Z = 0.13 (P = 0.89)



Footnotes
(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 15 mcg of antigen
(4) 30 mcg of antigen
(5) 12 mcg antigen
(6) 24 mcg antigen

Figure 2. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on all-cause mortality.



Footnotes
(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 15 mcg of antigen
(4) 30 mcg of antigen
(5) 300 mcg/dose aluminium
(6) 600 mcg/dose aluminium
(7) 12 mcg antigen
(8) 24 mcg antigen
(9) subunit vaccine
(10) whole virus vaccine

Figure 3. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the proportion of participants with one or more non-serious adverse events.

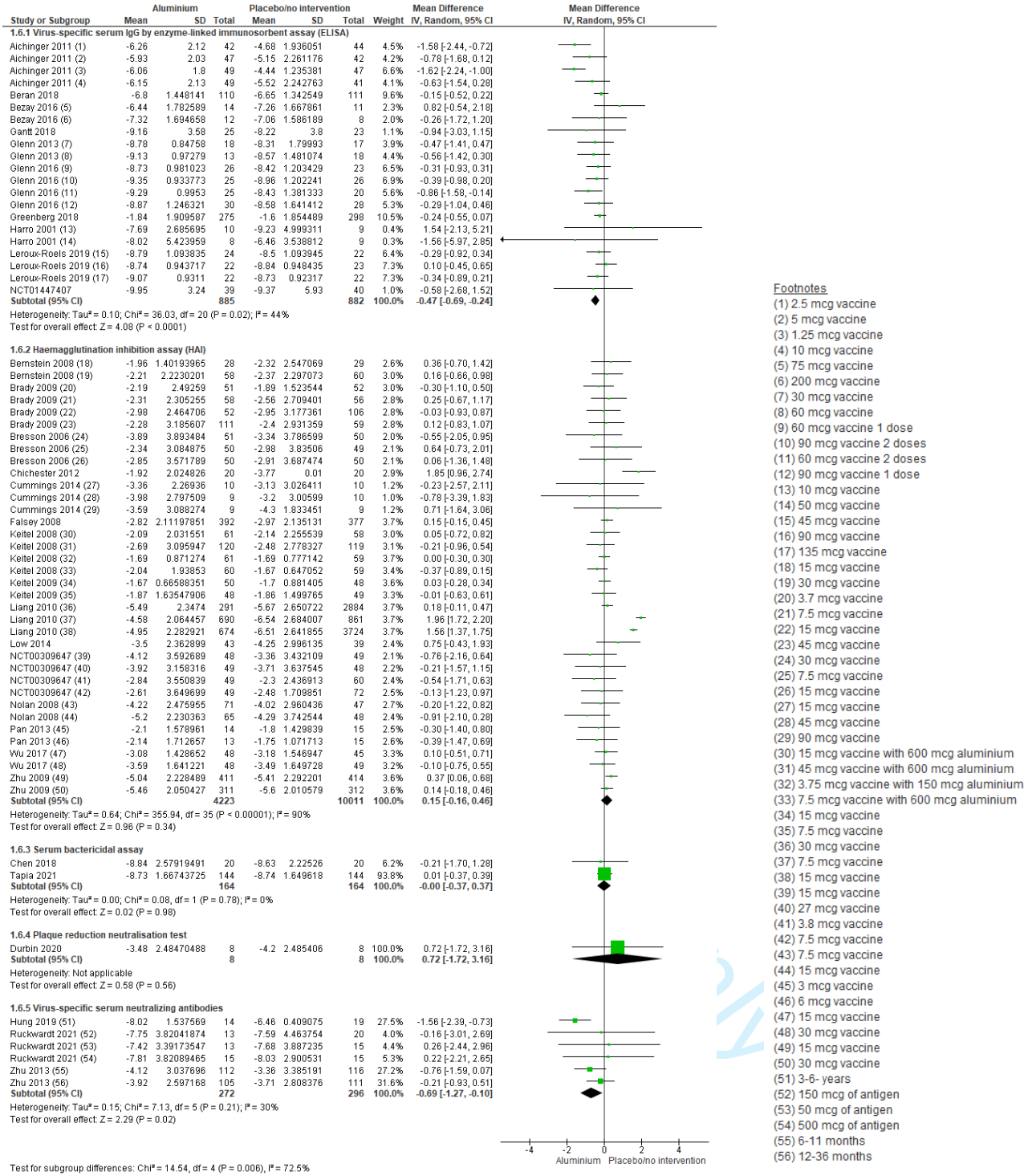


Figure 4. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean titres (GMT) grouped by analytical assay.

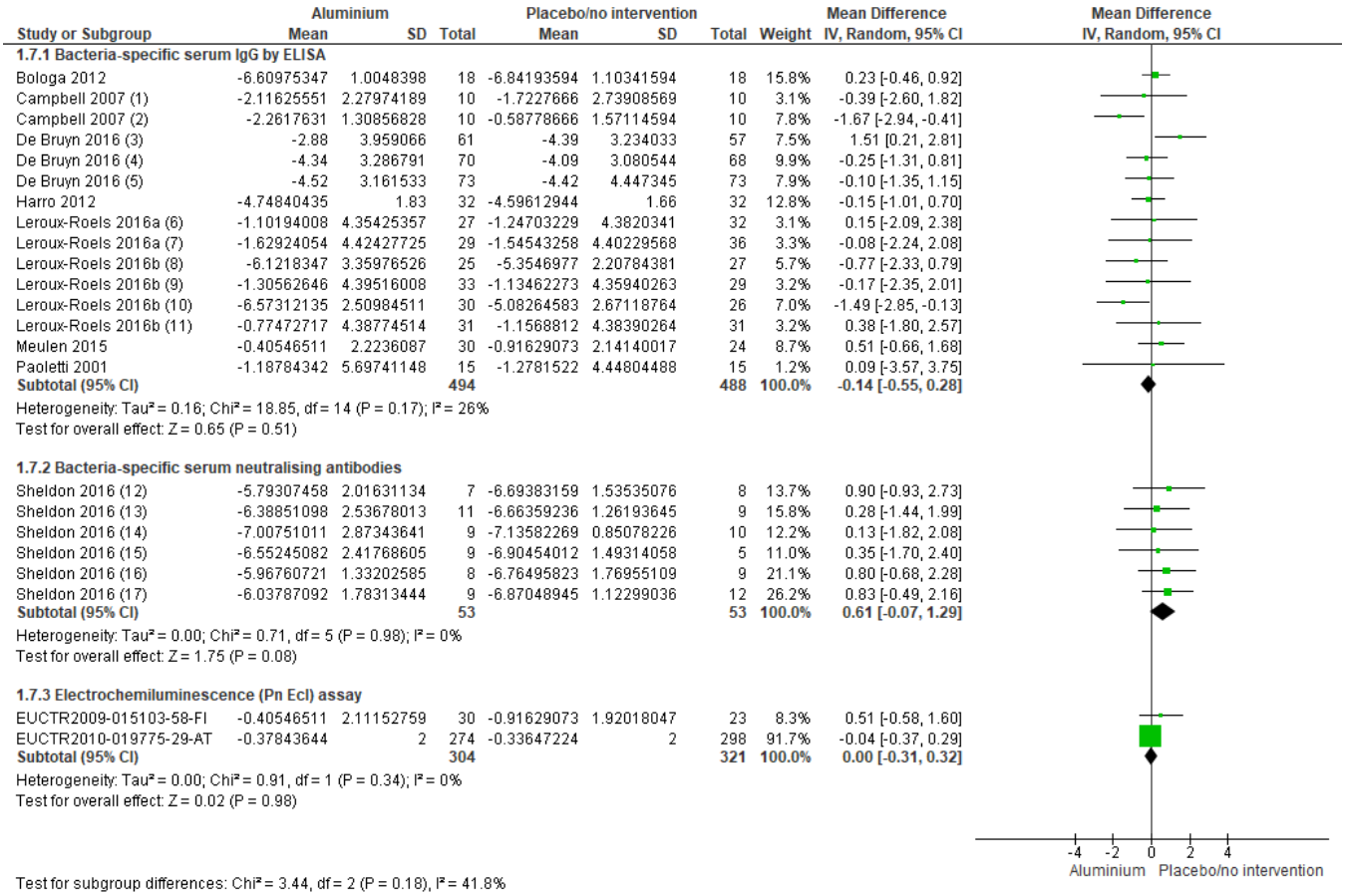
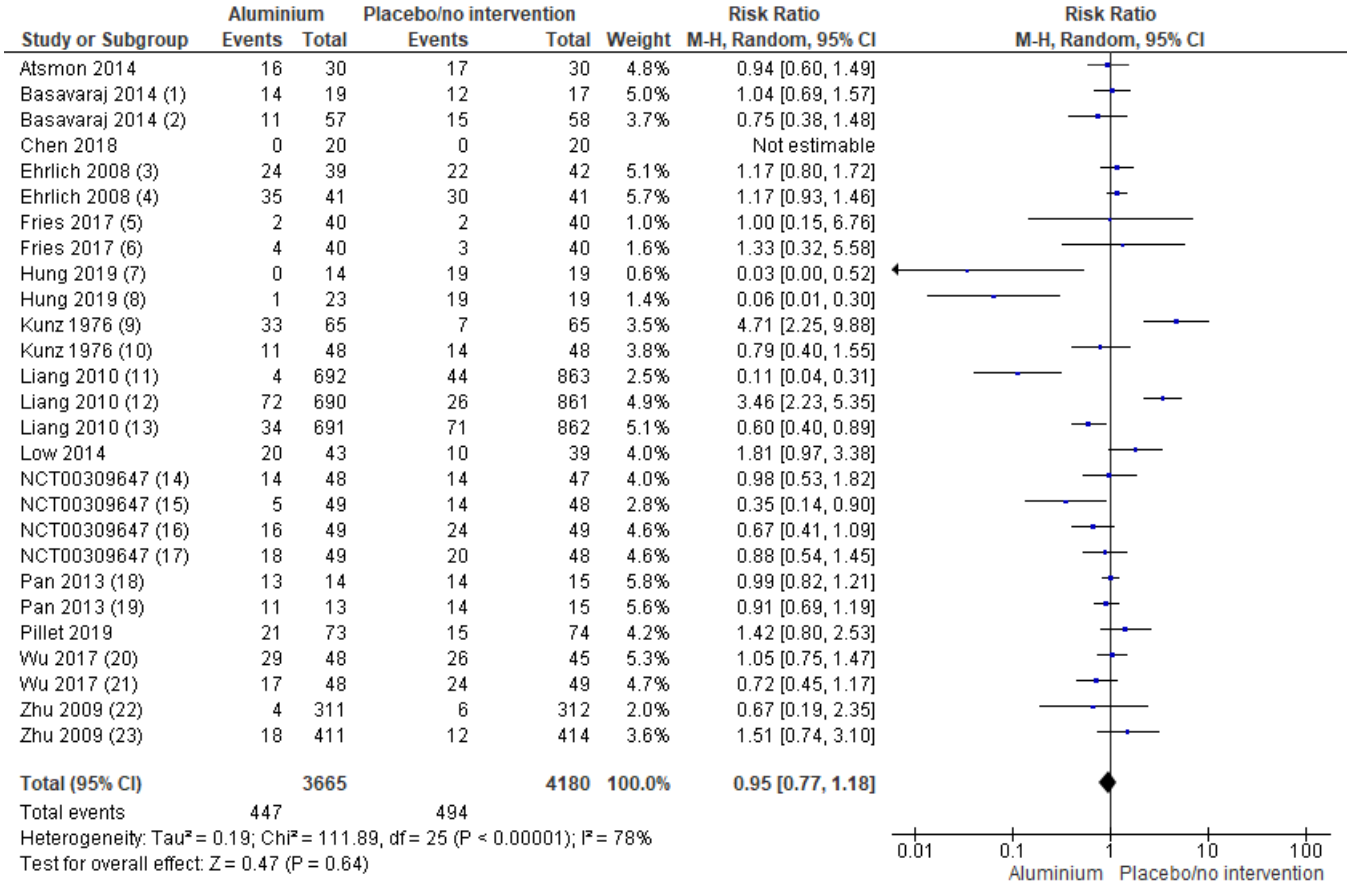


Figure 5. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean concentrations (GMC) grouped by analytical assay.



Study or Subgroup	Aluminium Events	Aluminium Total	Placebo/no intervention Events	Placebo/no intervention Total	Weight	Risk Ratio M-H, Random, 95% CI
Atsmon 2014	16	30	17	30	4.8%	0.94 [0.60, 1.49]
Basavaraj 2014 (1)	14	19	12	17	5.0%	1.04 [0.69, 1.57]
Basavaraj 2014 (2)	11	57	15	58	3.7%	0.75 [0.38, 1.48]
Chen 2018	0	20	0	20		Not estimable
Ehrlich 2008 (3)	24	39	22	42	5.1%	1.17 [0.80, 1.72]
Ehrlich 2008 (4)	35	41	30	41	5.7%	1.17 [0.93, 1.46]
Fries 2017 (5)	2	40	2	40	1.0%	1.00 [0.15, 6.76]
Fries 2017 (6)	4	40	3	40	1.6%	1.33 [0.32, 5.58]
Hung 2019 (7)	0	14	19	19	0.6%	0.03 [0.00, 0.52]
Hung 2019 (8)	1	23	19	19	1.4%	0.06 [0.01, 0.30]
Kunz 1976 (9)	33	65	7	65	3.5%	4.71 [2.25, 9.88]
Kunz 1976 (10)	11	48	14	48	3.8%	0.79 [0.40, 1.55]
Liang 2010 (11)	4	692	44	863	2.5%	0.11 [0.04, 0.31]
Liang 2010 (12)	72	690	26	861	4.9%	3.46 [2.23, 5.35]
Liang 2010 (13)	34	691	71	862	5.1%	0.60 [0.40, 0.89]
Low 2014	20	43	10	39	4.0%	1.81 [0.97, 3.38]
NCT00309647 (14)	14	48	14	47	4.0%	0.98 [0.53, 1.82]
NCT00309647 (15)	5	49	14	48	2.8%	0.35 [0.14, 0.90]
NCT00309647 (16)	16	49	24	49	4.6%	0.67 [0.41, 1.09]
NCT00309647 (17)	18	49	20	48	4.6%	0.88 [0.54, 1.45]
Pan 2013 (18)	13	14	14	15	5.8%	0.99 [0.82, 1.21]
Pan 2013 (19)	11	13	14	15	5.6%	0.91 [0.69, 1.19]
Pillet 2019	21	73	15	74	4.2%	1.42 [0.80, 2.53]
Wu 2017 (20)	29	48	26	45	5.3%	1.05 [0.75, 1.47]
Wu 2017 (21)	17	48	24	49	4.7%	0.72 [0.45, 1.17]
Zhu 2009 (22)	4	311	6	312	2.0%	0.67 [0.19, 2.35]
Zhu 2009 (23)	18	411	12	414	3.6%	1.51 [0.74, 3.10]
Total (95% CI)		**3665**		**4180**	**100.0%**	**0.95 [0.77, 1.18]**
Total events	447		494			

Heterogeneity: Tau² = 0.19; Chi² = 111.89, df = 25 (P < 0.00001); I² = 78%
Test for overall effect: Z = 0.47 (P = 0.64)

Footnotes
(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 7.5 mcg vaccine
(4) 15 mcg vaccine
(5) 60 mcg vaccine
(6) 90 mcg vaccine
(7) 3-6 years of age
(8) 2-35 months of age
(9) subunit vaccine
(10) whole virus vaccine
(11) 30 mcg vaccine
(12) 7.5 mcg vaccine
(13) 15 mcg vaccine
(14) 3.8 mcg vaccine
(15) 7.5 mcg vaccine
(16) 15 mcg vaccine
(17) 27 mcg vaccine
(18) 3 mcg vaccine
(19) 6 mcg vaccine
(20) 15 mcg vaccine
(21) 30 mcg vaccine
(22) 30 mcg vaccine
(23) 15 mcg vaccine

Figure 6. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on seroprotection.

Supplementary material

Aluminium adjuvants versus placebo or no intervention in vaccine development randomised clinical trials. A systematic review with meta-analysis and Trial Sequential Analysis

Sara Russo Krauss (0000-0002-5761-2118), research associate[1], Marija Barbateskovic, clinical research consultant [1], Sarah Louise Klingenberg, information specialist[1], Snezana Djurisic, clinical research consultant[1], Sesilje B Petersen, scientist,[2], Mette Kenfelt, chief executive officer,[3] De Zhao Kong, deputy chief physician and head of department[4,5], Janus C Jakobsen, professor and chief physician[1,6], Christian Gluud (0000-0002-8861-0799), professor and head of department[1,6]

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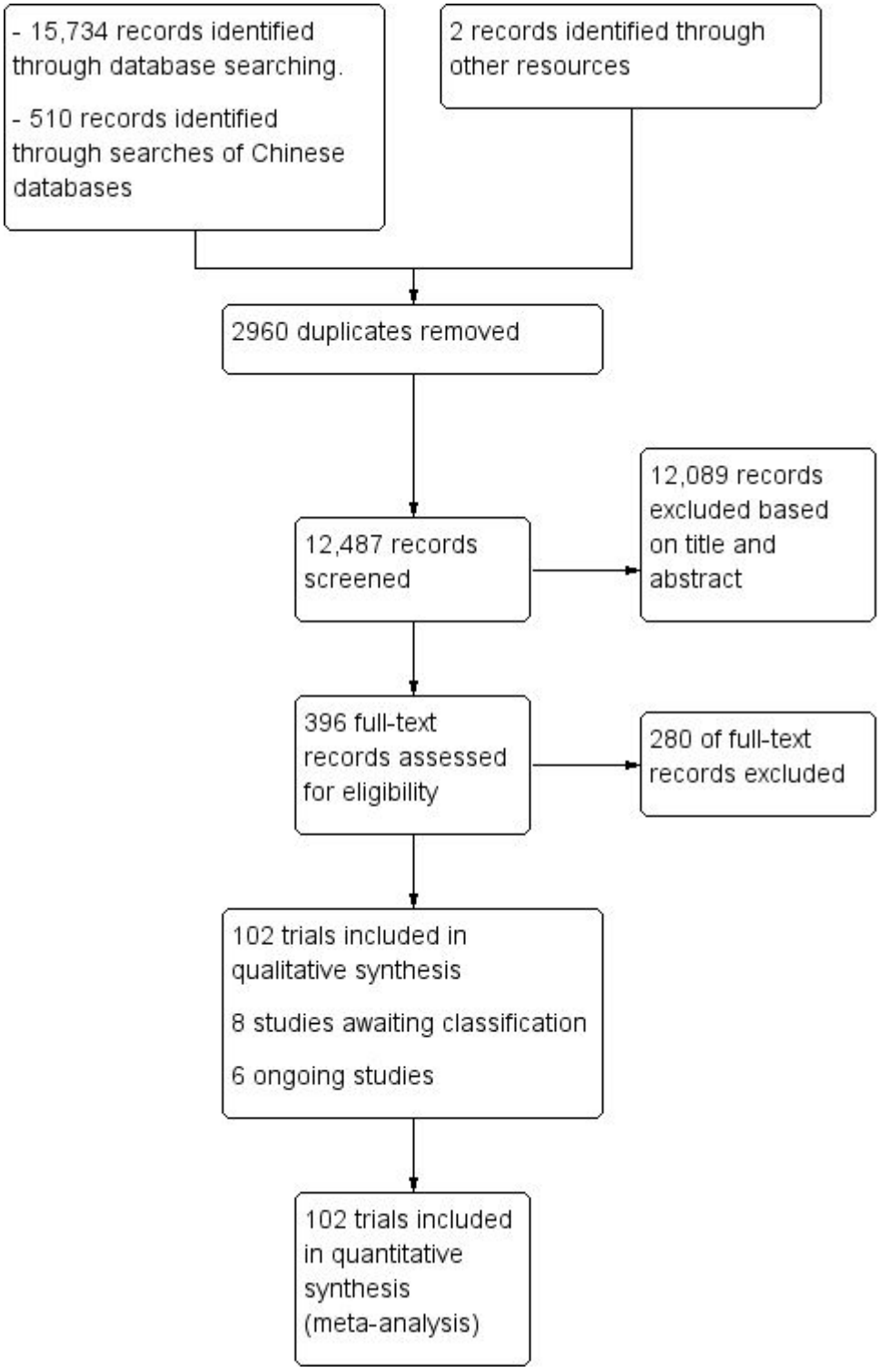
Database	Time span	Search strategy
Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library	2020, Issue 11	#1 MeSH descriptor: [Adjuvants, Immunologic] explode all trees#2 (immunologic* or alum* or vaccine*) and adjuvan*#3 #1 or #2#4 MeSH descriptor: [Vaccines] explode all trees#5 vaccin*#6 #4 or #5#7 #3 and #6
MEDLINE Ovid	1946 to July 2021	1. exp Adjuvants, Immunologic/2. ((immunologic* or alum* or vaccine*) and adjuvan*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]3. 1 or 24. exp Vaccines/5. vaccin*.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]6. 4 or 57. 3 and 68. (randomized controlled trial or controlled clinical trial).pt. or clinical trials as topic.sh. or trial.ti.9. (random* or blind* or placebo* or meta-analys*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]10. 7 and (8 or 9)11. limit 10 to humans
Embase Ovid	1974 to July 2021	1. exp immunological adjuvant/2. ((immunologic* or alum* or vaccine*) and adjuvan*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word]3. 1 or 24. exp vaccine/5. vaccin*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word]6. 4 or 57. 3 and 68. Randomized controlled trial/ or Controlled clinical study/ or trial.ti.9. (random* or blind* or placebo* or meta-analys*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word]10. 7 and (8 or 9)11. limit 10 to human
LILACS (Bireme)	1982 to to July 2021	(immunologic$ or alum$ or vaccine$) and adjuvan$ [Words] and vaccin$ [Words]

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BIOSIS (Web of Science)	1969 to July 2021	#5 #4 AND #3#4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*)#3 #2 AND #1#2 TS=vaccin*#1 TS=((immunologic* or alum* or vaccine*) and adjuvan*)
Science Citation Index Expanded (Web of Science)	1900 to July 2021	#5 #4 AND #3#4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*)#3 #2 AND #1#2 TS=vaccin*#1 TS=((immunologic* or alum* or vaccine*) and adjuvan*)
Conference Proceedings Citation Index – Science (Web of Science)	1990 to July 2021	#5 #4 AND #3#4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*)#3 #2 AND #1#2 TS=vaccin*#1 TS=((immunologic* or alum* or vaccine*) and adjuvan*)
CNKI	September, 2021	#1 **主**题检索：铝佐剂、硫酸铝、氢氧化铝、磷酸铝、羟基磷酸铝、铵明矾、钾明矾、硫酸铝铵、硫酸铝钾 #2 **主**题检索：疫苗 #3 **全文**检索：随机 #4 #1 AND #2 AND #3
Wanfang	September, 2021	**主**题:(铝佐剂 + **硫酸**铝 + 氢氧化铝 + **磷酸**铝 + 羟基磷酸铝 + 铵明矾 + 钾明矾 + **硫酸**铝铵 + **硫酸**铝钾)*主题:(**疫苗**)*全部:(**随机**)
VIP	September, 2021	**任意字段**：铝佐剂 OR **硫酸**铝 OR 氢氧化铝 OR **磷酸**铝 OR 羟基磷酸铝 OR 铵明矾 OR 钾明矾 OR **硫酸**铝铵 OR **硫酸**铝钾 **任意字段**：**疫苗** **注**：**加上随机之后**变成0
Sinomed	September, 2021	("**随机**"[**全部字段**:**智能**]) AND ("**疫苗**"[**全部字段**:**智能**]) AND ("铝佐剂"[**全部字段**:**智能**] OR "**硫酸**铝"[**全部字段**:**智能**] OR "氢氧化铝"[**全部字段**:**智能**] OR "**磷酸**铝"[**全部字段**:**智能**] OR "羟基磷酸铝"[**全部字段**:**智能**] OR "铵明矾"[**全部字段**:**智能**] OR "钾明矾"[**全部字段**:**智能**] OR "**硫酸**铝铵"[**全部字段**:**智能**] OR "**硫酸**铝钾"[**全部字段**:**智能**])

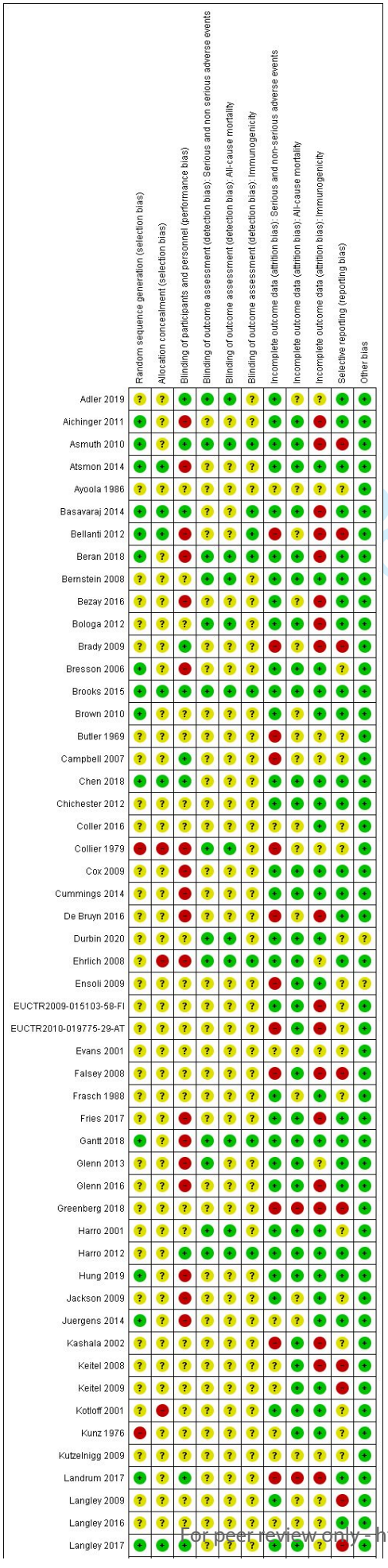
Supplementary Table 1. Search strategies

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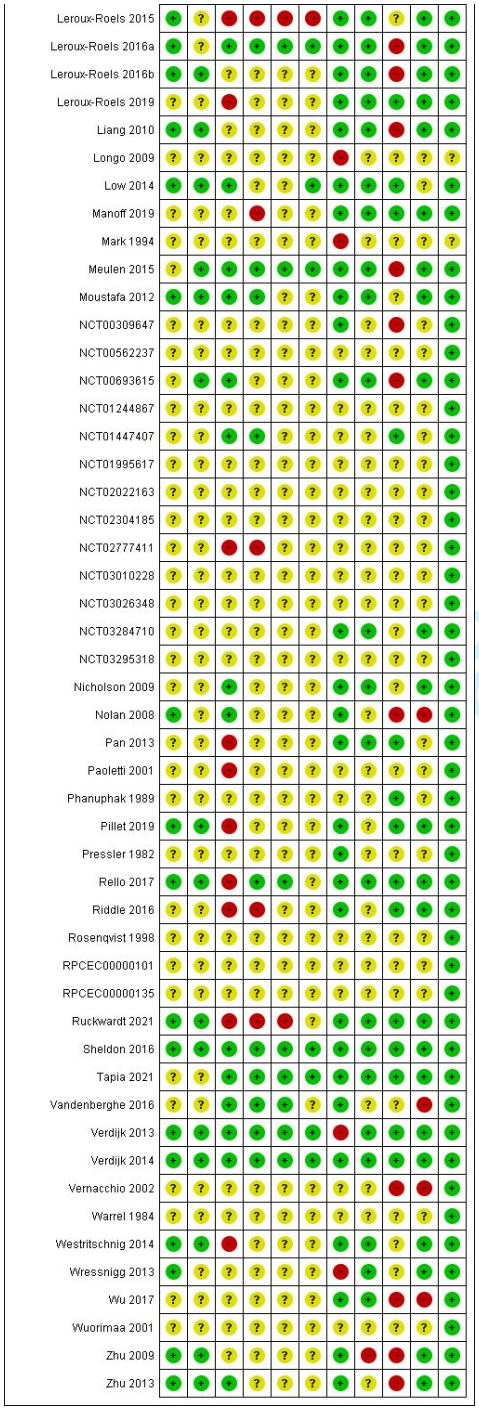
Supplementary Figure 1. Study flow diagram. Date of search June 29, 2021.

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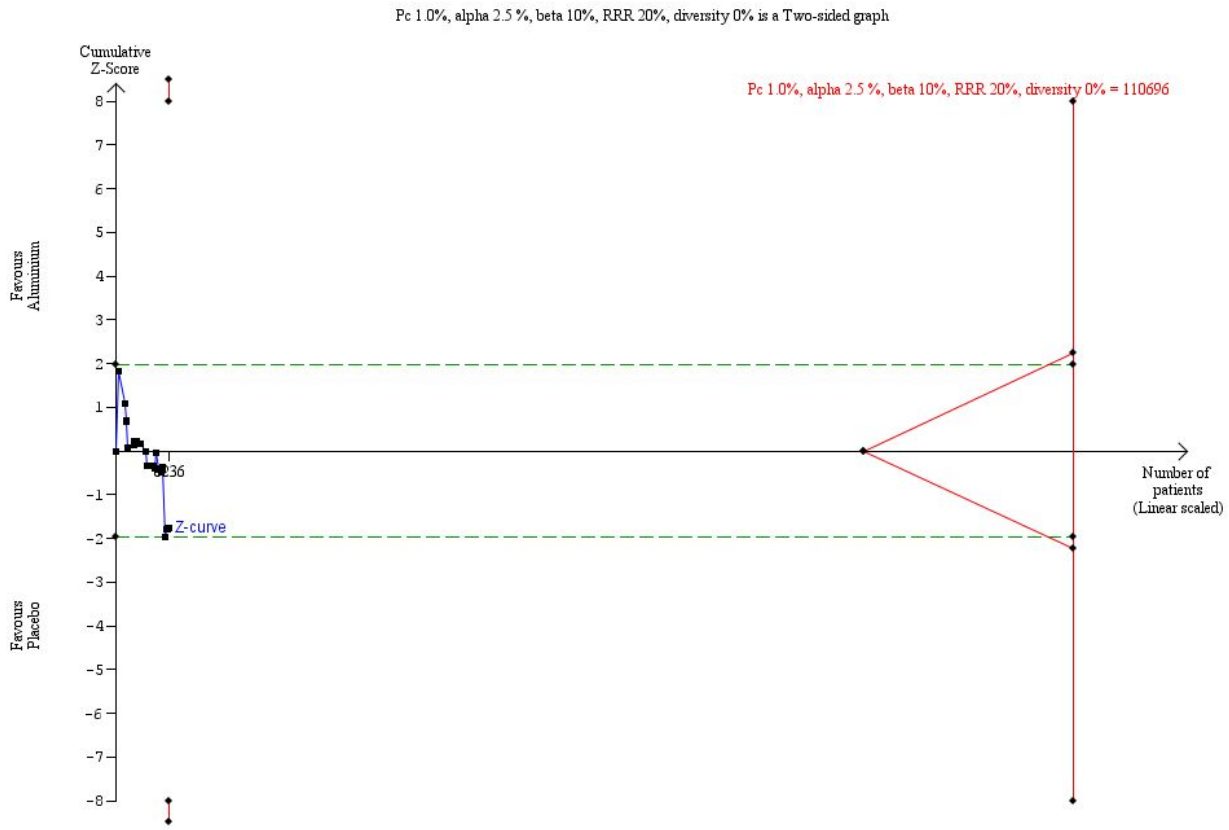
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Supplementary Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included trial.

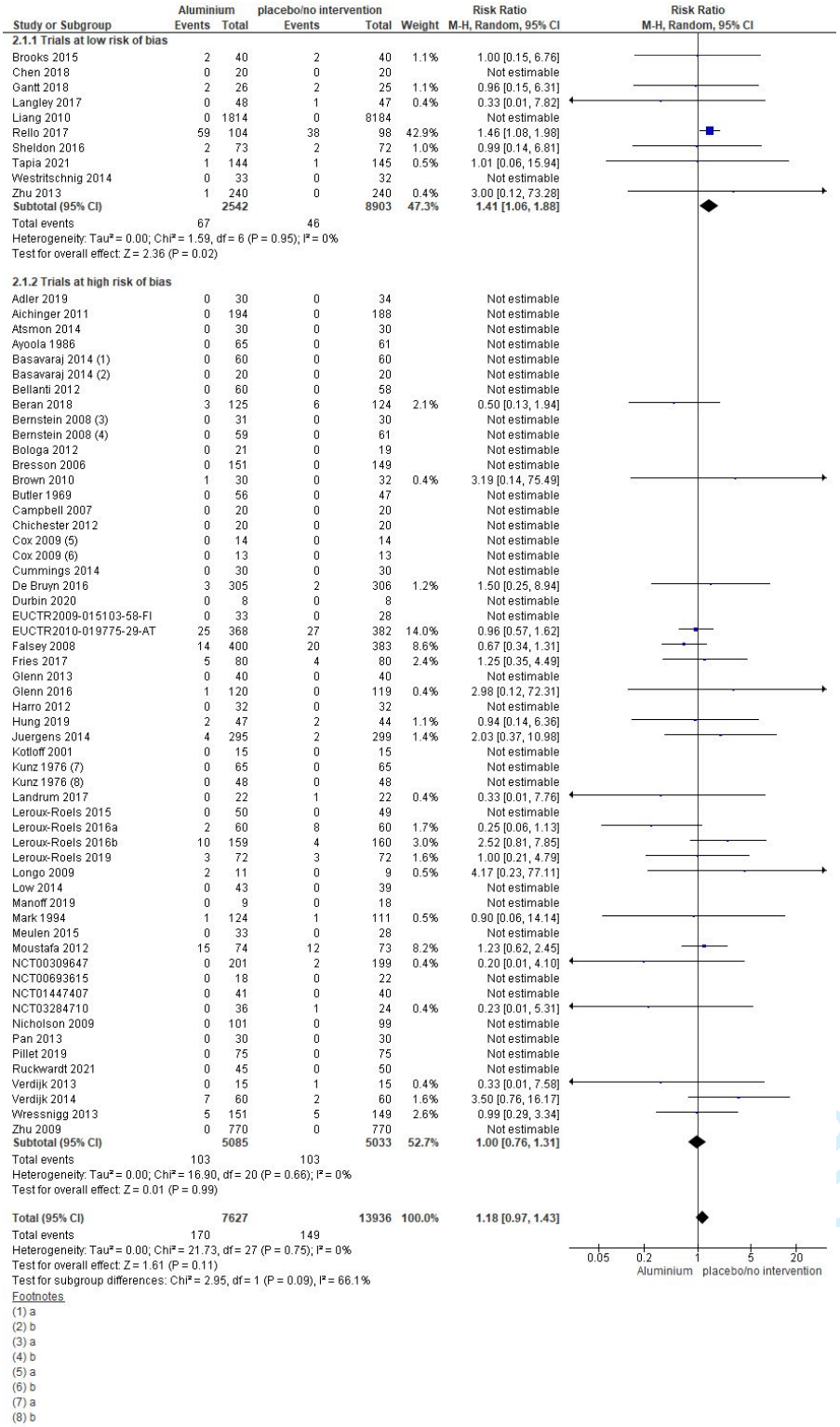
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Supplementary Figure 3. Trial Sequential Analysis using random-effects meta-analysis of the effects of aluminium adjuvants versus placebo or no intervention on the proportion of participants with serious adverse events. The analysis was based on a proportion in the control group (Pc) of 1.0 %, a relative risk reduction (RRR) of 20%, an alpha of 2.5%, a beta of 10%, and diversity of 0%. None of the trial sequential monitoring boundaries have been surpassed.

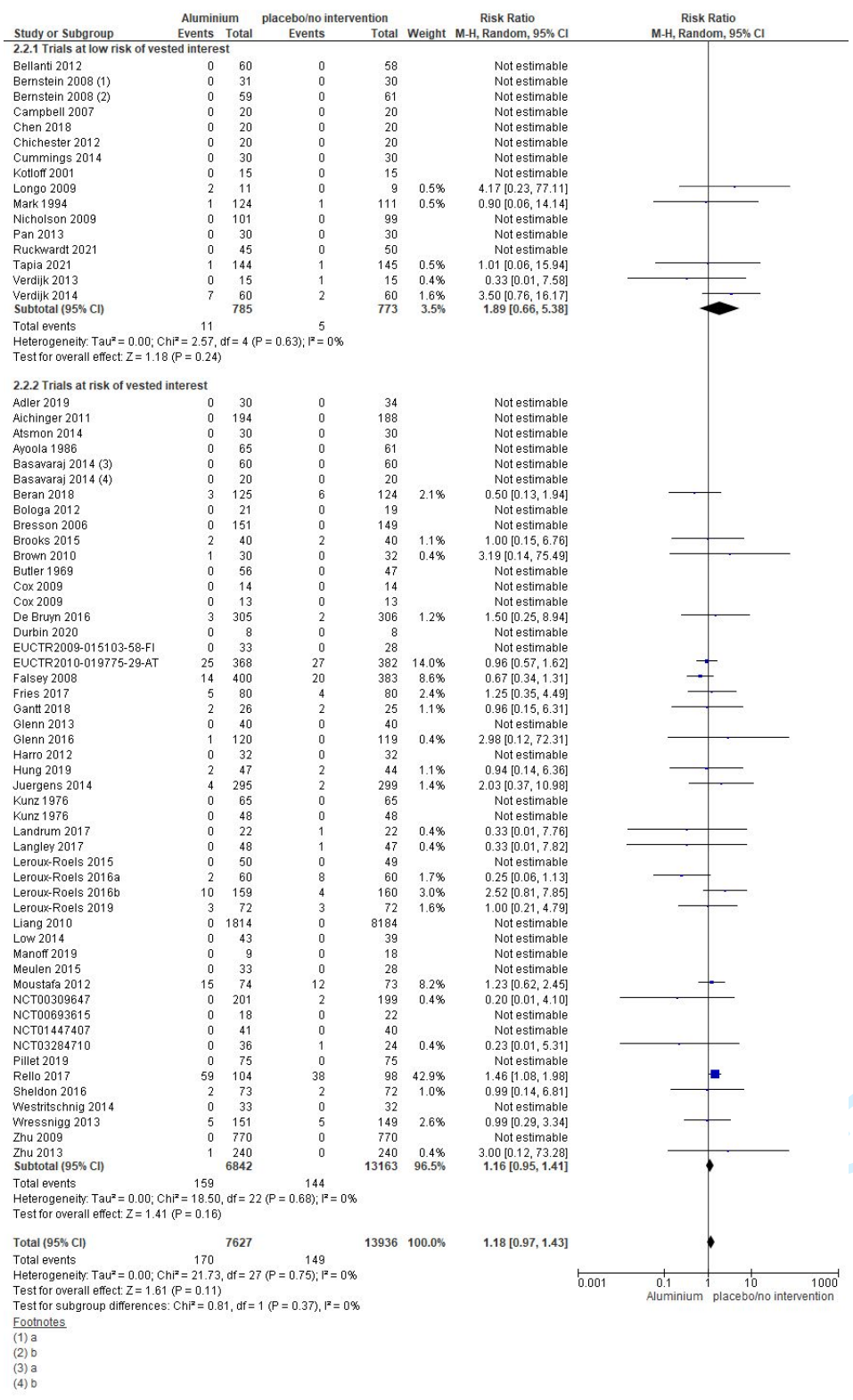
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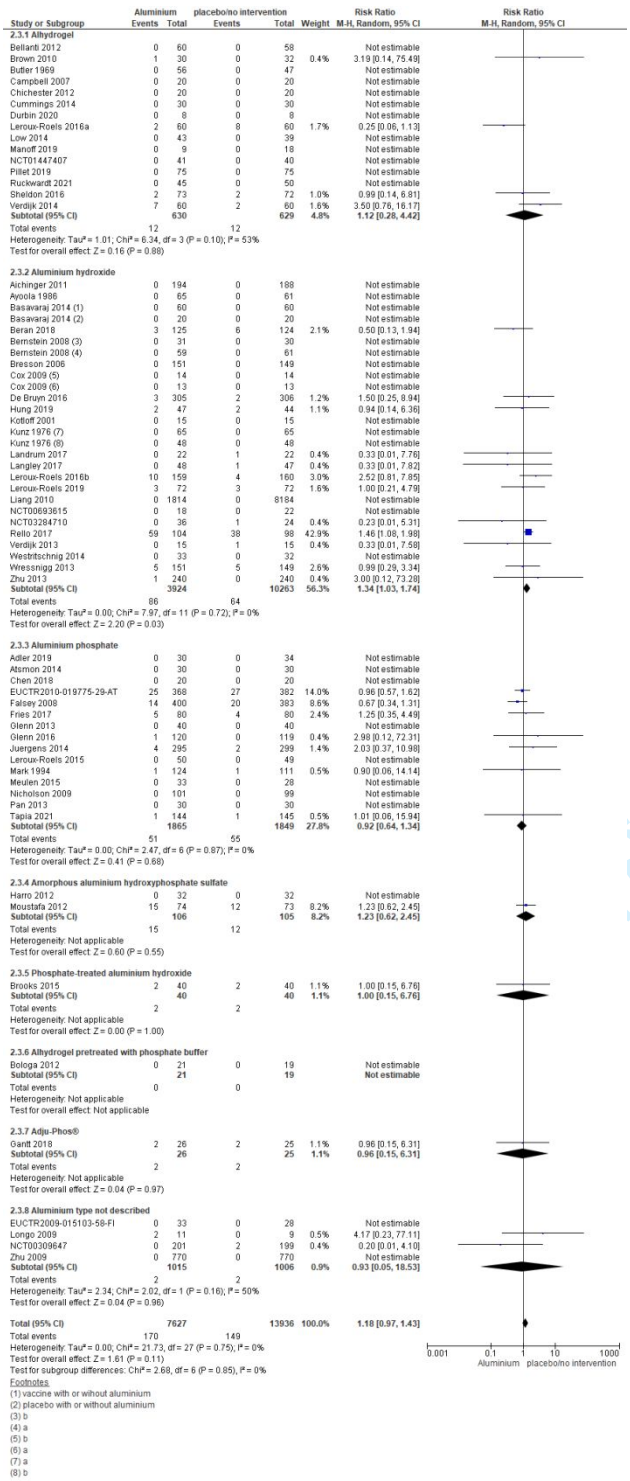
Supplementary Figure 4. Subgroup analysis for serious adverse events according to risk of bias.

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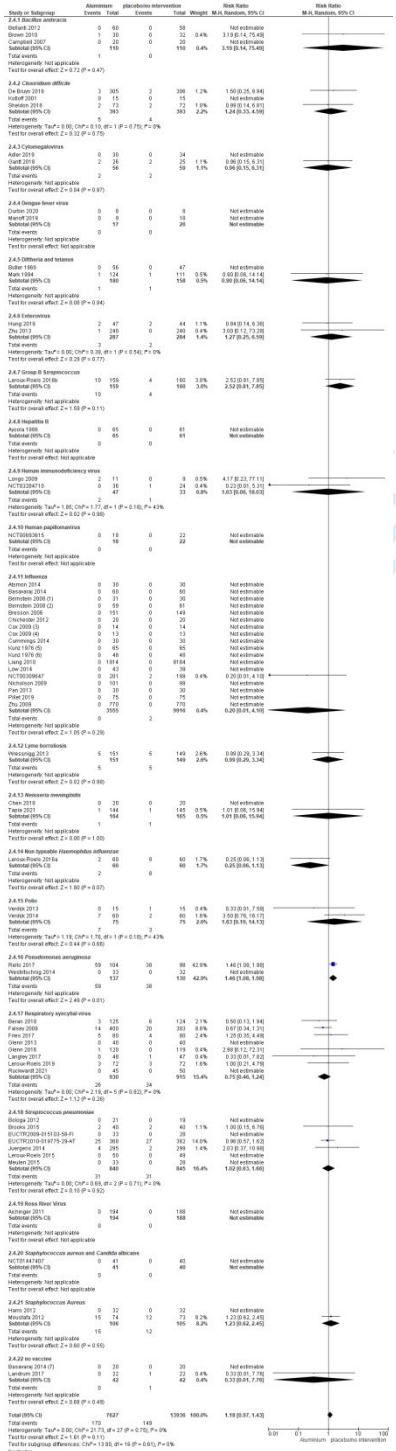
Supplementary Figure 5. Subgroup analysis of serious adverse events according to risk of vested interested.

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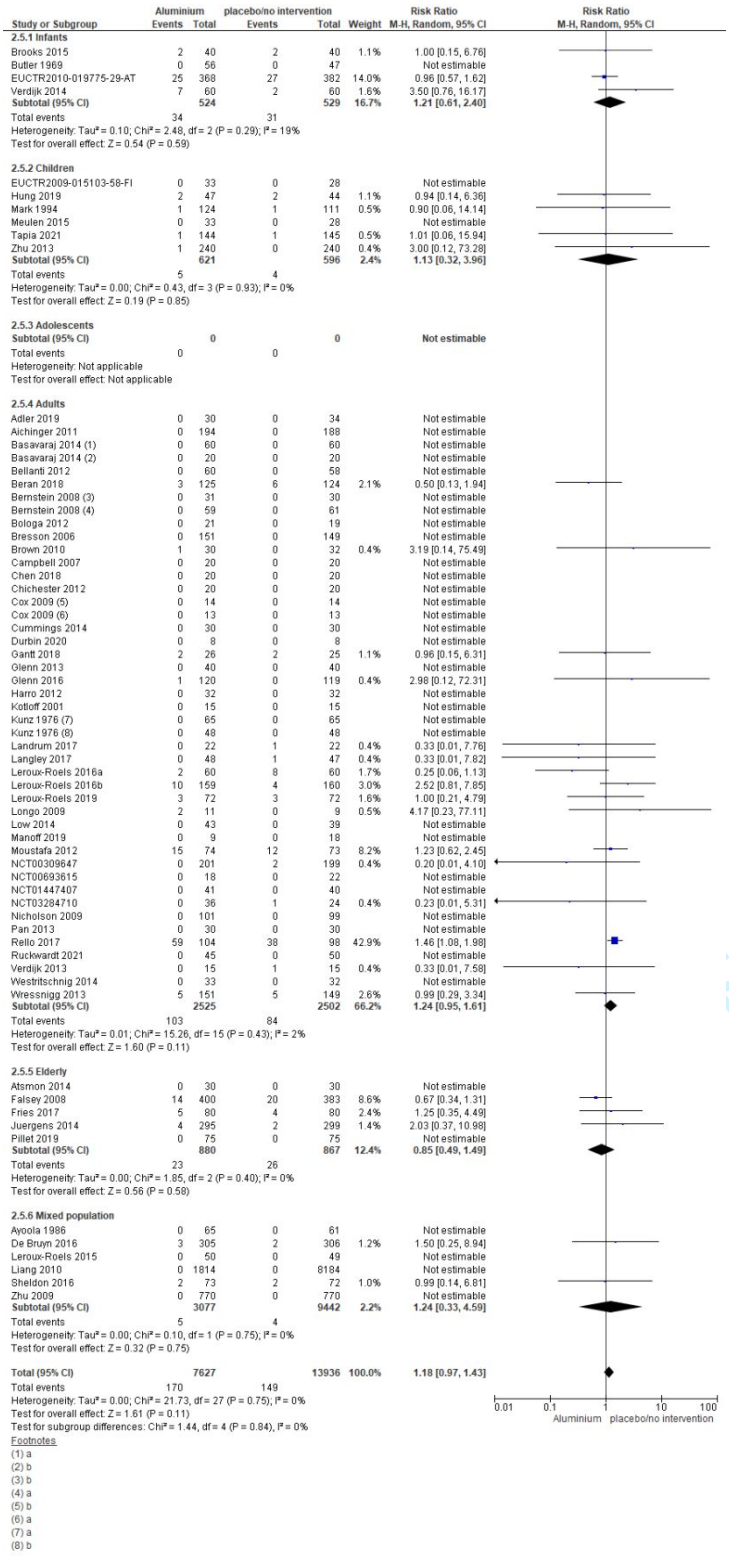
Supplementary Figure 6. Subgroup analysis of serious adverse events according to adjuvant type.

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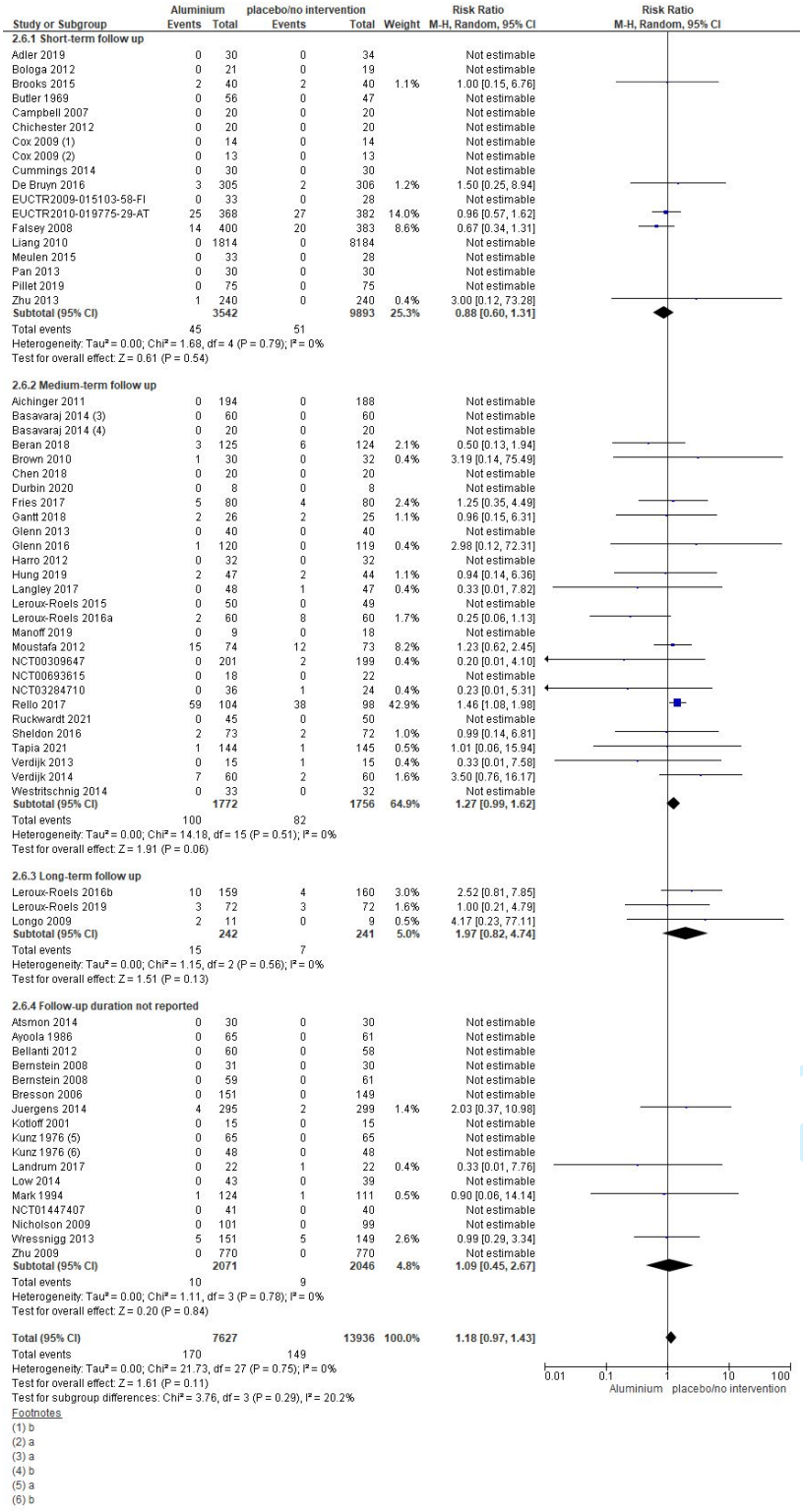
Supplementary Figure 7. Subgroup analysis of serious adverse events according to vaccine type.

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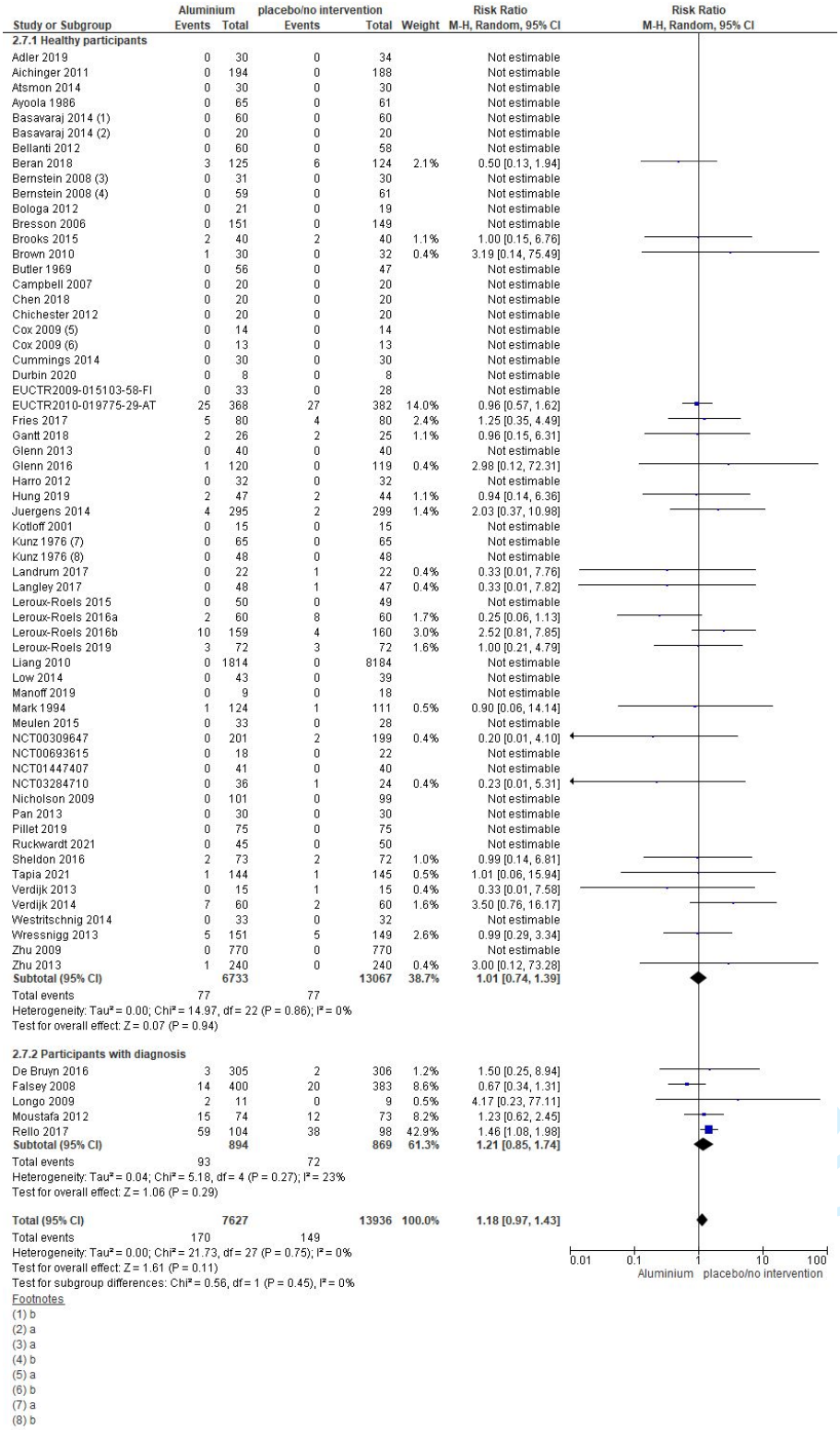
Supplementary Figure 8. Subgroup analysis of serious adverse events according to age.

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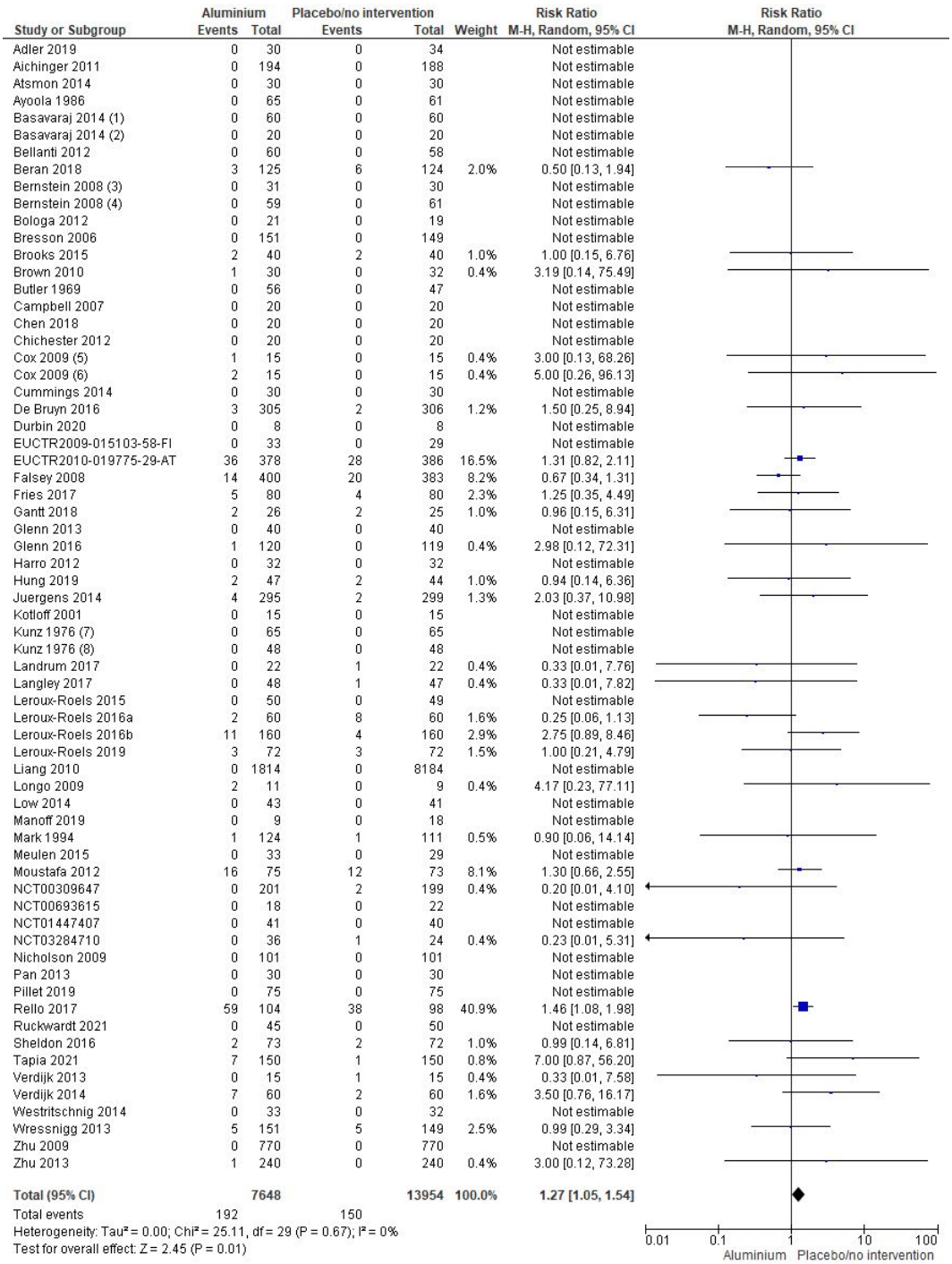
Supplementary Figure 9. Subgroup analysis of serious adverse events according to follow up duration.

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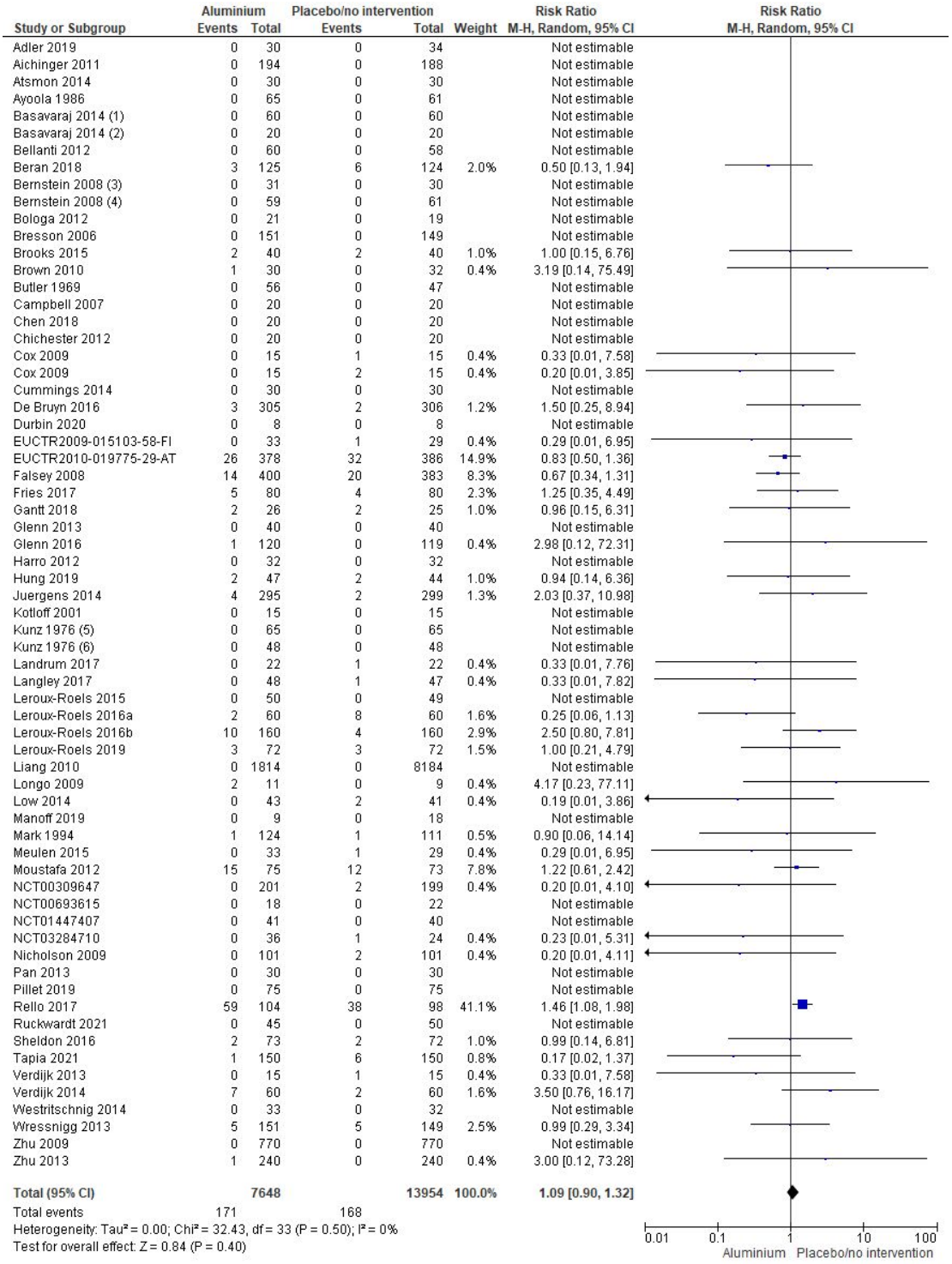
Supplementary Figure 10. Subgroup analysis of serious adverse events according to heath status.

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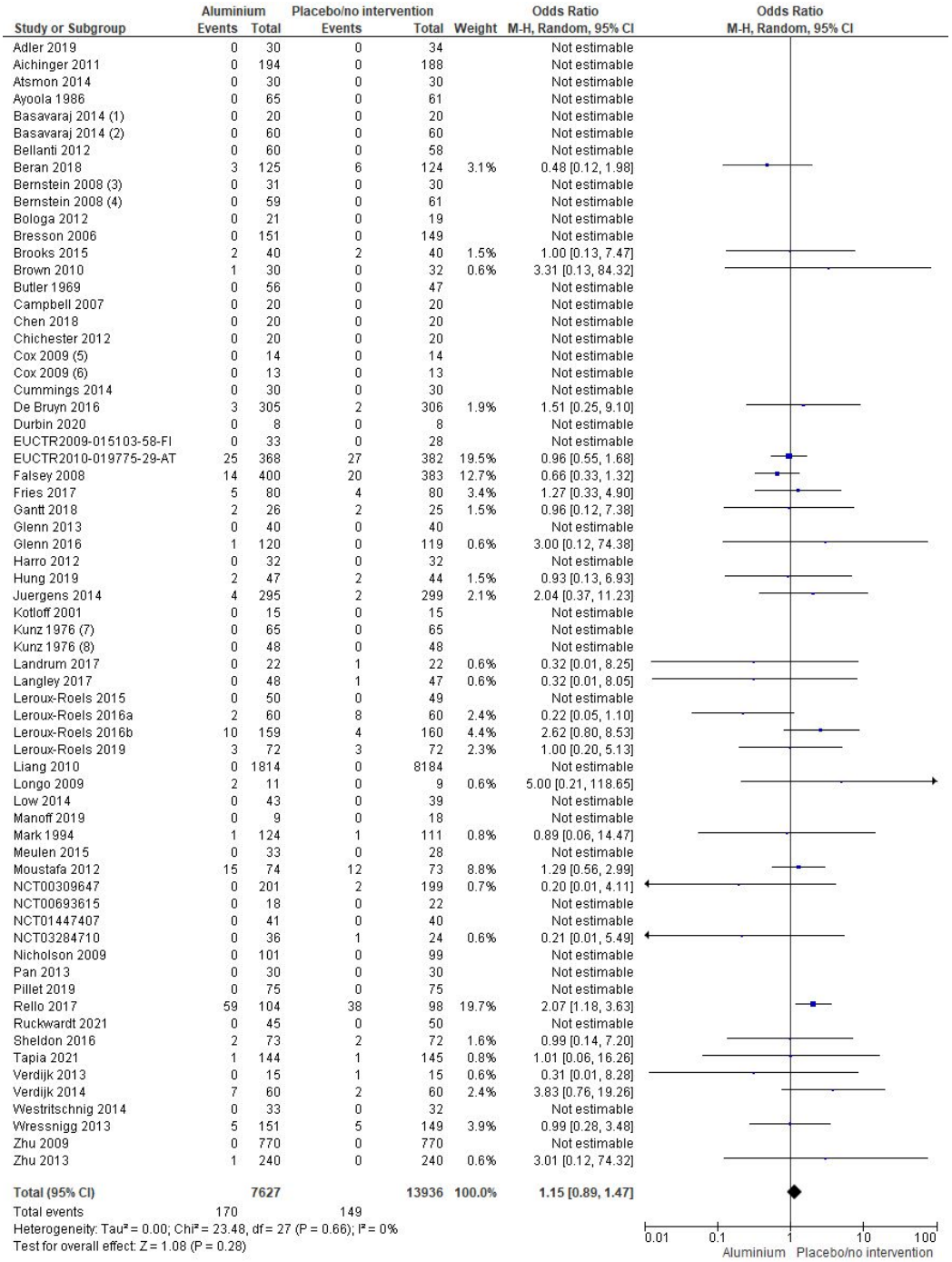
Supplementary Figure 11. Forest plot of 'worst best' sensitivity analysis of serious adverse events.

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Supplementary Figure 12. Forest plot of 'best worst' sensitivity analysis of serious adverse events.

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Supplementary Figure 13. Meta-analysis of serious adverse events with odds ratio.

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Supplementary Analysis 1.

Individual serious adverse events analysis

Meta-analysis showed no evidence of a difference between aluminium adjuvants versus control when assessing individual serious adverse events:

pneumonia (RR 0.59, 95% CI 0.16 to 2.12; trials = 6);

severe pulmonary embolism (RR 0.33, 95% CI 0.03 to 3.14; trials = 2);

gastroenteritis (RR 1.23, 95% CI 0.75 to 2.02; trials = 9);

pancreatitis (RR 0.99, 95% CI 0.10 to 9.44; trials = 2);

diarrhea (RR 0.82, 95% CI 0.17 to 3.86; trials = 4);

breast cancer (RR 0.33, 95% CI 0.05 to 2.08; trials = 3);

ligament rupture (RR 0.33, 95% CI 0.03 to 3.18; trials = 2);

depression (RR 0.33, 95% CI 0.03 to 3.08; trials = 2);

otitis (RR 0.22, 95% CI 0.03 to 1.92; trials = 2);

coronary artery disease (RR 2.98, 95% CI 0.31 to 28.28; trials = 2);

cardiac arrest (RR 1.61, 95% CI 0.08 to 31.86; trials = 2);

sepsis (RR 0.94, 95% CI 0.14 to 6.34; trials = 2).

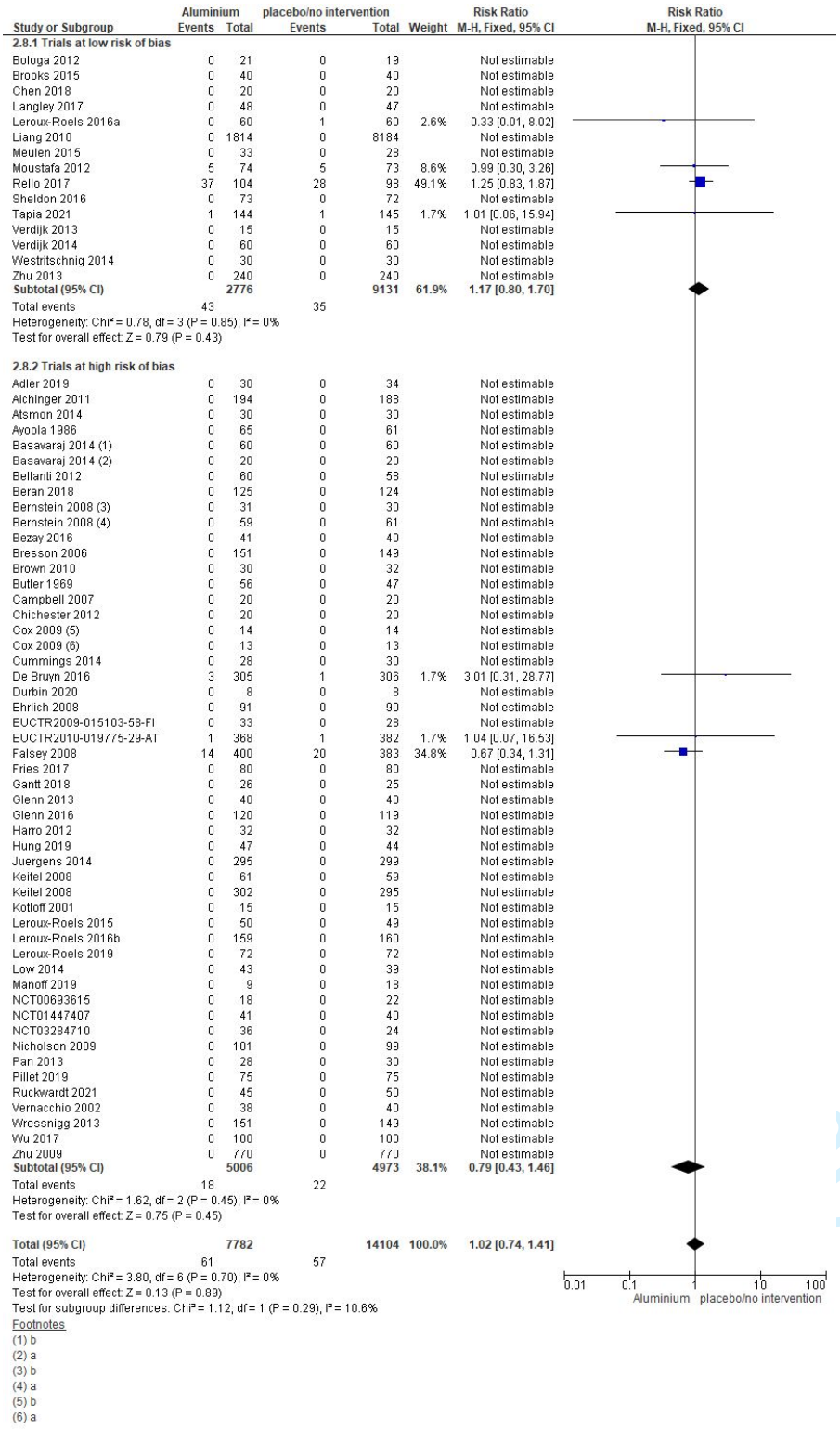
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Serious adverse events reported in the aluminum-adjuvanted group versus the placebo group		
Serious adverse event	**Aluminium event/total (study ID)**	**Placebo event/total (study ID)**
Astrocytoma	0/22 (Landrum2017)	1/22 (Landrum2017)
Congenital laryngomalacia	1/40 (Brooks 2015)	0/40 (Brooks 2015)
Acute appendicitis	1/30 (Glenn 2016)	0/29 (Glenn 2016)
Traumatic left knee dislocation	0/16 (Langley 2017)	1/16 (Langley 2017)
Tonsillitis	1/26 (Hung 2019)	1/24 (Hung 2019)
Cardiac failure	1/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Pulmonary hemorrhage	1/104 (Rello 2017)	0/98 (Rello 2017)
Shock	1/104 (Rello 2017)	0/98 (Rello 2017)
Prostate cancer	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Myocardial infarction	0/15 (Verdrijk 2013	1/15 (Verdrijk 2013
Moderate infective bursitis	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Severe cerebral metastasis and bleed	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Lung cancer	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Cubital tunnel syndrome	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Intravertebral disc protusion	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Pulmonary embolism and deep vein	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Bronchopneumonia	0/26 (Hung 2019)	1/24 (Hung 2019)
Viral encephalitis	1/120 (Zhu 2013)	0/120 (Zhu 2013)
Benign prostatic hyperplasia	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Elective abortion	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Musculoskeletal injuries	0/125 (Beran 2018)	1/124 (Beran 2018)
Spontaneous abortion	1/125 (Beran 2018)	0/124 (Beran 2018)
Transient ischemic attack	0/125 (Beran 2018)	1/124 (Beran 2018)
Umbilical hernia	1/125 (Beran 2018)	0/124 (Beran 2018)
Virally induced asthma	1/125 (Beran 2018)	0/124 (Beran 2018)
Colitis	1/125 (Beran 2018)	0/124 (Beran 2018)
Incisional hernia	1/125 (Beran 2018)	0/124 (Beran 2018)
Weber b fracture right ankle	1/125 (Beran 2018)	0/124 (Beran 2018)
Tibia fracture	1/125 (Beran 2018)	0/124 (Beran 2018)
Erysipelas cheek	1/125 (Beran 2018)	0/124 (Beran 2018)
Arrhythmia	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Cardiogenic Shock	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Gastric ulcer haemorrhage	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Gangrene	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Osteomyelitis bacteria	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Staphylococcal infection	1/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Ankle fracture	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Arteriovenous fistula occlusion	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Joint injury	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Diabetic ketoacidosis	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Azotaemia	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
COPD	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Pulmonary oedema	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Thrombosis	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Roseola infantum	0/26 (Hung 2019)	1/24 (Hung 2019)
Dehydration	0/26 (Hung 2019)	2/24 (Hung 2019)
Hand, foot, mouth disease	1/26 (Hung 2019)	0/24 (Hung 2019)
Clostridium difficile infection	1/26 (Hung 2019)	0/24 (Hung 2019)
Acute pyelonephritis	0/72 (Lerouz-Roels 2019)	1/72 (Lerouz-Roels 2019)
Cartilage injury	1/72 (Lerouz-Roels 2019)	0/72 (Lerouz-Roels 2019)
Head injury	0/72 (Lerouz-Roels 2019)	1/72 (Lerouz-Roels 2019)
Anaphylactic shock	1/72 (Lerouz-Roels 2019)	0/72 (Lerouz-Roels 2019)
Transient paralysis	1/25 (Bellino 2009)	0/22 (Bellino 2009)
Lipothymia-nausea	1/25 (Bellino 2009)	0/22 (Bellino 2009)
Laceration	0/26 (Gantt 2018)	1/25 (Gantt 2018)
Infections	1/124 (Mark 1994)	1/111 (Mark 1994)
Deep vein thrombosis	1/26 (Gantt 2018)	0/25 (Gantt 2018)

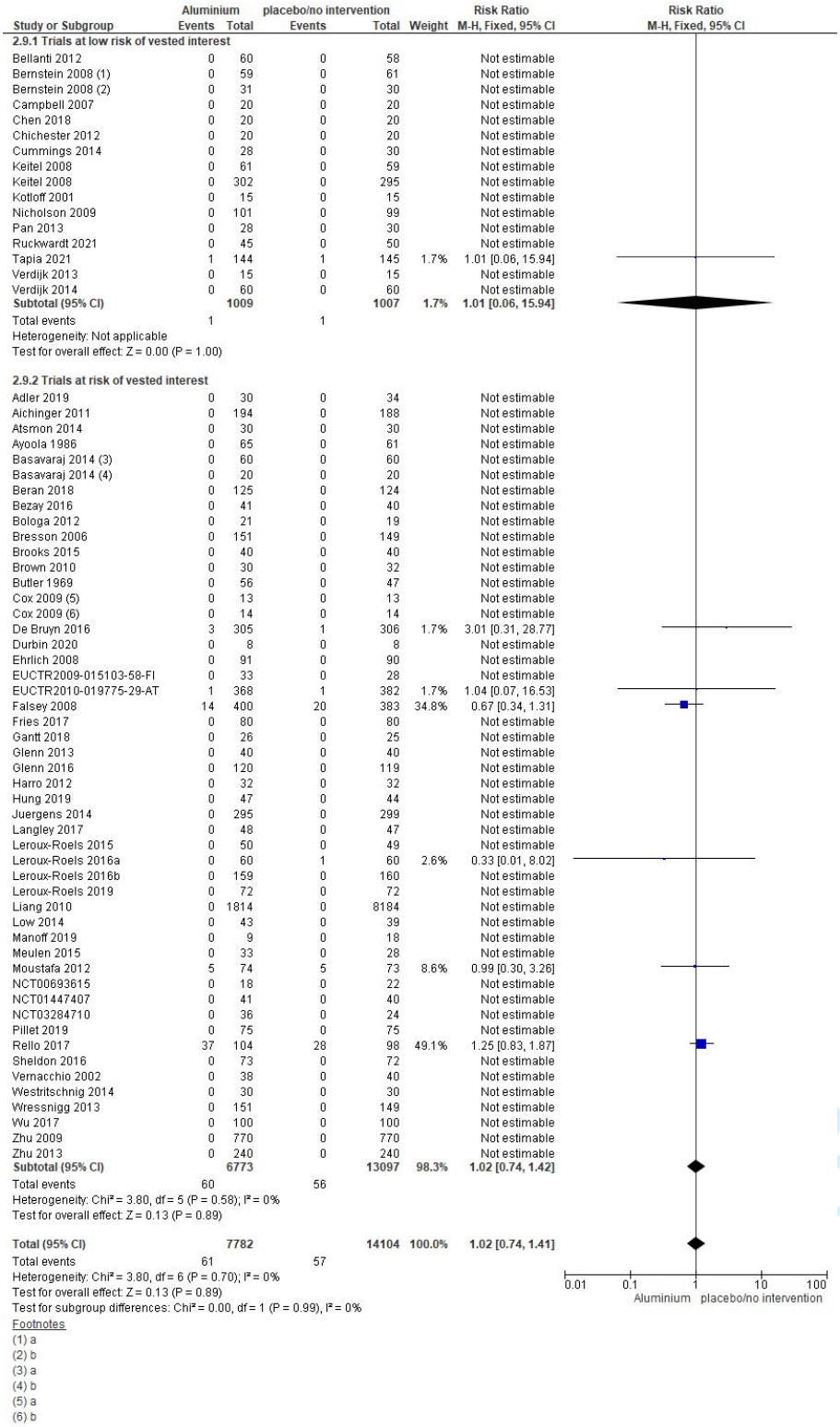
Supplementary Table 2. List of individual serious adverse events reported in only one trial. It was our aim to examine if the serious adverse events in the aluminium-adjuvanted groups of the included trials resembled the syndromes described in the literature related to aluminium toxicity. All included trials reporting serious adverse events were screened for detailed information about the events. However, the reporting was too sketchy to use the information for comparison with the syndromes. In general, the types of serious adverse events were not specified in the trials, and it was impossible to assess the total number of serious adverse events for each trial participant. We can therefore not evaluate whether the serious adverse event may be or maybe be not related to what is described as 'aluminium toxicity' in the literature.

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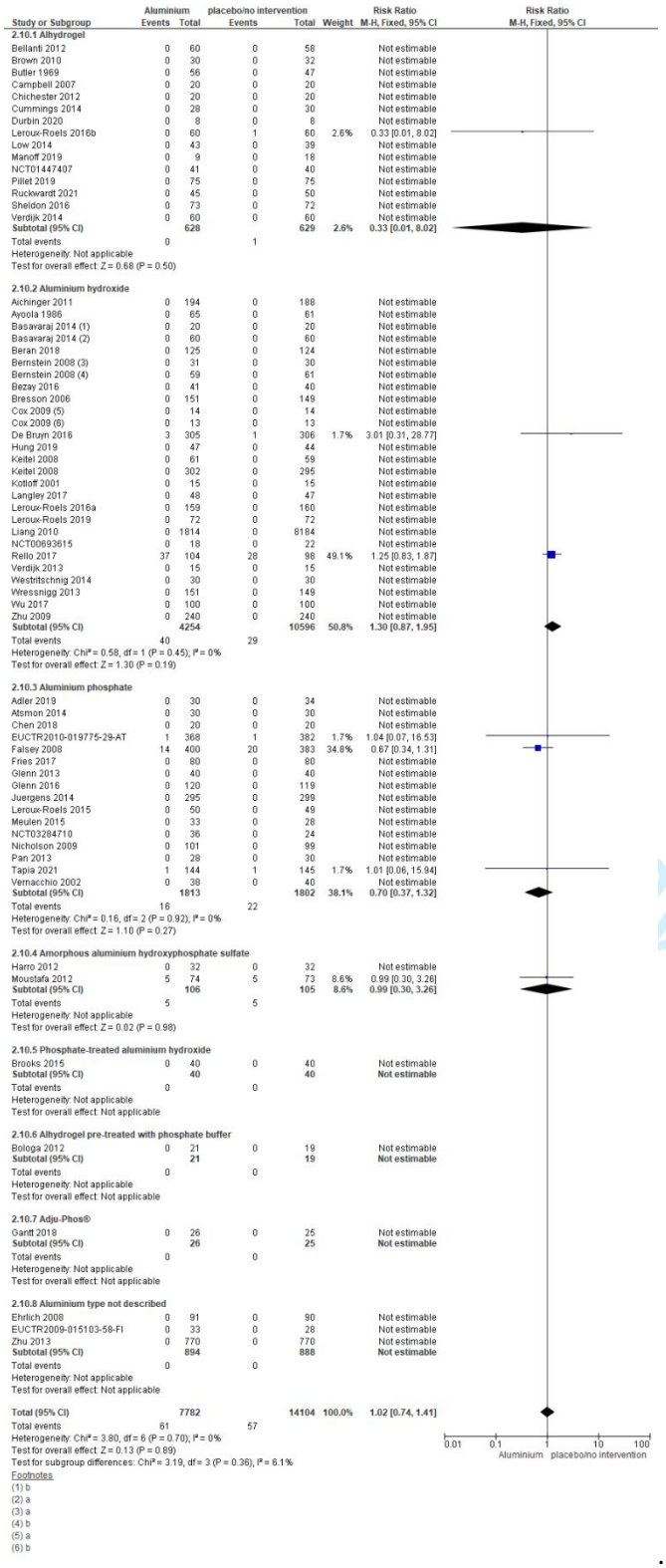
Supplementary Figure 14. Subgroup analysis of all-cause mortality according to risk of bias.

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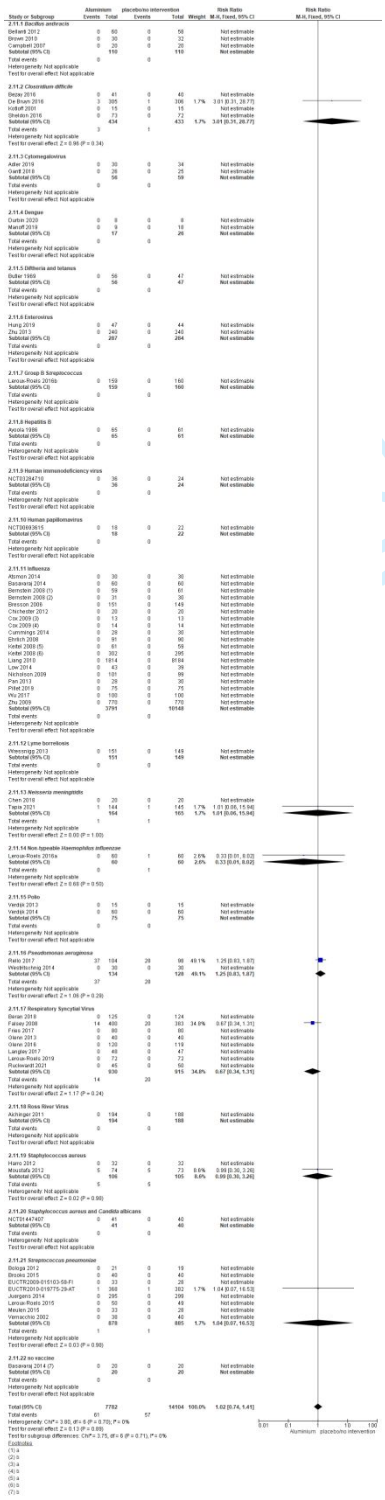
Supplementary Figure 15. Subgroup analysis of all-cause mortality according to risk of vested interest.

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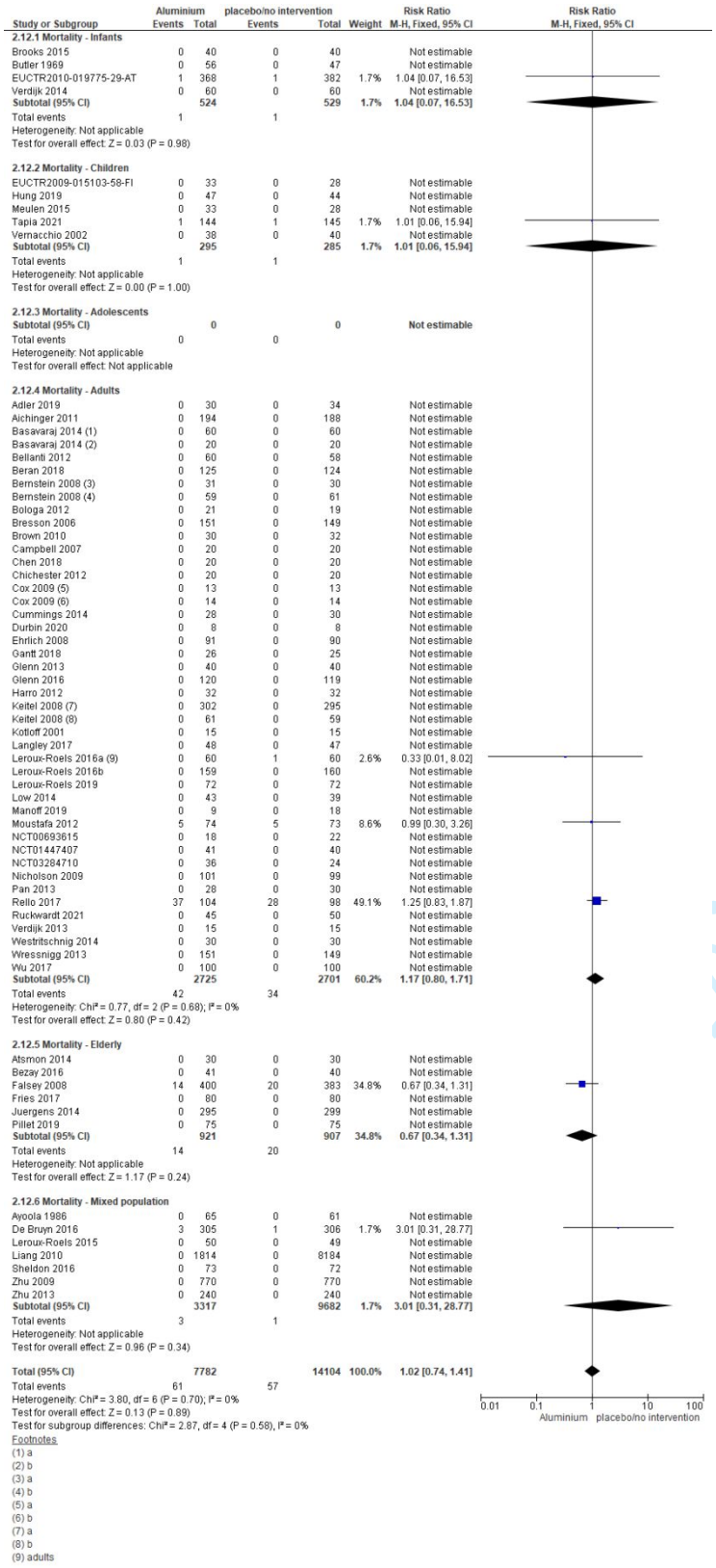
Supplementary Figure 16. Subgroup analysis of all-cause mortality according to aluminium adjuvant type.

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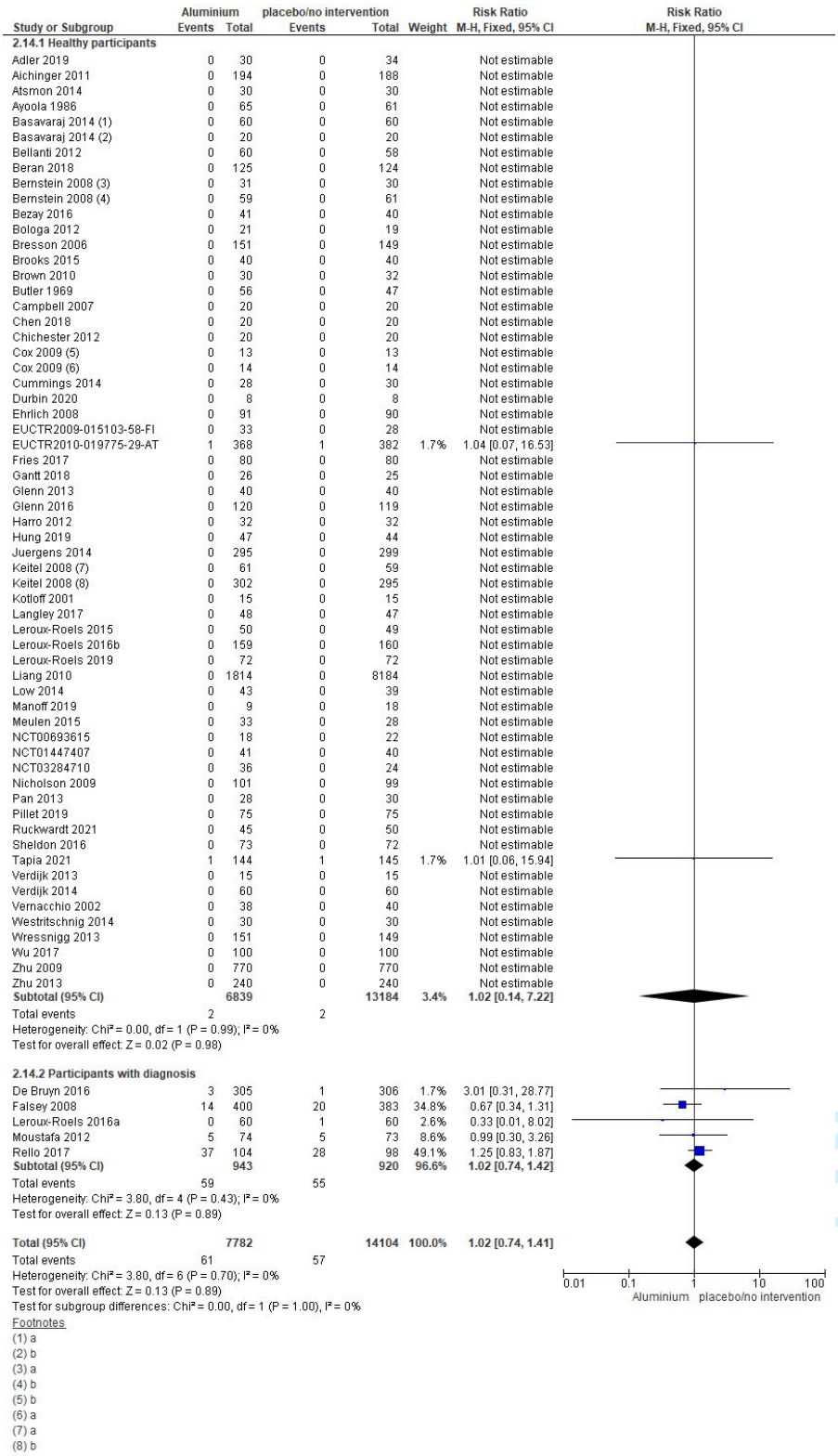
Supplementary Figure 17. Subgroup analysis of all-cause mortality according to vaccine type.

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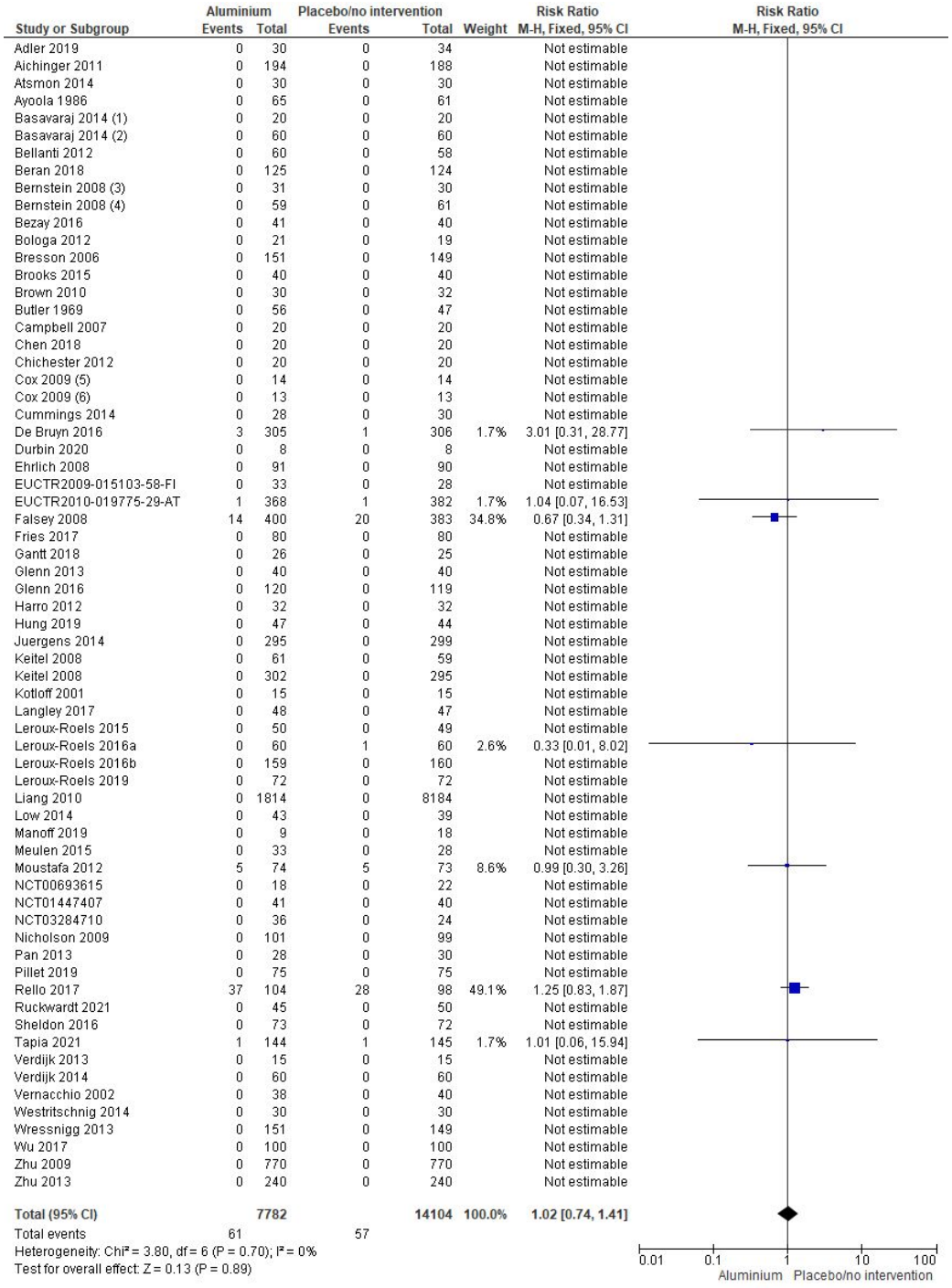
Supplementary Figure 18. Subgroup analysis of all-cause mortality according to age of participants.

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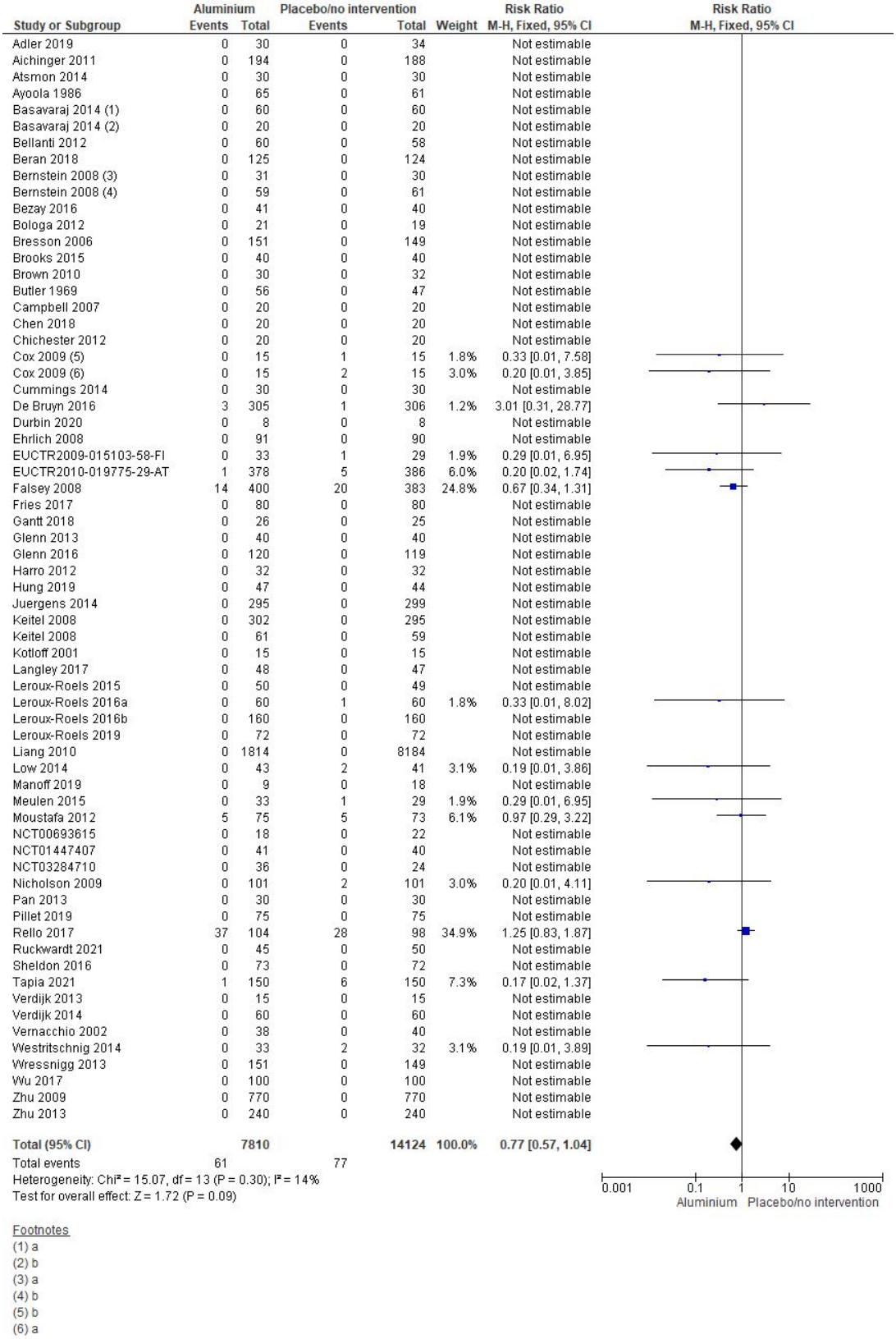
Supplementary Figure 19. Subgroup analysis of all-cause mortality according to health status.

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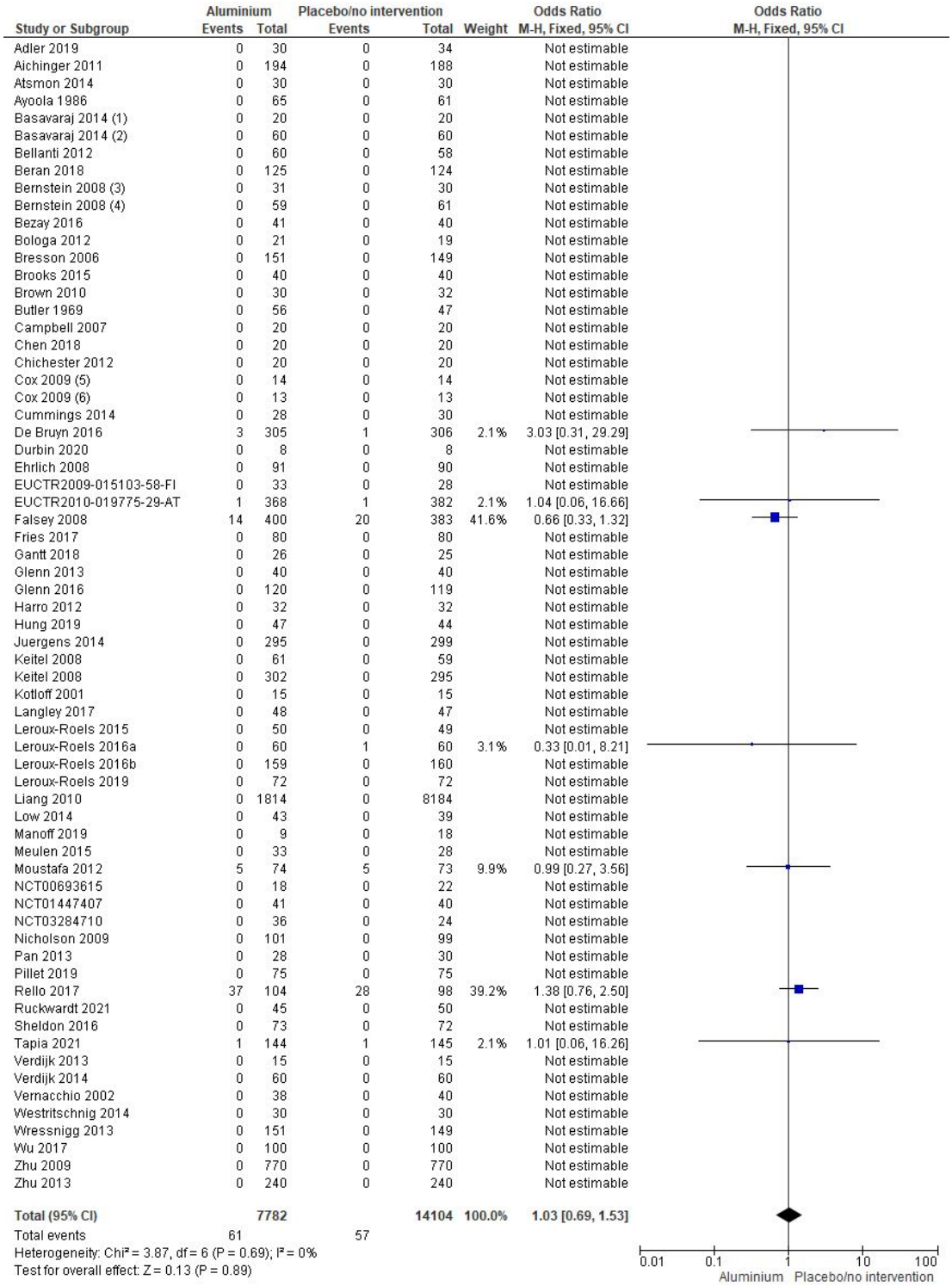
Supplementary Figure 20. Forest plot of 'worst best' sensitivity analysis of all-cause mortality.

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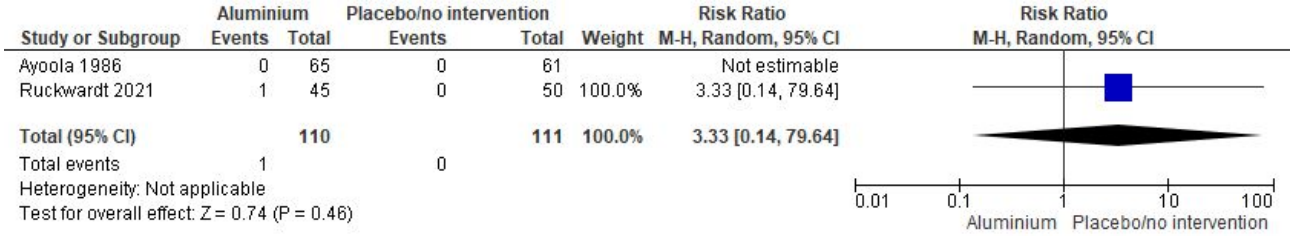
Supplementary Figure 21. Forest plot of 'best worst' sensitivity analysis of all-cause mortality.

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Supplementary Figure 22. Meta analysis of all-cause mortality with odds ratio.

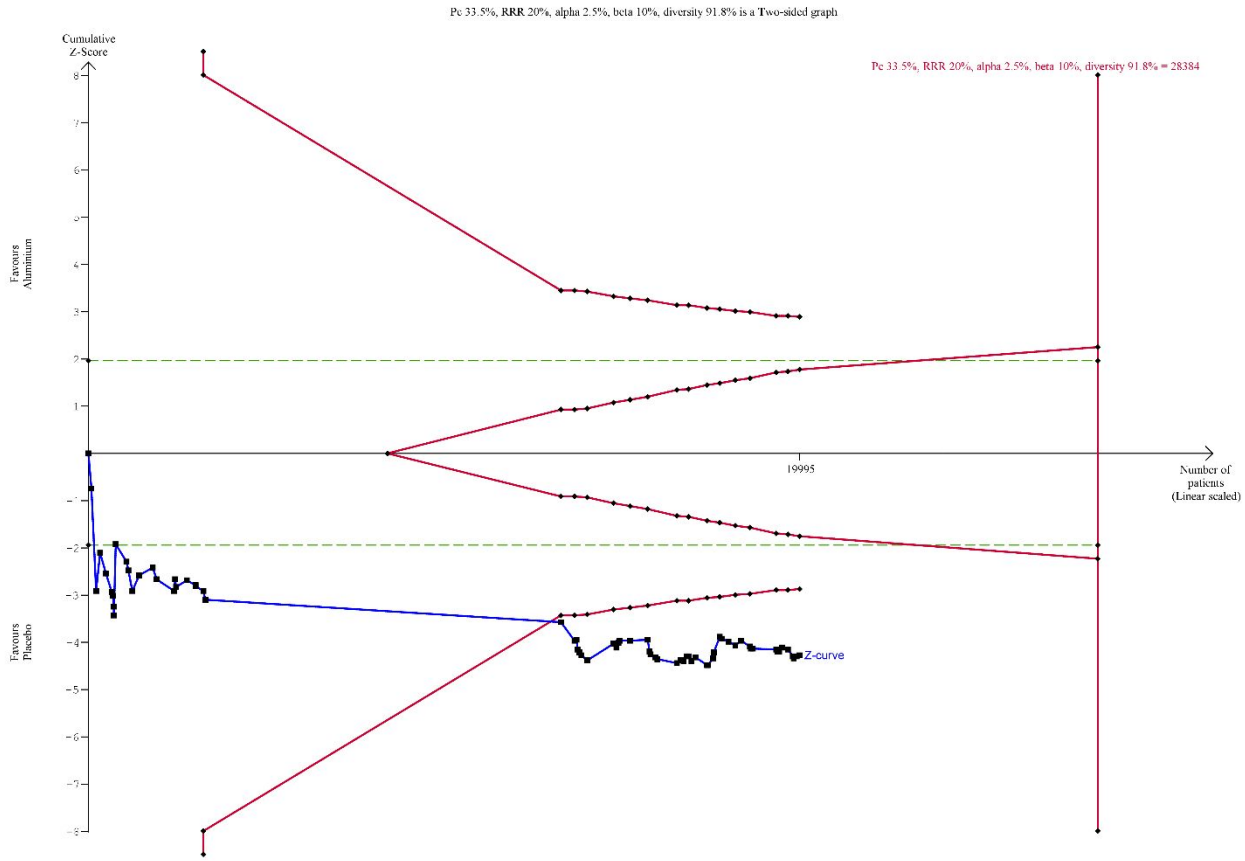
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Supplementary Figure 23. Meta analysis of the effect of aluminium adjuvants compared with placebo on the proportion of participants with disease.

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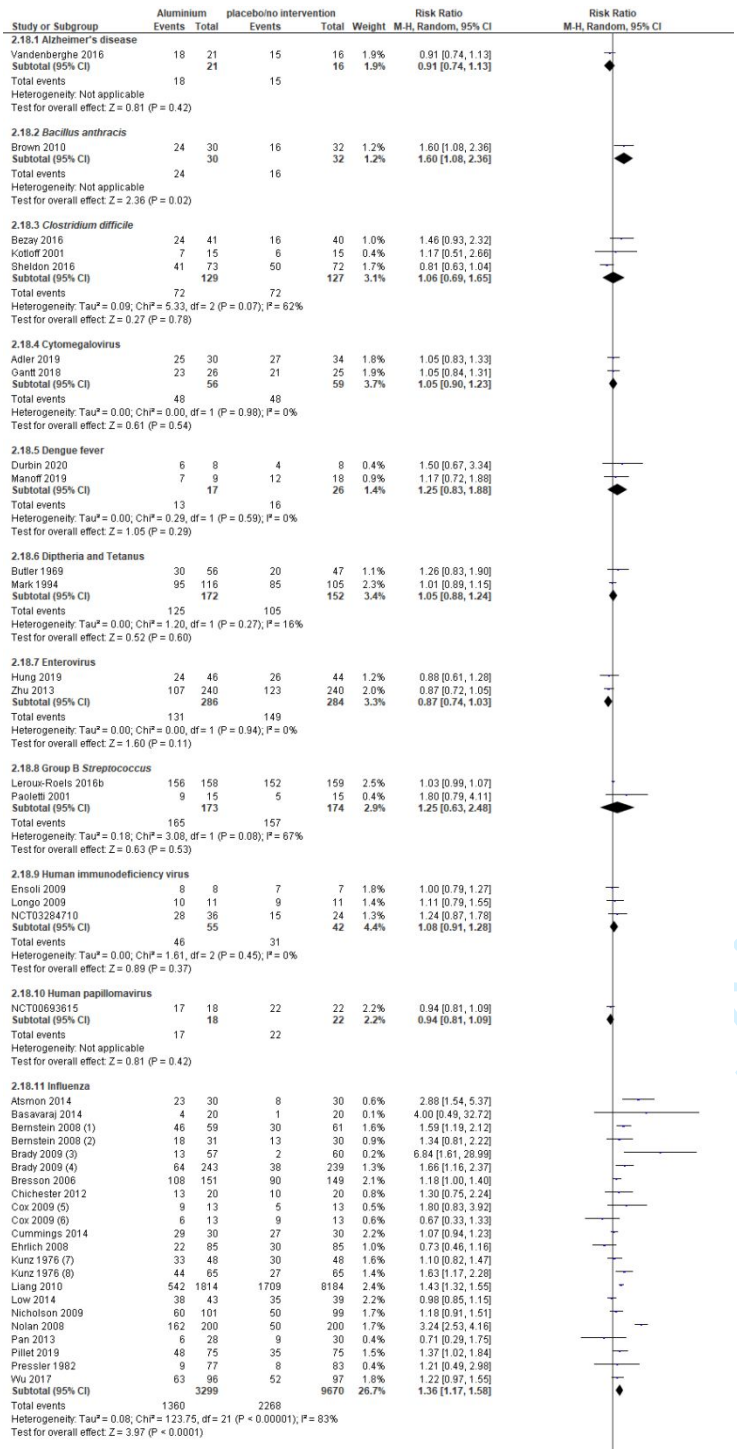
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Supplementary Figure 24. Trial Sequential Analysis using random-effects meta-analysis of the effects of aluminium adjuvants versus placebo or no intervention on the proportion of participants with adverse events considered not-serious. The analysis was based on a proportion in the control group (Pc) of 33.8%, a relative risk reduction (RRR) of 20%, an alpha of 2.5%, a beta of 10% and diversity of 92%. The graph shows that there was enough information to confirm that aluminium adjuvants compared with placebo increases the risk of one or more non-serious adverse events.

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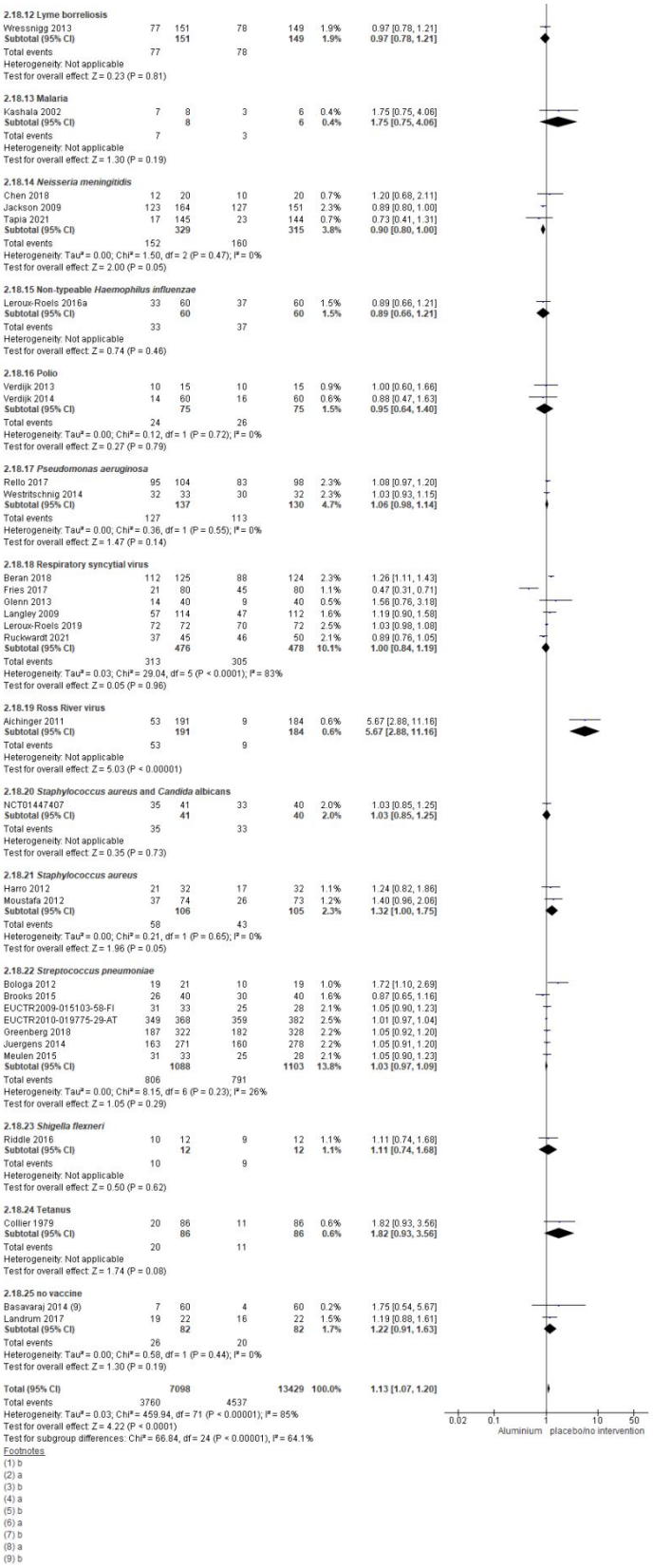
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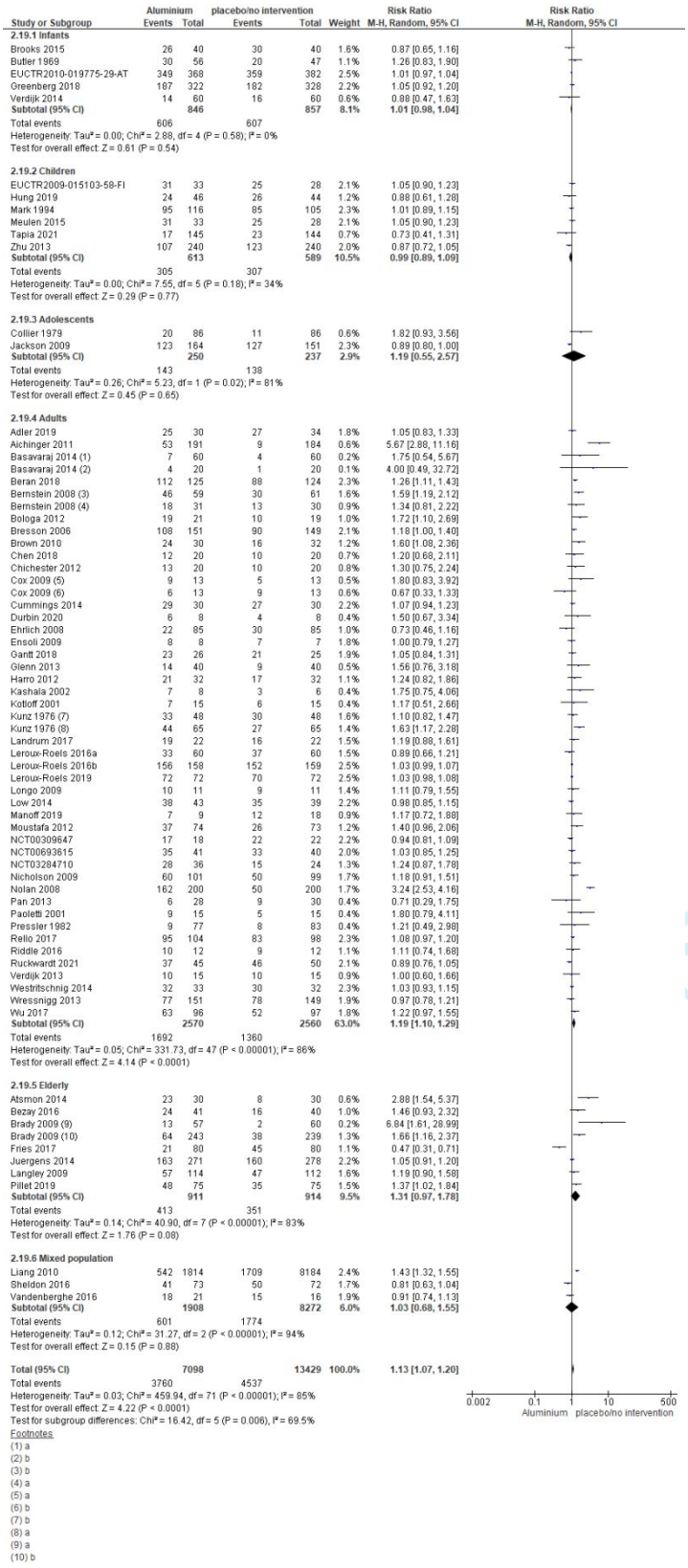
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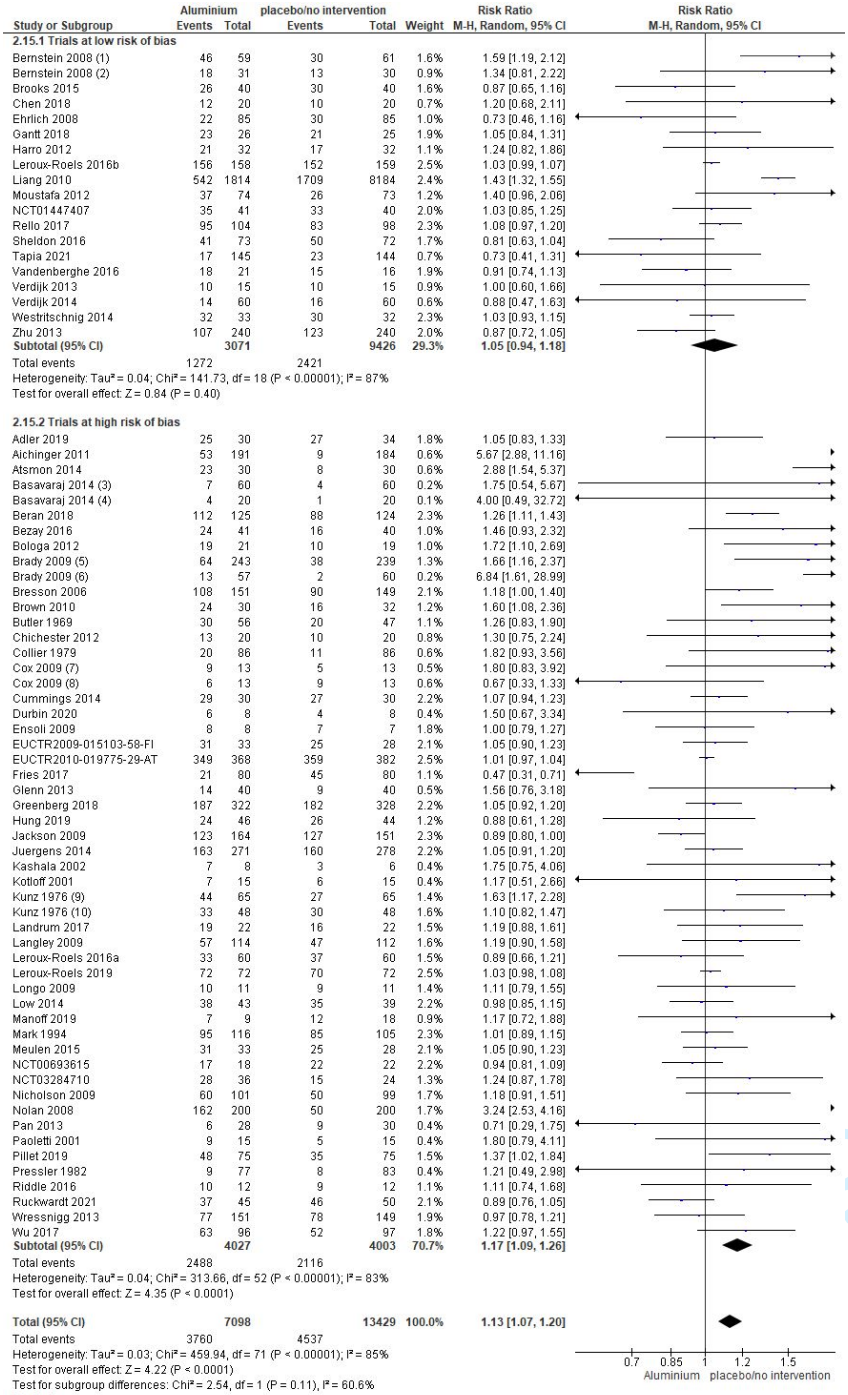
Supplementary Figure 25. Subgroup analyses of non-serious adverse events according to vaccine type.

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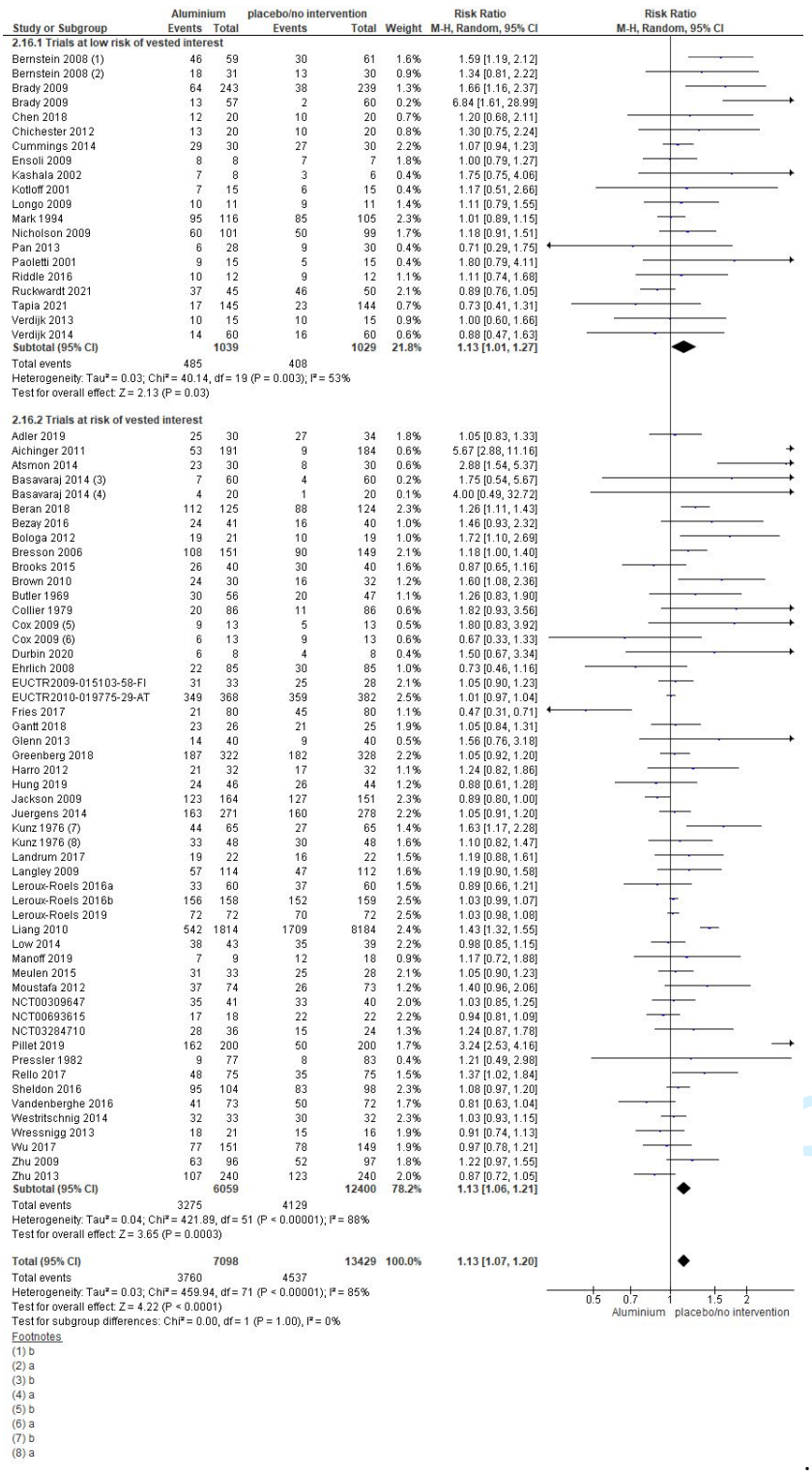
Supplementary Figure 26. Subgroup analyses of non-serious adverse events according to age.

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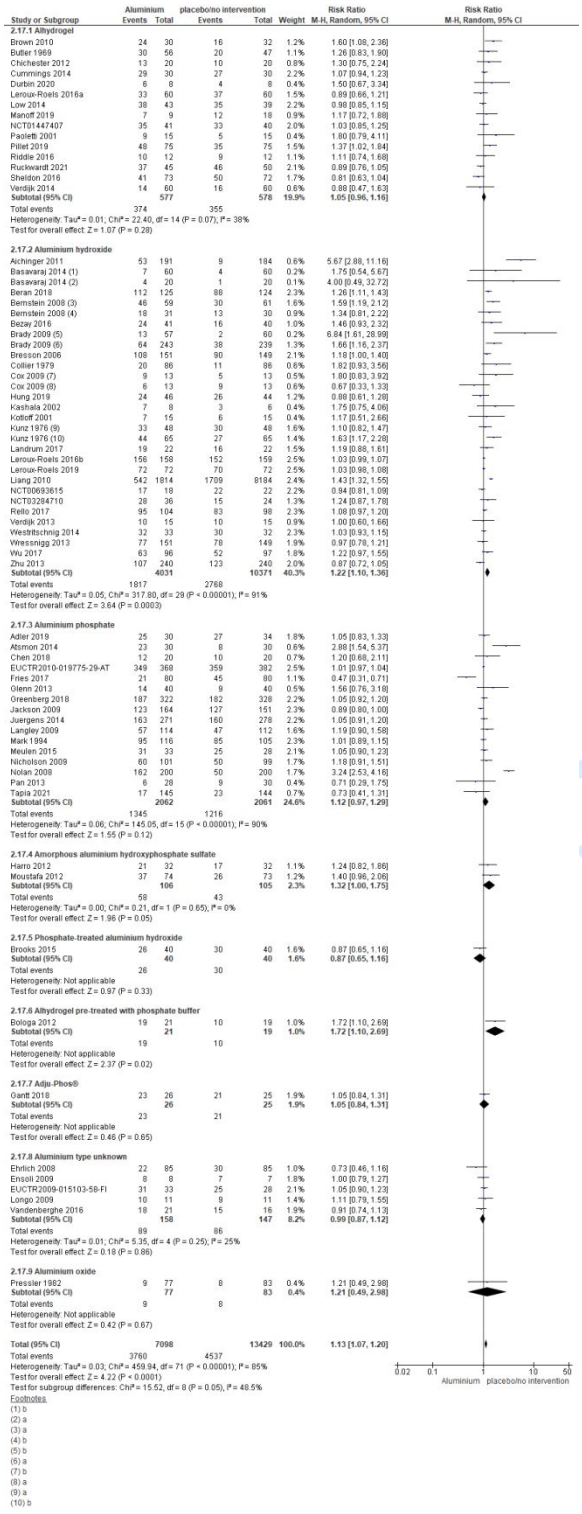
Supplementary Figure 27. Subgroup analyses of non-serious adverse events according to risk of bias.

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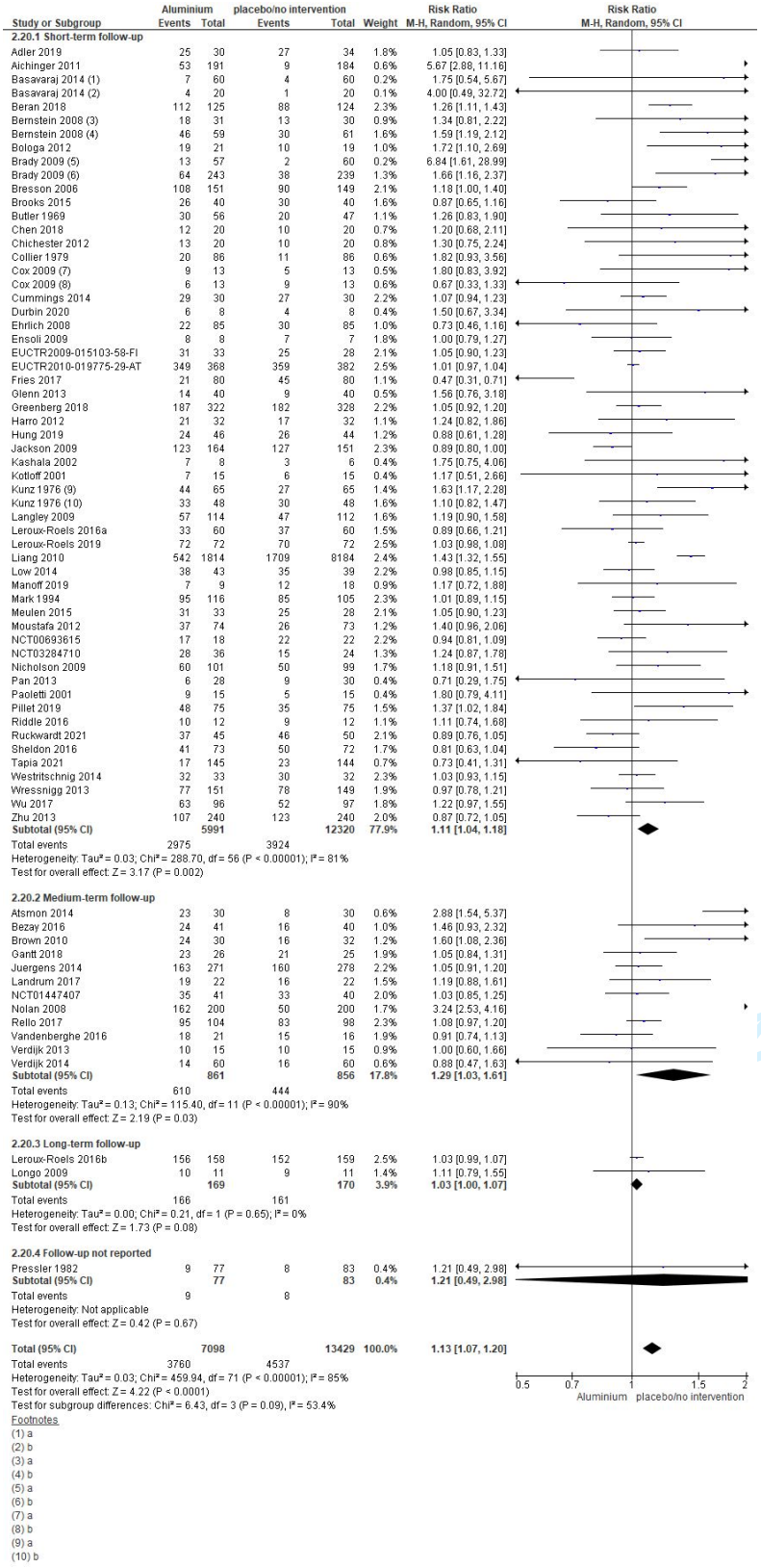
Supplementary Figure 28. Subgroup analyses of non-serious adverse events according to vested interest.

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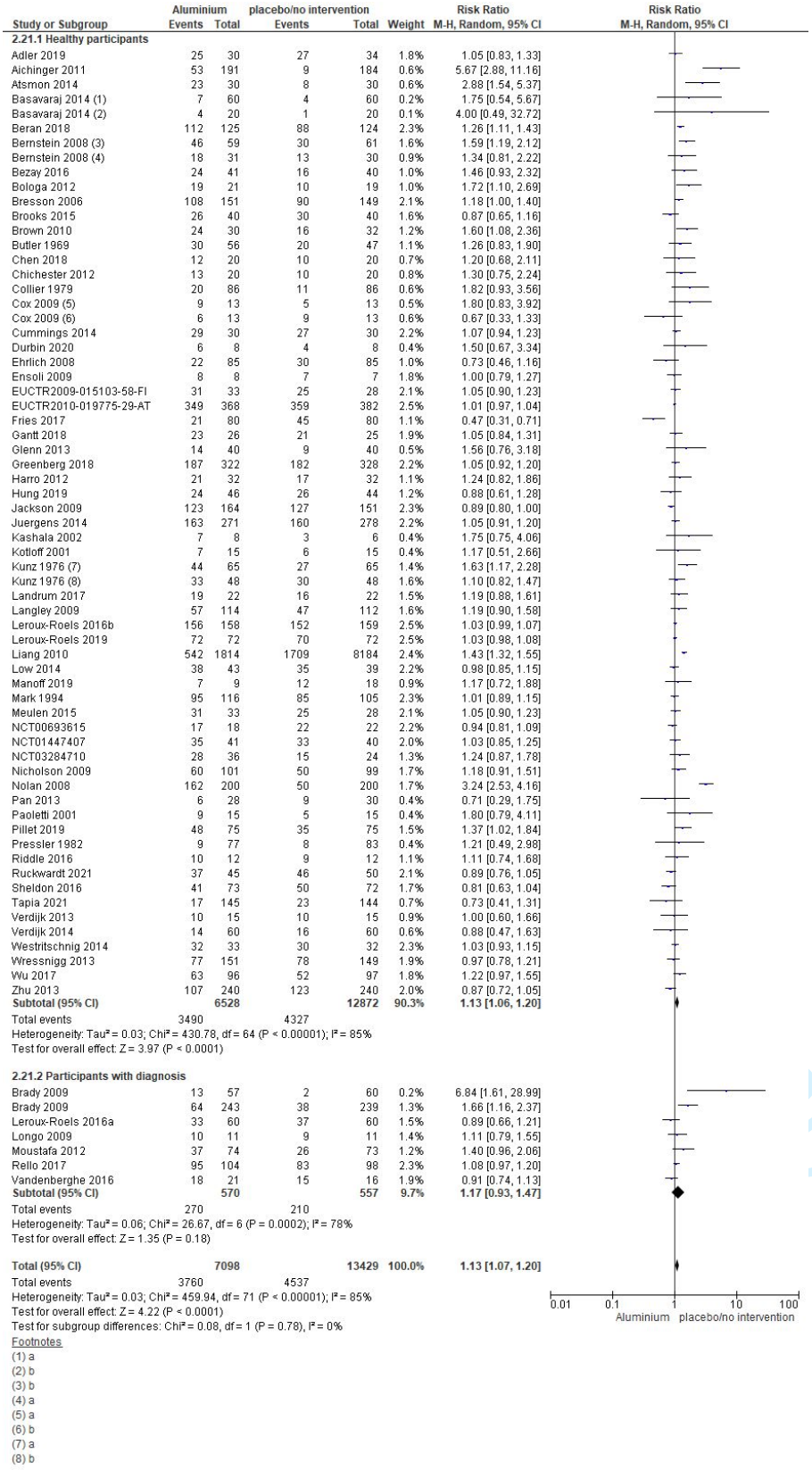
Supplementary Figure 29. Subgroup analyses of non-serious adverse events according to aluminium type.

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Supplementary Figure 30. Subgroup analyses of non-serious adverse events according to follow up duration.

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Supplementary Figure 31. Subgroup analyses of non-serious adverse events according to health status.

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Supplementary Analysis 2.

Individual adverse events considered non-serious analysis

Meta-analysis showed evidence of a harmful effect of aluminium adjuvants compared with placebo or no intervention when assessing the following adverse events considered non-serious:

swelling (RR 1.38, 95% CI 1.17 to 1.62);

injection site pain (RR 1.38, 95% CI 1.19 to 1.61);

tenderness (RR 1.67, 95% CI 1.28 to 2.17);

muscle pain (RR 1.30, 95% CI 1.05 to 1.61);

erythema (RR 1.23, 95% CI 1.02 to 1.50);

musculoskeletal and connective tissue disorders (RR 1.90, 95% CI 1.16 to 3.14);

pain (RR 1.32, 95% CI 1.19 to 1.46);

injection site reaction (RR 2.02, 95% CI 1.23 to 3.32);

myalgia proximal (RR 2.12, 95% CI 1.66 to 2.72); and

rash (RR 1.92, 95% CI 0.99 to 3.72).

Meta-analysis showed no evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing the following non-serious adverse events: induration, redness, fever onset within 7 days, fever, malaise, fatigue, headache, nausea, vomiting, joint pain, lymph node swelling, itching, sore throat, feeling hot, feverishness, myalgia, ecchymosis, shivering, loss of appetite, arthralgia, stayed home, upper respiratory tract infection, back pain, oropharyngeal pain, hypertension, bruising, cough, hypotension, anorexia, numbness, muscle stiffness, sweating, exanthema, rhinorrhoea, dizziness, nasopharyngitis, urinary tract infection, change in feeding pattern, more crying, infection, bronchitis, injury, poisoning and procedural complications, flu-like symptoms, joint sprain, warmth, insomnia, constipation, toothache, alanine aminotransferase increased, arthropod bite, aspartate aminotransferase increased, contusion, dysphonia, flushing, vertigo positional, vulvovaginal candidiasis, pruritus, pharyngitis, influenza-like illness, rhinitis, chills, gastrointestinal symptoms, blood and lymphatic disorders, gastrointestinal disorders, flatulence, injection site haemorrhage, acute sinusitis, conjunctivitis, herpes zoster, ligament sprain, blood creatinine increased, blood pressure increased, hemoglobin decreased, bursitis, neck pain, tendonitis, nervous system disorders, sinus congestion, skin and subcutaneous disorders, ear and labyrinth disorders, eye disorders, dental caries, food poisoning, gastritis, gingival disorder, edema peripheral, hepato-biliary disorders, renal and urinary disorders, blood bilirubin increased, blood urea increased, epistaxis/nose bleeding, nasal congestion, hardening, less appetite, local administration site disorders, cardio-vascular disorders, general disorders, kidney and urinary disorders, psychiatric disorders, asthma, dyspepsia, cystitis, influenza, oral herpes, hyperglycaemia, ear infection, reproductive system and breast disorders, swelling in the neck, asthenia, diarrhea, and gastroenteritis.

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Individual non-serious adverse events reported by only one trial that could not be meta-analysed are shown in the table below.

Non-serious adverse events reported in the aluminum-adjuvanted groups versus the placebo group or no intervention		
Non-serious adverse events	**Aluminium event/total (study ID)**	**Placebo event/total (study ID)**
Sinus headache	1/32 (Harro 2012)	0/32 (Harro 2012)
Body pain	2/60 (Basavaraj 2014)	1/60 (Basavaraj 2014)
Common cold	2/60 (Basavaraj 2014)	1/60 (Basavaraj 2014)
Allergy on face	1/60 (Basavaraj 2014)	0/60 (Basavaraj 2014)
Burning sensation	1/60 (Basavaraj 2014)	0/60 (Basavaraj 2014)
Lower limb pain	0/60 1/20 (Basavaraj 2014)	1/60 0/20 (Basavaraj 2014)
Systolic hpertension	1/80 (Fries 2017)	4/80 (Fries 2017)
Abnormal crying	9/40 (Brooks 2015)	9/40 (Brooks 2015)
Respiratory symptoms	5/14 0/13 (Cox 2009)	0/14 3/13 (Cox 2009)
Contralateral inflammation	2/8 (Kashala 2002)	1/6 (Kashala 2002)
Joint swelling	1/15 (Kotloff 2001)	0/15 (Kotloff 2001)
Limitation of movement	13/43 (Low 2014)	13/39 (Low 2014)
Nodule	10/33 (Meulen 2015)	3/28 (Meulen 2015)
Muscle rigidity	1/60 (Verdrijk 2014)	1/60 (Verdrijk 2014)
Change in stool	12/60 (Verdrijk 2014)	9/60 (Verdrijk 2014)
Change in sleeping pattern	14/60 (Verdrijk 2014)	16/60 (Verdrijk 2014)
Change in behavior	10/60 (Verdrijk 2014)	11/60 (Verdrijk 2014)
Pain in extremity	7/33 (Westritschnig 2014)	0/32 (Westritschnig 2014)
Eclampsia	1/120 (Zhu 2013)	2/120 (Zhu 2013)
Exanthema	2/46 (Hung 2019)	1/44 (Hung 2019)
Blood fibrinogen decreased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Blood lactate dehydrogenase increased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Bone contusion	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Excoriation	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Infected dermal cyst	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Platelet count increased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Rosacea	1/60 (Leroux-Roels 2016b)	0/60 (Leroux-Roels 2016b)
Injury	1/18 (NCT00693615)	6/22 (NCT00693615)
Infection viral	1/20 (Chen 2018)	0/20 (Chen 2018)
Stress	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Myalgia - distal	39 /200 (Nolan 2008)	35/200 (Nolan 2008)
Cardiac disorders	1/73 (Sheldon 2016)	2/72 (Sheldon 2016)
Palpitations	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Abdominal distension	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Infected bites	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Fall	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Periorbital hematoma	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Post-traumatic neck syndrome	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Aspartate aminotransferase increased	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Blood cholesterol increased	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Dyslipidemia	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Musculoskeletal and connective tissue disorders	5/73 (Sheldon 2016)	5/72 (Sheldon 2016)
Neoplasms benign malignant and unspecified (including cysts and polyps)	1/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Limb discomfort	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Heart valve stenosis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Vertigo postural	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Endocrine disorders	2/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Hypothyroidism	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Testicular failure	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Vitreous detachment	0/73 (Sheldon 2016)	2/72 (Sheldon 2016)
Cellulitis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Cholelithiasis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Folliculitis	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Hand fracture	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Osteoporosis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Urethral stenosis	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Prostatism	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Blood cortisol increased	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Local reactions	118/770 (Zhu 2009 all)	55/770 (Zhu 2009 all)
Neurologic	5/770 (Zhu 2009 all)	4/770 (Zhu 2009 all)
Systemic	65/770 (Zhu 2009 all)	60/770 (Zhu 2009 all)
Bacterial vaginosis	1/20 (Chen 2018)	0/20 (Chen 2018)
Blood pressure decreased	1/30 (Atsmon 2014)	0/30 (Atsmon 2014)
Herpangina	1/46 (Hung 2019)	0/44 (Hung 2019)
Papules	0/46 (Hung 2019)	1/44 (Hung 2019)
Lymph node pain	1/32 (Harro 2012)	0/32 (Harro 2012)

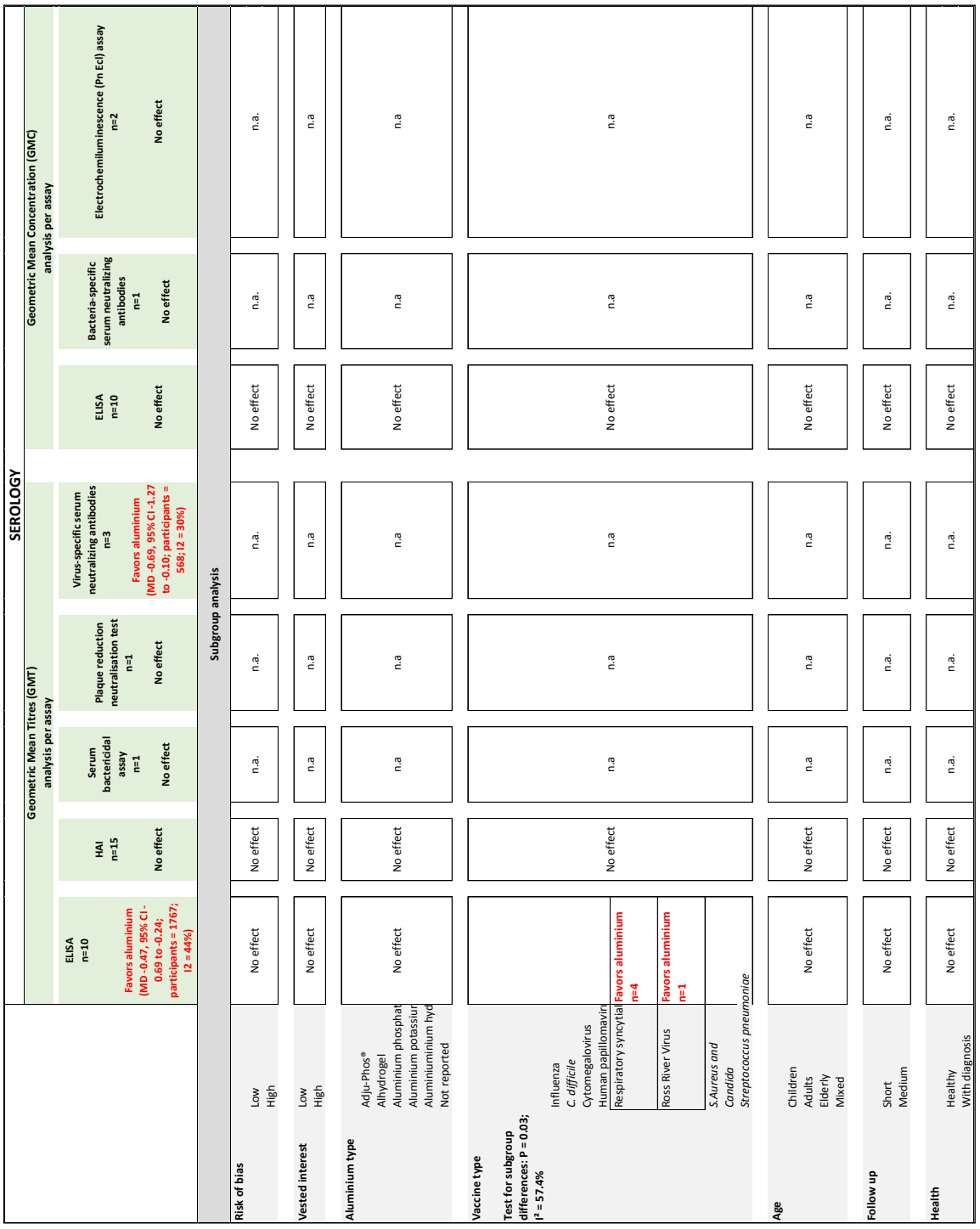
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Tooth disorder	2/72 (Leroux-Roels 2019)	0/72 (Leroux-Roels 2019)
Tinea Pedis	2/72 (Leroux-Roels 2019)	0/72 (Leroux-Roels 2019)
Wheezing	5/72 (Leroux-Roels 2019)	5/72 (Leroux-Roels 2019)
Hemoglobin abnormal	2/22 (Landrum 2017)	1/22 (Landrum 2017)
Blood LDH increase	1/22 (Landrum 2017)	1/22 (Landrum 2017)
Whole blood count increased	2/22 (Landrum 2017)	3/22 (Landrum 2017)
Neutrophil count increased	1/22 (Landrum 2017)	2/22 (Landrum 2017)
Hypoglycaemia	0/22 (Landrum 2017)	1/22 (Landrum 2017)
Blood sodium increased	2/22 (Landrum 2017)	1/22 (Landrum 2017)
Teething	0/33 (Meulen 2015)	1/28 (Meulen 2015)
Laryngitis	0/33 (Meulen 2015)	1/28 (Meulen 2015)
Respiratory syncytial virus bronchiolitis	1/33 (Meulen 2015)	0/28 (Meulen 2015)
Dysmenorrhea	0/18 (NCT00693615)	2/22 (NCT00693615)
Flu syndrome	1/18 (NCT00693615)	2/22 (NCT00693615)
Depression	0/18 (NCT00693615)	1/22 (NCT00693615)
Urogenital neoplasia	0/18 (NCT00693615)	1/22 (NCT00693615)
Vaginal moniliasis	0/18 (NCT00693615)	1/22 (NCT00693615)
Injection site vesicles	0/8 (Durbin 2020)	1/8 (Durbin 2020)
Armpit swelling	0/25 (Gantt 2018)	1/23 (Gantt 2018)
Gastroesophageal reflux	1/25 (Gantt 2018)	0/23 (Gantt 2018)
Herpes simplex	0/25 (Gantt 2018)	2/23 (Gantt 2018)
Laceration	0/25 (Gantt 2018)	1/23 (Gantt 2018)
Otitis	3/33 (Meulen 2015)	1/28 (Meulen 2015)
Head injury	1/33 (Meulen 2015)	0/28 (Meulen 2015)
Bacterial vaginosis	0/36 (NCT03284710)	2/24 (NCT03284710)
Orchitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Muscle twitching	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Dysuria	1/36 (NCT03284710)	0/24 (NCT03284710)
Otitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Discomfort	6/119 (Mark 1994)	4/105 (Mark 1994)
Neutropenia	3/45 (Ruckwardt 2021)	3/50 (Ruckwardt 2021)
Eosinophilia	1/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Leukopenia	1/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Lymphopenia	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Bradycardia	2/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Tachycardia	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Burns first degree	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Hepatic enzyme increased	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Abnormal dreams	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Laryngeal pain	0/45 (Ruckwardt 2021)	2/50 (Ruckwardt 2021)
Body tinea	0/36 (NCT03284710)	1/24 (NCT03284710)
Fungal skin infection	0/36 (NCT03284710)	1/24 (NCT03284710)
Furuncle	0/36 (NCT03284710)	1/24 (NCT03284710)
Genitourinary chlamydia infection	5/36 (NCT03284710)	0/24 (NCT03284710)
Genitourinary tract gonococcal infection	1/36 (NCT03284710)	1/24 (NCT03284710)
Pelvic inflammatory disease	1/36 (NCT03284710)	0/24 (NCT03284710)
Subcutaneous abscess	1/36 (NCT03284710)	0/24 (NCT03284710)
Tooth infection	0/36 (NCT03284710)	1/24 (NCT03284710)
Vulvovaginitis trichomonal	1/36 (NCT03284710)	0/24 (NCT03284710)
Arthropod sting	1/36 (NCT03284710)	0/24 (NCT03284710)
Limb injury	0/36 (NCT03284710)	1/24 (NCT03284710)
Skin abrasion	0/36 (NCT03284710)	1/24 (NCT03284710)
Sunburn	1/36 (NCT03284710)	0/24 (NCT03284710)
neutrophil count decreased	1/36 (NCT03284710)	0/24 (NCT03284710)
Abnormal loss of weight	0/36 (NCT03284710)	1/24 (NCT03284710)
Musculoskeletal pain	0/36 (NCT03284710)	1/24 (NCT03284710)
Glycosuria	1/36 (NCT03284710)	1/24 (NCT03284710)
Dysfunctional uterine bleeding	1/36 (NCT03284710)	0/24 (NCT03284710)
Prostatitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Vaginal discharge	1/36 (NCT03284710)	0/24 (NCT03284710)
Hand dermatitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash maculo-papular	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash papular	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash pruritic	1/36 (NCT03284710)	0/24 (NCT03284710)
Infection	4/36 (NCT03284710)	11/24 (NCT03284710)
Impetigo	1/145 (Tapia 2021)	1/144 (Tapia 2021)
Pharyngitis	5/145 (Tapia 2021)	4/144 (Tapia 2021)
Varicella	0/145 (Tapia 2021)	1/144 (Tapia 2021)
Thermal burn	1/145 (Tapia 2021)	0/144 (Tapia 2021)
erythema/induration	9/36 (NCT03284710)	6/24 (NCT03284710)
malaise and or fatigue	8/36 (NCT03284710)	3/24 (NCT03284710)
Max. Systemic Symptoms	13/36 (NCT03284710)	7/24 (NCT03284710)
Ear pain	0/36 (NCT03284710)	0/24 (NCT03284710)
Abdominal pain lower	0/36 (NCT03284710)	0/24 (NCT03284710)
Peripheral swelling	0/36 (NCT03284710)	0/24 (NCT03284710)
Vessel puncture site phlebitis	0/36 (NCT03284710)	0/24 (NCT03284710)
Lower respiratory tract infection	0/36 (NCT03284710)	0/24 (NCT03284710)
Respiratory tract infection	0/36 (NCT03284710)	0/24 (NCT03284710)
Urinary tract infection bacterial	0/36 (NCT03284710)	0/24 (NCT03284710)
Vulvovaginitis gonococcal	0/36 (NCT03284710)	0/24 (NCT03284710)
Eye injury	0/36 (NCT03284710)	0/24 (NCT03284710)
Human bite	0/36 (NCT03284710)	0/24 (NCT03284710)
Skin laceration	0/36 (NCT03284710)	0/24 (NCT03284710)
Investigation	14/36 (NCT03284710)	6/24 (NCT03284710)
Cardiac murmur	14/36 (NCT03284710)	6/24 (NCT03284710)
Blood bicarbonate decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
Platelet count decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
White blood cell count decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
Vulvovaginal rash	14/36 (NCT03284710)	6/24 (NCT03284710)
Rhinalgia	14/36 (NCT03284710)	6/24 (NCT03284710)
Dermatitis	14/36 (NCT03284710)	6/24 (NCT03284710)
Ingrowing nail	14/36 (NCT03284710)	6/24 (NCT03284710)
Musculoskeletal chest pain	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)

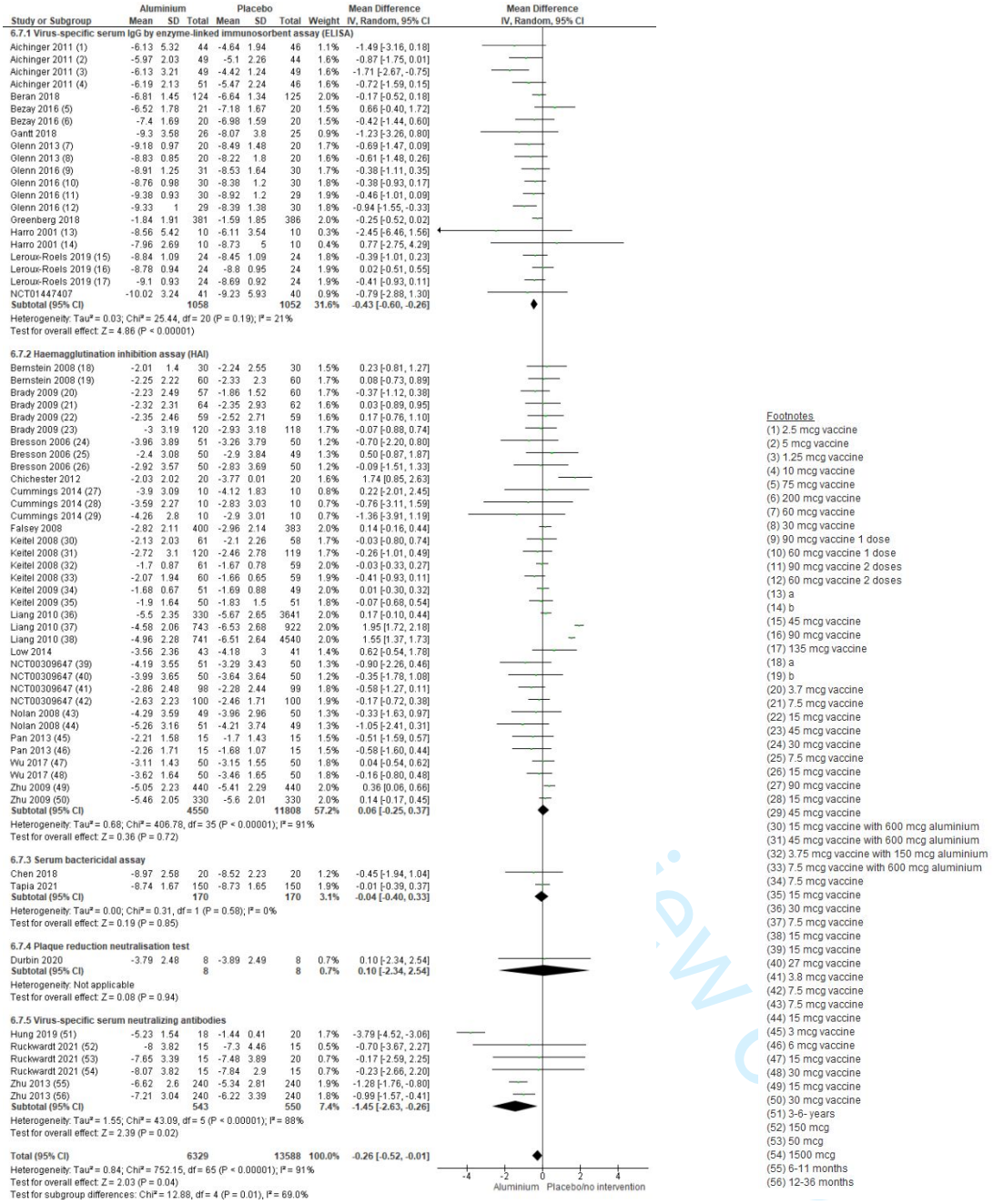
Supplementary Table 3. List of individual non-serious adverse events reported in only one trial.

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Supplementary Figure 32. Overview of the serology analysis with subgroup analysis. We performed subgroup analysis on the most frequently reported analytical assays: enzyme-linked immunosorbent assay (ELISA) and haemagglutination inhibition assay (HAI). We observed a beneficial effect of aluminium adjuvants on serological response in sporadic subgroups (based on vaccine types: respiratory syncytial virus and ross river virus).

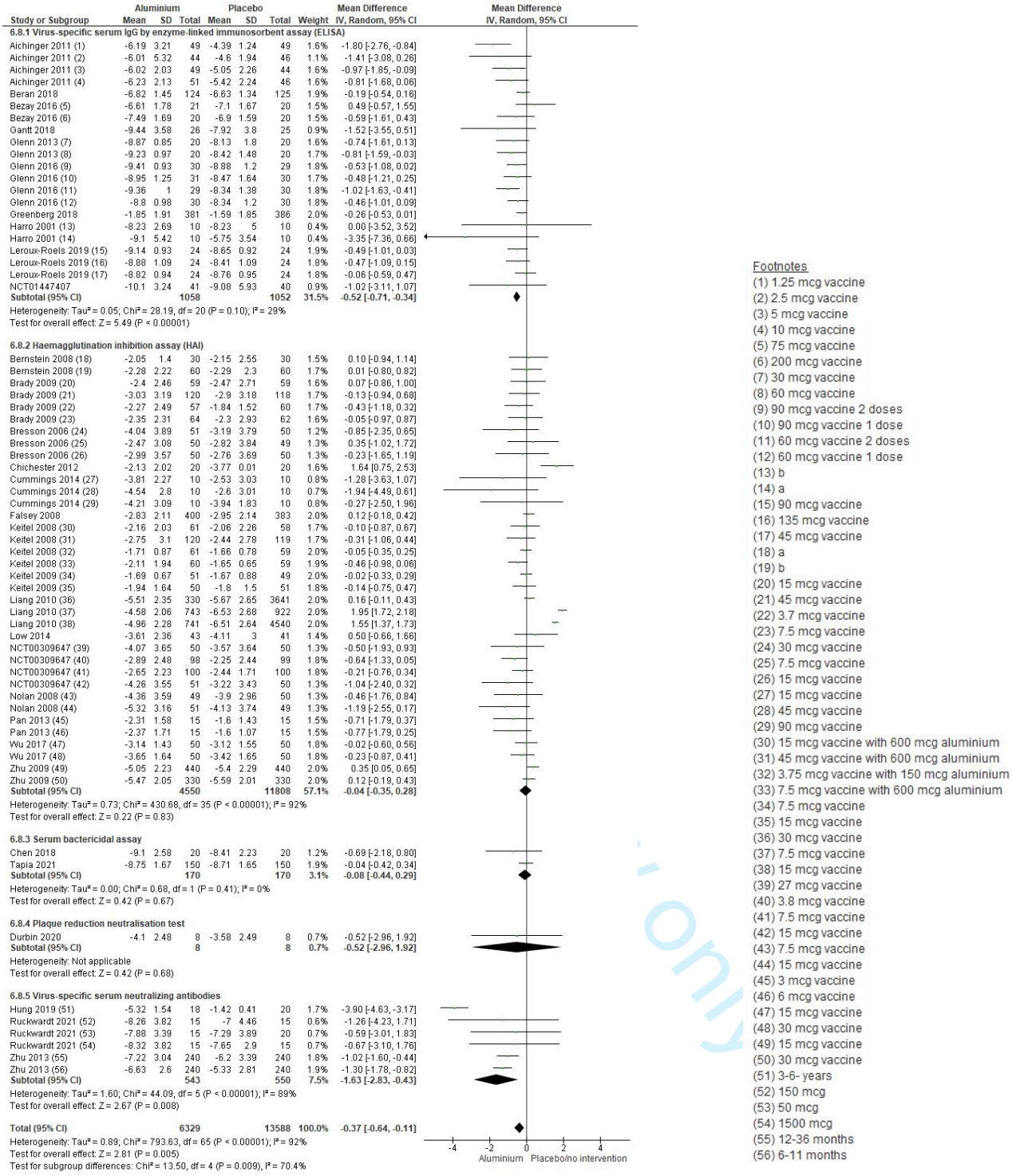
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Supplementary Figure 33. Sensitivity analysis. Best-worst case scenario with 1 standard deviation for GMT.

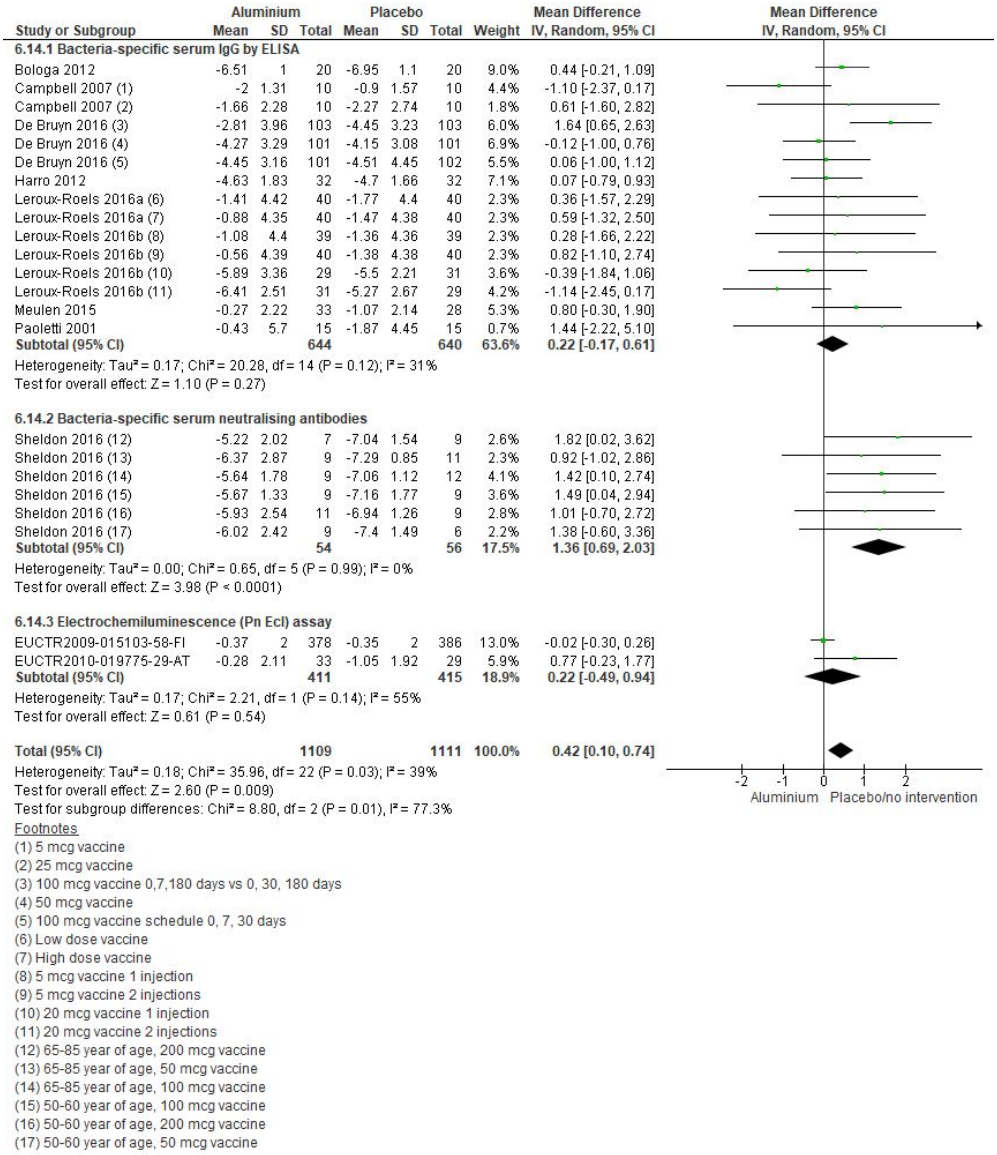
For serology analysed with geometric mean titre (GMT), 777/3950 (19.7%) participants in the intervention group and 2231/10,217 (21.8%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'best-worst' case scenario, so that there was a beneficial effect of aluminium adjuvants in random-effects meta-analysis (best worst with 1 standard deviation: MD -0.26, 95% CI -0.52 to -0.01; participants = 19 917; studies = 66; I^2 = 91%. Best worst with 2 standard deviations: MD -0.37, 95% CI -0.64 to -0.11; participants = 19 917; studies = 66; I^2 = 92%) Incomplete outcome data did not have the potential to influence the result in the 'worst-best' analysis with 1 or 2 standard deviations (data not shown).

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Supplementary Figure 34. Sensitivity analysis. Best-worst case scenario with 2 standard deviations for serology analysed with geometric mean titre (GMT).

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Supplementary Figure 35. Sensitivity analysis. Worst-best case scenario with 2 standard deviations for serology analysed with geometric mean concentration (GMC).

For serology analysed with GMC, 258/784 (32.9%) participants in the intervention group and 249/478 (52.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'worst-best' case scenario, so that there was a harmful effect of aluminium adjuvants in random-effects meta-analysis (2 standard deviations: MD 0.42, 95% CI 0.10 to 0.74; participants = 2220; studies = 23; I^2 = 39%). Incomplete outcome data did not have the potential to influence the result in the 'worst-best' analysis with 1 standard deviation. Incomplete outcome data did not have the potential to influence the result in the 'best-worst' analysis. Meta-analysis of trials with imputed SD showed no evidence of a difference between aluminium adjuvants versus placebo (GMT: MD -0.11, 95% CI -0.33 to 0.12; GMC: MD 0.04, 95% CI -0.17 to 0.26).

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Supplementary Analysis 3.

Subgroup and Sensitivity Analysis for Seroprotection

Test for subgroup differences showed no difference when comparing trials at high risk of bias to trials at low risk of bias; trials at risk of vested interest to trials at low risk of vested interest; and trials with different aluminium types.

Test for subgroup differences was statistically significant (P = 0.0005) in the subgroup analysis according to vaccine type.

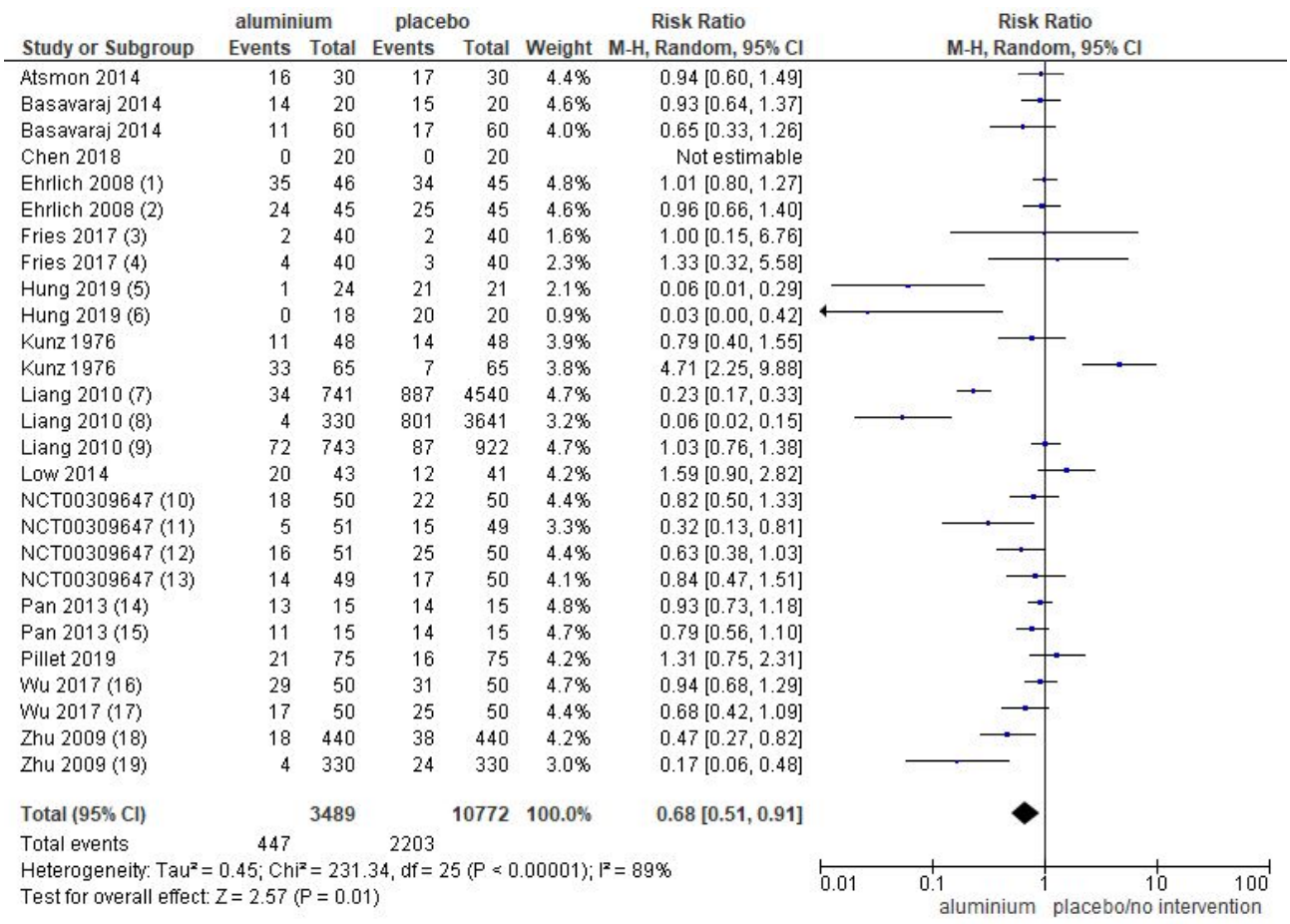
Test for subgroup differences was statistically significant (P = 0.0004) in the subgroup analysis according to age.

Trial Sequential Analysis showed that the accrued information for seroprotection was below 5% of the diversity-adjusted required information size (n = 52 586; Pc 11%, RRR 20%, alpha 2.5%, beta 10%, diversity 82.6%).

242/3165 (7.6%) participants in the intervention group and 1709/9986 (17.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'best-worst' case scenario. The 'best-worst' case analysis showed that incomplete outcome data did not have the potential to influence the result.

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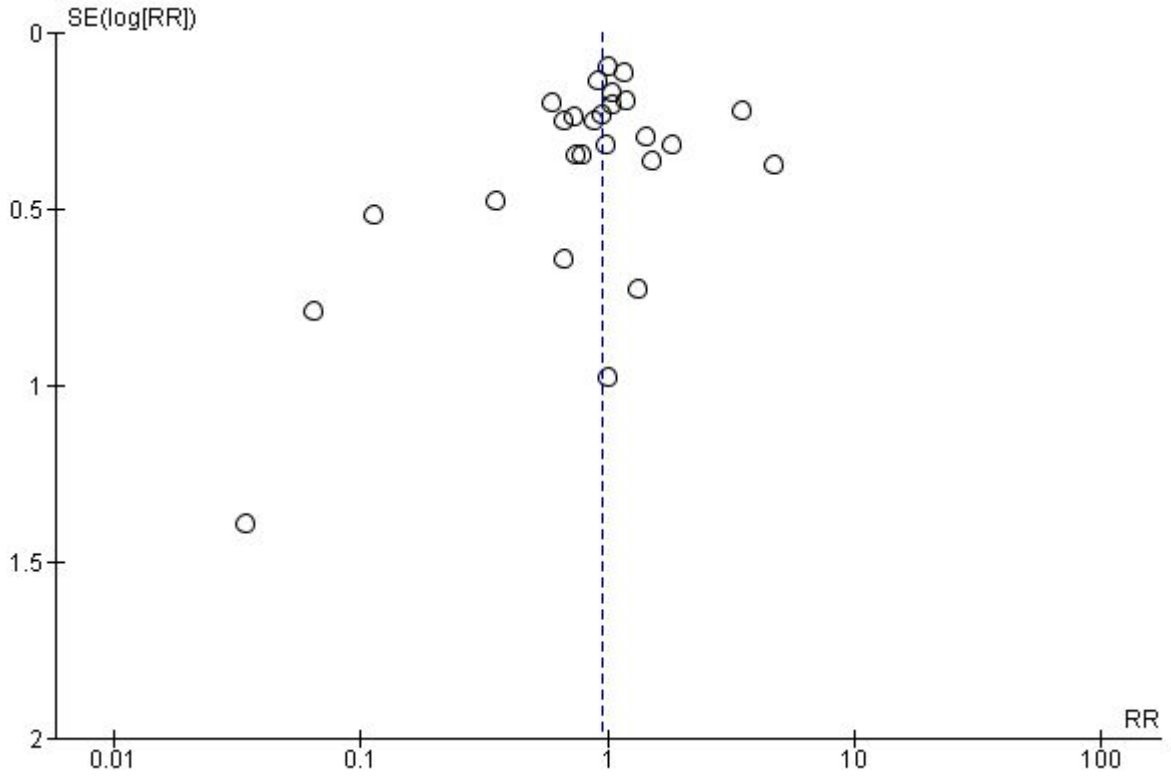
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Footnotes
(1) 7.5 mcg vaccine
(2) 15 mcg vaccine
(3) 90 mcg vaccine
(4) 60 mcg vaccine
(5) 3-6 years of age
(6) 2-35 months of age
(7) 15 mcg vaccine, adults
(8) 7.5 mcg vaccine, adults
(9) 15 mcg vaccine, elderly
(10) 7.5 mcg vaccine
(11) 27 mcg vaccine
(12) 15 mcg vaccine
(13) 3.8 mcg vaccine
(14) 3 mcg vaccine
(15) 6 mcg vaccine
(16) 15 mcg vaccine
(17) 30 mcg vaccine
(18) 15 mcg vaccine
(19) 30 mcg vaccine

Supplementary Figure 36. Sensitivity analysis of seroprotection. Best-worst case scenario for seroprotection. Incomplete outcome data did not have the potential to influence the result in the 'worst-best' (data not shown).

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Supplementary Figure 37. Funnel plot for seroprotection outcome.

Visual inspection of funnel plot may suggest potential publication bias for smaller trials reporting a harmful effect of aluminium adjuvants in the intervention group. Regression-based Harbord test showed no small-study effects (beta = 0.99).

review only

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Supplementary methods

For the serological response, we used the geometric mean concentration or titre (GMC or GMT) extracted from the first assay reported in the journal publication by the trialists. If a trial tested multiple antigens, we included data from the first antigen of the first assay as presented in the publication. For the meta-analysis, we transformed the GMT in the following way: we calculated the natural Log for both the GMT (or GMC) and for the CI.

If SD were not reported, we calculated them using data from the trial, if possible. When standard deviations (SDs) were not reported, we calculated the SD from the CI, we first calculated a t value from the number of participants (n) analysed for that outcome (t value=TINV(1-0.95;n-1), and then we calculated the SD as follows: (SQRT(n))*(LogCIhigh-LogCIlow) / t value). In the meta-analysis, we then plotted the GMT (or GMC) multiplied by -1 and the SD calculated as described above. When imputing the SD due to lack of reporting of CI by the trialists, an average SD was calculated from the other trials and used in the meta-analysis and presented in sensitivity analyses.

We did not impute missing values for any outcomes in our primary analysis. In our sensitivity analysis for dichotomous and continuous outcomes, we did impute data

Unit of analysis issues

We included data from trials where participants were individually randomised to two or more intervention groups. When trials had more than two intervention groups, we considered the following scenarios.

Experimental intervention	Control
Vaccine with different antigen concentrations and same aluminium concentration	Same vaccine with different antigen concentrations without aluminium
Vaccine with different antigen concentration and different aluminium concentration	Same vaccine with different antigen concentration without aluminium
Different aluminium concentration	Placebo

We analysed the scenarios above as follows:

- when the intervention groups had different antigen concentration but same aluminium concentration, we pooled the data from all the intervention groups for the safety analyses, and we analysed each intervention group separately for serology.
- when the intervention groups had different antigen concentration and different aluminium concentration, we analysed each intervention group separately both for safety and for serology.
- when we presented analyses from multiple groups from the same trial, the ID of the same trial appears multiple times in the forest plot together with a footnote describing the difference in the vaccine composition (i.e. antigen concentration) in that trial group.

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Differences between protocol and review

We did not specifically search for observational studies (quasi-randomised studies; cohort studies; and patient series), and we abandoned our potential plan to provide a narrative account of such data.
We originally planned to base our results, conclusions and 'Summary of findings' (SoF) table on trials at low risk of bias. however, due to lack of relevant data, we based our results, conclusions and SoF table on all trials regardless of the bias assessment.
We originally planned to perform the subgroup analysis on the effects of nanoparticle aluminium adjuvant compared to microparticle aluminium adjuvant regardless of type of aluminium salt. However, due to lack of data, we could not perform such analysis.
We originally planned to perform the subgroup analysis on the effect of total administration of aluminium between trials. However, this was replaced by a meta-regression analysis.
We originally planned to perform the subgroup analysis on the effects of aluminium adjuvant administration between trials with different maximal follow-up periods, by defining the follow up period 'from randomisation until the end of follow up'. However, we decided to change the definition of the follow up period to 'from end of the vaccination until the end of follow up'.

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Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Manuscript page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Manuscript page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Manuscript page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Manuscript page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Manuscript page 4; Supplementary page 50; Protocol
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Manuscript page 5; Supplementary Table 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary Table 1;
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 5,6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Manuscript page 5,6; Protocol
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Manuscript page 5,6; Protocol; Supplementary Page 51
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Protocol; Supplementary Page 50
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 6; Protocol
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Manuscript page 6; Supplementary Page 50; Protocol
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Manuscript page 4; Supplementary page 50; Protocol
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing	Supplementary page 50

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Section and Topic	Item #	Checklist item	Location where item is reported
		summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Supplementary Table2,3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Manuscript page 4,5; Supplementary page 50; Protocol
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Manuscript page 5,6; 7 Protocol
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Manuscript page 4,5; Supplementary page 50; Protocol
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Manuscript page 5; Supplementary page 50; Protocol
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Manuscript page 7; Protocol
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Manuscript Page 7; Supplementary Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Appendix 1
Study characteristics	17	Cite each included study and present its characteristics.	Appendix 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Manuscript Page 11 to 17
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	n.a.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Manuscript Page 11 to 17 Supplementary material for subgroup analysis
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Manuscript Page 11 to 17; Supplementary material for subgroup analysis
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Manuscript Page 11 to 17; Supplementary material for sensitivity analysis
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Manuscript Page 11 to 17;

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Section and Topic	Item #	Checklist item	Location where item is reported
			Supplementary material for sensitivity analysis
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Manuscript page 18 to 22
	23b	Discuss any limitations of the evidence included in the review.	Manuscript page 18 to 22
	23c	Discuss any limitations of the review processes used.	Manuscript page 18 to 22
	23d	Discuss implications of the results for practice, policy, and future research.	Manuscript page 18 to 22
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	abstract
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	abstract
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Supplementary page 51
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Manuscript page 23,24
Competing interests	26	Declare any competing interests of review authors.	Manuscript page 25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	n.a.

Supplementary Table 4. PRISMA checklist.

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Characteristics of studies

Characteristics of included studies

Adler 2019

Methods	2-part, double-blind, randomised, placebo controlled, dose escalation multicenter study
Participants	**Sex:** male and female **Age:** >18 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** body weight ≥50 kg and body mass index (BMI) 19–32 kg/m2 and serologically confirmed as CMV seronegative or seropositive **Exclusion criteria:** "subjects were excluded from the trial if they had previously received any CMV vaccine, had a history of allergic reaction or anaphylactic/anaphylactoid reaction that required medical attention to any vaccine component, or had a history of any severe allergic reaction. Other exclusion criteria included recent history of febrile illness (<72 hours); receipt of any live virus vaccine (4 weeks before through 1 month after study vaccine), inactivated vaccine (7 days before through 7 days after study vaccine), or immune globulin or blood product (within 90 days); known or suspected impairment of immunologic function; and recent receipt of immunosuppressive therapies".
Interventions	**Vaccine type:** defective human cytomegalovirus (CMV) vaccine **Experimental:** 100 or 30 Units/dose of V160 with aluminium phosphate adjuvant intramuscularly. n = 30 **Control:** 100 or 30 Units/dose of V160 intramuscularly. n=34 **Aluminium concentration:** 225 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** 3-dose regimen at day 1, month 1, and month 6
Outcomes	**Immunogenicity:** "geometric mean count of peripheral blood mononuclear cells secreting interferon-gamma, (time frame: 1 month after vaccination 3 (vaccination 3 is administered at Month 6)). Geometric mean concentration of interferon-gamma after stimulation of whole blood sample with pooled HCMV antigen peptides (time frame: 1 month after vaccination 3 (vaccination 3 is administered at Month 6)). Geometric mean titer of HCMV-specific neutralizing antibody (time frame: 1 month after vaccination 3 (vaccination 3 is administered at Month 6))". **Safety:** "percentage of participants with an adverse event, (time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6)). Percentage of participants with an injection-site adverse event, (time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6)). Percentage of participants with a systemic adverse event, (time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6)). Percentage of participants with a serious adverse event, time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6))".

Are there other groups in the trial not analysed in this review?	yes
Notes	**Contacts with authors:** on 12/02/2020 authors (luwy_musey@merck.com) were inquired about the followings: ● proportion of participants with any AE related and not related to the vaccine ● proportion of participants who developed CMV infection after the vaccination ● random sequence generation ● allocation concealment **Funding:** this work was supported by Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Kenilworth, NJ.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Subjects were randomised according to their CMV serostatus and gender.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	A study protocol was registered prior to randomisation (NCT01986010), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Aichinger 2011

Methods	Randomized, partially blinded (with respect to adjuvation) clinical trial
Participants	**Sex:** male and females **Age:** 18-40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects were eligible to participate if they were clinically healthy, understood the study procedures, provided written informed consent and agreed to keep a daily record of symptoms". **Exclusion criteria:** "subjects were not eligible if they reported a history of RRV exposure or travel to an RRV-endemic area, a history of any vaccine-related contraindicating event (e.g. anaphylaxis), a history of non-trauma-related arthritis, cardiovascular disease, immunodeficiency, autoimmune disease, inflammatory or degenerative neurological disease, a history of a hematological malignancy or an active neoplastic disease, or if they had a positive urine drug screen or a body mass index > 35, if they had a disease or were undergoing a form of treatment that could be expected to influence immune response. Smoking status was not recorded. Women were required to have a negative pregnancy test at screening and before each vaccination, and were not eligible if lactating".
Interventions	**Vaccine type:** inactivated whole virus Vero cell-derived Ross River virus vaccine **Experimental:** 1.25, 2.5, 5 or 10 mcg aluminium hydroxide-adjuvanted Ross River virus vaccine. n = 194 **Control:** 1.25, 2.5, 5 or 10 mcg Ross River virus vaccine. n = 188 **Aluminium concentration:** 0.2% aluminium hydroxide **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Baxter **Vaccination schedule:** the first two immunizations were administered 21 days apart (range 18–24), with a third immunization six months after the first immunization.
Outcomes	**Immunogenicity:** "the primary immunogenicity endpoint was defined as the RRV-specific IgG antibody titer determined 21 days after the second vaccine dose. Secondary immunogenicity endpoints included the number of subjects with a Ross River virus-specific seropositive IgG antibody response, number of subjects seroconverting to Ross River virus specific IgG antibody production, and the Ross River virus-specific neutralizing antibody response determined 21 days after each of the three vaccinations". **Safety:** "the primary safety outcome was the number of subjects with onset of fever within seven days of the first and second vaccination. Secondary safety outcomes were subjects with injection site reactions and related systemic adverse reactions after the first and second vaccination, any adverse event during the entire study period, and the number of subjects experiencing arthritis associated with one or more systemic symptoms commonly observed in patients with epidemic arthritis occurring at least three days after vaccination and lasting for more than 3 weeks".
Are there other groups in the trial not analysed in this review?	No

Notes	**Contacts with authors:** on 13/12/2018 authors (gerald_aichinger@baxter.com) were inquired about the followings: ● procedure for blinding of outcome assessment ● number of patients randomised per group ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) ● serological response per patient at maximum follow up No answer received. **Funding:** the study was designed and funded by Baxter. The trial only reports on vaccine-related serious adverse events.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomization was carried out by the study site using an electronic data capture system. Randomization was carried out in blocks with a block size >4 in Cohort 1 and >2 in cohorts 2 and 3 using the random number generator algorithm of Wichmann and Hill, as modified by McLeod.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Partially blinded (with respect to adjuvation). The study was double-blinded for use of adjuvant but open-label with respect to dose.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	A total of 382 subjects were randomised and received the first dose of vaccine. All vaccinated subjects were included in the safety analysis.

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	A total of 382 subjects were randomised and received the first dose of vaccine. All vaccinated subjects were included in the safety analysis.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Immunogenicity data were available from 373 subjects after the first vaccination, 375 subjects after the second vaccination, and 360 subjects after the third vaccination.
Selective reporting (reporting bias)	Low risk	The trial is registered at www.clinicaltrials.gov, NCT00717834. All outcomes assessed
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Asmuth 2010

Methods	Multicenter, blinded, randomised, placebo-controlled phase I trial
Participants	**Sex:** male and females **Age:** 18-55 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** healthy subjects between 18 and 55 years of age without evidence of HIV infection were assessed for eligibility. **Exclusion criteria:** "subjects were excluded if they were considered to be at high risk of acquiring HIV infection during the study or had a history of significant chronic medical conditions. Female subjects with childbearing potential were required to have a negative urine pregnancy test immediately before receiving any injections".
Interventions	**Vaccine type:** Human immunodeficiency virus (HIV)-1 gag **Experimental (a):** 5-mg DNA/AlPO4 + booster 5-mg DNA/AlPO4. n = 23 **Control (a):** 5 mg DNA/PBS + booster 5-mg DNA/PBS. n =55 **Experimental (b):** 5-mg DNA/AlPO4 + booster Ad5 gag. n = 56 **Control (b):** 5-mg DNA/PBS + booster Ad5 gag. n =19 **Aluminium concentration:** 700 mcg/mL **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** 4 dose regimen
Outcomes	**Immunogenicity:** neutralizing antibodies to Ad5 were measured. To assess vaccine immunogenicity, unfractionated IFN-g ELISpot assays using 15-mer clade B HIV-1 Gag peptide pools matched to the vaccine inserts were performed on PBMCs obtained at weeks 4, 8, 12, 26, 30, 34, and 42 at Merck Research Laboratories to detect HIV-1 gag-specific T cell responses after vaccination. **Safety:** clinic visits for safety evaluations were scheduled for 2–4 weeks after each dose. Study personnel monitored injection site, systemic, and laboratory adverse events.
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 13/12/2018 authors (david.asmuth@ucdmc.ucdavis.edu) were inquired about the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine. Answer received but no information or data added **Funding:** Financial support: Merck

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Subjects were randomised to receive vaccine or placebo according to allocation schedules generated by computer.
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias)	Low risk	Investigators, subjects, clinical safety monitors, and laboratory personnel performing the biological assays were blinded to treatment assignments.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	n.a.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	n.a.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Investigators, subjects, clinical safety monitors, and laboratory personnel performing the biological assays were blinded to treatment assignments.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.

Incomplete outcome data (attrition bias) Immunogenicity	High risk	Only subjects receiving all scheduled doses of vaccine and without major protocol violations were included in the analysis of immunogenicity. Drop outs more than 5%.
Selective reporting (reporting bias)	High risk	Safety and tolerability not reported. No protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias

Atsmon 2014

Methods	Two-center, randomised, placebo-controlled study
Participants	**Sex:** male and female **Age:** > 65 years **Ethnicity:** not reported **Diagnosis:** healthy but with some medical conditions **Inclusion criteria:** compatible with the Israeli guidelines for receiving the standard seasonal influenza vaccine, who were initially seronegative to at least one of the strains included in the seasonal vaccine against influenza for 2011–2012. **Exclusion criteria:** Any medical condition, clinically significant abnormality upon physical examination or laboratory safety tests,which in the investigator's judgment contraindicated administration of the vaccine, resulted in exclusion.
Interventions	**Vaccine type:** influenza Multimeric-001 (M-001) **Experimental:** influenza Multimeric-001 (M-001) + aluminium phosphate. n = 30 **Control:** influenza Multimeric-001 (M-001). n =30 **Aluminium concentration:** 2% **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Gentaur **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity:** blood was drawn for hemagglutinin inhibition evaluation at baseline (day 0), and 3 weeks after boosting with TIV. Whole blood samples were removed on day 0 (pre-immune) and 21 days after immunization. **Safety:** number of participants with adverse events, (time frame: 63 days). Safety evaluation included physical examination and measurement of vital signs on each vaccination day. Electrocardiogram and safety laboratory tests (blood chemistry and complete blood count) were performed before first vaccination and upon study termination, 3 weeks after TIV administration. Local reactions and systemic adverse events were continuously monitored.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (benyedidia@biondvax.com) were inquired on the followings: ● safety analysis up to 6 weeks (as stated in the NCT page) for serious and non-serious adverse events related and not related to the vaccine ● mortality ● final concentration of aluminium injected per dose ● number of patients who developed influenza after the vaccination

	Answer received but without extra information given **Funding:** the trial was funded by BiondVax Pharmaceuticals Ltd (Israel) and by the Office of Israeli Chief Scientist. The trial only reports on vaccine-related serious adverse events.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A computer-generated algorithm assigned the participants to 32 blocks of four
Allocation concealment (selection bias)	Low risk	An eligible participant was allocated a randomisation number by an independent, un-blinded, research coordinator who was not a member of the study teams. Due
Blinding of participants and personnel (performance bias)	High risk	The study was partially blinded.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	There were no drop outs during the study.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	There were no drop outs during the study.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	There were no drop outs during the study.
Selective reporting (reporting bias)	Low risk	A study protocol was registered prior to randomisation (NCT01419925), and all outcomes were reported on
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ayoola 1986

Methods	Randomised clinical trial
Participants	**Sex:** male and females **Age:** 1-45 years **Ethnicity:** African **Diagnosis:** healthy **Inclusion criteria:** domicile in the city of Ibadan, Nigeria. They were negative for HBV markers at the time of entry into the study. **Exclusion criteria:** not reported
Interventions	**Vaccine type:** hepatitis B Vaccine (Hevac B) **Experimental:** 2 mcg of Hevac B with aluminium hydroxide given subcutaneously. n = 65 **Control:** 2 mcg of Hevac B given intradermal. n = 61 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 doses given one month apart
Outcomes	**Immunogenicity:** hepatitis B surface antigen (HBsAg), antibody to HBsAg (anti-HBs), antibody to hepatitis B core antigen (anti-HBc) were detected by radioimmunoassay. Serum alanine aminotransferase was estimated by standard methods. **Safety:** untoward effects were documented by the completion of a specially designed questionnaires and clinical examinations at each visit.
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** no contact found **Funding:** no funding source reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised
Allocation concealment (selection bias)	Unclear risk	Not stated
Blinding of participants and personnel (performance bias)	Unclear risk	Not stated if blinded
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not stated

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not clear whether drop outs occurred
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not clear whether drop outs occurred
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Not clear whether drop outs occurred
Selective reporting (reporting bias)	Unclear risk	Not clear whether all outcome data is reported. No protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Basavaraj 2014

Methods	Randomized,double-blind, placebo-controlled, parallel assignment
Participants	**Sex:** male and females **Age:** 18-65 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female volunteers aged between 18 to 65 years inclusive Subjects of child-bearing potential must agree to practice adequate contraception that may include, but is not limited to, abstinence, barrier methods such as condoms, diaphragms, spermicides, intrauterine devices, and licensed hormonal methods during the study for at least 30 days following the last vaccination. The subject must be in good health as determined by medical history, physical examination and laboratory screening tests. Subject must be willing and able to comply with planned study procedures and be available for all study visits. Subject must provide written informed consent prior to initiation of any study procedures, and subject may provide written assent as appropriate. Subjects willing to stay for entire study duration and follow-up period in the current residential address". **Exclusion criteria:** "allergy to components of the vaccine including, thiomersal, and aluminium hydroxide. Has a history of previous severe allergic reaction with generalized urticaria, angioedema or anaphylaxis. Female with positive urine pregnancy test within 24 hours prior to vaccination or breastfeeding or unwilling to undergo pregnancy testing. History of smoking 20 pack-years or greater. Has received anticancer chemotherapy or radiation therapy (cytotoxic) within the past 36 months. Has an active neoplastic disease or a history of any hematologic malignancy. Known history of a bleeding disorder contraindicating intramuscular vaccination. History of schizophrenia, bipolar disease, suicidal attempt, major psychiatric illness or confinement for danger to self or others. History of receiving immunoglobulin, anti-cytokine antibodies, immunosuppressants, immune modifying agents or other blood product within the 3 months prior to vaccination in this study. Has received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 3 months prior to vaccination in this study or expected to receive an experimental agent during this study. Has received any live licensed vaccines within 4 weeks

	or inactivated licensed vaccines within 2 weeks prior to vaccination in this study, or plan receipt of such vaccines within 21 days following the second vaccination. Has an acute illness, including an oral temperature greater than >370C, 1 week before the vaccination. Has a chronic neurologic or autoimmune disorder. Has a history of Guillain-Barré Syndrome. Has an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe, or would interfere with the evaluation of vaccine response. Participated in a novel influenza H1N1 vaccine study in the past two years or has a history of novel influenza H1N1 2009 infection or treatment. Has known active HIV, Hepatitis B or Hepatitis C infection. Has a history of alcohol or drug addiction in the last 5 years".
Interventions	**Vaccine type:** H1N1 influenza vaccine HNVAC **Experimental (a):** 15 g of haemagglutinin protein per 0.5 mL adsorbed to 0.2% aluminium hydroxide in phosphate buffer. n = 60 **Control (a):** 15 g of haemagglutinin protein per 0.5 mL in phosphate buffer alone. n = 60 **Experimental (b):** 0.2% aluminium hydroxide in phosphate buffer. n = 20 **Control (b):** phosphate buffer alone. n = 20 **Aluminium concentration:** 0.2% aluminium hydroxide **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 1 dose
Outcomes	**Safety:** "subjects were monitored throughout the study for solicited and unsolicited local and systemic adverse events through diary cards as well as by the physician at the time of sampling. Subjects were monitored for local AEs for7 days and for systemic AEs for 21 days post-vaccination. Laboratory investigations such as haematological, biochemical and urine analyses were done at baseline (day 0) and at 21 days following immunization. Safety studies continued up to 6 weeks post-immunization. Symptoms of Guillian–Barre syndrome were evaluated throughout the study by questionnaire and by examination on days 7, 21 and 42. The subject's well being was enquired at regular intervals and a physician was on call throughout the study period, and any investigational procedures or examination was performed based on investigator's or medical officer's discretion at any time during the study". **Immunogenicity:** "blood was collected for serology studies at days 0 and 21. Serum samples were used to perform haemagglutination inhibition assay. Serological data were converted to seroconversion (SRC), sero-protection (SRP) and geometric mean titres (GMT)".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (hegden@ellafoundation.org) were inquired on the followings: ● number of patients who died/number of patients analysed ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** sponsored by Bharat Biotech International Limited

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Codes generated by the computer program Minitab 14 (Minitab Ltd., USA). The number assigned to a particular subject was unique, and the order in which the groups were assigned within the block of subjects was random.
Allocation concealment (selection bias)	Low risk	A copy of computer generated randomisation list of subject numbers and key were maintained by a non-study member located at the centre as well as the sponsor.
Blinding of participants and personnel (performance bias)	Low risk	Double blind, placebo-controlled. The group assignment was not disclosed to the subject, investigator or the laboratory staff performing the serologic assay.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind, placebo-controlled.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Safety assessment not described.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	The group assignment was not disclosed to the subject, investigator or the laboratory staff performing the serologic assay.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs accounted for
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs accounted for
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	Clinical Trial Registry Numbers: CTRI/2010/091/000152 (Phase I) andCTRI/2010/091/000661 (Phase II/III). All outcomes reported on
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bellanti 2012

Methods	Randomized, partially blinded
Participants	**Sex:** male and females **Age:** 18-45 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** healthy 18 to 45 year-olds of either sex **Exclusion criteria:** "patients receiving chronic medication, are pregnant or intend to become pregnant during the study. Individuals who are HIV positive, have abnormal liver function, had Bacillus anthracis, or received Bacillus anthracis vaccine, have received or intend to receive an experimental vaccine or medication within 30 days of injection of the experimental Bacillus anthracis vaccine".
Interventions	**Vaccine type:** bacillus anthracis **Experimental:** 10 or 20 mcg recombinant mutant protective antigen (rPA) adsorbed to 0.75 mg Alhydrogel and 0.035% formalin. n = 60 **Control:** 10 or 20 mcg recombinant mutant protective antigen (rPA) adsorbed to 0.035% formalin. n = 58 **Aluminium concentration:** 750 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag-Biosector, Frederiksund, Denmark **Vaccination schedule:** 4 doses at 0, 2, 4, and 16 months
Outcomes	**Immunogenicity:** "blood samples were taken before the 1st injection and 1 week after the 1st, 2 months after the 1st (i.e., pre-2nd), 2 months after the 2nd (pre-3rd), 2 weeks after the 3rd, 1 year after the 3rd (pre-4th), 2 weeks after the 4th, and 1 year after the 4th. Anti-rPA IgG was assayed by enzyme-linked immunosorbent assay. Anthrax lethal-toxin-neutralizing assay (TNA) was performed". **Safety:** "vaccinees were observed at the clinic for 30 min after injection and given a diary form and instructions for measuring and recording temperature and local and systemic symptoms at 6 h after injection and daily for 7 days. The vaccinees were also examined at the clinic one time following each injection, between 48 and 72 h post injection. Liver function tests were repeated 1 week after the 1st injection. Telephone inquiries about their health were made every 3 months by study nurses during the 24-month period from the 3rd injection through the year following the 4th".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (schneerr@mail.nih.gov, bellantj@georgetown.edu) were inquired on the followings: ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up

	● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** supported by the Divisions of Intramural Research, NICHD and NIAID, National Institutes of Health.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation schemes were prepared and kept by the Pharmacy Development Section, Pharmacy Department, CC, NIH, and given only to the members of the Data and Safety Monitoring Board
Allocation concealment (selection bias)	Low risk	The randomisation schemes were prepared and kept by the Pharmacy Development Section, Pharmacy Department, CC, NIH, and given only to the members of the Data and Safety Monitoring Board
Blinding of participants and personnel (performance bias)	High risk	Partially blinded; formalin-only-treated rPA was a clear fluid, distinguishable from alum-adsorbed or formalin-treated/alum-adsorbed rPA. The dosages were blinded.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All assays were performed in a blinded fashion.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	One hundred eighty-six volunteers received the 1st injection of the vaccine formulations, 175 the 2nd, 166 the 3rd, and 147 the 4th. An additional 52 volunteers were lost to the 1-year follow-up after the 4th injection.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	High risk	Clinical Trial registration number 0011462. No outcomes stated in the protocol

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Beran 2018

Methods	Rrandomised, observer-blinded, controlled study
Participants	**Sex:** male and females **Age:** 18-45 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects who, in the opinion of the investigator, can and will comply with the requirements of the protocol. Written informed consent obtained from the subject prior to performing any study specific procedure. Non-pregnant female between, and including, 18 and 45 years of age at the time of vaccination. Healthy subjects as established by medical history and clinical examination before entering into the study. Female subjects of non-childbearing potential may be enrolled in the study. Female subjects of childbearing potential may be enrolled in the study, if the subject:has practiced adequate contraception for 30 days prior to vaccination, and has a negative pregnancy test on the day of vaccination and has agreed to continue adequate contraception during the entire study period". **Exclusion criteria:** "use of any investigational or non-registered product other than the study vaccine within 30 days prior to study vaccination, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the subject has been or will be exposed to an investigational or a non-investigational vaccine/product. Chronic administration of immunosuppressants or other immune-modifying drugs, as well as administration of long-acting immune-modifying drugs during the period starting 6 months prior to study vaccination, or planned administration during the study period. Inhaled and topical steroids are allowed. Administration of immunoglobulins and/or any blood products during the period starting 3 months before study vaccination or planned administration during the study period. Planned administration/administration of a vaccine not foreseen by the study protocol in the period starting 30 days before and ending 30 days after the study vaccination, with the exception of any licensed influenza vaccine which may be administered ≥ 15 days before or after study vaccination. Previous experimental vaccination against RSV. Family history of congenital or hereditary immunodeficiency. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. History of or current auto-immune disease. Acute or chronic, clinically significant pulmonary, cardiovascular, hepatic or renal functional abnormality as determined by physical examination and/or Medical History. Lymphoproliferative disorder or malignancy within previous 5 years. History of hypersensitivity after a previous dose of any tetanus, diphtheria, or pertussis vaccine or to any component of Boostrix. History of encephalopathy of unknown aetiology occurring within 7 days following a previous vaccination with pertussis-containing vaccine. History of any neurological disorders or seizures. History of transient thrombocytopenia or neurological complications following a previous vaccination against diphtheria and/or tetanus. History of any reaction or hypersensitivity likely to be exacerbated by any component of the study vaccines. Hypersensitivity to latex. Any medical condition that in the judgment of

	the investigator would make intramuscular injection unsafe. Current chronic alcohol consumption and/or drug abuse. Acute disease and/or fever at the time of enrolment. Body mass index (BMI) > 40 kg/m2. Pregnant or lactating female. Planned move to a location that will prohibit participating in the trial until study end. Any other condition that the investigator judges may interfere with study procedures".
Interventions	**Vaccine type:** respiratory syncytial virus (RSV) vaccine **Experimental:** 60 µg of aluminium-adjuvanted RSV-PreF. n = 125 **Control:** 60 µg of non adjuvated RSV-PreF. n = 124 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** GSK **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity:** "RSV-A neutralizing antibodies on Vero cells were quantified using the RSV plaque-reduction neutralization assay. An indirect enzyme-linked immunosorbent assay format was used to quantify PreF total IgG and IgG subclass 1 antibodies". **Safety:** "number of subjects with abnormal biochemical laboratory values at day 7 and 30. Number of subjects with abnormal haematological laboratory at day 7 and at day 30. Number of subjects with abnormal biochemical laboratory parameter values by maximum grading from day 7 up to day 30. Number of subjects with solicited local symptoms during a 7-day follow-up period after vaccination. Assessed solicited local symptoms were pain, redness and swelling. Any = occurrence of the symptom regardless of intensity grade. Grade 3 pain = pain that prevented normal activity. Grade 3 redness/swelling = redness/swelling spreading beyond 100 millimetres (mm) of injection site. All solicited local symptoms are considered as related to the vaccination. Number of subjects with solicited general symptoms during a 7-day follow-up period after vaccination. Assessed solicited general symptoms were fatigue, temperature (defined as oral temperature equal to or above [≥] 37.5 degrees Celsius [°C] for oral, axillary or tympanic route), gastrointestinal symptoms (gastro) including nausea, vomiting, diarrhoea and/or abdominal pain; and headache. Any = occurrence of the symptom regardless of intensity grade and relationship to the vaccination. Grade 3 symptom = symptom that prevented normal activity. Grade 3 fever = fever ≥ 39.5 °C. Related = symptom assessed by the investigator as related to the vaccination. Number of subjects with unsolicited adverse events during a 30-day follow-up period after vaccination. An unsolicited AE covers any untoward medical occurrence in a clinical investigation subject temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product and reported in addition to those solicited during the clinical study and any solicited symptom with onset out-side the specified period of follow-up for solicited symptoms. Any was defined as the occurrence of any unsolicited AE regardless of intensity grade or relation to vaccination. Number of subjects with serious adverse events from vaccination (day 0) up to study end (day 30). Serious adverse events assessed include medical occurrences that result in death, are life threatening, require hospitalisation or prolongation of hospitalisation or result in disability/incapacity".
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 16/12/2019 authors (jiri.beran@vakcinace.cz) were inquired on the followings: ● which outcome assessors were blinded ● number of participants who were infected by RSV after the vaccination Answer received **Funding:** supported by GSK Biologicals SA, which was involved in all stages of the study conduct and analysis, and also took responsibility for all costs associated with the development and publishing of the present manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants in each study were randomised to study groups of equal size at the investigator sites, using a web-based program. The randomisation algorithm used a minimization procedure that addressed the age of the participant (18–32 years or 33–45 years)
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Only states observer blind
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	All outcome assessors were blinded.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	All outcome assessors were blinded.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All outcome assessors were blinded.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	The safety analysis was performed on the total vaccinated cohort in each study.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The safety analysis was performed on the total vaccinated cohort in each study.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	The analysis of vaccine immunogenicity in RSV F-020 was performed on the according-to-protocol cohort (all vaccinated subjects meeting eligibility criteria and protocol-defined procedures). Drop outs more than 5%
Selective reporting (reporting bias)	Low risk	Two study protocol was registered prior to randomisation (NCT02360475), and all outcomes were reported on.

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bernstein 2008

Methods	Multicenter randomised, double-blind, placebo-controlled trial conducted at 4 sites in the United States
Participants	**Sex:** male and females **Age:** 18-64 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy male or nonpregnant female (as indicated by a negative urine pregnancy test immediately prior to vaccine administration) between the ages of 18 and 64 years, inclusive. Women of childbearing potential (not surgically sterile or post menopausal for greater than or equal to one year) must agree to practice adequate contraception (i.e., barrier method, abstinence, and licensed hormonal methods) for the entire study period. Is in good health, as determined by vital signs (heart rate, blood pressure, oral temperature), medical history and a targeted physical examination based on medical history. Subjects should have normal safety laboratory values (Hgb, WBC, Plt, ALT, and creatinine) prior to the first immunization. Able to understand and comply with planned study procedures. Provides written informed consent prior to initiation of any study procedures". **Exclusion criteria:** "subjects were excluded if they had a known allergy to eggs or other vaccine components; had previously received any influenza A/H5 vaccine; were immunosuppressed; were known to have HIV, hepatitis B, or hepatitis C infection; had a history of Guillain-Barré syndrome; had received blood products in the past 3 months; had received killed vaccines in the past 2 weeks or live vaccines in the past 4 weeks; had an acute or chronic illness that would interfere with evaluations or increase risk; or had a history of severe reactions after influenza vaccination. Subjects were also excluded if they had laboratory values outside of the normal range on their screening blood tests, including hemoglobin level, white blood cell count, platelet count, alanine aminotransferase level, and creatinine level".
Interventions	**Vaccine type:** avian Influenza H5N1 vaccine **Experimental (a):** 15 mcg H5N1 combined with aluminium hydroxide. n = 31 **Control (a):** 15 mcg H5N1. n = 30 **Experimental (b):** 30 mcg H5N1 combined with aluminium hydroxide. n = 59 **Control (b):** 30 mcg H5N1. n = 61 **Aluminium concentration (a):** 175 mcg/dose. **(b)** 300 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Chiron, Novartis **Vaccination schedule:** each subject received 2 intramuscular injections of an identical formulation in the deltoid area, administered approximately 28 days apart.
Outcomes	**Immunogenicity:** "geometric mean titer and frequency of 4-fold or greater increases in serum hemagglutination inhibition (HAI) antibody titers in each group 28 days after receipt of the second dose of vaccine (time frame: approximately day 56). Proportion of subjects in each group achieving a serum neutralizing antibody titer ratio of 1:40 against the influenza A/H5N1 virus 28 days following second dose of vaccine, (time frame: approximately day 56).

	Proportion of subjects in each dose group achieving a serum hemagglutination (HAI) antibody titer of 1:40 against the influenza A/H5N1 virus 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Geometric mean titer (GMT) and frequency of 4-fold or greater increases in neutralizing antibody titers in each group 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Geometric mean titer and the frequency of 4-fold or greater increases in neutralizing antibody titers in each group 1 month after receipt of each dose, and 7 months after receipt of the first dose of vaccine, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization). Geometric mean titer and the frequency of 4-fold or greater increases in serum HAI antibody titers 1 month after receipt of each dose, and 7 months after receipt of the first dose of vaccine, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization). Development of serum antibody responses against antigenically drifted variants of H5N1influenza virus, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization)". **Safety:** "adverse event or SAE information (solicited in-clinic and via memory aids, concomitant medications, and periodic targeted physical assessments), (time frame: adverse events will be collected through 28 days following the second dose of vaccine (approximately day 56). Serious adverse events will be collected throughout the study through day 208)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (david.bernstein@cchmc.org) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** National Institutes of Health

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Eligible subjects were randomly assigned to 1 of 9 groups. No further information on how randomisation was achieved.
Allocation concealment (selection bias)	Unclear risk	Allocation concealment not described. After randomisation, an unblinded member of the study team who did not otherwise participate in the trial administered the injection.
Blinding of participants and personnel (performance bias)	Unclear risk	Only stated double blind but no information on how the blinding was achieved and maintained.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	The site staff who evaluated vaccine reactogenicity were fully blinded to the dose and type of vaccine given.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	The site staff who evaluated vaccine reactogenicity were fully blinded to the dose and type of vaccine given.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00280033), and all outcomes were reported on.
Other bias	Low risk	SAE reported only as related to the vaccine.

Bezay 2016

Methods	Open-label, partially randomised, dose-escalation Phase I trial in two parts
Participants	**Sex:** male and females **Age:** > 65 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "no clinically relevant pathological findings in any of the investigations at the Screening Visit including subjects with pharmacologically controlled conditions like hypercholesterolemia, hypertension, cardiovascular disease or non insulin-dependent diabetes mellitus. Minor deviations of laboratory values from the normal range may be accepted, if judged by the

investigator to have no clinical relevance. Weight: ≥ 45.5 kg and <= 150 kg at Visit 0 (Screening Visit). White blood cells ≥2,500/mm3 and <11,000/mm3 at Visit 0.
Platelets within normal limits at Visit 0. Written informed consent obtained from the subject prior to any study related procedures".
Exclusion criteria: "use of any other investigational or non-registered medicinal product within 30 days prior to IC84 vaccination at Visit 1 (Day 0) and throughout the entire study period.
Active or passive vaccination four weeks before first vaccination at Visit 1 and during the entire study period. Immunodeficiency including status post-organ-transplantation or immuno-suppressive therapy, and a family history of congenital or hereditary immunodeficiency. Infection with the human immunodeficiency virus (HIV, a negative test result within 30 days before screening is acceptable), Hepatitis B virus (HBV, Hepatitis B surface antigen [HBsAg]) or Hepatitis C virus (HCV). History of severe hypersensitivity reactions and anaphylaxis.
History of allergic bronchial asthma and severe allergic rhinoconjunctivitis. Known hypersensitivity or allergic reactions to one of the components of the vaccine. History of autoimmune disease, including Type I Diabetes mellitus. Subjects with vitiligo or thyroid disease taking thyroid hormone replacement are not excluded. Any malignancy in the past 5 years. If treatment for cancer was successfully completed more than 5 years ago and the malignancy is considered to be cured, the subject may be enrolled. Clinically significant hematological, renal, hepatic, pulmonary, central nervous, cardiovascular or gastrointestinal disorders, which are not adequately controlled by medical treatment within the last 12 weeks before IC84 vaccination at Visit 1 (Day 0) as judged by the site's Principal Investigator. Systemic antibiotic use within four weeks prior to first vaccination and during treatment period (until Day 28). Clinically significant diseases as judged by the investigator. Administration of chronic (defined as longer than 14 days) immunosuppressants or other immune-modifying drugs within 30 days prior to IC84 vaccination at Visit 1 (Day 0) and during the study until Visit 5 (Day 28). (For corticosteroids this means prednisone or equivalent >= 0.05 mg/kg/day; topical and inhaled steroids are allowed.). Periodic steroid injections, e.g., intra-articular, are not allowed within 30 days prior to first IC84 vaccination at Visit 1 (Day 0) and until Visit 5 (Day 28). Intake of NSAID within three days prior to and within three days after all three vaccinations (Day 0, 7 and 21). Acute febrile infections or exacerbation of chronic infection within four weeks prior to first vaccination at Visit 1 and during treatment period (until Day 21). Body temperature > 37°C (oral) immediately prior to each vaccination. Drug addiction within 6 months prior to Visit 0 [including alcohol dependence, i.e. more than approximately 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day] indicated by an elevated MCV above normal value at the Screening Visit. Inability or unwillingness to avoid more than the usual intake of alcohol [i.e., not more then 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day] during the 48 hours after vaccination. Pregnancy (positive pregnancy test during screening), lactation or unreliable contraception in female subjects with child-bearing potential (and unreliable contraception in male subjects. Not able to understand and comply with protocol requirements, instructions and protocol-stated restrictions. Blood donation within 4 weeks prior to first vaccination. Clinically significant mental disorder not adequately controlled by medical treatment. History of Guillain-Barré-Syndrome (GBS). Any condition which might interfere with study objectives or would limit the subject's ability to

	complete the study in the opinion of the investigator. Inability or unwillingness to provide informed consent. Persons who are committed to an institution (by virtue of an order issued either by the judicial or the administrative authorities)".
Interventions	**Vaccine type:** VLA84 vaccine against *Clostridium difficile* **Experimental:** VLA84 75 or 200mcg with aluminium hydroxide. n =41 **Control:** VLA84 75 or 200 mcg. n = 40 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** unclear **Vaccination schedule:** 4 doses at day 0, 7, 28, 56. Immunizations were injected intramuscularly into the deltoid of the non-dominant arm.
Outcomes	**Immunogenicity:** "determination of vaccine (IC84) specific IgG levels on Day 0, 7, 14, 21, 28, 113 and 201 after the first vaccination using an Enzyme-Linked Immunosorbent Assay, (time frame: day 201). Determination of *Clostridium difficile* Toxin A- and Toxin B-specific IgG levels on Day 0, 7, 14, 21, 28, 113 and 201 after the first vaccination using an Enzyme-Linked Immunosorbent Assay, (time frame: day 201). Determination of vaccine-induced C. difficile Toxin A- and Toxin B-neutralizing antibody levels on Day 0, 7, 14, 21, 28, 113 and 201 after the first vaccination using a Toxin Neutralization Assay, (time frame: day 201)". **Safety:** "rate of subjects with any serious adverse event possibly, probably or certainly related to the study vaccine at any time during the study, (time frame: day 201). Rate of subjects with any unsolicited or solicited Grade 3 or higher adverse event possibly, probably or certainly related to the study vaccine at any time during the study, (time frame: day 201). Rate of subjects with any unsolicited or solicited Grade 3 or higher adverse event possibly, probably or certainly related to the study vaccine during treatment phase (i.e. until day 28), (time frame: day 201). Rate of subjects with solicited local adverse events (injection site pain, tenderness, redness, swelling, induration, itching) within 1 week (Day 0-6) after each vaccination: Severity and duration, (time frame: 6 days after vaccination). Rate of subjects with solicited systemic adverse event (headache, muscle pain, fever, flu-like symptoms, nausea, vomiting, rash, excessive fatigue) within 1 week (Day 0 6) after each vaccination: Severity and duration, (time frame: 6 days after vaccination). Rate of subjects with unsolicited non-serious adverse events (including safety laboratory parameters (hematology, serum chemistry, urinalysis)) within 6 months after last vaccination: Severity and causality, (time frame: day 201). Rate of subjects with unsolicited non-serious adverse events (including safety laboratory parameters (hematology, serum chemistry, urinalysis) during treatment phase (i.e., until day 28): Severity and causality, (time frame: day 28)".
Are there other groups in the trial not analysed in this review?	Yes.
Notes	**Contacts with authors:** on 16/12/2019 authors (bernd.jilma@meduniwien.ac.at, nicole.bezay@valneva.com) were inquired on the followings: ● final aluminium concentration per dose ● aluminium manufacturer ● Method of allocation concealment ● randomization sequence generation

	● total number of serious adverse event related and not related to the vaccine ● number of participants who got Clostridium difficile infection after the vaccine Answer received but no further data added **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only stated randomised. No further information on how randomisation was achieved
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No dropouts
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	No dropouts
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01296386), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bologa 2012

Methods	A phase I, single-center, randomised, placebo controlled, repeated administration, dose-escalating, observer blinded study
Participants	**Sex:** male and females **Age:** 18-50 years **Ethnicity:** not stated **Diagnosis:** healthy **Inclusion criteria:** "aged 18 to 50 years on the day of inclusion, iInformed consent form has been signed and dated, able to attend all scheduled visits and comply with all trial procedures Subject is healthy, as determined by medical history and physical examination, for a woman of child-bearing potential, use of an effective method of contraception or abstinence from at least 4 weeks prior to first vaccination until 4 weeks after the last vaccination". **Exclusion criteria:** "known pregnancy or positive serum/urine pregnancy test, currently breastfeeding a child, chronic illness, that in the opinion of the investigator, is at a stage that could interfere with trial conduct or completion, participation in another clinical trial investigating a vaccine, drug, medical device, or a medical procedure in the 4 weeks preceding the first trial vaccination, planned participation in another clinical trial during the present trial period, known or suspected congenital or acquired immunodeficiency, or receipt of immunosuppressive therapy such as anti-cancer chemotherapy or radiation therapy within the preceding 6 months; or long-term systemic corticosteroid therapy (prednisone or equivalent for more than 2 consecutive weeks within the past 3 months), known systemic hypersensitivity to any of the vaccine components or history of a life threatening reaction to the trial vaccine(s) or to a vaccine containing any of the same substances, current alcohol abuse or drug addiction that may interfere with the subject's ability to comply with trial procedures. Receipt of blood or blood-derived products in the past 3 months, that might interfere with the assessment of immune response, receipt of any vaccine in the 4 weeks preceding the first trial vaccination, planned receipt of any vaccine in the 4 weeks following the trial vaccination, known seropositivity for Human Immunodeficiency Virus (HIV), Hepatitis B or Hepatitis C. Laboratory confirmed / self-reported thrombocytopenia contraindicating intramuscular vaccination, deprived of freedom by an administrative or court order, or in an emergency setting, or hospitalized involuntarily, identified as a study site employee who is involved in the protocol and may have direct access to trial-related data, previous vaccination against pneumococcal disease (in the previous 5 years), history of pneumococcal infection (confirmed either clinically, serologically, or microbiologically) within 5 years, at high risk for pneumococcal infection during the trial, living in a household with children < 5 years of age".
Interventions	**Vaccine type:** pneumococcal protein vaccine PhtD-Pcp **Experimental:** 25 mcg PhtD 25 mcg PcpA adjuvated with aluminium hydroxide. n = 21 **Control:** 25 mcg PhtD 25 mcg PcpA. n = 19 **Aluminium concentration:** 280 mcg/dose **Aluminium type:** aluminium hydroxide Alhydrogel pre-treated with phosphate buffer **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** 2 injections 30 days apart

Outcomes	**Safety:** number and percentage of participants reporting solicited injection site reactions, solicited systemic reactions, unsolicited systemic reactions, and serious adverse events occurring throughout the trial, (time frame: 30 days post-vaccination). Solicited injection site reactions: pain, erythema, and swelling. Solicited systemic reactions: fever, headache, malaise, and myalgia. **Immunogenicity:** immunogenicity of investigational pneumococcal vaccines after 2 vaccinations, (time frame: days 0 and 30 post-vaccination). Evaluation of immune responses to antigen component of the investigational vaccine by enzyme linked immunosorbent assay.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (monica@mbpharma.ca, sanjay.Gurunathan@sanofipasteur.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no information or data added **Funding:** funded by Sanofi Pasteur, which also contributed to the study design, data collection, analysis and interpretation, review of the manuscript and decision to publish.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further information on how randomisation was achieved.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Observer blinded study. Unclear how blinding was achieved and maintained.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Observer blinded study. Blinded safety reviews by a Sanofi Pasteur safety review committee.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Observer blinded study. Blinded safety reviews by a Sanofi Pasteur safety review committee.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Both full analysis set and per protocol set (only results from per protocol presented)
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	No drop outs
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01444339), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Brady 2009

Methods	Multicenter, randomised, dose-ranging clinical trial
Participants	**Sex:** male and females **Age:** >65 **Ethnicity:** mixed **Diagnosis:** ambulatory and judged to be medically stable for any underlying conditions **Inclusion criteria:** "participants are ambulatory adults aged 65 and older, are in good health, as determined by vital signs (heart rate less than 100 bpm, blood pressure [systolic less than or equal to 160 mm Hg and diastolic less than or equal to 90 mm Hg] oral temperature less than 100.0°F), medical history to ensure stable medical condition, and targeted physical examination based on medical history, are able to understand and comply with planned study procedures, provide written informed consent prior to initiation of any study procedures, stable medical condition - no change in prescription medication, dose, or frequency of medication in the last 3 months and health outcomes of the specific disease are considered to be within acceptable limits in the last 6 months. Any change that is due to change of health care provider, insurance company, etc, or that is done for financial reasons, as long as in the same class of medication will not be considered a violation of this inclusion criterion. Any change in prescription medication due to improvement of a disease outcome will not be considered a violation of this inclusion criterion." **Exclusion criteria:** "have a known allergy to eggs or other components of the vaccine (including gelatin, formaldehyde, octoxinol, thimerosal, aluminium hydroxide, and chicken protein). Have immunosuppression as a result of an underlying illness or treatment, or used anticancer chemotherapy or radiation therapy within the preceding 36 months. Have any malignancy (excluding nonmelanotic skin cancer) or lymphoproliferative disorder diagnosed or treated actively during the past 5 years. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (> 800 micrograms/day of beclomethasone dipropionate or equivalent) within the preceding 6 months (Nasal and topical steroids are allowed.) Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to

	self or others. Are receiving psychiatric drugs (aripiprazole, clozapine, ziprasidone, haloperidol, molindone, loxapine, thioridazine, thiothixene, pimozide, fluphenazine, risperidone, mesoridazine, quetiapine, trifluoperazine, chlorprothixene, chlorpromazine, perphenazine, trifluopromazine, olanzapine, carbamazepine, divalproex sodium, lithium carbonate or lithium citrate.) Subjects who are receiving a single antidepressant drug and stable for at least 3 months prior to enrollment, without de-compensating symptoms will be allowed to be enrolled in the study. Have a history of receiving immunoglobulin or other blood product within the 3 months prior to enrollment in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to enrollment in this study. Have an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses. These conditions include, but are not limited to: history of significant renal impairment (dialysis and treatment for kidney disease, including diabetic and hypertensive kidney disease); subjects with diabetes mellitus, well-controlled with oral agents may enroll as long there has been no dose adjustment within the past 6 months; insulin-dependent diabetes is excluded; cardiac insufficiency, if heart failure is present (New York Heart Association Functional Class III or IV); an arteriosclerotic event during the 6 months prior to enrollment (e.g., history of myocardial infarction, stroke, recanalization of femoral arteries, or transient ischemic attack). Have a history of severe reactions following immunization with influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week prior to vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product or medication) within 1 month prior to enrollment in this study, or expects to receive an experimental agent during the 7- to 8-month study period. Have any condition that would, in the opinion of the site investigator, place the subject at an unacceptable risk of injury or render the subject unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Planned to travel outside of the USA in the time between the first vaccination and 56 days following the first vaccination. Have any diagnosis of dementia or associated concomitant medications (e.g., Aricept) used for treating dementia. Have a history of Guillain Barre syndrome. Have any condition that the investigator believes may interfere with successful completion of the study".
Interventions	**Vaccine type:** subvirion Inactivated Influenza A/H5N1 vaccine **Experimental** **(a):** 3.75 mcg of A/H5N1 vaccine with aluminium hydroxide. n = 57 **Control (a):** 3.75 mcg of A/H5N1 vaccine. n = 60 **Experimental (b):** 7.5, 15 or 45 mcg of A/H5N1 vaccine with aluminium hydroxide. n = 243 **Control (b):** 7.5, 15 or 45 mcg of A/H5N1 vaccine. n = 239 **Aluminium concentration: (a)** 300 mcg/dose. **(b)** 600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** sanofi pasteur, Swiftwater, PA **Vaccination schedule:** two doses administered approximately 28 days apart into the deltoid muscle

Outcomes	**Immunogenicity:** "geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Proportion of subjects in each dose group achieving a serum hemagglutination inhibition (HAI) antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Proportion of subjects in each dose group achieving a serum neutralizing antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Development of serum antibody responses against antigenically drifted variants of H5N1 influenza virus, (time frame: day 0, and at days 28, 56, and 208 after the first immunization). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group 1 month and 7 months after receipt of the first dose of vaccine, (time frame: day 0, and at days 28, 56, and 208 after the first immunization). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group 1 month and 7 months after receipt of the first dose of vaccine, (time frame: day 0, and at days 28, 56, and 208 after the first immunization)". **Safety:** "adverse event or serious adverse event information (solicited in the clinic and via memory aids, concomitant medications, and periodic targeted physical assessment), (time frame: adverse events will be collected through 28 days following the 2nd dose of vaccine (day 56). Serious adverse events will be collected throughout the study through day 208)".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (hhill@emmes.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** research supported by the authors and summarized in this report was supported by Public Health Service Contracts N01-AI-25459, N01-AI-25465, N01-AI-25460, N01-AI-25461, N01-AI-45250, and N01-AI-30068 from the National Institute of Allergy and Infectious Diseases. Partial support was also provided by the University of Maryland General Clinical Research Center Grant

	M01-RR-16500, General Clinical Research Centers Program, National Center for Research Resources, National Institutes of Health. Wilbur Chen, M.D. receives support from K12RR023250.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further information on how randomisation was achieved.
Allocation concealment (selection bias)	Unclear risk	No description of allocation concealment.
Blinding of participants and personnel (performance bias)	Low risk	All subjects as well as site and laboratory staff, except the vaccinators, were blinded to the vaccines administered. The vaccinators were not involved in the assessment of responses after immunization.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	600 enrolled, 599 received 1 vaccination, 545 received 2 vaccination. The subject who received an incorrect dose as the first vaccine was excluded from the safety and immunogenicity analyses. Two subjects who did not receive the second vaccine and did not provide post-vaccination blood samples were excluded from all immunogenicity analyses. One subject did not return for a blood sample 28 days after the first vaccination and was consequently not included in the immunogenicity analyses for this time point, but this subject was included in all other analyses for which data were available.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	High risk	600 enrolled, 599 received 1 vaccination, 545 received 2 vaccination. The subject who received an incorrect dose as the first vaccine was excluded from the safety and immunogenicity analyses. Two subjects who did not receive the second vaccine and did not provide post-vaccination blood samples were excluded from all immunogenicity analyses. One subject did not return for a blood sample 28 days after the first vaccination and was consequently not

		included in the immunogenicity analyses for this time point, but this subject was included in all other analyses for which data were available.
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00294099), and all outcomes were reported on. However, SAE reported only as related to the vaccine. Thirty-seven serious adverse events, including two deaths, were reported during the study. None of these SAEs was considered to be associated with vaccination. Not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bresson 2006

Methods	Randomised, open-label, non-controlled phase I trial
Participants	**Sex:** male and females **Age:** 18–40 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "age 18–40 years on day of enrolment, provision of written informed consent, ability to attend all scheduled visits and to comply with all trial procedures, entitled to national social security, and registered in the French file of healthy volunteers in clinical trials. Women had to be unable to bear children or able to produce a negative urine pregnancy test, having used an effective method of contraception or been abstinent for at least 4 weeks before the first vaccination (to be continued for at least 4 weeks after the second vaccination)". **Exclusion criteria:**" current febrile illness; immunodeficiency; recent receipt of blood, blood derive products, or immunosuppressive treatment; current long-term systemic corticosteroid therapy; recent (previous 4 weeks) vaccination or vaccination planned in next month; and pregnancy".
Interventions	**Vaccine type:** inactivated split-virion influenza A/Vietnam/1194/2004 (H5N1) vaccine **Experimental:** 7.5, 15 or 30 mcg H5N1 vaccine combined with aluminium hydroxide and 45 mcg of thiomerosal. n = 151 **Control:** 7.5, 15 or 30 mcg vaccine and 45 mcg of thiomerosal. n = 149 **Aluminium concentration:**600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** unclear **Vaccination schedule:** two intramuscular injections, 21 days apart, of the assigned formulation into the deltoid
Outcomes	**Immunogenicity:** "serum samples obtained from individuals on day 0 (before first vaccination), 21, and 42 were assayed at the same time to detect haemagglutination-inhibition activity and neutralising activity against the vaccine seed virus (NIBRG14). Haemagglutination-inhibition assay was used. Neutralising antibody activity according was performed". **Safety:** "participants attended three trial visits (on day 0, 21, and 42) for blood sampling, vaccination (day 0 and 21 only), and safety data collection. Iindividuals were kept under observation for 30 min after vaccination, and were given safety diaries, digital

	thermometers, and rulers to assess and record adverse events. From days 0 to 7, diaries included a list of solicited injection site and systemic adverse events, including those recommended for the assessment of interpandemic vaccines by the CHMP. The intensity of non-quantifiable reactions—e.g., pain—was assessed with a severity scale: mild, noticeable but does not interfere with daily activities; moderate, interferes with daily activities; severe, prevents daily activities. At the next visit (day 21 and 42), patients were interviewed, adverse events transcribed onto case report forms, and unsolicited adverse events were assessed whether vaccine-related. By convention, all solicited adverse events we judged vaccine-related".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (maria.zambon@hpa.org.uk) were inquired about the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further information or data added **Funding:** funded by a grant from sanofi pasteur. Sanofi pasteur had a role, through its project team, in study design and monitoring, and in writing of the report.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation lists was stratified by centre and created with the block method, using decreasing block sizes of 18, 12, and six so that a similar number of people was enrolled into each group at any given time. The sequence was generated at sanofi pasteur by a statistician who was not involved in the rest of the trial.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described

Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	All 300 individuals received the two planned vaccinations, attended all visits, completed the study up to day 42, gave three blood samples, and were included in analyses.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	All 300 individuals received the two planned vaccinations, attended all visits, completed the study up to day 42, gave three blood samples, and were included in analyses.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	All 300 individuals received the two planned vaccinations, attended all visits, completed the study up to day 42, gave three blood samples, and were included in analyses.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Brooks 2015

Methods	Phase I, single-center, randomised, placebo-controlled, observer-blind study
Participants	**Sex:** male and females **Age:** infants 42-49 days of age **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "aged 6 to 7 weeks (42 to 49 days) for infants and 12 to 13 months for toddlers on the day of inclusion. Born at full term of pregnancy (≥ 37 weeks) and having a weight ≥ 60% of the median weight for Bangladesh as measured by the weight-to-age index · Written informed consent obtained from the parents or legal guardians of infants and toddlers. Subject and parent/guardian was able to attend all scheduled visits and to comply with all trial procedures". **Exclusion criteria:** "participation or planned participation in another clinical trial investigating a vaccine, drug, medical device, or medical procedure in the 4 weeks preceding the first trial vaccination except if non-interventional follow-up for an earlier study (e.g., long-term surveillance). Receipt of any vaccine in the 4 weeks preceding the first trial vaccination or planned receipt of any other vaccine than study vaccines in the 4 weeks following any trial vaccination. Previous vaccination against *S. pneumoniae.* History of *S. pneumoniae* infection, confirmed microbiologically. Receipt of blood or blood-derived products in the past 30 days (infants) that might interfere with assessment of the immune response. Known or suspected congenital or acquired immunodeficiency; receipt of immunosuppressive therapy such as anti-cancer chemotherapy or radiation therapy; or long-term systemic corticosteroid therapy (prednisone or equivalent for more than 2 consecutive weeks. Known systemic hypersensitivity to any of the vaccine components, or history of a

	life-threatening reaction to the vaccine used in the trial or to a vaccine containing any of the same substances. Bleeding disorder, including thrombocytopenia contraindicating intramuscular vaccination, or receipt of anticoagulants in the 3 weeks preceding inclusion. History of seizures. Chronic illness that, in the opinion of the investigator, was at a stage where it could have interfered with trial conduct or completion. Parents or guardians identified as employees of the investigator or study center with direct involvement in the proposed study or other studies under the direction of that investigator or study center, as well as family members (i.e., immediate, husband, wife and their children, adopted or natural) of the employees or the investigator. Febrile illness (temperature ≥ 38.0°C) or moderate or severe acute illness/infection (according to Investigator judgment) on the day of vaccination. Receipt of oral or injected antibiotic therapy within 72 h prior to any blood draw".
Interventions	**Vaccine type:** trivalent recombinant PcpA, PhtD,and PlyD1 Pneumococcal protein vaccines (PPrVs) **Experimental:** 25 mcg PPrV + phosphate-treated aluminium hydroxide. n = 40 **Control:** 25 mcg PPrV. n = 40 **Aluminium concentration:** 280 mcg/dose **Aluminium type:** phosphate-treated aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** infants received three injections at 6, 10, and 14 weeks of age
Outcomes	**Immunogenicity**: "serum anti-PhtD, anti-PcpA, and anti-PlyD1 concentrations at day 0 and month 1 for adults and toddlers and at day 0, month 2, and month 3 for infants. Enzyme-linked immunosorbent assays were used used to measure anti-PcpA, anti-PhtD and anti-PlyD1 antibody concentrations". **Safety:** "occurrence,time to onset, duration, action taken, and intensity of solicited reactions; occurrence, nature, and intensity of all unsolicited AEs; and occurrence, nature, time to onset, duration, seriousness criteria, relationship to vaccination, and outcome for SAEs. For systemic unsolicited AEs, relationship to the vaccine, action taken, and whether the event occurred within 30 min after vaccination were also recorded. Solicited reactions were recorded daily for 7 days following each vaccination. AEs and SAEs were defined as described in the International Conference for Harmonization E2A Guidelines. Unsolicited AEs were recorded up to 30 days after each vaccination. SAEs were recorded throughout the study. Immediate AEs were defined as those occurring within 30 min after the vaccination".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (robert.hopfer@sanofipasteur.com) were inquired about the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs

	related and not related to the vaccine) No answer received **Funding:** funded by Sanofi Pasteur. Sanofi Pasteur participated in study design; the collection, analysis and interpretation of data; writing of the report; and the decision to submit the article for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Treatments were assigned using scratch-off lists, which were centrally generated by non-stratified randomisation with a fixed block size within cohorts.
Allocation concealment (selection bias)	Low risk	Centrally generated scratch-off lists
Blinding of participants and personnel (performance bias)	Low risk	Observer-blind study. The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	One drop out in experimental group and 2 drop outs in control group.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	One drop out in experimental group and 2 drop outs in control group.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (WHO Universal Trial Number U1111-1117-7316; NCT01446926), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Brown 2010

Methods	Randomized, double-blind, placebo controlled
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "citizens of the U.S., age 18 to 40 years, for women, a negative serum pregnancy test will be required at study entry and within 24 hours prior to each vaccination, as well as verbal assurance that adequate birth control measures are applied prior to initial vaccination and for 3 months after the last vaccination. Good health as determined by medical history, physical examination, and clinical judgment, normal baseline clinical laboratory values at screening, availability for at least 13 months of follow-up from the time of the screening visit, successful completion of the Test of Understanding defined as 90% correct with three opportunities to take test, commitment for trial participation and signature of the approved consent form". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** rPA Anthrax vaccine **Experimental:** 5, 25, 50 or 100 mcg rPA with Alhydrogel. n = 30 **Control:** 5, 25, 50 or 100 mcg rPA. n = 32 **Aluminium concentration:** 704 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** BioSector (Frederikssund, Denmark) **Vaccination schedule:** two intramuscular injections in the left deltoid four weeks apart
Outcomes	**Immunogenicity:** "longer term safety and immune response data was collected at visits 6, 10, 16, 26, 36, and 52 weeks after the initial injection. Enzyme-linked immunosorbent assay, lymphocyte proliferation assay and toxin neutralization assay were performed". **Safety:** "safety data was analysed and reported on every subject at every available time point immediately after initial injection. These include immediate reactions and at 48–72 hours and 2 weeks post each injection. Longer term safety and immune response data was collected at visits 6, 10, 16, 26, 36, and 52 weeks after the initial injection. Serious adverse events were solicited at all time points through the 52 week visit. Diary of local reactions such as pain, tenderness, or warmth and systemic reactions such as fatigue, headache, or diarrhea for one week post-immunization which was used as memory aid to identify solicited adverse events (post injection reactions). The post injection reactions were recorded separate from other adverse events unless a post injection reaction persisted beyond one week post-injection".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (mrobb@hivresearch.org) were inquired about the followings: ● procedure for blinding of participants and personnel ● funding source ● final concentration of aluminium injected ● aluminium manufacturer

	● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● serological response per patient at maximum follow up No answer received **Funding:** funded by the Chemical Biological Medical Systems-Joint Vaccine Acquisition Program. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript. SAE reported only as related to the vaccine.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The study statistician prepared a randomisation list using the RANUNI function in SAS Version 8 (SAS Institute, Cary, NC USA); only the study pharmacist and the statistician had access to this randomisation list, and study personnel were unblinded only after study completion.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind (study personnel were unblinded only after study completion)
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Immunogenicity data was analysed and reported on all subjects who provided a sample at the time point being assessed. Two subjects were excluded from the immunogenicity analyses; one subject who was randomised to receive placebo inadvertently was administered vaccine product (50 mg of rPA with PBS) at the second inoculation and thus was included in safety analysis, but not immunogenicity analysis. One subject, randomised to the 100 mg rPA with PBS arm, had prior undisclosed Bacillus anthracis vaccine exposure as determined by detectable levels of anti-rPA antibodies prior to injection of vaccine. This volunteer denied Bacillus anthracis vaccine exposure at entry and after unblinding reported never being aware of receiving an Bacillus anthracis vaccine. This subject was also included

		in safety analysis, but excluded from the immunogenicity analysis.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00057525), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Butler 1969

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 3-6 months old **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** combined diphtheria, tetanus, and pertussis vaccines **Experimental:** vaccine with Alhydrogel. n = 56 **Control:** vaccine without aluminium. n = 47 **Aluminium concentration:** 2500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 injections every 4 weeks
Outcomes	**Immunogenicity** and **safety**
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Funding**: not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs more than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not deacribed
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Campbell 2007

Methods	Phase I, randomised study
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** rPA anthrax vaccine **Experimental:** 5 or 25 mcg rPA with Alhydrogel. n = 20 **Control:** 5 or 25 mcg rPA. n = 20 **Aluminium concentration:** 800 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injection at day 0 and day 28
Outcomes	**Immunogenicity:** "serum was collected immediately prior to immunization and at post-vaccination days 14, 28, 42, 120 and 210 for evaluation of the humoral immune response. Investigators performed two serologic assays: an enzyme-linked immunosorbent assay and a toxin neutralization assay. The main serologic endpoints for each dose group were the following: (1) the proportion of volunteers in each vaccine group whose day 42 anti-PA IgG antibody concentrations measured by enzyme-linked immunosorbent assay were greater than or equal to four times their day zero concentrations and no less than 10 mg/mL (four times the lower limit of quantification of the assay), (2)

	the geometric mean concentration of anti-PA antibody on day 42 for volunteers in each group, (3) the proportion of volunteers in each group with a four-fold increase in ED50 from day 0 to day 42, and (4) the geometric mean titer of neutralizing antibodies, measured by ED50, for each group on day 42". **Safety:** "investigators observed volunteers for 30 minutes following vaccination to record and treat any immediate side effects. Investigators surveyed volunteers at 30 minutes post-vaccination using a pain assessment tool on which is recorded the severity, quality and length of time of injection site pain. Volunteers recorded reactogenicity data, including an oral temperature, in a diary on the day of each vaccination and the six subsequent days. Local and systemic effects were recorded. Adverse events and serious adverse events were evaluated and recorded for the entire study period. Safety laboratory tests (complete blood count, chemistries, liver and renal function tests) were done prior to vaccination and at post-vaccination days 7 and 35 (data not shown)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (jcampbel@medicine.umaryland.edu) were inquired about the followings: ● aluminium manufacturer ● method of allocation concealment ● randomization sequence generation ● total number of participants with serious and non-serious adverse event related and not related to the vaccine. ● number of participants who got Bacillus anthracis infection after the vaccine No answer received **Funding:** this work was supported by the National Institutes of Health Vaccine and Treatment Evaluation Unit, Contract N01-AI-25461, and University of Maryland General Clinical Research Center Grant M01 RR 16500, General Clinical Research Centers Program, National Center for Research Resources (NCRR), NIH.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Ranomized. No further information on how this was achieved.
Allocation concealment (selection bias)	Unclear risk	Allocation to vaccine within each group was randomly assigned. Unclear how this was achieved.
Blinding of participants and personnel (performance bias)	Low risk	Neither the investigators nor the volunteers were made aware of the vaccine assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported

Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	No drop outs
Selective reporting (reporting bias)	Unclear risk	No pre-registered protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Chen 2018

Methods	Single-centre, double-blind, randomised controlled trial
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "age 18-45 years, written informed consent of volunteers, healthy as established by medical history, laboratory evaluation and screening evaluations, participants are able to understand and comply with planned study procedures and be available for all study visits, female subjects must be of non-childbearing potential (defined as surgically sterile or postmenopausal for more than 1 year), or if of childbearing potential must be practicing abstinence or using an effective licensed method of birth control". **Exclusion criteria:** "previous vaccination against Neisseria meningitidis. Known exposure to Neisseria meningitidis in the past. History of meningitis or seizures or any neurological or psychiatric disorder. Administration of any other vaccine within 30 days prior or after administration of study vaccines. Use of any investigational or non-registered drug or vaccine within 30 days prior to the administration of study vaccines or planned during the study. History of allergic disease or known hypersensitivity to any component of the three study vaccines. History of Serious Adverse Reactions following administration of Tetanus Toxoid, Diphtheria Toxoid or CRM containing vaccines. History of Guillan-Barré syndrome. Confirmed or suspected immunosuppressive or immune-deficient condition. 10. A family history of congenital or hereditary immunodeficiency. Chronic administration (defined as more than 14 days) of immune-suppressants or other immune-modifying agents within six months prior to administration of study vaccine. 12. Laboratory confirmed infection of either hepatitis B virus (HBs Ag positive on ELISA), hepatitis C virus (anti-HCV positive on ELISA as well as PCR) or human immunodeficiency virus (HIV on ELISA). Major congenital defects or serious chronic illness. Acute or chronic, clinically significant pulmonary, cardiovascular, hepatic, or renal functional

	abnormality, as determined by medical history, physical examination or laboratory screening tests. Known bleeding disorders. Administration of immunoglobulins and/or any blood products within the three months preceding the administration of study vaccines or planned administration during the vaccine period. History (within the past year) or signs of alcohol or substance abuse. Pregnancy or lactation. A Body Mass Index (BMI) equal to or above 30. Any other condition, which in the opinion of the investigator, might interfere with the study objectives, jeopardize the safety or rights of the participant or making it unlikely the participant could complete the protocol".
Interventions	**Vaccine type:** NnmCV-5 pentavalent meningococcal conjugate vaccine **Experimental:** aluminium-phosphate adjuvated NmCV-5. n = 20 **Control:** NmCV-5. n = 20 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium-phosphate **Aluminium manufacturer:** unclear **Vaccination schedule:** one intramuscular injection
Outcomes	**Immunogenicity:** "percentage of participants with ≥4-fold increase in post-immunization rabbit complement serum bactericidal activity, (time frame: 28 days post vaccination); the percentage of participants with an serum bactericidal activity titer of 1:8 and 1:128, (time frame: 28 days post vaccination). Geometric mean titers of Meningococcal Polysaccharide A, C, Y, W, and X specific antibodies, (time frame: 28 days post vaccination)". **Safety:** "reactogenicity assessed by occurrence, severity of solicited local and systemic reactions, (time frame: day 7 post vaccination); occurrence of unsolicited adverse events by severity and relatedness to vaccination, (time frame: 28 days post vaccination); occurrence of serious adverse events, (time frame: 180 days post vaccination)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (wchen@som.umaryland.edu) were inquired about the followings: ● aluminium manufacturer ● method of allocation concealment ● randomization sequence generation ● blinding of outcome assessors ● total number of participants with non-serious adverse event related and not related to the vaccine. ● number of participants who got meningococcal disease after the vaccine No answer received **Funding:** UK Department for International Development

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation sequence was generated by the trial statistician in SAS (version 9.3) using a permuted block design with randomly chosen block sizes of three and six. Upon completion of entry of a participant's enrolment

		information, a concealed treatment assignment number was provided in the electronic data entry system.
Allocation concealment (selection bias)	Low risk	Upon completion of entry of a participant's enrolment information, a concealed treatment assignment number was provided in the electronic data entry system. The number was used by the unmasked research pharmacist to prepare the allocated product for each participant, using a treatment code list that was maintained by the pharmacist in a secure location, to maintain blinding. The vaccines were filled into identical-appearing syringes and labelled in a manner such that each syringe could be matched to the study participant but did not provide unblinding information, and they were administered with procedures to ensure participants and prospectively assigned study personnel remained masked to treatment
Blinding of participants and personnel (performance bias)	Low risk	Double blind. Participants and prospectively assigned study personnel remained masked to treatment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. Unclear in regards to outcome assessors
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. Unclear in regards to outcome assessors
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind. Unclear in regards to outcome assessors
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02810340), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Chichester 2012

Methods	Phase 1 randomised, double-blind, placebo-controlled, dose-escalation clinical study
Participants	**Sex:** male and female **Age:** 18–49 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "male or female aged 18 to 49 years inclusive; able to give written informed consent to participate; healthy as determined by medical history, physical examination, vital signs, and clinical safety laboratory examinations at baseline; females should fulfil one of the following criteria: at least 1 year post-menopausalSurgically sterileWill use oral, implantable, transdermal, or injectable contraceptives for 30 days prior to first vaccination and until 28 days after each vaccination, willing to abstain from sexual intercourse or use another reliable form of contraception approved by the investigator (e.g., intrauterine device [IUD], female condom, diaphragm with spermicide, cervical cap, use of condom by the sexual partner, or a sterile sexual partner) for study duration and until 28 days after vaccination. Women of childbearing potential must have a negative urine pregnancy test within 24 hours preceding receipt of each dose. Comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits and to be contacted by telephone throughout the follow-up period". **Exclusion criteria:** "prior receipt of any influenza vaccine containing H5; presence of significant uncontrolled medical or psychiatric illness (acute or chronic); includes institution of new medical or surgical treatment or a significant dose alteration for uncontrolled symptoms or drug toxicity within 3 months of screening and reconfirmed on day 1 prior to vaccination Positive serology for human immunodeficiency virus (HIV 1 or HIV 2), hepatitis B surface antigen. or hepatitis C; cancer or treatment for cancer within 3 years excluding basal cell carcinoma or squamous cell carcinoma which are allowed; presence of any medical condition that may be associated with impaired immune responsiveness, including diabetes mellitus Presently receiving (or has a history of receiving) during the preceding 3 month period any medications or other treatments that may adversely affect the immune system, such as allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs; other drugs known to be frequently associated with significant major organ toxicity; or systemic (oral or injectable) corticosteroids (inhaled and topical corticosteroids allowed); receipt or planned administration of a non study vaccine within 30 days prior to vaccination and during the study (Immunization on an emergency basis with Tetanus Toxoids Adsorbed for adult use vaccine—tetanus and diphtheria or tetanus, diphtheria, and pertussis up to 8 days before or at least 8 days after a dose of study vaccine will be allowed, and administration of study vaccine injection can be delayed if a non study vaccine has been administered and will be given as soon as acceptable [as described above] provided the vaccine is not administered within 2 weeks prior to study enrollment.); history of significant allergic reactions (anaphylactic-type reactions, respiratory difficulties, or angioedema) to injected vaccines or to any of the study vaccine components; history of drug or chemical abuse in the year before the study; positive urine drug screen at screening; receipt of any investigational product or non registered drug within the 30 days prior to vaccination or currently enrolled in any investigational drug study or intends to enroll in such a study within the ensuing study period; receipt of blood

	or blood products 8 weeks prior to vaccination or planned administration during the study period; donation of blood or blood products within 8 weeks prior to vaccination or at any time during the study; acute disease within 72 hours prior to vaccination defined as the presence of a moderate or severe illness (as determined by the investigator through medical history and physical examination) with or without fever, or a fever >38ºC orally (Study vaccine can be administered to persons with a minor illness, such as diarrhea, or mild upper respiratory tract infection with or without low-grade fever, and vaccination can be delayed until the subject has recovered.); any condition that, in the opinion of the investigator, might interfere with the primary study objectives or the subject's safety".
Interventions	**Vaccine type:** plant-produced recombinant hemagglutinin-based influenza vaccine (HAI-05) Derived from A/Indonesia/05/2005 (H5N1) influenza virus **Experimental:** 90 mcg HAI-05 + Alhydrogel. n = 20 **Control:** 90 mcg HAI-05. n = 20 **Aluminium concentration:** 750 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Denmark **Vaccination schedule:** two injections three weeks apart intramuscularly
Outcomes	**Immunogenicity:** "sera for immunogenicity assessments were collected on days 1 (pre-vaccination; baseline), 22 (pre-vaccination; results for the first vaccination) and 43 (results for the second vaccination). Immunogenicity assessments included determination of 1) geometric mean titres of HAI and MN antibody titers; 2) the proportion of subjects in each group who seroconverted on Days 22 and 43 based on HAI and MN antibody titers; and 3) the proportion of subjects in each group who achieved HAI titers ≥40 on Days 22 and 43". **Safety:** "reactogenicity events, adverse events, clinical laboratory tests (serum chemistry, hematology and urinalysis), vital signs, and abnormal findings upon physical examination. Any unsolicited symptoms were recorded at 30 minutes after vaccination by the clinical site staff, and thereafter through day 43 by subjects in a supplemental diary".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (vyusibov@fraunhofer-cmb.org, jchichester@fraunhofer-cmb.org) were inquired about the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** funded by The Bill and Melinda Gates Foundation (grant Nr. 49840)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further description
Allocation concealment (selection bias)	Unclear risk	No details in regards to allocation concealment
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further description
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not stated
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No drop outs
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	No drop outs
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	No drop outs
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01250795), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Coller 2016

Methods	Abstract: phase I, randomised, placebo-controlled, double-blind study
Participants	**Sex:** not reported **Age:** not reported **Ethnicity:** not reported **Diagnosis:** not reported. Subjects who had previously received one or two doses of live-attenuated tetravalent vaccine **Inclusion criteria:** not reported **Exclusion criteria:** not reported

Interventions	**Vaccine type:** dengue vaccine V180 **Experimental:** V180 adjuvated with Alhydrogel. n = 8 **Control:** V180. n = 8 **Aluminium concentration:** not reported **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity:** plaque reduction neutralization test at Day 0, Day 14, Day 28, and Month 6 relative to vaccination. **Safety:** solicited and unsolicited adverse events were assessed using a Vaccination Report Card for 28 days following vaccination. Serious adverse events were captured from the time of informed consent through the final study visit at 6 months postvaccination.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (beth-ann.coller@merck.com) were inquired about the followings: ● aluminium manufacturer ● final aluminium concentration injected per dose ● method of allocation concealment ● randomization sequence generation ● blinding of outcome assessors ● blinding of participants and personnel ● immunogenicity response expressed as geometric mean titre or concentration per intervention arm ● total number of participants with non-serious adverse event related and not related to the vaccine ● number of participants who got dengue disease after the vaccine Answer received but without extra information **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not reported
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported

Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Collier 1979

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** not reported **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** all children participating were accepted by their school as having had at least the primary course of diphtheria-tetanus-pertussis vaccine or diphtheria tetanus vaccine in infancy. **Exclusion criteria:** not reported
Interventions	**Vaccine type:** tetanus vaccine BP **Experimental:** aluminium hydroxide-adsorbed Tetanus Vaccine. n = 134 **Control:** plain tetanus Vaccine. n = 132 **Aluminium concentration:** 400 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** unclear **Vaccination schedule:** 1 dose
Outcomes	**Safety and immunogenicity:** "the investigation was designed to determine (1) the incidence, nature, and severity of reactions to reinforcing doses of tetanus toxoid given to school-leavers; (2) the relationship of reactions with the tetanus antibody titre at the time of inoculation; (3) whether reactions can be lessened by using plain formol toxoid (FT), and if so, whether the antibody response to such a vaccine is adequate; and (4) the necessity for a booster dose of tetanus vaccine at the time of leaving school".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** no contact found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Randomised given alternately to the children
Allocation concealment (selection bias)	High risk	Allocated at random by giving them alternately as the children presented themselves
Blinding of participants and personnel (performance bias)	High risk	Not blinded
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Examinations were shared by S.P. and L.H.C. who were unaware of which vaccines the children had received.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Examinations were shared by S.P. and L.H.C. who were unaware of which vaccines the children had received.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Not all participants included in all outcomes in all time points
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Cox 2009

Methods	Open label, phase I, dose escalating clinical trial
Participants	**Sex:** male and female **Age:** 19–39 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** all subjects aged 19–39 years having normal haematology, biochemistry and immunology parameters **Exclusion criteria:** "a history of anaphylaxis or hypersensitivity to vaccines, an oral temperature >38 ○C (in the preceding 72 h) or an acute respiratory infection (up to 7 days) prior to immunisation, pregnancy, breast-feeding women, recent

	treatment with blood products, vaccination with any vaccine during the 4 weeks preceding the first trial vaccination, known or suspected immunosuppressive or chronic illness, and the use of systemic immunosuppressive drugs or immunostimulant therapy.Women were required to use reliable contraception from 4 weeks before the first vaccination until 4 weeks after the second vaccination and have a negative urine pregnancy test prior to administration of each dose of vaccine".
Interventions	**Vaccine type:** split H7N1 influenza virus vaccine **Experimental (a):** 12 mcg haemagglutinin influenza vaccine with aluminium hydroxide. n = 15 **Control (a):** 12 mcg haemagglutinin influenza vaccine. n = 15 **Experimental (b):** 24 mcg haemagglutinin influenza vaccine with aluminium hydroxide. n = 15 **Control (b):** 24 mcg haemagglutinin influenza vaccine. n = 15 **Aluminium concentration: (a)** 300 mcg/dose. **(b)** 600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injections into the deltoid muscle separated by 21 (±1) sos
Outcomes	**Immunogenicity:** "blood samples were collected pre, 3, 7 (±1), 14 (±2) and 21 days after each vaccine dose, and 6 (±1) months after the second vaccination. Lymphocytes were immediately analysed for antibody secreting cells and cytokine responses. Pre- and post-vaccination serum samples from each individual were run blinded in duplicate on at least two occasions in the haemagglutination inhibition and microneutralisation". **Safety:** "injection site reactions were evaluated at the end of a 45min observation period and any area of erythema or induration recorded. Volunteers were provided with a diary card on which they recorded the incidence of local (erythema/redness, local itching, oedema/local swelling, pain at the injection site, ecchymosis, hardness/induration) and systemic reactions (fever,malaise, shivering/chills, fatigue, headache, sweating, myalgia, arthralgia, respiratory symptoms and diarrhoea). Any use of antipyretic/relief medication and occurrence of intercurrent illness was also recorded. The diary card was reviewed at each visit to the clinic".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (rebecca.cox@gades.uib.no) were inquired about the followings: ● method for random sequence generation ● method for allocation concealment ● Fifnal concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● serological response per patient at maximum follow up No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further description
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	5/60 participants withdrew (reasons reported). These participants were included in the immunogenicity and safety analyses up to the point of withdrawal from the study.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	5/60 participants withdrew (reasons reported). These participants were included in the immunogenicity and safety analyses up to the point of withdrawal from the study.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	5/60 participants withdrew (reasons reported). These participants were included in the immunogenicity and safety analyses up to the point of withdrawal from the study.
Selective reporting (reporting bias)	Low risk	The trial was registered in the WHO Initiative for Vaccine Research and the EudraCT Database (EudraCT no. 2006-001582-42).
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Cummings 2014

Methods	First-in-human, phase 1, single-center, randomised, placebo-controlled, single-blind, dose-escalation clinical study
Participants	**Sex:** male and female **Age:** 18–50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "male or female aged 18 - 50 years inclusive; able to give written informed consent to participate; score of at least 80% correct on a 10 question multiple-choice quiz (2 attempts); healthy, as determined by medical

	history, physical examination, weight, vital signs, and clinical safety laboratory examinations at baseline; females must fulfil one of the following criteria: At least one year post-menopausal; surgically sterile or have a surgically sterile partner; Willing to abstain from sexual intercourse; willing to use a reliable form of contraception approved by the investigator (e.g., oral, implantable, transdermal or injectable contraceptives, intrauterine device (IUD), female condom, diaphragm with spermicide, cervical cap, or male condoms) for 30 days prior to first vaccination through 3 months after second vaccination; women of childbearing potential must have a negative urine pregnancy test within 24 hours preceding receipt of each dose; comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits and to be contacted by telephone throughout the follow-up period". **Exclusion criteria:** "prior receipt of 2010-2011 seasonal influenza vaccine containing A/California/04/09-like virus; screening H1N1 titer of > 1:40; presence of significant uncontrolled medical or psychiatric illness (acute or chronic) including institution of new medical or surgical treatment or a significant dose alteration for uncontrolled symptoms or drug toxicity within 3 months of screening; presence of any medical condition that may be associated with impaired immune responsiveness, including diabetes mellitus; cancer, or treatment for cancer, within the previous 3 years, excluding basal cell carcinoma or squamous cell carcinoma; presently receiving or history of receiving, during the preceding 3-month period, any medications or other treatments that may adversely affect the immune system: This includes allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs or other drugs known to be frequently associated with significant major organ toxicity, or systemic corticosteroids (oral or injectable); Inhaled and topical corticosteroids will be allowed; receipt of blood or blood products 8 weeks prior to vaccination or planned administration during the study period; donation of blood or blood products within 8 weeks prior to vaccination or at any time during the study; receipt or planned administration of a non study vaccine within 30 days prior to vaccination; Immunization on an emergency basis with Tetanus Toxoids Adsorbed for adult use (Td or Tdap) vaccine up to 8 days before or at least 8 days after a dose of study vaccine will be allowed; history of anaphylactic type reaction to injected vaccines; receipt of any investigational product or non registered drug within the 30 days prior to vaccination or currently enrolled in any investigational drug study or intends to enrol in such a study within the ensuing study period; history of drug or chemical abuse in the year before the study; positive serology for HIV-1 or HIV-2, or HBsAg or HCV antibodies; unwilling to allow storage of specimens for future use; acute disease within 72 hours prior to vaccination: Acute disease is defined as the presence of a moderate or severe illness (as determined by the Investigator through medical history and physical examination) with or without fever (>38ºC />100.4ºF), or an oral temperature of >38ºC orally; study vaccine can be administered to persons with a minor illness; any condition that, in the opinion of the investigator, might interfere with interpretation of data supporting the primary study objectives".
Interventions	**Vaccine type:** hemagglutinin-based influenza vaccine derived from influenza A (H1N1) pdm09 virus. **Experimental:** 15, 25 or 90 mcg H1N1 with Alhydrogel. n = 30 **Control:** 15, 25 or 90 mcg H1N1. n = 30 **Aluminium concentration:** 750 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Denmark

	Vaccination schedule: two doses in the deltoid muscle of the same non-dominant arm at Day 0 and Day 21
Outcomes	**Immunogenicity:** "sera for immunogenicity assessments were collected on day 0 (pre-vaccination), day 21 (pre-vaccination) and day 56, and sent to the Influenza Division, CDC for analysis by hemagglutination inhibition assay and microneutralization assays. Immunogenicity assessments included determination of (1) geometric mean titer of hemagglutination inhibition assay and microneutralization antibody titers; (2) the proportion of subjects in each group who seroconverted on days 21 and 56; and (3) the proportion of subjects in each group who achieved an hemagglutination inhibition titer of ≥40 or an microneutralization titer of ≥80 or ≥160 on days 21 and 56". **Safety:** "proportion of subjects in each treatment group reporting one or more local or systemic solicited and unsolicited reactions within 7 days after vaccination (days 0–6 and 21–27), and the occurrence of any serious adverse events, defined as any untoward events resulting in hospitalization or death,during the study. Clinical safety laboratory assessments were performed at baseline, day of vaccination, and 7 days after each vaccination. The presence of solicited local and general signs and symptoms, including measurement of oral temperature, were assessed after each vaccination and at 1, 2, 3, 7 and 14 days post-vaccination. Also, the investigators recorded any other adverse events occurring on the first administration of the vaccine through day 56 as unsolicited adverse events".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (james.f.cummings6.mil@mail.mil, Jamesfcummings@gmail.com) were inquired about the followings: ● Method of allocation concealment ● Randomization sequence generation ● Blinding of outcome assessors ● Total number of participants with non-serious adverse event related and not related to the vaccine ● Number of participants who got influenza after the vaccine No answer received **Funding:** the HAC1 clinical trial and serologic testing were supported by the Defense Advanced Research Projects Agency. The cGMP manufacture of HAC1 was supported by DARPA under contract number # HR0011-10-C-0051.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Single blind

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	2/80 lost to follow up or discontinued
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01177202), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

De Bruyn 2016

Methods	Randomized, placebo controlled, Phase 2 study
Participants	**Sex:** male and female **Age:** 40-75 years **Ethnicity:** mixed **Diagnosis:** at risk of *Clostridium difficile* infection **Inclusion criteria:** "aged 40 to 75 years on the day of inclusion; informed consent form has been signed and dated; able to attend all scheduled visits and to comply with all trial procedures; for a woman of childbearing potential, use of an effective method of contraception or abstinence from at least 4 weeks prior to the first vaccination until at least 4 weeks after the last vaccination; at risk for developing Clostridium difficile infection during the trial because of impending elective surgery or hospitalization within 60 days of enrollment, or current or impending residence in a long-term care facility or rehabilitation facility". **Exclusion criteria:** "known pregnancy, or a positive urine pregnancy test; currently breastfeeding a child; participation in another clinical trial investigating a vaccine, drug, medical device, or medical procedure in the 4 weeks preceding the first trial vaccination; planned participation in another clinical trial during the present trial period; receipt of any vaccine in the 4 weeks preceding the first trial vaccination, except for influenza (seasonal or pandemic) and pneumococcal vaccine; planned receipt of any vaccine in the 4 weeks following any trial vaccination, except for influenza (seasonal or pandemic) and pneumococcal

	vaccines; previous vaccination against Clostridium difficile with either the trial vaccine or another vaccine; current or prior Clostridium difficile infection episode; receipt of blood or blood-derived products in the past 3 months, which might interfere with assessment of the immune response; known or suspected congenital or acquired immunodeficiency; or receipt of immunosuppressive therapy such as anti-cancer chemotherapy or radiation therapy within the preceding 6 months; or long-term systemic corticosteroid therapy (prednisone or equivalent for more than 2 consecutive weeks within the past 3 months); self-reported seropositivity for Human Immunodeficiency Virus, hepatitis B, or hepatitis C; anticipated or current receipt of kidney dialysis treatment; known systemic hypersensitivity to any of the vaccine components, or history of a life-threatening reaction to a vaccine containing any of the same substances; self-reported thrombocytopenia, contraindicating intramuscular vaccination; bleeding disorder, or receipt of anticoagulants in the 3 weeks preceding inclusion, contraindicating IM vaccination; deprived of freedom by an administrative or court order, or in an emergency setting, or hospitalized involuntarily; current alcohol abuse or drug addiction that might interfere with the ability to comply with trial procedures; chronic illness that, in the opinion of the investigator, is at a stage where it might interfere with trial conduct or completion; identified as a study site employee who is involved in the protocol and may have direct access to trial-related data; subjects who have any history of intestinal diverticular bleeding; subjects who have had surgery within the past three months for gastrointestinal malignancy".
Interventions	**Vaccine type:** *Clostridium difficile* vaccine **Experimental:** 50 or 100 mcg *Clostridium difficile* with aluminium hydroxide. n = 305 **Control:** 50 or 100 mcg *Clostridium difficile*. n = 306 **Aluminium concentration:** 400 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3-dose schedule on either days 0, 7, 180 or days 0, 30, 180 or on days 0, 7, 30
Outcomes	**Immunogenicity:** "blood samples were collected on days 0, 7, 14, 30, 60, 180, and 210 and analysed for (i) serum anti-toxin IgG to Clostridium difficile toxins A and B enzyme-linked immunosorbent assay, and (ii) anti-toxin A and B neutralizing capacity". **Safety:** "pre-defined (solicited) adverse events were automatically considered to be related to vaccination: solicited injection site (pain, erythema, swelling) and systemic(fever, headache, malaise, myalgia, arthralgia) adverse reactions were collected on the day of vaccination and for 6 days after each vaccination. Unsolicited adverse events were collected for 30 days after each vaccination.Serious adverse events were collected throughout the study.The relationship of unsolicited AEs and SAEs to vaccination was assessed by the investigator".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (guy.debruyn@sanofipasteur.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment

	● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further data or information added. The number of participants with adverse events are unclear; thus, we have not included these data in the meta-analyses. **Funding:** supported by Sanofi Pasteur

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised
Allocation concealment (selection bias)	Unclear risk	No information on allocation concealment
Blinding of participants and personnel (performance bias)	High risk	Open-label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	A safety analysis set, including all participants who received at least one trial vaccination, was used for the safety analyses. Dropouts > 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	High risk	A per protocol analysis set was defined for immunogenicity based on meeting inclusion/exclusion criteria, receipt of the correct number of doses,receipt of the correct formulation administered according to the protocol and in the proper time window, provision of pre- or post-dose 1 serology samples according to protocol-defined time windows, protocol

		deviations, lack of a valid serology result pre-or post-dose 1. Dropouts > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01230957), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Durbin 2020

Methods	Phase I randomised, placebo-controlled, double-blind study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** adult males or females in generally good health between 18 and 50 years of age, inclusive, who had previously received the TV003 or TV005 LATV dengue vaccine formulation successfully completed the NIH study of TV003 or TV005 (NCT01436422 orNCT01072786), with no safety concerns, and had seroconverted for ≥ 3 serotypes. **Exclusion criteria:** Participants were excluded if they had potential or known exposure to Flavivirus by infection or vaccination in the interim since the receipt of the last dose of the experimental dengue vaccine (TV003 or TV005) and planned travel to a DENVendemic area in the 30 days before or anytime during the current study. Other exclusion criteria included evidence of clinically significant neurologic, cardiac, pulmonary, hepatic, rheumatologic, autoimmune, or renal disease; poorly controlled diabetes mellitus; recent receipt of blood products; febrile illness £ 72 hours before the first dose; and receipt of any licensed vaccine within 14 or 30 days, respectively, before vaccination in this study.
Interventions	**Vaccine type:** tetravalent recombinant subunit dengue vaccine **Experimental:** V180 with Alhydrogel. n = 8 **Control:** V180. n = 8 **Aluminium concentration:** 230 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** one intramuscular injection
Outcomes	**Immunogenicity:** "the primary immunogenicity objective of the trial was to evaluate if at least one of the V180 formulations induced a booster-neutralizing antibody response, as measured by the PRNT50, to at least three of the four DENV serotypes at day 28 after vaccination. Exploratory immunogenicity objectives included analysis of the kinetics and durability of the virus neutralizing response using the PRNT50 performed at days 15 and 180 as well as analysis of the neutralization response using the focus reduction neutralization test (FRNT50) at days 28 and 180 after vaccination". **Safety:** "the overall safety and tolerability of the tested V180 formulations were assessed and reported as a summary of serious and non-serious solicited and unsolicited AEs".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 26/01/2021 authors (swhitehead@niaid.nih.gov) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of participants and personnel **Fundings**: this study was funded by Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Kenilworth, NJ, and in part by the Intramural Research Program of the NIAID, NIH. Serology data were available up to day 180 but were extracted in this review at day 28 due to fewer drop outs at this time-point.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only stated randomised
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. An external, independent, and blinded data safety monitoring board reviewed the trial design before initiation and reviewed data during the trial to evaluate the safety, study progress, and conduct of the study.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Double-blind. An external, independent, and blinded data safety monitoring board reviewed the trial design before initiation and reviewed data during the trial to evaluate the safety, study progress, and conduct of the study.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	An external, independent, and blinded data safety monitoring board reviewed the trial design before initiation and reviewed data during the trial to evaluate the safety, study progress, and conduct of the study.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No drop outs
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	No drop outs
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	No drop outs

Selective reporting (reporting bias)	Unclear risk	The study protocol was registered prior to randomisation (NCT02450838), but safety outcome not described in the protocol.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ehrlich 2008

Methods	Randomized, partially blinded (between groups) clinical trial
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects are 18 to 45 years of age, inclusive, on the day of screening; have an understanding of the study, agree to its provisions, and give written informed consent prior to study entry; are clinically healthy, as determined by medical history and physical examination; agree to keep a daily record of symptoms; if female and capable of bearing children, have a negative urine pregnancy test result at study entry and agree to employ adequate birth control measures for the duration of the study". **Exclusion criteria:** "subjects are at high risk of contracting H5N1 influenza infection (e.g. poultry workers); suffer from a significant neurological, cardiac, pulmonary (including asthma), hepatic, rheumatic, autoimmune, hematological or renal disorder; are unable to lead an independent life as a result of either physical or mental handicap; suffer from any kind of immunodeficiency; suffer from a disease or are undergoing a form of treatment that can be expected to influence immune response. Such treatment includes, but is not limited to, systemic or high dose inhaled (>800µg/day of beclomethasone dipropionate or equivalent) corticosteroids, radiation treatment or other immunosuppressive or cytotoxic drugs; have a history of inflammatory or degenerative neurological disease (e.g. Guillain Barré); have a history of severe allergic reactions or anaphylaxis; have a rash, dermatologic condition or tattoos which may interfere with injection site reaction rating; have received a blood transfusion or immunoglobulins within 90 days of study entry; have donated blood or plasma within 30 days of study entry; have received any live vaccine within 4 weeks or inactivated vaccine within 2 weeks prior to vaccination in this study; have undergone systemic corticoid therapy within 30 days prior to study entry; have a functional or surgical asplenia; have a known or suspected problem with alcohol or drug abuse; were administered an investigational drug within six weeks prior to study entry or are concurrently participating in a clinical study that includes the administration of an investigational product; are a member of the team conducting this study or are in a dependent relationship with the study investigator. Dependent relationships include close relatives (i.e., children, partner/spouse, siblings, parents) as well as employees of the investigator; if female, are pregnant or lactating".
Interventions	**Vaccine type:** whole-virus H5N1 influenza vaccine **Experimental:** 7.5 or 15 mcg H5N1 with aluminium. n = 91 **Control:** 7.5 or 15 mcg H5N1. n = 90 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported

	Vaccination schedule: two injections into the deltoid muscle at an interval of 21 days
Outcomes	**Immunogenicity:**" number of subjects with hemagglutination-inhibition and neutralizing antibodies to the vaccine strain (A/Vietnam/1203/2004) 21 days after the first and second doses of vaccine". **Safety:** any systemic reaction during the first 7 days after each dose of vaccine
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** 14/12/2018 authors (hartmut.ehrlich@abivax.com) were inquired about the followings ● method for random sequence generation ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further data or information added **Funding:** supported by Baxter The trial only reports on vaccine-related serious adverse events.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	High risk	Subject assignment to a particular cohort was in order of enrollment.
Blinding of participants and personnel (performance bias)	High risk	Subjects and investigators were unaware of the dose of vaccine administered within the subgroups. Investigators were partially blinded, meaning that they were blinded within the respective dosing cohorts.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Unblinded safety data from Cohorts 1a, 1b, 2 and 3 were provided to the independent Data Safety Monitoring Board (DSMB) while it was ensured that study monitors, data managers and other staff involved in the conduct of the study as well as laboratory personnel remained blinded.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Unblinded safety data from Cohorts 1a, 1b, 2 and 3 were provided to the independent Data Safety Monitoring Board (DSMB) while it was ensured that study monitors, data managers and other staff involved in the conduct of the study

		as well as laboratory personnel remained blinded.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Laboratory personnel remained blinded.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	<5 % drop-outs.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	<5 % drop-outs.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Drop outs>5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00349141), and all outcomes were reported on .
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ensoli 2009

Methods	Randomized, double blind, placebo-controlled clinical trial
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** Caucasian **Diagnosis:** healthy **Inclusion criteria:** "negative pregnancy test for women of childbearing potential within 3 days prior to baseline evaluation and use of an acceptable means of contraception (condom, hormonal or mechanical method) for one month prior to immunization and for all the duration of the study; complete blood count and differential defined as: -hematocrit >=30% for women, >= 38% for men -hemoglobin >= 9.5 g/dL -white cell counts >= 4,000 and <= 9,500 cells/mm3 -total lymphocyte count >=1000 cells/mm3 -CD4+ T cell count > 500 cells/microL based on 2 separate determinations (Hannet, 1992) -platelets (100,000-550,000/ mm3) -differential within institutional normal limits or approval of site physician; normal ALT (as defined by the range of the clinical site laboratory) and Creatinine (< 1.6 mg/dl). Normal urine dipstick with esterase and nitrite; normal thyroid function; negative ELISA for HIV-1/HIV-2 and HIV-1 viral load (plasma viremia) < 50 copies/mL within 1 month of immunization; availability for follow-up for planned duration of at least 12 months and willing to have further brief evaluations at 24 and 36 months". **Exclusion criteria:** "identifiable high-risk behavior for HIV-1 infection, including a history of injection drug use within 12 months prior to enrollment or higher-risk sexual behavior; history of neoplastic diseases, encephalopathy, neuropathy or

	unstable CNS pathology, immunodeficiency, autoimmune disease, angina or cardiac arrhythmias, or any other clinically significant medical problems; chest radiography showing evidence of active or acute cardiac or pulmonary disease; history of anaphylaxis or serious adverse reactions to vaccines as well as serum IgE levels exceeding 1000 U.I./ml; history of serious allergic reaction to any substance, requiring hospitalization or emergent medical care (e.g. Steven-Johnson syndrome, bronchospasm, or hypotension); active syphilis [NOTE. If the serology is documented to be a false positive or due to an adequately treated infection, the volunteer is eligible]; active tuberculosis [NOTE: Volunteers with a positive PPD and a normal chest X-ray showing no evidence of TB and not requiring isoniazid (INH) therapy are eligible]; medical or psychiatric condition or occupational responsibilities which preclude subject compliance with the protocol. Specifically excluded are persons with psychotic disorders, major affective disorders, suicidal ideation; current use of psychotropic drugs; participation in another experimental protocol within six months prior to pre-study screening; current or prior therapy with immunomodulators or immunosuppressive drugs and anticoagulant drugs within 30 days prior to study medication administration; any unstable cardiovascular disease (e.g. unstable hypersensitive disease needing modification or introduction of an anti-hypersensitive treatment); live attenuated vaccines within 60 days of study [NOTE: Medically indicated sub-unit or killed vaccines (e.g., influenza, pneumococcal, hepatitis A and B) are not exclusionary, but should be given at least 4 weeks away from HIV immunizations]; use of investigational agents within 90 days prior to study; receipt of blood products or immunoglobulin in the past 6 months; prior receipt of HIV-1 vaccine in a previous HIV vaccine trial; positivity for HBV antigens (HBs Ag, HBe Ag), and HCV, HTLV-I and HTLV-II antibodies; pregnant or lactating women".
Interventions	**Vaccine type:** human immunodeficiency virus (HIV)-1 Tat-based vaccine **Experimental:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine with aluminium. n = 8 **Control:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine. n = 7 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** 5 doses (weeks 0, 4, 8, 12, 16) either subcute in alum or intradermic without adjuvant
Outcomes	**Immunogenicity**: "volunteers are monitored for anti-Tat specific antibodies, anti-Tat proliferative response and in vitro gamma-IFN and IL-4 (or IL-10) production in response to Tat (Elispot)". **Safety**: "clinical observation and monitoring of haematological, biochemical, virological and immunological parameters. Safety is evaluated by monitoring of volunteers for local and systemic adverse reactions during the trial".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 12/02/2020 authors (barbara.ensoli@iss.it) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results

	• proportion of participants with any AE related and not related to the vaccine • serological response GMT for each study arm at maximum follow up • proportion of participants who developed HIV after the vaccination • random sequence generation • allocation concealment • blinding of participants and personnel • blinding of outcome assessment No answer received. **Funding:** funded by "ISS special funding for Tat vaccine trials", the Italian Concerted Action on HIV-AIDS Vaccine Development (ICAV, National AIDS Program) and the Italian Ministry of Health

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs > 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	The study protocol for one trial was registered prior to randomisation (NCT00529698), and all outcomes were reported on.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

EUCTR2009-015103-58-FI

Methods	A multicenter, randomised, double-blind, active control trial
Participants	**Sex:** male and female **Age:** 20-45 years / 12-15 months **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** afebrile healthy toddlers 12-15 months of age documented to have completed a full 3-dose infant series of Prevnar 4, and 6 months of age **Exclusion criteria:** not reported
Interventions	**Vaccine type:** pneumococcal conjugate vaccine (V114). **Experimental:** V114 aluminium adjuvated. n = 33 **Control:** V114. n = 29 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** Merck **Vaccination schedule:** one intramuscular dose
Outcomes	**Immunogenicity:** "serotype-specific IgG concentrations were measured using the Meso-Scale Discovery electrochemiluminescence assay on sera collected prior to vaccination and 30 days postvaccination. A multiplex opsonophagocytic assay assay was used to measure opsonophagocytic assay activity in the pre- and post vaccination sera. The percentage of subjects with postvaccination serotype-specific IgG concentrations corresponding to a level of ≥0.35 µg/mL in the WHO ELISA was evaluated. Other endpoints of interest were the geometric mean concentrations of IgG, the percentage of subjects with opsonophagocytic killing activity as defined by functional antibacterial opsonophagocytic assay titer ≥1:8 and the geometric mean opsonophagocytic assay titers for the serotypes contained in the V114 vaccine". **Safety:** "adverse experiences were collected using a validated vaccine report card. All adverse experiences were graded for severity. **Adult and Toddler Stages**: 1) Study subjects were observed for 30 minutes postvaccination for any immediate adverse experiences. 2) Solicited injection site adverse experiences (redness, swelling, nodules, and pain/tenderness) and solicited systemic adverse experiences (muscle pain, joint pain, and tiredness) were collected day 1 to day 14 after vaccination. 3) Any other systemic or injection site adverse experiences were collected day 1 to day 14 after vaccination. 4) Serious adverse experiences from the time the consent form was signed through completion of the subject's participation in the study at day 30 after receipt of study vaccine. 5) Body temperature was measured during day 1 to day 7 after vaccination. If fever was suspected, body temperature was also measured during day 8 to day 14. **Adults Only:** 1) CBC with differential including platelets, serum creatinine, ALK, ALT, bilirubin, and urinalysis was performed at day 1 and day 14; 2) Severity of adverse events and safety laboratory abnormalities were assessed according to a toxicity grading scale as mild (Grade I), moderate (Grade II), severe (Grade III), and potentially life threatening (Grade IV)".
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** authors not shown **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

EUCTR2010-019775-29-AT

Methods	Multicenter, randomised, double-blind, worldwide, active control trial
Participants	**Sex:** male and female **Age:** 6 to 12 weeks **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** Afebrile healthy infant subjects 6 to 12 weeks (42 to 89 days) old **Exclusion criteria:** not reported

Interventions	**Vaccine type:** pneumococcal conjugate vaccine (V114) **Experimental:** V114 aluminium adjuvated. n = 378 **Control:**V114. n = 386 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** Merck **Vaccination schedule:** 4 intramuscular doses at 2, 4, 6, and 12 to15 months of age
Outcomes	**Immunogenicity:** "serotype-specific IgG concentrations were measured using the Meso-Scale Discovery (MSD) electrochemiluminescence (ECL) assay and a multiplex opsonophagocytic assay was used to measure functional opsonophagocytic assay. The percentage of subjects with postvaccination serotype-specific IgG concentrations corresponding to a level of ≥0.35 µg/mL in the WHO ELISA was evaluated. Other endpoints of interest were the geometric mean concentrations of IgG, the percentage of subjects with opsonophagocytic killing activity as defined by functional antibacterial opsonophagocytic assay titer ≥1:8 and the geometric mean opsonophagocytic assay titers for the serotypes contained in the V114 vaccine". **Safety:** "adverse experiences were collected using a validated vaccination report card. All adverse experiences were graded for severity. (1) Study subjects were observed for 30 minutes postvaccination for any immediate adverse experiences. (2) Solicited injection-site adverse experiences (redness, swelling, hard lump, and pain/tenderness) and solicited systemic adverse experiences (irritability, drowsiness, appetite lost, and hives or welts) were collected day 1 to day 14 after vaccination. (3) Any other systemic or injection-site adverse experiences were collected day 1 to day 14 after vaccination. 4) Serious adverse experiences were collected from the time the consent form was signed through completion of the subject's adverse experiences participation in the study at day 30 after receipt of last dose of study vaccine. 5) Body temperature was measured during day 1 to day 7 after vaccination. If fever was suspected, body temperature was also measured during day 8 to day 14".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** authors not shown **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further information

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop-outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop-outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop-outs >5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Evans 2001

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-55 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "had no evidence of serious medical illness as evaluated by history, physical examination, or screening laboratory studies. The subjects were all seronegative for HIV-1 and were at low or intermediate risk of HIV infection as judged by the earlier described AIDS Vaccine Evaluation Group criteria. Women of childbearing age had negative pregnancy tests before each immunization and agreed to use birth control measures throughout the study period and follow-up". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** human immunodeficiency virus (HIV) **Experimental:** different doses of QS-21 (50 and 100 mg) with or without rsgp120 with aluminium. n =77 **Control:** different doses of QS-21 (50 and 100 mg) with or without rsgp120. n = 79 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 intramuscular injections at months 0, 1, 10 or 0, 1, 6 or 0, 1, 2

Outcomes	**Safety** and **immunogenicity**
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 25/02/2020 authors (tomevansmd@gmail.com) were inquired on the followings: ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any non-serious and serious adverse event related and not related to the vaccine for each intervention arm ● serological response GMT for each study arm at maximum follow up ● proportion of participants who developed HIV after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment Answer received about randomisation **Funding:** sponsored by the NIH Division of AIDS

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a

Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Falsey 2008

Methods	Randomized, double-blind, placebo-controlled study
Participants	**Sex:** male and female **Age:** >65 years **Ethnicity:** mixed **Diagnosis:** cardiopulmonary diseases **Inclusion criteria:** "persons with cardiopulmonary disease. Study subjects were required to be 65 years old and to have a physician-confirmed diagnosis of congestive heart failure(CHF) or active pulmonary disease that required medication to control symptoms. Active pulmonary diseases included chronic obstructive pulmonary disease (COPD), chronic bronchitis, emphysema, asthma, interstitial fibrosis, and environmental lung disease". **Exclusion criteria:** known immunodeficiency, autoimmune disease, or active malignancy, or were taking immunosuppressive medications
Interventions	**Vaccine type:** respiratory syncytial virus (RSV) vaccine **Experimental:** RSV vaccine with aluminium phosphate. n = 400 **Control:** RSV vaccine. n = 383 **Aluminium concentration:** 1500 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Sanofi Pasteur **Vaccination schedule:** one dose intramuscularly into the deltoid muscle. All participants received licensed inactivated trivalent influenza vaccine (Fluzone; sanofi pasteur) at the same visit in the opposite arm.
Outcomes	**Immunogenicity:** "assessed at 1 month and 12 months after the year 1 and year 2 vaccinations. Levels of serum antibody to RSV were assessed by a plaque reduction neutralization assay and enzyme immunoassay. Hemagglutinin-inhibition antibody assays for H1N1, H3N2, and influenza B vaccine strains were performed". **Safety:** "immediate safety was assessed on site for 30 min after receipt of vaccination. Adverse events, both local and systemic, were solicited for 7 days after vaccination by use of diary cards. Non-solicited adverse events were recorded for 4 weeks after vaccination, and serious adverse events were recorded throughout the entire study period. Surveillance for acute respiratory illness began 8 days after vaccination (starting in late September) and continued through the end of April in the following year. Weekly phone calls to subjects ensured that illnesses were evaluated. Acute respiratory illness was defined as new onset or worsening of either 2 respiratory symptoms (sore throat, nasal congestion, or discharge, hoarseness, cough, sputum, wheezing, dyspnea, or pleuritic chest pain) or 1 respiratory symptom and 1 systemic symptom (feverishness, fatigue, anorexia, headache, or myalgia). If an illness met the acute respiratory illness criteria, a nasal swab sample and blood sample were collected during the acute phase of infection. Participants filled out diary

	cards during the illness and a convalescent-phase blood sample was collected approximately 4 weeks after the illness".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (ann.falsey@viahealth.org) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** Sanofi Pasteur

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	High risk	No pre-published protocol. Serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Frasch 1988

Methods	Ransomised, double-blind study
Participants	**Sex:** male and female **Age:** 22-43 years **Ethnicity:** not reported **Diagnosis:** cardiopulmonary diseases **Inclusion criteria:** medical students and employees from Baylor College of Medicine within the Texas Medical Center in Houston **Exclusion criteria:** candidates were excluded if they had an acute or chronic illness, a strong allergic history (especially to a vaccine component), a history of unusual or severe reactions to any vaccination, or a history of meningococcal disease or if they were pregnant.
Interventions	**Vaccine type:** neisseria meningitidis serotype 2b protein-group B polysaccharide **Experimental:** aluminium hydroxide-adsorbed Neisseria meningitidis vaccine. n = 12 **Control:** neisseria meningitidis vaccine. n = 12 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 doses at day 0 and 42 intramuscularly
Outcomes	**Immunogenicity:** blood specimen obtained at vaccination, three and six weeks after the first dose, and four weeks after the second immunization. Antibody response measured by ELISA. Bactericidial assay 3 weeks after the first and four weeks after the second immunization. **Safety:** volunteers took their own temperatures 6 h after vaccination and returned at 24 and 48 h for temperature measurement, an examination of the injection site, and a general clinical exam done by study clinicians.
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** no contact found **Funding:** supported in part by contract AI-22672 from the National Institute of Allergy and Infectious Diseases

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	n.a
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Fries 2017

Methods	Phase 1, randomised, observer-blinded, placebo controlled trial
Participants	**Sex:** male and female **Age:** >60 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "adult males and females, ≥ 60 years of age, without symptomatic cardiopulmonary disease. The following are examples of subjects who may bear cardio-pulmonary diagnoses but who would remain

	eligible:subjects on stable (no change in ≥ 2 months) therapy for findings (e.g., hypertension or hyperlipidemia) that are not associated with current symptoms or disability.Subjects who receive intermittent prophylaxis for risks associated with asymptomatic findings (e.g., antibiotic prophylaxis prior to dental procedures in a subject with asymptomatic mitral valve prolapse).Other clinically insignificant findings, not deemed to be associated with increased risk due to respiratory viral infections as determined by the Investigator. Free of other illnesses which are believed to increase the risk of influenza or influenza related complications including: diabetes mellitus, congenital or acquired blood dyscrasias, renal or hepatic dysfunction, and morbid obesity. Willing and able to give informed consent prior to study enrollment. Able to comply with study requirements". **Exclusion criteria:** "subjects who have any functional limitation or symptoms related to cardiac and/or pulmonary disease (including asthma or other episodic symptoms), or who receive ongoing therapy to control symptoms or functional limitation, are not eligible. Participation in research involving investigational product (drug / biologic / device) within 45 days before planned date of first vaccination and/or planned participation at any time during the study. History of a serious reaction to any prior vaccination or known allergy to constituents of licensed TIV (e.g., egg proteins). History of Guillain-Barré Syndrome within 6 weeks following a previous influenza vaccine. Receipt of any influenza vaccine within the preceding 3 months. Receipt of any vaccine in the 4 weeks preceding the study vaccination and planned receipt of a licensed vaccine any time prior to Day 56. Receipt of an RSV vaccine at any time. Any known or suspected immunosuppressive condition, acquired or congenital, as determined by medical history and/or physical examination. Chronic administration (defined as more than 14 continuous days) of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. An immunosuppressant dose of glucocorticoid will be defined as a systemic dose ≥10 mg of prednisone per day or equivalent. The use of topical, inhaled, and nasal glucocorticoids will be permitted provided these are not administered for diagnoses inconsistent with the inclusion criteria. The use of inhaled glucocorticoids, although typically not associated with system absorption, will generally indicate the presence of a diagnosis inconsistent with inclusion criteria 1 or 2. Administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or during the study. Acute disease at the time of enrollment (defined as the presence of a moderate or severe illness with or without fever, or an oral temperature >38.0°C on the planned day of vaccine administration). Known disturbance of coagulation. Suspicion or recent history (within one year of planned vaccination) of alcohol or other substance abuse. Any condition that in the opinion of the Investigator would pose a health risk to the subject if enrolled or could interfere with evaluation of the vaccine or interpretation of study results (including neurologic or psychiatric conditions deemed likely to impair the quality of safety reporting)".
Interventions	**Vaccine type:** respiratory syncytial virus fusion protein (RSV F) nanoparticle vaccine **Experimental:** 60 or 90 mcg RSV F with aluminium phosphate. n = 80 **Control:** 60 or 90 mcg RSV F. n = 80 **Aluminium concentration:** 1200 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark

	Vaccination schedule: one intramuscular dose into the left deltoid
Outcomes	**Safety:** "time frame: day 0 to day 364. Number (and percentage) of subjects with solicited local and systemic Adverse events over the seven days post injection; all adverse events, solicited and unsolicited over 56 days post-first injection. Significant New Medical Conditions, Medically Attended Events and Serious Adverse Events will be collected for one year. Diaries were provided to all subjects to document any adverse events (both solicited and unsolicited) and to record body temperature, concomitant medications, and medical visits and procedures, starting on each vaccination day and continuing for 6 days thereafter". **Immunogenicity:** "serum IgG antibody titers for the F-Protein antigen (time frame: day 0 to day 364). Immunogenicity will be measured using derived / calculated endpoints based on: geometric mean titer, geometric mean ratio, seroconversion rate, seroresponse rate. Serum HAI titers specific for the influenza antigens contained in the seasonal vaccine, (time frame: day 0 to day 56). Subjects provided blood samples on days 0, 7, 28, 56, 119, 182, and 364. Antibody concentrations/titers for the palivizumab-competitive antibody and linear antigenic Site II peptide ELISAs; and the RSV/A and RSV/B microneutralization (MN) assays were determined. Day 0, 28, 56, and 182 sera were used for HAI titer determination for all three antigens contained in the TIV".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (vshinde@novavax.com) were inquired about the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** funded and sponsored by Novavax, Inc., 20 Firstfield Road, Gaithersburg, MD 20878, US

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Observer blind, no further details

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	The safety population was used for all safety analyses and was defined as all study subjects who provided consent, were randomised, and received any study test article.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The safety population was used for all safety analyses and was defined as all study subjects who provided consent, were randomised, and received any study test article.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	The Per protocol population was used for all immunogenicity analyses and was defined as all subjects in the safety population who received the assigned dose of the investigational vaccine according to protocol and as randomised, had RSV serology results at least at days 0, 28, and 56, and had no major protocol deviations affecting the primary immunogenicity outcomes as determined by Novavax prior to unblinding. More than 5% drop outs in the per protocol population.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01709019), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Gantt 2018

Methods	Phase 1 randomised, observer-blind, placebo-controlled
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "generally healthy adult female and male 18 to 40 years of age, inclusive; serologically confirmed to be CMV seronegative at screening, female volunteers must agree to use an adequate contraception method as deemed appropriate by the investigator". **Exclusion criteria:** "history of or current clinically significant medical illness or any other illness that in the opinion of the investigator interferes with the interpretation of the study results, clinically significant abnormal physical examination, vital signs, or clinically significant abnormal values for hematology, clinical chemistry or urinalysis at screening as determined by the investigator, previous receipt of any cytomegalovirus vaccine, history of allergic reactions or

	anaphylactic reaction to any vaccine component, pregnant or breastfeeding or plans to conceive from two weeks before the study start through six months after the last dose of study vaccine, known or suspected impairment of immunological function, including but not limited to autoimmune diseases, splenectomy, or HIV/AIDS, chronic administration (defined as more than 14 days in total) of immune-suppressive or other immune-modifying drug with six months prior to the product dose (for corticosteroids, this is defined as prednisone ≥20 mg/day or equivalent). Inhaled and topical steroids are allowed, participation in another clinical study within 30 days or plans to participate in another treatment based clinical study during the conduct of the present study, any skin abnormality or tattoo that would limit post-vaccination injection site assessment, any history of cancer requiring chemotherapy or radiation within 5 years of randomization or current disease, are family members of study center staff".
Interventions	**Vaccine type:** human cytomegalovirus **Experimental:** 1 mcg of CMV envelope glycoprotein B (gB) with aluminium. n =26 **Control:** 1 mcg of CMV envelope glycoprotein B (gB). n = 25 **Aluminium concentration:** not reported **Aluminium type:** Adju-Phos®® **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 injections on days 0, 56, and 168
Outcomes	**Safety:** "number of participants with local and systemic adverse events during seven-day follow-up period [time frame: day of vaccine administration (days 0, 56, 168) and six subsequent days]. Number of participants with any adverse event [time frame: following each of the 3 injections of study vaccine, the occurrence of adverse events was captured during a 28-day follow-up period as well as through day 336 or early withdrawal.]. Number of participants with any serious adverse event [time frame: through day 336 or early withdrawal]. Number of participants with any hematological or biochemical laboratory abnormality [time frame: through day 336 or early withdrawal]". **Immunogenicity:** "geometric mean titer of antibody binding to CMV gB [time frame: through day 336 or early withdrawal]. Geometric mean titer of antibody avidity index value against gB [time frame: through day 336 or early withdrawal]. To measure the avidity of responses against CMV gB protein, a standard ELISA assay using recombinant gB protein which did or did not include treatment with 5M urea for 30 minutes of samples after sera had been incubated with recombinant protein. The reported value, or Avidity Index, represents the percent of signal measured in ELISA after treatment with urea relative to samples not exposed to urea. Geometric mean titer of neutralizing antibody against CMV infection of fibroblast cells [time frame: through day 196 or early withdrawal]. Geometric mean titer of neutralizing antibody against CMV infection of epithelial cells [time frame: through day 336 or early withdrawal]".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 30/01/2020 authors (Joanne.Langley@Dal.Ca) were inquired about the followings: ● random sequence generation ● allocation concealment ● final aluminium concentration per dose

	● aluminium manufacturer Answer received and information added **Fundings:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A modified blocked randomization stratified by study center ensured that each study center had at least one subject randomised to every dose group. Participants were assigned to a specified treatment based on the randomization schedule.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the

		study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02826798), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Glenn 2013

Methods	Phase 1, placebo-controlled, observer-blinded, dose-escalating study
Participants	**Sex:** male and female **Age:** 18–49 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male or female aged 18 to 49 years inclusive; ability to provide written informed consent to participate; healthy, as determined by medical history, physical examination, vital signs, and clinical safety laboratory examinations at baseline; females are required to fulfill one of the following criteria:At least 1 year post-menopausalSurgically sterileWilling to use oral, implantable, transdermal or injectable contraceptives for 30 days prior to first vaccination and until 28 days after each vaccinationWilling to abstain from sexual intercourse or use another reliable form of contraception approved by the Investigator (eg, intrauterine device, female condom, and diaphragm with spermicide, cervical cap, use of condom by the sexual partner, or a sterile sexual partner) for study duration and until 28 days after vaccination; all female subjects must have a negative urine pregnancy test within 48 hours preceding receipt of each vaccination. Comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits and be contacted by telephone throughout the follow-up period". **Exclusion criteria:** "presence of significant uncontrolled medical or psychiatric illness (acute or chronic). This includes institution of new medical or surgical treatment, or a significant dose alteration for uncontrolled symptoms or drug toxicity within 3 months of screening and reconfirmed on Day 1 prior to vaccination; positive serology for HIV-1 or HIV-2, or HBsAg or HCV antibodies; pregnant or lactating female; females who plan to become pregnant or plan to discontinue contraceptive precautions within 30 days prior to first vaccination

	and 28 days after each vaccination; cancer, or treatment for cancer, within 3 years, excluding basal cell carcinoma or squamous cell carcinoma, which is allowed; presence of any medical condition that may be associated with impaired immune responsiveness, including diabetes mellitus; receipt (or history of receipt), during the preceding 3-month period, of any medications or other treatments that may adversely affect the immune system such as allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs or other drugs known to be frequently associated with significant major organ toxicity, or systemic corticosteroids (oral or injectable). Inhaled and topical corticosteroids will be allowed. receipt or planned administration of a non study vaccine within 30 days prior to vaccination or during the study. If a non study vaccine has been administered, administration of study vaccine injection can be delayed and given as soon as allowable within the 30-day window, provided the non study vaccine is not administered within 2 weeks prior to study enrollment. Immunization with Tetanus Toxoids Adsorbed for adult use (Td or Tdap) vaccine, on an emergency basis, up to 8 days before or at least 8 days after a dose of study vaccine will be allowed. History of illicit drug or alcohol abuse within the previous 1 year or positive drug or alcohol screen; history of anaphylactic type reaction to injected vaccines; receipt of any investigational product or non registered drug within the 30 days prior to vaccination or current enrollment in any investigational drug study or intent to enroll in such a study within the ensuing study period; receipt or donation of blood or blood products 8 weeks prior to vaccination or planned receipt or donation during the study period; acute disease, defined as the presence of a moderate or severe illness (as determined by the Investigator through medical history and physical examination) with or without fever, or an oral temperature of ≥38ºC, within 72 hours prior to vaccination. Study vaccine can be administered to persons with a minor illness, such as diarrhea or mild upper respiratory tract infection with or without low-grade fever. Vaccination can be delayed until the subject has recovered. Any condition that, in the opinion of the Investigator, would interfere with the primary study objectives".
Interventions	**Vaccine type:** recombinant respiratory syncytial virus (RSV) fusion (F) protein nanoparticle vaccine **Experimental:** 30 or 60 mcg RSV F with aluminium phosphate. n = 40 **Control:** 30 or 60 mcg RSV F. n =40 **Aluminium concentration:** 1200 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** 2 doses intramuscularly on day 0 and 30
Outcomes	**Immunogenicity:** "neutralizing antibody and total anti-F antibody of the 6 RSV-F protein particle vaccine formulations; time frame: day 60 (2 months) assessed in the following manner: neutralizing antibody against RSV measured in a PRNT assayTotal anti-F IgG measured by enzyme-linked immunosorbent assay". **Safety:** "time frame: day 60 (2 months): immediate adverse events, solicited adverse events, all serious adverse events and SNMCs, vital signs, laboratory assessments. Study subjects used diary cards to record the occurrence and intensity of solicited reactions for 7 days post each vaccination. Unsolicited events including serious adverse events were recorded at each visit through study end (day 210). A clinical chemistry panel, CBC, differential and platelet count were evaluated at day 0 and on study days 30 and 60. Tolerability of each dose level was monitored by a Safety Monitoring Committee after the first

	injection before proceeding to the next higher dose. The criteria used to evaluate safety included: immediate adverse events reported within 30 min of vaccination; solicited and unsolicited adverse events, serious adverse events; and clinical laboratory safety measurements".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (gglenn@novavax.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Observer-blind. The site study personnel, Safety Monitoring Committee and study participants were blinded to treatment assignments. Study personnel responsible for vaccine preparation were unblinded, but had no involvement in vaccine administration or evaluation of the safety or immunogenicity results
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Blinded Safety Monitoring Committee
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Drop outs less than 5%
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Safety analysis was on total randomised population.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop-outs <5%.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Immunogenicity analysis was on the per protocol population. Drop-outs > than 5%.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01290419), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Glenn 2016

Methods	Phase 2 randomised, observer-blind, placebo-controlled study
Participants	**Sex:** female **Age:** ≥ 18 and ≤ 35 years of age **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy adult females, ≥ 18 and ≤ 35 years of age. "Healthy" shall be defined by the absence of any illness, acute or chronic, that requires ongoing systemic therapy for the control of symptoms or prevention of disability. Subjects on stable (no change in ≥ 3 months) therapy for findings (e.g., hypertension or hyperlipidemia) that are not associated with symptoms or disability are eligible, as are users of hormonal contraceptives. Subjects who receive intermittent prophylaxis for risks associated with asymptomatic findings (e.g., antibiotic prophylaxis prior to dental procedures in a subject with mitral valve prolapse) are eligible. Ongoing therapy will be defined as continuous or, if intermittent, more frequent than once every 3 months (e.g., use of an inhaled bronchodilator for exercise-induced bronchospasm more than once every 3 months). Persons being treated for illnesses or conditions that would become acutely symptomatic or disabling in the absence of treatment are not eligible. Willing and able to give informed consent prior to study enrollment. Able to comply with study requirements. Women who are not surgically sterile must have a negative urine pregnancy test prior to each vaccination; will be advised through the Informed Consent process to avoid becoming pregnant over the duration of the study, and must assert that they will employ an effective form of birth control for the duration of the study. Acceptable forms of birth control are: credible history of continuous abstinence from heterosexual activity, hormonal contraceptives (oral, injectable, implant, patch, ring), double-barrier contraceptives (condom or diaphragm, with spermicide), and IUD". **Exclusion criteria:** "participation in research involving investigational product (drug / biologic / device) within 45 days before planned date of first vaccination. History of a serious reaction to any prior vaccination. Received any vaccine in the 4 weeks preceding the study vaccination; or any RSV vaccine at any time. Any known or suspected immunosuppressive condition, acquired or congenital, as determined by history and/or physical examination. Chronic administration

	(defined as more than 14 continuous days) of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. An immunosuppressant dose of glucocorticoid will be defined as a systemic dose ≥10mg of prednisone per day or equivalent. The use of topical, inhaled, and nasal glucocorticoids will be permitted. Administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or during the study. Acute disease at the time of enrollment (defined as the presence of a moderate or severe illness with or without fever, or an oral temperature >38.0°C on the planned day of vaccine administration). Known disturbance of coagulation. Women who are pregnant or breastfeeding, or plan to become pregnant during the study. Suspicion or recent history (within one year of planned vaccination) of alcohol or other substance abuse. Any condition that in the opinion of the investigator would pose a health risk to the subject if enrolled or could interfere with evaluation of the vaccine or interpretation of study results (including neurologic or psychiatric conditions deemed likely to impair the quality of safety reporting)".
Interventions	**Vaccine type:** recombinant respiratory syncytial virus (RSV) fusion (F) protein nanoparticle vaccine **Experimental:** 60 or 90 mcg RSV F with aluminium phosphate. n =120 **Control:** 60 or 90 mcg RSV F. n =119 **Aluminium concentration:** 1200 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** one intramuscular injections into the deltoid on day 0 and a placebo dose consisted of isotonic saline on day 28 to preserve the blind or two intramuscular injections into the deltoid, on day 0 and day 28 (±2 days) in opposite arms
Outcomes	**Immunogenicity:** "serum IgG antibody titers specific for the F-Protein antigen across treatment groups, (time frame: day 0 to day 112). Immunogenicity will be measured using derived / calculated endpoints based on: geometric mean titer, geometric mean ratio, seroconversion rate, seroresponse rate". **Safety:** "time frame: day 0 to day 182: number (and percentage) of subjects with solicited local and systemic adverse events over the seven days post-injections; all adverse events, solicited and unsolicited over 56 days post-first injection. Significant new medical conditions, medically attended events and serious adverse events will be collected for six months".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (gglenn@novavax.com) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of outcome assessment ● total number of participants with non-serious adverse events ● number of patients who got respiratory syncytial virus infection after the vaccination. No answer received **Funding:** supported in part by PATH (formerly Program for Appropriate Technology in Health), Seattle, Washington

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Observer blind
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Safety analysis was on total randomised population.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Immunogenicity analysis was on the per protocol population. Drop-outs > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01704365), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Greenberg 2018

Methods	A phase 2 randomised, double-blind, multicenter study
Participants	**Sex:** male and female **Age:** 6 weeks to 12 weeks **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy infants ≥ 42 days to ≤ 89 days. Participant's parent/legal guardian understands the study procedures, alternate treatments available and risks involved with the study, and voluntarily agrees to allow the child to participate by giving written informed consent. Afebrile, with a rectal

	temperature <38.1°C (<100.5°F) or axillary temperature <37.8°C (<100.0°F) on day of vaccination. Participant's parent/legal guardian is able to read, understand, and complete study questionnaires (i.e., the Vaccination Report Card). Participant is able to attend all scheduled visits and to comply with the study procedures. Participant's parent/legal guardian has access to a telephone". **Exclusion criteria:** "prior administration of any pneumococcal vaccine. Known hypersensitivity to any component of the pneumococcal conjugate vaccine. Known or suspected impairment of immunological function. Participant has a history of congenital or acquired immunodeficiency (e.g., splenomegaly). Participant or his/her mother has documented human immunodeficiency virus infection. Functional or anatomic asplenia. History of autoimmune disease including multiple sclerosis, systemic lupus, polymyositis, inclusion body myositis, dermatomyositis, Hashimoto's thyroiditis, Sjogren's syndrome, rheumatoid arthritis, other autoimmune disorders. Known neurologic or cognitive behavioral disorders including multiple sclerosis, MS-like disease, encephalitis/myelitis, acute disseminating encephalomyelitis, pervasive developmental disorder, and related disorders. Use of any immunosuppressive therapy (Note: topical and inhaled/nebulized steroids are permitted). Participants on intramuscular, oral, or intravenous corticosteroid treatment should be excluded if they are receiving or expected to receive in the period from 30 days prior to Visit 1 through Visit 6 (30 days post-dose 4) more than 2 mg/kg per day of prednisone (or its equivalent), or more than 20 mg/d if they weigh more than 10 kg and are not otherwise immunocompromised. Excluded immunosuppressive therapies also include chemotherapeutic agents used to treat cancer or other conditions, and treatments associated with organ or bone marrow transplantation, or autoimmune disease. Participant has received other licensed non-live vaccines within the 14 days before receipt of study vaccine. Participant has received a licensed live virus vaccine within the 30 days prior of receipt of study vaccine. Prior receipt of a blood transfusion or blood products, including immunoglobulins. Investigational drugs or vaccines received within the 2 months before receipt of study vaccine. Participation in another clinical study within 42 days before the beginning or anytime during the duration of the current clinical study. History of invasive pneumococcal disease (positive blood culture, positive cerebrospinal fluid culture, or other sterile site) or known history of other culture positive pneumococcal disease. A recent febrile illness (rectal temperature ≥38.1°C [≥100.5°F]) occurring within 72 hours before receipt of study vaccine. History of failure to thrive. Participant has a coagulation disorder contraindicating IM vaccination. Participant and his/her mother have documented hepatitis B surface antigen-positive. Any infant who cannot be adequately followed for safety according to the protocol plan. Participant's parent/legal guardian is unlikely to adhere to study procedures, keep appointments, or is planning to relocate during the study. Any other reason that in the opinion of the investigator may interfere with the evaluation required by the study".
Interventions	**Vaccine type:** 15-valent pneumococcal conjugate vaccine (PCV15) **Experimental:** aluminium-adjuvanted-PCV15. n = 378 **Control:** PCV15. n = 386 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** alum phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** 4-dose regimen at 2, 4, 6, and 12–15 months of age

Outcomes	**Immunogenicity:** "percentage of participants achieving the Immunoglobulin G serotype-specific ≥0.35 µg/mL threshold value for postvaccination 3, (time frame: one month postvaccination 3). Percentage of participants meeting the serotype-specific IgG reference level (an antibody concentration measured by the pneumococcal polysaccharide electrochemiluminescence assay corresponding to the World Health Organization enzyme-linked immunosorbent assay ≥ 0.35 µg/mL) (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. IgG Geometric mean concentrations for Postvaccination 3, (time frame: one month postvaccination 3). The IgG geometric mean concentrations were evaluated as measured in the pneumococcal polysaccharide electrochemiluminescence assay, for recipients of adjuvated V114, non-adjuvanted V114, and Prevnar 13® (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. IgG geometric mean concentrations for postvaccination 4, (time frame: one month postvaccination 4). The IgG geometric mean concentrations were evaluated as measured in the pneumococcal polysaccharide electrochemiluminescence assay, for recipients of adjuvated V114, non-adjuvanted V114, and Prevnar 13® (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Percentage of participants with opsonophagocytic killing activity titer ≥ 8 as measured by fourfold multiplexed opsonization assay for postvaccination 3, (time frame: one month postvaccination 3). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity antibody responses. Percentage of participants with opsonophagocytic killing activity titer ≥ 8 as measured by fourfold multiplexed opsonization assay for postvaccination 4 (time frame: one month postvaccination 4). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity antibody responses. Opsonophagocytic killing activity geometric mean titers as measured by fourfold multiplexed opsonization assay for postvaccination 3 (time frame: one month postvaccination 3). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. The opsonophagocytic killing activity antibody responses included the percentage of participants with opsonophagocytic killing activity titer and the geometric mean titers. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity

	antibody responses.Opsonophagocytic killing activity geometric mean titers as measured by fourfold multiplexed opsonization assay for postvaccination 4 (time frame: one month postvaccination 4). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity antibody responses. Percentage of participants achieving the IgG Serotype-specific Threshold Value of ≥0.35 µg/mL for Postvaccination 4, (time frame: one month postvaccination 4). Percentage of participants achieving World Health Organization predefined antibody threshold as measured by the pneumococcal polysaccharide electrochemiluminescence assay corresponding to the World Health Organization enzyme-linked immunosorbent assay value of ≥ 0.35µg/mL (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Percentage of participants Achieving the IgG serotype-specific threshold value of ≥1.0 µg/mL for Postvaccination 4 [(time frame: one month postvaccination 4). Percentage of participants achieving World Health Organization predefined antibody threshold as measured by the pneumococcal polysaccharide electrochemiluminescence assay corresponding to the World Health Organization enzyme-linked immunosorbent assay value of ≥ 1.0 µg/mL (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®". **Safety:** "number of participants with an adverse event (time frame: up to day 14 postvaccination). An adverse event is defined as any unfavorable and unintended change in the structure, function, or chemistry of the body temporally associated with the use of the Sponsor's product, whether or not considered related to the use of the product. Any worsening (i.e., any clinically significant adverse change in frequency and/or intensity) of a preexisting condition which is temporally associated with the use of the Sponsor's product, is also an adverse experience. Number of participants with an injection-site adverse event (time frame: up to day 14 postvaccination). Injection-site adverse event reported by > 0% of participants in one or more vaccination groups were assessed. Number of participants with a systemic adverse event (time frame: up to day 14 postvaccination). Systemic adverse events reported by > 0% of participants in one or more vaccination groups were assessed. Number of participants with a serious adverse event, (time frame: up to one month after last dose of study vaccine). A serious adverse event is an adverse event that results in death, is life threatening, requires or prolongs an existing hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is another important medical event deemed such by medical or scientific judgment. Number of Participants who discontinued the study due to an adverse event, (time frame: up to day 14 postvaccination). Any worsening (i.e., any clinically significant adverse change in frequency and/or intensity) of a preexisting condition which is temporally associated with the use of the Sponsor's product, is also an adverse experience".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** the email address orf the corresponding author provided in the publication was not valid. **Funding:** provided by Merck & Co., Inc., Kenilworth, NJ, USA

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop-outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	High risk	Drop-outs >5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop-outs >5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT01215188), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Harro 2001

Methods	Double-blind, randomised, placebo-controlled, phase I safety and immunogenicity trial
Participants	**Sex:** male and female **Age:** 18- to 29-years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy, human immunodeficiency virus-1-seronegative, 18- to 29-year-old volunteers. Subjects were determined by history to be at low risk for HPV16 exposure". **Exclusion criteria:** "individuals were not eligible to participate if they had a history of more than four lifetime sexual partners or more than two sexual partners within the preceding 6 months. History of abnormal cervical cytology, immunodeficiency, anaphylaxis to medicines or vaccines, receipt of blood products within 3 months of enrollment, current pregnancy or lactation, and any other condition that might interfere with the study objectives".
Interventions	**Vaccine type:** human papillomavirus 16 L1 virus-like particle vaccine (HPV16 L1 VLP) **Experimental (a):** 10 mcg HPV16 L1 VLP with aluminium potassium sulfate. n = 10 **Control (a):** 10 mcg HPV16 L1 VL. n = 10 **Experimental (b):** 50 mcg HPV16 L1 VLP with aluminium potassium sulfate. n = 10 **Control (b):** 50 mcg HPV16 L1 VL. n =10 **Aluminium concentration: (a)** 1200 mcg/dose. **(b)** 6000 mcg/dose **Aluminium type:** aluminium potassium sulfate **Aluminium manufacturer:** Merck **Vaccination schedule:** 3-doses; intramuscular injection in the deltoid region at months 0, 1, and 4
Outcomes	**Immunogenicity:** "subjects were evaluated clinically, and blood was collected for clinical and immunologic tests before each injection (months 0, 1, and 4) and 1 month after each injection. Clinical laboratory tests done at each of these visits included the following: CBC, platelet count, serum ALT, serum creatinine, urine protein, and urine hemoglobin measurements. IgG-specific HPV16 L1 VLP-based enzyme-linked immunosorbent assays were performed. In vitro HPV16 pseudovirion neutralization assays were performed". **Safety:** "after each injection, oral temperatures were recorded after 30 minutes, 6 hours, and daily for 6 days. Adverse reactions were monitored by study staff in the clinic at 30 minutes and 2 days after each injection and by telephone for the following 6 days".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (lowyd@mail.nih.gov) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel

	● number of drop outs (patients randomised but not analysed) ● funding source ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events /number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) ● serological response per patient at maximum follow up No answer received **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	To maintain blinding, the individual who prepared and administered the vaccine was not involved in data collection or clinical evaluation of volunteers.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	To maintain blinding, the individual who prepared and administered the vaccine was not involved in data collection or clinical evaluation of volunteers.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%

Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Harro 2012

Methods	Phase-1, randomised, controlled, double-blind trial
Participants	**Sex:** male and female **Age:** 18–70 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "participant was in good physical health. Participant was able to understand study procedures and agreed to participate in the study by providing written informed consent. Participant was willing and able to participate in the entire study duration planned for up to 12 months (~360 days). Female participants of reproductive potential were required to have a negative urine pregnancy test immediately prior to study vaccination. Female participants of reproductive potential must have used an acceptable method of birth control for 2 weeks prior to enrollment, and agreed to use an acceptable method of birth control for 1 month after vaccination". **Exclusion criteria:** "participant suffered from a chronic skin condition that predisposed the individual to the development of chronic skin or soft-tissue infections (e.g., psoriasis, chronic granulomatous disease, atopic dermatitis). Participant developed a serious infection (e.g., bacteremia, pneumonia, mediastinitis) attributed to S. aureus in the 12 months prior to screening. Participant had a history of anaphylaxis to aluminium-containing adjuvant or other vaccine components. Participant had a temperature of ≥100.4ºF (≥38.0ºC), oral equivalent, within 48 hours prior to receipt of V710. Participant had received a live virus vaccine within 30 days prior to receipt of V710 or was scheduled to receive vaccination with a live virus vaccine within 30 days following study entry. Participant had received any other licensed vaccine (including non-live virus vaccines) within 14 days prior to receipt of V710 or was scheduled to receive any other licensed vaccine (including non-live virus vaccines) within 30 days following study entry. Participant had received V710 in a prior clinical study (Merck V710 Protocol 001). Participant was administered immunoglobulin or blood product within 90 days prior to receipt of V710 or was scheduled to receive such products within 30 days following study entry. Participant had received treatment with systemic (intramuscular, oral, or intravenous) corticosteroids or another immunosuppressive medication (e.g., calcineurin inhibitors, mycophenolate, azathioprine) or biological agents (e.g., rituximab) in the 14 days prior to receipt of V710 or was anticipated to receive such medications for a chronic medical condition during the course of the study. Participant had a condition that required active medical intervention or monitoring to avert serious danger to the participant's health or well-being, such as diabetes mellitus, autoimmune disease, or clinically significant chronic medical conditions that were considered progressive, including but not limited to: coronary artery disease, congestive heart failure, cardiomyopathy, progressive valvular heart disease, chronic obstructive pulmonary disease, pulmonary fibrosis, active peptic ulcer disease, chronic renal disease, chronic hepatic disease, multiple sclerosis, progressive neuropathies, or seizure disorder requiring therapy in the past 3 years. Participant had known or

	suspected impairment of immunologic function including, but not limited to, the following conditions: autoimmune disease, diabetes mellitus, endstage renal disease, hepatic insufficiency/cirrhosis, splenectomy, or human immunodeficiency virus/acquired immune deficiency syndrome. Participant had a condition in which repeated venipuncture or injections posed more than minimal risk for the subject, such as hemophilia, other severe coagulation disorders, or significantly impaired venous access. Participant was pregnant or breastfeeding, or planning to conceive within the 12-month study duration period. Participant had clinically significant abnormalities based on the participant or physical examination. Participant had recent history (within the past 5 years) or current evidence of drug or alcohol abuse. Participant had a major psychiatric illness, including any history of schizophrenia or severe psychosis, bipolar disorder requiring therapy, or any subject with suicidal ideation within the previous 3 years. Participant was legally or mentally incapacitated. Participant had participated in another clinical study in the past 4 weeks, or participant planned to participate in a treatment-based study or study in which an invasive procedure was to be performed during the first 3 months of this study (through Day 84). Participant had a history of any condition which, in the opinion of the investigator, posed an additional risk for the subject or confounded the results of the study".
Interventions	**Vaccine type:** staphylococcus aureus vaccine (V710) **Experimental:** aluminium-adjuvanted V710. n = 32 **Control:** V710. n = 32 **Aluminium concentration:** not reported **Aluminium type:** amorphous aluminium hydroxyphosphate sulfate (AAHS) **Aluminium manufacturer:** Merck **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "number of participants with a positive immune response, defined as a change in antibody level greater than or equal to a 2-fold-rise at day 14 compared to baseline, (time frame: baseline and day 14 postvaccination). Immunoglobulin G antibodies were measured by a LUMINEX assay. Geometric mean antibody concentrations, (time frame: day 14 postvaccination). Immunoglobulin G antibodies were measured by a LUMINEX assay. Change in antibody concentration (titer) at day 14 compared to baseline, expressed as the geometric mean fold-rise (time frame: day 14 postvaccination). Immunoglobulin G antibodies were measured by a LUMINEX assay.The geometric mean fold-rise is the ratio of the antibody concentration at day 14 to the antibody concentration at baseline". **Safety:** "number of vaccine-related serious adverse experiences, (time frame: up to day 360 postvaccination). Investigators were instructed to determine the seriousness and causality (relatedness to test vaccine) of each adverse experiences based on criteria defined in the protocol: a serious adverse event is any adverse experiences that: results in death, is life threatening, results in a persistent or significant disability/incapacity, results in or prolongs an existing inpatient hospitalization, is a congenital anomaly/birth defect, is a cancer, is an overdose, or is another important medical event that may require medical or surgical intervention to prevent one of the outcomes listed above".
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 14/12/2018 authors (charro@jhsph.edu, nicholaskartsonis@merck.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received Results from the trial is also available here: clinicaltrials.gov/ct2/show/results/NCT01324440. There are some minor discrepancies in the number of adverse events reported in the trial report compared to the data available at clinicaltrials.gov. We used the data from clinicaltrials.gov when meta-analysing AEs as more adverse events were available here. **Funding:** funded by Merck Sharp & Dohme, a subsidiary of Merck & Co., Inc. The study sponsor was involved in the study design; data collection, analysis, and interpretation of data; in writing the report; and the decision to submit the paper for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	"Double-blind. Because of the differences in appearance between V710 with and without adjuvant, and between V710 and saline placebo, one unblinded member of the study site staff was responsible for preparing and administering the study vaccine or placebo. This person had no further involvement with any subsequent study procedure postvaccination, including safety follow-up procedures. All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained blinded to treatment group throughout the study".
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	"Double-blind. Because of the differences in appearance between V710 with and without adjuvant, and between V710 and saline placebo, one unblinded member of the study site staff was responsible for preparing and administering the study vaccine or placebo. This person had no further involvement with any subsequent study procedure postvaccination, including safety follow-up procedures. All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained

		blinded to treatment group throughout the study".
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained blinded to treatment group throughout the study
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained blinded to treatment group throughout the study
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	All patients in both studies were included in the primary analyses.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	All patients in both studies were included in the primary analyses.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	All patients in both studies were included in the primary analyses.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01324440), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Hung 2019

Methods	Open-label, multicenter, phase II, randomised studies
Participants	**Sex:** male and female **Age:** 2 months to 6 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "healthy children aged from 2 months to 6 years old at the time of first vaccination. Subject's guardians are able and willing to comply with study procedures and provide the signed informed consent. Subject is able and can comply with the requirements of the protocol. Subject with body temperature ≤38°C". **Exclusion criteria:** "subject with previous known exposure to Enterovirus 71 (EV71). Subject with a history of herpangina, hand-foot-mouth disease,and acute hemorrhagic conjunctivitis associated with enterovirus infection in the past 3 months. Subject with gestation < 37 weeks. Subject with birth weight <2.5 kg. Subject with a history of hypersensitivity to vaccines, or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Family history of seizures or progressive neurological disease. Family history of congenital or hereditary immunodeficiency. Severe malnutrition or dysgenopathy. Major congenital defects or serious chronic illness, including perinatal brain damage. Subject diagnosed of having autoimmune disease (e.g., celiac disease, type I diabetes, lupus, juvenile dermatomyositis, scleroderma, juvenile idiopathic arthritis, immune (or idiopathic) thrombocytopenia purpura). Bleeding disorder diagnosed by a doctor or significant bruising or hemostatic difficulties with IM injections or blood draws. Any acute infections 7 days prior to administrating the first vaccination. Use of any investigational product (including

	drug, vaccine) within 30 days prior to vaccination or planned use during the study period. Administration of any vaccines within 14 days prior to randomisation. Use of immunoglobulins or any blood products within 3 months prior to vaccination or planned use during the study period. Chronic administration (defined as > 14 days) of immunosuppressants or other immunomodulators or systemic corticosteroids within 6 months prior to vaccination or planned use during the study period. Subjects who had ever received investigational EV-71 vaccine prior to randomisation. Under anti-tuberculosis prevention or therapy. Any condition that in the opinion of the investigator may interfere with the evaluation of study objectives".
Interventions	**Vaccine type:** inactivated enterovirus A71 vaccine **Experimental:** 1.0 mcg EV-A71 with aluminium hydroxide. n = 47 **Control:** 1.0 mcg EV-A71. n =44 **Aluminium concentration:** 150 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** one intramuscular injection
Outcomes	**Immunogenicity:** "serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 56). To evaluate the immunogenicity change of serum neutralizing antibody titers induced by the EV71 vaccine from baseline on day 56. Seroconversion rate based on neutralizing antibody titers, (time frame: day 56). to evaluate the immunogenicity change of seroconversion rate from baseline on day 56. Serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 28).To evaluate the immunogenicity change of serum neutralizing antibody titers induced by the EV71 vaccine from baseline on day 28. Serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 196). To evaluate the immunogenicity change of serum neutralizing antibody titers induced by the EV71 vaccine from baseline on Day 196. Seroconversion rate based on neutralizing antibody titers, (time frame: day 28).To evaluate the immunogenicity change of seroconversion rate from baseline on Day 28. Seroconversion rate based on neutralizing antibody titers, (time frame: day 196). To evaluate the immunogenicity change of seroconversion rate from baseline on day 196. Serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 364). T evaluate the immunogenicity of serum neutralization antibody titer induced by the EV 71 vaccine on Day 364". **Safety:** "solicited adverse events, (time frame: 7 days after each vaccination). Unsolicited adverse events, (time frame: 28 days after each vaccination). The occurrence of overall adverse events and serious adverse event, (time frame: day 0 to day 196). The occurrence of EV 71 breakthrough infection after Visit 3, (time frame: day 57 to day 364)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (mchung@vghtpe.gov.tw, kgwu@vghtpe.gov.tw) were inquired on the followings: ● aluminium manufacturer. ● allocation concealment. ● total number of participants with non-serious adverse event related and not related to the vaccine ● number of participants who got Enterovirus infection after the vaccine.

	No answer received **Funding:** supported by Enimmune Corporation

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation list was generated using SAS software (version 9.4) and contained sequential coding which was assigned to group A1, A2 and A3 in the ratio of 1:1:1.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open-label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03268083), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Jackson 2009

Methods	Randomized, controlled, single-blind, multicenter phase II study
Participants	**Sex:** male and female **Age:** 11–17 years **Ethnicity:** Asian **Diagnosis:** healthy

	Inclusion criteria: healthy adolescents between and including 11-17 years of age, who provide written informed consent. **Exclusion criteria:** subjects with a previous or suspected disease caused by N. meningitidis; or previous immunization with a meningococcal vaccine or vaccine containing meningococcal antigen(s); any serious acute, chronic or progressive disease.
Interventions	**Vaccine type:** quadrivalent meningococcal glycoconjugate vaccine (MenACWY-CRM) **Experimental:** MenACWY-CRM with aluminium phosphate. n = 164 **Control:** MenACWY-CRM. n = 151 **Aluminium concentration:** 300 mcg **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Novartis **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "percentages of subjects with N.Meningitidis human serum bactericidal activity titers≥ 1:4, After one dose of MenACWY-CRM Vaccine, with Adjuvant or without adjuvant, (time frame:1 month after vaccination). Immune response of a single dose of MenACWY-CRM conjugate vaccine, with adjuvant or without adjuvant, one month after administration to subjects aged 11 to 17 years, as measured by the percentage of subjects with human serum bactericidal activity titers >1:4 directed against N meningitidis serogroups A, C, W and Y. Human serum bactericidal activity geometric mean titers after one dose of MenACWY-CRM Vaccine, with adjuvant or without adjuvant (time frame: 1 month after vaccination). Immune response of one dose of MenACWY-CRM conjugate vaccine, with adjuvant or without adjuvant one month after administration in subjects 11 to 17 years of age, as measured by human serum bactericidal activity geometric mean titers against N meningitidis serogroups A, C, W, and Y. Percentages of subjects with human serum bactericidal activity titers≥ 1:4, after one dose of MenACWY-CRM(Ad-) (time frame: 12 months after vaccination). Human serum bactericidal activity geometric mean titers after one dose of MenACWY-CRM(Ad-) (time frame: 12 months after vaccination)". **Safety:** "numbers of subjects 11 to 17 years of age who reported solicited local and systemic adverse events after the vaccination (time frame: day 1 to day 7). Safety was assessed as the number of subjects 11 to 17 years of age who reported solicited local and systemic adverse events from day 1 up to and including day 7 after the vaccination of MenACWY-CRM conjugate vaccine, with adjuvant or without adjuvant".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (jackson.l@ghc.org, Lisa.A.Jackson@kp.org) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events/number of patients analysed at maximum follow up (please note that we are interested in

	SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** supported by grants from Novartis Vaccines, Siena, Italy

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Single blind
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	A total of 13 serious adverse events occurred but were not assigned per arm.

Juergens 2014

Methods	Randomized, open-label, multicenter trial
Participants	**Sex:** male and female **Age:** ≥65 years **Ethnicity:** African **Diagnosis:** healthy

	Inclusion criteria: "male and female adults 65 years of age and older. Determined by medical history, physical examination, and clinical judgment to be eligible for the study. Expected to be available for the duration of the trial (up to approximately 13 months)". **Exclusion criteria:** "received previous immunization with 23vPS. Serious chronic disorders including metastatic malignancy. Known or suspected hypersensitivity to any vaccine or vaccine components".
Interventions	**Vaccine type:** 13-valent pneumococcal conjugate vaccine (PCV13) **Experimental:** PCV13 with aluminium phosphate. n = 295 **Control:** PCV13. n =299 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** one dose intramuscularly into the deltoid muscle of the arm
Outcomes	**Immunogenicity:** "PCV13 serotype-specific immune responses were measured in a central laboratory by ELISA measuring IgG binding antibodies25 or by standardized microcolony OPA assays measuring functional antibody titers". **Safety:** "participants recorded local reactions (redness, swelling, pain, and limitation of arm movement) and systemic events (oral temperature to measure fever, fatigue, headache, chills, rash, vomiting, decreased appetite, and new and aggravated generalized muscle and joint pain) daily for 14 days after vaccination in an e-diary. Other AEs were collected by the investigator at each visit (29–43 days after each vaccination) based on clinical evaluation as well as in response to nonspecific questions".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (Beate.Schmoele-Thoma@Pfizer.com) were inquired on the followings: ● aluminium manufacturer ● allocation concealment ● number of participants with mortality and non-serious adverse event related and not related to the vaccine ● number of participants who got Streptococcus pneumoniae infection after the vaccine No answer received **Funding:** funded by Pfizer Inc

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation schedule was determined through an interactive voice response telephone system.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open-label

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00269672), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kashala 2002

Methods	Randomised, double-blinded, placebo-controlled phase I trial
Participants	**Sex:** male **Age:** 18–23 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "participants were recruited from the Colombian navy personnel in Tumaco by non-coercive means under international scientific and ethical standards embodied in the Declaration of Helsinki". **Exclusion criteria:** "subjects were excluded if they had a history of allergic reactions to vaccination, vaccination with live virus within 4 weeks of the beginning of the trial; had undergone splenectomy; or had a history of malaria, malaria parasitemia, or malaria vaccination, atopy, immunosuppressive drug therapy such as steroids; abnormal haematology and/or blood chemistry values. Volunteers were also excluded if they had positive markers of active auto-immunity; had cardiovascular, renal, and hepatic dysfunction or any other medical or psychiatric condition that in the opinion of the protocol chairman would compromise the patient's ability to tolerate the treatment. Furthermore, they were excluded if they were unwilling to give informed consent, or were willing to withdraw from the study".

Interventions	**Vaccine type:** plasmodium falciparum malaria peptide vaccine SPf66 **Experimental:** SPf66 with aluminium and 50 mcg of QS-21. n = 20 **Control:** SPf66 with 50 mcg of QS-21. n =19 **Aluminium concentration:** 1250 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Superfos Biosector (Vedvaek, Denmark) **Vaccination schedule:** three doses on days 0, 30 and 180
Outcomes	**Immunogenicity:** "sera were obtained at baseline, on day 20 following each immunisation, and at 4 months after the second and third doses, respectively for measurement of vaccine-induced immune responses. The antibody response against SPf66 has been characterised in previous reports based on end-point titres, using two-fold increments, as low/no response (enzyme-linked immunosorbent assay titres <100), intermediate (enzyme-linked immunosorbent assay titre: 200–800) or high (enzyme-linked immunosorbent assay titre =1600)". **Safety:** "evaluation of the reactogenicity, tolerability and toxicity. Reactogenicity was assessed by observing the incidence and severity of local and systemic reactions within 1, 24–48, 72 h and 20 days following the injections. Assessment of local reactions included erythema, tenderness/induration, pain, local pruritus, arm motion limitation related to pain and presence of local adenopathy. Erythema, tenderness/induration were scored as none, mild (<30mm in the largest diameter), moderate (30–120mm in the largest diameter) and severe (>120mm in the largest diameter). The limitation of the range of motion was considered a surrogate measurement of local pain, and was used to determine the severity of the local pain. Pain was scored as none or mild, if the range of motion was unchanged, and the subject could raise his arm to greater than a 90○ angle, moderate if the subject could raise his arm to no more than 45–90○ angle, and severe if the subject could not raise his/her arm to a 45○ angle. Systemic reactions were also rated, including allergic reactions such as rash/hives, generalised pruritus, bronchospasm, with or without hypotension, fever (≲38 ○C), anorexia, vomiting, abdominal pain, malaise/dizziness, chills, diarrhoea, headache, hypotension, cyst/nodules, abscess, necrosis, axillary adenopathy, myalgia, arthralgia and arthritis. To monitor toxicity, the blood specimens and sera were obtained from each subject at baseline, 48 h and 20 days post vaccination for haematology and blood chemistry".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (oscar.kashala@emdpharmaceuticals.com, gerd.pluschke@swisstph.ch) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in

	SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received, but no further information or data added **Funding:** supported by the Government of Colombia, and UNDP/World Bank/WHO Special Programme for Research and Training in Tropical Diseases

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blinded, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Keitel 2008

Methods	Multicenter, randomised, double-blind, placebo-controlled clinical trial
Participants	**Sex:** male and female **Age:** 18–49 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "males or nonpregnant females between the ages of 18 and 49 years, inclusive. Agree to practice adequate contraception (i.e., barrier methods, abstinence, intrauterine devices, and licensed hormonal methods) for the entire study period if they are females of childbearing potential (not surgically sterile or postmenopausal for greater than or equal to 1 year). Are in good health as determined by vital signs, medical history to ensure stable medical condition, and targeted physical examination based on medical history. Are able to understand and comply with planned study procedures. Provide written informed consent prior to initiation of any study procedures. Stable medical condition - no change in prescription medication, dose, or frequency of medication in the last 3 months and health outcomes of the specific disease are considered to be within acceptable limits in the last 6 months. Any change that is due to change of health care provider, insurance company etc, or that is done for financial reasons, as long as in the same class of medication will not be considered a violation of this inclusion criterion. Any change in prescription medication due to improvement of a disease outcome will not be considered a violation of this inclusion criterion". **Exclusion criteria:** "have a known allergy to eggs or other components of the vaccine (including gelatin, formaldehyde, octoxinol, thimerosal, aluminium hydroxide, and chicken protein). Have a positive urine or serum pregnancy test prior to vaccination (if female of childbearing potential) or are women who are breastfeeding. Have immunosuppression as a result of an underlying illness or treatment, or use of anticancer chemotherapy or radiation therapy within the preceding 36 months. Have an active neoplastic disease or a history of any hematologic malignancy. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (>800 mcg/day of beclomethasone dipropionate or equivalent) within the preceding 6 months (Nasal and topical steroids are allowed.). Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to self or others. Are receiving psychiatric drugs (aripiprazole, clozapine, ziprasidone, haloperidol, molindone, loxapine, thioridazine, thiothixene, pimozide, fluphenazine, risperidone, mesoridazine, quetiapine, trifluoperazine, chlorprothixene, chlorpromazine, perphenazine, trifluopromazine, olanzapine, carbamazepine, divalproex sodium, lithium carbonate or lithium citrate). Subjects who are receiving a single antidepressant drug and stable for at least 3 months prior to enrollment, without de-compensating symptoms will be allowed to be enrolled in the study. Have a history of receiving immunoglobulin or other blood product within the 3 months prior to vaccination in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to vaccination in this study. Have an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses (This includes, but is not limited to, known chronic liver disease, significant renal disease, unstable or progressive neurological disorders, diabetes mellitus, and transplant

	recipients.). Have a history of severe reactions following immunization with contemporary influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week of vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 1 month prior to vaccination in this study or expect to receive an experimental agent during the 7-month study period. Have any condition that would, in the opinion of the site investigator, place them at an unacceptable risk of injury or render them unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Planned to travel outside of the USA in the time between the first vaccination and 56 days following the first vaccination. Have a history of Guillain Barre syndrome. Have any condition that the investigator believes may interfere with successful completion of the study".
Interventions	**Vaccine type:** inactivated Influenza A/H5N1 vaccine **Experimental (a):** 3.75 mcg H5N1 with aluminium hydroxide. n = 61 **Control (a):** 3.75 mcg H5N1. n = 59 **Experimental (b):** 15 or 45 mcg H5N1 with aluminium hydroxide. n = 302 **Control (b):** 15 or 45 mcg H5N1. n =295 **Aluminium concentration: (a)** 300 mcg/dose. **(b)** 600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Sanofi **Vaccination schedule:** 2 vaccinations 1 month apart
Outcomes	**Immunogenicity:** "proportion of subjects in each group achieving a serum neutralizing antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus, (time frame: 28 days following second dose of vaccine). Proportion of subjects in each dose group achieving a serum Hemagglutination Inhibition antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus, (time frame: 28 days following second dose of vaccine). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group, (time frame: 28 days following second dose of vaccine). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group, (time frame: 28 days following second dose of vaccine). Geometric mean titer and the frequency of 4-fold or greater increases in serum HAI antibody titers, (time frame: 1 month after receipt of each dose, and 7 months after receipt of the first dose of vaccine). Development of serum antibody responses against antigenically drifted variants of H5N1 influenza virus, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization)". **Safety:** "adverse event or serious adverse event information (solicited in the clinic and via memory aids, concomitant medications, and periodic targeted physical assessments), (time frame: adverse events will be collected through 28 days following the second dose of vaccine (approximately day 56). Serious adverse events will be collected throughout the study through day 208)".
Are there other groups in the trial not analysed in this review?	No

Notes	**Contacts with authors:** on 14/12/2018 authors (wkeitel@bcm.edu) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● Final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** financial support: Division of Microbiology and Infectious Diseases, National Institute of Allergy and Infectious Diseases, National Institutes of Health (contracts AI-25465, AI-25461, and AI-25460); University of Maryland General Clinical Research Center (grant M01 RR165001); University of Rochester Clinical and Translational Science Institute (grant UL1 RR024160 from the National Center for Research Resources, National Institutes of Health).

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not reported

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00296634), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Keitel 2009

Methods	Multicenter, randomised, double-blind, placebo-controlled clinical trial
Participants	**Sex:** male and female **Age:** 18 and 40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "stage 1 cohort: healthy males and females aged 18 to 40 years, inclusive; females of childbearing potential must agree to practice adequate contraception for the duration of the study (not surgically sterile or Aftermenopausal for at least 1 year); good health as determined by screening medical history and physical examination; on no medications with the exception of licensed hormonal contraceptives and vitamins; willingness to comply with study procedures; ability to provide informed consent; availability for follow-up for 6 months after second vaccination; normal screening laboratory values for total WBC, platelet count, hemoglobin, creatinine, and alanine aminotransferase (Stage 1 only). Stage 2 Cohort: healthy males and non-pregnant females aged 18 to 40 years, inclusive. Females of childbearing potential must agree to practice adequate contraception for the duration of the study. Good health as determined by screening medical history and physical examination. Willingness to comply with study procedures. Ability to provide informed consent. Availability for follow-up for 6 months after second vaccination". **Exclusion criteria:** "stage 1 cohort: have a positive urine or serum pregnancy test in the 24 hours prior to vaccination (if female of childbearing potential) or are women who are breastfeeding. Have immunosuppression as a result of an underlying illness or treatment, or use of anticancer chemotherapy or radiation therapy within the preceding 36 months. Have an active neoplastic disease or a history of any hematologic malignancy. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (>800 mcg/day of beclomethasone dipropionate or equivalent) within the preceding 6 months. (Nasal and topical steroids are allowed.) Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to self or others. Have a history of receiving immunoglobulin or other blood product within the 3 months prior to vaccination in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to vaccination in this study. Have an acute or chronic medical

condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses (This includes, but is not limited to, known chronic liver disease, significant renal disease, unstable or progressive neurological disorders, diabetes mellitus, and transplant recipients.) Have a history of severe reactions following immunization with contemporary influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week of vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 1 month prior to vaccination in this study or expect to receive an experimental agent during the 7-month study period. Have any condition that would, in the opinion of the site investigator, place them at an unacceptable risk of injury or render them unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Have a history of Guillain Barre Syndrome. Have any condition that the investigator believes may interfere with successful completion of the study. Stage 2 Cohort: have a positive urine or serum pregnancy test prior to vaccination (if female of childbearing potential) or are women who are breastfeeding. Have immunosuppression as a result of an underlying illness or treatment, or use of anticancer chemotherapy or radiation therapy within the preceding 36 months. Have an active neoplastic disease or a history of any hematologic malignancy. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (>800 mcg/day of beclomethasone dipropionate or equivalent) within the preceding 6 months (nasal and topical steroids are allowed.). Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to self or others. Are receiving psychiatric drugs*. Subjects who are receiving a single antidepressant drug and stable for at least 3 months prior to enrollment, without de-compensating symptoms will be allowed to be enrolled in the study (aripiprazole, clozapine, ziprasidone, haloperidol, molindone, loxapine, thioridazine, molindone, thiothixene, pimozide, fluphenazine, fluphenazine, risperidone, mesoridazine, quetiapine, trifluoperazine, chlorprothixene, chlorpromazine, perphenazine, trifluopromazine, olanzapine, carbamazepine, divalproex sodium, lithium carbonate or lithium citrate). Have a history of receiving immunoglobulin or other blood product within the 3 months prior to vaccination in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to vaccination in this study. Have an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses (this includes, but is not limited to, known chronic liver disease, significant renal disease, unstable or progressive neurological disorders, diabetes mellitus, and transplant recipients.). Have a history of severe reactions following immunization with contemporary influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week of vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 1 month prior to vaccination in this study or expect to receive an experimental agent during the 7-month study period. Have any condition that would, in the opinion of the site investigator, place them at an unacceptable risk of injury or render them unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does

	not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Have a history of Guillain Barre Syndrome. Have any condition that the investigator believes may interfere with successful completion of the study".
Interventions	**Vaccine type:** inactivated whole virus Influenza A/H5N1 vaccine **Experimental:** 7.5 or 15 mcg H5N1 with aluminium hydroxide. n = 101 **Control:** 7.5 or 15 mcg H5N1. n =100 **Aluminium concentration:** 350 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses of vaccine in the deltoid muscle approximately 28 days apart
Outcomes	**Immunogenicity:** "proportion of subjects in each dose group achieving a serum neutralizing antibody titer of greater than or equal to 40 against the influenza A/H5N1 virus, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Proportion of subjects in each dose group achieving a serum hemagglutination inhibition antibody titer of greater than or equal to 40 against the influenza A/H5N1 virus, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group, (time frame: 1 month and 7 months after receipt of the 1st dose of vaccine). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group, (time frame: 1 month and 7 months after receipt of the 1st dose of vaccine)". **Safety:** "adverse event or serious adverse event information (solicited in the clinic and via memory aids, concomitant medications, periodic targeted physical assessment, and laboratory safety evaluations for a subset of subjects, (time frame: duration of study)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (wkeitel@bcm.edu) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up

	● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** National Institute of Allergy and Infectious Diseases; Division of Microbiology and Infectious Diseases (contracts NO1-AI-25465, NO1-AI-25461, NO1-AI-25462, NO1-AI-25463, NO1-AI25462); the Stanford University General Clinical Research Center (grant M01 RR-00070); the University of Maryland General Clinical Research Center (grant M01 RR165001); and the Vanderbilt University General Clinical Research Center (grant M01 RR-00095)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00382980), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kotloff 2001

Methods	Phase 1, randomised study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy adults 18 to 50 years of age were recruited from the Baltimore-Washington community and the University of Maryland College Park campus. Women were required to have a negative pregnancy test within 7 days of each vaccination". **Exclusion criteria:** "after medical history determination, physical examination, and a battery of laboratory tests, individuals meeting the following criteria were excluded: significant medical or psychiatric abnormalities, a history of gastrointestinal disease including previous antibiotic-associated diarrhea, a positive fecal *C. difficile* stool cytotoxin assay, antibiotic therapy during the previous month, and failure to achieve a passing grade (70%) on a written examination which tested the volunteer's comprehension of the trial".
Interventions	**Vaccine type:** *Clostridium difficile* toxoid vaccine **Experimental:** 6.25, 25 or 100 mcg Clostridium difficile vaccine with aluminium hydroxide. n = 15 **Control:** 6.25, 25 or 100 mcg Clostridium difficile vaccine. n = 15 **Aluminium concentration:** 723 to 757 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Reheis **Vaccination schedule:** four inoculations administered by intramuscular injection into the deltoid muscle (alternating arms) on study days 1, 8, 30, and 60
Outcomes	**Immunogenicity:** "peripheral blood mononuclear cells were collected before and 7 days after administration of each dose of vaccine to measure circulating immunoglobulin A and IgG antibody secreting cells recognizing toxin A and toxin B, by enzyme-linked immune absorbent spot assay. Sera were collected pre-vaccination and 8, 15, 30, 37, 60, 67, and 90 days after immunization and tested for antibodies to toxins A and B by enzyme-linked immunosorbent assay and by neutralization assay. Specific fecal immunoglobulin A antibody responses were measured before and 7 days after each dose of vaccine by enzyme-linked immunosorbent assay". **Safety:** "subjects were monitored at the study site for 30 min after receiving vaccine to detect immediate reactions. For 7 days after each inoculation, they kept a diary in which they recorded their evening oral temperature; the time, date, and consistency (loose or formed) of all stools passed; and the following signs and symptoms: anorexia, malaise, joint swelling, numbness or tingling in fingers or toes, rash, abdominal pain, joint pain, injection site pain, and other, unsolicited complaints. Subjects graded their pain as mild (aware of symptom but symptom easily tolerated), moderate (symptom interferes with usual activity), or severe (symptom prevents work or ability to perform usual activity). In addition, a medical history was obtained at each visit to the study site and by telephone interview 3 days after each inoculation. A panel of clinical laboratory tests (complete blood count with differential and platelet counts,

	creatinine, alanine aminotransferase, and urinalysis) was performed 1 week after each inoculation. Study personnel inspected the injection site 1 and 7 days after each inoculation for the presence of local reactions (swelling, erythema, or heat)".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** 14/12/2018 authors (kkotloff@som.umaryland.edu) were inquired on the followings: ● method for random sequence generation ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** supported by a contract from the National Institute of Allergy and Infectious Diseases to Myron Levine (NO1-AI-45251)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	High risk	Volunteers were sequentially assigned, in groups of 10, to receive one of three increasing doses of vaccine
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kunz 1976

Methods	Field trial
Participants	**Sex:** male and female **Age:** adults, 18-65 years old **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** volunteers, previously received detailed information about the trial, male and female, adults, 18-65 years old **Exclusion criteria:** not reported
Interventions	**Vaccine type:** H3N2 influenza **Experimental (a):** subunit - 700 IU of strain a/port chalmers/73 in 0.5 ml with aluminium. n = 15 **Control (a):** subunit - 700 IU of strain a/port chalmers/73 in 0.5 ml. n = 15 **Experimental (b):** whole virus with aluminium. n = 48 **Control (b):** whole virus. n = 48 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Sandoz **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity**: "geometric mean of hemagglutination inhibition assay (HIA) before and after vaccination; number of participants with seroconversion in HIA (4 fold increase in titer); number of participants that reach a titer over 1:64 in HIA (assumed immunity against disease level); geometric mean of antibodies in neuraminidase inhibition assay before and after vaccination". **Safety:** "questionnaire where participants record local and systemic side effects for 3 days after vaccination, participants are asked about side effects at visits on day 28 and 90".
Are there other groups in the trial not analysed in this review?	No
Notes	**Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Randomization using a continuous list
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kutzelnigg 2009

Methods	Two phase I randomised clinical studies
Participants	**Sex:** not reported **Age:** not reported **Ethnicity:** not reported **Diagnosis:** probable Alzheimer's disease of mild to moderate severity **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** Ab Affitopes vaccines for Alzheimer's disease **Experimental:**Ab Affitopes with aluminium hydroxide. n = 12 **Control:** Ab Affitopes. n = 12 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 4 vaccinations at 4-week intervals

Outcomes	Immunogenicity and safety
Are there other groups in the trial not analysed in this review?	No
Notes	Abstract **Funding**: not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Landrum 2017

Methods	Randomized, double-blind, placebo-controlled dose-escalation study
Participants	**Sex:** male and female **Age:** 18-55 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy adult males or females, DoD beneficiaries, including active duty members, 18-55 years of age. Negative urine pregnancy test for female subjects of child bearing potential (negative test within 24 hours prior to investigational product injection) or documented surgical sterility. Female subjects of child-bearing potential must use an acceptable method of birth control, as determined by the PI. Willingness to participate in this study as evidenced by written informed consent". **Exclusion criteria:** "prior receipt of S. aureus rAT or rLukS-PV; known S. aureus infection requiring medical treatment within the 3 months prior to study injection; known active viral or bacterial infection or symptoms/signs consistent with such an infection within the 2 weeks prior to study injection (mild intercurrent viral illness with a temperature of 100.3°F (37.9°C) or less did not require exclusion, if in the judgment of the investigator or sub-investigator this illness would not interfere with the safety evaluation); seropositivity for HIV infection at the time of screening; known or suspected abuse of prescribed or illicit drugs, or alcohol in the past year; use of any new medications (except oral contraceptives, over-the-counter medications or vitamin supplements) within the 7 days prior to IP injection; use of investigational drugs, vaccines, or devices within the 30 days prior to each dose of study injection or anticipated use of such items during the study (licensed vaccines could be administered at any time with the exception of 14 days prior to and 14 days after administration of each study injection); use of systemic steroids (any dose) or high daily dose of inhaled steroids within the last month (use of low or medium daily dose inhaled, intranasal, or low potency topical steroid creams/ointments was allowed unless such medication was begun within the 7 days prior to study injection); history of a bleeding or coagulation disorder, or use of anticoagulant medications within 7 days prior to study injection; actively breastfeeding; presence of grade 1 or higher abnormality in laboratory or vital signs parameter at time of screening; or presence of any condition which, in the opinion of the investigator, placed the subject at undue risk or potentially jeopardized the quality of the data to be generated".
Interventions	**Vaccine type:** no vaccine. Only placebo arm extracted in this review **Experimental:** 0.9% saline placebo with aluminium hydroxide. n = 22 **Control:** 0.9% saline placebo. n = 22 **Aluminium concentration:** 800 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Nabi Biopharmaceuticals **Vaccination schedule:** one dose
Outcomes	**Safety:** "clinical examinations, clinical laboratory results, self-reported diary reactogenicity data and adverse event reports, (time frame: up to 6 months). Adverse events, local reactogenicity, and systemic reactogenicity were assessed through clinical examination by study providers, clinical lab results, as well as review of subject-completed diary". **Immunogenicity:** "geometric mean concentrations after first injection, complete population, (time frame: up to 3 months). Immunogenicity is the ability of a particular substance, such as an antigen or epitope, to provoke an immune

	response in the body of a human or animal. Immunogenicity was determined on the basis of anti-rAT and anti-rLukS-PV IgG concentrations assessed by enzyme-linked immunosorbent assay in sera from blood samples collected on days 0 (baseline), 14, 28 and 84 for those receiving a single dose of vaccine. For those receiving a second dose of vaccine, immunogenicity assessments were also conducted on days 98 and 112. Immunogenicity was evaluated using the following metrics: geometric mean concentrations, geometric mean fold increase and seroresponse status. Seroresponse variables are normally defined in terms of exceeding a threshold".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (tlalani@idcrp.org) were inquired on the followings: ● allocation concealment. ● blinding of outcome assessment ● number of participants who got Staphylococcus aureus infection after the vaccine. No answer received **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Web-based randomisation system
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Study personnel and subjects were double-blinded to active vaccine vs. placebo within each treatment cohort, but not to volume of vaccine/placebo injected
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs more than 5%

Incomplete outcome data (attrition bias) All-cause mortality	High risk	Drop outs more than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs more than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01011335), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Langley 2009

Methods	Blinded, randomised, dose-ranging, placebo-controlled study
Participants	**Sex:** male and female **Age:** ≥65 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** participants were enrolled from the community in five urban areas of Canada (Halifax, NS; Toronto, ON; Edmonton, AB; Surrey and Port Coquitlam, BC) Eligible persons were ≥65 years of age with informed consent. **Exclusion criteria:** "immunosuppressive condition (secondary to underlying disease or to immunosuppressive therapy in the preceeding 6-months, or currently on immunosuppressive or immunomodulating medication), known impairment of neurologic function or unstable seizure disorder, receipt of blood products or immunoglobulins within the previous 3-months or planned receipt in 60 days post-immunization, known chemical dependency associated with immune suppression, previous immunization against RSV, known or suspected hypersensitivity to a previous vaccine intended for study use, or to its components, recent receipt of other vaccines (killed vaccines <14 days; live <30 days) or planned receipt of other vaccines in 32 days post-study immunization, receipt of an investigational product within a month of study enrollment, known or suspected acute infectious respiratory tract infection at time of study enrollment, acute febrile illness, other conditions which might interfere with vaccine evaluation or pose a health risk to participant according to the investigator's judgement or previous treatment with a hormone of pituitary origin".
Interventions	**Vaccine type:** respiratory syncytial virus subtype A vaccine (RSV-A) **Experimental:** 100 mcg RSV-A with aluminium. n = 114 **Control:** 100 mcg RSV-A. n = 112 **Aluminium concentration:** 1500 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** one intramuscular injection. All participants were administered influenza vaccine (Fluzone® or Vaxigrip ®) by intramuscular injection on day 32.
Outcomes	**Safety:** "study participants were observed for 30min after immunization in the clinic. Telephone contact was made by a study team member on days 1, 2 and 4 after immunization to assess solicited local and systemic adverse events. All adverse events were recorded in the participant diary. Adverse event data was

	also collected on days 120, 180 and 360 during visits. Severe/serious adverse events were assessed throughout the study. Solicited local (redness, swelling, pain, tenderness) and systemic AEs (fever, chills, sore throat, sinus/nasal congestion, cough, headache, nausea, vomiting, diarrhea, general aches, tiredness, sore joints) were categorized as mild, moderate and severe by the participants. Serious adverse events were defined as any untoward medical occurrence, which may be related to the study product, resulting in any of the following: death, is life-threatening, persistent or significant disability/incapacity, requires inpatient hospitalisation or is a congenital anomaly or birth defect. All events regardless of relationship to vaccine were recorded". **Immunogenicity:** "plaque reduction neutralization assay [RSV-A long strain and RSVB strain W/V/1467/*85 [B-1 wild type]] and enzyme immunoassay. The RSVneutralizing antibody titre was performed. The proportion of participants with RSV-A and -B NA titres ≥4-fold the baseline value were determined at day 32 (1-month post-vaccination), days 60, 180 and 360. RSV F and G glycoprotein specific antibodies and RSV M protein-specific antibodies were also assayed using the enzyme-linked immunoassay on these days. Hemagglutinin-inhibition antibody assays for H1N1 and H3N2 influenza A, and influenza B vaccine strains were performed".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 30/01/2020 authors (joanne.langley@dal.ca) were inquired on the followings: ● Method for random sequence generation ● Procedure for blinding of outcome assessment ● Funding source ● Final concentration of aluminium injected ● Aluminum manufacturer ● Number of patients who died/number of patients analysed ● Number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● Number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● Total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** no source of funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized. Participants were assigned by within-center block randomisation to one of five study arms.
Allocation concealment (selection bias)	Unclear risk	Not described

Blinding of participants and personnel (performance bias)	Unclear risk	Blinded. Because the un-adjuvanted preparation is a clear colorless liquid and the other alum-containing test articles were cloudy white, all study vaccines were over-labelled with a blinding label to conceal allocation.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop-outs < 5%.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	High risk	No pre-published protocol. Serious adverse events not considered to be related to the vaccine were not assigned by arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Langley 2016

Methods	Phase I randomised, observer-blind, controlled, dose escalation trial
Participants	**Sex:** male and female **Age:** 50–64 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "age 50-64 years, inclusive. Good general health status, as determined by history and physical examination no greater than 30 days prior to administration of the test article. Participants who, in the opinion of the investigator, can and will comply with the requirements of the protocol (e.g. completion of Diary Cards, return for follow-up visits). Written informed consent obtained from the participant. If female of child-bearing potential and heterosexually active, has practiced adequate contraception for 30 days prior to injection, has a negative pregnancy test on the day of injection, and has agreed to continue adequate contraception until 180 days after injection. (Please refer to the glossary for the definition of child-bearing potential and adequate contraception)".

	Exclusion criteria: "use of any investigational or non-registered product (drug or vaccine) other than the study product within 28 days preceding the dose of study product, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the participant has been or will be exposed to an investigational or a non-investigational vaccine/product (pharmaceutical product or device). Planned administration/ administration of a vaccine/product not foreseen by the study protocol within the period starting 28 days before injection of a study vaccine and ending 84 days after. Administration of immunoglobulins and/or any blood products within the 3 months preceding the dose of study product or planned administration during the study period. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. (Laboratory testing for HIV, Hepatitis C and Hepatitis B will be performed during the screening visit). Chronic administration (defined as more than 14 days in total) of immunosuppressants or other immune-modifying drug within 6 months prior to the product dose (for corticosteroids, this will mean prednisone ≥ 20 mg/day, or equivalent). Inhaled and topical steroids are allowed. Family history of congenital or hereditary immunodeficiency. History of or current autoimmune disease. History of hypersensitivity to any test article constituent or products used during the course of study procedures. Known or suspected hypersensitivity to any ingredient in the formulation or component of the container. Pregnant or lactating female. Female planning to become pregnant or planning to discontinue contraceptive precautions within 180 days of study vaccine receipt. Any hematological (hemoglobin level, white blood cell, and platelet count) and biochemical (alanine aminotransferase, aspartate aminotransferase, blood urea nitrogen and creatinine) abnormality as per local laboratory normal values considered clinically significant by the transient mild laboratory abnormalities may be rescreened and the participant will be deemed eligible if the laboratory repeat test is normal as per local laboratory normal values and investigator assessment. Any acute or chronic, clinically significant disease, as determined by physical examination or laboratory screening tests. Malignancies within previous 5 years (excluding non-melanic skin cancer) and lymphoproliferative disorders. Current alcoholism and/or drug abuse. Acute disease and/or fever at the time of Screening ≥ 38°C. Planned move to a location that will prohibit participating in the trial until study end. Any other condition that the investigator judges may interfere with study procedures (e.g. drawing blood) or findings (e.g. immune response)".
Interventions	**Vaccine type:** respiratory syncytial virus vaccine (RSV) **Experimental:** 10 or 25 mcg RSV with aluminium **Control:** 10 or 25 mcg RSV **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** not reported
Outcomes	**Immunogenicity**: blood samples for IgG Ab to SH Ag and hematologic or biochemical parameters were collected on day 0, 7, 28, 56, 63, 84 and 236 **Safety**: solicited local and general adverse events were collected on a memory aide day 0-6 post-injection and unsolicited adverse events through to day 28 post-injection. Serious adverse events, medically attended adverse events, and potentially immune mediated disease were collected to day 236.

Are there other groups in the trial not analysed in this review?	Yes
Notes	Abstract **Contacts with authors:** on 26/01/2021 authors (joanne.langley@dal.ca) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment ● number of drop outs per outcome ● aluminium concentration per dose ● aluminium manufacturer ● safety and immunogenicity results No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Participants and observers were blinded to allocation.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Participants and observers were blinded to allocation.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Participants and observers were blinded to allocation.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Participants and observers were blinded to allocation.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.

Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02472548), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Langley 2017

Methods	Randomized, controlled, observer-blinded, first in- humans, phase 1 clinical trial
Participants	**Sex:** male **Age:** 18–44 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects who, in the opinion of the investigator, can and will comply with the requirements of the protocol. Written informed consent obtained from the trial participants. A male between, and including, 18 and 44 years of age at the time of vaccination. Healthy subjects as established by medical history and clinical examination before entering into the study". **Exclusion criteria:** "use of any investigational or non-registered product other than the study vaccines within 30 days preceding the dose of study vaccine, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the subject has been or will be exposed to an investigational or a non-investigational vaccine/product. Planned administration/ administration of a vaccine not foreseen by the study protocol within the period starting 30 days before and ending 30 days after vaccination, with the exception of any licensed influenza vaccine which may be administered ≥ 15 days before or after vaccination. Previous vaccination against RSV. Chronic administration of immunosuppressants or other immune-modifying drugs within 6 months prior to the vaccine dose. Inhaled and topical steroids are allowed. Administration of long-acting immune-modifying drugs at any time during the study period. Administration of immunoglobulins and/or any blood products within the 3 months preceding the dose of study vaccine or planned administration during the study period. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. Family history of congenital or hereditary immunodeficiency. History of or current autoimmune disease. History of any reaction or hypersensitivity likely to be exacerbated by any component of the vaccines. Hypersensitivity to latex. Any clinically significant hematological (hemoglobin level, white blood cell [WBC], lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine aminotransferase, aspartate aminotransferase and creatinine) abnormality as per the opinion of the investigator, based on the local laboratory normal ranges.Subjects with hematological/ biochemical values out of normal range which are expected to be temporary may be re-screened at a later date. Any acute or chronic, clinically significant disease, as determined by physical examination or laboratory screening tests. Malignancies within previous 5 years (excluding non-melanoma skin cancer) and lymphoproliferative disorders. Current alcoholism and/or drug abuse. Acute disease and/or fever at the time of Screening.Fever is defined as temperature ≥ 37.5°C /99.5°F for oral, axillary or tympanic route, or ≥ 38.0°C /100.4°F on rectal route. The preferred route for recording temperature in this study will be oral.Subjects with a minor illness (such as mild diarrhea) without fever may be enrolled at the discretion of the investigator.Subjects with acute

	disease and/ or fever at the time of Screening may be re screened at a later date. Planned move to a location that will prohibit participating in the trial until study end. Any other condition that the investigator judges may interfere with study procedures (e.g. drawing blood) or findings (e.g. immune response)".
Interventions	**Vaccine type:** respiratory syncytial virus vaccine (RSV) **Experimental:** 10, 30 or 60 mcg RSV with aluminium. n = 48 **Control:** 10, 30 or 60 mcg RSV. n = 47 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** GSK **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "humoral immune response in terms of neutralizing antibody titers to the investigational RSV vaccines, in all subjects, in all groups, (time frame: at pre-vaccination (day 0) and post-vaccination (day 7, day 30 and day 60)). Persistence of the humoral immune response in terms of neutralizing antibody titers to the investigational RSV vaccines, in all subjects, in all groups, (time frame: at day 180 and day 360)". **Safety:** "occurrence of each solicited local and general adverse event, (time frame: during the 7 days (Days 0-6) follow-up period after vaccination). Occurrence of any hematological (hemoglobin level, white blood cells lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine amino-transferase, aspartate amino-transferase and creatinine) laboratory abnormality, (time frame: at day 0, 7 and 30). Occurrence of any hematological (hemoglobin level, whole blood cells, lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine amino-transferase, aspartate amino-transferase and creatinine) laboratory abnormality, (time frame: at day 60). Occurrence of any unsolicited adverse events, (time frame: during a 30-day (days 0-29) follow-up period after vaccination). Occurrence of any serious adverse events, (time frame: from day 0 to day 60). Occurrence of any hematological (hemoglobin level, whole blood cells, lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine amino-transferase, aspartate amino-transferase and creatinine) laboratory abnormality, (time frame: at day 180 and day 360). Occurrence of any serious adverse events, (time frame: from day 60 to the study conclusion (i.e. day 360))".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (joanne.langley@dal.ca) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received **Funding:** supported by GlaxoSmithKline Biologicals SA. GlaxoSmithKline Biologicals SA was involved in all stages of the study conduct and analysis.

	GlaxoSmithKline Biologicals SA paid for open access of the article.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation sequence was generated using MATerial Excellence, a software program developed for use in Statistical Analysis System (SAS®) (Cary, North Carolina) by GSK (Rixensart, Belgium). A randomisation blocking scheme was used to ensure balance between study groups in each step, and a minimization procedure accounted for age category (18–32 or 33–44 years).
Allocation concealment (selection bias)	Low risk	Study group assignment was allocated by an Internet-based central randomisation system.
Blinding of participants and personnel (performance bias)	Low risk	Participants and all study personnel, except for an unblinded nurse, whose sole role was to prepare and administer study vaccines, were not aware of treatment assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT02247726), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2015

Methods	Two phase I/II, single-blind, parallel-group studies
Participants	**Sex:** male and female **Age:** >18 **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects who the investigator believes will comply with the requirements of the protocol. A male or female ≥ 18 years at the time of the first vaccination. Written informed consent obtained from the subject. Free of obvious health problems as established by medical history and clinical examination before entering into the study. If the subject is female, she must be of non-childbearing potential, i.e., either surgically sterilized or one year post-menopausal; or, if of childbearing potential, she must be abstinent or have used adequate contraceptive precautions for 30 days prior to vaccination, have a negative pregnancy test and must agree to continue such precautions for two months after completion of the vaccination series". **Exclusion criteria:** "use of any investigational or non-registered product (drug or vaccine) other than the study vaccine(s) within 30 days preceding the first dose of study vaccine, or planned use during the study period or participation to another pharmaceutical/vaccine study. Chronic administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within six months prior to the first vaccine dose. Use of any anticoagulants. Planned administration/ administration of a vaccine not foreseen by the study protocol within 2 weeks of the first dose of vaccines. Previous vaccination against Streptococcus pneumoniae. Bacterial pneumonia within 3 years prior to 1st vaccination. Any confirmed or suspected immunosuppressive or immunodeficient condition, including human immunodeficiency virus (HIV) infection. History of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Current serious neurologic or mental disorders. Currently smoking > 25 cigarettes per day. Inflammatory processes such as known chronic active infections. All malignancies (excluding non-melanic skin cancer) and lymphoproliferative disorders diagnosed or treated actively during the past 5 years. History of administration of an experimental vaccine containing MPL or QS21. Acute disease at the time of enrolment. Acute or chronic, clinically significant pulmonary, cardiovascular, hepatic or renal functional abnormality, as determined by physical examination or laboratory screening tests, at the discretion of the investigator. History of chronic alcohol consumption and/or intravenous drug abuse".
Interventions	**Vaccine type:** pneumococcal protein PhtD vaccine **Experimental:** 30 mcg PhtD with aluminium. n = 50 **Control:** 30 mcg PhtD. n = 49 **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injections into the deltoid region of the upper right arm at 0 and 2 months. Influenza vaccine (FluarixTM; GlaxoSmithKline,Rixensart, Belgium) was offered
Outcomes	**Safety**: "occurrence, intensity and relationship of any solicited local and general signs and symptoms, (time frame: during a 7-day follow up period after each vaccine dose). Occurrence, intensity and relationship to vaccination of

	unsolicited local and general signs and symptoms, (time frame: during a 30-day follow up period after each vaccine dose). Occurrence of all serious adverse events, (time frame: during the entire study period). Number and percentage of subjects with normal or abnormal values for biochemical assessments and for haematological analysis, (time frame: at each scheduled time point (month 0, 1, 3, 12, 24 and 36))". **Immunogenicity**: "anti- PhtD antibody concentration, (time frame: one month after the first injection). Anti-PhtD antibody concentration, (time frame: one month after 2 injections). Anti- PhtD antibody concentration, (time frame: at 12, 24 and 36 months after the first vaccination). Anti-PhtD antibody avidity, (time frame: at month 0, 1 and 3). Evaluation of protection afforded by passive transfer of anti PhtD antibodies sera pooled from all individuals, (time frame: at month 0, 1 and 3). Frequency of PhtD specific plasma cells generated by in vitro cultivated memory B-cells, in a subset of subjects, (time frame: at month 0, 1 and 3). Frequency of CD4 and/or CD8 T cells that produce cytokines (IL-2, IL-4, IFNg, CD40L and/or GM-CSF, and TNFα), upon PhtD re-stimulation in vitro, to evaluate the T-cell response, in a subset of subjects, (time frame: at month 0, 1, 12, 24 and 36). Anti-polysaccharide total IgG concentration in Group A for all vaccine pneumococcal serotypes, (time frame: at month 0, 1, 12, 24 and 36). Anti-PS antibody avidity for 5 serotypes in Group A, (time frame: at month 0, 1). Deposition of complement components on the surface of different bacterial strains 3 strains (GSK/CDC, OPA, isogenic TIGR4) of 5 serotypes in Group A, (time frame: at month 0, 1). Opsonophagocytic activity titres in Group A to all vaccine pneumococcal serotypes, (time frame: at month 0, 1 and 12). Frequency of PS-specific plasma cells generated by in vitro cultivated memory B-cells in Group A in a subset of subjects, (time frame: at month 0, 1)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (isabel.lerouxroels@ugent.be, yves.horsmans@uclouvain.be) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further data or information added **Funding:** supported by GlaxoSmithKline (GSK) Biologicals SA Only vaccine-related serious adverse events were reported.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Centralised randomisation system on the Internet
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	The studies were single-blind, meaning that participants were unaware of the vaccination assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	The studies were single-blind, meaning that participants were unaware of the vaccination assignment.
Blinding of outcome assessment (detection bias) All-cause mortality	High risk	Single-blind
Blinding of outcome assessment (detection bias) Immunogenicity	High risk	Single-blind
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00307528), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2016a

Methods	Study 2 was a phase I, randomised,observer-blind, placebo-controlled, single-center, dose-escalation study
Participants	**Sex:** male and female **Age:** 18-70 years **Ethnicity:** Caucasian **Diagnosis:** previous or current smokers **Inclusion criteria:** "subjects who, in the opinion of the investigator, can and will comply with the requirements of the protocol. A male or female between, and including, 50 and 70 years of age at the time of the first vaccination. Written informed consent obtained from the subject. Subjects without medical history, clinical finding or laboratory finding which in the opinion of the investigator could pose a safety concern or interfere with the protocol. Current or former smokers.

	A smoking history of at least 10 pack-years. Female subjects of non-childbearing potential may be enrolled in the study. Female subjects of childbearing potential may be enrolled in the study, if the subject:has practiced adequate contraception for 30 days prior to vaccination, and has a negative pregnancy test on the day of vaccination, and has agreed to continue adequate contraception during the entire treatment period and for 2 months after completion of the vaccination series". **Exclusion criteria:** "use of any investigational or non-registered product other than the study vaccines within 30 days preceding the first dose of study vaccine, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the subject has been or will be exposed to an investigational or a non-investigational vaccine/product. Planned administration/ administration of a vaccine not foreseen by the study protocol in the period starting 30 days before the first dose and ending 30 days after the last dose of vaccine, with the exception of any influenza vaccine which may be administered ≥ 15 days preceding or following any study vaccine dose. Previous vaccination with any vaccine containing NTHi-antigens. Chronic administration of immunosuppressants or other immune-modifying drugs within six months prior to the first vaccine dose. Topical steroids are allowed. Administration of immunoglobulins and/or any blood products within the 3 months preceding the first dose of study vaccine or planned administration during the study period. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. History of or current autoimmune disease. Post-bronchodilator Forced Expiratory Volume in 1 second (FEV1) < 80% of predicted normal value. Diagnosed with a respiratory disorder. Please note that subjects with mild pulmonary obstruction can be enrolled. Laboratory evidence of clinically significant haematological abnormalities at Screening. Acute disease and/or fever at the time of enrolment. Current alcoholism and/or drug abuse. Has significant disease, in the opinion of the investigator, likely to interfere with the study and/or likely to cause death within the study duration. History of or current condition preventing intramuscular injection as bleeding or coagulation disorder. Has contraindication for spirometry testing. Malignancies within previous 5 years and lymphoproliferative disorders. Pregnant or lactating female. Female planning to become pregnant or planning to discontinue contraceptive precautions. Any other condition that the investigator judges may interfere with study findings".
Interventions	**Vaccine type:** non-typeable Haemophilus influenzae (NtHi) protein vaccine **Experimental:** 10 or 30 mcg NtHi with aluminium. n = 60 **Control:** 10 or 30 mcg NtHi. n = 60 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** two doses at months 0, 2 and 6
Outcomes	**Immunogenicity:** "humoral immune response to the components of the NtHi vaccine formulations, in all subjects, in all groups, in terms of antibody concentrations, (time frame: prior to each vaccination, 30 days after each vaccination, and at study conclusion (day 420)). Cell-mediated immune responses to components of the NtHi vaccine formulations, in a sub-cohort of subjects, in all groups (time frame: prior to each vaccination, 30 days after each vaccination, and at study conclusion (day 420))". **Safety:** "occurrence of each solicited local and general adverse event, in all

	subjects, in all groups, (time frame: during a 7-day follow-up period (from day 0 to day 6) after each vaccination). Occurrence of any unsolicited adverse event, in all subjects, in all groups, (time frame: during a 30-day follow-up period (from day 0 to day 29) after each vaccination). Occurrence of haematological and biochemical laboratory abnormalities, in all subjects, in all groups, (time frame: prior to each vaccination). Occurrence of haematological and biochemical laboratory abnormalities, in all subjects, in all groups, (time frame: 7 days after each vaccination). Occurrence of haematological and biochemical laboratory abnormalities, in all subjects, in all groups, (time frame: at study conclusion (day 420)). Occurrence of any serious adverse event, in all subjects, in all groups, (time frame: from first vaccination (day 0) to study conclusion (day 420)). Occurrence of any potential immune-mediated disease in all subjects, in all groups, (time frame: from first vaccination (day 0) to study conclusion (day 420))".
Are there other groups in the trial not analysed in this review?	Yes.
Notes	**Contacts with authors:** on 14/12/2018 authors (Geert.LerouxRoels@UGent.be, Mathieu.peeters@yahoo.co.uk) were inquired on the followings: ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no further data or information added **Funding:** GlaxoSmithKline Biologicals SA was the funding source and was involved in all stages of the clinical trial conduct and analysis. GlaxoSmithKline Biologicals SA also took in charge all costs associated with the development and the publishing of the manuscript. Only vaccine-related serious adverse events were reported.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomized using a centralized randomisation system on internet (SBIR)
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Observer-blind, i.e. the vaccine recipients and those responsible for the evaluation of any study endpoint were blinded to the administered vaccines. Due to differences in the appearance of the study vaccines, vaccines were prepared and administered by authorized medical personnel who did not participate in any of the study clinical evaluations or assays.

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Data were collected in an observer-blind manner.Those responsible for the evaluation of any study endpoint were blinded to the administered vaccines.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Data were collected in an observer-blind manner.Those responsible for the evaluation of any study endpoint were blinded to the administered vaccines.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Data were collected in an observer-blind manner.Those responsible for the evaluation of any study endpoint were blinded to the administered vaccines.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	The safety analysis was performed on the total vaccinated cohort which included all vaccinated participants
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The safety analysis was performed on the total vaccinated cohort which included all vaccinated participants
Incomplete outcome data (attrition bias) Immunogenicity	High risk	. The immunogenicity analysis was based on the according-to-protocol cohort for immunogenicity Drop outs more than 5%.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01678677), and all outcomes were reported on www.gsk-clinicalstudyregister.com (GSK studies 116018 and 116647)-GlaxoSmithKline (GSK) 2838497A.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2016b

Methods	Phase Ib, randomised, observer-blind and placebo-controlled study
Participants	**Sex:** female **Age:** 18-40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** healthy females 18-40 years of age inclusive **Exclusion criteria:** "individuals who are pregnant or nursing. Individuals who have had a previous immunization with a vaccine containing streptococcus antigens or any vaccine within 30 days of enrollment through 30 days of study completion. Individuals with a history of severe allergic reactions after previous vaccination. Individuals with designated blood tests that are not within normal range".
Interventions	**Vaccine type:** trivalent group B Streptococcus vaccine (GBS) **Experimental:** 5 or 20 mcg GBS with aluminium. n = 160 **Control:** 5 or 20 mcg GBS. n = 160 **Aluminium concentration:** 1000 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** one or two injections, administered one month apart

Outcomes	**Immunogenicity:**" percentage of subjects achieving specific thresholds of antibody concentrations for subjects in enrollment group, (time frame: day 1 and day 61). The percentage of subjects achieving a specific threshold of antibody concentrations of ≥ 0.5 µg/mL as determined by enzyme-linked immunosorbent assay at day 61 across cohorts 1 and 2 for potential use in pregnant women. Geometric mean concentrations of Serum Anti-GBS IgG antibody levels by serotype at day 61, (time frame: day 1 and day 61]. Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody levels in healthy subjects for potential use in pregnant woman was assessed in terms of geometric mean concentrations at day 61 in cohorts 1 and 2. Geometric mean ratios of serum group B streptococcus IgG antibody levels by serotype, (time frame: day 61/day 1). Geometric mean ratios relative to baseline (day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels measured at day 61 for subjects across cohorts 1 and 2. Percentage of subjects achieving specific thresholds of antibody concentrations for subjects in enrollment groups 1 & 2, (time frame: day 1 and day 361). The percentage of subjects achieving a specific threshold of antibody concentrations of ≥ 0.5 µg/mL at day 361 across all the cohorts for potential use in non-pregnant women. Geometric mean concentrations of serum anti-group B streptococcus IgG antibody levels by serotype at day 361, (time frame: day 1 and day 361). Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody levels in healthy subjects for potential use in non-pregnant woman was assessed in terms of geometric mean concentrations at day 361 across all cohorts. Geometric mean ratios of serum group B streptococcus IgG Antibody levels by serotype at day 361, (time frame: day 1 and day 361). Geometric mean ratios relative to baseline (day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels in healthy subjects for potential use in non-pregnant women, measured at day 361 across all cohorts. Persistence of antibody response in terms of geometric mean concentrations by serotype for subjects in enrollment group 1, (time frame: day 1 to day 721). The persistence of Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody response was assessed by comparing enzyme-linked immunosorbent assay responses in terms of geometric mean concentrations at day 721 after the first injection for subjects enrolled in cohorts 1 and 2. Persistence of antibody response in terms of geometric mean concentrations by serotype for subjects in enrollment group 2, (time frame: day 1 to day 721). The persistence of Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody response assessed by comparing enzyme-linked immunosorbent assay responses in terms of geometric mean concentrations at day 721 after the first injection for subjects enrolled in cohort 3 and 4. Geometric mean ratios of serum group B streptococcus IgG antibody levels by serotype for subjects in enrollment group 1, (time frame: day 1 to day 721). Geometric mean ratios was calculated at Day 721 relative to baseline (Day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels across all cohorts 1 and 2. Geometric mean ratios of serum group B streptococcus IgG antibody levels by serotype for subjects in enrollment group 2, (time frame: day 1 to day 721). Geometric mean ratio was calculated at Day 721 relative to baseline (day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels across cohorts 3 and 4". **Safety:** "number of participants reporting solicited adverse events after receiving the group B streptococcus trivalent vaccine in enrollment group 1, (time frame: day 1 to day 7). Safety was assessed in terms of number of participants reporting solicited local and systemic adverse events after receiving trivalent GBS vaccine from day 1 to day 7 for subjects across cohorts 1 and 2.

	Number of participants reporting solicited adverse events after receiving the group B streptococcus trivalent vaccine in enrollment group 2, (time frame: day 1 to day 7). Safety was assessed in terms of number of participants reporting solicited local and systemic adverse events after receiving GBS vaccine from day 1 to day 7 for subjects in cohorts 3 and 4. Number of participants reporting unsolicited adverse events after receiving group B streptococcus trivalent vaccine in enrollment group 1, (time frame: day 1 to day 721). Safety was assessed in terms of number of participants with unsolicited adverse events after receiving group B streptococcus trivalent vaccine, reported from day 1 to day 721 across subjects in cohorts 1 and 2. Number of participants reporting unsolicited adverse events after receiving group B streptococcus trivalent vaccine in enrollment group 2, (time frame: day 1 to day 721). Safety was assessed in terms of number of participants with unsolicited adverse events after receiving group B streptococcus trivalent vaccine, reported from day 1 to day 721 across subjects in cohorts 3 and 4".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (Geert.LerouxRoels@UGent.be, isa.x.bedell@gsk.com) were inquired on the followings: ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no further data or information added **Funding:** supported by Novartis Vaccines and DiagnosticsDivision, now part of the GSK group of companies

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A web-based randomisation system was used to randomly assign subjects into each treatment arm within the two Enrolment Groups
Allocation concealment (selection bias)	Low risk	A web-based randomisation system was used to randomly assign subjects into each treatment arm within the two Enrolment Groups
Blinding of participants and personnel (performance bias)	Unclear risk	Observer-blind, no further details

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01150123), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2019

Methods	Phase I, observer-blind, randomised, placebo-controlled, dosage-escalation study
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** Caucasian **Diagnosis:** healthy **Inclusion criteria:** "healthy males and non-pregnant females 18 to 45 years of age at time of enrollment. Individuals who have given written consent after the nature of the study has been explained according to local regulatory requirements. Individuals in good health as determined by the outcome of the medical history, physical examination and clinical judgment of the investigator. Individuals who can comply with the study procedures and are available for follow up". **Exclusion criteria:** "individuals with any severe chronic or acute disease. Individuals with a history of illness or with an ongoing illness that may pose additional risk to the subject if he/she participates in the study, including the following: history of any chronic respiratory illness, including current diagnosis of asthma within 2 years, exercise induced wheezing, reactive airway disease, emphysema, chronic bronchitis, cystic fibrosis or chronic obstructive pulmonary disease.Any respiratory illness within 7 days prior to receiving the first study injection.Any active pulmonary infection or other inflammatory conditions, even in the absence of febrile episodes, within 14 days prior to the first study

	injection.Hepatitis B or hepatitis C infection. Individuals who have had a malignancy or lymphoproliferative disorder within the past 5 years. Individuals with known or suspected impairment of the immune system including but not limited to HIV, autoimmune disorders, immunosuppressive therapy, and diabetes mellitus. Individuals with any history of progressive or severe neurologic disorder, seizure disorder or Guillian-Barré syndrome. Individuals with a known bleeding diathesis, or any condition that may be associated with a prolonged bleeding time. Individuals with a BMI > 35 kg/m2. BMI is to be calculated by the following formula: subject weight at baseline divided by subject height in meters multiplied by the subject height in meters. The numerical result will be rounded to the nearest 0.1. Individuals who are allergic to any of the vaccine components, or with a history of anaphylaxis after vaccination. Individuals who during the 90 days prior to enrollment receive any medications or other treatments that may adversely affect the immune system such as allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs or other drugs known to be frequently associated with significant major organ toxicity. Individuals who receive systemic immunosuppressive agents including steroids. Prior corticosteroid therapy should be discontinued 28 days prior to enrollment. Individuals using inhaled or topical corticosteroids will be permitted. Receipt or donation of blood or blood products 8 weeks prior to vaccination or planned receipt or donation during the study period. Individuals participating in any clinical trial with another investigational product 28 days prior to receiving the first study vaccination or intent to participate in another clinical study at any time during the conduct of this study. Individuals who have received any vaccine 28 days prior to enrollment in this study, or who plan to receive any non-study vaccines within 28 days of the second dose of study vaccine. Individuals with any clinically significant abnormal safety laboratory result, as judged by the investigator. If female, 'of childbearing potential', sexually active and has not used any of the 'acceptable contraceptive methods' for at least two months prior to study entry.Childbearing potential is defined as status post onset of menarche and not meeting any of the following conditions: menopausal for at least two years; sterile status after bilateral tubal ligation for at least one year, immediately after bilateral oophorectomy or after hysterectomy.Acceptable methods of birth control are defined as one or more of the following: Hormonal contraceptives.Barrier each and every time during intercourse.Intrauterine device (IUD).Monogamous relationship with vasectomized partner. Partner must have been vasectomized for at least six months prior to subject's study entry. If female subject of childbearing potential and have a positive urine pregnancy test prior to study vaccinations, or are currently lactating. If female of childbearing potential and sexually active, refusal to use an 'acceptable contraceptive method' through to three weeks after last study vaccination. Individuals with behavioral or cognitive impairment or psychiatric disease that, in the opinion of the investigator, may interfere with the subject's ability to participate in the study. Individuals with a history of drug or alcohol abuse within the past 2 years. Individuals who are acting as study personnel or immediate family members or the spouse of study personnel. Individuals with a body temperature ≥38 °C (≥100.4○F) within 3 days of intended study vaccination. Individuals with any condition that, in the opinion of the investigator, would interfere with the primary study objectives".

Interventions	**Vaccine type:** respiratory syncytial virus fusion glycoprotein F (RSV F) subunit vaccine **Experimental:** 45, 90 or 135 mcg RSV F with aluminium. n = 72 **Control:** 45, 90 or 135 mcg RSV F. n = 72 **Aluminium concentration:** 1000 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses 28 days apart
Outcomes	**Immunogenicity:** geometric mean titers of the serum anti-RSV neutralizing antibody titers, (time frame: at day 57). Immunogenicity was measured in terms of the geometric mean titers of the serum anti-RSV neutralizing antibody titers at day 57 (28 days after the second dose). Percentage of subjects with a ≥ 4-fold Increase in serum anti-RSV neutralizing antibody titers, (time frame: at day 57). Immunogenicity was measured in terms of percentage of subjects with a ≥ 4-fold increase in serum anti-RSV neutralizing antibody titers, from day 1 (baseline) to day 57 (28 days after the second dose). **Safety:** number of subjects with any solicited local symptoms, (time frame: within 30 minutes after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited local symptoms, (time frame: from day 1 (6 hour) through day 3 after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited local symptoms, (time frame: from day 4 through day 7 after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited local symptoms, (time frame: from day 1 (6 hours) to Day 7 after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of Subjects with any solicited systemic symptoms and other Indicators of reactogenicity, (time frame: within 30 minutes after each vaccination]Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited systemic symptoms and other indicators of reactogenicity, (time frame: from day 1 (6 hours) through day 3 after each vaccination). Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited systemic symptoms and other indicators of reactogenicity, (time frame: from day 4 through day 7 after each

vaccination). Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Subjects with any solicited systemic symptoms and other indicators of reactogenicity, (time frame: from day 1 (6 hours) to day 7 after each vaccination). Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Number of subjects with unsolicited adverse events, (time frame: from day 1 to day 28 after each vaccination). An unsolicited adverse events covers any untoward medical occurrence in a clinical investigation subject temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product and reported in addition to those solicited during the clinical study and any solicited symptom with onset outside the specified period of follow-up for solicited symptoms. Any was defined as the occurrence of any unsolicited adverse events regardless of intensity grade or relation to vaccination. Number of subjects with serious adverse events and other significant adverse events, (time frame: from study start (day 1) until study completion (day 394)]. Serious adverse events assessed include medical occurrences that resulted in death, were life-threatening, required hospitalization or prolongation of hospitalization or resulted in disability/incapacity, congenital anomaly or birth defect. Any adverse events = occurrence of the serious adverse events regardless of intensity grade. Possibly or probably related serious adverse events = serious adverse events assessed by the investigator as possibly or probably related to the study vaccination.Other significant adverse events assessed include unsolicited medically attended adverse events, unsolicited adverse events leading to study withdrawal, new onset of chronic diseases and adverse events of special interest. Medically attended adverse events = an adverse event that leads to an unscheduled visit to a healthcare practitioner. New onset of chronic diseases = an adverse event that represents a new diagnosis of a chronic medical condition that was not present or suspected in a subject prior to study enrollment.

Immunogenicity: geometric mean titers of the serum anti-RSV neutralizing antibody titers, (time frame: at day 1, day 29 and day 181). Immunogenicity was measured in terms of geometric mean titers of the serum anti-RSV neutralizing antibody titers at day 1, day 29 (28 days after the first dose) and day 181 (six months after the first dose). Percentage of subjects with a ≥ 4-fold Increase in serum anti-RSV neutralizing antibody titers, (time frame: at day 1, day 29 and day 181). Immunogenicity was measured in terms of percentage of subjects with a ≥ 4-fold increase in serum anti-RSV neutralizing antibody titers from day 1 to day 29 (28 days after the first dose) and from day 1 to day 181 (six months after the first dose). Percentage of subjects with serum anti-RSV neutralizing antibody titers greater than the 3rd quartile of serum anti-RSV neutralizing antibody titers at day 1, (time frame: at day 29, day 57 and day 181). Immunogenicity was measured in terms of the percentage of subjects at day 29 (28 days after the first dose), day 57 (28 days after the second dose) and day 181 (six months after the first dose) with serum anti-RSV neutralizing antibody titers greater than the 3rd quartile of serum anti-RSV neutralizing antibody titers at day 1 (baseline). Geometric mean titers of the serum total binding antibody to each of the RSV Proteins F, G and N, (time frame: at day 1, day 29, day 57 and

	day 181). Immunogenicity was measured in terms of the geometric mean titers of the serum total binding antibody to each of the RSV proteins F, G and N at day 1 (baseline), day 29 (28 days after the first dose), day 57 (28 days after the second dose) and day 181 (six months after the first dose). Percentage of subjects with a ≥ 4-fold increase in serum total binding antibody to each of the RSV Proteins F, G and N, (time frame: at day 1, day 29, day 57 and day 181). Immunogenicity was measured in terms of the percentage of subjects with a ≥ 4-fold increase in serum total binding antibody to each of the RSV Proteins F, G and N from day 1 (baseline) to all time points (day 29 [28 days after the first dose], day 57 [28 days after the second dose], and day 181 [six months after the first dose]). Percentage of subjects with serum total binding antibody titers to each of the RSV Proteins F, G, and N greater than the 3rd quartile of serum total binding antibody titers to RSV Protein F at day 1, (time frame: at day 1, day 29, day 57 and day 181). Immunogenicity was measured in terms of the percentage of subjects at day 29 (28 days after the first dose), day 57 (28 days after the second dose) and day 181 (six months after the first dose), with serum total binding antibody titers to each of the RSV proteins F, G, and N greater than the 3rd quartile of serum total binding antibody titers to RSV protein F at day 1 (baseline). Ratio of RSV F serum neutralizing antibody titers to each of the RSV F serum total binding antibody titers to RSV Proteins F, G and N, (time frame: at day 1, day 29, day 57 and day 181). Immunogenicity was measured in terms of the ratio of RSV F serum Nab titers to each of the RSV F serum total binding antibody titers to RSV proteins F, G and N at days 1, 29, 57 and 181.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (Geert.LerouxRoels@UGent.be) were inquired on the followings: ● aluminium manufacturer. ● allocation concealment. ● blinding of outcome assessment. ● number of participants who got respiratory syncytial virus infection after the vaccine. No answer received **Funding:** sponsored by Novartis Vaccines Division. On 02 March 2015, Novartis non-influenza Vaccines Business was acquired by the GlaxoSmithKline group of companies. The Sponsor was involved in all stages of the study conduct and analysis and took charge of all costs associated with developing and publishing this manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Observer-blind

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Dropouts <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Dropouts <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Dropouts <5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02298179), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Liang 2010

Methods	Multicentre, double-blind, placebo-controlled, randomised trial
Participants	**Sex:** male and female **Age:** >3 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female aged >3 years; able to show legal identity card for the sake of recruitment; volunteers or their guardians are able to understand and sign the informed consent". **Exclusion criteria:** "cases, cured cases and close contact of influenza A (H1N1) virus. Women of pregnancy, lactation or about to be pregnant in 60 days. Subject that has a medical history of any of the following: allergic history, or allergic to any ingredient of vaccine, such as egg, egg protein, etc. Serious adverse reactions to vaccines such as anaphylaxis, hives, respiratory difficulty, angioedema, or abdominal pain. Autoimmune disease or immunodeficiency. Asthma that is unstable or required emergent care, hospitalization or intubation during the past two years or that required the use of oral or intravenous corticosteroids. Diabetes mellitus (type I or II), with the exception of gestational diabetes. History of thyroidectomy or thyroid disease that required medication within the past 12 months. Serious angioedema episodes within the previous 3 years or requiring medication in the previous two years. Bleeding disorder diagnosed by a doctor (e.g. factor deficiency, coagulopathy, or platelet disorder

	requiring special precautions) or significant bruising or bleeding difficulties with IM injections or blood draws. Active malignancy or treated malignancy for which there is not reasonable assurance of sustained cure or malignancy that is likely to recur during the period of study. Seizure disorder other than: febrile seizures under the age of two years old, seizures secondary to alcohol withdrawal more than 3 years ago, or a singular seizure not requiring treatment within the last 3 year. Asplenia, functional asplenia or any condition resulting in the absence or removal of the spleen. Guillain-Barre Syndrome. Any history of immunosuppressive medications or cytotoxic medications or inhaled corticosteroids within the past six months (with the exception of corticosteroid nasal spray for allergic rhinitis or topical corticosteroids for an acute uncomplicated dermatitis). History of any blood products or seasonal influenza vaccine administration within 3 months before the dosing. Administration of any other investigational research agents within 30 days before the dosing. Administration of any live attenuated vaccine within 30 days before the dosing. Administration of subunit or inactivated vaccines, e.g., pneumococcal vaccine, or allergy treatment with antigen injections, within 14 days before the dosing. Be receiving anti-TB prophylaxis or therapy currently. Axillary temperature > 37.0 centigrade at the time of dosing. Psychiatric condition that precludes compliance with the protocol: past or present psychoses, past or present bipolar disorder requiring therapy that has not been well controlled on medication for the past two years, disorder requiring lithium, suicidal ideation occurring within five years prior to enrollment. Any medical, psychiatric, social condition, occupational reason or other responsibility that, in the judgment of the investigator, is a contraindication to protocol participation or impairs a volunteer's ability to give informed consent".
Interventions	**Vaccine type:** 2009 pandemic influenza A H1N1 vaccine **Experimental:** 7.5, 15 or 30 mcg H5N1 with aluminium hydroxide. n = 1814 **Control:** 7.5, 15 or 30 mcg H5N1. n =8184 **Aluminium concentration:** 1000 mcg/dose or 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Sinovac Biotech **Vaccination schedule:** two doses 21 days apart
Outcomes	**Immunogenicity:** hemagglutination inhibition antibody titer, (time frame: day 0, day 21, day 35) **Safety:** local and systemic adverse reaction after vaccination, (time frame: day 0, day 42)
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/!2/2019 authors (wangyu@chinacdc.cn) were inquired on the followings: ● blinding of participants and personnel ● blinding of outcome assessment. ● number of participants who got influenza after the vaccination. No answer received **Funding:** the vaccine manufacturers funded the trials (Sinovac Biotech, Hualan Biological Bacterin, China National Biotec Group, Beijing Tiantan Biological Products, Changchun Institute of Biological Products, Changchun Changsheng Life Sciences, Jiangsu Yanshen Biological Technology Stock, Zhejiang

	Tianyuan Bio-Pharmaceutical, Lanzhou Institute of Biological Products, Shanghai Institute of Biological Products, and Dalian Aleph Biomedical).

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Vaccine and placebo used for each age-group were labelled with sequential numbers according to a random number table generated with the block method by a statistician not involved in the rest of the trial. Block size varied by age group and centre.
Allocation concealment (selection bias)	Low risk	The same sequential numbers were also assigned to participants according to their sequence of enrolment. Participants received vaccine or placebo according to their sequential numbers. Masking was achieved by covering all labels with aluminium foil.
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further info
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind, no further info
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind, no further info
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind, no further info
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00956111 and NCT00975572), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Longo 2009

Methods	Randomized, double blind, placebo-controlled phase I study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** not reported **Diagnosis:** HIV infected **Inclusion criteria:** "negative pregnancy test for women of childbearing potential within 3 days prior to baseline evaluation and use of an acceptable means of contraception (condom, hormonal or mechanical method) for one month prior to immunization and the duration of the study; clinically asymptomatic HIV-1 infected individuals (CDC Clinical category A), as determined by two positive Enzyme-linked immunosorbent assay (ELISA) and a confirmatory Western Blot; mean CD4+ T cell count >= 400 cells/microL based on 2 separate determinations, at least 2 weeks apart, within the four weeks pre-study screening period. [Note: If one of the two values is < 400 the patient will be excluded. Patients ever having had a value of CD4+ T cell number < 250 will be excluded]; plasma HIV-1 viremia levels <= 50,000 copies/mL; complete blood count and differential defined as: - Hematocrit >= 30% for women, >= 38% for men - Hemoglobin >= 9.5 g/dL - White cell counts >= 4,000 cells/mm3 - Total lymphocyte count >= 1000 cells/mm3 - platelets >= 100,000/mm3 Differential within institutional normal limits or approval of site physician. Normal ALT (as defined by the range of the clinical site laboratories) and Creatinine (<= 1.6 mg/dL). Normal urine dipstick with esterase and nitrite. Normal thyroid function. Availability for follow-up for planned duration of at least 12 months and willing to have further brief evaluations at 24 and 36 months. Signed informed consent". **Exclusion criteria:** "history of AIDS-related opportunistic or neoplastic disease; history of encephalopathy, neuropathy, or unstable CNS pathology (HIV or non-HIV related); History of non-HIV related neoplastic diseases, autoimmune diseases, angina or cardiac arthymias, or any other clinically significant medical problems; chest radiography showing evidence of active or acute cardiac or pulmonary disease; history of anaphylaxis or serious adverse reactions to vaccines as well as serum IgE levels exceeding 1000 U.I./mL; history of serious allergic reaction to any substance, requiring hospitalization or emergent medical care (e.g. Steven-Johnson syndrome, bronchospasm, or hypotension); active syphilis [NOTE. If the serology is documented to be a false positive or due to an adequately treated infection, the volunteer is eligible]; active tuberculosis [NOTE: Volunteers with a positive PPD and a normal chest X-ray showing no evidence of TB and not requiring isoniazid (INH) therapy are eligible]; medical or psychiatric condition or occupational responsibilities which preclude subject compliance with the protocol. Specifically excluded are persons with psychotic disorders, major affective disorders, suicidal ideation; current use of psychotrophic drugs; use of antiretroviral therapy within 3 months of pre-study screening. Use of any experimental HIV therapy or participation in another experimental protocol within three (3) months of pre-study screening; current or prior therapy with immunomodulators or immunosuppressive drugs and anticoagulant drugs within 30 days prior to study medication administration; any

	unstable cardio-vascular disease (e.g unstable hypertensive disease needing modification or introduction of an anti-hypertensive treatment); live attenuated vaccines within 60 days of study [NOTE: Medically indicated sub-unit or killed vaccines (e.g., influenza, pneumococcal, hepatitis A and B) are not exclusionary, but should be given at least 4 weeks away from HIV immunizations]; receipt of blood products or immunoglobulin in the past year; prior receipt of HIV-1 vaccine in a previous HIV vaccine trial; positivity for HBV antigens (HBs Ag, HBe Ag), and HCV, HTLV-I and HTLV-II antibodies; Positivity for HHV-8 antibodies and plasmaviremia (volunteers will be screened for anti-HHV-8 antibodies and positivity confirmed by PCR, only the individuals confirmed positive by PCR will be excluded); pregnant or lactating women".
Interventions	**Vaccine type:** human immunodeficiency virus (HIV)-1 Tat-based vaccine **Experimental:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine with aluminium **Control:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** 5 doses (weeks 0, 4, 8, 12, 16) either subcute in alum or intradermic without adjuvant
Outcomes	**Immunogenicity**: "volunteers are monitored for anti-Tat specific antibodies, anti-Tat proliferative response and in vitro gamma-IFN and IL-4 (or IL-10) production in response to Tat (Elispot)". **Safety**: "clinical observation and monitoring of haematological, biochemical, virological and immunological parameters. Safety is evaluated by monitoring of volunteers for local and systemic adverse reactions during the trial".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** 12/02/2020 authors (barbara.ensoli@iss.it) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any AE related and not related to the vaccine ● serological response GMT for each study arm at maximum follow up ● proportion of participants who developed HIV after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment **Funding:** funded by “ISS special funding for Tat vaccine trials”, the Italian Concerted Action on HIV-AIDS Vaccine Development (ICAV, National AIDS Program) and the Italian Ministry of Health

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	The study protocol for one trial was registered prior to randomisation (NCT00505401), and all outcomes were reported on.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

Low 2014

Methods	Double-blinded, randomised, Phase 1 study
Participants	**Sex:** male and female **Age:** 21-64 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** participants, between 21 and 64 years of age, with satisfactory baseline medical assessment and laboratory values within the normal ranges **Exclusion criteria:** presence of acute infection during 14 days preceding the first vaccination, a temperature ≥38○C at the date of the first vaccination, and

	the receipt of immunoglobulins or blood products within 9 months prior to enrolment or during the study. Additional exclusion criteria were receipt of seasonal influenza vaccine in the past 2 years,or any licensed vaccine within 30 days prior to the first injection or HAI titers >1:40 at screening. Concomitant medications (except other vaccines) were not restricted. Women of child-bearing potential had to have a negative pregnancy test at each visit.
Interventions	**Vaccine type:** virus-like particle pandemic influenza A (H1N1) 2009 vaccine **Experimental:** 100 mcg H1N1 with Alhydrogel. n =43 **Control:** 100 mcg H1N1. n = 41 **Aluminium concentration:** 1000 mcg/dose or 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag-Biosector, Denmark **Vaccination schedule:** two intramuscular injections 21 days apart
Outcomes	**Immunogenicity:** "seroconversion against A/California/07/2009 (H1N1) by HAI on Day 21 and 42, defined as either a pre-vaccination HAI titer <10 and a post vaccinationHAI titer ≥40, or a pre-vaccination HAI titer ≥10 and minimum four-fold rise in post-vaccination HAI titer. In addition,geometric mean titers and the percentage of subjects achieving seroprotection (HAI titer ≥40), were calculated". **Safety:** "adverse events and severe adverse events were summarized by treatment group withe each subject counted once per adverse event category with the highest severity of treatment emergent adverse event (Day 0-Day 42). The following factors assessed: vital signs: (blood pressure, pulse, respiratory rates and body temperature) at all visits (Days 0, 21 and 42) and on vaccination days pre- and 60 min post-vaccination, hematology and blood biochemistry at all visits, solicited and unsolicited local/systemic events (symptom diaries) 21 days following each vaccination".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (vdiermayr@d3.a-star.edu.sg, veronica@diermayr.com) were inquired on the followings: ● procedure for blinding of outcome assessment ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received **Funding:** funded by A*Star, but the funding body was not scientifically involved in the clinical study or the decision to submit this article for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Independent statistician from the Singapore Clinical Research Institute generated the randomisation list. An unblinded research coordinator randomly assigned subjects1:1 to the adjuvated or the non-adjuvanted group. A computer-generated list (SAS®software, NC, USA) with randomly permutated block sizes of 4 and 6 was provided by SCRI.

Allocation concealment (selection bias)	Low risk	An unblinded research coordinator randomly assigned subjects1:1 to the adjuvated or the non-adjuvanted group. A computer-generated list (SAS®software, NC, USA) with randomly permutated block sizes of 4 and 6 was provided by SCRI.
Blinding of participants and personnel (performance bias)	Low risk	Double-blinded. Study personnel and participants were blinded to the treatment allocation, except for the independent statistician from the CRO, generating the randomisation list and the unblinded clinical research coordinator, mixing the vaccine with alhydrogel or diluent prior to injection. The 500 µL for injection were drawn up in a masked syringe by the unblended study coordinator prior to injection.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Immunogenicity assessments on sera of recruited volunteers from baseline (Day 0), Day 21 and Day 42 were performed on blinded samples. Unblinding was performed by the statistician at the CRO.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Manoff 2019

Methods	Randomized, placebo controlled, phase I dose-escalation, first in human, proof of principle trial
Participants	**Sex:** male and female **Age:** 18-49 years **Ethnicity:** not reported **Diagnosis:** healthy

	Inclusion criteria: "in good health. Voluntarily agrees to participate by giving written informed consent. Able to read, understand, and complete study questionnaires. Able to complete all scheduled visits and comply with study procedures. Access to a telephone. Agrees to avoid unusual, vigorous exercise from 72 hours before any dose of study vaccine/placebo through 15 days after that dose. Weighs ≥110 pounds (50 kg) and has a body mass index (BMI) of 19 to 32 kg/m^2. No fever (temperature ≥100.4°F/38.0°C) for 72 hours prior to vaccination. Females of reproductive potential agree to remain abstinent or to use 2 acceptable methods of birth control from enrollment through 6 weeks after the last dose of study vaccine/placebo". **Exclusion criteria:** "history of receiving any flavivirus vaccine (e.g. Japanese encephalitis, tick-borne encephalitis, or yellow fever) or planned receipt of any such vaccine during the study period. History of any flavivirus infection or serologic evidence of any flavivirus infection, including West Nile, dengue, yellow fever, Saint Louis encephalitis (if available), Kunjin, Murray Valley encephalitis, and Japanese encephalitis. History of residence for a cumulative period of >1 year in a country where dengue, Japanese encephalitis virus, or yellow fever virus is common. Planned travel to an area where dengue is common through 28 days after receiving the last dose of study vaccine/placebo. Known hypersensitivity to any component of the dengue vaccine. Abuse of drugs or alcohol within 12 months prior to screening. Pregnant or breastfeeding, or expecting to conceive in the time from enrollment through 6 weeks after the last dose of study vaccine/placebo. Positive serum test for human immunodeficiency virus (HIV), hepatitis B surface antigen (HBsAg), and/or hepatitis C antibody. Known, suspected, or a history of immunocompromise. History of malignancy within 5 years prior to enrollment. Poorly controlled diabetes mellitus. Use of any immunosuppressive therapy (except topical and inhaled/nebulized steroids). Receipt of any licensed non-live vaccine within 14 days prior to the first dose of study vaccine/placebo or plans to receive a licensed non-live vaccine during the time between receiving the first dose and 28 days after receiving the last dose of study vaccine/placebo. Receipt of any licensed live vaccine within 30 days prior to the first dose of study vaccine/placebo or plans to receive a licensed live vaccine during the time between receiving the first dose and 28 after receiving the last dose of study vaccine/placebo. Received investigational drugs or vaccines within 2 months prior to the first dose of study vaccine/placebo. History of receiving 1 or more doses of an investigational dengue vaccine. Participation in another clinical study within 42 days prior to enrollment, or plans to participate in another clinical study from enrollment through 1 year after the last dose of study vaccine/placebo. Planned donation of eggs or sperm from the time of enrollment through 28 days after the last dose of study vaccine/placebo. Prior receipt of a blood transfusion or blood products within 6 months prior to the first dose of study vaccine/placebo. Hospitalization for acute illness within 3 months prior to the first dose of vaccine/placebo".
Interventions	**Vaccine type:** recombinant subunit vaccine for dengue V180 **Experimental:** V180 with Alhydrogel. n = 9 **Control:** V180. n = 18 **Aluminium concentration:** 225 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** three injections at months 0, 1, and 2

Outcomes	**Immunogenicity:** seroconversion rate for each serotype, (time frame: 28 days post-dose 3 (day 84). Geometric mean titer of virus neutralizing antibodies for each serotype, (time frame: 28 days post-dose 3 (day 84) **Safety:** serious adverse events were followed for 1 year after the last injection. Adverse events were collected until 14 days from the last injection.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (beth-ann.coller@merck.com) were inquired on the followings: ● aluminium manufacturer ● allocation concealment ● random sequence generation ● blinding of participants and personnel ● blinding of outcome assessment ● number of patient analysed per outcome (i.e. number of drop-outs) ● number of participants with and non-serious adverse event related and not related to the vaccine ● number of participants who got Dengue infection after the vaccination No answer received **Funding:** funded by Merck

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	The safety was monitored and advised at pre-specified time points by an unblinded independent external Data Monitoring Committee.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01477580), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Mark 1994

Methods	Randomized, double-blind, clinical trial
Participants	**Sex:** male and female **Age:** 10 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "10 years of age due for their booster DT vaccination.". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** Diptheria Toxoid vaccine **Experimental:** DT vaccine with aluminium phosphate. n = 124 **Control:** DT vaccine. n = 111 **Aluminium concentration:** 250 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** one booster dose
Outcomes	**Safety and Immunogenicity**
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** no contact found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described.
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. No further details

Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a
Selective reporting (reporting bias)	Unclear risk	No protocol pre-published.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

Meulen 2015

Methods	Randomized, double-blind, multicenter clinical trial
Participants	**Sex:** male and female **Age:** 12-15 months **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy toddlers, 12-15 months of age who have previously completed a documented full 3 dose infant series of Prevnar™ at 2, 4, and 6 months of age. Participant's parent/legal guardian understands the study procedures, alternate treatments available and risks involved with the study, and voluntarily agrees to participate by giving written informed consent. Afebrile, with a rectal temperature <38.1°C (<100.5°F) or axillary temperature <37.8°C (<100.0°F) on day of vaccination. Participant's parent/legal guardian is able to read, understand, and complete study questionnaires (i.e., the Vaccination Report Card). Participant is able to attend all scheduled visits and to comply with the study procedures. Participant's parent/legal guardian has access to a telephone". **Exclusion criteria:** "have received less than the full 3-dose infant series of Prevnar™ or 3rd dose less than 2 months before study vaccine. Known hypersensitivity to any component of the pneumococcal conjugate vaccine. Known or suspected impairment of immunological function. Participant has a history of congenital or acquired immunodeficiency (e.g., splenomegaly). Participant or his/her mother has documented HIV infection. Functional or anatomic asplenia. History of autoimmune disease including multiple sclerosis, systemic lupus, polymyositis, inclusion body myositis, dermatomyositis, Hashimoto's thyroiditis, Sjogren's syndrome, rheumatoid arthritis, other autoimmune disorders. Known neurologic or cognitive behavioral disorders including multiple sclerosis, multiple sclerosis-like disease, encephalitis/myelitis,

	acute disseminating encephalomyelitis, pervasive developmental disorder, and related disorders. Receipt of intramuscular, oral, or intravenous corticosteroid treatment within the 2 weeks prior to vaccination. (Note: Toddlers on topical and inhaled/nebulized steroids may participate in the study.) Use of systemic steroids are only permitted when the participant is receiving less than 2 mg/kg per day of prednisone (or its equivalent), or less than 20 mg/d if they weigh more than 10 kg and are not otherwise immunocompromised. Participant has received other licensed non-live vaccines administered within the 14 days before receipt of study vaccine. Participant has received a licensed live virus vaccine within 30 days prior of receipt of study vaccine (Exception: Influenza virus vaccine given according to recommended guidelines within 7 days of receiving study vaccine). Prior receipt of a blood transfusion or blood products, including immunoglobulins. Investigational drugs or vaccines received within the 2 months before receipt of study vaccine. Participation in another clinical study within 42 days before the beginning or anytime during the duration of the current clinical study. History of invasive pneumococcal disease (positive blood culture, positive cerebrospinal fluid culture, or other sterile site) or known history of other culture positive pneumococcal disease. A recent (<72 hours) febrile illness (rectal temperature ≥38.1°C [≥100.5°F]) occurring within 48 hours before receipt of study vaccine. History of failure to thrive. Participant has a coagulation disorder contraindicating IM vaccination. Participant and his/her mother is documented hepatitis B surface antigen-positive. Any toddler who cannot be adequately followed for safety according to the protocol plan. Participant's parent/legal guardian is unlikely to adhere to study procedures, keep appointments, or is planning to relocate during the study. Any other reason that in the opinion of the investigator may interfere with the evaluation required by the study".
Interventions	**Vaccine type:** 15-valent Pneumococcal Conjugate Vaccine PCV15 **Experimental:** PCV15 with aluminium. n = 33 **Control:** PCV15. n = 29 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** Merck aluminium phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** one dose administered intramuscularly on day 1
Outcomes	**Immunogenicity:** "percentage of participants achieving the Immunoglobulin G serotype-specific Threshold value of ≥ 0.35µg/mL, (time frame: day 30 after vaccination). Immunoglobulin G for the serotypes contained in V114 was determined using an electrochemiluminescence assay. Geometric mean concentration of Pneumococcal Serotype Immunoglobulin G antibodies, (time frame: day 30 after vaccination). Pneumococcal serotype-specific IgG was measured in serum using an electrochemiluminescence assay. Percentage of participants achieving opsonophagocytic activity ≥1:8, (time frame: day 30 after vaccination). Opsonophagocytic activity for the serotypes contained in V114 was determined using a multiplex opsonophagocytic assay. Geometric mean titer of pneumococcal serotype-specific opsonophagocytic activity, (time frame: day 30 after vaccination). Opsonophagocytic activity for the serotypes contained in V114 was determined using a multiplex opsonophagocytic assay". **Safety:** "percentage of participants with any adverse event, (time frame: up to day 14 after vaccination). An adverse event is defined as any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment. An adverse even can therefore be any

	unfavourable and unintended sign, symptom, or disease temporally associated with the use of a medicinal product or protocol-specified procedure, whether or not considered related to the medicinal product or protocol-specified procedure. Any worsening of a preexisting condition that is temporally associated with the use of the Sponsor's product, is also an adverse even. Percentage of participants with any serious adverse even, (time frame: up to day 14 after vaccination). A serious adverse event is an adverse event that results in death, is life threatening, requires or prolongs an existing hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is another important medical event deemed such by medical or scientific judgment. percentage of participants with any vaccine-related adverse event, (time frame: up to day 14 after vaccination). An adverse event is defined as any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment. An adverse event can therefore be any unfavourable and unintended sign, symptom, or disease temporally associated with the use of a medicinal product or protocol-specified procedure, whether or not considered related to the medicinal product or protocol-specified procedure. Any worsening of a preexisting condition that is temporally associated with the use of the Sponsor's product, is also an adverse event. Relatedness of the adverse event to study vaccine was determined by the investigator".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/ 2018 authors (luwy_musey@merck.com, ajokestm@uw.edu) were inquired on the followings: ● method for random sequence generation ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** provided by Merck & Co., Inc

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Low risk	Subjects were allocated to treatment assignment using a randomised schedule.
Blinding of participants and personnel (performance bias)	Low risk	Double-blind. All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01215175), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Moustafa 2012

Methods	Randomized, blinded, placebo-controlled, phase II immunogenicity study
Participants	**Sex:** male and female **Age:** 18 to 80 years **Ethnicity:** mixed **Diagnosis:** end-stage renal disease on chronic hemodialysis **Inclusion criteria:** "participant is scheduled to undergo cardiothoracic surgery involving a full median sternotomy (not including cardiac transplantation surgery) within 14 to 60 days after vaccination. Female participants of reproductive potential are required to have a negative urine or serum pregnancy test immediately prior to study vaccination and must use an acceptable form of birth control". **Exclusion criteria:** "participants had an invasive Staphylococcus aureus infection within the past 3 months prior to study entry. A realistic (>50%) possibility that cardiothoracic surgery will be necessary sooner than 10 days after vaccination. Participant is planning to undergo cardiac transplantation surgery or sternal debridement to remedy an infection resulting from a prior cardiothoracic surgery. Participant has any type of ventricular-assist device in place at the time of study entry. Participant has a history of anaphylaxis to any of the vaccine components. Participant received V710 vaccine, any other investigational Staphylococcus

	aureus vaccine, or investigational Staphylococcus aureus antibodies. Participant has a temperature of ≥100.4ºF (≥38.0ºC), oral equivalent, within 48 hours prior to study vaccination. Participant has impairment of the immune system. Participant has a medical condition in which the expected survival is less than 90 days".
Interventions	**Vaccine type:** Staphylococcus aureus Vaccine V710 **Experimental:** "3a": V710 60 mcg with AAHSA x3, or "3b": V710 60 mcg with AAHSA + V710 60 µg with AAHSA + Placebo, or "4a": V710 60 mcg with AAHSA + Placebo + V710 60 mcg with AAHSA or "4b": V710 60 mcg with AAHSA + Placebo + Placebo. n = 75 **Control:** "1a": V710 60 mcg, or "1b": V710 60 mcg + V710 60 + Placebo, or "2a": V710 60 mcg + Placebo + V710 60 mcg, or "2b": V710 60 mcg + Placebo + Placebo. n = 73 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxyphosphate sulfate adjuvant (AAHSA) **Aluminium manufacturer:** not reported **Vaccination schedule:** all patients received their priming doses on day 1 and day 28, with a booster dose (equally divided between V710 and placebo) on day 180
Outcomes	**Immunogenicity:** "number of participants with invasive Staphylococcus aureus Infection, (time frame: up to 90 days after surgery). Diagnosis of the Staphylococcus aureus infections employed standardized definitions adapted from the Centers for Disease Control guidelines for nosocomial infections. An invasive Staphylococcus infection included bacteremia, deep sternal wound infection, deep-tissue organ/space infection at another surgical site, or any other deep-tissue infection. Number of participants with surgical-site Staphylococcus aureus infection, (time frame: up to 90 days after surgery). Diagnosis of the Staphylococcus aureus infections employed standardized definitions adapted from the Centers for Disease Control guidelines for nosocomial infections. A Staphylococcus infection surgical-site infection included any superficial incisional, deep incisional, or organ/space infection at the sternal site, the vascular harvest (donor) site, or any other site at which the surgery was performed". **Safety:** "number of participants with staphylococcus aureus bacteremia and/or deep sternal wound infection, (time frame: up to 90 days after surgery). Diagnosis of the Staphylococcus aureus infections employed standardized definitions adapted from the Centers for Disease Control guidelines for nosocomial infections (Garner JS, Jarvis WS, Emori TG, et al. CDC definitions for nosocomial infections. APIC Infect Control App Epidemiol 1996;A1-20). Bacteremia was defined as ≥1 positive blood culture for S. aureus regardless of the presence of clinical symptoms. A Staphylococcus aureus deep sternal wound infection included mediastinitis or a deep incisional surgical-site infection involving the sternal wound. Incidence rate of vaccine-related serious adverse experiences, (time frame: up to 360 days after surgery). Vaccine-related adverse experiences were those deemed by the investigator to be possibly, probably, or definitely vaccine related. A serious adverse experience was any adverse experience occurring at any dose that 1) resulted in death, 2) was life threatening, 3) resulted in a persistent or significant disability/incapacity, 4) resulted in or prolonged an existing inpatient hospitalization, 5) was a congenital anomaly/birth defect, 6) was a cancer, 7) was an overdose, or 8) jeopardized the participant and required medical or surgical intervention".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (dalya_guris@merck.com) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** funded by Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Whitehouse Station, NJ

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomization and allocation were assigned using an interactive voice response system. Numbered packaging was used to implement allocation.
Allocation concealment (selection bias)	Low risk	Randomization and allocation were assigned using an interactive voice response system. Numbered packaging was used to implement allocation.
Blinding of participants and personnel (performance bias)	Low risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study

Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00572910), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT00309647

Methods	Phase I partially-blind randomised multi-centric study
Participants	**Sex:** male and female **Age:** 18-60 years **Ethnicity:** Caucasian **Diagnosis:** healthy **Inclusion criteria:** "male or female between, and including, 18 and 60 years of age at the time of the first vaccination. Healthy subjects as established by medical history and clinical examination before entering into the study. If the subject is female, she must be of non-childbearing potential". **Exclusion criteria:** "administration of any vaccine during the period starting 15 days before the first administration of the study vaccine and ending 21 after the second one. Administration of an influenza vaccine other than the study vaccines during the entire study period. Chronic administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within six months prior to the first administration of the study vaccine. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. History of hypersensitivity to vaccines. History of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Acute clinically significant pulmonary, cardiovascular, hepatic or renal functional abnormality, as determined by physical examination or laboratory screening tests. Acute disease at the time of enrolment. Administration of immunoglobulins and/or any blood products within the three months preceding the first administration of the study vaccine or during the study. Lactating women. Use of any investigational or non-registered product (drug or vaccine) other than the study vaccine(s) within 30 days prior to the first vaccination, or planned use

	during the study period".
Interventions	**Vaccine type:** influenza H5N1 vaccine **Experimental:** 27, 15, 7.5 or 3.8 mcg H5N1 with aluminium. n = 201 **Control:** 27, 15, 7.5 or 3.8 mcg H5N1. n = 199 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular administrations at day 0 and Day 21 in the deltoid region of the non-dominant arm
Outcomes	**Immunogenicity**: "the following parameters were calculated with 95 % confidence intervals: from the anti-heamagglutinin antibody titres: geometric mean titres pre- and post-vaccinations (days 0, 21, and 42), seroconversion rates and seroconversion factors at days 21 and 42, seroprotection rates at days 0, 21 and 42. From the neutralisation antibody titres: geometric mean titres pre- and post-vaccinations (days 0, 21 and 42), seroconversion rates at days 21 and 42. In order to evaluate the cell-mediated immune response, the following parameter was calculated:frequency of influenza-specific cytokine-positive CD4/CD8 T-cells at days 0, 21 and 42". **Safety**: "recording of solicited local (pain, redness, swelling, induration, ecchymosis) and general (fatigue, headache, fever, shivering, sweating increase, myalgia and arthralgia) adverse events during a 7 days follow-up period after vaccination. Recording of unsolicited adverse events occurring during a 21-day follow-up period after the first vaccination and a 30-day follow-up period after the second vaccination. Recording of serious adverse events occurring during the entire study".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** 30/01/2020 authors (dk-info@gsk.com) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any AE related and not related to the vaccine ● proportion of participants who developed influenza after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment No answer received **Funding**: GSK

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described

Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Partially blinded
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Partially blinded
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Partially blinded
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Partially blinded
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Unclear risk	The study protocol was registered prior to randomisation (NCT00309647), and all outcomes were reported on.
Other bias	Low risk	The study was not published in a peer-reviewed journal but only on the company website.

NCT00562237

Methods	Randomized, observer-blind, placebo-controlled study
Participants	**Sex:** male and female **Age:** 18-49 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "being healthy and ≥ 18 and ≤ 49 years of age; willing and able to give informed consent". **Exclusion criteria:** "having participated in an influenza H5 vaccine trial in the past; known to be allergic to any constituent of the vaccine; serious adverse reactions to previous (influenza) vaccination; currently participating in another clinical trial or having participated in any clinical trial in the month preceding the start of the study; using medication that influences the immune system".
Interventions	**Vaccine type:** influenza H5N1 vaccine **Experimental:** S205 10 or 30 mcg HA with AlOH **Control:** S205 10 or 30 mcg HA **Aluminium concentration:** 500 mcg/dose

	Aluminium type: aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 intramuscular injections
Outcomes	**Immunogenicity**: hemagglutination inhibition titers, virus neutralization,anti-HA antibody level kinetics (time frame: one year). **Safety**: Not stated. Time frame: one year
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Solvay Pharmaceutical) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT00693615

Methods	Phase II, double-blind, randomised study
Participants	**Sex:** female **Age:** 18-30 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "females 18 through 30 years of age (must not have reached the 31st birthday). Written informed consent obtained from the volunteer. Unless previously surgically sterilized, agrees to use an effective method of birth control beginning 30 days before the first study injection and continuing through 60 days after the final study injection. Must have a negative serum pregnancy test within 21 days of study entry and must not be breast feeding. Healthy by medical history and physical examination. Seronegative for HPV-16 and HPV-18 antibody by ELISA within 21 days of study entry. Cervical specimen negative for high-risk HPV DNA using the Digene Hybrid Capture® II HPV test (high-risk types Probe B) within 21 days of study entry. Normal Pap smear, using the Cytyc ThinPrep® Pap Test, within 21 days of study entry. No evidence of anogenital HPV lesions or physical findings suggestive of other gynaecologic pathogens on pelvic examination within 21 days of study entry. Agrees to no other vaccines or experimental therapy until 30 days after the last study injection". **Exclusion criteria:** "acute illness or fever (oral temperature ≥ 99.5°F [37.5°C]) at start of the study. History or clinical manifestations of significant medical or psychiatric disorder. Use of immunosuppressive medication (inhaled and topical corticosteroids are permitted) within the previous 90 days or history of immunodeficiency. History of cancer. History of alcohol or drug abuse within the past 2 years. Abnormal laboratory blood values at screening. Other abnormal laboratory values in the screening panel which in the opinion of the principal investigator are judged to be clinically significant. Receipt of immunoglobulin or blood products within 90 days prior to study entry. History of abnormal Pap smear (other than a single prior report of ASCUS or indeterminate Pap smear with a subsequent normal report). Genital herpes disease involving the cervix or with disease characterized on examination or by history by extensive external lesions. Volunteers with a history of recurrent genital herpes disease characterized by limited external lesions are eligible to participate in the study. Positive tests for hepatitis C antibody, hepatitis B surface antigen, or HIV-1 antibody. Any prior receipt of any vaccine (experimental or otherwise) for treatment or prophylaxis of genital warts or other papillomavirus related condition. Any treatment of genital warts or other papillomavirus related condition within 6 months of randomization (local therapy for common skin and/or plantar warts is allowed) Previous administration of any components of the investigational vaccine. Receipt of any experimental vaccine within 90 days prior to entry into this study. Receipt of any experimental drug therapy within 30 days or five half-lives of the experimental drug (if the half-life is known), whichever is longer".
Interventions	**Vaccine type:** MEDI-517 HPV vaccine **Experimental:** MEDI-517 HPV vaccine with aluminium. n = 18 **Control:** MEDI-517 HPV vaccine. n = 22 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported

	Vaccination schedule: 3 intramuscular injections into the deltoid at day 0, 30, 180
Outcomes	**Immunogenicity**: "serum enzyme-linked immunosorbent assay antibody titers against HPV-16 and HPV-18 at 30 days after the third dose (day 210). Serum enzyme-linked immunosorbent assay antibody titers against HPV-16 and HPV-18 on study days 0, 7, 30, 60, 180 and 360, and at 18, 24, 36, 48 months. Neutralization antibody titers against HPV-16 and HPV-18 on study days 0, 60, 210 and 360. Inhibitory ELISAs at 18, 24, 36 and 48 months. Lymphoproliferation, interleukin-5 and interferon-gamma assays at study days 0, 60, 210 and 360, and at 18, 24 months".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/01/2020 authors (dk-info@gsk.com) were inquired on the followings: ● safety and immunogenicity results Answer received **Funding**: GSK Note: The serology outcome data are available up to 4 years. However, in this meta-analysis, we use the 360 days time point. At the maximum follow up there were a too high drop out rate.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Low risk	The investigators assigned a patient identification number. A randomization list provided to the study pharmacist by MedImmune contained assignment to treatment groups. The pharmacist did not reveal treatment assignment to anyone.
Blinding of participants and personnel (performance bias)	Low risk	The investigator, all clinical staff (except the study pharmacist), the volunteers and the sponsor's staff (other than the study statistician) were unaware of treatment assignment
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs < 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs < 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs more than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00693615), and all outcomes were reported on.
Other bias	Low risk	The study was not published in a peer-reviewed journal but only on the company web site.

NCT01244867

Methods	Phase 2, randomised, observer-blind, single center, dose-ranging study
Participants	**Sex:** male and female **Age:** 18-60 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male and female adults, 18 to 60 years of age; healthy as judged by the Principal Investigator (PI) and determined by medical history, physical examination, vital signs, screening laboratories and medical history conducted no more than 30 days prior to study vaccine administration; BMI of ≥18 and ≤ 32; comprehension of the study requirements, expressed availability for the required study period and ability to attend scheduled visits; accessible by telephone on a consistent basis; In the opinion of the Investigator, competence and willingness to provide written, informed consent for participation after reading the informed consent form. The subject must have adequate opportunity to discuss the study with an Investigator or qualified designee; If female and capable of child-bearing, have a negative serum pregnancy test result at study entry, has been consistently using effective birth control for the 28 days prior to study entry and agree to continue employing adequate birth control measures for the duration of the study". **Exclusion criteria:** "presence of significant acute or chronic, uncontrolled medical or neuropsychiatric illness. Any medical or neuropsychiatric condition which, in the Investigator's opinion, would render the subject incompetent to provide informed consent or unable to provide valid safety observations and reporting; confirmed or suspected immunosuppressive condition or immunodeficiency including history of human immunodeficiency virus (HIV) infection or Hepatitis B or C or presence of lymphoproliferative disease; Presence of any febrile illness, oral temperature of >38.0˚C within 24 hours of test article administration. Such subjects may be re-evaluated for enrolment after resolution of illness; History of autoimmune disease; Administration of any vaccine (including any other influenza vaccine) within a 30 day period prior to study enrolment, or planned administration within the period from the first vaccination up to blood sampling at Day 42 or within 30 days prior to blood sampling at Day 228. Immunization on an emergency basis of a tetanus and

	diphtheria toxoids adsorbed for adult use (Td) will be allowed provided the vaccine is not administered within two weeks prior to test article administration. Receipt of any other emergency immunizations (e.g. rabies) will result in a case-by-case review of continued participation; Use of any investigational or non-registered product within 30 days prior to study enrolment or planned use during the study period. Subjects may not participate in any other drug study while participating in this study; Treatment with systemic glucocorticoids at a dose exceeding ≥ 10 mg of prednisone per day, or equivalent for more than 7 consecutive days or for 10 or more days in total, within one month of first test article administration, or any other cytotoxic or immunosuppressant drug or any immune globulin preparation within three months of vaccination. Nasal or inhaled glucocorticoids are allowed; any significant disorder of coagulation or treatment with Coumadin derivatives or heparin. Persons receiving prophylactic anti-platelet medications, e.g., low-dose aspirin, and without a clinically apparent bleeding tendency are eligible; history of previous H5N1 vaccination or a history of exposure to H5N1 virus; history of allergy to any of the constituents of H5 VLP (H5N1) study vaccine, Alhydrogel® (aluminium hydroxide), or the phosphate buffer; history of severe allergic reactions or anaphylaxis; history of tobacco allergy; history of anti-histaminics used continuously for 4 weeks or more at any time in the past year; have a rash, dermatological condition or tattoos which may interfere with injection site reaction rating; have received a blood transfusion or immunoglobulins within 90 days of study entry; if female, and of childbearing potential, has not been consistently using effective birth control for the 28 days prior to study entry. An example of highly effective birth control is oral contraceptives, hormone implants, abstinence (confirmed by Investigator), or male condom plus spermicide. All female subjects, regardless of birth control history must provide a serum sample for pregnancy screening. Effective birth control must be used for the duration of the study for female of child bearing potential must have no plan to become pregnant during the study period.; among female subjects, either known pregnancy or urine beta-human chorionic gonadotropin (ß-hCG) test results consistent with pregnancy prior to test article administration on Day 0; female subjects who are lactating; Vital sign abnormalities: systolic blood pressure ≥150 mmHg, diastolic blood pressure ≥90 mmHg, resting pulse rate <40 bpm or >100 bpm or according to the Investigator's opinion; Cancer or treatment for cancer within 3 years of test article administration. Persons with a history of cancer who are disease-free without treatment for 3 years or more are eligible. Persons with treated and uncomplicated basal cell carcinoma of the skin are eligible".
Interventions	**Vaccine type:** plant-made H5 VLP influenza vaccine **Experimental:** 45 mcg VLP with aluminium **Control:** 45 mcg VLP **Aluminium concentration:** not reported **Aluminium type:** alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** two consecutive doses 21 days apart
Outcomes	**Immunogenicity**: "geometric mean titres of hemagglutination inhibition antibody on days 0, 21 and 42. Follow-up serology samples for geometric mean titres will be taken at Day 228". **Safety**: "rate and severity of solicited and unsolicited adverse events post-vaccination. A 6-month follow-up period will be performed".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Medicago) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT01447407

Methods	Partially-blind, placebo controlled Phase 1b study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "informed of the nature of the study and have agreed to and

		are able to read, review, and sign the informed consent document prior to screening. The informed consent document will be written in English, therefore the volunteer must have the ability to read and communicate in English. Completed the screening process (as described in this protocol) within 28 days prior to dosing. Healthy male and female volunteers 18-50 years of age, inclusive, at the time of dosing. No clinically significant deviation from normal as judged by the investigator(s) in the medical history, physical examination (including but may not be limited to an evaluation of the cardiovascular, gastrointestinal, respiratory and central nervous systems), vital sign assessments, 12-lead electrocardiogram (ECG), clinical laboratory assessments, and by general observations. Fmale volunteers must be one of the following: of childbearing potential and practicing an acceptable method of birth control as judged by the Investigator(s), naturally postmenopausal (no menses) for at least 1 year and has a documented FSH level ≥ 40 mIU/mL surgically postmenopausal (bilateral oophorectomy or hysterectomy), sterile (surgically [bilateral tubal ligation] or the Essure® Procedure) Female volunteers that are surgically sterile or surgically postmenopausal must provide documentation of the bilateral tubal ligation, bilateral oophorectomy, or hysterectomy prior to dosing or the volunteer must agree to use a medically acceptable method of birth control. The Essure® Procedure must have been inserted at least 3 months prior with documentation of the Essure® confirmation test prior to Period I dosing. If the procedure was inserted less than 3 months prior to Period I dosing or proper documentation of the confirmation test is not provided, the volunteer must agree to use an additional medically acceptable method of birth control". **Exclusion criteria:** "reports receiving any investigational drug, investigational vaccine, or investigational device within 30 days prior to dosing. Reports any presence or history of a clinically significant disorder involving the cardiovascular, respiratory, renal, gastrointestinal, immunologic, hematologic, endocrine, or neurologic system(s) or psychiatric disease as determined by the Investigator(s). Clinical laboratory test values outside the accepted range. When confirmed upon additional testing, demonstrates a reactive screen for hepatitis B surface antigen, hepatitis C antibody, or HIV antibody. Demonstrates a positive drug screen for non-prescription drugs. Reports a clinically significant illness during the 28 days prior to dosing (as determined by the Investigator[s]). Reports a history of allergic response(s) to nickel or anaphylaxis (or other serious reactions) to aluminium. Reports receiving any live attenuated vaccine including FluMist® within 6 weeks prior to dosing or any licensed inactivated vaccine within 3 weeks prior to dosing. Reports the use of any immunosuppressive drugs, including systemic corticosteroids, within 4 weeks prior to dosing. Reports the use of any medications or treatments that may alter immune responses to the study vaccine within 3 weeks prior to dosing (eg, cyclosporine, tacrolimus, cytotoxic drugs, immune globulin, Bacillus Calmette-Guerin [BCG], monoclonal antibodies, radiation therapy). Reports a history of clinically significant allergies including food or drug allergies or anaphylaxis (or other serious reactions) to vaccines. Reports a history of drug or alcohol addiction or abuse within the past year. Reports receiving any blood products within 3 months prior to dosing and throughout the study. Reports donating blood within 28 days prior to dosing. All subjects will be advised not to donate blood for four weeks after completing the study. Reports donating plasma (e.g. plasmapheresis) within 14 days prior to dosing. All subjects will be advised not to donate plasma for four weeks after completing the study. Demonstrates, in the opinion of study staff, veins unsuitable for repeated

	venipuncture (e.g. veins difficult to locate, access, or puncture; veins with a tendency to rupture during or after puncture). Pregnant, lactating, breastfeeding, or intends to become pregnant over the course of the study. Demonstrates a positive pregnancy screen. Reports smoking or using tobacco products or is currently using nicotine products (patches, gums, etc). Thirty (30) days abstinence prior to dosing is required. Any other medical and/or social (e.g. uncooperative or non-compliant) reason which, in the opinion of the investigator(s), would prevent participation in the study".
Interventions	**Vaccine type:** NDV-3, directed against *Staphylococcus aureus* and Candida sp **Experimental:** NDV-3 vaccine with alum. n =41 **Control:** NDV-3 vaccine without alum. n = 40 **Aluminium concentration:** 1 mcg/dose **Aluminium type:** alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** one dose administered IM
Outcomes	**Immunogenicity**: "to compare the serum IgG immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Serum IgG will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the serum IgA1 immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Serum IgA1 will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the cervicovaginal wash IgG immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Cervicovaginal wash IgG will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the cervicovaginal wash IgA1 immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Cervicovaginal wash IgA1 will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the cellular immune response for Als3-specific production of IFN-g from PBMCs between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. The IFN-g cellular immune responses will be evaluated by ELISpot using approximately 200,000 PBMCs per well. A positive response was defined as a sample with greater than 20 spot forming units per 10^6 PBMCs. To compare the cellular immune response for Als3-specific production of IL-17A from PBMCs between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. The IL-17A cellular immune responses will be evaluated by ELISpot using approximately 200,000 PBMCs per well. A positive response was defined as a sample with greater than 20 spot forming units per 10^6 PBMCs. Time Frame: Baseline, Day 7, Day 14, Day 28, Day 90/Exit". **Safety**: "safety of a single dose of NDV-3 vaccine, administered either IM with or without alum adjuvant at one dose level or ID at a lower dose level, compared to placebo. Clinical evaluations will be assessed on each subject at selected time points up to 90 days post-vaccination".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/01/2020 author (john_hennessey@novadigm.net) contacted. Study results uploaded on clinicaltrial.gov after request **Funding:** NovaDigm Therapeutics

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Unclear risk	The study protocol for one trial was registered prior to randomisation (NCT01447407), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT01995617

Methods	Phase 1, randomised, double-blind, placebo-controlled, dose escalation study
Participants	**Sex:** male and female **Age:** 18-55 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "males and non-pregnant females, ages 18 to 55 years inclusive. Willing and able to provide written informed consent. Willing to perform and comply with all study procedures including attending clinic visits as scheduled. Willing to practice adequate contraception that may include, but is not limited to, abstinence, monogamous relationship with vasectomized partner, vasectomy, barrier methods such as condoms, diaphragms, intrauterine devices, and licensed hormonal methods for 28 days before and 90 days after receiving Study Drug". **Exclusion criteria:** "immunocompromised individuals, including those receiving immunosuppressive doses of corticosteroids (more than 20 mg of prednisone given daily or on alternative days for 2 weeks or more within 6 months prior to the first dose of Study Drug, any dose of corticosteroids within 30 days of the first dose of Study Drug, or high dose inhaled corticosteroids [> 960 µg/day of beclomethasone dipropionate or equivalent]) or other immunosuppressive agents. Presence or history of autoimmune disease, regardless of current treatment. Insulin dependent diabetes. Positive serologic test for HIV-1 or hepatitis C infection; positive hepatitis B surface antigen (HBsAg). Screening serum chemistry or hematology abnormality ≥ Grade 1. Screening urinalysis abnormality ≥ Grade 2. Positive urine drug screen or alcohol Breathalyzer test at Screening or on Study Day 1. Other active, uncontrolled co-morbidities that, in the opinion of the Investigator would make the subject unsuitable for the study or unable to comply with the study requirements.NOTE: Subjects who are taking a medication to control an underlying co-morbidity may be enrolled if there have been no changes to their medication within 60 days prior to the first dose of Study Drug. Any acute illness including, fever (>100.4 degrees F [> 38 degrees C]) within 3 days prior to the first dose of Study Drug. Pregnant or nursing women. Receipt of any investigational drug within 30 days prior to the first dose of Study Drug. Receipt of blood products within 90 days prior to the first dose of Study Drug. Donation of blood or plasma within 56 days prior to Screening. Receipt of a live vaccine within 28 days prior or a subunit vaccine within 14 days prior to the first dose of Study Drug or planned vaccination within 30 days following the last dose of Study Drug. Prior vaccination with pneumococcal vaccine. History of hypersensitivity to any component of the vaccine or history of an allergic reaction to an immunization. History of invasive pneumococcal disease (i.e., sepsis, meningitis or pneumonia with bacteremia). History of drug or alcohol abuse that, in the opinion of the Investigator, would interfere with the subject's ability to comply with the study requirements. Any other condition that, in the opinion of the Investigator would make the subject unsuitable for the study or unable to comply with the study requirements".
Interventions	**Vaccine type:** GEN-004, a *Streptococcus Pneumoniae* protein subunit vaccine **Experimental:** low, mid or high dose vaccine with aluminium **Control:** low, mid or high dose vaccine **Aluminium concentration:** 350 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported

	Vaccination schedule: 3 doses at 4 week intervals
Outcomes	**Immunogenicity** and **safety** up to 12 months after the last dose
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Genocea Biosciences) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT02022163

Methods	Phase I, randomised trial
Participants	**Sex:** male and female **Age:** 18-60 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male and female adults, 18 to 60 years of age, inclusive. Healthy as judged by the Investigator or designee and determined by medical history, complete general history/symptom-directed physical examination, vital signs, screening laboratories, and medical history conducted no more than 30 days prior to study vaccine administration. BMI of ≥18 and ≤32. Comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits. Accessible by phone on a consistent basis Give his/her consent to participate in this study (by signing the ICF). In the opinion of the Investigator, competence and willingness to provide written, informed consent for participation after reading the informed consent form. The subject must have adequate opportunity to discuss the study with an Investigator or qualified designee. If female, have a negative pregnancy test result prior to first immunization. Female of childbearing potential (except subjects in a monogamous same sex relationship), must use an effective birth control for the 28 days prior to immunization and must agree to continue employing adequate birth control measures from Day 0 (first immunization) until at least 60 days post-second immunization and must have no plan to become pregnant from Day 0 (first immunization) until at least 60 days post-second immunization. Highly effective birth control includes hormonal contraceptives (e.g., injectable, topical [patch], estrogenic vaginal ring, etc.), intra-uterine device (IUD), abstinence (confirmed by Investigator), or male condom plus spermicide. Abstinent subjects should be asked what method(s) they would use, should their circumstances change, and subjects without a well-defined plan should be excluded". **Exclusion criteria:** "presence of significant acute or chronic, uncontrolled medical or neuropsychiatric illness. "Uncontrolled" is defined as: requiring a new medical or surgical treatment within one month prior to study vaccine administration, requiring a change in medication dosage in one month prior to study vaccine administration due to uncontrolled symptoms or drug toxicity (elective dosage adjustments in stable subjects are acceptable). Hospitalization or an event fulfilling the definition of a serious adverse event within one month prior to study vaccine administration. Any medical or neuropsychiatric condition or any history of excessive alcohol use or drug abuse which, in the Investigator's opinion, would render the subject incompetent to provide informed consent or unable to provide valid safety observations and reporting. Any confirmed or suspected immunosuppressive condition or immunodeficiency including history of human immunodeficiency virus (HIV) infection, Hepatitis B or C, or the presence of lymphoproliferative disease. Presence of any febrile illness, oral temperature of >38.0˚C within 24 hours prior to immunization. Such subjects may be re-evaluated for enrolment after resolution of illness. History of autoimmune disease. Administration of any vaccine (including any other influenza vaccine) within 30 days prior to study enrolment or planned administration within the period from the vaccination up to blood sampling at Day 42 or within 30 days prior to blood sampling at Day 228. Immunization on

	an emergency basis of a tetanus and diphtheria toxoids adsorbed for adult use (Td) will be allowed provided the vaccine is not administered within two weeks prior to study vaccine administration. Receipt of any other emergency immunizations (e.g., rabies) will result in a case-by-case review by the medical monitor of continued participation. Administration of any adjuvated or investigational influenza vaccine other than a 'simple' seasonal Trivalent Inactivated Vaccine (TIV) or Quadrivalent Inactivated Vaccine (QIV) within 1 year prior to study enrolment or planned administration prior to the end of this trial (Day 228). Use of any investigational or non-registered product within 30 days prior to study enrolment or planned use during the study period. Subjects may not participate in any other investigational or marketed drug study while participating in this study. Treatment with systemic glucocorticoids at a dose exceeding 10 mg of prednisone per day, or equivalent for more than 7 consecutive days or for 10 or more days in total, within one month of study vaccine administration, any other cytotoxic or immunosuppressant drug, or any globulin preparation within 3 months of vaccination. Low doses of nasal or inhaled glucocorticoids are generally allowed. Use of high dose inhaled steroids or oral and parenteral high dose steroid medications. Nasal steroids are allowed. Any significant disorder of coagulation or treatment with warfarin derivatives or heparin. Persons receiving prophylactic anti-platelet medications, e.g., low-dose aspirin [≤ 325 mg/day (1 regular adult aspirin) or ≤ 81 mg/day (1 baby aspirin)], and without a clinically apparent bleeding tendency are eligible. History of previous H7N9 vaccination or a history of exposure to H7N9 virus. Any subject that was enrolled on previous H7N9 studies (except the ones that received placebo) would not be eligible. Are at high risk of contracting H7N9 influenza infection (e.g. poultry workers). History of allergy to any of the constituents of the H7 VLP (H7N9) study vaccine, Alhydrogel® (aluminium hydroxide), or to the phosphate-buffered saline (PBS) (used as placebo). History of severe allergic reactions (including anaphylaxis) to any food, medication or bee sting or previous severe asthma. History of tobacco allergy. Continuous use of anti-histamines in the last 4 weeks prior to first immunization or use of anti-histamines 48 hours prior to each study immunization. Have a rash, dermatological condition, tattoos, or muscle mass at injection site which may interfere with injection site reaction rating. Have received a blood transfusion within 90 days prior to study vaccination. If female, positive pregnancy test results at screening, and prior to immunizations on Day 0 and Day 21. Female subjects who are lactating. Any vital sign abnormalities: systolic blood pressure, diastolic blood pressure, resting heart rate not well controlled or according to the Investigator's opinion. Cancer or treatment for cancer within 3 years of study vaccine administration. Persons with a history of cancer who are disease-free without treatment for 3 years or more are eligible. Persons with treated and uncomplicated basal cell carcinoma of the skin are eligible".
Interventions	**Vaccine type:** plant-made H7 virus-like particle (VLP) influenza vaccine **Experimental:** high dose of H7 VLP vaccine + Alhydrogel **Control:** high dose of H7 VLP vaccine **Aluminium concentration:** not reported **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 intramuscular injections

Outcomes	**Immunogenicity**: "geometric mean titers of hemagglutination inhibition (HI antibody on days 0, 21 and 42 to measure geometric mean fold rise and seroconversion rates). Follow-up serology samples for geometric mean titers will be taken at day 228". **Safety**: "rate, severity and relationship to vaccination of solicited and unsolicited adverse events post-vaccinations. A 6-month follow-up period will be performed".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Medicago) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT02304185

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "participant must be healthy on the basis of physical examination, medical history, electrocardiogram, laboratory criteria, and vital signs measurement performed at Screening. Participants are negative for human immunodeficiency virus (HIV) infection at Screening (negative United States Food and Drug Administration-approved HIV diagnostic blood test). All female participants of childbearing potential must have a negative serum (beta human chorionic gonadotropin) at Screening, and a negative urine pregnancy test pre-dose on Day 1 and Day 29. A woman must agree not to donate eggs (ova, oocytes) for the purposes of assisted reproduction until 3 months after receiving the last dose of study vaccine. A man must agree not to donate sperm until 3 months after receiving the last dose of study vaccine. Participants are assessed by the clinic staff as being at low risk for HIV infection". **Exclusion criteria:** "participant has chronic active hepatitis B or active hepatitis C, active syphilis infection, chlamydia, gonorrhea, or trichomonas. Active syphilis documented by exam or serology unless positive serology is due to past treated infection. In the 12 months prior to enrollment, participant has a history of newly acquired syphilis, gonorrhea, non-gonococcal urethritis, herpes simplex virus type 2 (HSV2), chlamydia, pelvic inflammatory disease, trichomonas, mucopurulent cervicitis, epididymitis, proctitis, lymphogranulomavenereum, chancroid, or hepatitis B. Participant has any clinically significant acute or chronic medical condition that in the opinion of the Investigator would preclude participation (for example, history of seizure disorders, bleeding/clotting disorder, autoimmune disease, active malignancy, poorly controlled asthma, active tuberculosis or other systemic infections). Participant has had major surgery within the 4 weeks prior to study entry or planned major surgery through the course of the study. Participant has had a thyroidectomy, or thyroid disease requiring medication during the last 12 months".
Interventions	**Vaccine type:** trimeric glycoprotein140 (gp140) **Experimental:** gp140, 50 mcg with aluminium phosphate **Control:**gp140, 50 mcg **Aluminium concentration:** 425 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 intramuscular injections on day 1 and day 29
Outcomes	**Immunogenicity:** "envelop binding antibody titer at week 8" **Safety**: "number of participants with local and systemic reactogenicity events over 8 days after first and second vaccination. Number of participants with adverse events or serious adverse events (time frame: week 52)"
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** 17/01/2020 authors (RA-RNDUS-JANSSENRND@its.jnj.com) were inquired on the safety and immunogenicity results. Answer received. **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT02777411

Methods	Randomised, open-label, dose-finding, phase II study
Participants	**Sex:** male and female **Age:** 6 months to 6 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy children aged from 3 to 6 years old (i.e. ≥ 3 years old and < 7 years old) for Part A; from 6 to 35 months old (i.e. ≥ 6 months old

	and < 36 months old) for Part B at the time of first vaccination. Subject's guardians were able and willing to comply with study procedures and give written informed consent. Subject was able and could comply with the requirements of the protocol. Subject with body temperature ≤38°C" **Exclusion criteria:** "subject with previous known exposure to Enterovirus 71 (EV71). Subject with a history of herpangina, hand-foot-mouth disease, acute hemorrhagic conjunctivitis associated with enterovirus infection in the past 3 months. Subject with gestation < 37 weeks. Subject with birth weight <2.5 kg. Subject with a history of hypersensitivity to vaccines, or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Family history of seizures or progressive neurological disease. Family history of congenital or hereditary immunodeficiency. Severe malnutrition or dysgenopathy. Major congenital defects or serious chronic illness, including perinatal brain damage. Subject diagnosed of having autoimmune disease (e.g., celiac disease, type I diabetes, lupus (SLE), juvenile dermatomyositis, scleroderma, juvenile idiopathic arthritis (JIA), immune (or idiopathic) thrombocytopenia purpura). Bleeding disorder diagnosed by a doctor or significant bruising or bleeding difficulties with IM injections or blood draws. Any acute infections 7 days prior to administrate the first vaccination. Use of any investigational product (including drug, vaccine) within 30 days prior to vaccination or planned use during the study period.Administration of any attenuated live vaccine within 7 days prior to vaccination. Use of immunoglobulins or any blood products within 3 months prior to vaccination or planned use during the study period. Chronic administration (defined as > 14 days) of immunosuppressants or other immunomodulators or systemic corticosteroids within 6 months prior to vaccination. Under anti-tuberculosis prevention or therapy. Any condition that in the opinion of the investigator may interfere with the evaluation of study objectives".
Interventions	**Vaccine type:** enterovirus EV71 **Experimental:** 1 or 2 mcg of EV71 with aluminium phosphate **Control:** 1 or 2 mcg of EV71 **Aluminium concentration:** 150 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two vaccinations at 28 days apart
Outcomes	**Immunogenicity**: "immunoglobulin G titers induced by the EV71 vaccine (determined by ELISA), time frame: day 28, day 56, day 196; serum neutralizing antibody titers induced by the EV71 vaccine; seroconversion rate based on neutralizing antibody titers; change in the laboratory results based on hematology tests in each visit; change in the laboratory results based on biochemistry tests in each visit (time frame: day 28, day 56, day 196)" **Safety**: "solicited adverse events, time frame: 7 days after each vaccination. Unsolicited adverse events, time frame: 28 days after each vaccination. The occurrence of overall adverse events and serious adverse event, time frame: day 0 to day 196. Change in heart rates at each visit; change in blood pressures at each visit (for the 3 to 6 years old group only)(time frame: day 28, day 56, day 196)"
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** company (Enimmune Corporation) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	Open label
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Open label
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Open label
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03010228

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-39 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy adults ≥18 to <40 years of age (for US healthy adults ≥ 19 years to <40 years) at the time of screening. Health status is assessed by investigator at time of screening based on medical history, physical

examination, and laboratory parameters.
Written informed consent obtained from the subject prior to any study related procedures.
BMI ≥18.5 and <30 at Visit 0 (Screening Visit). Men or women; women require a negative pregnancy test at screening. Women with childbearing potential must agree to use an adequate contraception during the entire study".
Exclusion criteria: "pathological findings in any of the investigations (i.e. medical history, physical examination) as deemed clinically relevant by the investigator or any abnormal laboratory parameter of hematology, clinical chemistry, coagulation, RF (Rheumatoid Factor) or ACPA (Anti-citrullinated protein antibodies) at the Screening Visit. Medical history of severe cardiovascular, respiratory (including asthma), metabolic, neurological, hepatic, rheumatic, hematological, gastrointestinal, renal disorders. Medical history of or current musculoskeletal disorders as deemed clinically relevant by the investigator, arthritis or chronic musculoskeletal pain. Previous vaccination against Lyme borreliosis with any (investigational) vaccine. Use of any other investigational or non-registered medicinal product within 30 days prior to VLA15 vaccination at Visit 1 (Day 0) and throughout the entire study period. Chronic illness related to Lyme borreliosis (LB), a history of or active symptomatic LB as suspected or diagnosed by a physician. Subjects with a positive serology test result for Borrelia burgdorferi sensu lato (s.l.) antibodies at screening are excluded. Tick bite within 3 weeks prior to screening, or tick bite during vaccination period (i.e. Day 0 to Day 56). Known active infection with Babesia microti (babesiosis) or Anaplasma phagocytophilum (ehrlichiosis). Active or passive immunization four weeks before first vaccination at Visit 1 and up to Day 84 (i.e. 4 weeks after the last VLA15 immunization). Afterwards, vaccinations should be avoided, except for influenza (seasonal or pandemic) vaccines which may be administered after Day 84 (i.e. 4 weeks after the last VLA15 immunization). Subjects susceptible to require a vaccine during the study period (e.g. due to planned travel) should be excluded at screening. Known congenital, hereditary or acquired immunodeficiency, including infection with human immunodeficiency virus (HIV), status post organ transplantation or immuno-suppressive therapy within 30 days prior to Day 0 and up to Day 84. Immuno-suppressive therapy is defined as administration of chronic (longer than 14 days) prednisone or equivalent ≥ 0.05 mg/kg/day. Topical and inhaled steroids are allowed.
Planned intake of NSAID (Nonsteroidal anti-inflammatory drug) within three days prior and within seven days after any VLA15 vaccination. History of severe hypersensitivity reactions and anaphylaxis. History of allergic bronchial asthma and severe allergic rhinoconjunctivitis. Known hypersensitivity or allergic reactions to one of the components of the vaccine. History of autoimmune disease, including Type I Diabetes mellitus. Subjects with vitiligo or thyroid disease taking thyroid hormone replacement are not excluded. Any malignancy in the past 5 years. If treatment for cancer was successfully completed more than 5 years ago and the malignancy is considered to be cured, the subject may be enrolled. Acute febrile infections within 4 weeks prior to first vaccination and body temperature >37.8 C (oral) prior to each vaccination. Known or suspected alcohol abuse, alcohol dependence, i.e. an average of more than approximately 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day] or illicit drug use within the last year. Inability or unwillingness to avoid more than the usual intake of alcohol (i.e. not more than 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day) during the 48 hours after vaccination. Pregnancy (positive pregnancy test), lactation or inadequate contraception in

	women with childbearing potential. Inability or unwillingness to provide informed consent or not able to understand and comply with protocol requirements, instructions and protocol-stated restrictions. Donation of blood or blood-derived products (e.g. plasma) within 4 weeks prior to Visit 0 (Screening Visit) and during the entire study. Receipt of blood or blood-derived products in the past 3 months prior to Visit 0 (Screening Visit) or anticipation of such products during the entire study. Mental disorder as deemed clinically relevant by the investigator. History of Guillain-Barré-Syndrome (GBS). Any condition which might interfere with study objectives or that would limit the subject's ability to complete the study in the opinion of the investigator. Persons who are committed to an institution (by virtue of an order issued either by the judicial or the administrative authorities). Persons who are in a dependent relationship with the sponsor, an investigator or other study team members, or the study center. Dependent relationships include close relatives and household members (i.e. children, partner/spouse, siblings, parents) as well as employees of the investigator or study center personnel".
Interventions	**Vaccine type:** VLA15 vaccine against Lyme borreliosis **Experimental:** VLA15 with aluminium **Control:** VLA15 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 intramuscular vaccinations into the deltoid region on days 0, 28 and 56
Outcomes	**Immunogencity:** "GMTs (Geometric Mean Titre) for IgG against each OspA serotype ST1 to ST6, determined by ELISA, (time frame: day 0, 28, 56, 84, 180, 236 and 365). SCRs (Seroconversion Rate, defined based on fold increase of each OspA serotype specific IgG (ST1 to ST6) as compared to baseline), (time frame: day 28, 56, 84, 180, 236 and 365). GMFR (Geometric Mean of the fold rise as compared to baseline) for IgG against each OspA serotype ST1 to ST6, determined by ELISA,(time frame: day 28, 56, 84, 180, 236 and 365)" **Safety**: "rate of SAEs, rate of any solicited or unsolicited Grade 3 or Grade 4 events to day 84. Rate of solicited, unsolicited local and systemic AEs within 7 days after each and after any vaccination up to day 84. Rate of SAEs during the entire study period, up to day 365. Rate of any solicited or unsolicited Grade 3 or Grade 4 AEs during the entire study period".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** company (Valneva Austria GmbH) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described

Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	n.a.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03026348

Methods	Randomized, observer-blind, trial
Participants	**Sex:** male and female **Age:** 60-80 years **Ethnicity:** not reported **Diagnosis:** clinically stable for underling conditions **Inclusion criteria:** "males and females 60 through 80 years of age who are ambulatory and live in the community or in an assisted-living facility that provides minimal assistance, such that the subject is primarily responsible for self-care and activities of daily living. Subjects may have one or more chronic medical diagnoses, but should be clinically stable as assessed by: absence of changes in medical therapy within one month due to treatment failure or toxicity (dose adjustments of ongoing therapies for optimal effect, or replacements within a class of drugs due to convenience or cost, will be deemed acceptable),Absence of medical events qualifying as SAEs within one month of the planned vaccination on Day 0, andAbsence of known, current, and life-limiting diagnoses which, in the opinion of the investigator, render survival to completion of the protocol unlikely. Willing and able (on both a physical and cognitive basis) to give informed consent prior to study enrollment. To complete the consent process, all qualifying subjects will correctly answer at least 4 out of 5 questions of the informed consent form (ICF) comprehension assessment in no more than 2 attempts.Able to comply with study requirements. As the

	protocol procedures involve telephone contacts for safety ascertainment, eligible subjects must have a reliable access to a telephone". **Exclusion criteria:** "received any prior RSV vaccine. Participation in research involving any additional investigational product (drug / biologic / device) within 45 days before planned date of first vaccination. History of a serious reaction to any prior vaccination or a history of Guillain-Barré syndrome (GBS) within 6 weeks of any prior influenza immunization. Receipt of inactivated influenza vaccine within 14 days prior to the Day 0 dose of test article or any other vaccine within the 4 weeks prior to the Day 0 dose of test article. Any known or suspected immunosuppressive condition, acquired or congenital, as determined by history and/or physical examination. Chronic administration (defined as more than 14 continuous days) of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. An immunosuppressant dose of glucocorticoid will be defined as a systemic dose ≥ 10 mg of prednisone per day or equivalent. The use of topical, inhaled, and nasal glucocorticoids will be permitted. Administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or during the study. Acute disease at the time of enrollment (defined as the presence of a moderate or severe illness with or without fever, or an oral temperature ≥ 38.0°C on the planned day of vaccine administration). Known disturbance of coagulation. Potential subjects receiving aspirin, clopidogrel, prasugrel, dipyridamole, dabigatran, apixaban, rivaroxaban, or warfarin under good control for cardiovascular prophylaxis or prophylaxis of thromboembolic disease or stroke in the setting of atrial fibrillation will NOT be excluded. Suspicion or recent history (within one year of planned vaccination) of alcohol or other substance abuse. Any condition that in the opinion of the investigator would pose a health risk to the subject if enrolled or could interfere with evaluation of the vaccine or interpretation of study results (including neurologic, cognitive, or psychiatric conditions deemed likely to impair the quality of study compliance or safety reporting)".
Interventions	**Vaccine type:** RSV F vaccine **Experimental:** RSV F vaccine with aluminium **Control:** RSV F vaccine **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two injections at day 1 and 21
Outcomes	**Immunogenicity**: "neutralizing antibody titers to at least one RSV/A strain, (time frame: day 0, 21, 28). Serum concentrations of antibodies competitive with palivizumab for binding to the RSV F protein, (time Frame: day 0, 21, 28, 56, 119, 385). Serum IgG antibody concentrations as ELISA units specific for the F protein antigen, (time frame: day 0, 21, 28, 56, 119, 385). Counts of IFN-γ spot forming units following in vitro stimulation of day 0, day 7, and day 28 PBMC isolates with RSV F peptides, (time frame: day 0, 7, 28). Counts and proportions of Day 0, Day 7, and Day 28 peripheral blood T cells positive by intracellular staining for IL-2, IFN-γ, or TNF-α production (alone or any combination thereof) following in vitro stimulation with RSV F peptides, (time frame: day 0, 7, 28)" **Safety**: "subjects with solicited local and systemic AEs occurring within the 7-day period following dosings on Day 0 and Day 21 and all adverse events, solicited and unsolicited, occurring within the 56-day period of Day 0, (time frame: Day 0 - Day 6, Day 21 - Day 27; Day 0 - Day 56)"

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Novavax) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03284710

Methods	Phase 1/2a Partially Double-blinded, Randomized Clinical Trial
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** black **Diagnosis:** healthy

Inclusion criteria: "Age of 18 to 40 years. Access to a participating HVTN CRS and willingness to be followed for the planned duration of the study
Ability and willingness to provide informed consent. Assessment of understanding: volunteer demonstrates understanding of this study; completes a questionnaire prior to first vaccination with verbal demonstration of understanding of all questionnaire items answered incorrectly.Agrees not to enroll in another study of an investigational research agent. Good general health as shown by medical history, physical exam, and screening laboratory. testsHIV-Related Criteria: Willingness to receive HIV test results. Willingness to discuss HIV infection risks and amenable to HIV risk reduction counseling. Assessed by the clinic staff as being at "low risk" for HIV infection and committed to maintaining behavior consistent with low risk of HIV exposure through the last required protocol clinic visit.Laboratory Inclusion ValuesHemogram/Complete blood count (CBC). Hemoglobin ≥ 11.0 g/dL for volunteers who were born female, ≥ 13.0 g/dL for volunteers who were born male. White blood cell count = 3,300 to 12,000 cells/mm^3. Total lymphocyte count ≥ 800 cells/mm^3
Remaining differential either within institutional normal range or with site physician approval. Platelets = 125,000 to 550,000/mm^3. Chemistry panel: ALT, AST, and ALP < 1.25 times the institutional upper limit of normal; creatinine ≤ institutional upper limit of normal.Virology. Negative HIV-1 and -2 blood test: Sites may use locally available assays that have been approved by HVTN Laboratory Operations. Negative Hepatitis B surface antigen (HBsAg). Negative anti-Hepatitis C virus antibodies (anti-HCV), or negative HCV polymerase chain reaction (PCR) if the anti-HCV is positive Urine. Normal urine: negative urine glucose, and negative or trace urine protein, andNegative or trace urine hemoglobin (if trace hemoglobin is present on dipstick, a microscopic urinalysis with red blood cells levels within institutional normal range). Volunteers who were born female: negative serum or urine beta human chorionic gonadotropin (β-HCG) pregnancy test performed prior to vaccination on the day of initial vaccination. Persons who are NOT of reproductive potential due to having undergone total hysterectomy or bilateral oophorectomy (verified by medical records), are not required to undergo pregnancy testing. A volunteer who was born female must: Agree to consistently use effective contraception for sexual activity that could lead to pregnancy from at least 21 days prior to enrollment through the last required protocol clinic visit. Effective contraception is defined as using condoms (male or female), or diaphragm or cervical cap, PLUS 1 of the following methods: Intrauterine device (IUD), Hormonal contraception (in accordance with Republic of South Africa: National Contraception Clinical Guidelines), successful vasectomy in the male partner (considered successful if a volunteer reports that a male partner has [1] documentation of azoospermia by microscopy, or [2] a vasectomy more than 2 years ago with no resultant pregnancy despite sexual activity postvasectomy); or any other contraceptive method approved by the HVTN 107 Protocol Safety Review TeamOr not be of reproductive potential, such as having reached menopause (no menses for 1 year) or having undergone hysterectomy, bilateral oophorectomy, or tubal ligation;Or be sexually abstinent. Volunteers who were born female must also agree not to seek pregnancy through alternative methods, such as artificial insemination or in vitro fertilization until after the last required protocol clinic visit. Volunteers who were born female consenting to provide cervical samples: pap smear within the 3 years prior to enrollment, with the latest result reported as normal or ASCUS (atypical squamous cells of undetermined significance); for those 21 years and older that have not had a

pap smear within the last 3 years prior to enrollment, must be willing to undergo a pap smear with the result reported as normal or ASCUS prior to sample collection."

Exclusion criteria: "Blood products received within 120 days before first vaccination. Investigational research agents received within 30 days before first vaccination. Body mass index (BMI) ≥ 40; or BMI ≥ 35 with 2 or more of the following: systolic blood pressure > 140 mm Hg, diastolic blood pressure > 90 mm Hg, current smoker, known hyperlipidemia. Intent to participate in another study of an investigational research agent or any other study that requires non-HVTN HIV antibody testing during the planned duration of the HVTN 107 study. Pregnant or breastfeedingVaccines and other Injections
HIV vaccine(s) received in a prior HIV vaccine trial. For volunteers who have received control/placebo in an HIV vaccine trial, the HVTN 107 PSRT will determine eligibility on a case-by-case basis. Non-HIV experimental vaccine(s) received within the last 5 years in a prior vaccine trial. Exceptions may be made for vaccines that have subsequently undergone licensure in a volunteer's country of residence. For volunteers who have received control/placebo in an experimental vaccine trial, the HVTN 107 PSRT will determine eligibility on a case-by-case basis. For volunteers who have received an experimental vaccine(s) greater than 5 years ago, eligibility for enrollment will be determined by the HVTN 107 PSRT on a case-by-case basis.Live attenuated vaccines other than influenza vaccine received within 30 days before first vaccination or scheduled within 14 days after injection (eg, measles, mumps, and rubella [MMR]; oral polio vaccine [OPV]; varicella; yellow fever). Influenza vaccine or any vaccines that are not live attenuated vaccines and were received within 14 days prior to first vaccination (eg, tetanus, pneumococcal, Hepatitis A or B). Allergy treatment with antigen injections within 30 days before first vaccination or that are scheduled within 14 days after first vaccinationImmune System. Immunosuppressive medications received within 168 days before first vaccination. (Not exclusionary: [1] corticosteroid nasal spray; [2] inhaled corticosteroids; [3] topical corticosteroids for mild, uncomplicated dermatitis; or [4] a single course of oral/parenteral corticosteroids at doses < 2 mg/kg/day and length of therapy < 11 days with completion at least 30 days prior to enrollment.) Serious adverse reactions to vaccines or to vaccine components such as eggs, egg products, or neomycin, including history of anaphylaxis and related symptoms such as hives, respiratory difficulty, angioedema, and/or abdominal pain. (Not excluded from participation: a volunteer who had a nonanaphylactic adverse reaction to pertussis vaccine as a child.). Immunoglobulin received within 60 days before first vaccination. Autoimmune disease. ImmunodeficiencyClinically significant medical conditions. Untreated or incompletely treated syphilis infection. Clinically significant medical condition, physical examination findings, clinically significant abnormal laboratory results, or past medical history with clinically significant implications for current health. A clinically significant condition or process includes but is not limited to: Any medical, psychiatric, occupational, or other condition that, in the judgment of the investigator, would interfere with, or serve as a contraindication to, protocol adherence, assessment of safety or reactogenicity, or a volunteer's ability to give informed consent. Psychiatric condition that precludes compliance with the protocol. Specifically excluded are persons with psychoses within the past 3 years, ongoing risk for suicide, or history of suicide attempt or gesture within the past 3 years. Current anti-tuberculosis (TB) prophylaxis or therapy
Asthma other than mild, well-controlled asthma. (Symptoms of asthma severity as defined in the most recent National Asthma Education and Prevention

	Program (NAEPP) Expert Panel report).Exclude a volunteer who uses a short-acting rescue inhaler (typically a beta 2 agonist) daily, orUses moderate/high dose inhaled corticosteroids, orIn the past year has either of the following: greater than 1 exacerbation of symptoms treated with oral/parenteral corticosteroids;Needed emergency care, urgent care, hospitalization, or intubation for asthma. Diabetes mellitus type 1 or type 2, including cases controlled with diet alone. (Not excluded: history of isolated gestational diabetes.). Thyroidectomy, or thyroid disease requiring medication during the last 12 months Hypertension: if a person has been found to have elevated blood pressure or hypertension during screening or previously, exclude for blood pressure that is not well controlled. Well-controlled blood pressure is defined as consistently ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic, with or without medication, with only isolated, brief instances of higher readings, which must be ≤ 150 mm Hg systolic and ≤ 100 mm Hg diastolic. For these volunteers, blood pressure must be ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic at enrollment.If a person has NOT been found to have elevated blood pressure or hypertension during screening or previously, exclude for systolic blood pressure ≥ 150 mm Hg at enrollment or diastolic blood pressure ≥ 100 mm Hg at enrollment. Bleeding disorder diagnosed by a doctor (eg, factor deficiency, coagulopathy, or platelet disorder requiring special precautions) Malignancy (Not excluded from participation: Volunteer who has had malignancy excised surgically and who, in the investigator's estimation, has a reasonable assurance of sustained cure. or who is unlikely to experience recurrence of malignancy during the period of the study) Seizure disorder: History of seizure(s) within past three years. Also exclude if volunteer has used medications in order to prevent or treat seizure(s) at any time within the past 3 years. Asplenia: any condition resulting in the absence of a functional spleen. Current anti-tuberculosis (TB) prophylaxis or therapy Asthma other than mild, well-controlled asthma. (Symptoms of asthma severity as defined in the most recent National Asthma Education and Prevention Program (NAEPP) Expert Panel report).Exclude a volunteer who: uses a short-acting rescue inhaler (typically a beta 2 agonist) daily, orUses moderate/high dose inhaled corticosteroids, orIn the past year has either of the following: Greater than 1 exacerbation of symptoms treated with oral/parenteral corticosteroids;Needed emergency care, urgent care, hospitalization, or intubation for asthma. Diabetes mellitus type 1 or type 2, including cases controlled with diet alone. (Not excluded: history of isolated gestational diabetes.). Thyroidectomy, or thyroid disease requiring medication during the last 12 months Hypertension: If a person has been found to have elevated blood pressure or hypertension during screening or previously, exclude for blood pressure that is not well controlled. Well-controlled blood pressure is defined as consistently ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic, with or without medication, with only isolated, brief instances of higher readings, which must be ≤ 150 mm Hg systolic and ≤ 100 mm Hg diastolic. For these volunteers, blood pressure must be ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic at enrollment.If a person has NOT been found to have elevated blood pressure or hypertension during screening or previously, exclude for systolic blood pressure ≥ 150 mm Hg at enrollment or diastolic blood pressure ≥ 100 mm Hg at enrollment. Bleeding disorder diagnosed by a doctor (eg, factor deficiency, coagulopathy, or platelet disorder requiring special precautions). Malignancy (Not excluded from participation: Volunteer who has had malignancy excised surgically and who, in

	the investigator's estimation, has a reasonable assurance of sustained cure. or who is unlikely to experience recurrence of malignancy during the period of the study). Seizure disorder: History of seizure(s) within past three years. Also exclude if volunteer has used medications in order to prevent or treat seizure(s) at any time within the past 3 years. Asplenia: any condition resulting in the absence of a functional spleen. History of hereditary angioedema, acquired angioedema, or idiopathic angioedema.
Interventions	**Vaccine type:** Clade C ALVAC and gp120 HIV Vaccine **Experimental:** ALVAC-HIV + gp120/Al(OH)3 **Control:** ALVAC-HIV + gp120/ **Aluminium concentration:** 625mcg/dose **Aluminium type:** aluminum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** participants will receive the ALVAC-HIV vaccine in the left deltoid at months 0, 1, 3, 6, and 12, and Bivalent Subtype C gp120 in the right deltoid at months 3, 6, and 12.
Outcomes	**Immunogenicity**: "occurrence of Vaccine-induced Systemic IgG Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 2 [Time Frame: Measured at Month 6.5]Serum IgG responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:50 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Samples from post-enrollment visits have positive responses if they meet three criteria: (1) net MFI greater than or equal to an antigen-specific response threshold (defined as the maximum of 100 and the 95th percentile of baseline net MFI), (2) net MFI values are greater than 3 times baseline net MFI, and (3) experimental antigen MFI values are greater than 3 times baseline MFI. Level of Vaccine-induced Systemic IgG Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 2 [Time Frame: Measured at Month 6.5]Serum IgG responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:50 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Comparisons were performed among positive responders only (positivity criteria are described in Outcome 1). Occurrence of Vaccine-induced Serum IgA Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 3 [Time Frame: Measured at Month 6.5]Serum IgA responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:10 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus

reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Samples from post-enrollment visits have positive responses if they meet three criteria: (1) net MFI greater than or equal to an antigen-specific response threshold (defined as the maximum of 100 and the 95th percentile of baseline net MFI), (2) net MFI values are greater than 3 times baseline net MFI, and (3) experimental antigen MFI values are greater than 3 times baseline MFI.
Level of Vaccine-induced Serum IgA Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 3 [Time Frame: Measured at Month 6.5]Serum IgA responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:10 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Comparisons were performed among positive responders only.
Safety: number of Participants Reporting Local Reactogenicity Signs and Symptoms: Pain and/or Tenderness [Time Frame: Measured through Month Measured through 3 days after each vaccine dose at Months 0, 1, 3, 6, and 12]Graded according to the Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events, Version 2.1 [July 2017]. The maximum grade observed for each symptom over the time frame is presented.
Number of Participants Reporting Local Reactogenicity Signs and Symptoms: Erythema and/or Induration [Time Frame: Measured through Month Measured through 3 days after each vaccine dose at Months 0, 1, 3, 6, and 12]Graded according to the Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events, Version 2.1 [July 2017]. The maximum grade observed for each symptom over the time frame is presented.
Number of Participants Reporting Systemic Reactogenicity Signs and Symptoms [Time Frame: Measured through Month Measured through 3 days after each vaccine dose at Months 0, 1, 3, 6, and 12]Graded according to the Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events, Version 2.1 [July 2017]. The following symptoms are considered as systemic reactogenicity if the onset date was within the periods of assessment specified in the protocol: malaise and/or fatigue, myalgia, headache, nausea, vomiting, chills, arthralgia, and body temperature. The item Max. Systemic Symptoms is the maximum of the individual systemic reactogenicities excluding body temperature for a participant. Number of Participants Reporting Adverse Events (AEs), by Relationship to Study Product [Time Frame: Measured through 30 days after each vaccine dose at Months 0, 1, 3, 6, and 12]For participants reporting multiple AEs over the time frame, the maximum relationship is counted. Number of Participants Reporting Adverse Events (AEs), by Severity Grade [Time Frame: Measured through 30 days after each vaccine dose at Months 0, 1, 3, 6, and 12]For participants reporting multiple AEs over the time frame, the maximum severity grade is counted.
Number of Participants Reporting Serious Adverse Events (SAEs) [Time Frame: Measured through Month 18]Measured as outlined in Version 2.0 (January 2010) of the Manual for Expedited Reporting of Adverse Events to DAIDS (DAIDS EAE Manual). Number of Participants Reporting Adverse

	Events of Special Interest (AESIs) [Time Frame: Measured through Month 18]There were no adverse events of special interest reported by any participant. Number of Participants Reporting New Chronic Conditions (Requiring Medical Intervention for >= 30 Days) [Time Frame: Measured through Month 18]. Chemistry and Hematology Laboratory Measures - ALT (SGPT), AST, Alkaline Phosphatase [Time Frame: Measured during screening, Days 7, 42, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Creatinine [Time Frame: Measured during screening, Days 7, 42, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Hemoglobin [Time Frame: Measured during screening, Days 7, 42, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Lymphocytes, Neutrophils [Time Frame: Measured during screening, Days 0, 1, 3, 7, 42, 84, 85, 87, 91, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Platelets, WBC [Time Frame: Measured during screening, Days 0, 1, 3, 7, 42, 84, 85, 87, 91, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Numbers of Participants With Grade 1 or Higher Local Laboratory Results [Time Frame: Measured during screening, Days 0, 1, 3, 7, 42, 84, 85, 87, 91, 98, 182, 378, and 455]The number (percentage) of participants with local laboratory values recorded as meeting Grade 1 AE criteria or above as specified in the DAIDS AE Grading Table were tabulated by treatment arm for each post-vaccination time point. Number of Participants With Early Study Termination Associated With an AE or Reactogenicity [Time Frame: Measured through Month 18]There were no early study terminations associated with an AE or reactogenicity reported by any participant. Number of Participants With Study Product Discontinuation Associated With an AE or Reactogenicity [Time Frame: Measured through Month 18]From the study product discontinuation form, study product discontinuation reasons are tabulated by treatment arm.".
Are there other groups in the trial not analysed in this review?	Yes.
Notes	**Contacts with authors:** on 23/08/2021 authors (paulg@uab.edu; kmngadi@auruminstitute.org) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of participants, personnel and outcome assessors ● aluminium manufacturer ● proportion of participants per group who developed HIV ● geometric mean titres after vaccination per group No answer received **Funding:** funded by National Institute of Allergy and Infectious Diseases (NIAID)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized. No further information
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind. No further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. No further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. No further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind. No further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03284710), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03295318

Methods	Phase 2, observer-blind, randomised, controlled study
Participants	**Sex:** male and female **Age:** 12-16 months **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male and female children between 12 months and 16 months old inclusive (minimum 365 days of age and maximum 16 months plus 29 days of age); for whom parent(s)/legal guardian(s) have given written informed consent after the nature of the study has been explained according to local regulatory requirements; who the investigator believes that their

	parent(s)/guardian(s) will be available for all the subject visits and would comply with the requirements of the protocol (e.g. timely reporting of adverse events, availability for study site visits and home visits); individuals in good health as determined by the outcome of medical history, physical examination and clinical judgment of the investigator. Individuals who completed their local infant EPI schedule through 9 months of age (except MenAfriVac dose). A birth dose of OPV is not required)". **Exclusion criteria:** "history of any meningococcal vaccine administration. Current or previous, confirmed or suspected disease caused by N. meningitidis. Household contact with and/or intimate exposure to an individual with any laboratory confirmed N. meningitidis infection within 60 days of enrolment. History of severe allergic reactions after previous vaccinations or hypersensitivity to any study vaccine component including tetanus, diphtheria and diphtheria toxoid (CRM197). Acute or chronic, clinically significant pulmonary, cardiovascular, metabolic, neurological, hepatic, or renal functional abnormality, as determined by medical history or physical examination. Any confirmed or suspected condition with impaired/altered function of immune system (immunodeficient or autoimmune conditions). Have any bleeding disorder which is considered as a contraindication to intramuscular injection or blood draw. Severe acute malnutrition. A family history of congenital or hereditary immunodeficiency. History of either hepatitis B or hepatitis C virus infection or human immunodeficiency virus infection. Major congenital defects. Chronic administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within three months prior to the study vaccination or planned use throughout the study period. (For corticosteroids, this means prednisone, or equivalent, ≥ 0.5 mg/kg per day. Inhaled, intranasal and topical steroids are allowed). Administration of blood, blood products and/or plasma derivatives or any parenteral immunoglobulin preparation in the past 3 months or planned use throughout the study period. Administration of any vaccine within 28 days prior to enrolment in the study or planned administration of any vaccine within 14 days before or after any study vaccination. Use of any investigational or non-registered drug or vaccine within 30 days prior to the administration of study vaccines or planned during the study. Malaria infection as confirmed by a Rapid Diagnostic Test. Individuals who are close family members of individuals conducting this study. Have experienced a moderate or severe acute infection and/or fever (defined as temperature ≥ 37.5°C) within 3 days prior to enrolment. Have received systemic antibiotic treatment within 3 days prior to enrolment. Non-residence in the study area or intent to move out within six months. Any condition which, in the opinion of the investigator, might pose additional risk to the subject due to participation in the study".
Interventions	**Vaccine type:** meningococcal ACYWX conjugate vaccine NmCV-5 **Experimental:** NmCV-5 with aluminium **Control:** NmCV-5 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** two dose series separated by at least 84 days
Outcomes	**Immunogenicity**: "seroprotective rSBA titres, (time frame: 112 days). Percentage of subjects with rSBA titer ≥ 8 against serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Long term protective rSBA titres, (time frame: 112 days). Percentage of subjects with rSBA titer ≥ 128 against

	serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Rise in rSBA titres, (time frame: 112 days). Percentage of subjects with fourfold rise in rSBA titers against serogroups A, C, W, Y and X at Visits Day 28 and Day 112. Geometric mean of rSBA titres, (time frame: 112 Days). rSBA GMT for serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112" **Safety**: percentage of subjects with at least one severe solicited AE within 7 days after any study vaccination (Days 0-6 and Days 84-90). Solicited local and systemic AEs reported during the 7 days after each vaccination (Days 0-6 and Days 84-90); unsolicited AEs reported during 28 days after each vaccination (Days 0-27 and Days 84-111); AEs leading to premature withdrawal during the entire study period; SAEs reported during the entire study period
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Serum Institute of India Pvt. Ltd) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.

Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Nicholson 2009

Methods	Randomized, phase I trial
Participants	**Sex:** male and female **Age:** >18 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "mentally competent adults, who have signed an informed consent form after having received a detailed explanation of the study protocol. Male or female subjects over 18 years who are either healthy or have a stable medical condition. Able to understand and comply with all study procedures and to complete study diaries. Individuals who can be contacted throughout the study and are available for all study visits. Females should either be using secure contraceptive precautions including a) the oral contraceptive pill, b) condom/barrier contraception c) partner has had a vasectomy, d) be surgically sterilised, or e) post- (defined as at least two years since the last menstrual period)" **Exclusion criteria:** "any clinically significant concurrent illness or unstable medical condition including: malignant tumours, autoimmune illnesses (including rheumatoid arthritis), acute or progressive renal or hepatic pathology, chronic obstructive pulmonary disease requiring oxygen therapy, and any active neurological disorder. Individuals with a history of anaphylaxis or serious reactions to vaccines; hypersensitivity to eggs, chicken protein, chicken feathers, influenzal viral protein, neomycin or polymixin, or products containing mercury. Persons with known immunosuppressive disease or who use systemic immunosuppressive drugs or other drugs listed in section 8 of the British National Formulary (BNF) or chloroquine, gold or penicillamine or other drugs listed in section 10.1.3 of the BNF to suppress a chronic disease process, or have received in the last 6 months radiotherapy or chemotherapy. Subjects who are at high risk of developing illnesses of the immune system. Individuals who are taking immunostimulant therapy or interferon. Individuals who have received blood products or immunoglobulins parenterally during the preceding three months. Women should not be pregnant or lactating. Women who refuse to use a reliable contraceptive method throughout the study. Known or suspected drug abuse. Individuals who have received another vaccine or experimental (investigational) drug in the preceding 4 weeks. Individuals who have previously received H9N2 vaccine. Unable to lead an independent life either physically or mentally. Regularly drink more than 40 units of alcohol weekly. Individuals who have had acute respiratory pathology or infections requiring systemic antibiotic or antiviral therapy during the preceding 7 days (chronic antibiotic therapy for prevention of urinary tract infections is acceptable). Individuals who had a temperature over 38 degrees C in the preceding 3 days. Individuals who, in the opinion of the investigator, have conditions that might complicate interpretation of the study results".
Interventions	**Vaccine type:** whole-virus influenza A/H9N2 vaccine **Experimental:** 1.7, 5, 15 or 45 mcg H9N2 with aluminium. n = 101 **Control:** 1.7, 5, 15 or 45 mcg H9N2. n = 101

	Aluminium concentration: 250 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses of vaccine, 21 days apart, either by intramuscular injection into the deltoid muscle of the non-dominant arm, or by intradermal injection into skin overlying the deltoid
Outcomes	**Immunogenicity:** "geometric mean antibody titres of antibody to influenza H9 by HI and neutralising antibody assays, (time frame: pre-vaccination, 21 days and 42 days). Seroprotective antibody titres to influenza H9 by HI and neutralising antibody, (time frame: pre-vaccination, 21 days and 42 days)" **Safety:** "local and systemic reactogenicity of influenza H9 vaccine, (time frame: within 7 days of vaccination)"
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (karlgnicholson@doctors.org.uk) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further information or data added **Funding:** the funder, the UK Department of Health approved the study design, but took no part in data collection, data analysis, data interpretation, or writing of the report Only vaccine-related serious adverse events were reported.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	A vaccine randomisation list with a randomisation block size of 14 was used to maintain group balance.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Vaccine injections were given by a nurse who was not involved in further assessment. The contents of the syringe were shielded from the volunteers' view.

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00814229), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Nolan 2008

Methods	Randomised, dose comparison, parallel assignment, multicentre trial
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy adults. Men or women aged 18–45 years, and for women of childbearing potential, a negative urine pregnancy test and willingness to use adequate methods of contraception". **Exclusion criteria:** "hypersensitivity to vaccine components (including eggs, chicken protein, neomycin, polymixin, aluminium or thiomersal); immunosuppression; administration of immunoglobulins and/or any blood products (within 90 days); participation in a clinical study or use of an investigational compound (within 90 days); recent vaccination with a registered vaccine (within 30 days); known history of Guillain–Barre´ Syndrome; active neurological disease or current treatment with an anticoagulant".
Interventions	**Vaccine type:** inactivated split-virus influenza A (H5N1) vaccine **Experimental:** 7.5 or 15 mcg H5N1 with aluminium + 50 mcg thiomerosal. n = 200 **Control:** 7.5 or 15 mcg H5N1 + 50 mcg thiomerosal. n = 200 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium phosphate

	Aluminium manufacturer: not reported **Vaccination schedule:** two single doses by intramuscular or deep subcutaneous injection into the deltoid muscle, 21 days apart
Outcomes	**Immunogenicity:** "haemagglutinin inhibition with horse erythrocytes and virus microneutralisation assays were conducted. Blood samples (10mL) were collected immediately before each vaccine dose, on Day 0, Day 21, Day 42, Month 6". **Safety:** "participants were observed for 30min post-vaccination and were given diary cards to record solicited local and systemic adverse events on the evening of vaccination and 6 days thereafter; 21-day diary cards were issued for any unsolicited adverse events. Local reaction measurement cards were used to measure redness, swelling and ecchymosis at the injection site and a digital thermometer was provided for measuring oral temperature. Serious adverse events were to be reported within 12 months of the last dose of study vaccine".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (terry.nolan@mcri.edu.au, t.nolan@unimelb.edu.au) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received, but no further information or data added **Funding:** CSL Limited, Australia

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation code was prepared before each trial commenced using a SAS database (Version 9.1.3, SAS Institute,Inc., Cary, NC).
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Participants and investigators were blinded to treatment allocation in both trials. In the phase I trial there was a visible difference between vaccines with and without adjuvant. Therefore, personnel who prepared and administered the

		study vaccine had no further involvement in the study.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00136331), and all outcomes were reported on. 23 SAEs reported in the phase I trial but data not shown.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Pan 2013

Methods	Prospective, randomised, open label, observer-blind, single-center study
Participants	**Sex:** male and female **Age:** 20-60 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "healthy male and female individuals (age 20 and 60 years at the time of enrolment). The female participants were: (1) of non childbearing potential, i.e. surgically sterilized (defined as having undergone hysterectomy and/or bilateral oophorectomy and/or bilateral salpingectomy); (2) 1 year postmenopausal; or (3) had agreed to practice abstinence or use adequate contraceptive precautions for 30 days prior to the study and for 2 months after study completion. Female participants were also required to have a negative pregnancy test and to not be breastfeeding for the duration of the study". **Exclusion criteria:** "previous known or potential exposure to avian influenza virus or any adjuvated or unadjuvanted H5N1 HA antigen vaccine; vaccination with any influenza vaccine within 6 months prior to study enrolment or at any time during the study period; presence of confirmed or suspected abnormal immune function, immunosuppressive, or immunodeficient condition, including human immunodeficiency virus infection; history of hypersensitivity to vaccines

	or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine; chronic administration (defined as >14 days) of immunosuppressants or other".
Interventions	**Vaccine type:** influenza A/H5N1 vaccine **Experimental:** 3 or 6 mcg H5N1 with aluminium. n = 30 **Control:** 3 or 6 mcg H5N1. n = 30 **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular doses 21 days apart
Outcomes	**Immunogenicity:** serum samples were obtained prior to vaccination, as well as on days 21 and 42 after vaccination. Antibody titers were measured by hemagglutination-inhibition assays and virus neutralization assays. **Safety:** all study participants were monitored for self-reported as well as investigator-assessed local and general adverse events during the 6 weeks following the first vaccination. Complete blood count, serum biochemistry, adverse events, and serious adverse events were evaluated.
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (hsmaids@hotmail.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** supported by National Health Research Institution (NSC grant: 98A1-VCSP01-014) and National Taiwan University Hospital (DOH-101-TD-B-111-001)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open label

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Paoletti 2001

Methods	Phase I, randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** group B Streptococcal type III conjugate vaccine (GBS-III) **Experimental:** 12.5 mcg GBS-III with aluminium. n = 15 **Control:** 12.5 mcg GBS-III. n = 15 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Superfos Biosector a/s, Vedbæk, Denmark **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "GBS type III CPS-specific IgG in sera from all subjects was measured by an enzyme-linked immunosorbent assay". **Safety:** "subjects in the first trial were instructed regarding the completion of a vaccine diary form, how to record oral temperature, and regarding the occurrence of systemic symptoms or injection site symptoms or signs daily for 8 days. For the second trial, the vaccine diary was completed for 3 days unless symptoms existed, which were monitored until resolution. Any volunteer with a temperature of 100.0°F or with grade 3 symptoms within 48 h of vaccination came to the clinic for examination by a study physician".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (lawrance_paoletti@hms.harvard.edu, lpaoletti@channing.harvard.edu) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** supported by NIH-NIAID contracts AI-25152 and AI-75326

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Dropouts <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Dropouts <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Phanuphak 1989

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 19-52 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "none of the subjects had a history of previous rabies vaccination. None were taking antimalarials, corticosteroids or immunosuppressive drugs" **Exclusion criteria:** not reported
Interventions	**Vaccine type:** combined rabies and tetanus vaccine (PVRV) **Experimental:** PVRV with aluminium. n = 15 **Control:** PVRV. n = 15 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses intramuscularly on both deltoid areas on day 0, and one dose on days 7 and 21
Outcomes	**Immunogenicity:** "neutralizing antirabies antibody was determined by rapid fluorescent focus inhibition test. Antitetanus antibody was determined by enzyme-linked immunosorbent assay according, cell-mediated immune response to rabies antigen was determined by antigen-stimulated lymphocyte transformation test". **Safety:** "during each follow-up visit, the subject was asked about any possible side-effects from the preceding vaccination session according to a set of checklist questions".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** contact not found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Pillet 2019

Methods	Randomized, phase II, clinical trial
Participants	**Sex:** male and female **Age:** > 50 years **Ethnicity:** white **Diagnosis:** healthy **Inclusion criteria:** "male and female subjects, 50 years of age or older. BMI of ≥18 and ≤32. Give his/her consent to participate in this study (by signing the informed consent form). In the opinion of the Investigator, competence and willingness to provide written, informed consent for participation after reading the informed consent form. The subject must have adequate opportunity to discuss the study with an Investigator or qualified designee. Healthy as judged by the Investigator or designee and determined by general physical examination, vital signs, clinical laboratory tests, and medical history conducted

	no more than 90 days prior to study vaccine administration. Subjects with a pre-existing chronic disease will be allowed to participate if the disease is stable and, according to the Investigator's judgment, the condition is unlikely confound the results of the study or pose additional risk to the subject by participating in the study. Stable disease is defined as no new onset of exacerbation of pre-existing chronic disease 6 months prior to immunization. Based on Investigator's judgment, a subject with more recent stabilisation of a disease may still be eligible. Comprehension of the study requirements, expressed availability for the required study period, ability to attend scheduled visits, accessible by phone on a consistent basis. If female, have a negative serum pregnancy test result at screening and negative urine pregnancy test on Day 0 prior to immunization. Female of childbearing potential must use an effective method of contraception for 1 month prior to immunization and agrees to continue employing adequate birth control measures for at least 60 days post-immunization. Moreover, she must have no plan to become pregnant for at least 2 months post-immunization. Abstinent subjects should be asked what method(s) they would use, should their circumstances change, and subjects without a well-defined plan should be excluded.The following relationship or methods of contraception are considered to be effective: hormonal contraceptives (e.g., injectable, topical [patch], estrogenic vaginal ring, etc.);Intra-uterine device (IUD) with or without hormonal release;Male partner using a condom plus spermicide or sterilized partner (at least 1 year prior to immunization);History of credible abstinence (self-reported);Heterosexual abstinence at least 60 days post-immunization;Female partner. Non-childbearing females defines as: surgically-sterile (defined as bilateral tubal ligation or hysterectomy performed more than 1 month prior to immunization);Post-menopausal (absence of menses for 24 consecutive months and age consistent with natural cessation of ovulation)". **Exclusion criteria:** "according to Investigator's opinion, presence of significant acute or chronic, uncontrolled medical or neuropsychiatric illness. "Uncontrolled" is defined as:Requiring a new medical or surgical treatment within one month prior to study vaccine administration which would interfere with vaccine evaluation or study completion (subject requiring medical or surgical treatment that would not interfere with evaluation [e.g., minor knee surgery, cataracts, etc.] would be eligible);Requiring a change in medication dosage in one month prior to study vaccine administration due to uncontrolled symptoms or drug toxicity (elective dosage adjustments in stable subjects are acceptable). Any medical or neuropsychiatric condition or any history of excessive alcohol use or drug abuse which, in the Investigator's opinion, would render the subject unable to provide informed consent or unable to provide valid safety observations and reporting. Any autoimmune disease or any confirmed or suspected immunosuppressive condition or immunodeficiency including history of human immunodeficiency virus (HIV) infection, Hepatitis B or C, or the presence of lymphoproliferative disease. Administration of any vaccine (including any other influenza vaccine) within 30 days prior to study enrolment or planned administration within the period from the vaccination up to blood sampling at Day 21 or within 30 days prior to blood sampling at Day 201. Immunization on an emergency basis will be evaluated in a case-by-case by the Investigator. Administration of any adjuvated or investigational influenza vaccine within 1 year prior to study enrolment or planned administration prior to the end of this trial (Day 201). Administration of any 'standard', not adjuvated influenza vaccine (e.g., live attenuated Trivalent Inactivated Influenza Vaccine (TIV) or Quadrivalent Inactivated Vaccine (QIV) vaccine IN or split TIV or QIV vaccine by

	either intra-dermal or intramuscular route) prior to the 30-day exclusion period mentioned above would be acceptable. Use of any investigational or non-registered product within 30 days prior to study enrolment or planned use during the study period. Subjects may not participate in any other investigational or marketed drug study while participating in this study (approximately 7-12 months [depending when subjects undergo screening session and Day 201 visit]). Treatment with systemic glucocorticoids at a dose exceeding 10 mg of prednisone per day, or equivalent for more than 7 consecutive days or for 10 or more days in total, within one month of study vaccine administration, any other cytotoxic or immunosuppressant drug, or any immunoglobulin preparation within 3 months of vaccination. Routine use of standard doses of nasal or inhaled glucocorticoids is allowed. Any significant disorder of coagulation or treatment with warfarin derivatives or heparin. Persons receiving prophylactic anti-platelet medications, e.g., low-dose aspirin (no more than 325 mg/day), and without a clinically apparent bleeding tendency are eligible. Subjects treated with new generation drugs that will not increase risk of intramuscular bleeding (such as clopidogrel) are also eligible. History of allergy to any of the constituents of the VLP quadrivalent study vaccine, Alhydrogel® (aluminium hydroxide) or to the phosphate buffered saline (PBS; used as placebo), or tobacco allergy. History of anaphylaxis reaction to any food, medication or bee sting. Any history of severe asthma (e.g., status asthmatic, hospitalization for asthma control) or recurrent asthma episodes requiring medical attention in the last 3 years (≥ 1 episode/year). Continuous use of anti-histamines in the last 4 weeks prior to immunization or use of anti-histamines 48 hours prior to study immunization. Have a rash, dermatological condition, tattoos, muscle mass, or any other abnormalities at injection site which may interfere with injection site reaction rating. Have received a blood transfusion within 90 days prior to study vaccination. If female, have a positive or doubtful pregnancy test prior to immunization or lactating females. Vital sign abnormalities (systolic blood pressure and/or diastolic blood pressure, heart rate). Although a vital signs measurement is out of the acceptable ranges, a subject may be included in the study based on Investigator's judgment. Presence of any febrile illness (including oral temperature (OT) ≥38.0˚C within 24 hours prior to immunization). Such subjects may be re-evaluated for enrolment after resolution of illness. Cancer or treatment for cancer within 3 years of study vaccine administration. Persons with a history of cancer who are disease-free without treatment for 3 years or more are eligible. Persons with treated and uncomplicated basal cell carcinoma of the skin are eligible. Person with non-treated, non-disseminated local prostate cancer are eligible. Identified as an Investigator or employee of the Investigator or study center with direct involvement in the proposed study, or identified as an immediate family member (i.e., parent, spouse, natural or adopted child) of the Investigator or employee with direct involvement in the proposed study".
Interventions	**Vaccine type:** quadrivalent plant-derived virus like particle influenza vaccine candidate **Experimental:** 15 mcg QVLP with aluminium. n = 75 **Control:** 15 mcg QVLP. n = 75 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Canada Inc, Toronto **Vaccination schedule:** 1 intramuscular injection

Outcomes	**Immunogenicity**: "(time frame: 21 days after injection) geometric mean titers of hemagglutination inhibition antibodies against the vaccine strains on days 0 and 21 and assessed by measuring geometric mean fold rise, seroconversion rate and seroprotection rate. Follow-up serology samples for geometric mean titers will be collected at day 201. Immunogenicity will be evaluated by geometric mean titers of microneutralization antibodies (days 0 and 21) against the vaccine strains and assessed by measuring geometric mean fold rise ans seroconversion rate. Cross-reactivity of antibodies induced by a single dose of the quadrivalent VLP vaccine will also be assessed and measured by HI and microneutralization antibody titers for heterologous influenza strains. This will be assessed using the measure of geometric mean fold rise, seroconversion rate and seroprotection rate (for hemagglutination inhibition only)". **Safety**: "solicited systemic and local reactions (time frame: 21 days after injection). Safety and tolerability will be assessed by the rate, severity and relationship to vaccination of solicited and unsolicited adverse events post-vaccination. A 6-month follow-up period will be performed".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 29/01/2020 authors (landryn@medicago.com) were inquired on the followings: ● blinding of safety assessors ● total number of patients with any adverse events/number of patients analysed at 6 months follow up (please note that we are interested in AEs related and not related to the vaccine). ● total number of patients with any serious adverse events/number of patients analysed at 6 months follow up (please note that we are interested in SAEs related and not related to the vaccine). ● proportion of participants who developed influenza after the vaccination ● GMT values for 15 mcg VLP with and without aluminium No answer received **Funding:** this work was supported in part by awards to BJW and SP from Canadian Institutes of Health Research (http://www.cihr-irsc.gc.ca/e/193. html) and the Ministère de l'Economie, de la Science et de l'Innovation du Quebec (https://www. economie.gouv.qc.ca/accueil/), and by Medicago Inc. Additionally, Medicago Inc. provided support for this study in the form of salaries for NL, JC, and SP. Caprion Biosciences Inc. also provided support for this study in the form of salaries for JFP and BYD. BG has received honoraria from Medicago Inc. The specific roles of these authors are articulated in the 'author contributions' section. As well, Medicago Inc. designed the protocols of the two clinical trials reported in this paper, decided to publish the results, and was involved in the preparation of the manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation is performed using random block permutations obtained with the PLAN procedure of SAS.

Allocation concealment (selection bias)	Low risk	The site will be provided a pre-determined sequence of randomisation number according to the randomisation code.
Blinding of participants and personnel (performance bias)	High risk	Vaccine or placebo is administer by unblinded staff.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind. Immunogenicity assessors were blinded. No further description for safety assessors.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind. Immunogenicity assessors were blinded. No further description for safety assessors.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind. Immunogenicity assessors were blinded. No further description for safety assessors.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02236052), and all outcomes were reported on. However 6-month follow-up period for safety not reported.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Pressler 1982

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18–50 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** H3N2 influenza vaccine **Experimental:** Influenza vaccine with aluminium. n = 94 **Control:** Influenza vaccine. n = 95 **Aluminium concentration:** 2400 mcg/dose **Aluminium type:** aluminium oxide **Aluminium manufacturer:** not reported **Vaccination schedule:** one intramuscular injection

Outcomes	**Immunogenicity**: haemagglutiantion inhibition assay and neuraminidase inhibition test at day 0, 30, 90 and 180 **Safety**: only most severe reactions assessed
Are there other groups in the trial not analysed in this review?	No
Notes	**Funding**: not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Rello 2017

Methods	Phase II, randomised, placebo-controlled, partially blinded, parallel-group, multicenter study
Participants	**Sex:** male and female **Age:** 18–80 years **Ethnicity:** mixed **Diagnosis:** intensive care unit patients on mechanical ventilation **Inclusion criteria:** "male or female patients aged between 18 and 80 years. Patients admitted to an ICU with a need for mechanical ventilation for more than 48 hours at visit 0. At high risk for acquiring infection against P. aeruginosa at visit 0. Patients who, as determined by the investigator, have a high probability of survival for at least 48 hours. In females, either childbearing potential terminated by surgery or 1 year post menopausal, or a negative pregnancy test and the willingness of practicing a reliable methods of contraception. Written informed consent or waiver according to the national regulations". **Exclusion criteria:** "known use of any other investigational or non-registered drug within 30 days prior to IC43 vaccination at Visit 0. Low severity of illness defined by an acute physiology score < 8 at visit 0. Patients < 6 months post organ transplantation. Severe thrombocytopenia or other coagulopathy which in the opinion of the investigator makes the patient unsuitable for intramuscular injection. Pregnancy, lactation. Persons who have been committed involuntarily to an institution, e.g. mental health facility or prison, will not participate in the study".
Interventions	**Vaccine type:** *Pseudomonas aeruginosa* vaccine IC43 **Experimental:** 100 mcg IC43 with aluminium. n = 104 **Control:** 100 mcg IC43. n = 98 **Aluminium concentration:** 1400 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** the authors do not want to disclose this information (email conversation with the author). **Vaccination schedule:** two doses in a 7-day interval
Outcomes	**Immunogenicity:** OprF/I specific IgG antibody titer measured by enzyme-linked immunosorbent assay, (time frame: day 7 and 14) **Safety:** rate of serious adverse events, safety laboratory parameters, systemic tolerability, local tolerability, (time frame: day 7)
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (jrello@crips.es) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received **Funding:** financed by Valneva Austria GmbH

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation lists were generated by a statistician at a contract research organization (Assign Data Management and Biostatistics GmbH, Innsbruck, Austria), using nQuery Advisor.
Allocation concealment (selection bias)	Low risk	Each subject was assigned the next free medication kit (randomisation) number for the corresponding study center in ascending order. Sealed envelopes containing details of treatment allocation (one per patient) were provided to study sites in case emergency un-blinding was required.
Blinding of participants and personnel (performance bias)	High risk	Partially blinded. IC43 without adjuvant (i.e. purified OprF/I fusion protein) was administered un-blinded, and IC43 with adjuvant and placebo were administered blinded because of different appearance and storage conditions.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	A blinded independent Clinical Endpoint Committee reviewed and confirmed diagnosis of P. aeruginosa infections. A blinded independent DSMB evaluated safety data and provided recommendations to the sponsor.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	A blinded independent Clinical Endpoint Committee reviewed and confirmed diagnosis of P. aeruginosa infections. A blinded independent DSMB evaluated safety data and provided recommendations to the sponsor.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Primary immunogenicity and safety analyses were based on the intention to treat population.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Primary immunogenicity and safety analyses were based on the intention to treat population.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Primary immunogenicity and safety analyses were based on the intention to treat population.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00876252), and all outcomes were reported on.
Other bias	Low risk	"After 137 patients had been enrolled, data became available from a phase-I trial, in which both 100 µg IC43 without adjuvant and 100 µg IC43 with adjuvant had a favorable safety profile and a similar immunogenicity profile. Based on the newly available data and following the recommendation of a Data Safety Monitoring Board (DSMB), a protocol amendment was introduced to add a fourth treatment group

		in this phase II study, to assess IC43 100 µg without aluminium hydroxide adjuvant. From this point in the study, randomisation was performed in a 2:1:1:1 ratio (double the number of patients randomised to 100 µg IC43 without adjuvant to achieve equal numbers per group)".

Riddle 2016

Methods	Randomised, single-blind study
Participants	**Sex:** male and female **Age:** 18–50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female volunteers, age of 18 to 50 years (inclusive) at the time of enrollment. Signed informed consent form. Completion and review of comprehension test (achieved >70% accuracy). Available for the required follow-up period and scheduled clinic visits. Women: negative pregnancy test with understanding (through informed consent process) to not become pregnant or or breast feed during the study or within twelve (12) weeks after the last vaccine dose". **Exclusion criteria:** "health condition that, in the opinion of the investigator, may interfere with optimal participation in the study or place the volunteer at increased risk of adverse Events (AEs). Study clinicians, in consultation with the PI, will use clinical judgment on a case by-case basis to assess safety risks under this criterion. The PI will consult with the Research Monitor as appropriate. Clinically significant abnormalities on physical examination. Clinically significant abnormalities on basic laboratory screening. Presence of significant unexplained laboratory abnormalities that, in the opinion of the PI, may potentially confound the analysis of the study results. Regular use of constipation, antacid or anti-diarrheal medications or treatments. Abnormal stool pattern (fewer than 3 stools per week or more than 3 per day) or loose/liquid stools more than occasionally. Use of immunosuppressive drugs such as corticosteroids or chemotherapeutics that may influence antibody development. Women currently nursing. Participation in research involving another investigational product (defined as receipt of investigational product or exposure to invasive investigational device) within 30 days of planned date of first vaccination or anytime throughout the duration of the study. Positive blood test for HBsAg, hepatitis C Virus (HCV), HIV-1. Positive blood test for HLA-B27. Immunosuppressive illness or immunoglobulin deficiency (serum immunoglobulin A level < 7 mg/dL or limit of detection of assay). Family history of congenital or hereditary immunodeficiency. Treatment with immunoglobulins or blood products within 3 months from first candidate vaccine injection. History of microbiologically confirmed Shigella infection. Personal or family history of inflammatory arthritis. Personal or family history of irritable bowel syndrome. Received previous experimental Shigella vaccine or live Shigella challenge. Have had diarrhea while travelling outside the United States or lived for 2 or more months during the past 3 years in a country with potentially higher Shigella infection rates, including Africa, South America, Central America, and Asia (except Japan). Occupation involving handling of Shigella bacteria currently, or in the past 3 years. History of allergy to any vaccine. History of allergy to aluminium hydroxide. Serum immunoglobulin G endpoint titer ≥ 2500 to Shigella Lipopolysaccharide".

Interventions	**Vaccine type:** bioconjugate vaccine against Shigella flexneri 2a (Flexyn2a) **Experimental:** 10 mcg Flexyn2a with alhydrogel. n = 12 **Control:** 10 mcg Flexyn2a. n =12 **Aluminium concentration:** not reported **Aluminium type:** alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** two injections 4 weeks apart
Outcomes	**Immunogenicity:** "evaluation of antigen-specific antibodies between baseline (day 0) and after injection for all groups, (time frame: until day 56). Immunogenicity will be evaluated after the first and second injection and compared to pre-immune levels. Evaluation of antigen-specific antibodies between subjects receiving the candidate vaccine with and without adjuvant, (time frame: until day 56). Immunogenicity will be compared between subjects receiving candidate vaccine with and without adjuvant after the first and second injection". **Safety:** "occurrence and severity of adverse events, (time frame: until day 56). Number and severity of local site injection and general adverse events will be collected and compared between the different arms of the study".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (mark.s.riddle10.mil@mail.mil, veronica.gambillara@lmtbio.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** funded by Wellcome Trust (100527/B/12/Z)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described

Blinding of participants and personnel (performance bias)	High risk	Single-blind. Only volunteers were blinded to their group assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	Single-blind
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Single-blind
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Single-blind
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02388009), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Rosenqvist 1998

Methods	Randomized, phase II trial
Participants	**Sex:** male and female **Age:** not reported **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy adult volunteers. Subjects were laboratory personnel or medical students from Oslo and Bergen". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** group B neisseria meningitidis outer membrane vesicle vaccine (PC) **Experimental:** PC with aluminium **Control:** PC **Aluminium concentration:** 3300 mcg/ml **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Superfos Biosector, Vedbiek, Denmark **Vaccination schedule:** intramuscularly in the upper arm twice, six weeks apart

Outcomes	**Immunogenicity:** antibody response was evaluated by enzyme-linked immunosorbent assay. Bactericidal assay was performed **Safety:** local and systemic side effects were reported on a standard form for the first two days after vaccination
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** contact not found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

RPCEC00000101

Methods	Randomized controlled, double Blind, parallel phase 2 clinical trial
Participants	**Sex:** male and female **Age:** 15-18 months **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "1) Voluntary participation (signature of informed consent by parents). 2) Healthy children of both sexes, aged 15-18 months of age, with history and normal clinical examination. 3) Nutritional evaluation over 10 percentile. 4) History of full primary immunization series against Hib (2-4-6 months)" **Exclusion criteria:** "1) Having received the booster dose against Haemophilus influenzae type b. 2) Submit acute infectious disease at the time of vaccination or within three days prior to the administration of it. 3) Higher temperature at 37oC at the time of the visit to administer the vaccine or from the previous 24 hours. 4) History of chronic diseases eg. Bronchial Asthma, Congenital or acquired Cardiopathies, neuropathies, haematological diseases, etc. 5) Thiomersal allergy or allergy to any components of the vaccine 6) History of convulsive or non-convulsive encephalopathy 7) Underlying immunosuppressive disease, intake of immunosuppressor drugs (including steroids) in the six month period prior to study entry"
Interventions	**Vaccine type:** haemophilus influenzae type b infection **Experimental:** alum phosphate adjuvated Quimi-Hib, 10 mcg **Control:** Quimi-Hib, 10 mcg **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** intramuscular route
Outcomes	**Immunogenicity**: "IgG antiPRP geometric mean titre, Short-term (=0,15 mcg/mL) and long-term (=1 mcg/mL) seroprotection percentages, seroconversion rate (increase in =2 folds in baseline IgG antiPRP antibody titer measured at day 30 after administration of the last vaccine dose)" **Safety**: percentage of local and systemic adverse events within 72 hours, and at 7 and 30 days after vaccination
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 94/04/2020 authors (biomed@cigb.edu.cu) were inquired on the followings: • type of aluminium used • concentration of aluminium injected per dose • aluminium manufacturer • all-cause mortality results • proportion of participants with any AE related and not related to the vaccine • proportion of participants with any SAE related and not related to the vaccine • serological response GMT for each study arm at maximum follow up • proportion of participants who developed influenza after the vaccination • random sequence generation • allocation concealment

	• blinding of participants and personnel • blinding of outcome assessment No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-punished protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

RPCEC00000135

Methods	Multicenter, controlled, randomized, doubled blind clinical trial
Participants	**Sex:** male and female **Age:** 6-7 weeks **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "1.Signing of the written inform consent Act by parents/legal guardians. 2.6 weeks old healthy infants, both sexes, with normal clinical examination and medical history. 3.Nutritional evaluation over 10 percentile

	4.History of no primary immunization series against Hib or DTP. 5. Administration of the scheduled Hepatitis B vaccine dose at birth" **Exclusion criteria:** "1. Administration of other Investigation Medicinal Product (IMP) different to IMP under study in the 30 days previous to administration of the 1st vaccine dose or its intention of use during the period planned for the clinical trial. 2. Acute infectious illness or body temperature over 37?C at the moment of vaccination visit. 3. History of preterm delivery (birth before 36th weeks of pregnancy). 4. Major congenital defects or serious chronic disease. 5. Infants with history of convulsive or non-convulsive encephalopathy. 6. Chronic medication (defined as more than 21 days of treatment) with immune-suppressor or immuno-modulatory drugs prior to administration of the 1st vaccine dose. 7. History of invasive Hib disease. 8. Any suspected or diagnosed immune-suppression or immunodeficiency (congenital or secondary), including HIV infection, based on the medical history and physical examination. 9. History of allergic reactions or allergic disease with probability to be exacerbated by any component of the vaccine under study (including thiomersal allergy). 10. Gammaglobulin administration or other blood –derived components during the period prior to administration of the 1st vaccine dose or the intention of administration during the planned trial period. 11. Maternal history for HIV or HBV infection, based on the medical records"
Interventions	**Vaccine type:** haemophilus influenzae type b infection **Experimental:** alum phosphate adjuvated Quimi-Hib, 10 mcg **Control:** Quimi-Hib, 10 mcg **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** intramuscular route, 6-10-14 weeks schedule
Outcomes	**Immunogenicity**: IgG antiPRP geometric mean titre, Short-term (=0,15 mcg/mL) and long-term (=1 mcg/mL) seroprotection percentages, seroconversion rate (increase in =2 folds in baseline IgG antiPRP antibody titer measured at day 30 after administration of the last vaccine dose) **Safety**: percentage of local and systemic adverse events at 1h, 24h, 48h, 72h, Days 7 and 30 post-vaccination in each dose
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 04/02/2020 authors (biomed@cigb.edu.cu) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any AE related and not related to the vaccine ● proportion of participants with any SAE related and not related to the vaccine ● serological response GMT for each study arm at maximum follow up ● proportion of participants who developed influenza after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment No answer received

	Funding: not reported No results

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind.No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind.No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind.No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind.No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ruckwardt 2021

Methods	Randomised, open-label, phase 1 clinical trial.
Participants	**Sex:** male and female **Age:** adults **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "18 to 50 years of age. Willing and able to complete the informed consent process. Available for clinic visits through 44 weeks after enrollment. Able to provide proof of identity to the satisfaction of the study clinician completing the enrollment process. Willing to donate blood and mucosal samples to be stored and used for future research. In good general

	health without clinically significant medical history. Physical examination and laboratory results without clinically significant findings and a Body Mass Index (BMI) less than or equal to 40 within the 56 days prior to enrollment. Laboratory criteria within 56 days prior to enrollment: White Blood Cell (WBC) and differential either within institutional normal range or accompanied by Principal Investigator (PI) or designee approval. Platelets = 125,000-500,000/mm^3. Hemoglobin within institutional normal range. Creatinine less than or equal to 1.1 x upper limit of normal (ULN). Alanine aminotransferase (ALT) less than or equal to 1.25 x ULN. Negative for HIV infection by an FDA approved method of detection.Criteria applicable to women of childbearing potential: Negative result on a human chorionic gonadotropin pregnancy test on day of enrollment before receiving study product. Agree to use effective means of birth control from at least 21 days before enrollment through 4 weeks after the last injection." **Exclusion criteria:** breast-feeding or planning to become pregnant through 4 weeks after the last injection.Subject has received any of the following: More than 10 days of systemic immunosuppressive medications or cytotoxic medications within the 4 weeks prior to enrollment or any within the 14 days prior to enrollment.Blood products within 16 weeks prior to enrollment. Live attenuated vaccines within 4 weeks prior to enrollment. Inactivated vaccines within 2 weeks prior to enrollment. Investigational research agents within 4 weeks prior to enrollment or planning to receive investigational products while on the study. Current allergen immunotherapy with antigen injections, unless on maintenance schedule. Current anti-tuberculosis(TB) prophylaxis or therapy.Subject has any of the following: Serious reactions to vaccines that preclude receipt of study injections as determined by the investigator. Hereditary angioedema, acquired angioedema, or idiopathic forms of angioedema. Asthma that is not well controlled. Diabetes mellitus (type I or II), with the exception of gestational diabetes. Thyroid disease that is not well controlled. Hypertension that is not well controlled. Evidence of autoimmune disease or immunodeficiency. Idiopathic urticaria within the past year. Bleeding disorder diagnosed by a doctor (e.g. factor deficiency, coagulopathy, or platelet disorder requiring special precautions) or significant bruising or bleeding difficulties with IM injections or blood draws. Malignancy that is active or history of malignancy that is likely to recur during the study. Seizure disorder other than: 1) febrile seizures, 2) seizures secondary to alcohol withdrawal more than 3 years ago, or 3) seizures that have not required treatment within the last 3 years. Asplenia, functional asplenia or any condition resulting in the absence or removal of the spleen. Psychiatric condition that precludes compliance with the protocol; past or present psychoses; or within 5 years prior to enrollment, a history of suicide plan or attempt. Any medical, psychiatric, or social condition that, in the judgment of the investigator, is a contraindication to protocol participation or impairs a subject s ability to give informed consent.
Interventions	**Vaccine type:** respiratory syncytial virus prefusion F subunit vaccine DS-Cav1 **Experimental:** 50 µg, 150 µg, or 500 µg DS-Cav1 with aluminium hydroxide **Control:** 50 µg, 150 µg, or 500 µg DS-Cav1 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Brenntag-Biosector **Vaccination schedule:** administered in the deltoid muscle by needle and syringe at weeks 0 and 12.

Outcomes	**Safety**: number of subjects reporting local and systemic reactogenicity signs and symptoms within 7 days of the first, second or any dose of the product administration of DS-Cav1 alone or with alum adjuvant. Number of subjects with abnormal laboratory measures of safety up tp day 308. Number of subjects with one or more unsolicited non-serious adverse events (time frame: day 0 through day 28 after product administration). Number of subjects with serious adverse events (time frame: day 0 through day 308). Number of subjects who had respiratory syncytial virus infection following product sdministration (time frame: day 0 through day 308). **Immunogenicity**: respiratory syncytial virus subtype A (RSV A) antigen-specific neutralizing antibody geometric mean titers at 4 weeks after the first or second product administration of DS-Cav1 alone or with alum adjuvant.
Are there other groups in the trial not analysed in this review?	No
Notes	**Funding:** this work was supported with intramural funding from the NIAID. The funding source had no involvement with the study design, execution, or preparation of the manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomized using permuted block randomisation with block sizes of 4 or 6, chosen at random.
Allocation concealment (selection bias)	Low risk	The protocol statistician generated the randomisation sequence which was programmed into the database. Participants were assigned to a group upon electronic enrolment by a study clinician.
Blinding of participants and personnel (performance bias)	High risk	Open-label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	Open-label
Blinding of outcome assessment (detection bias) All-cause mortality	High risk	Open-label
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Open-label
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03049488), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Sheldon 2016

Methods	Phase 1, first-in-human, dose-escalating, placebo-controlled, randomised, observer-blinded study
Participants	**Sex:** male and female **Age:** 50-85 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "evidence of a personally signed and dated informed consent document. Healthy male and female adults aged 50 to 85 years at enrollment as determined by medical history, physical examination, and the clinical judgment of the investigator to be eligible for the study. Subjects with preexisting chronic medical conditions determined to be stable may be included. Male subjects who, in the opinion of the investigator, are biologically capable of fathering children, and who are sexually active with women of childbearing potential must agree to use a highly effective method of contraception from the time of informed consent through at least 28 days after the last dose of investigational product (through Visit 9 at Month 7). Female subjects who are not of childbearing potential (ie, meet at least one of the following criteria): have undergone hysterectomy or bilateral oophorectomy;Have medically confirmed ovarian failure orAre medically confirmed to be postmenopausal. Availability for the entire duration of the study, and able to comply with scheduled visits, treatment plan, laboratory tests, and other study procedures including completion of the electronic diary (eDiary) from day 1 to day 7 following each vaccination. Ability to be contacted by telephone during study participation". **Exclusion criteria:** "previous administration of an investigational C. difficile vaccine or C. difficile monoclonal antibody therapy. Proven or suspected prior episode of CDAD. Unstable chronic medical condition or disease requiring significant change in therapy or hospitalization for worsening disease within 12 weeks before receipt of study vaccine. Serious chronic medical disorders including metastatic malignancy, severe chronic obstructive pulmonary disease (COPD) , end stage renal disease with or without dialysis, clinically unstable cardiac disease, or any other disorder that in the investigator's opinion precludes the subject from participating in the study. Donation of blood volume of 250 mL or greater, or donation of plasma within 3 months prior to enrollment or during the conduct of the study. Bleeding diathesis or condition associated with prolonged bleeding time that may contraindicate intramuscular injection or blood draw including subjects taking anticoagulant, antiplatelet and/or antithrombotic agents except for low dose daily aspirin (≤325mg per day) within 30 days before enrollment through completion of Visit 9 (Month 7). Any

	contraindication to vaccination or vaccine components. "Immunocompromised persons or subjects currently on immunosuppressive therapy or with a history of immunosuppressive therapy, including chemotherapy agents for treatment of diseases including, but not limited to cancer, inflammatory bowel disease or autoimmune disease. Recent history (within the past 6 months) of long term (7 days or longer) systemic corticosteroid use." Subjects who received oral or parenteral antibiotics within 1 month before enrollment. Topical antibiotics are allowed. Receipt of blood products or immunoglobulins (including monoclonal antibodies) within 6 months before enrollment through conclusion of the study. Participation in other investigational or interventional studies within 30 days before the current study begins and/or during study participation. Participation in purely observational studies is acceptable. Subjects who are investigational site staff members or relatives of those site staff members, or subjects who are Pfizer employees directly involved in the conduct of the trial. Females of childbearing potential; males of childbearing potential not using highly effective contraception or not agreeing to continue highly effective contraception for at least 28 days after last dose of investigational product (through Visit 9 at Month 7). Females receiving exogenous estrogen therapy. Residence in a nursing home, long-term care facility, requirement for semiskilled nursing care or assisted living. An ambulatory subject who lives in an autonomous manner in a retirement home or village is eligible for the trial. Any abnormality in screening hematology and/or blood chemistry laboratory values according to the toxicity grading scale. Subjects with stable Grade 1 abnormalities for hemoglobin, leukocyte count, and platelets may be considered eligible at the discretion of the investigator. A positive screening test or known infection with human immunodeficiency virus (HIV), hepatitis B virus (HBV) and/or hepatitis C virus (HCV). Other severe acute or chronic medical or psychiatric condition or laboratory abnormality that may increase the risk associated with study participation or investigational product administration or may interfere with the interpretation of study results and, in the judgment of the investigator, would make the subject inappropriate for entry into this study".
Interventions	**Vaccine type:** *Clostridium difficile* vaccine **Experimental:** 50, 100 or 200 mcg *Clostridium difficile* vaccine with aluminium. n = 73 **Control:** 50, 100 or 200 mcg *Clostridium difficile* vaccine. n = 72 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag **Vaccination schedule:** 3-dose vaccine regimen administered on day 1, Month 1,and Month 6 by intramuscular injection into the upper deltoid muscle
Outcomes	**Immunogenicity:** "neutralizing antibody levels at Month 2, (time frame: month 2). Neutralizing antibody levels after each vaccination dose, (time frame: baseline, day 14, month 1, day 37, month 6, day 187, month 7). Number of subjects in each treatment group with 4 and higher fold-rises in neutralizing antibody levels after each vaccination dose, (time frame: baseline, day 14, month 1, day 37, month 6, day 187, month 7). Number of subjects in each treatment group with neutralizing antibody levels ≥ a specified threshold after each vaccination dose., (time frame: baseline, day 14, month 1, day 37, month 6, day 187, month 7)". **Safety:** "number subjects reporting local reactions (pain, erythema, and induration) and their severity, as self reported on eDiaries for 7 days following each vaccination, (time frame: 7 days). Number of subjects reporting systemic

	reactions (fever, vomiting, diarrhea, headache, fatigue, new or worsening muscle pain, and new or worsening joint pain) and their severity, as self reported on eDiaries for 7 days following each vaccination, (time frame: 7 days). Number of subjects reporting adverse events from day 1 (at the time of vaccination) up to 28 days post dose 3 (visit 9, month 7) and serious adverse events throughout the study period, (time frame: 7 months (adverse events) 12 months (serious adverse events)). Number of subjects with abnormal hematology and blood chemistry laboratory assessments after each vaccination dose, (time frame: day 3, day 14, month 1, day 37, month 6, day 187)"
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2028 authors (nicholas.kitchin@pfizer.com) were inquired on the followings: ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received **Funding:** sponsored by Pfizer Inc. The sponsor was responsible for the design of the study, monitoring, analysis, and interpretation of the data, as well as the decision to submit the data for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Subjects were randomised via an interactive response technology system.
Allocation concealment (selection bias)	Low risk	Allocation to investigational product was via an interactive response technology system
Blinding of participants and personnel (performance bias)	Low risk	Observer-blinded study. The study staff dispensing and administering the vaccine were unblinded. The subject, all other study site members, including the investigator and study site coordinator,remained blinded.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Those performing the clinical assessments including the investigator and study site coordinator, and the laboratory staff performing the immunogenicity testing, remained blinded.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Those performing the clinical assessments including the investigator and study site coordinator, and the laboratory staff performing the immunogenicity testing, remained blinded.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Those performing the clinical assessments including the investigator and study site coordinator, and the laboratory staff performing the immunogenicity testing, remained blinded.

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01706367), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Tapia 2021

Methods	Phase 2, observer-blind, randomized, controlled Study
Participants	**Sex:** male and female **Age:** 12 months to 16 months (child) **Ethnicity:** African **Diagnosis:** healthy **Inclusion criteria:** "male and female children between 12 months and 16 months old inclusive (minimum 365 days of age and maximum 16 months plus 29 days of age); for whom parent(s)/legal guardian(s) have given written informed consent after the nature of the study has been explained according to local regulatory requirements; who the investigator believes that their parent(s)/ guardian(s) will be available for all the subject visits and would comply with the requirements of the protocol (e.g., timely reporting of adverse events, availability for study site visits and home visits); Individuals in good health as determined by the outcome of medical history, physical examination and clinical judgment of the investigator.Individuals who completed their local infant EPI schedule through 9 months of age (except MenAfriVac dose). A birth dose of OPV is not required)". **Exclusion criteria:** "history of any meningococcal vaccine administration. Current or previous, confirmed or suspected disease caused by N. meningitidis. Household contact with and/or intimate exposure to an individual with any laboratory confirmed N. meningitidis infection within 60 days of enrolment. History of severe allergic reactions after previous vaccinations or hypersensitivity to any study vaccine component including tetanus, diphtheria and diphtheria toxoid (CRM197). Acute or chronic, clinically significant pulmonary, cardiovascular, metabolic, neurological, hepatic, or renal functional abnormality, as determined by medical history or physical examination. Any confirmed or suspected condition with impaired/altered function of immune system (immunodeficient or autoimmune conditions). Have any bleeding disorder which is considered as a contraindication to intramuscular injection or blood draw. Severe acute malnutrition. A family history of congenital or hereditary immunodeficiency. History of either hepatitis B or hepatitis C virus infection or human immunodeficiency virus infection. Major congenital defects. Chronic

	administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within three months prior to the study vaccination or planned use throughout the study period. (For corticosteroids, this means prednisone, or equivalent, ≥ 0.5 mg/kg per day. Inhaled, intranasal and topical steroids are allowed). Administration of blood, blood products and/or plasma derivatives or any parenteral immunoglobulin preparation in the past 3 months or planned use throughout the study period. Administration of any vaccine within 28 days prior to enrolment in the study or planned administration of any vaccine within 14 days before or after any study vaccination. Use of any investigational or non-registered drug or vaccine within 30 days prior to the administration of study vaccines or planned during the study. Malaria infection as confirmed by a Rapid Diagnostic Test. Individuals who are close family members of individuals conducting this study. Have experienced a moderate or severe acute infection and/or fever (defined as temperature ≥ 37.5°C) within 3 days prior to enrolment. Have received systemic antibiotic treatment within 3 days prior to enrolment. Non-residence in the study area or intent to move out within six months. Any condition which, in the opinion of the investigator, might pose additional risk to the subject due to participation in the study.
Interventions	**Vaccine type:** meningococcal ACYWX conjugate vaccine **Experimental:** NmCV-5 with aluminium phosphate **Control:** NmCV-5 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** intramuscularly in two doses 12 weeks apart
Outcomes	**Safety**: percentage of subjects with at least one severe solicited AE within 7 days after any study vaccination. Solicited local and systemic AEs reported during the 7 days after each vaccination. Unsolicited AEs reported during 28 days after each vaccination. AEs leading to premature withdrawal during the entire study period. SAEs reported during the entire study period. **Immunogenicity**: percentage of subjects with rSBA titer ≥ 8 against serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Percentage of subjects with rSBA titer ≥ 128 against serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Percentage of subjects with fourfold rise in rSBA titers against serogroups A, C, W, Y and X at Visits Day 28 and Day 112. Geometric mean of rSBA titres.
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 23/08/2021 authors (drpsk@seruminstitute.com') were inquired on the followings: ● aluminium manufacturer ● proportion of participants per group who developed meningitis ● total number of participants per group who experienced any adverse events Answer received but no data provided. **Funding:** U.K. Foreign, Commonwealth, and Development Office.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03295318), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Vandenberghe 2016

Methods	Phase 2, 90-week, randomised, double-blind, placebo-controlled, multicenter study
Participants	**Sex:** male and female **Age:** <85 years **Ethnicity:** mixed **Diagnosis:** mild Alzheimer's disease **Inclusion criteria:** "male and/or female patients below 85 years of age (inclusive) Diagnosis of mild Alzheimer's Disease Mini-Mental State Examination (MMSE) 20 to 26 (inclusive) at screening, untreated or on stable dose of cholinesterase inhibitor or memantine over the last 4 weeks prior to

	clinical assessments" **Exclusion criteria:** "previously participated in an AD vaccine study and received active treatment. History or presence of an active autoimmune disease. History or presence of seizure disorder. Presence of significant coronary heart disease and/or cerebrovascular disease. Presence of other neurodegenerative disease and/or psychiatric disorders (with the exception of successfully treated depression). Advanced, severe, progressive or unstable disease that might interfere with the safety, tolerability and pharmacodynamic assessments of the patient Other protocol-defined inclusion/exclusion criteria may apply".
Interventions	**Vaccine type:** CAD106 Alzheimer's disease vaccine **Experimental:** 450 mcg CAD106 with aluminium. n = 21 **Control:** 450 mcg CAD106. n = 16 **Aluminium concentration:** 450 mcg/dose **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** seven intramuscular injections at weeks 0, 6, 12, 24, 36, 48, and 60
Outcomes	**Immunogenicity:** "amyloid beta-specific and Qβ carrier-specific antibody response to CAD106 in serum and cerebrospinal fluid, (time frame: screening and through the end of the study to week 90). Amyloid beta-specific and Qβ carrier-specific T-cell response to CAD106 using peripheral blood mononuclear cells, (time frame: screening and at week 8). Changes over time of the concentrations of disease related markers (Aβ1-40 and Aβ1-42 in plasma; Aβ1-40, Aβ1-42, total-tau, phospho-tau in CSF, or other markers) in patients with mild Alzheimer's disease receiving CAD106 compared to placebo, (time frame: screening and through the end of the study to week 90)". **Safety:** "physical/neurological examinations, electrocardiogram, vital signs, standard and special laboratory evaluations, magnetic resonance imaging, adverse events / serious adverse events monitoring, (time frame: screening and through the end of the study to week 90)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (rik.vandenberghe@uz.kuleuven.ac.be) were inquired on the followings: ● aluminium type ● aluminium manufacturer ● allocation concealment ● random sequence generation ● blinding of immunogenicity outcome assessment ● number of participants with mortality, serious and non-serious adverse event related and not related to the vaccine per study arm No answer received **Funding:** sponsored by Novartis Pharma AG, Basel, Switzerland

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Patients were centrally randomised.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Double-blind. CAD106 and adjuvant doses were prepared by an unblinded pharmacist and administered by an unblinded study nurse. Patients, investigators, and the site personnel were masked to the treatment assigned.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	An independent unblinded data monitoring committee monitored safety data including adverse events, vital signs, weight, electrocardiograms, and safety laboratory measurements regularly throughout the study. Image analysis blinded to treatment allocation was performed by Avid Radiopharmaceuticals Inc.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	An independent unblinded data monitoring committee monitored safety data including adverse events, vital signs, weight, electrocardiograms, and safety laboratory measurements regularly throughout the study. Image analysis blinded to treatment allocation was performed by Avid Radiopharmaceuticals Inc.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No drop outs
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT01097096), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Verdijk 2013

Methods	Randomized, controlled, double blind, phase I 'proof-of-concept' trial
Participants	**Sex:** male **Age:** 18-49 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "age 18 to 49, inclusive at the time of enrolment; in good health as determined by the outcome of medical history, physical examination screening/baseline labs and clinical judgment of the investigator. Male must have received poliovirus vaccinations with OPV according to the Polish National Immunization Program as a child based on vaccination cards or information given by subjects during interview with investigator. Preferred: number (and date) of poliovirus vaccinations known. Willingness and ability to adhere to the study regimen. Having a signed informed consent form" **Exclusion criteria:** "IPV or OPV booster dose after the age of 12 years. Positive for HIV, Hepatitis B or Hepatitis C. Known or suspected allergy against any of the vaccine components History of unusual or severe reactions to any previous vaccination. Known or suspected disease or use of medication that may influence the immune system. Known or suspected immune deficiency. Systemic treatment with corticosteroids within one month before screening. Administration of plasma (including immunoglobulins) or blood products three months prior to the study. Blood donation within one month before screening. Any vaccination within three months before screening and during the study until the last visit. History of any neurological disorder including epilepsy or febrile seizures. Evidence of excessive alcohol use or drug use. Any infectious disease at the time of screening and/or inclusion. Participation in another clinical trial within three months before screening. Abnormal pre-treatment laboratory parameters which are clinically relevant according to the investigator. Bleeding disorders or the usage of anticoagulants".
Interventions	**Vaccine type:** inactivated poliovirusvirus vaccine Sabin-IPV **Experimental:** Sabin-IPV with aluminium. n = 15 **Control:** Sabin-IPV. n = 15 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag, Denmark **Vaccination schedule:** a single intramuscular injection
Outcomes	**Immunogencity:** "immunogenicity of Sabin-IPV and adjuvated Sabin-IPV was evaluated by virus neutralizing titers against type , type 2 and type 3 poliovirusviruses before and after vaccination" **Safety:** "subjects were observed for 30 min after vaccination. Subjects were required to complete a diary on the vaccination day and during the first three days following the day of vaccination in which solicited and unsolicited adverse events were to be recorded. When adverse reactions were still present after these three days, subjects were to continue to monitor these reactions until they had disappeared. Blood samples were taken before vaccination and on day 28 (range 25–31) after vaccination. A follow-up phone call was made 6 months after vaccination to assess whether any serious adverse event had occurred in the period in between".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (pauline.verdijk@intravacc.nl, Wilfried.bakker@intravacc.nl) were inquired on the followings: ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received **Funding:** supported by the World Health Organization using funds provided by a Grant from the Bill and Melinda Gates Foundation. The World Health Organization was involved in the design of the clinical trial.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The statistician prepared a randomisation list with a block size of three using SAS version 9.1.3. Randomization numbers were printed on the label of each vial with IMP.
Allocation concealment (selection bias)	Low risk	Randomization was performed using SAS version 9.1.3. SAS procedure plan was used to randomly assign treatment to subject numbers with an appropriate block size. The output of this procedure was used to prepare the randomisation list, the emergency list and emergency envelopes. Subject numbers were printed on the label of each vial with IMP. Each subject receives a vial with his own subject number.
Blinding of participants and personnel (performance bias)	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.

Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	No drop outs. Six months after vaccination, a follow-up call was performed to verify whether any serious adverse events had occurred. Two subjects could not be contacted - not stated which arm.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01708720), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Verdijk 2014

Methods	Double-blind, randomised,controlled, phase I/IIa dose-escalation trial in the target population
Participants	**Sex:** male and female **Age:** infants aged 56–63 days **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "age 8 weeks (56-63 days) at the time of first vaccination. Infants in good general health, eligible to be vaccinated according to the Polish national vaccination program. The same health criteria apply as used in well-baby clinics when a child receives a vaccination, e.g. also children with a small increase in temperature (< 38.5° C) or with a common cold (runny nose etc) are seen as children with normal health. The parent(s)/legally representative(s) have to be willing and able to allow their child to participate in the trial according to the described procedures. Presence of a signed informed consent in which the parent(s)/legally representative(s) have given written informed consent after receiving oral and written information (signature from

	one parent in case of single-parent family)". **Exclusion criteria:** "any IPV or OPV dose. Known or suspected allergy against any of the vaccine components. History of unusual or severe reactions to any previous vaccination administration of plasma (including immunoglobulins) or blood products prior and during the study. Any vaccination less than 14 days before or after each vaccination with the IMP History of any neurological disorder including epilepsy or febrile seizures. Any infectious disease at the time of screening and/or inclusion that might interfere with the results of the study. Present evidence of serious disease(s) demanding immunosuppressive medical treatment, like cytostatics and prednisolones, that might interfere with the results of the study Any known or suspected primary or secondary immunodeficiency. Communication problems interfering in the study realization according to the judgment of the investigator Bleeding disorders and use of anticoagulants. Premature birth (<37 weeks). Participation in another clinical trial".
Interventions	**Vaccine type:** inactivated poliovirusvirus vaccine Sabin-IPV **Experimental:** low, middle or high dose sIPV with aluminium. n = 60 **Control:** low, middle or high dose sIPV. n = 60 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag, Denmark **Vaccination schedule:** one dose of 0.5 mL intramuscularly in the right thigh. Infants participating in the trial also received the regular booster dose at 15–18 months with wIPV
Outcomes	**Immunogenicity:** level of virus neutralizing titers in serum, (time frame: 28 days after last vaccination) **Safety:** number of adverse reactions following vaccination, (time frame: 5 days or until adverse reactions have resolved). A follow-up telephone call was made 6 months after the last vaccination with the IMP to assess whether any serious adverse event had occurred during that period.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (pauline.verdijk@intravacc.nl, Wilfried.bakker@intravacc.nl) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of participants and personnel ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received

	Funding: supported by the World Health Organization using funds provided by a grant from the Bill and Melinda Gates Foundation. The World Health Organization was involved in the design of the clinical trial.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Inclusion and randomisation was performed in three steps according to a randomisation list prepared by the statistician.
Allocation concealment (selection bias)	Low risk	At the sites vaccines with lowest tracking/identification number was given to the patients in chronological order of recruitment date. The investigator and other site staff were blinded during the conduct of the study
Blinding of participants and personnel (performance bias)	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%

Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01709071), and all outcomes were reported on.
Other bias	Low risk	Serious adverse events occurred but were not assigned per arm in the publication. Data added after contact with authors.

Vernacchio 2002

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** infants aged 15-36 months **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** healthy toddlers aged 15–36 months **Exclusion criteria:** not reported
Interventions	**Vaccine type:** pneumococcal–CRM197 conjugate vaccine (PCV9) **Experimental:** 10 or 25 mcg PCV9 with aluminium. n = 38 **Control:** 10 or 25 mcg PCV9. n = 40 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses 60 days apart
Outcomes	**Immunogenicity:** "serum IgG antibodies to pneumococcal capsular polysaccharides were measured by enzyme-linked immunosorbent assay. Anti-CRM197 IgG concentrations were measured by standard enzyme-linked immunosorbent assay. Microneutralization assay was performed". **Safety:** "following each immunization, parents were asked to complete a 7-day diary card and record daily a single rectal temperature, local site reactions and systemic side effects. On day 7 after each immunization (range 7–10 days), study subjects were examined by trained research nurses and the recorded safety data were reviewed with the parents. Blood samples were obtained before the first immunization and on days 7 and 30 following the second immunization for safety and immunogenicity analyses. The safety evaluation consisted of monitoring a complete blood count, aspartate aminotransferase, alanine aminotransferase and creatinine concentrations".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (louis.vernacchio@childrens.harvard.edu) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed

	● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs more than 5%
Selective reporting (reporting bias)	High risk	No pre-published protocol. Four serious adverse events occurred but were not assigned by arm because considered as not related to the vaccine.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Warrel 1984

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 9–69 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** at very low risk of rabies infection **Exclusion criteria:** not reported
Interventions	**Vaccine type:** HDCSV rabies vaccine **Experimental:** HDCSV with aluminium. n = 20 **Control:** HDCSV without aluminium. n = 20 **Aluminium concentration:** 980 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** unclear
Outcomes	**Immunogenicity** and **safety**
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Funding**: the Wellcome Trust of Great Britain

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.

Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Westritschnig 2014

Methods	Phase I, randomised, placebo-controlled, observer blinded, multicenter trial
Participants	**Sex:** male and female **Age:** 18–65 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "written informed consent obtained prior to study entry; healthy adults aged between 18 and 65 years; no clinically relevant pathological findings in any of the investigations at the Screening visit. Minor deviations of laboratory values from the normal range may be accepted, if judged by the investigator to have no clinical relevance. In female subjects either childbearing potential terminated by surgery or a negative serum pregnancy test during screening and the willingness not to become pregnant during the entire study period by practicing reliable methods of contraception". **Exclusion criteria:** "history of autoimmune diseases and malignancies. Active or passive vaccination 4 weeks before and during the entire study protocol. Use of any other investigational or non-registered drug or vaccine in addition to the study vaccine during the study period or within 30 days preceding the first dose of study vaccine. History of severe hypersensitivity reactions and anaphylaxis. Known hypersensitivity or allergic reactions to one of the components of the vaccine. Clinically significant diseases as judged by the investigator. Immunodeficiency due to immunosuppressive therapy. A family history of congenital or hereditary immunodeficiency".
Interventions	**Vaccine type:** pseudomonas aeruginosa hybrid outer membrane protein OprF/I vaccine (IC43) **Experimental:** 100 mcg IC43 with aluminium. n = 33 **Control:** 100 mcg IC43. n = 32 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injections 7 days apart
Outcomes	**Immunogenicity:** "measurement of fOprF/I-specific IgG antibodies by enzyme-linked immunosorbent assay at day 14 (7 days after second vaccination), measurement of antibody avidity on days 7 and 14, measurement of anti-histidine antibodies on days 7, 14, 90, and 180. Functional antibodies; avidity of OprF/I-specific IgG antibodies; anti-histidine IgG antibodies; peripheral blood mononuclear cell analysis were performed". **Safety:** rate of serious adverse events during vaccination period until 6 months after first vaccination, safety laboratory parameters at intervals up to day 180,

	systemic and local tolerability at intervals up to day 180
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (bernd.jilma@meduniwien.ac.at) were inquired on the followings: ● Final concentration of aluminium injected ● Aluminum manufacturer ● Number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no further information or data added **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomised list was created using nQuery Advisor by a statistician at a contract research organization (Assign Data Management and Biostatistics Gmb).
Allocation concealment (selection bias)	Low risk	Subjects were randomly allocated. Each subject was assigned the next free medication kit (randomisation) number for the corresponding study center in ascending order.
Blinding of participants and personnel (performance bias)	High risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.

Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00778388), and all outcomes were reported on.
Other bias	Low risk	"Major protocol deviations included: incorrect study treatment dosage (21 subjects [65.6%] in the IC43 50 µg group, and 2 subjects [6.1%] in the IC43 100 µg with adjuvant group); forbidden concomitant medication (1 subject [3.0%] in the IC43 100 µg with adjuvant group); treatment kit number issue (1 subject [3.1%] in the IC43 100 µg without adjuvant group); and second vaccination not given (2 subjects [6.1%] in the IC43 200 µg group, 1 subject [3.1%] in the IC43 50 µg group, and 1 subject [3.0%] in the placebo group)".

Wressnigg 2013

Methods	Double blind, randomised, dose-escalation, phase 1/2 study
Participants	**Sex:** male and female **Age:** 18–70 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subject is 18 to 70 years old at the time of screening. Subject has an understanding of the study, agrees to its provisions, and gives written informed consent prior to study entry. Subject is generally healthy, as determined by the investigator's clinical judgment through collection of medical history and the performance of a physical examination. If female of childbearing potential, presents with a negative urine pregnancy test, and agrees to employ adequate birth control measures for the duration of the study. Additional inclusion criterion for seropositive subjects in Section 2 only: subject is seropositive for Borrelia burgdorferi sensu lato antibodies at study entry". **Exclusion criteria:** "subject has a physician-diagnosed chronic illness related to Lyme borreliosis (LB) or active LB. Subject has been treated for LB with antibiotics within 3 months of study entry. Subject had a tick bite within 3 weeks prior to screening or first vaccination. Subject has a history or active infection with Babesia microti (babesiosis) or Anaplasma. phagocytophilum (ehrlichiosis). Subject currently has or has a history of significant cardiovascular, respiratory (including asthma), metabolic, neurological, hepatic, rheumatic, autoimmune, hematological, gastrointestinal or renal disorder. Subject has clinically significant abnormal laboratory values at screening. Subject currently has or has a history of immunodeficiency.Subject tests positive for human immunodeficiency virus, hepatitis B surface antigen or hepatitis C virus. Subject has a disease or is currently undergoing a form of treatment or was undergoing a form of treatment within 30 days prior to study entry that could be expected to influence immune response. Subject has a history of anaphylaxis or severe allergic reactions. Subject has received any live vaccine within 4 weeks or inactivated vaccine within 2 weeks prior to vaccination in this study. Subject is pregnant or lactating at the time of study enrollment. Additional exclusion criterion for subjects in Section 1 and seronegative subjects in Section 2: subject is seropositive for Borrelia burgdorferi sensu lato antibodies at study entry".

Interventions	**Vaccine type:** multivalent OspA vaccine against Lyme borreliosis **Experimental:** 30, 60 or 90 mcg OspA with aluminium. n = 151 **Control:** 30, 60 or 90 mcg OspA. n = 149 **Aluminium concentration:** 1000 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** three intramuscular immunisations with intervals of 28 days, and a booster 9–12 months after the first dose
Outcomes	**Immunogenicity:** "antibody response to the vaccine, (time frame: 28 days after the third vaccination (= day 85)). Antibody response, (time frame: at baseline, 28 days after each vaccination (i.e. days 29, 57 and 85), 180 and 270 days after the first vaccination (day 181, day 271) and 180 days after the booster vaccination (day 361 or day 451 - 546)). Fold increase in antibody titer compared to baseline, (time frame: 28 days after each vaccination, 180 and 270 days after the first vaccination and 180 days after the booster vaccination). Seroconversion rate (at least 4-fold increase of each rOspA type-specific Immunoglobulin G titer) as compared to baseline, (time frame: 28 days after each vaccination, 180 and 270 days after the first vaccination and 180 days after the booster vaccination)". **Safety:** "frequency and severity of injection site and systemic reactions; (time frame: within 7 days after each vaccination (i.e. days 8, 36 and 64))"
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (noel_barrett@baxter.com) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** funded by Baxter. Baxter employees were responsible for study design, data collection, data analysis, data interpretation, writing of the report, and the decision to submit for publication

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was done centrally via an electronic data capture system.
Allocation concealment (selection bias)	Unclear risk	Participants were recruited sequentially and randomly assigned to one of six study groups in equal ratios.
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further information

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01504347), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Wu 2017

Methods	Phase I/II randomised trial
Participants	**Sex:** male and female **Age:** 20–60 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subject within the age limit range, is free of obvious health problems as judged by the investigator before entering the study. Female subjects must be either of non-childbearing potential, or one year post-menopausal; or, if of childbearing potential, must be abstinent or have used adequate contraceptive precautions for 30 days prior to receiving the study vaccination, have a negative pregnancy test and must agree to continue such precautions for two months after completion of the vaccination series. Subject is willing and able to comply with all required study visits and follow-up required by this protocol. Subject must provide written informed consent or the Subjects' legal representative must understand and consent to the procedure". **Exclusion criteria:** "subjects with previous known or potential exposure to avian influenza virus H7N9 HA antigen and any other avian influenza including H5N1. Subjects who have had seasonal influenza vaccine within 6 months prior to enrolment or who will have seasonal influenza vaccine at any time during the study period. Subjects administered with any other vaccine during the period within 6 weeks before the first administration of study vaccine. Subjects with

	confirmed or suspected abnormal immune function, immunosuppressive or immunodeficient condition, including human immunodeficiency virus (HIV) infection, based on medical history and physical examination. Subjects with a history of hypersensitivity to vaccines or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Subjects with a history of inflammatory or degenerative neurological disease. Subjects receiving chronic administration of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. Inhaled and topical steroids are allowed. Known HIV, hepatitis B (HBsAg) or hepatitis C seropositivity. Subjects, judged by the investigator, with any clinically significant medical illness including acute pulmonary, cardiovascular, hepatic or renal functional abnormality, as determined by medical history, physical examination and/or laboratory screening tests. The subjects would not be suitable for the vaccination if the screening laboratory tests show any of the follows: ALT or AST > upper limit of normal range; serum creatinine > upper limit of normal range; any significant laboratory abnormality as judged by the investigator. Subjects receiving administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or at any time during the study. Subjects with acute disease at the time of enrolment. Use of any investigational or non-registered product other than the study vaccine within 30 days prior to the first vaccination, or planned use of the abovementioned product during the entire study period. Psychiatric, addictive, or any disorder, which may compromise the ability to give a truly informed consent for participation of this study or adequate compliance. Any clinically relevant disease and/or abnormal laboratory findings (, which, in the opinion of the Investigator, may either put the patient at risk because of participation in the study, or influence the results of the study, or the patient's ability to participate in the study. Breast feeding or pregnant women or refusal to submit to a urine test to rule out pregnancy prior to enrolment and during the study".
Interventions	**Vaccine type:** inactivated cell culture-derived H7N9 influenza vaccine **Experimental:** 15 or 30 mcg H7N9 vaccine with aluminium. n = 100 **Control:** 15 or 30 mcg H7N9. n = 100 **Aluminium concentration:** 300 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 doses at 21-day intervals
Outcomes	**Immunogenicity:** "HI and neutralizing antibody titers, (time frame: 1 year from the start of study enrolment). Change of antibody titers from baseline to day 22 and day 43. Geometric mean titer of both serum HI titers and neutralizing antibody titers, pre-vaccination and day 22 and day 43 post-vaccination" **Safety:** "adverse event, (time frame: 1 year from the start of study enrolment). Percentage, intensity and relationship to vaccination of solicited local and general signs and symptoms during a 7-day follow-up period after the first and second dose of administered vaccine. Percentage, intensity and relationship to vaccination of unsolicited local and general signs and symptoms during a 21-day follow-up period after each administered vaccine. Occurrence of overall adverse events and serious adverse events during the entire period of study"
Are there other groups in the trial not analysed in this review?	No

Notes	**Contacts with authors:** on17/12/2019 authors (changsc@ntu.edu.tw) were inquired on the followings: ● aluminium manufacturer ● allocation concealment ● random sequence generation ● blinding of participants and personnel ● blinding of outcome assessment ● number of participants with any serious and non-serious adverse event related and not related to the vaccine. Please kindly provide this number out of total number of participants analysed per arm No answer received **Funding:** funded by Medigen Vaccine Biologics Co., Taiwan

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized using the blocking factor to ensure that the balance between the 4 study groups was maintained.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT02436928), and all outcomes were reported on. One serious adverse event was not assigned to any arm.

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Wuorimaa 2001

Methods	Randomised clinical study
Participants	**Sex:** male and female **Age:** infant **Ethnicity:** Finnish and Israeli **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** 11-valent pneumococcal conjugate vaccine **Experimental:** 11-valent pneumococcal conjugate vaccine with aluminium **Control:** 11-valent pneumococcal conjugate vaccine **Aluminium concentration:** 300 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 4 doses at the age of 2, 4, 6, and 12 months
Outcomes	**Immunogenicity**: "functional activity of antibodies to five capsular serotypes (4, 6B, 14, 19F, and 23F). OPA of antibodies against serotypes 4, 6B, 14, 19F, and 23F were analysed. Serum samples obtained 1 month after the third dose (at 7 months), and before (at 12 months) and 1 month after (at 13 months) the fourth dose was used. Enzyme immunoassay to measure IgG antibody concentrations to capsular polysaccharides was used. The OPA of sera was determined with the viable cell killing assay".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on14/12/2018 authors (helena.kayhty@thl.fi) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● Final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no information added. **Funding:** Sanofi Pasteur provided the financial support

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further information
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Zhu 2009

Methods	Randomized, double-blind, placebo-controlled, clinical trial
Participants	**Sex:** male and female **Age:** 3 years to 90 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female aged 3 and older. Be able to show legal identity card for the sake of recruitment. Volunteers or their guardians are able to understand and sign the informed consent". **Exclusion criteria:** "cases, cured cases and close contact of influenza A (H1N1) virus. Women of pregnancy, lactation or about to be pregnant in 60 days. Subject that has a medical history of any of the following: allergic history, or allergic to any ingredient of vaccine, such as egg, egg protein, etc. Serious

	adverse reactions to vaccines such as anaphylaxis, hives, respiratory difficulty, angioedema, or abdominal pain. Autoimmune disease or immunodeficiency. Asthma that is unstable or required emergent care, hospitalization or intubation during the past two years or that required the use of oral or intravenous corticosteroids. Diabetes mellitus (type I or II), with the exception of gestational diabetes. History of thyroidectomy or thyroid disease that required medication within the past 12 months. Serious angioedema episodes within the previous 3 years or requiring medication in the previous two years. Bleeding disorder diagnosed by a doctor (e.g. factor deficiency, coagulopathy, or platelet disorder requiring special precautions) or significant bruising or bleeding difficulties with IM injections or blood draws. Active malignancy or treated malignancy for which there is not reasonable assurance of sustained cure or malignancy that is likely to recur during the period of study. Seizure disorder other than: febrile seizures under the age of two years oldSeizures secondary to alcohol withdrawal more than 3 years ago, or a singular seizure not requiring treatment within the last 3 years. Asplenia, functional asplenia or any condition resulting in the absence or removal of the spleen Guillain-Barre Syndrome. Any history of immunosuppressive medications or cytotoxic medications or inhaled corticosteroids within the past six months (with the exception of corticosteroid nasal spray for allergic rhinitis or topical corticosteroids for an acute uncomplicated dermatitis). History of any blood products or seasonal influenza vaccine administration within 3 months before the dosing. Administration of any other investigational research agents within 30 days before the dosing. Administration of any live attenuated vaccine within 30 days before the dosing. Administration of subunit or inactivated vaccines, e.g., pneumococcal vaccine, or allergy treatment with antigen injections, within 14 days before the dosing. Be receiving anti-TB prophylaxis or therapy currently. Axillary temperature > 37.0 centigrade at the time of dosing. Psychiatric condition that precludes compliance with the protocol: past or present psychoses; past or present bipolar disorder requiring therapy that has not been well controlled on medication for the past two years; disorder requiring lithium; suicidal ideation occurring within five years prior to enrollment. Any medical, psychiatric, social condition, occupational reason or other responsibility that, in the judgment of the investigator, is a contraindication to protocol participation or impairs a volunteer's ability to give informed consent".
Interventions	**Vaccine type:** Influenza A (H1N1) vaccine **Experimental:** 15 or 30 mcg H1N1 with aluminium. n = 770 **Control:** 15 or 30 mcg H1N1. n = 770 **Aluminium concentration:** 1200 mcg/ml **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** two vaccinations intramuscularly, in the deltoid muscle, 21 days apart
Outcomes	**Immunogenicity:** proportions of subjects with an increase in the hemagglutination- inhibition titer by a factor of 4 or more on day 21 after the first dose and on day 35 (14 days after the second dose) **Safety:** presence of any systemic reaction or injection-site reaction 21 days after the first dose and 14 days after the second dose
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 14/12/2018 authors (jszfc@vip.sina.com) were inquired on the followings: ● procedure for blinding of participants and personnel ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** sponsored by Hualan Biological Bacterin Company Serious adverse events are reported only as vaccine-related.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation schedule was created with the use of SAS software (version 9.1). The block size was defined by an independent statistician at Southeast University (Nanjing, China).Randomization-code numbers were assigned to subjects in chronological order by the investigator.
Allocation concealment (selection bias)	Low risk	Randomization-code numbers were assigned to subjects in chronological order by the investigator.
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	All samples were assayed in a blinded manner, in duplicate, and were double-checked by two investigators at the Chinese National Institute for the Control of Pharmaceuticals and Biological Products.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs >5%

Incomplete outcome data (attrition bias) All-cause mortality	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00975572), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Zhu 2013

Methods	Randomised, double-blind, placebo-controlled, phase 2 trial
Participants	**Sex:** male and female **Age:** 6-36 months old **Ethnicity:** Chinese **Diagnosis:** healthy **Inclusion criteria:** "healthy children aged from 6 to 36 months old as established by medical history and clinical examination. The subjects' guardians are able to understand and sign the informed consent. Had never received the vaccine against EV71. Subjects who can and will comply with the requirements of the protocol. Subjects with temperature <=37.0°C on axillary setting". **Exclusion criteria:** "subject who has a medical history of HFMD. <= 37 weeks gestation. Subjects with a birth weight <2.5 kg. Subject that has a medical history of any of the following: allergic history, or allergic to any ingredient of vaccine. Family history of seizures or progressive neurological disease. Family history of congenital or hereditary immunodeficiency. Severe malnutrition or dysgenopathy. Major congenital defects or serious chronic illness, including perinatal brain damage. Autoimmune disease. Bleeding disorder diagnosed by a doctor or significant bruising or bleeding difficulties with IM injections or blood draws. Any acute infections in last 7 days. Any prior administration of immunodepressant or corticosteroids in last month. Any prior administration of blood products in last 3 month. Any prior administration of other research medicines in last 1 month. Any prior administration of attenuated live vaccine in last 28 days. Any prior administration of subunit or inactivated vaccines in last 14 days, such as pneumococcal vaccine. Under the anti-TB prevention or therapy. Any condition that in the opinion of the investigator, may interfere with the evaluation of study objectives. Exclusion Criteria for the second dose: had any Grade 3 or Grade 4 adverse reaction within 7 days after first dose. Any situation meet the exclusion criteria stated in the exclusion criteria for first dose. Had any SAE related to first dose during the following-up of first dose. Any condition that in the opinion of the investigator, or IRB".
Interventions	**Vaccine type:** enterovirus 71 vaccine (EV71) **Experimental:** 640 U EV71 with aluminium. n = 240 **Control:** 640 U EV71. n = 240 **Aluminium concentration:** 180 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses given on day 0 and day 28

Outcomes	**Immunogenicity:** "geometric mean titre of anti-EV71 antibodies in serum after first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination). Seroconversion rate of anti-EV71 antibodies in serum after first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination). The clinical abnormality of hematological examination, blood biochemical test and urinalysis after first vaccination in children, (time frame: 3 days after first vaccination) and the second vaccination, (time frame: 3 days after second vaccination). The persistence of immunogenicity of the EV71vaccine after two doses in children and infants,(time frame: 6 months after blood collection at day 56)". **Safety:** "frequency of systemic and local adverse reactions after the first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination). Frequency of adverse events and any SAE after the first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination)"
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (hen54@x263.net, wangjz@nicpbp.org.cn) were inquired on the followings: ● aluminium manufacturer. ● blinding of immunogenicity outcome assessment. ● number of participants who got enterovirus infection after the vaccination. Email address wangjz@nicpbp.org.cn not valid **Funding:** The National Science and Technology Major Project (2011ZX10004-902) of the Chinese Ministry of Science and Technology, China's 12–5 National Major Infectious Disease Program (2012ZX10002-001), and Beijing Vigoo Biological. One of the sponsors of the study (Beijing Vigoo Biological) was involved in trial design, but none of the sponsors were involved in data collection, statistical analysis, interpretation, or writing of the report.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	"We stratified eligible participants by age (6–11 months [infants] and 12–36 months [children]) and assigned each a sequential number according to their sequence of enrolment. Participants were then randomly allocated (1:1:1:1:1) to receive either placebo, EV71 vaccine of 160 U with adjuvant, 320 U with adjuvant, 640 U with adjuvant, or 640 U without adjuvant, according to a blocked (block size of 20) randomisation list generated by the statistical analysis program SAS".
Allocation concealment (selection bias)	Low risk	The four vaccine formulations and placebo had identical packaging and were blindly labelled with the sequential numbers, which were the only identifiers.

Blinding of participants and personnel (performance bias)	Low risk	Double-blind. Individuals involved in randomisation and masking did not participate in any other process of the trial. All participants, their guardians, and investigators were masked to the treatment allocation
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	An independent data and safety monitoring committee was set up to oversee the recording of serious adverse events and provide final decisions of causality between SAEs and vaccination before unbinding. The committee also supervised the updating of the safety data within 28 days after each injection.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	An independent data and safety monitoring committee was set up to oversee the recording of serious adverse events and provide final decisions of causality between SAEs and vaccination before unbinding. The committee also supervised the updating of the safety data within 28 days after each injection.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01399853), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Footnotes

Characteristics of excluded studies

August 2017

Reason for exclusion	Saline placebo without a vaccine

Bravo 2020

Reason for exclusion	Wrong comparator

Brown 2014

Reason for exclusion	Placebo contained aluminium

De Carvalho 2009

Reason for exclusion	Placebo contained aluminium

De Carvalho 2010

Reason for exclusion	Placebo contained aluminium

Garland 2009

Reason for exclusion	Wrong comparator

Garland 2013

Reason for exclusion	Wrong comparator

Giannini 2006

Reason for exclusion	Placebo contained aluminium

GlaxoSmithKline Vaccine HPV-007 Study 2009

Reason for exclusion	Placebo contained aluminium

Harper 2004

Reason for exclusion	Placebo contained aluminium

Lindgren 2017

Reason for exclusion	Wrong comparator

Naud 2014

Reason for exclusion	Placebo contained aluminium

NCT00851643

Reason for exclusion	Wrong comparator

Paavonen 2007

Reason for exclusion	Wrong comparator

Rivera 2018

Reason for exclusion	Wrong comparator

Roteli-Martins 2012

Reason for exclusion	Placebo contained aluminium

Saez-Llorens 2020

Reason for exclusion	Wrong comparator

Skinner 2014

Reason for exclusion	Placebo contained aluminium

Sow 2013

Reason for exclusion	Placebo contained aluminium

Sullivan 2018

Reason for exclusion	Placebo alone without vaccine

Szarewski 2009

Reason for exclusion	Wrong comparator

Teixeira 2017

Reason for exclusion	Placebo contained aluminium

Van Damme 2014

Reason for exclusion	Placebo contained aluminium

Villa 2006

Reason for exclusion	Placebo contained aluminium

Wheeler 2016

Reason for exclusion	Placebo contained aluminium

Zhu 2014

Reason for exclusion	Placebo contained aluminium

Footnotes

Characteristics of studies awaiting classification

Chua 2019

Methods	Phase 1A randomised controlled trial
Participants	HIV-1 uninfected healthy volunteers aged 18 to 45 years
Interventions	GP120-CD4 chimeric subunit vaccine
Outcomes	Safety and immunogenicity
Notes	Palcebo not described. Author (jchua@ihv.umaryland.edu) contacted

EUCTR2004-001896-21-FI

Methods	A Phase II, Randomized, Observer Blind, Multi-Center, Active Controlled Study
Participants	Healthy Children Aged 12-59 Months
Interventions	MenACWY with or without adjuvant
Outcomes	Safety and immunogenicity
Notes	Adjuvant not described. Company (Novartis) contacted

Lin 2006

Methods	Phase I randomised controlled trial
Participants	Healthy individuals aged 18–60 years in and around Beijing
Interventions	Two doses of placebo or an inactivated whole-virion influenza A (H5N1) vaccine with 1·25 μg, 2·5 μg, 5 μg, or 10 μg haemagglutinin per dose with aluminium hydroxide adjuvant on day 0 and 28
Outcomes	Safey and immunogenicity
Notes	Placebo content not described. Authors (yinwd@sinovac.com) contacted 3 times but no answer received

NCT00001053

Methods	A Phase I Safety and Immunogenicity Study
Participants	Healthy Volunteers
Interventions	HIV p17/p24:Ty-VLP (virus-like particles) vaccine with or without aluminium
Outcomes	No information
Notes	Company (protocol@tech-res.com) contacted about more information on study arms

NCT00100724

Methods	A Phase 2 Multi-Center, Randomized Dose-Finding Trial
Participants	Healthy Volunteers
Interventions	rPA102 vaccine given intramuscularly at 0 and 4 weeks over 2 dose ranges and 4 adjuvant levels.
Outcomes	Safety and immunogenicity.
Notes	Adjuvants not described.

NCT01594320

Methods	Phase 1 Randomized, Observer-Blinded, Dose-Ranging Study.
Participants	Healthy Young Adults.
Interventions	Monovalent A/Indonesia/05/05 (H5N1) Virus-Like Particle Avian Influenza Antigen (Recombinant) With and Without Adjuvant 1
Outcomes	Immunogenicity and Safety
Notes	Adjuvants and arms not described. Company (ir@novavax.com) contacted

NCT01596725

Methods	Phase 1 randomised, observer-blinded, dose-ranging study
Participants	Healthy young adults
Interventions	Monovalent A/Indonesia/05/05 (H5N1) Virus-Like Particle Avian Influenza Antigen (Recombinant) With and Without Adjuvant 2
Outcomes	Immunogenicity and Safety
Notes	Adjuvants and arms not described. Company (ir@novavax.com) contacted

NCT03572062

Methods	A phase1/2, placebo-controlled, randomised, observer-blind, dose-fining, first-in-human study
Participants	65 years to 85 years (Older Adult)
Interventions	62 subjects will be randomised 1:1 to receive a dose of high dose adjuvated RSV vaccine or placebo followed by a second dose 2 months later
Outcomes	Immunogenicity and Safety
Notes	Adjuvant not described. Company (ClinicalTrials.gov_Inquiries@pfizer.com) contacted

Footnotes

Characteristics of ongoing studies

NCT03049488

Study name	Dose, Safety, Tolerability and Immunogenicity of a Stabilized Prefusion RSV F Subunit Protein Vaccine, VRC-RSVRGP084-00-VP (DS-Cav1), Alone or With Alum Adjuvant, in Healthy Adults
Methods	Phase I Randomized, Open-Label Clinical Trial
Participants	Healthy Adults
Interventions	VRC-RSVRGP084-00-VP (DS-Cav1) was developed by VRC, NIAID and is composed of the respiratory syncytial virus (RSV) fusion (F) glycoprotein ectodomain. Adjuvant is an aluminium hydroxide suspension (alum) provided in a sterile, pyrogenfree suspension at a concentration of 5 mg/mL in 3 mL glass vials filled to 0.7 +/- 0.10 mL. The alum dose is 500 mcg ● Group: 1; Subjects: 15; Dose: 50mcg; Day 0: DS-Cav1; Week 12 [1]: DS-Cav1 ● Group: 2; Subjects: 15; Dose: 50mcg; Day 0: DS-Cav1 + alum; Week 12 [1]: DS-Cav1 + alum ● Group: 3; Subjects: 15; Dose: 150mcg; Day 0: DS-Cav1; Week 12 [1]: DS-Cav1 ● Group: 4; Subjects: 15; Dose: 150mcg; Day 0: DS-Cav1 + alum; Week 12 [1]: DS-Cav1 + alum ● Group: 5; Subjects: 15; Dose: 500mcg; Day 0: DS-Cav;1 Week 12 [1]: DS-Cav1 ● Group: 6; Subjects: 15; Dose: 500mcg; Day 0: DS-Cav1 + alum; Week 12 [1]: DS-Cav1 + alum
Outcomes	Safet and immunogenicity
Starting date	February 9, 2017
Contact information	Grace L Chen, M.D. National Institute of Allergy and Infectious Diseases (NIAID)
Notes	

NCT03776994

Study name	Venezuelan Equine Encephalitis Monovalent Virus-Like Particle Vaccine (VEEV)
Methods	Phase 1 dose escalation radnomised study
Participants	Healthy volunteers between 18 and 50 years
Interventions	Non-adjuvanted and adjuvated monovalent VEE VLP Vaccine administered via intramuscular injection at escalating doses of 2 µg, 10 µg, and 20 µg as a 2-dose primary series (Day 0, Day 28) with a Day 140 booster dose
Outcomes	Safety and immunogenicity
Starting date	December 17, 2018
Contact information	SRI Biosciences Clinical Trials Unit, Plymouth, Michigan, United States, 48170. Gloria Jen, MS 734-527-4201 gloria.jen@sri.com
Notes	

NCT03879603

Study name	VRC 313: A Trivalent Virus-like Particle (VLP) Encephalitis Vaccine (WEVEE) in Healthy Adults
Methods	Phase 1, randomised, open-label, dose-escalation study
Participants	Healthy Adults
Interventions	WEVEE vaccine alone (Group 1, Group 3, and Group 5) or, WEVEE vaccine mixed with alum (Group 2, Group 4, and Group 6, respectively)
Outcomes	Safety and immunogenicity
Starting date	March 19, 2019
Contact information	Julie E Ledgerwood, DO, National Institute of Allergy and Infectious Diseases (NIAID)
Notes	

NCT04009824

Study name	Evaluating the Safety and Immunogenicity of AGS-v PLUS, a Universal Mosquito-Borne Disease and Mosquito Control Vaccine
Methods	Randomized, Double-Blind, Placebo-Controlled, Single-Center, Phase 1 Study
Participants	Healthy Volunteers
Interventions	Participants in Group 1 will receive placebo on Days 1 and 22. Participants in Group 2 will receive unadjuvanted AGS-v PLUS vaccine on Days 1 and 22. Participants in Group 3 will receive Montanide ISA-51 adjuvated AGS-v PLUS vaccine on Day 1 and placebo on Day 22. Participants in Group 4 will receive Montanide ISA-51 adjuvated AGS-v PLUS vaccine on Days 1 and 22. Participants in Group 5 will receive Alhydrogel® adjuvated AGS-v PLUS vaccine on Days 1 and 22.
Outcomes	Safety and immunogenicity
Starting date	July 5, 2019
Contact information	Matthew B. Laurens, MD, MPH, Matthew J. Memoli, MD, MS, National Institute of Allergy and Infectious Diseases (NIAID)
Notes	

NCT04056117

Study name	Safety and Immunogenicity of a Shigella-Tetravalent Bioconjugate Vaccine: A Phase 1/2 Randomized Controlled and Age Descending Study Including Dose Finding in 9 Month Old Infants
Methods	Phase 1/2 trial is a multicentre, randomised, controlled, double blind, with two steps; a step 1 with a safety cohort and a step 2 with a dose-finding cohort
Participants	8 Months to 50 Years (Child, Adult)
Interventions	Tetravalent bioconjugate candidate vaccine Shigella4V. Adults will receive a 2 dose schedule, children and infants will receive a 3 dose schedule. For each vaccine dose, formulation with and without Aluminium adjuvant will be tested.

Outcomes	Safety and immunogenicity
Starting date	August 14, 2019
Contact information	Mainga Hamaluba, MD Mainga Hamaluba <MHamaluba@kemri-wellcome.org>
Notes	

NCT04563533

Study name	Clinical Trial of Quadrivalent Recombinant Norovirus Vaccine (Pichia Pastoris)
Methods	A single-center, randomised, blinded, multi-cohort, placebo-controlled trial
Participants	6 Weeks and older (Child, Adult, Older Adult)
Interventions	Tetravalent recombinant Norovirus vaccine There are 10 cohorts, randomly divided into experimental group (low-dose or high-dose), placebo group (containing aluminium), and the random ratio of each entry stage is 2:1. Among them, infants and the elderly will be added a placebo group without aluminium adjuvant, test group (low dose or high dose), placebo group (aluminium), placebo (aluminum-free), the random ratio is 2:1:1; Dose expansion stage: It is planned to enroll 240 subjects in Rong'an County. The elderly, infants, and infants will be divided into 3 cohorts, and each age group will be randomly divided into low-dose or high-dose groups, with a random ratio of 1:1.
Outcomes	Safety and immunogenicity
Starting date	September 24, 2020
Contact information	huangt24@126.com Liuzhou Center for Disease Control and Prevention
Notes	

Footnotes

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Manuscript page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Manuscript page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Manuscript page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Manuscript page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Manuscript page 4; Supplementary page 50; Protocol
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Manuscript page 5; Supplementary Table 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary Table 1;
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 5,6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Manuscript page 5,6; Protocol
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Manuscript page 5,6; Protocol; Supplementary Page 51
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Protocol; Supplementary Page 50
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 6; Protocol
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Manuscript page 6; Supplementary Page 50; Protocol
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Manuscript page 4; Supplementary page 50; Protocol
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing	Supplementary page 50

For peer review only

Section and Topic	Item #	Checklist item	Location where item is reported
		summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Supplementary Table2,3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Manuscript page 4,5; Supplementary page 50; Protocol
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Manuscript page 5,6; 7 Protocol
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Manuscript page 4,5; Supplementary page 50; Protocol
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Manuscript page 5; Supplementary page 50; Protocol
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Manuscript page 7; Protocol
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Manuscript Page 7; Supplementary Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Appendix 1
Study characteristics	17	Cite each included study and present its characteristics.	Appendix 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Manuscript Page 11 to 17
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	n.a.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Manuscript Page 11 to 17 Supplementary material for subgroup analysis
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Manuscript Page 11 to 17; Supplementary material for subgroup analysis
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Manuscript Page 11 to 17; Supplementary material for sensitivity analysis
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Manuscript Page 11 to 17; Supplementary material

For peer review only

Section and Topic	Item #	Checklist item	Location where item is reported
			for sensitivity analysis
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Manuscript page 18 to 22
	23b	Discuss any limitations of the evidence included in the review.	Manuscript page 18 to 22
	23c	Discuss any limitations of the review processes used.	Manuscript page 18 to 22
	23d	Discuss implications of the results for practice, policy, and future research.	Manuscript page 18 to 22
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	abstract
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	abstract
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Supplementary page 51
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Manuscript page 23,24
Competing interests	26	Declare any competing interests of review authors.	Manuscript page 25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	n.a.

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Aluminium adjuvants versus placebo or no intervention in vaccine randomised clinical trials. A systematic review with meta-analysis and Trial Sequential Analysis

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Aluminium adjuvants versus placebo or no intervention in vaccine randomised clinical trials. A systematic review with meta-analysis and Trial Sequential Analysis

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Abstract

Objectives: To assess the benefits and harms of aluminium adjuvants versus placebo or no intervention in randomised clinical trials in relation to human vaccine development.

Design: Systematic review with meta-analysis and Trial Sequential Analysis (TSA) assessing the certainty of evidence with Grading of Recommendations Assessment, Development, and Evaluation (GRADE).

Data sources: We searched CENTRAL, MEDLINE, Embase, LILACS, BIOSIS, Science Citation Index Expanded, and Conference Proceedings Citation Index – Science until June 29, 2021, and Chinese databases until September 2021.

Eligibility criteria: Randomised clinical trials irrespective of type, status, and language of publication, with trial participants of any sex, age, ethnicity, diagnosis, comorbidity, and country of residence.

Data extraction and synthesis: Two independent reviewers extracted data and assessed risk of bias with Cochrane's RoB tool 1. Dichotomous data were analysed as risk ratios (RR) and continuous data as mean differences. We explored both fixed-effect and random-effects models, with 95% confidence interval (CI). Heterogeneity was quantified with I^2 statistic. GRADE assessed the certainty of the evidence.

Results: We included 102 randomised clinical trials (26 457 participants). Aluminium adjuvants versus placebo or no intervention may have no effect on serious adverse events (RR 1.18, 95% CI 0.97 to 1.43; very low certainty). TSA showed that the required information size was not reached. Aluminium adjuvants versus placebo or no intervention may have no effect on all-cause mortality (RR 1.02, 95% CI 0.74 to 1.41; very low certainty). TSA showed that the required information size was not reached. Only two trials reported on the proportion of participants with relevant disease being vaccinated against. No trial reported on quality of life. Aluminium adjuvants versus placebo or no intervention may increase adverse events (RR 1.13, 95% CI 1.07 to 1.20; very low certainty). TSA showed that there was sufficient information to confirm a harmful effect of aluminium adjuvants. We found no or little evidence of a difference between aluminium adjuvants versus placebo or no intervention

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when assessing geometric mean titres or concentrations. Aluminium adjuvants versus placebo or no intervention may have no effect on participants' seroprotection (RR 0.95, 95% CI 0.77 to 1.18; very low certainty).

Conclusions: Based on evidence at very low certainty, we were unable to identify benefits of aluminium adjuvants, which may be associated with adverse events considered non-serious.

Protocol: PROSPERO CRD42017083013.

Strengths and limitations of this study

- We seem to be the first to assess the benefits and harms of aluminium adjuvants by conducting a systematic review comparing aluminium adjuvants versus placebo or no intervention in combination with all types of vaccines.
- We included 102 randomised clinical trials from a comprehensive search with no language limitations or restrictions on outcomes reported in the trials, type of aluminium adjuvant or type of vaccine.
- The certainty of evidence is very low and this makes it difficult to draw firm conclusions.

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Introduction

Vaccination is one of the major triumphs of modern medicine.[1 2] Vaccination prevents infectious diseases, and the worldwide eradication of the deadly smallpox and the restriction of diseases such as poliovirus, measles, and tetanus can largely be ascribed to the numerous successful mass vaccination programs launched since the 1960s.[1 2] Similar successes are presently unfolding during the COVID-19 pandemic.[3] It is estimated that after the roll out of the COVID-19 vaccines, nearly half a million deaths have been averted in one year thanks to the vaccination programs.[4] In addition to its intended effect, a vaccine may be accompanied by one or more harmful effects upon administration. Harms may be considered non-serious (e.g. mild, transient headache) or serious (e.g. causing hospitalisation or death) and they may appear shortly after vaccine administration (e.g. pain at the injection site) or belated (e.g. autoimmune responses).

The human papilloma virus (HPV) vaccination program was launched in the USA in 2006 in order to prevent HPV infection, one of the causes of cervical cancer and the second most common cancer in women.[5] Despite the HPV vaccines have been assessed for efficacy (immunogenicity) in clinical trials, and approved on the basis of their ability to raise a potent immune response against HPV and their ability to prevent persistent HPV infections[6] concerns have been raised about adverse events possibly related to the HPV vaccines formulation.[7 8] Both the national vaccine adverse events reporting system in the USA and the European Union, the European Medical Agency (EMA), have received reports on a high number of adverse events suspected to relate to the HPV vaccination.[8] However, no scientific evidence for an association was found.[9] Several observational studies also failed to identify associations with clinical diagnoses.[10-14] However, reasons to oppose these findings have been proposed.[7 15 16]

Vaccine toxicity, efficacy, and effectiveness may originate from, or depend on, a plethora of factors, including the vaccine components (e.g. the antigen itself, the excipient, or the adjuvant); interaction between different vaccine components; vaccine manufacture; overall vaccine composition; route of administration; dose; and number of

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booster vaccinations.[17] Aluminium salts are widely used adjuvants, as aluminium phosphate, aluminium hydroxide, aluminium potassium sulphate, and amorphous aluminium hydroxyphosphate sulphate.[18] They have been the standard adjuvant in vaccines against diphtheria, tetanus, and pertussis, haemophilus influenza type B, pneumococcus conjugates, hepatitis A, and hepatitis B.[19] More recently, aluminium was co-formulated with vaccines against HPV in the form of Adjuvant System 04 (aluminum hydroxide and monophosphoryl lipid A), aluminium hydroxide or amorphous aluminium hydroxyphosphate sulfate as well as in the worlds most used COVID-19 vaccines CoronaVac®[20] and Sinopharm BIBP COVID-19 vaccine[21] in the form of aluminium hydroxide.

The mechanism of action of aluminium, like for most adjuvants, is only partially understood. Its biological or physiological role is unknown. While aluminium is generally considered safe and is regularly ingested in food and water, it can be toxic based on the concentration, chemical form, and the environment.[22] Aluminium seems to have an impact on the immune system, which has rendered it useful as a vaccine adjuvant.[19 23] Aluminium is believed to exert its adjuvant effects by stimulating Th2-type cell responses and antibody production through B cells activation,[24 25] by activating the complement system, and by recruiting immune cells to the site of injection.[24 26 27] At the injection site, aluminium promotes antigen uptake by specialised antigen-presenting immune cells, termed dendritic cells, as well as dendritic cell maturation.[23 28 29] The consensus within the scientific community is that aluminium affects antigen uptake, induces danger signals, recruits various types of immune cells and elicits Th2 responses.[30]

One previous attempt to assess the effects of aluminium adjuvants with a review was undertaken in 2004 by Jefferson and colleagues.[31] The review covered existing evidence of adverse events to the aluminium-containing DTP vaccine, but it did not assess benefits.[31] In 2018, Lin and colleagues conducted the first meta-analysis on the efficacy of aluminium salts as an adjuvant for prepandemic influenza vaccines.[32] Their results showed inferior seroprotection after aluminium-adjuvanted H5N1 vaccines compared with that conferred by non-adjuvanted counterparts, however, these findings only related to the prepandemic influenza vaccines. New adjuvants are being introduced continuously and FDA and WHO do not require genotoxicity or cardiotoxicity studies of new

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aluminium adjuvants.[33] The theory that aluminium adjuvant is responsible for symptoms following specific vaccine formulation is impossible to refute or prove based on the data from current clinical trials. For example, aluminium adjuvant has been administered to both experimental and control group in the vast majority of randomised clinical trials on HPV vaccines, thus masking its potentially harmful effect.[34] Aluminium adjuvants, new or old, should be evaluated for benefits and harms on their own merits. While the consequences of adding aluminium to vaccines have been discussed broadly, no systematic review has been conducted to assess the effect of aluminium adjuvants across different types of vaccines.

The objectives of this review are to assess the benefits and harms of aluminium adjuvants versus placebo or no intervention in randomised clinical trials in relation to human vaccine development. Our aim is not to analyse the benefits and harms of any type of vaccine formulations for prevention of a specific disease. The results of our systematic reviews could influence future vaccine formulation and bring upon changes among policymakers and vaccine manufacturers to secure safe and efficient vaccines to people.

Methods

Detailed description of our methodology is in our prepublished protocol,[35] PROSPERO protocol (CRD42017083013) and our supplementary material.

Criteria for considering studies for this review

We searched for randomised clinical trials irrespective of type, status, date, and language of publication. We included vaccine development trials comparing any type of aluminium adjuvant versus placebo or no intervention.

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We accepted any co-interventions of vaccines if planned to be delivered equally to the intervention groups. We used the trial results reported at maximum follow-up.

Types of outcome measures

Primary outcomes were serious adverse events,[36] all-cause mortality and proportion of participants with the disease being vaccinated against. Secondary outcomes were health-related quality of life, non-serious adverse events and serological response.

Search methods for identification of studies

We searched the Cochrane Central Register of Controlled Trials (2021, Issue 7) in The Cochrane Library, MEDLINE Ovid (1946 to July 2021), Embase Ovid (1974 to July 2021), LILACS (Bireme; 1982 to July 2021), BIOSIS (Web of Science; 1969 to July 2021), Science Citation Index Expanded (Web of Science; 1900 to July 2021), and Conference Proceedings Citation Index – Science (Web of Science; 1990 to July 2021). In addition, we searched (September 2021) the Chinese Biomedical Literature Database (CBM), China Network Knowledge Information (CNKI), Chinese Science Journal Database (VIP), and Wanfang Database (**supplementary Table S1**). We also searched Google Scholar, The Turning Research into Practice (TRIP) Database, ClinicalTrials.gov (www.clinicaltrials.gov/), European Medicines Agency (EMA; www.ema.europa.eu/ema/), WHO International Clinical Trial Registry Platform (www.who.int/ictrp), The Food and Drug Administration (FDA; www.fda.gov) and pharmaceutical company sources for ongoing or unpublished trials (until March 2021). We applied EMA, FDA and several national medicines agencies (Australia, China, India, Japan, United Kingdom, Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain and Sweden) for clinical study reports on trials fulfilling our inclusion criteria.

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Data collection and analysis

Three review authors (SRK, SLK and MB) independently and in pairs screened titles and abstracts for inclusion of potentially eligible trials using Covidence (www.covidence.org). Following any unsolved disagreements, we asked a third author to arbitrate (JCJ or CG). The review author pair collected full-text trial reports/publications, and independently screened the full-texts and identified trials for inclusion. SRK extracted all data on all trials. SLK and MB each independently extracted half of the data. Extractions were compared and validated by SRK, SLK, and MB and in case of disagreement, the same review authors consulted JCJ or CG.

Assessment of risk of bias in included studies

The review author pair (SRK, SLK and MB) independently assessed the risk of bias (RoB 1) of each included trial according to the recommendations in the Cochrane Handbook for Systematic Reviews of Interventions.[37] We used the following bias risk domains: ‘allocation sequence generation’; ‘allocation concealment’; ‘blinding of participants and treatment providers’; 'blinding of outcome assessment’; ‘incomplete outcome data’; ‘selective outcome reporting’; and 'other bias’. We assessed the domains 'blinding of outcome assessment', 'incomplete outcome data' and 'selective outcome reporting' for each outcome. The trial was classified at overall 'low risk of bias' only if all the bias domains described in the above paragraphs were classified at low risk of bias, or at 'high risk of bias' if any of the bias risk domains described above were classified at 'unclear' or 'high risk of bias'.

We used the GRADE system to assess the certainty of the body of evidence associated with each of the outcomes. [38] We constructed the 'Summary of findings' tables using the GRADEpro software.[39] The GRADE system appraises the certainty of evidence based on the extent to which one can be confident that an estimate of effect or association reflects the item being assessed.

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Measures of treatment effect and data synthesis

We visually investigated forest plots to assess the risk of statistical heterogeneity. We also assessed the presence of statistical heterogeneity using the Chi^2 test (threshold $P < 0.1$) and measured the quantities of heterogeneity using the I^2 statistic.[40 41] We assessed reporting bias using funnel plots where ten or more trials per comparison were included.

When the total proportion of participants experiencing any serious and non-serious adverse event was not reported, we extracted data from the highest proportion of participants experiencing an individual adverse event.

We performed subgroup analyses on A: outcomes at low risk of bias compared to outcomes at high risk of bias or unclear risk of bias (collectively termed high risk of bias); B: trials at low risk of vested interests compared to trials at high risks of vested interests;[42] C: according to aluminium adjuvants type; D: according to different vaccines; E: according to age groups; F: according to different maximal follow-up periods; G: according to participants' health and H: according to vaccines against extracellular or intracellular pathogens.

We assessed the potential impact of missing data with the 'best-worst' and 'worst-best case' scenario.

Intervention effects were assessed with both random-effects model[43] and fixed-effect model[44] meta-analyses. The more conservative point estimate of the two (the analysis with the highest P value) was reported primarily. For analysis of three primary outcomes, a P value < 0.025 was considered statistically significant[45] because this would secure a family-wise error rate (FWER) below 0.05.

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We analysed all primary and secondary outcomes using Trial Sequential Analysis to control random errors.[35]

We included aluminium concentration (as described by trialist or manufacturer) as a covariate in meta-regression to assess whether the concentration influences the effect of aluminium adjuvant administration on outcomes.

Patient and public involvement

A patient and a representative of the public were involved in formulating the research question and the outcomes at the protocol stage. They were both involved in the interpretation and writing up of results. There are plans to disseminate the results of the research to the public and the relevant patient communities.

Results

Supplementary Figure S1 shows the flow of records obtained through the electronic searches. We identified 15 734 records through database searching. We obtained 396 full-text reports that were assess for eligibility. We excluded 280 records. We identified eight trials awaiting classification and six ongoing trials (**Appendix 1**).

We identified 102 randomised clinical trials including a total of 26 457 participants that fulfilled our inclusion criteria. Characteristics of included and excluded studies are given in **Appendix 1**. We were unable to identify any clinical study report from regulatory authorities that was eligible for inclusion in this review. We approached all corresponding authors to request missing information or explanations on unclear information and received some additional information from seven authors.

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The 102 included trials were published between 1969 and 2021; 35 were conducted in the USA;[46-80] 15 in more than one country;[81-95] six in Canada;[96-101] four in China;[102-105] four in Belgium;[106-109] four in Africa;[110-113] three each in the UK,[114-116] Taiwan,[117-119] and Australia;[90 120 121] two each in Thailand,[122 123] Poland,[124 125] Norway,[126 127] Italy,[128 129] Germany,[93 130] Cuba,[131 132] and Austria; [133 134] and one each in Switzerland,[135] Sweden,[136] Singapore,[137] Mali,[138] Israel,[139] India,[140] France,[141] Colombia,[142] Chile,[143] and Bangladesh.[144] Three trials did not report a country.[145-147]

Trial participants

Type of participants

The trials randomised different types of participants. Ninety trials randomised healthy participants; nine trials randomised participants with a disease diagnosis; [50 57 60 71 90-92 129 134] and three trials did not describe the inclusion criteria of the participants.[55 99 100]

Age of participants

In regards of age, the trials randomised: infants (six trials[86 114 125 132 144 146]); children (11 trials[85 87 95 102 104 117 132 138 147-149]); adolescents (two trials[66 115]); elderly (nine trials[50 60 62 83 90 98 101 112 139]); and mixed populations (eight trials[57 78 92 103 105 109 113 123]). Two trials did not specify the population type.[55 134] The remaining 65 trials randomised adult participants.

Interventions and comparisons

Types of aluminium adjuvants

The included trials assessed different types of aluminium adjuvants: aluminium hydroxide (37 trials[49 50 57 59 67-70 72 74 81-83 88 89 91 93 94 97 102-104 106 107 111 113 115 119 122-124 126 133 134 140-142 145]); aluminium phosphate (27 trials[46 47 53 60 62 63 66 75 80

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[86 87 98 109 110 112 116-118 121 131 132 136 139 146-148 150]); alhydrogel (21 trials[48 51 52 54-56 58 73 76-79 99-101 108 114 120 125 127 137]); amorphous aluminium hydroxyphosphate sulfate (two trials[64 71]); aluminium fluoride (one trial[95]); phosphate-treated aluminium hydroxide (one trial[144]); alhydrogel pretreated with phosphate buffer (one trial[135]); Adju-Phos® (aluminium phosphate gel) (one trial[96]); aluminium potassium sulphate (one trial[65]); aluminium chloride (one trial[61]); and aluminium oxide (one trial[143]). Eight trials did not describe the type of aluminium adjuvant used. [84 85 92 105 128-130 138]

Vaccines against different virus, bacteria, toxins, or diseases

The included trials assessed the effects of vaccines against different virus, bacteria, toxins, or diseases: influenza (25 trials[49 50 54 56 67 68 84 88 99-101 103 105 116 118 119 121 126 130 133 137 139-141 143]); Streptococcus pneumoniae (11 trials[74 85-87 95 109 112 135 144 146 147]); respiratory syncytial virus (11 trials[60 62 63 79 80 82 90 97 98 106 145]); human immunodeficiency virus (six trials[47 59 75 111 128 129]); Neisseria meningitidis (six trials[53 61 66 110 127 138]); Clostridium difficile (four trials[57 69 78 83]); dengue fever virus (four trials[55 58 120 151]); enterovirus (three trials[102 104 117]); Bacillus anthracis (three trials[48 51 52]); diphtheria and tetanus (two trials[114 136]); human papillomavirus (two trials[65 72]); Lyme borreliosis (two trials[89 94]); Haemophilus influenzae type B (two trials[131 132]); groups B Streptococcus (two trials[76 108]); Staphylococcus aureus (two trials[64 71]); poliovirus (two trials[124 125]); Pseudomonas aeruginosa (two trials[91 93]); Alzheimer's disease (two trials[92 134]); cytomegalovirus (two trials[46 96]); tetanus (one trial[115]); non-typeable Haemophilus influenzae (one trial[108]); Ross River virus (one trial[81]); hepatitis B (one trial[113]); malaria (one trial[142]); rabies and tetanus (one trial[122]); rabies (one trial[123]); Shigella flexneri (one trial[77]), and Staphylococcus aureus and Candida albicans (one trial[73]).

Overall, 41 trials assessed vaccines against extracellular pathogens (bacteria or toxins) and 61 trials assessed vaccines against intracellular pathogens (viruses).

Vaccine doses

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The included trials administered different numbers of vaccine doses: 24% of the trials administered 1 dose;[53 55 58 60 64 66 70 73 76 82 87 97 98 101 108 112 115 124 131 133 136 139 140 143] 40% of the trials administered two doses;[49-52 54 56 61-63 67 68 75 77 79 80 84 88 90 91 93 99 100 102-106 109 110 116-119 121 126 127 130 135 137 138 141 147] 21% of the trials administered three doses;[46 47 57 59 65 71 72 74 78 81 85 89 96 107 111 113 114 120 123 125 132 142 144] 10% of the trials administered four doses;[69 83 86 94 95 134 146] two trials administered five doses;[128 129] and one trial administered seven doses.[92] Two trials did not specify the number of doses administered.[122 145]

Aluminium concentrations

The included trials used different aluminium concentrations ranging from 125 mcg/dose to 6000 mcg/dose.

Control groups

Two comparisons (from two trials) extracted in this review did not involve a vaccine (i.e. comparison between saline placebo with or without aluminium).[70 140] All the other control groups contained the same vaccine as the intervention group but without aluminium adjuvant.

Risk of bias within individual trials

Based on the information collected from the published reports and from authors, only 3/102 trials were at overall low risk of bias (all outcomes reported at low risk of bias). The remaining trials were at overall high risk of bias (**supplementary Figure S2**).

Effects of aluminium adjuvants

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Serious adverse events

170/7627 (2.2%) participants who received aluminium adjuvants with or without vaccines suffered a serious adverse event versus 149/13 936 (1.1%) participants receiving no aluminium adjuvants with or without vaccines (RR 1.18, 95% CI 0.97 to 1.43; 21 563 participants; 62 trials; I^2 0%; Bayes factor 146.8; very low certainty of evidence. **Figure 1, Table 1, supplementary Figure S3**). Visual inspection of the forest plot and I^2 reveal no statistical heterogeneity. Trial Sequential Analysis showed that the cumulative Z-curve (blue full line with quadratic squares indicating of each trial) touches the traditional boundary for harm. However, none of the trial sequential monitoring boundaries (etched curves above and below the traditional horizontal lines for statistical significance) were surpassed. The result is inconclusive, and the required information size has not been achieved. The TSA-adjusted confidence interval is 0.53 to 2.69 (Pc (proportion with an outcome in the control group) 1.0%, RR reduction or increase (RRR) 20%, alpha 2.5%, beta 10%, diversity 0%; diversity-adjusted required information size (DARIS) 110 696 participants) (**supplementary Figure S4**).

Subgroup analyses

Test for subgroup differences showed no difference when comparing the effects of aluminium adjuvants in trials at high risk of bias to trials at low risk of bias; in trials at risk of vested interest to trials at low risk of vested interest; trials according to different aluminium types; trials with different vaccines; trials with different participants' ages; trials with different follow up durations; trials with participants with different diagnoses compared to healthy participants; and trials assessing vaccines against different pathogens types (**supplementary Figures S5-S12**).

Sensitivity analysis

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21/7648 (0.3%) participants in the intervention group versus 18/13 954 (0.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'worst-best' case scenario (**supplementary Figure S13**). The 'best-worst' case analysis showed that incomplete outcome data did not have the potential to influence the result (**supplementary Figure S14**).

Meta-regression showed that the proportion of participants with serious adverse events was not affected by the aluminium concentration used in the vaccine (P = 0.28).

Due to several trials with zero events, we performed meta-analysis also with OR. The results did not change (**supplementary Figure S15).**

Individual serious adverse events analyses

Meta-analyses showed no evidence of a difference between aluminium adjuvants versus control when assessing individual serious adverse events (**supplementary Analysis S1).** Individual serious adverse events reported only in one trial that were not possible to meta-analyze are shown in **supplementary Table S2.**

All-cause mortality

61/7782 (0.8%) aluminium participants died compared with 57/14 104 (0.4%) control participants (RR 1.02, 95% CI 0.74 to 1.41; 21 886 participants; 62 trials; I^2 0%; Bayes factor 2.35; very low certainty evidence. **Figure 2, Table 1, supplementary Figure S16**). Visual inspection of the forest plot and I^2 indicated no statistical heterogeneity. Funnel plot showed no publication bias. Trial Sequential Analysis showed that the accrued information for all-

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cause mortality was below 5% of the DARIS (Pc 0.4%, RRR 20%, alpha 2.5%, beta 10%, diversity 0%; DARIS 278 247).

Subgroup analyses

Test for subgroup differences showed no difference when comparing trials at high risk of bias to trials at low risk of bias; trials at risk of vested interest to trials at low risk of vested interest; trials with no vaccine cointervention to trials with vaccine cointervention; trials with different aluminium types; trials with different vaccines; trials with different participants' ages; trials with different participants' diagnoses; and trial assessing vaccines against different pathogens types (**supplementary Figures S17-S23**). Due to lack of relevant data, it was not possible to conduct the subgroup analyses on trials with different follow up durations.

Sensitivity analysis

28/7909 (0.35%) participants in the aluminium group versus 20/14 173 (0.14%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence our result in the 'worst-best' case scenario showing a harmful effect of aluminium adjuvants compared with placebo (**supplementary Figure S24**). The 'best-worst' case analysis showed that incomplete outcome data did not have the potential to influence our result (**supplementary Figure S25**).

Meta-regression showed that the proportion of participants with all-cause mortality was not affected by the aluminium concentration used in the vaccine (P = 0.88).

Due to several trials with zero events, we performed meta-analysis also with OR. The results did not change (**supplementary Figure S26**).

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Participants with disease

Only two trials (one event) reported on the proportion of participants that developed the disease they were vaccinated against (**supplementary Figure S27**).

Adverse events considered non-serious

Out of the 67 trials reporting adverse events considered non-serious, 34 trials reported the overall proportion of participants with one or more adverse events considered non-serious. From the remaining 33 trials reporting on adverse events considered non-serious, we extracted data from the highest proportion of participants experiencing an individual adverse event.

3760/7098 (52.9%) aluminium participants experienced one or more non-serious adverse event compared with 4537/13 429 (33.8%) control participants. Meta-analysis of these trials showed that aluminium adjuvants compared with placebo or no intervention may increase the proportion of participants with one or more adverse events considered non-serious (RR 1.13, 95% CI 1.07 to 1.20; participants = 20 527; trials = 67; I^2 = 85%; Bayes factor 2.64*1084; the evidence was very uncertain. **Figure 3, Table 1, supplementary Figure S28**). Trial Sequential Analysis of non-serious adverse events shows that the cumulative Z curve crosses the boundary for harm, indicating that there was enough information to confirm that aluminium adjuvants compared with placebo or no intervention increases the risk of one or more non-serious adverse events (TSA Pc 33.5%, RRR 20%, alpha 2.5%, beta 10%, diversity 78%; DARIS 28 384. TSA adjusted CI 1.06 to 1.23. **supplementary Figure S29**).

Visual inspection of the funnel plot and regression-based Harbord test showed no publication bias or small-study effects (beta = 0.99).

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Subgroup analyses

Test for subgroup differences was statistically significant in the subgroup analysis according to according to vaccine type (P < 0.00001; **supplementary Figure S30)** and age (P = 0.007; **supplementary Figure S31).**

Test for subgroup differences showed no difference when comparing the effect of aluminium adjuvants in trials at low risk of bias to trials at high risk of bias; in trials at low risk of vested interest to trials at risk of vested interest; in trials with different aluminium salts; in trials with different follow-up durations; in trials with participants with different health status; and trials assessing vaccines against different pathogen types **(supplementary Figures S32-S37**).

Sensitivity analysis

195/7392 (2.6%) participants in the aluminium group versus 186/13 341 (1.4%) participants in the control group were lost to follow up. Incomplete outcome data did not have the potential to influence our results.

We included aluminium concentration (as described by trialists) as a covariate in meta-regression to assess whether aluminium concentration has an impact on the effect sizes of the proportion of participants with adverse events considered non-serious. Meta-regression showed that the proportion of participants with adverse events considered non-serious was not affected by the aluminium concentration used in the vaccine (P = 0.68)

Individual non-serious adverse events

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We performed meta-analysis on each of the 145 reported individual adverse event considered non-serious. Mainly local injection site reactions were increased in the aluminium group (**supplementary Analysis S2 and supplementary Table S3)**.

Serological response

Geometric mean titers (GMT) and geometric mean concentrations (GMC)

Serological response was assessed by different analytical assays and was reported as either geometric mean titre (GMT, 31 trials) or geometric mean concentration (GMC, 11 trials).

Meta-analysis showed no or little evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing geometric mean titres or geometric mean concentrations (**Figure 4**, **Figure 5, supplementary Figure S38 and S39**).

Subgroup and sensitivity analysis for serology

Subgroup and sensitivity analysis for the serological response is reported in **supplementary figure S40-S43**).

Seroprotection

Meta-analysis showed that there was no evidence of a difference between aluminium adjuvants compared with placebo or no intervention when assessing seroprotection (RR 0.95, 95% CI 0.77 to 1.18; trials = 27; I^2 78%. Bayes factor 36.1; low certainty of evidence **Figure 6, supplementary Figure S44**). Visual inspection of the forest plot and I^2-statistics indicated high heterogeneity (I^2 78%).

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Subgroup and sensitivity analysis for seroprotection

Subgroup and sensitivity analysis for seroprotection is reported in **supplementary text and Figure S45 and S46**.

Discussion

This review included 102 randomised clinical trials assessing a total of 26 457 participants. Aluminium adjuvants versus placebo or no adjuvants may have no effect on the proportion of participants with one or more serious adverse events and on all-cause mortality, but the evidence was very uncertain. Two trials reported on the proportion of participants with the disease they were vaccinated against. However, only one event was reported. None of the trials reported on quality of life. Aluminium adjuvants versus placebo or no adjuvants seems to increase the proportion of participants with one or more adverse events considered non-serious, but the evidence was very uncertain. We found no or little evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing geometric mean titres or concentrations. Aluminium adjuvants versus placebo or no intervention may have no effect on participants without seroprotection, but the evidence was very uncertain.

Strengths and weaknesses

We seem to be the first to conduct a systematic review comparing aluminium adjuvants versus placebo or no intervention in any type of vaccine. We followed our peer-reviewed protocol which was published before the literature search began,[35] and we conducted the review using the methods recommended by Cochrane.[37 152] We reported our review according to the PRISMA statement[153] (**supplementary Table S4**).

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Our systematic review has several limitations. Despite our inclusion criteria being broad, we could only find phase I or II trials that met our inclusion criteria. This limitation is because phase III or IV trials of marketed vaccines are mainly designed with an active comparator (another vaccine or alleged 'placebo' with aluminium), and therefore these trial designs do not match the inclusion criteria of our review.

Another limitation of the applicability of our results is that we chose maximum follow-up as our time point of primary interest. This approach does not allow us to make conclusions on the effect of aluminium adjuvants on safety and immunogenicity after each individual vaccine dose (for those trials having multiple vaccine injections).

Another limitation is that we identified high clinical heterogeneity, especially within the immunogenicity outcome. Included trials did not use the same assays to assess the serological response and most of the trials had multiple assays performed. We chose to analyse only the geometric mean titre or concentration reflecting the data from the first assay presented in the publication, however, we are aware that this might limit the strength of our findings.

We chose to merge multiple groups in those trials that used vaccines with different antigen concentration. In so doing, we were unable to conclude whether the effect of aluminium adjuvants is correlated to the effect of different antigen concentrations.

Only two trials (one event in total) reported on the proportion of participants with disease they were vaccinated against and none of the authors provided us with such data when contacted by email. Therefore, our conclusion regarding the effect of aluminium adjuvants on the immunogenicity is based on the surrogate outcome of the

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serological response to vaccine measured by different assays and on the seroprotection values as defined by the trialists.

Of the 62 trials that reported on serious adverse events, eight trials reported only vaccine-related serious adverse events.[49 51 81 101 105 108 109 116] Of the 62 trials that reported on serious adverse events, 14 trials reported that serious adverse events occurred but these were not assigned per intervention group because they were considered unrelated to the vaccine (a total of 107 serious adverse events reported to having occurred but not described per intervention group.[50 66-68 77 83 84 86 92 98 119 121 134 147] Only 7/102 authors contacted provided us with all or some of the data requested (see Characteristics of included studies in the **Appendix 1**).

Out of the 67 trials that reported on adverse events considered non-serious, 34 trials reported the overall proportion of participants with one or more adverse event considered non-serious.[54 56 58 64 66 69 70 72 77-79 82 83 85 87 91 93 96 98 103 104 106-108 111 114 118 124 126 133 137 139 146 150 154] From the remaining 33/67 trials reporting on adverse events considered non-serious, we extracted data from the highest proportion of participants experiencing an individual adverse event.[46 49-51 53 62 76 80 81 84 86 92 94 101 102 110 112 115 116 119-121 124 125 128 129 135 136 140-144] A substantial number of trials reported only solicited adverse events instead of solicited and unsolicited adverse events as a combined outcome. This limitation may have resulted in an underestimation of the unsolicited adverse events that might have occurred but were not reported.

Agreements and disagreements with other studies or reviews

In 2004, Jefferson and colleagues reviewed evidence of adverse events after exposure to aluminium-containing vaccines against diphtheria, tetanus, and pertussis, alone or in combination, compared with identical vaccines,

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either without aluminium or containing aluminium in different concentrations.[155] They included three randomised trials, four semi-randomised trials, and one cohort study. They found that in young children, vaccines with aluminium hydroxide caused significantly more erythema and induration than plain vaccines and significantly fewer reactions of all types. In older children there was an association with local pain lasting up to 14 days. Despite a lack of good-quality evidence, the authors surprisingly recommend against any further research on this topic.

In 2018, Lin and colleagues conducted the first meta-analysis on the efficacy of aluminium salts as an adjuvant for pre-pandemic influenza vaccines.[32] They included a total of nine randomised clinical trials (published during 2006 to 2013), including 22 comparisons in 2467 participants that compared aluminium-adjuvanted H5N1 vaccines versus non-adjuvanted counterparts.[32] Their results showed an inferior seroprotection after aluminium-adjuvanted H5N1 vaccines compared with that conferred by non-adjuvanted counterparts. Furthermore, H5N1 vaccines with aluminium adjuvants were associated with a significantly higher risk of pain/tenderness at the injection site during the seven days after the first vaccination and after the second dose versus the non-adjuvanted counterparts.

In 2020, Jørgensen and colleagues set out to assess the benefits and harms of the human papillomavirus (HPV) vaccines in clinical study reports obtained from the European Medicines Agency and GlaxoSmithKline from 2014 to 2017.[156] They included 24 randomised clinical trials comparing an aluminium-adjuvanted HPV vaccine versus a placebo or active comparator in healthy participants of all ages. They found that at four years follow-up, the HPV vaccines decreased HPV-related precursors to cervical cancer and treatment procedures but increased serious nervous system disorders (exploratory analysis) and general harms.[131] As the trials included in their review were primarily designed to assess benefits and not adequately designed to assess harms, the extent to which the benefits outweigh the harms was unclear.

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In agreement with Lin,[32] our systematic review does not find an increased serological response of aluminium-adjuvanted vaccines compared with that conferred by non-adjuvanted counterparts. Also, in agreement with both Jefferson[155] and Lin[32] we find an increase in local injection site reactions after administration of aluminium-adjuvanted vaccines.

Implications for practice and research

Considering the lack of good quality evidence to assess beneficial and harmful effects of adding aluminium to vaccines as presented here, relevance of this adjuvant should be investigated in future studies. Questions on aluminium form, concentration, and size remain unanswered due to scarcity or lack of data. Questions on the effects of aluminium adjuvants on vaccine effectiveness also remain unanswered.

Future randomised clinical trials in humans should be conducted according to the ethical principles that have their origin in the Declaration of Helsinki, and that are consistent with the ICH Good Clinical Practice and the applicable regulatory requirement(s).[157 158] Such trials should be designed in accordance with guidelines for clinical trials (SPIRIT)[159] and reported in accordance with the Consolidated Standards of Reporting Trials (CONSORT).[160]

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All authors approved of the current version (for publication).

Declarations of interest

SRK: no known conflict of interest

MB: no known conflict of interest

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CG: no known conflict of interest

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Figures

Table 1. Summary of findings table

Figure 1. Meta-analysis of the effect of aluminium adjuvant compared with placebo or no intervention on the proportion of participants with one or more serious adverse events.

Figure 2. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on all-cause mortality.

Figure 3. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the proportion of participants with one or more non-serious adverse events.

Figure 4. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean titres (GMT) grouped by analytical assay.

Figure 5. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean concentrations (GMC) grouped by analytical assay.

Figure 6. Meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on seroprotection.

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Table 1. Summary of findings table.

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Summary of findings table

Aluminium adjuvants compared with placebo or no intervention in vaccines

Patient or population: any population

Setting: any settings

Intervention: aluminium adjuvants

Comparison: placebo or no intervention

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo or no intervention	**Risk with aluminium adjuvants**				
Proportion of participants with one or more serious adverse events	11 per 1,000	**12 per 1,000** (9 to 15)	RR 1.10 (0.88 to 1.38)	21563 (58 RCTs)	⨁◯◯◯ VERY LOW [a,b,c]	Aluminium adjuvants versus placebo or no adjuvants may have no effect on the proportion of participants with one or more serious adverse events, but the evidence was very uncertain. Imprecision and indirectness were considered 'very serious' and therefore downgraded twice
All-cause mortality	4 per 1,000	**4 per 1,000** (3 to 6)	RR 1.02 (0.74 to 1.42)	21886 (60 RCTs)	⨁◯◯◯ VERY LOW [d,e]	Aluminium adjuvants versus placebo or no adjuvants may have no effect on all-cause mortality, but the evidence was very uncertain Imprecision was considered 'very serious' and therefore downgraded twice

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Proportion of participants with one or more adverse events considered non-serious	338 per 1,000	**385 per 1,000** (362 to 409)	RR 1.14 (1.07 to 1.21)	20527 (64 RCTs)	⨁◯◯◯ VERY LOW [f,g,h]	Aluminium adjuvants may increase the proportion of participants with one or more adverse events considered non-serious but the evidence is very uncertain. Indirectness was considered 'very serious' and therefore downgraded twice.
Participants without seroprotection	118 per 1,000	**112 per 1,000** (91 to 139)	RR 0.95 (0.77 to 1.18)	7845 (14 RCTs)	⨁◯◯◯ VERY LOW [i,j,k]	Aluminium adjuvants versus placebo or no intervention may have no effect on participants without seroprotection, but the evidence was very uncertain. Publication bias was considered 'strongly suspected' and therefore downgraded twice.
Health-related quality of life	0 per 1,000	**0 per 1,000** (0 to 0)	not estimable	(studies)	-	no data
Participants with disease being vaccinated against	0 per 1,000	**0 per 1,000** (0 to 0)	not estimable	221 (2 studies)	-	too few data

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio

GRADE Working Group grades of evidence
High certainty: We are very confident that the true effect lies close to that of the estimate of the effect
Moderate certainty: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low certainty: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect
Very low certainty: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

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Explanations

a. Downgraded for risk of bias. Overall: 89% of the trials were at high risk of bias for this outcome and 11% of the trials were at low risk of bias for this outcome. Specifically, 17% of the trials reporting on serious adverse events were at high risk of bias in blinding of outcome assessors and 54% of the trials reporting on serious adverse events had unclear risk of bias in blinding of outcome assessors. 17% of the trials reporting on serious adverse events were at high risk of bias in missing outcome data and 27% of the trials reporting on serious adverse events were at unclear risk of bias in missing outcome data. 4% of the trials reporting on serious adverse events were at high risk of bias in selective outcome reporting and 27% of the trials reporting on serious adverse events were at unclear risk of bias in selective outcome reporting.

b. Differences in outcomes measures: of the 61 trials that reported on serious adverse events, 8 trials reported only vaccine-related serious adverse events without giving information on the serious adverse events that potentially occurred but were not related to the vaccine. Other 14 trials reported that serious adverse events occurred but these were not described per intervention group because they were considered unrelated to the vaccine (a total of 107 serious adverse events reported to having occurred but not described per intervention group).

c. Downgraded for imprecision. Optimal information size (n = 110,696) not reached.

d. Downgraded for risk of bias. Overall: 79% of the trials were at high risk of bias for this outcome and 21% of the trials were at low risk of bias for this outcome. Specifically, 16% of the trials reporting on mortality had high risk of bias in blinding of outcome assessors and 43% of the trials reporting on mortality had unclear risk of bias in blinding of outcome assessors. 9% of the trials reporting on mortality had high risk of bias in missing outcome data and 22% of the trials reporting on mortality had unclear risk of bias in missing outcome data. 1% of the trials reporting on mortality had high risk of bias in selective outcome reporting and 21% of the trials reporting on mortality had unclear risk of bias in selective outcome reporting. However, the overall limitations were unlikely to influence this outcome.

e. Downgraded for imprecision. Optimal information size (n = 278,247) not reached.

f. Downgraded for risk of bias. Overall: 75% of the trials were at high risk of bias for this outcome and 25% of the trials were at low risk of bias for this outcome. Specifically, 14% of the trials reporting on non-serious adverse events were at high risk of bias in blinding of outcome assessors and 49% of the trials reporting on serious adverse events were at unclear risk of

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bias in blinding of outcome assessors. 13% of the trials reporting on non-serious adverse events were at high risk of bias in missing outcome data and 9% of the trials reporting on non-serious adverse events were at unclear risk of bias in missing outcome data. 1% of the trials reporting on non-serious adverse events were at high risk of bias in selective outcome reporting and 26% of the trials reporting on serious adverse events were at unclear risk of bias in selective outcome reporting.

g. Downgraded for inconsistency. I^2 = 85%. Visual inspection of funnel plot may suggest potential publication bias for smaller trials reporting a harmful effect of aluminium adjuvants in the placebo group. Regression-based Harbord test showed no small-study effects (beta = 0.99)

h. Downgraded for indirectness. Differences in outcomes measures: a number of studies report only number of solicited or unsolicited adverse events, rather than total number of participants experiencing any adverse events. Out of the 66 trials reporting non-serious adverse events, only 34 trials reported the overall proportion of participants with one or more non-serious adverse event. From the remaining 32 out of the 66 trials reporting on non-serious adverse events we extracted data from the highest proportion of participants experiencing an individual adverse event.

i. Overall: 100% of the trials were at high risk of bias for this outcome. Specifically, 0% of the trials reporting on seroprotection were at high risk of bias in blinding of outcome assessors and 68% of the trials reporting on serious adverse events were at unclear risk of bias in blinding of outcome assessors. 74% of the trials reporting on seroprotection were at high risk of bias in missing outcome data and 0% of the trials reporting on serious adverse events were at unclear risk of bias in missing outcome data. 0% of the trials reporting on seroprotection were at high risk of bias in selective outcome reporting and 23% of the trials reporting on serious adverse events were at unclear risk of bias in selective outcome reporting.

j. Downgraded for inconsistency. I^2 = 78%.

k. Visual inspection of funnel plot may suggest potential publication bias for smaller trials reporting an harmful effect of aluminium adjuvants in the intervention group. Regression-based Harbord test showed no small-study effects (beta = 0.99).

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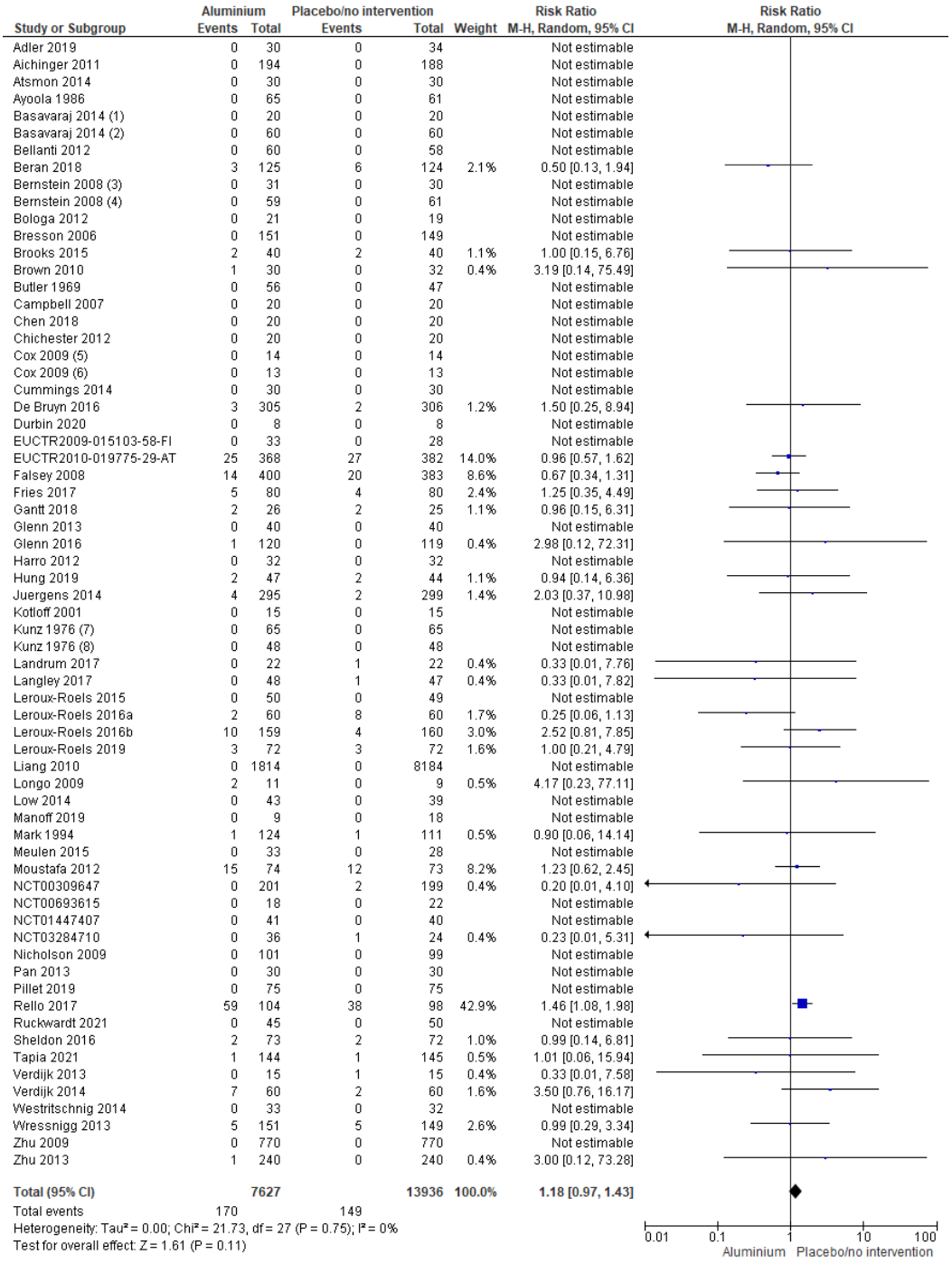


Figure 1.

Random-effects meta-analysis of the effect of aluminium adjuvant compared with placebo or no intervention on the proportion of participants with one or more serious adverse events.

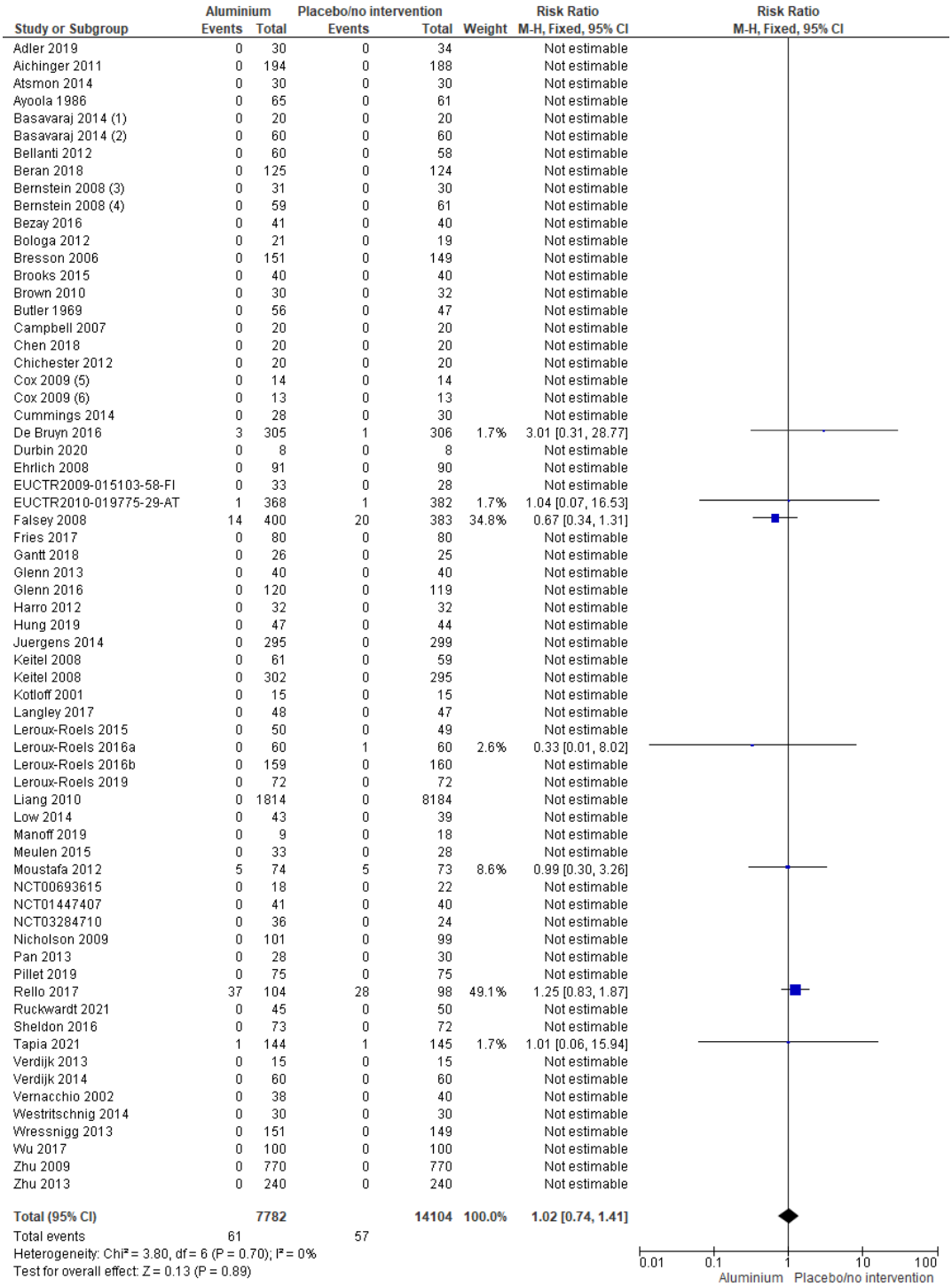


Figure 2.

Fixed-effect meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on all-cause mortality.

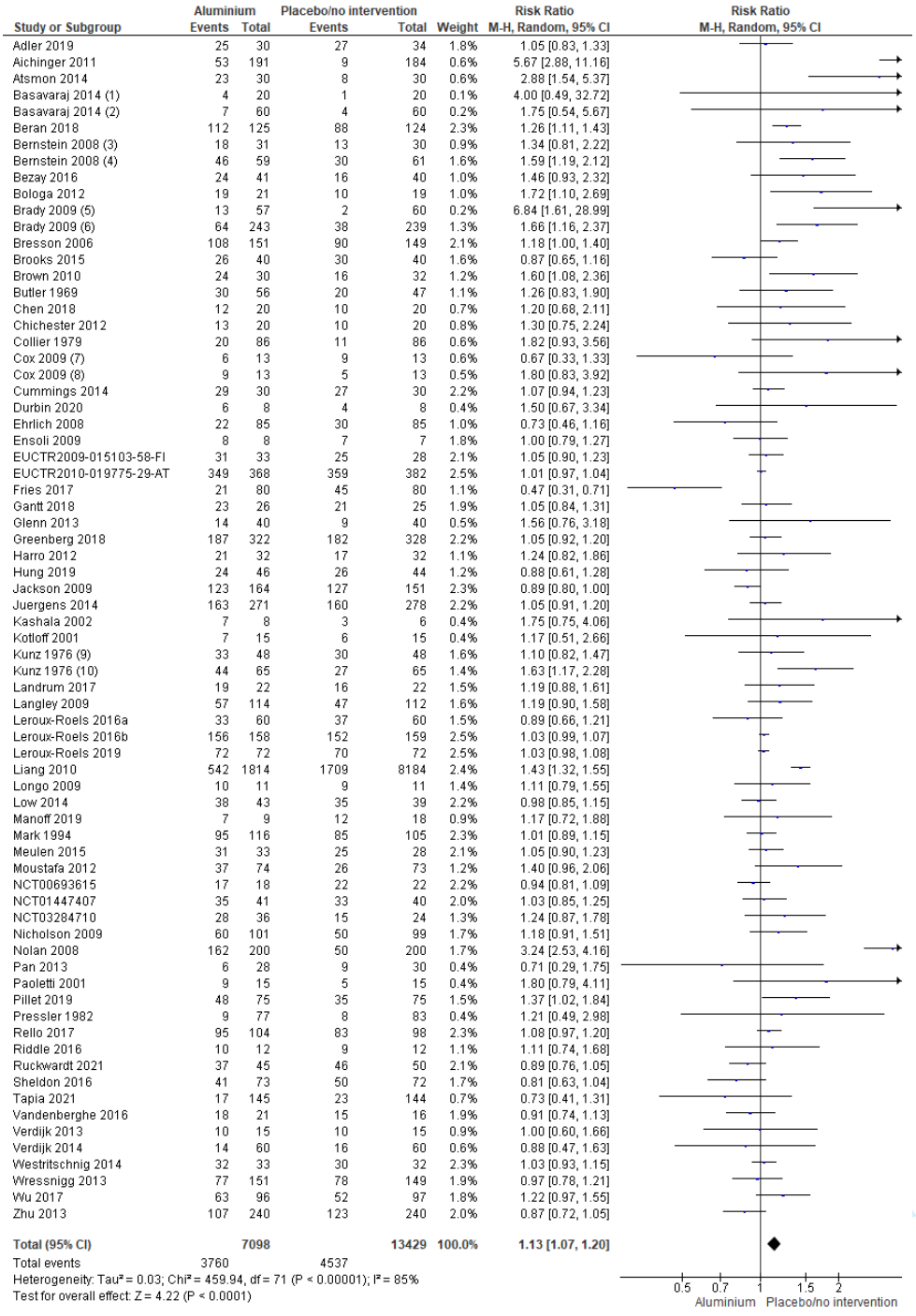


Figure 3.

Random-effects meta analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the proportion of participants with one or more non-serious adverse events.

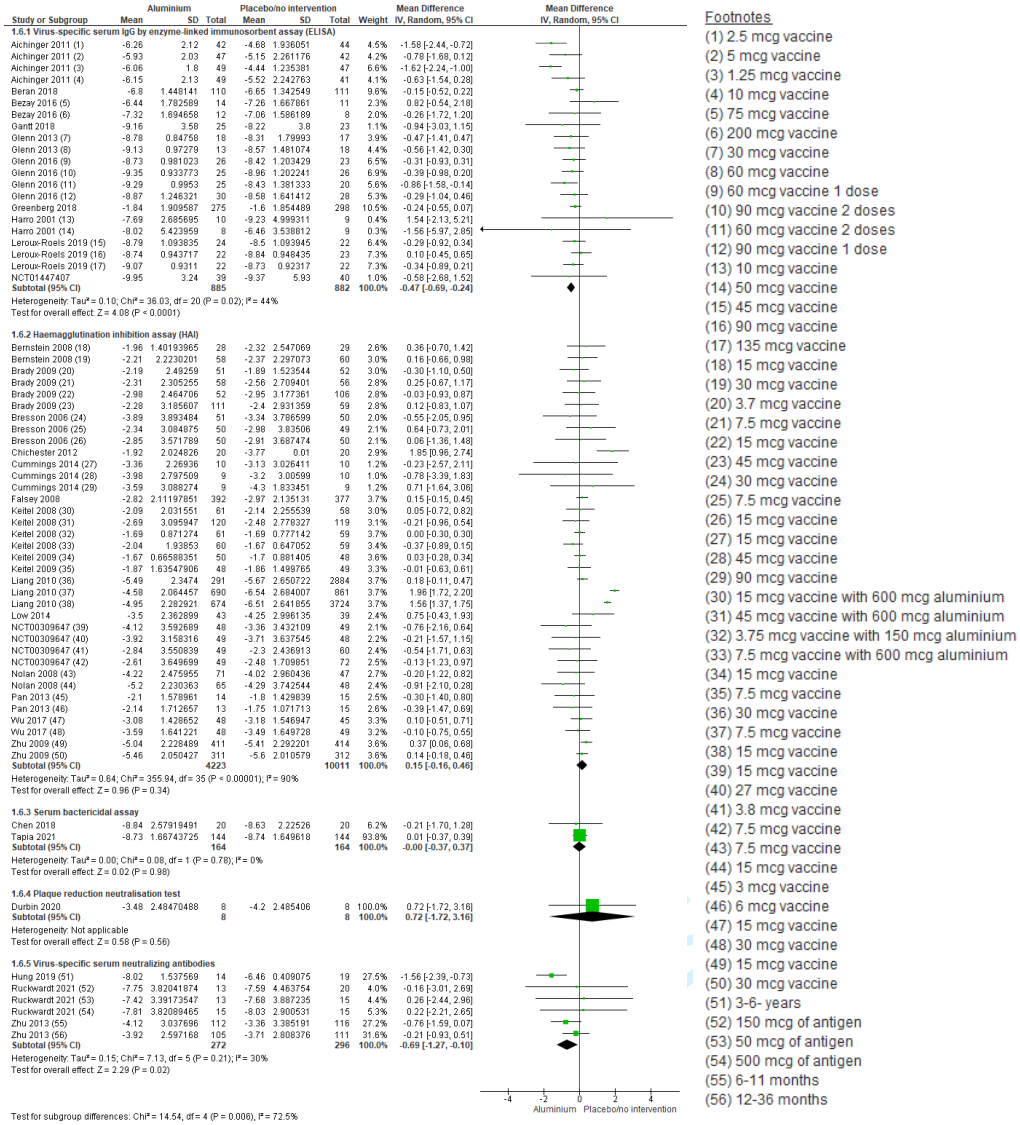


Figure 4.

Random-effects meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean titres (GMT) grouped by analytical assay.

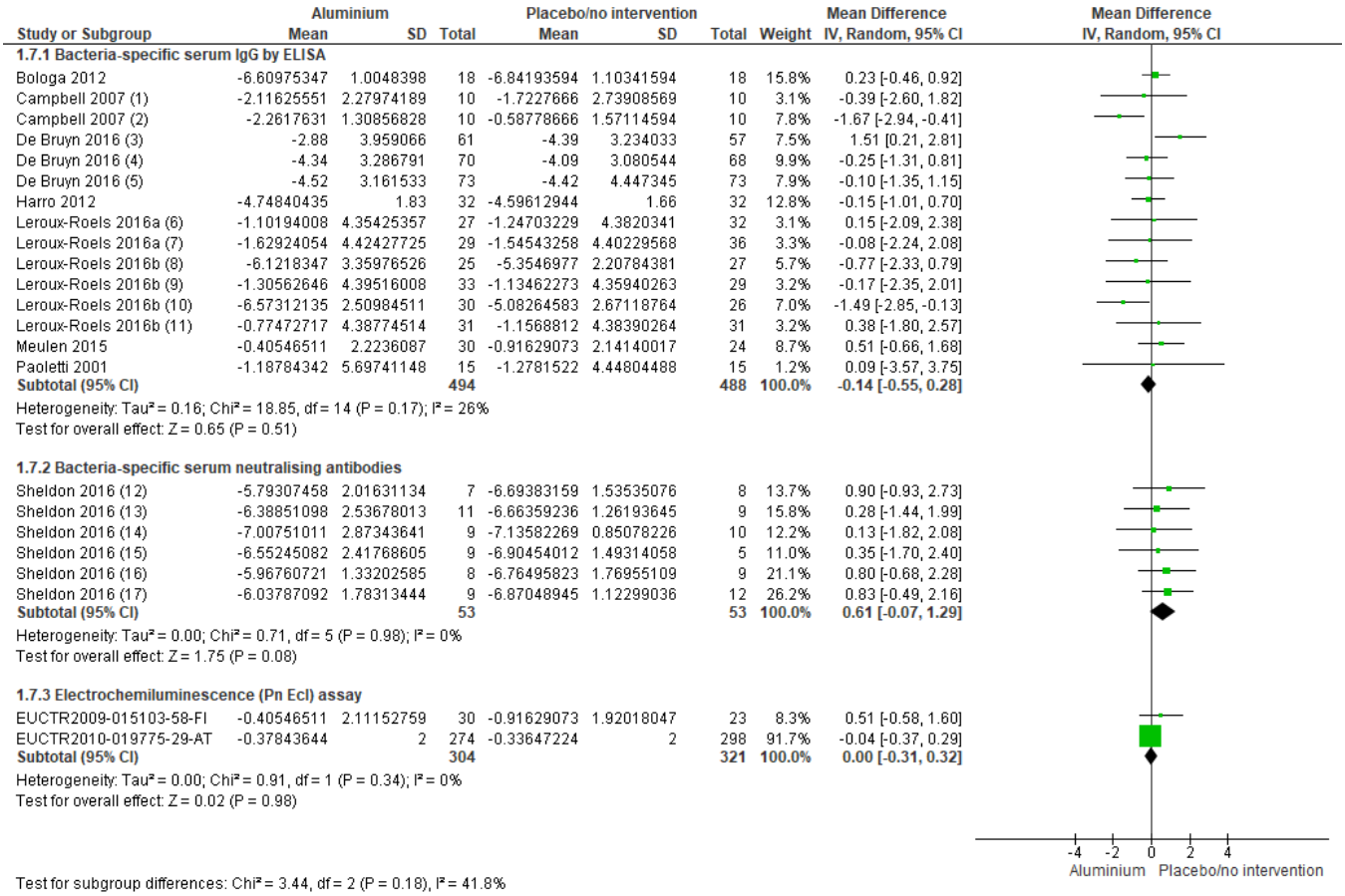
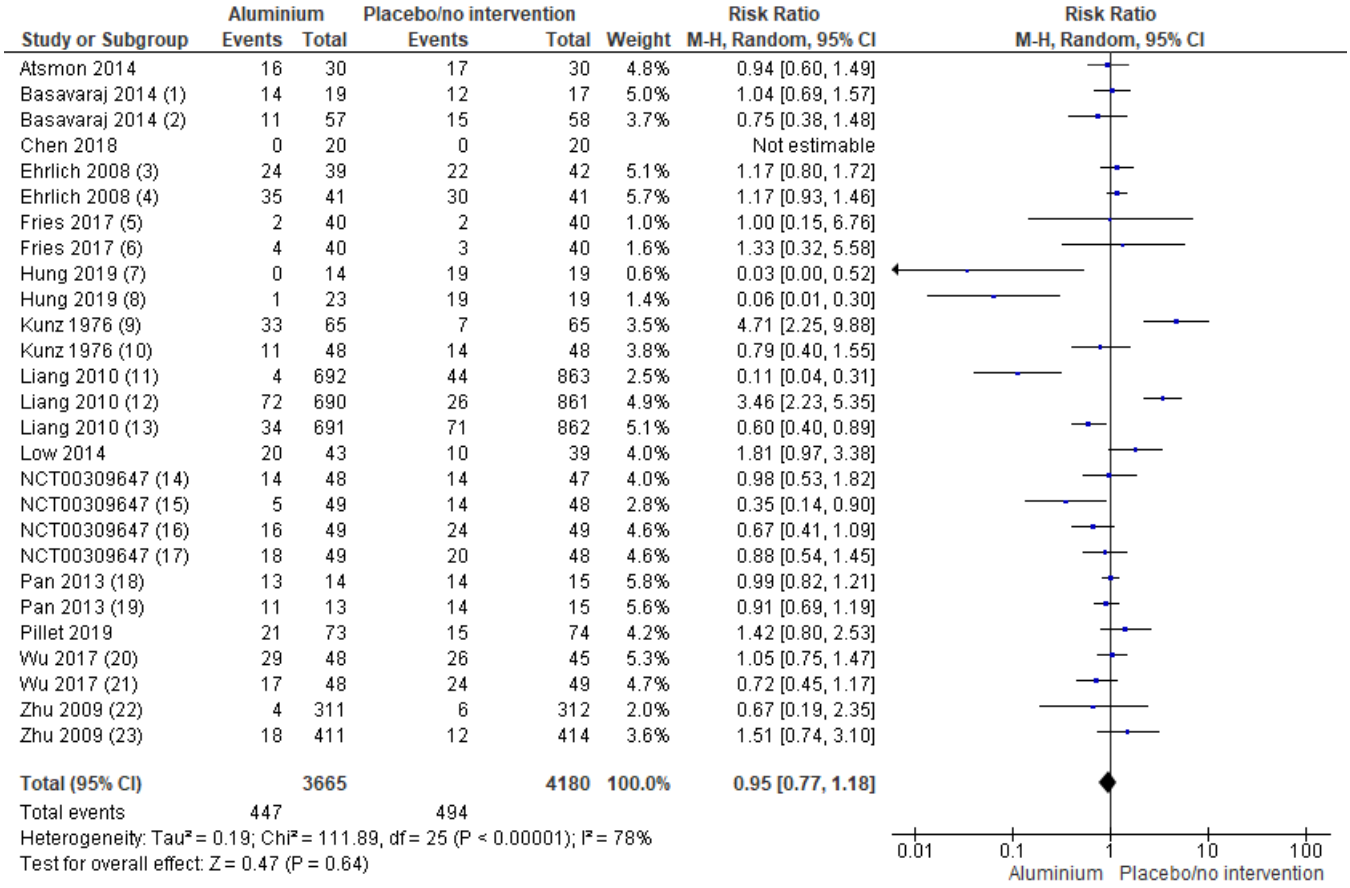


Figure 5.

Random-effects meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean concentrations (GMC) grouped by analytical assay.



Footnotes

(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 7.5 mcg vaccine
(4) 15 mcg vaccine
(5) 60 mcg vaccine
(6) 90 mcg vaccine
(7) 3-6 years of age
(8) 2-35 months of age
(9) subunit vaccine
(10) whole virus vaccine
(11) 30 mcg vaccine
(12) 7.5 mcg vaccine
(13) 15 mcg vaccine
(14) 3.8 mcg vaccine
(15) 7.5 mcg vaccine
(16) 15 mcg vaccine
(17) 27 mcg vaccine
(18) 3 mcg vaccine
(19) 6 mcg vaccine
(20) 15 mcg vaccine
(21) 30 mcg vaccine
(22) 30 mcg vaccine
(23) 15 mcg vaccine

Figure 6.

Random-effects meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on seroprotection.

Characteristics of studies

Characteristics of included studies

Adler 2019

Methods	2-part, double-blind, randomised, placebo controlled, dose escalation multicenter study
Participants	**Sex:** male and female **Age:** >18 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** body weight ≥50 kg and body mass index (BMI) 19–32 kg/m2 and serologically confirmed as CMV seronegative or seropositive **Exclusion criteria:** "subjects were excluded from the trial if they had previously received any CMV vaccine, had a history of allergic reaction or anaphylactic/anaphylactoid reaction that required medical attention to any vaccine component, or had a history of any severe allergic reaction. Other exclusion criteria included recent history of febrile illness (<72 hours); receipt of any live virus vaccine (4 weeks before through 1 month after study vaccine), inactivated vaccine (7 days before through 7 days after study vaccine), or immune globulin or blood product (within 90 days); known or suspected impairment of immunologic function; and recent receipt of immunosuppressive therapies".
Interventions	**Vaccine type:** defective human cytomegalovirus (CMV) vaccine **Experimental:** 100 or 30 Units/dose of V160 with aluminium phosphate adjuvant intramuscularly. n = 30 **Control:** 100 or 30 Units/dose of V160 intramuscularly. n=34 **Aluminium concentration:** 225 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** 3-dose regimen at day 1, month 1, and month 6
Outcomes	**Immunogenicity:** "geometric mean count of peripheral blood mononuclear cells secreting interferon-gamma, (time frame: 1 month after vaccination 3 (vaccination 3 is administered at Month 6)). Geometric mean concentration of interferon-gamma after stimulation of whole blood sample with pooled HCMV antigen peptides (time frame: 1 month after vaccination 3 (vaccination 3 is administered at Month 6)). Geometric mean titer of HCMV-specific neutralizing antibody (time frame: 1 month after vaccination 3 (vaccination 3 is administered at Month 6))". **Safety:** "percentage of participants with an adverse event, (time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6)). Percentage of participants with an injection-site adverse event, (time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6)). Percentage of participants with a systemic adverse event, (time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6)). Percentage of participants with a serious adverse event, time frame: from the time of vaccination up to 14 days after any vaccination (vaccinations are administered on day 1, month 1 and month 6))".

Are there other groups in the trial not analysed in this review?	yes
Notes	**Contacts with authors:** on 12/02/2020 authors (luwy_musey@merck.com) were inquired about the followings: ● proportion of participants with any AE related and not related to the vaccine ● proportion of participants who developed CMV infection after the vaccination ● random sequence generation ● allocation concealment **Funding:** this work was supported by Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Kenilworth, NJ.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Subjects were randomised according to their CMV serostatus and gender.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	The trial was described as double-blind. Masking: Triple (Participant, Investigator, Outcomes Assessor).
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	A study protocol was registered prior to randomisation (NCT01986010), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Aichinger 2011

Methods	Randomized, partially blinded (with respect to adjuvation) clinical trial
Participants	**Sex:** male and females **Age:** 18-40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects were eligible to participate if they were clinically healthy, understood the study procedures, provided written informed consent and agreed to keep a daily record of symptoms". **Exclusion criteria:** "subjects were not eligible if they reported a history of RRV exposure or travel to an RRV-endemic area, a history of any vaccine-related contraindicating event (e.g. anaphylaxis), a history of non-trauma-related arthritis, cardiovascular disease, immunodeficiency, autoimmune disease, inflammatory or degenerative neurological disease, a history of a hematological malignancy or an active neoplastic disease, or if they had a positive urine drug screen or a body mass index > 35, if they had a disease or were undergoing a form of treatment that could be expected to influence immune response. Smoking status was not recorded. Women were required to have a negative pregnancy test at screening and before each vaccination, and were not eligible if lactating".
Interventions	**Vaccine type:** inactivated whole virus Vero cell-derived Ross River virus vaccine **Experimental:** 1.25, 2.5, 5 or 10 mcg aluminium hydroxide-adjuvanted Ross River virus vaccine. n = 194 **Control:** 1.25, 2.5, 5 or 10 mcg Ross River virus vaccine. n = 188 **Aluminium concentration:** 0.2% aluminium hydroxide **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Baxter **Vaccination schedule:** the first two immunizations were administered 21 days apart (range 18–24), with a third immunization six months after the first immunization.
Outcomes	**Immunogenicity:** "the primary immunogenicity endpoint was defined as the RRV-specific IgG antibody titer determined 21 days after the second vaccine dose. Secondary immunogenicity endpoints included the number of subjects with a Ross River virus-specific seropositive IgG antibody response, number of subjects seroconverting to Ross River virus specific IgG antibody production, and the Ross River virus-specific neutralizing antibody response determined 21 days after each of the three vaccinations". **Safety:** "the primary safety outcome was the number of subjects with onset of fever within seven days of the first and second vaccination. Secondary safety outcomes were subjects with injection site reactions and related systemic adverse reactions after the first and second vaccination, any adverse event during the entire study period, and the number of subjects experiencing arthritis associated with one or more systemic symptoms commonly observed in patients with epidemic arthritis occurring at least three days after vaccination and lasting for more than 3 weeks".
Are there other groups in the trial not analysed in this review?	No

Notes	**Contacts with authors:** on 13/12/2018 authors (gerald_aichinger@baxter.com) were inquired about the followings: ● procedure for blinding of outcome assessment ● number of patients randomised per group ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) ● serological response per patient at maximum follow up No answer received. **Funding:** the study was designed and funded by Baxter. The trial only reports on vaccine-related serious adverse events.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomization was carried out by the study site using an electronic data capture system. Randomization was carried out in blocks with a block size >4 in Cohort 1 and >2 in cohorts 2 and 3 using the random number generator algorithm of Wichmann and Hill, as modified by McLeod.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Partially blinded (with respect to adjuvation). The study was double-blinded for use of adjuvant but open-label with respect to dose.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	A total of 382 subjects were randomised and received the first dose of vaccine. All vaccinated subjects were included in the safety analysis.

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	A total of 382 subjects were randomised and received the first dose of vaccine. All vaccinated subjects were included in the safety analysis.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Immunogenicity data were available from 373 subjects after the first vaccination, 375 subjects after the second vaccination, and 360 subjects after the third vaccination.
Selective reporting (reporting bias)	Low risk	The trial is registered at www.clinicaltrials.gov, NCT00717834. All outcomes assessed
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Asmuth 2010

Methods	Multicenter, blinded, randomised, placebo-controlled phase I trial
Participants	**Sex:** male and females **Age:** 18-55 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** healthy subjects between 18 and 55 years of age without evidence of HIV infection were assessed for eligibility. **Exclusion criteria:** "subjects were excluded if they were considered to be at high risk of acquiring HIV infection during the study or had a history of significant chronic medical conditions. Female subjects with childbearing potential were required to have a negative urine pregnancy test immediately before receiving any injections".
Interventions	**Vaccine type:** Human immunodeficiency virus (HIV)-1 gag **Experimental (a):** 5-mg DNA/AlPO4 + booster 5-mg DNA/AlPO4. n = 23 **Control (a):** 5 mg DNA/PBS + booster 5-mg DNA/PBS. n =55 **Experimental (b):** 5-mg DNA/AlPO4 + booster Ad5 gag. n = 56 **Control (b):** 5-mg DNA/PBS + booster Ad5 gag. n =19 **Aluminium concentration:** 700 mcg/mL **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** 4 dose regimen
Outcomes	**Immunogenicity:** neutralizing antibodies to Ad5 were measured. To assess vaccine immunogenicity, unfractionated IFN-g ELISpot assays using 15-mer clade B HIV-1 Gag peptide pools matched to the vaccine inserts were performed on PBMCs obtained at weeks 4, 8, 12, 26, 30, 34, and 42 at Merck Research Laboratories to detect HIV-1 gag-specific T cell responses after vaccination. **Safety:** clinic visits for safety evaluations were scheduled for 2–4 weeks after each dose. Study personnel monitored injection site, systemic, and laboratory adverse events.
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 13/12/2018 authors (david.asmuth@ucdmc.ucdavis.edu) were inquired about the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine. Answer received but no information or data added **Funding:** Financial support: Merck

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Subjects were randomised to receive vaccine or placebo according to allocation schedules generated by computer.
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias)	Low risk	Investigators, subjects, clinical safety monitors, and laboratory personnel performing the biological assays were blinded to treatment assignments.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	n.a.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	n.a.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Investigators, subjects, clinical safety monitors, and laboratory personnel performing the biological assays were blinded to treatment assignments.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.

Incomplete outcome data (attrition bias) Immunogenicity	High risk	Only subjects receiving all scheduled doses of vaccine and without major protocol violations were included in the analysis of immunogenicity. Drop outs more than 5%.
Selective reporting (reporting bias)	High risk	Safety and tolerability not reported. No protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias

Atsmon 2014

Methods	Two-center, randomised, placebo-controlled study
Participants	**Sex:** male and female **Age:** > 65 years **Ethnicity:** not reported **Diagnosis:** healthy but with some medical conditions **Inclusion criteria:** compatible with the Israeli guidelines for receiving the standard seasonal influenza vaccine, who were initially seronegative to at least one of the strains included in the seasonal vaccine against influenza for 2011–2012. **Exclusion criteria:** Any medical condition, clinically significant abnormality upon physical examination or laboratory safety tests,which in the investigator's judgment contraindicated administration of the vaccine, resulted in exclusion.
Interventions	**Vaccine type:** influenza Multimeric-001 (M-001) **Experimental:** influenza Multimeric-001 (M-001) + aluminium phosphate. n = 30 **Control:** influenza Multimeric-001 (M-001). n =30 **Aluminium concentration:** 2% **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Gentaur **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity:** blood was drawn for hemagglutinin inhibition evaluation at baseline (day 0), and 3 weeks after boosting with TIV. Whole blood samples were removed on day 0 (pre-immune) and 21 days after immunization. **Safety:** number of participants with adverse events, (time frame: 63 days). Safety evaluation included physical examination and measurement of vital signs on each vaccination day. Electrocardiogram and safety laboratory tests (blood chemistry and complete blood count) were performed before first vaccination and upon study termination, 3 weeks after TIV administration. Local reactions and systemic adverse events were continuously monitored.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (benyedidia@biondvax.com) were inquired on the followings: ● safety analysis up to 6 weeks (as stated in the NCT page) for serious and non-serious adverse events related and not related to the vaccine ● mortality ● final concentration of aluminium injected per dose ● number of patients who developed influenza after the vaccination

	Answer received but without extra information given **Funding:** the trial was funded by BiondVax Pharmaceuticals Ltd (Israel) and by the Office of Israeli Chief Scientist. The trial only reports on vaccine-related serious adverse events.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A computer-generated algorithm assigned the participants to 32 blocks of four
Allocation concealment (selection bias)	Low risk	An eligible participant was allocated a randomisation number by an independent, un-blinded, research coordinator who was not a member of the study teams. Due
Blinding of participants and personnel (performance bias)	High risk	The study was partially blinded.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	There were no drop outs during the study.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	There were no drop outs during the study.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	There were no drop outs during the study.
Selective reporting (reporting bias)	Low risk	A study protocol was registered prior to randomisation (NCT01419925), and all outcomes were reported on
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ayoola 1986

Methods	Randomised clinical trial
Participants	**Sex:** male and females **Age:** 1-45 years **Ethnicity:** African **Diagnosis:** healthy **Inclusion criteria:** domicile in the city of Ibadan, Nigeria. They were negative for HBV markers at the time of entry into the study. **Exclusion criteria:** not reported
Interventions	**Vaccine type:** hepatitis B Vaccine (Hevac B) **Experimental:** 2 mcg of Hevac B with aluminium hydroxide given subcutaneously. n = 65 **Control:** 2 mcg of Hevac B given intradermal. n = 61 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 doses given one month apart
Outcomes	**Immunogenicity:** hepatitis B surface antigen (HBsAg), antibody to HBsAg (anti-HBs), antibody to hepatitis B core antigen (anti-HBc) were detected by radioimmunoassay. Serum alanine aminotransferase was estimated by standard methods. **Safety:** untoward effects were documented by the completion of a specially designed questionnaires and clinical examinations at each visit.
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** no contact found **Funding:** no funding source reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised
Allocation concealment (selection bias)	Unclear risk	Not stated
Blinding of participants and personnel (performance bias)	Unclear risk	Not stated if blinded
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not stated

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not clear whether drop outs occurred
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not clear whether drop outs occurred
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Not clear whether drop outs occurred
Selective reporting (reporting bias)	Unclear risk	Not clear whether all outcome data is reported. No protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Basavaraj 2014

Methods	Randomized,double-blind, placebo-controlled, parallel assignment
Participants	**Sex:** male and females **Age:** 18-65 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female volunteers aged between 18 to 65 years inclusive Subjects of child-bearing potential must agree to practice adequate contraception that may include, but is not limited to, abstinence, barrier methods such as condoms, diaphragms, spermicides, intrauterine devices, and licensed hormonal methods during the study for at least 30 days following the last vaccination. The subject must be in good health as determined by medical history, physical examination and laboratory screening tests. Subject must be willing and able to comply with planned study procedures and be available for all study visits. Subject must provide written informed consent prior to initiation of any study procedures, and subject may provide written assent as appropriate. Subjects willing to stay for entire study duration and follow-up period in the current residential address". **Exclusion criteria:** "allergy to components of the vaccine including, thiomersal, and aluminium hydroxide. Has a history of previous severe allergic reaction with generalized urticaria, angioedema or anaphylaxis. Female with positive urine pregnancy test within 24 hours prior to vaccination or breastfeeding or unwilling to undergo pregnancy testing. History of smoking 20 pack-years or greater. Has received anticancer chemotherapy or radiation therapy (cytotoxic) within the past 36 months. Has an active neoplastic disease or a history of any hematologic malignancy. Known history of a bleeding disorder contraindicating intramuscular vaccination. History of schizophrenia, bipolar disease, suicidal attempt, major psychiatric illness or confinement for danger to self or others. History of receiving immunoglobulin, anti-cytokine antibodies, immunosuppressants, immune modifying agents or other blood product within the 3 months prior to vaccination in this study. Has received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 3 months prior to vaccination in this study or expected to receive an experimental agent during this study. Has received any live licensed vaccines within 4 weeks

	or inactivated licensed vaccines within 2 weeks prior to vaccination in this study, or plan receipt of such vaccines within 21 days following the second vaccination. Has an acute illness, including an oral temperature greater than >370C, 1 week before the vaccination. Has a chronic neurologic or autoimmune disorder. Has a history of Guillain-Barré Syndrome. Has an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe, or would interfere with the evaluation of vaccine response. Participated in a novel influenza H1N1 vaccine study in the past two years or has a history of novel influenza H1N1 2009 infection or treatment. Has known active HIV, Hepatitis B or Hepatitis C infection. Has a history of alcohol or drug addiction in the last 5 years".
Interventions	**Vaccine type:** H1N1 influenza vaccine HNVAC **Experimental (a):** 15 g of haemagglutinin protein per 0.5 mL adsorbed to 0.2% aluminium hydroxide in phosphate buffer. n = 60 **Control (a):** 15 g of haemagglutinin protein per 0.5 mL in phosphate buffer alone. n = 60 **Experimental (b):** 0.2% aluminium hydroxide in phosphate buffer. n = 20 **Control (b):** phosphate buffer alone. n = 20 **Aluminium concentration:** 0.2% aluminium hydroxide **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 1 dose
Outcomes	**Safety:** "subjects were monitored throughout the study for solicited and unsolicited local and systemic adverse events through diary cards as well as by the physician at the time of sampling. Subjects were monitored for local AEs for7 days and for systemic AEs for 21 days post-vaccination. Laboratory investigations such as haematological, biochemical and urine analyses were done at baseline (day 0) and at 21 days following immunization. Safety studies continued up to 6 weeks post-immunization. Symptoms of Guillian–Barre syndrome were evaluated throughout the study by questionnaire and by examination on days 7, 21 and 42. The subject's well being was enquired at regular intervals and a physician was on call throughout the study period, and any investigational procedures or examination was performed based on investigator's or medical officer's discretion at any time during the study". **Immunogenicity:** "blood was collected for serology studies at days 0 and 21. Serum samples were used to perform haemagglutination inhibition assay. Serological data were converted to seroconversion (SRC), sero-protection (SRP) and geometric mean titres (GMT)".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (hegden@ellafoundation.org) were inquired on the followings: ● number of patients who died/number of patients analysed ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** sponsored by Bharat Biotech International Limited

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Codes generated by the computer program Minitab 14 (Minitab Ltd., USA). The number assigned to a particular subject was unique, and the order in which the groups were assigned within the block of subjects was random.
Allocation concealment (selection bias)	Low risk	A copy of computer generated randomisation list of subject numbers and key were maintained by a non-study member located at the centre as well as the sponsor.
Blinding of participants and personnel (performance bias)	Low risk	Double blind, placebo-controlled. The group assignment was not disclosed to the subject, investigator or the laboratory staff performing the serologic assay.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind, placebo-controlled.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Safety assessment not described.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	The group assignment was not disclosed to the subject, investigator or the laboratory staff performing the serologic assay.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs accounted for
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs accounted for
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	Clinical Trial Registry Numbers: CTRI/2010/091/000152 (Phase I) andCTRI/2010/091/000661 (Phase II/III). All outcomes reported on
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bellanti 2012

Methods	Randomized, partially blinded
Participants	**Sex:** male and females **Age:** 18-45 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** healthy 18 to 45 year-olds of either sex **Exclusion criteria:** "patients receiving chronic medication, are pregnant or intend to become pregnant during the study. Individuals who are HIV positive, have abnormal liver function, had Bacillus anthracis, or received Bacillus anthracis vaccine, have received or intend to receive an experimental vaccine or medication within 30 days of injection of the experimental Bacillus anthracis vaccine".
Interventions	**Vaccine type:** bacillus anthracis **Experimental:** 10 or 20 mcg recombinant mutant protective antigen (rPA) adsorbed to 0.75 mg Alhydrogel and 0.035% formalin. n = 60 **Control:** 10 or 20 mcg recombinant mutant protective antigen (rPA) adsorbed to 0.035% formalin. n = 58 **Aluminium concentration:** 750 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag-Biosector, Frederiksund, Denmark **Vaccination schedule:** 4 doses at 0, 2, 4, and 16 months
Outcomes	**Immunogenicity:** "blood samples were taken before the 1st injection and 1 week after the 1st, 2 months after the 1st (i.e., pre-2nd), 2 months after the 2nd (pre-3rd), 2 weeks after the 3rd, 1 year after the 3rd (pre-4th), 2 weeks after the 4th, and 1 year after the 4th. Anti-rPA IgG was assayed by enzyme-linked immunosorbent assay. Anthrax lethal-toxin-neutralizing assay (TNA) was performed". **Safety:** "vaccinees were observed at the clinic for 30 min after injection and given a diary form and instructions for measuring and recording temperature and local and systemic symptoms at 6 h after injection and daily for 7 days. The vaccinees were also examined at the clinic one time following each injection, between 48 and 72 h post injection. Liver function tests were repeated 1 week after the 1st injection. Telephone inquiries about their health were made every 3 months by study nurses during the 24-month period from the 3rd injection through the year following the 4th".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (schneerr@mail.nih.gov, bellantj@georgetown.edu) were inquired on the followings: ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up

	● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** supported by the Divisions of Intramural Research, NICHD and NIAID, National Institutes of Health.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation schemes were prepared and kept by the Pharmacy Development Section, Pharmacy Department, CC, NIH, and given only to the members of the Data and Safety Monitoring Board
Allocation concealment (selection bias)	Low risk	The randomisation schemes were prepared and kept by the Pharmacy Development Section, Pharmacy Department, CC, NIH, and given only to the members of the Data and Safety Monitoring Board
Blinding of participants and personnel (performance bias)	High risk	Partially blinded; formalin-only-treated rPA was a clear fluid, distinguishable from alum-adsorbed or formalin-treated/alum-adsorbed rPA. The dosages were blinded.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All assays were performed in a blinded fashion.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	One hundred eighty-six volunteers received the 1st injection of the vaccine formulations, 175 the 2nd, 166 the 3rd, and 147 the 4th. An additional 52 volunteers were lost to the 1-year follow-up after the 4th injection.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	High risk	Clinical Trial registration number 0011462. No outcomes stated in the protocol

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Beran 2018

Methods	Rrandomised, observer-blinded, controlled study
Participants	**Sex:** male and females **Age:** 18-45 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects who, in the opinion of the investigator, can and will comply with the requirements of the protocol. Written informed consent obtained from the subject prior to performing any study specific procedure. Non-pregnant female between, and including, 18 and 45 years of age at the time of vaccination. Healthy subjects as established by medical history and clinical examination before entering into the study. Female subjects of non-childbearing potential may be enrolled in the study. Female subjects of childbearing potential may be enrolled in the study, if the subject:has practiced adequate contraception for 30 days prior to vaccination, and has a negative pregnancy test on the day of vaccination and has agreed to continue adequate contraception during the entire study period". **Exclusion criteria:** "use of any investigational or non-registered product other than the study vaccine within 30 days prior to study vaccination, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the subject has been or will be exposed to an investigational or a non-investigational vaccine/product. Chronic administration of immunosuppressants or other immune-modifying drugs, as well as administration of long-acting immune-modifying drugs during the period starting 6 months prior to study vaccination, or planned administration during the study period. Inhaled and topical steroids are allowed. Administration of immunoglobulins and/or any blood products during the period starting 3 months before study vaccination or planned administration during the study period. Planned administration/administration of a vaccine not foreseen by the study protocol in the period starting 30 days before and ending 30 days after the study vaccination, with the exception of any licensed influenza vaccine which may be administered ≥ 15 days before or after study vaccination. Previous experimental vaccination against RSV. Family history of congenital or hereditary immunodeficiency. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. History of or current auto-immune disease. Acute or chronic, clinically significant pulmonary, cardiovascular, hepatic or renal functional abnormality as determined by physical examination and/or Medical History. Lymphoproliferative disorder or malignancy within previous 5 years. History of hypersensitivity after a previous dose of any tetanus, diphtheria, or pertussis vaccine or to any component of Boostrix. History of encephalopathy of unknown aetiology occurring within 7 days following a previous vaccination with pertussis-containing vaccine. History of any neurological disorders or seizures. History of transient thrombocytopenia or neurological complications following a previous vaccination against diphtheria and/or tetanus. History of any reaction or hypersensitivity likely to be exacerbated by any component of the study vaccines. Hypersensitivity to latex. Any medical condition that in the judgment of

	the investigator would make intramuscular injection unsafe. Current chronic alcohol consumption and/or drug abuse. Acute disease and/or fever at the time of enrolment. Body mass index (BMI) > 40 kg/m2. Pregnant or lactating female. Planned move to a location that will prohibit participating in the trial until study end. Any other condition that the investigator judges may interfere with study procedures".
Interventions	**Vaccine type:** respiratory syncytial virus (RSV) vaccine **Experimental:** 60 µg of aluminium-adjuvanted RSV-PreF. n = 125 **Control:** 60 µg of non adjuvated RSV-PreF. n = 124 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** GSK **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity:** "RSV-A neutralizing antibodies on Vero cells were quantified using the RSV plaque-reduction neutralization assay. An indirect enzyme-linked immunosorbent assay format was used to quantify PreF total IgG and IgG subclass 1 antibodies". **Safety:** "number of subjects with abnormal biochemical laboratory values at day 7 and 30. Number of subjects with abnormal haematological laboratory at day 7 and at day 30. Number of subjects with abnormal biochemical laboratory parameter values by maximum grading from day 7 up to day 30. Number of subjects with solicited local symptoms during a 7-day follow-up period after vaccination. Assessed solicited local symptoms were pain, redness and swelling. Any = occurrence of the symptom regardless of intensity grade. Grade 3 pain = pain that prevented normal activity. Grade 3 redness/swelling = redness/swelling spreading beyond 100 millimetres (mm) of injection site. All solicited local symptoms are considered as related to the vaccination. Number of subjects with solicited general symptoms during a 7-day follow-up period after vaccination. Assessed solicited general symptoms were fatigue, temperature (defined as oral temperature equal to or above [≥] 37.5 degrees Celsius [°C] for oral, axillary or tympanic route), gastrointestinal symptoms (gastro) including nausea, vomiting, diarrhoea and/or abdominal pain; and headache. Any = occurrence of the symptom regardless of intensity grade and relationship to the vaccination. Grade 3 symptom = symptom that prevented normal activity. Grade 3 fever = fever ≥ 39.5 °C. Related = symptom assessed by the investigator as related to the vaccination. Number of subjects with unsolicited adverse events during a 30-day follow-up period after vaccination. An unsolicited AE covers any untoward medical occurrence in a clinical investigation subject temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product and reported in addition to those solicited during the clinical study and any solicited symptom with onset out-side the specified period of follow-up for solicited symptoms. Any was defined as the occurrence of any unsolicited AE regardless of intensity grade or relation to vaccination. Number of subjects with serious adverse events from vaccination (day 0) up to study end (day 30). Serious adverse events assessed include medical occurrences that result in death, are life threatening, require hospitalisation or prolongation of hospitalisation or result in disability/incapacity".
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 16/12/2019 authors (jiri.beran@vakcinace.cz) were inquired on the followings: ● which outcome assessors were blinded ● number of participants who were infected by RSV after the vaccination Answer received **Funding:** supported by GSK Biologicals SA, which was involved in all stages of the study conduct and analysis, and also took responsibility for all costs associated with the development and publishing of the present manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants in each study were randomised to study groups of equal size at the investigator sites, using a web-based program. The randomisation algorithm used a minimization procedure that addressed the age of the participant (18–32 years or 33–45 years)
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Only states observer blind
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	All outcome assessors were blinded.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	All outcome assessors were blinded.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All outcome assessors were blinded.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	The safety analysis was performed on the total vaccinated cohort in each study.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The safety analysis was performed on the total vaccinated cohort in each study.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	The analysis of vaccine immunogenicity in RSV F-020 was performed on the according-to-protocol cohort (all vaccinated subjects meeting eligibility criteria and protocol-defined procedures). Drop outs more than 5%
Selective reporting (reporting bias)	Low risk	Two study protocol was registered prior to randomisation (NCT02360475), and all outcomes were reported on.

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bernstein 2008

Methods	Multicenter randomised, double-blind, placebo-controlled trial conducted at 4 sites in the United States
Participants	**Sex:** male and females **Age:** 18-64 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy male or nonpregnant female (as indicated by a negative urine pregnancy test immediately prior to vaccine administration) between the ages of 18 and 64 years, inclusive. Women of childbearing potential (not surgically sterile or post menopausal for greater than or equal to one year) must agree to practice adequate contraception (i.e., barrier method, abstinence, and licensed hormonal methods) for the entire study period. Is in good health, as determined by vital signs (heart rate, blood pressure, oral temperature), medical history and a targeted physical examination based on medical history. Subjects should have normal safety laboratory values (Hgb, WBC, Plt, ALT, and creatinine) prior to the first immunization. Able to understand and comply with planned study procedures. Provides written informed consent prior to initiation of any study procedures". **Exclusion criteria:** "subjects were excluded if they had a known allergy to eggs or other vaccine components; had previously received any influenza A/H5 vaccine; were immunosuppressed; were known to have HIV, hepatitis B, or hepatitis C infection; had a history of Guillain-Barré syndrome; had received blood products in the past 3 months; had received killed vaccines in the past 2 weeks or live vaccines in the past 4 weeks; had an acute or chronic illness that would interfere with evaluations or increase risk; or had a history of severe reactions after influenza vaccination. Subjects were also excluded if they had laboratory values outside of the normal range on their screening blood tests, including hemoglobin level, white blood cell count, platelet count, alanine aminotransferase level, and creatinine level".
Interventions	**Vaccine type:** avian Influenza H5N1 vaccine **Experimental (a):** 15 mcg H5N1 combined with aluminium hydroxide. n = 31 **Control (a):** 15 mcg H5N1. n = 30 **Experimental (b):** 30 mcg H5N1 combined with aluminium hydroxide. n = 59 **Control (b):** 30 mcg H5N1. n = 61 **Aluminium concentration (a):** 175 mcg/dose. **(b)** 300 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Chiron, Novartis **Vaccination schedule:** each subject received 2 intramuscular injections of an identical formulation in the deltoid area, administered approximately 28 days apart.
Outcomes	**Immunogenicity:** "geometric mean titer and frequency of 4-fold or greater increases in serum hemagglutination inhibition (HAI) antibody titers in each group 28 days after receipt of the second dose of vaccine (time frame: approximately day 56). Proportion of subjects in each group achieving a serum neutralizing antibody titer ratio of 1:40 against the influenza A/H5N1 virus 28 days following second dose of vaccine, (time frame: approximately day 56).

	Proportion of subjects in each dose group achieving a serum hemagglutination (HAI) antibody titer of 1:40 against the influenza A/H5N1 virus 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Geometric mean titer (GMT) and frequency of 4-fold or greater increases in neutralizing antibody titers in each group 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Geometric mean titer and the frequency of 4-fold or greater increases in neutralizing antibody titers in each group 1 month after receipt of each dose, and 7 months after receipt of the first dose of vaccine, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization). Geometric mean titer and the frequency of 4-fold or greater increases in serum HAI antibody titers 1 month after receipt of each dose, and 7 months after receipt of the first dose of vaccine, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization). Development of serum antibody responses against antigenically drifted variants of H5N1influenza virus, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization)". **Safety:** "adverse event or SAE information (solicited in-clinic and via memory aids, concomitant medications, and periodic targeted physical assessments), (time frame: adverse events will be collected through 28 days following the second dose of vaccine (approximately day 56). Serious adverse events will be collected throughout the study through day 208)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (david.bernstein@cchmc.org) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** National Institutes of Health

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Eligible subjects were randomly assigned to 1 of 9 groups. No further information on how randomisation was achieved.
Allocation concealment (selection bias)	Unclear risk	Allocation concealment not described. After randomisation, an unblinded member of the study team who did not otherwise participate in the trial administered the injection.
Blinding of participants and personnel (performance bias)	Unclear risk	Only stated double blind but no information on how the blinding was achieved and maintained.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	The site staff who evaluated vaccine reactogenicity were fully blinded to the dose and type of vaccine given.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	The site staff who evaluated vaccine reactogenicity were fully blinded to the dose and type of vaccine given.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00280033), and all outcomes were reported on.
Other bias	Low risk	SAE reported only as related to the vaccine.

Bezay 2016

Methods	Open-label, partially randomised, dose-escalation Phase I trial in two parts
Participants	**Sex:** male and females **Age:** > 65 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "no clinically relevant pathological findings in any of the investigations at the Screening Visit including subjects with pharmacologically controlled conditions like hypercholesterolemia, hypertension, cardiovascular disease or non insulin-dependent diabetes mellitus. Minor deviations of laboratory values from the normal range may be accepted, if judged by the

investigator to have no clinical relevance. Weight: ≥ 45.5 kg and <= 150 kg at Visit 0 (Screening Visit). White blood cells ≥2,500/mm3 and <11,000/mm3 at Visit 0.
Platelets within normal limits at Visit 0. Written informed consent obtained from the subject prior to any study related procedures".
Exclusion criteria: "use of any other investigational or non-registered medicinal product within 30 days prior to IC84 vaccination at Visit 1 (Day 0) and throughout the entire study period.
Active or passive vaccination four weeks before first vaccination at Visit 1 and during the entire study period. Immunodeficiency including status post-organ-transplantation or immuno-suppressive therapy, and a family history of congenital or hereditary immunodeficiency. Infection with the human immunodeficiency virus (HIV, a negative test result within 30 days before screening is acceptable), Hepatitis B virus (HBV, Hepatitis B surface antigen [HBsAg]) or Hepatitis C virus (HCV). History of severe hypersensitivity reactions and anaphylaxis.
History of allergic bronchial asthma and severe allergic rhinoconjunctivitis. Known hypersensitivity or allergic reactions to one of the components of the vaccine. History of autoimmune disease, including Type I Diabetes mellitus. Subjects with vitiligo or thyroid disease taking thyroid hormone replacement are not excluded. Any malignancy in the past 5 years. If treatment for cancer was successfully completed more than 5 years ago and the malignancy is considered to be cured, the subject may be enrolled. Clinically significant hematological, renal, hepatic, pulmonary, central nervous, cardiovascular or gastrointestinal disorders, which are not adequately controlled by medical treatment within the last 12 weeks before IC84 vaccination at Visit 1 (Day 0) as judged by the site's Principal Investigator. Systemic antibiotic use within four weeks prior to first vaccination and during treatment period (until Day 28). Clinically significant diseases as judged by the investigator. Administration of chronic (defined as longer than 14 days) immunosuppressants or other immune-modifying drugs within 30 days prior to IC84 vaccination at Visit 1 (Day 0) and during the study until Visit 5 (Day 28). (For corticosteroids this means prednisone or equivalent >= 0.05 mg/kg/day; topical and inhaled steroids are allowed.). Periodic steroid injections, e.g., intra-articular, are not allowed within 30 days prior to first IC84 vaccination at Visit 1 (Day 0) and until Visit 5 (Day 28). Intake of NSAID within three days prior to and within three days after all three vaccinations (Day 0, 7 and 21). Acute febrile infections or exacerbation of chronic infection within four weeks prior to first vaccination at Visit 1 and during treatment period (until Day 21). Body temperature > 37°C (oral) immediately prior to each vaccination. Drug addiction within 6 months prior to Visit 0 [including alcohol dependence, i.e. more than approximately 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day] indicated by an elevated MCV above normal value at the Screening Visit. Inability or unwillingness to avoid more than the usual intake of alcohol [i.e., not more then 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day] during the 48 hours after vaccination. Pregnancy (positive pregnancy test during screening), lactation or unreliable contraception in female subjects with child-bearing potential (and unreliable contraception in male subjects. Not able to understand and comply with protocol requirements, instructions and protocol-stated restrictions. Blood donation within 4 weeks prior to first vaccination. Clinically significant mental disorder not adequately controlled by medical treatment. History of Guillain-Barré-Syndrome (GBS). Any condition which might interfere with study objectives or would limit the subject's ability to

	complete the study in the opinion of the investigator. Inability or unwillingness to provide informed consent. Persons who are committed to an institution (by virtue of an order issued either by the judicial or the administrative authorities)".
Interventions	**Vaccine type:** VLA84 vaccine against *Clostridium difficile* **Experimental:** VLA84 75 or 200mcg with aluminium hydroxide. n =41 **Control:** VLA84 75 or 200 mcg. n = 40 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** unclear **Vaccination schedule:** 4 doses at day 0, 7, 28, 56. Immunizations were injected intramuscularly into the deltoid of the non-dominant arm.
Outcomes	**Immunogenicity:** "determination of vaccine (IC84) specific IgG levels on Day 0, 7, 14, 21, 28, 113 and 201 after the first vaccination using an Enzyme-Linked Immunosorbent Assay, (time frame: day 201). Determination of *Clostridium difficile* Toxin A- and Toxin B-specific IgG levels on Day 0, 7, 14, 21, 28, 113 and 201 after the first vaccination using an Enzyme-Linked Immunosorbent Assay, (time frame: day 201). Determination of vaccine-induced C. difficile Toxin A- and Toxin B-neutralizing antibody levels on Day 0, 7, 14, 21, 28, 113 and 201 after the first vaccination using a Toxin Neutralization Assay, (time frame: day 201)". **Safety:** "rate of subjects with any serious adverse event possibly, probably or certainly related to the study vaccine at any time during the study, (time frame: day 201). Rate of subjects with any unsolicited or solicited Grade 3 or higher adverse event possibly, probably or certainly related to the study vaccine at any time during the study, (time frame: day 201). Rate of subjects with any unsolicited or solicited Grade 3 or higher adverse event possibly, probably or certainly related to the study vaccine during treatment phase (i.e. until day 28), (time frame: day 201). Rate of subjects with solicited local adverse events (injection site pain, tenderness, redness, swelling, induration, itching) within 1 week (Day 0-6) after each vaccination: Severity and duration, (time frame: 6 days after vaccination). Rate of subjects with solicited systemic adverse event (headache, muscle pain, fever, flu-like symptoms, nausea, vomiting, rash, excessive fatigue) within 1 week (Day 0 6) after each vaccination: Severity and duration, (time frame: 6 days after vaccination). Rate of subjects with unsolicited non-serious adverse events (including safety laboratory parameters (hematology, serum chemistry, urinalysis)) within 6 months after last vaccination: Severity and causality, (time frame: day 201). Rate of subjects with unsolicited non-serious adverse events (including safety laboratory parameters (hematology, serum chemistry, urinalysis) during treatment phase (i.e., until day 28): Severity and causality, (time frame: day 28)".
Are there other groups in the trial not analysed in this review?	Yes.
Notes	**Contacts with authors:** on 16/12/2019 authors (bernd.jilma@meduniwien.ac.at, nicole.bezay@valneva.com) were inquired on the followings: ● final aluminium concentration per dose ● aluminium manufacturer ● Method of allocation concealment ● randomization sequence generation

	● total number of serious adverse event related and not related to the vaccine ● number of participants who got Clostridium difficile infection after the vaccine Answer received but no further data added **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only stated randomised. No further information on how randomisation was achieved
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No dropouts
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	No dropouts
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01296386), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bologa 2012

Methods	A phase I, single-center, randomised, placebo controlled, repeated administration, dose-escalating, observer blinded study
Participants	**Sex:** male and females **Age:** 18-50 years **Ethnicity:** not stated **Diagnosis:** healthy **Inclusion criteria:** "aged 18 to 50 years on the day of inclusion, iInformed consent form has been signed and dated, able to attend all scheduled visits and comply with all trial procedures Subject is healthy, as determined by medical history and physical examination, for a woman of child-bearing potential, use of an effective method of contraception or abstinence from at least 4 weeks prior to first vaccination until 4 weeks after the last vaccination". **Exclusion criteria:** "known pregnancy or positive serum/urine pregnancy test, currently breastfeeding a child, chronic illness, that in the opinion of the investigator, is at a stage that could interfere with trial conduct or completion, participation in another clinical trial investigating a vaccine, drug, medical device, or a medical procedure in the 4 weeks preceding the first trial vaccination, planned participation in another clinical trial during the present trial period, known or suspected congenital or acquired immunodeficiency, or receipt of immunosuppressive therapy such as anti-cancer chemotherapy or radiation therapy within the preceding 6 months; or long-term systemic corticosteroid therapy (prednisone or equivalent for more than 2 consecutive weeks within the past 3 months), known systemic hypersensitivity to any of the vaccine components or history of a life threatening reaction to the trial vaccine(s) or to a vaccine containing any of the same substances, current alcohol abuse or drug addiction that may interfere with the subject's ability to comply with trial procedures. Receipt of blood or blood-derived products in the past 3 months, that might interfere with the assessment of immune response, receipt of any vaccine in the 4 weeks preceding the first trial vaccination, planned receipt of any vaccine in the 4 weeks following the trial vaccination, known seropositivity for Human Immunodeficiency Virus (HIV), Hepatitis B or Hepatitis C. Laboratory confirmed / self-reported thrombocytopenia contraindicating intramuscular vaccination, deprived of freedom by an administrative or court order, or in an emergency setting, or hospitalized involuntarily, identified as a study site employee who is involved in the protocol and may have direct access to trial-related data, previous vaccination against pneumococcal disease (in the previous 5 years), history of pneumococcal infection (confirmed either clinically, serologically, or microbiologically) within 5 years, at high risk for pneumococcal infection during the trial, living in a household with children < 5 years of age".
Interventions	**Vaccine type:** pneumococcal protein vaccine PhtD-Pcp **Experimental:** 25 mcg PhtD 25 mcg PcpA adjuvated with aluminium hydroxide. n = 21 **Control:** 25 mcg PhtD 25 mcg PcpA. n = 19 **Aluminium concentration:** 280 mcg/dose **Aluminium type:** aluminium hydroxide Alhydrogel pre-treated with phosphate buffer **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** 2 injections 30 days apart

Outcomes	**Safety:** number and percentage of participants reporting solicited injection site reactions, solicited systemic reactions, unsolicited systemic reactions, and serious adverse events occurring throughout the trial, (time frame: 30 days post-vaccination). Solicited injection site reactions: pain, erythema, and swelling. Solicited systemic reactions: fever, headache, malaise, and myalgia. **Immunogenicity:** immunogenicity of investigational pneumococcal vaccines after 2 vaccinations, (time frame: days 0 and 30 post-vaccination). Evaluation of immune responses to antigen component of the investigational vaccine by enzyme linked immunosorbent assay.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (monica@mbpharma.ca, sanjay.Gurunathan@sanofipasteur.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no information or data added **Funding:** funded by Sanofi Pasteur, which also contributed to the study design, data collection, analysis and interpretation, review of the manuscript and decision to publish.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further information on how randomisation was achieved.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Observer blinded study. Unclear how blinding was achieved and maintained.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Observer blinded study. Blinded safety reviews by a Sanofi Pasteur safety review committee.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Observer blinded study. Blinded safety reviews by a Sanofi Pasteur safety review committee.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Both full analysis set and per protocol set (only results from per protocol presented)
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	No drop outs
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01444339), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Brady 2009

Methods	Multicenter, randomised, dose-ranging clinical trial
Participants	**Sex:** male and females **Age:** >65 **Ethnicity:** mixed **Diagnosis:** ambulatory and judged to be medically stable for any underlying conditions **Inclusion criteria:** "participants are ambulatory adults aged 65 and older, are in good health, as determined by vital signs (heart rate less than 100 bpm, blood pressure [systolic less than or equal to 160 mm Hg and diastolic less than or equal to 90 mm Hg] oral temperature less than 100.0°F), medical history to ensure stable medical condition, and targeted physical examination based on medical history, are able to understand and comply with planned study procedures, provide written informed consent prior to initiation of any study procedures, stable medical condition - no change in prescription medication, dose, or frequency of medication in the last 3 months and health outcomes of the specific disease are considered to be within acceptable limits in the last 6 months. Any change that is due to change of health care provider, insurance company, etc, or that is done for financial reasons, as long as in the same class of medication will not be considered a violation of this inclusion criterion. Any change in prescription medication due to improvement of a disease outcome will not be considered a violation of this inclusion criterion." **Exclusion criteria:** "have a known allergy to eggs or other components of the vaccine (including gelatin, formaldehyde, octoxinol, thimerosal, aluminium hydroxide, and chicken protein). Have immunosuppression as a result of an underlying illness or treatment, or used anticancer chemotherapy or radiation therapy within the preceding 36 months. Have any malignancy (excluding nonmelanotic skin cancer) or lymphoproliferative disorder diagnosed or treated actively during the past 5 years. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (> 800 micrograms/day of beclomethasone dipropionate or equivalent) within the preceding 6 months (Nasal and topical steroids are allowed.) Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to

	self or others. Are receiving psychiatric drugs (aripiprazole, clozapine, ziprasidone, haloperidol, molindone, loxapine, thioridazine, thiothixene, pimozide, fluphenazine, risperidone, mesoridazine, quetiapine, trifluoperazine, chlorprothixene, chlorpromazine, perphenazine, trifluopromazine, olanzapine, carbamazepine, divalproex sodium, lithium carbonate or lithium citrate.) Subjects who are receiving a single antidepressant drug and stable for at least 3 months prior to enrollment, without de-compensating symptoms will be allowed to be enrolled in the study. Have a history of receiving immunoglobulin or other blood product within the 3 months prior to enrollment in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to enrollment in this study. Have an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses. These conditions include, but are not limited to: history of significant renal impairment (dialysis and treatment for kidney disease, including diabetic and hypertensive kidney disease); subjects with diabetes mellitus, well-controlled with oral agents may enroll as long there has been no dose adjustment within the past 6 months; insulin-dependent diabetes is excluded; cardiac insufficiency, if heart failure is present (New York Heart Association Functional Class III or IV); an arteriosclerotic event during the 6 months prior to enrollment (e.g., history of myocardial infarction, stroke, recanalization of femoral arteries, or transient ischemic attack). Have a history of severe reactions following immunization with influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week prior to vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product or medication) within 1 month prior to enrollment in this study, or expects to receive an experimental agent during the 7- to 8-month study period. Have any condition that would, in the opinion of the site investigator, place the subject at an unacceptable risk of injury or render the subject unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Planned to travel outside of the USA in the time between the first vaccination and 56 days following the first vaccination. Have any diagnosis of dementia or associated concomitant medications (e.g., Aricept) used for treating dementia. Have a history of Guillain Barre syndrome. Have any condition that the investigator believes may interfere with successful completion of the study".
Interventions	**Vaccine type:** subvirion Inactivated Influenza A/H5N1 vaccine **Experimental** **(a):** 3.75 mcg of A/H5N1 vaccine with aluminium hydroxide. n = 57 **Control (a):** 3.75 mcg of A/H5N1 vaccine. n = 60 **Experimental (b):** 7.5, 15 or 45 mcg of A/H5N1 vaccine with aluminium hydroxide. n = 243 **Control (b):** 7.5, 15 or 45 mcg of A/H5N1 vaccine. n = 239 **Aluminium concentration: (a)** 300 mcg/dose. **(b)** 600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** sanofi pasteur, Swiftwater, PA **Vaccination schedule:** two doses administered approximately 28 days apart into the deltoid muscle

Outcomes	**Immunogenicity:** "geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Proportion of subjects in each dose group achieving a serum hemagglutination inhibition (HAI) antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Proportion of subjects in each dose group achieving a serum neutralizing antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus 28 days after receipt of the second dose of vaccine, (time frame: approximately day 56). Development of serum antibody responses against antigenically drifted variants of H5N1 influenza virus, (time frame: day 0, and at days 28, 56, and 208 after the first immunization). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group 1 month and 7 months after receipt of the first dose of vaccine, (time frame: day 0, and at days 28, 56, and 208 after the first immunization). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group 1 month and 7 months after receipt of the first dose of vaccine, (time frame: day 0, and at days 28, 56, and 208 after the first immunization)". **Safety:** "adverse event or serious adverse event information (solicited in the clinic and via memory aids, concomitant medications, and periodic targeted physical assessment), (time frame: adverse events will be collected through 28 days following the 2nd dose of vaccine (day 56). Serious adverse events will be collected throughout the study through day 208)".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (hhill@emmes.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** research supported by the authors and summarized in this report was supported by Public Health Service Contracts N01-AI-25459, N01-AI-25465, N01-AI-25460, N01-AI-25461, N01-AI-45250, and N01-AI-30068 from the National Institute of Allergy and Infectious Diseases. Partial support was also provided by the University of Maryland General Clinical Research Center Grant

	M01-RR-16500, General Clinical Research Centers Program, National Center for Research Resources, National Institutes of Health. Wilbur Chen, M.D. receives support from K12RR023250.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further information on how randomisation was achieved.
Allocation concealment (selection bias)	Unclear risk	No description of allocation concealment.
Blinding of participants and personnel (performance bias)	Low risk	All subjects as well as site and laboratory staff, except the vaccinators, were blinded to the vaccines administered. The vaccinators were not involved in the assessment of responses after immunization.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	600 enrolled, 599 received 1 vaccination, 545 received 2 vaccination. The subject who received an incorrect dose as the first vaccine was excluded from the safety and immunogenicity analyses. Two subjects who did not receive the second vaccine and did not provide post-vaccination blood samples were excluded from all immunogenicity analyses. One subject did not return for a blood sample 28 days after the first vaccination and was consequently not included in the immunogenicity analyses for this time point, but this subject was included in all other analyses for which data were available.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	High risk	600 enrolled, 599 received 1 vaccination, 545 received 2 vaccination. The subject who received an incorrect dose as the first vaccine was excluded from the safety and immunogenicity analyses. Two subjects who did not receive the second vaccine and did not provide post-vaccination blood samples were excluded from all immunogenicity analyses. One subject did not return for a blood sample 28 days after the first vaccination and was consequently not

		included in the immunogenicity analyses for this time point, but this subject was included in all other analyses for which data were available.
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00294099), and all outcomes were reported on. However, SAE reported only as related to the vaccine. Thirty-seven serious adverse events, including two deaths, were reported during the study. None of these SAEs was considered to be associated with vaccination. Not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Bresson 2006

Methods	Randomised, open-label, non-controlled phase I trial
Participants	**Sex:** male and females **Age:** 18–40 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "age 18–40 years on day of enrolment, provision of written informed consent, ability to attend all scheduled visits and to comply with all trial procedures, entitled to national social security, and registered in the French file of healthy volunteers in clinical trials. Women had to be unable to bear children or able to produce a negative urine pregnancy test, having used an effective method of contraception or been abstinent for at least 4 weeks before the first vaccination (to be continued for at least 4 weeks after the second vaccination)". **Exclusion criteria:**" current febrile illness; immunodeficiency; recent receipt of blood, blood derive products, or immunosuppressive treatment; current long-term systemic corticosteroid therapy; recent (previous 4 weeks) vaccination or vaccination planned in next month; and pregnancy".
Interventions	**Vaccine type:** inactivated split-virion influenza A/Vietnam/1194/2004 (H5N1) vaccine **Experimental:** 7.5, 15 or 30 mcg H5N1 vaccine combined with aluminium hydroxide and 45 mcg of thiomerosal. n = 151 **Control:** 7.5, 15 or 30 mcg vaccine and 45 mcg of thiomerosal. n = 149 **Aluminium concentration:**600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** unclear **Vaccination schedule:** two intramuscular injections, 21 days apart, of the assigned formulation into the deltoid
Outcomes	**Immunogenicity:** "serum samples obtained from individuals on day 0 (before first vaccination), 21, and 42 were assayed at the same time to detect haemagglutination-inhibition activity and neutralising activity against the vaccine seed virus (NIBRG14). Haemagglutination-inhibition assay was used. Neutralising antibody activity according was performed". **Safety:** "participants attended three trial visits (on day 0, 21, and 42) for blood sampling, vaccination (day 0 and 21 only), and safety data collection. Iindividuals were kept under observation for 30 min after vaccination, and were given safety diaries, digital

	thermometers, and rulers to assess and record adverse events. From days 0 to 7, diaries included a list of solicited injection site and systemic adverse events, including those recommended for the assessment of interpandemic vaccines by the CHMP. The intensity of non-quantifiable reactions—e.g., pain—was assessed with a severity scale: mild, noticeable but does not interfere with daily activities; moderate, interferes with daily activities; severe, prevents daily activities. At the next visit (day 21 and 42), patients were interviewed, adverse events transcribed onto case report forms, and unsolicited adverse events were assessed whether vaccine-related. By convention, all solicited adverse events we judged vaccine-related".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (maria.zambon@hpa.org.uk) were inquired about the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further information or data added **Funding:** funded by a grant from sanofi pasteur. Sanofi pasteur had a role, through its project team, in study design and monitoring, and in writing of the report.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation lists was stratified by centre and created with the block method, using decreasing block sizes of 18, 12, and six so that a similar number of people was enrolled into each group at any given time. The sequence was generated at sanofi pasteur by a statistician who was not involved in the rest of the trial.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described

Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	All 300 individuals received the two planned vaccinations, attended all visits, completed the study up to day 42, gave three blood samples, and were included in analyses.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	All 300 individuals received the two planned vaccinations, attended all visits, completed the study up to day 42, gave three blood samples, and were included in analyses.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	All 300 individuals received the two planned vaccinations, attended all visits, completed the study up to day 42, gave three blood samples, and were included in analyses.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Brooks 2015

Methods	Phase I, single-center, randomised, placebo-controlled, observer-blind study
Participants	**Sex:** male and females **Age:** infants 42-49 days of age **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "aged 6 to 7 weeks (42 to 49 days) for infants and 12 to 13 months for toddlers on the day of inclusion. Born at full term of pregnancy (≥ 37 weeks) and having a weight ≥ 60% of the median weight for Bangladesh as measured by the weight-to-age index · Written informed consent obtained from the parents or legal guardians of infants and toddlers. Subject and parent/guardian was able to attend all scheduled visits and to comply with all trial procedures". **Exclusion criteria:** "participation or planned participation in another clinical trial investigating a vaccine, drug, medical device, or medical procedure in the 4 weeks preceding the first trial vaccination except if non-interventional follow-up for an earlier study (e.g., long-term surveillance). Receipt of any vaccine in the 4 weeks preceding the first trial vaccination or planned receipt of any other vaccine than study vaccines in the 4 weeks following any trial vaccination. Previous vaccination against *S. pneumoniae.* History of *S. pneumoniae* infection, confirmed microbiologically. Receipt of blood or blood-derived products in the past 30 days (infants) that might interfere with assessment of the immune response. Known or suspected congenital or acquired immunodeficiency; receipt of immunosuppressive therapy such as anti-cancer chemotherapy or radiation therapy; or long-term systemic corticosteroid therapy (prednisone or equivalent for more than 2 consecutive weeks. Known systemic hypersensitivity to any of the vaccine components, or history of a

	life-threatening reaction to the vaccine used in the trial or to a vaccine containing any of the same substances. Bleeding disorder, including thrombocytopenia contraindicating intramuscular vaccination, or receipt of anticoagulants in the 3 weeks preceding inclusion. History of seizures. Chronic illness that, in the opinion of the investigator, was at a stage where it could have interfered with trial conduct or completion. Parents or guardians identified as employees of the investigator or study center with direct involvement in the proposed study or other studies under the direction of that investigator or study center, as well as family members (i.e., immediate, husband, wife and their children, adopted or natural) of the employees or the investigator. Febrile illness (temperature ≥ 38.0°C) or moderate or severe acute illness/infection (according to Investigator judgment) on the day of vaccination. Receipt of oral or injected antibiotic therapy within 72 h prior to any blood draw".
Interventions	**Vaccine type:** trivalent recombinant PcpA, PhtD,and PlyD1 Pneumococcal protein vaccines (PPrVs) **Experimental:** 25 mcg PPrV + phosphate-treated aluminium hydroxide. n = 40 **Control:** 25 mcg PPrV. n = 40 **Aluminium concentration:** 280 mcg/dose **Aluminium type:** phosphate-treated aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** infants received three injections at 6, 10, and 14 weeks of age
Outcomes	**Immunogenicity**: "serum anti-PhtD, anti-PcpA, and anti-PlyD1 concentrations at day 0 and month 1 for adults and toddlers and at day 0, month 2, and month 3 for infants. Enzyme-linked immunosorbent assays were used used to measure anti-PcpA, anti-PhtD and anti-PlyD1 antibody concentrations". **Safety:** "occurrence,time to onset, duration, action taken, and intensity of solicited reactions; occurrence, nature, and intensity of all unsolicited AEs; and occurrence, nature, time to onset, duration, seriousness criteria, relationship to vaccination, and outcome for SAEs. For systemic unsolicited AEs, relationship to the vaccine, action taken, and whether the event occurred within 30 min after vaccination were also recorded. Solicited reactions were recorded daily for 7 days following each vaccination. AEs and SAEs were defined as described in the International Conference for Harmonization E2A Guidelines. Unsolicited AEs were recorded up to 30 days after each vaccination. SAEs were recorded throughout the study. Immediate AEs were defined as those occurring within 30 min after the vaccination".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (robert.hopfer@sanofipasteur.com) were inquired about the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs

	related and not related to the vaccine) No answer received **Funding:** funded by Sanofi Pasteur. Sanofi Pasteur participated in study design; the collection, analysis and interpretation of data; writing of the report; and the decision to submit the article for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Treatments were assigned using scratch-off lists, which were centrally generated by non-stratified randomisation with a fixed block size within cohorts.
Allocation concealment (selection bias)	Low risk	Centrally generated scratch-off lists
Blinding of participants and personnel (performance bias)	Low risk	Observer-blind study. The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	The subject, investigator, study staff who collected the safety data and samples and the laboratory personnel who analysed the blood samples were blinded to the vaccine administered.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	One drop out in experimental group and 2 drop outs in control group.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	One drop out in experimental group and 2 drop outs in control group.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (WHO Universal Trial Number U1111-1117-7316; NCT01446926), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Brown 2010

Methods	Randomized, double-blind, placebo controlled
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "citizens of the U.S., age 18 to 40 years, for women, a negative serum pregnancy test will be required at study entry and within 24 hours prior to each vaccination, as well as verbal assurance that adequate birth control measures are applied prior to initial vaccination and for 3 months after the last vaccination. Good health as determined by medical history, physical examination, and clinical judgment, normal baseline clinical laboratory values at screening, availability for at least 13 months of follow-up from the time of the screening visit, successful completion of the Test of Understanding defined as 90% correct with three opportunities to take test, commitment for trial participation and signature of the approved consent form". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** rPA Anthrax vaccine **Experimental:** 5, 25, 50 or 100 mcg rPA with Alhydrogel. n = 30 **Control:** 5, 25, 50 or 100 mcg rPA. n = 32 **Aluminium concentration:** 704 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** BioSector (Frederikssund, Denmark) **Vaccination schedule:** two intramuscular injections in the left deltoid four weeks apart
Outcomes	**Immunogenicity:** "longer term safety and immune response data was collected at visits 6, 10, 16, 26, 36, and 52 weeks after the initial injection. Enzyme-linked immunosorbent assay, lymphocyte proliferation assay and toxin neutralization assay were performed". **Safety:** "safety data was analysed and reported on every subject at every available time point immediately after initial injection. These include immediate reactions and at 48–72 hours and 2 weeks post each injection. Longer term safety and immune response data was collected at visits 6, 10, 16, 26, 36, and 52 weeks after the initial injection. Serious adverse events were solicited at all time points through the 52 week visit. Diary of local reactions such as pain, tenderness, or warmth and systemic reactions such as fatigue, headache, or diarrhea for one week post-immunization which was used as memory aid to identify solicited adverse events (post injection reactions). The post injection reactions were recorded separate from other adverse events unless a post injection reaction persisted beyond one week post-injection".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 13/12/2018 authors (mrobb@hivresearch.org) were inquired about the followings: ● procedure for blinding of participants and personnel ● funding source ● final concentration of aluminium injected ● aluminium manufacturer

	● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● serological response per patient at maximum follow up No answer received **Funding:** funded by the Chemical Biological Medical Systems-Joint Vaccine Acquisition Program. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript. SAE reported only as related to the vaccine.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The study statistician prepared a randomisation list using the RANUNI function in SAS Version 8 (SAS Institute, Cary, NC USA); only the study pharmacist and the statistician had access to this randomisation list, and study personnel were unblinded only after study completion.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind (study personnel were unblinded only after study completion)
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Immunogenicity data was analysed and reported on all subjects who provided a sample at the time point being assessed. Two subjects were excluded from the immunogenicity analyses; one subject who was randomised to receive placebo inadvertently was administered vaccine product (50 mg of rPA with PBS) at the second inoculation and thus was included in safety analysis, but not immunogenicity analysis. One subject, randomised to the 100 mg rPA with PBS arm, had prior undisclosed Bacillus anthracis vaccine exposure as determined by detectable levels of anti-rPA antibodies prior to injection of vaccine. This volunteer denied Bacillus anthracis vaccine exposure at entry and after unblinding reported never being aware of receiving an Bacillus anthracis vaccine. This subject was also included

		in safety analysis, but excluded from the immunogenicity analysis.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00057525), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Butler 1969

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 3-6 months old **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** combined diphtheria, tetanus, and pertussis vaccines **Experimental:** vaccine with Alhydrogel. n = 56 **Control:** vaccine without aluminium. n = 47 **Aluminium concentration:** 2500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 injections every 4 weeks
Outcomes	**Immunogenicity** and **safety**
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Funding**: not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs more than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not deacribed
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Campbell 2007

Methods	Phase I, randomised study
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** rPA anthrax vaccine **Experimental:** 5 or 25 mcg rPA with Alhydrogel. n = 20 **Control:** 5 or 25 mcg rPA. n = 20 **Aluminium concentration:** 800 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injection at day 0 and day 28
Outcomes	**Immunogenicity:** "serum was collected immediately prior to immunization and at post-vaccination days 14, 28, 42, 120 and 210 for evaluation of the humoral immune response. Investigators performed two serologic assays: an enzyme-linked immunosorbent assay and a toxin neutralization assay. The main serologic endpoints for each dose group were the following: (1) the proportion of volunteers in each vaccine group whose day 42 anti-PA IgG antibody concentrations measured by enzyme-linked immunosorbent assay were greater than or equal to four times their day zero concentrations and no less than 10 mg/mL (four times the lower limit of quantification of the assay), (2)

	the geometric mean concentration of anti-PA antibody on day 42 for volunteers in each group, (3) the proportion of volunteers in each group with a four-fold increase in ED50 from day 0 to day 42, and (4) the geometric mean titer of neutralizing antibodies, measured by ED50, for each group on day 42". **Safety:** "investigators observed volunteers for 30 minutes following vaccination to record and treat any immediate side effects. Investigators surveyed volunteers at 30 minutes post-vaccination using a pain assessment tool on which is recorded the severity, quality and length of time of injection site pain. Volunteers recorded reactogenicity data, including an oral temperature, in a diary on the day of each vaccination and the six subsequent days. Local and systemic effects were recorded. Adverse events and serious adverse events were evaluated and recorded for the entire study period. Safety laboratory tests (complete blood count, chemistries, liver and renal function tests) were done prior to vaccination and at post-vaccination days 7 and 35 (data not shown)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (jcampbel@medicine.umaryland.edu) were inquired about the followings: ● aluminium manufacturer ● method of allocation concealment ● randomization sequence generation ● total number of participants with serious and non-serious adverse event related and not related to the vaccine. ● number of participants who got Bacillus anthracis infection after the vaccine No answer received **Funding:** this work was supported by the National Institutes of Health Vaccine and Treatment Evaluation Unit, Contract N01-AI-25461, and University of Maryland General Clinical Research Center Grant M01 RR 16500, General Clinical Research Centers Program, National Center for Research Resources (NCRR), NIH.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Ranomized. No further information on how this was achieved.
Allocation concealment (selection bias)	Unclear risk	Allocation to vaccine within each group was randomly assigned. Unclear how this was achieved.
Blinding of participants and personnel (performance bias)	Low risk	Neither the investigators nor the volunteers were made aware of the vaccine assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported

Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	No drop outs
Selective reporting (reporting bias)	Unclear risk	No pre-registered protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Chen 2018

Methods	Single-centre, double-blind, randomised controlled trial
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "age 18-45 years, written informed consent of volunteers, healthy as established by medical history, laboratory evaluation and screening evaluations, participants are able to understand and comply with planned study procedures and be available for all study visits, female subjects must be of non-childbearing potential (defined as surgically sterile or postmenopausal for more than 1 year), or if of childbearing potential must be practicing abstinence or using an effective licensed method of birth control". **Exclusion criteria:** "previous vaccination against Neisseria meningitidis. Known exposure to Neisseria meningitidis in the past. History of meningitis or seizures or any neurological or psychiatric disorder. Administration of any other vaccine within 30 days prior or after administration of study vaccines. Use of any investigational or non-registered drug or vaccine within 30 days prior to the administration of study vaccines or planned during the study. History of allergic disease or known hypersensitivity to any component of the three study vaccines. History of Serious Adverse Reactions following administration of Tetanus Toxoid, Diphtheria Toxoid or CRM containing vaccines. History of Guillan-Barré syndrome. Confirmed or suspected immunosuppressive or immune-deficient condition. 10. A family history of congenital or hereditary immunodeficiency. Chronic administration (defined as more than 14 days) of immune-suppressants or other immune-modifying agents within six months prior to administration of study vaccine. 12. Laboratory confirmed infection of either hepatitis B virus (HBs Ag positive on ELISA), hepatitis C virus (anti-HCV positive on ELISA as well as PCR) or human immunodeficiency virus (HIV on ELISA). Major congenital defects or serious chronic illness. Acute or chronic, clinically significant pulmonary, cardiovascular, hepatic, or renal functional

	abnormality, as determined by medical history, physical examination or laboratory screening tests. Known bleeding disorders. Administration of immunoglobulins and/or any blood products within the three months preceding the administration of study vaccines or planned administration during the vaccine period. History (within the past year) or signs of alcohol or substance abuse. Pregnancy or lactation. A Body Mass Index (BMI) equal to or above 30. Any other condition, which in the opinion of the investigator, might interfere with the study objectives, jeopardize the safety or rights of the participant or making it unlikely the participant could complete the protocol".
Interventions	**Vaccine type:** NnmCV-5 pentavalent meningococcal conjugate vaccine **Experimental:** aluminium-phosphate adjuvated NmCV-5. n = 20 **Control:** NmCV-5. n = 20 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium-phosphate **Aluminium manufacturer:** unclear **Vaccination schedule:** one intramuscular injection
Outcomes	**Immunogenicity:** "percentage of participants with ≥4-fold increase in post-immunization rabbit complement serum bactericidal activity, (time frame: 28 days post vaccination); the percentage of participants with an serum bactericidal activity titer of 1:8 and 1:128, (time frame: 28 days post vaccination). Geometric mean titers of Meningococcal Polysaccharide A, C, Y, W, and X specific antibodies, (time frame: 28 days post vaccination)". **Safety:** "reactogenicity assessed by occurrence, severity of solicited local and systemic reactions, (time frame: day 7 post vaccination); occurrence of unsolicited adverse events by severity and relatedness to vaccination, (time frame: 28 days post vaccination); occurrence of serious adverse events, (time frame: 180 days post vaccination)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (wchen@som.umaryland.edu) were inquired about the followings: ● aluminium manufacturer ● method of allocation concealment ● randomization sequence generation ● blinding of outcome assessors ● total number of participants with non-serious adverse event related and not related to the vaccine. ● number of participants who got meningococcal disease after the vaccine No answer received **Funding:** UK Department for International Development

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation sequence was generated by the trial statistician in SAS (version 9.3) using a permuted block design with randomly chosen block sizes of three and six. Upon completion of entry of a participant's enrolment

		information, a concealed treatment assignment number was provided in the electronic data entry system.
Allocation concealment (selection bias)	Low risk	Upon completion of entry of a participant's enrolment information, a concealed treatment assignment number was provided in the electronic data entry system. The number was used by the unmasked research pharmacist to prepare the allocated product for each participant, using a treatment code list that was maintained by the pharmacist in a secure location, to maintain blinding. The vaccines were filled into identical-appearing syringes and labelled in a manner such that each syringe could be matched to the study participant but did not provide unblinding information, and they were administered with procedures to ensure participants and prospectively assigned study personnel remained masked to treatment
Blinding of participants and personnel (performance bias)	Low risk	Double blind. Participants and prospectively assigned study personnel remained masked to treatment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. Unclear in regards to outcome assessors
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. Unclear in regards to outcome assessors
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind. Unclear in regards to outcome assessors
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02810340), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Chichester 2012

Methods	Phase 1 randomised, double-blind, placebo-controlled, dose-escalation clinical study
Participants	**Sex:** male and female **Age:** 18–49 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "male or female aged 18 to 49 years inclusive; able to give written informed consent to participate; healthy as determined by medical history, physical examination, vital signs, and clinical safety laboratory examinations at baseline; females should fulfil one of the following criteria: at least 1 year post-menopausalSurgically sterileWill use oral, implantable, transdermal, or injectable contraceptives for 30 days prior to first vaccination and until 28 days after each vaccination, willing to abstain from sexual intercourse or use another reliable form of contraception approved by the investigator (e.g., intrauterine device [IUD], female condom, diaphragm with spermicide, cervical cap, use of condom by the sexual partner, or a sterile sexual partner) for study duration and until 28 days after vaccination. Women of childbearing potential must have a negative urine pregnancy test within 24 hours preceding receipt of each dose. Comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits and to be contacted by telephone throughout the follow-up period". **Exclusion criteria:** "prior receipt of any influenza vaccine containing H5; presence of significant uncontrolled medical or psychiatric illness (acute or chronic); includes institution of new medical or surgical treatment or a significant dose alteration for uncontrolled symptoms or drug toxicity within 3 months of screening and reconfirmed on day 1 prior to vaccination Positive serology for human immunodeficiency virus (HIV 1 or HIV 2), hepatitis B surface antigen. or hepatitis C; cancer or treatment for cancer within 3 years excluding basal cell carcinoma or squamous cell carcinoma which are allowed; presence of any medical condition that may be associated with impaired immune responsiveness, including diabetes mellitus Presently receiving (or has a history of receiving) during the preceding 3 month period any medications or other treatments that may adversely affect the immune system, such as allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs; other drugs known to be frequently associated with significant major organ toxicity; or systemic (oral or injectable) corticosteroids (inhaled and topical corticosteroids allowed); receipt or planned administration of a non study vaccine within 30 days prior to vaccination and during the study (Immunization on an emergency basis with Tetanus Toxoids Adsorbed for adult use vaccine—tetanus and diphtheria or tetanus, diphtheria, and pertussis up to 8 days before or at least 8 days after a dose of study vaccine will be allowed, and administration of study vaccine injection can be delayed if a non study vaccine has been administered and will be given as soon as acceptable [as described above] provided the vaccine is not administered within 2 weeks prior to study enrollment.); history of significant allergic reactions (anaphylactic-type reactions, respiratory difficulties, or angioedema) to injected vaccines or to any of the study vaccine components; history of drug or chemical abuse in the year before the study; positive urine drug screen at screening; receipt of any investigational product or non registered drug within the 30 days prior to vaccination or currently enrolled in any investigational drug study or intends to enroll in such a study within the ensuing study period; receipt of blood

	or blood products 8 weeks prior to vaccination or planned administration during the study period; donation of blood or blood products within 8 weeks prior to vaccination or at any time during the study; acute disease within 72 hours prior to vaccination defined as the presence of a moderate or severe illness (as determined by the investigator through medical history and physical examination) with or without fever, or a fever >38ºC orally (Study vaccine can be administered to persons with a minor illness, such as diarrhea, or mild upper respiratory tract infection with or without low-grade fever, and vaccination can be delayed until the subject has recovered.); any condition that, in the opinion of the investigator, might interfere with the primary study objectives or the subject's safety".
Interventions	**Vaccine type:** plant-produced recombinant hemagglutinin-based influenza vaccine (HAI-05) Derived from A/Indonesia/05/2005 (H5N1) influenza virus **Experimental:** 90 mcg HAI-05 + Alhydrogel. n = 20 **Control:** 90 mcg HAI-05. n = 20 **Aluminium concentration:** 750 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Denmark **Vaccination schedule:** two injections three weeks apart intramuscularly
Outcomes	**Immunogenicity:** "sera for immunogenicity assessments were collected on days 1 (pre-vaccination; baseline), 22 (pre-vaccination; results for the first vaccination) and 43 (results for the second vaccination). Immunogenicity assessments included determination of 1) geometric mean titres of HAI and MN antibody titers; 2) the proportion of subjects in each group who seroconverted on Days 22 and 43 based on HAI and MN antibody titers; and 3) the proportion of subjects in each group who achieved HAI titers ≥40 on Days 22 and 43". **Safety:** "reactogenicity events, adverse events, clinical laboratory tests (serum chemistry, hematology and urinalysis), vital signs, and abnormal findings upon physical examination. Any unsolicited symptoms were recorded at 30 minutes after vaccination by the clinical site staff, and thereafter through day 43 by subjects in a supplemental diary".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (vyusibov@fraunhofer-cmb.org, jchichester@fraunhofer-cmb.org) were inquired about the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** funded by The Bill and Melinda Gates Foundation (grant Nr. 49840)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further description
Allocation concealment (selection bias)	Unclear risk	No details in regards to allocation concealment
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further description
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not stated
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not stated
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No drop outs
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	No drop outs
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	No drop outs
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01250795), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Coller 2016

Methods	Abstract: phase I, randomised, placebo-controlled, double-blind study
Participants	**Sex:** not reported **Age:** not reported **Ethnicity:** not reported **Diagnosis:** not reported. Subjects who had previously received one or two doses of live-attenuated tetravalent vaccine **Inclusion criteria:** not reported **Exclusion criteria:** not reported

Interventions	**Vaccine type:** dengue vaccine V180 **Experimental:** V180 adjuvated with Alhydrogel. n = 8 **Control:** V180. n = 8 **Aluminium concentration:** not reported **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity:** plaque reduction neutralization test at Day 0, Day 14, Day 28, and Month 6 relative to vaccination. **Safety:** solicited and unsolicited adverse events were assessed using a Vaccination Report Card for 28 days following vaccination. Serious adverse events were captured from the time of informed consent through the final study visit at 6 months postvaccination.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (beth-ann.coller@merck.com) were inquired about the followings: ● aluminium manufacturer ● final aluminium concentration injected per dose ● method of allocation concealment ● randomization sequence generation ● blinding of outcome assessors ● blinding of participants and personnel ● immunogenicity response expressed as geometric mean titre or concentration per intervention arm ● total number of participants with non-serious adverse event related and not related to the vaccine ● number of participants who got dengue disease after the vaccine Answer received but without extra information **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not reported
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported

Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Collier 1979

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** not reported **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** all children participating were accepted by their school as having had at least the primary course of diphtheria-tetanus-pertussis vaccine or diphtheria tetanus vaccine in infancy. **Exclusion criteria:** not reported
Interventions	**Vaccine type:** tetanus vaccine BP **Experimental:** aluminium hydroxide-adsorbed Tetanus Vaccine. n = 134 **Control:** plain tetanus Vaccine. n = 132 **Aluminium concentration:** 400 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** unclear **Vaccination schedule:** 1 dose
Outcomes	**Safety and immunogenicity:** "the investigation was designed to determine (1) the incidence, nature, and severity of reactions to reinforcing doses of tetanus toxoid given to school-leavers; (2) the relationship of reactions with the tetanus antibody titre at the time of inoculation; (3) whether reactions can be lessened by using plain formol toxoid (FT), and if so, whether the antibody response to such a vaccine is adequate; and (4) the necessity for a booster dose of tetanus vaccine at the time of leaving school".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** no contact found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Randomised given alternately to the children
Allocation concealment (selection bias)	High risk	Allocated at random by giving them alternately as the children presented themselves
Blinding of participants and personnel (performance bias)	High risk	Not blinded
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Examinations were shared by S.P. and L.H.C. who were unaware of which vaccines the children had received.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Examinations were shared by S.P. and L.H.C. who were unaware of which vaccines the children had received.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Not all participants included in all outcomes in all time points
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Cox 2009

Methods	Open label, phase I, dose escalating clinical trial
Participants	**Sex:** male and female **Age:** 19–39 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** all subjects aged 19–39 years having normal haematology, biochemistry and immunology parameters **Exclusion criteria:** "a history of anaphylaxis or hypersensitivity to vaccines, an oral temperature >38 ○C (in the preceding 72 h) or an acute respiratory infection (up to 7 days) prior to immunisation, pregnancy, breast-feeding women, recent

	treatment with blood products, vaccination with any vaccine during the 4 weeks preceding the first trial vaccination, known or suspected immunosuppressive or chronic illness, and the use of systemic immunosuppressive drugs or immunostimulant therapy.Women were required to use reliable contraception from 4 weeks before the first vaccination until 4 weeks after the second vaccination and have a negative urine pregnancy test prior to administration of each dose of vaccine".
Interventions	**Vaccine type:** split H7N1 influenza virus vaccine **Experimental (a):** 12 mcg haemagglutinin influenza vaccine with aluminium hydroxide. n = 15 **Control (a):** 12 mcg haemagglutinin influenza vaccine. n = 15 **Experimental (b):** 24 mcg haemagglutinin influenza vaccine with aluminium hydroxide. n = 15 **Control (b):** 24 mcg haemagglutinin influenza vaccine. n = 15 **Aluminium concentration: (a)** 300 mcg/dose. **(b)** 600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injections into the deltoid muscle separated by 21 (±1) sos
Outcomes	**Immunogenicity:** "blood samples were collected pre, 3, 7 (±1), 14 (±2) and 21 days after each vaccine dose, and 6 (±1) months after the second vaccination. Lymphocytes were immediately analysed for antibody secreting cells and cytokine responses. Pre- and post-vaccination serum samples from each individual were run blinded in duplicate on at least two occasions in the haemagglutination inhibition and microneutralisation". **Safety:** "injection site reactions were evaluated at the end of a 45min observation period and any area of erythema or induration recorded. Volunteers were provided with a diary card on which they recorded the incidence of local (erythema/redness, local itching, oedema/local swelling, pain at the injection site, ecchymosis, hardness/induration) and systemic reactions (fever,malaise, shivering/chills, fatigue, headache, sweating, myalgia, arthralgia, respiratory symptoms and diarrhoea). Any use of antipyretic/relief medication and occurrence of intercurrent illness was also recorded. The diary card was reviewed at each visit to the clinic".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (rebecca.cox@gades.uib.no) were inquired about the followings: ● method for random sequence generation ● method for allocation concealment ● Fifnal concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● serological response per patient at maximum follow up No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further description
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	5/60 participants withdrew (reasons reported). These participants were included in the immunogenicity and safety analyses up to the point of withdrawal from the study.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	5/60 participants withdrew (reasons reported). These participants were included in the immunogenicity and safety analyses up to the point of withdrawal from the study.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	5/60 participants withdrew (reasons reported). These participants were included in the immunogenicity and safety analyses up to the point of withdrawal from the study.
Selective reporting (reporting bias)	Low risk	The trial was registered in the WHO Initiative for Vaccine Research and the EudraCT Database (EudraCT no. 2006-001582-42).
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Cummings 2014

Methods	First-in-human, phase 1, single-center, randomised, placebo-controlled, single-blind, dose-escalation clinical study
Participants	**Sex:** male and female **Age:** 18–50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "male or female aged 18 - 50 years inclusive; able to give written informed consent to participate; score of at least 80% correct on a 10 question multiple-choice quiz (2 attempts); healthy, as determined by medical

	history, physical examination, weight, vital signs, and clinical safety laboratory examinations at baseline; females must fulfil one of the following criteria: At least one year post-menopausal; surgically sterile or have a surgically sterile partner; Willing to abstain from sexual intercourse; willing to use a reliable form of contraception approved by the investigator (e.g., oral, implantable, transdermal or injectable contraceptives, intrauterine device (IUD), female condom, diaphragm with spermicide, cervical cap, or male condoms) for 30 days prior to first vaccination through 3 months after second vaccination; women of childbearing potential must have a negative urine pregnancy test within 24 hours preceding receipt of each dose; comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits and to be contacted by telephone throughout the follow-up period". **Exclusion criteria:** "prior receipt of 2010-2011 seasonal influenza vaccine containing A/California/04/09-like virus; screening H1N1 titer of > 1:40; presence of significant uncontrolled medical or psychiatric illness (acute or chronic) including institution of new medical or surgical treatment or a significant dose alteration for uncontrolled symptoms or drug toxicity within 3 months of screening; presence of any medical condition that may be associated with impaired immune responsiveness, including diabetes mellitus; cancer, or treatment for cancer, within the previous 3 years, excluding basal cell carcinoma or squamous cell carcinoma; presently receiving or history of receiving, during the preceding 3-month period, any medications or other treatments that may adversely affect the immune system: This includes allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs or other drugs known to be frequently associated with significant major organ toxicity, or systemic corticosteroids (oral or injectable); Inhaled and topical corticosteroids will be allowed; receipt of blood or blood products 8 weeks prior to vaccination or planned administration during the study period; donation of blood or blood products within 8 weeks prior to vaccination or at any time during the study; receipt or planned administration of a non study vaccine within 30 days prior to vaccination; Immunization on an emergency basis with Tetanus Toxoids Adsorbed for adult use (Td or Tdap) vaccine up to 8 days before or at least 8 days after a dose of study vaccine will be allowed; history of anaphylactic type reaction to injected vaccines; receipt of any investigational product or non registered drug within the 30 days prior to vaccination or currently enrolled in any investigational drug study or intends to enrol in such a study within the ensuing study period; history of drug or chemical abuse in the year before the study; positive serology for HIV-1 or HIV-2, or HBsAg or HCV antibodies; unwilling to allow storage of specimens for future use; acute disease within 72 hours prior to vaccination: Acute disease is defined as the presence of a moderate or severe illness (as determined by the Investigator through medical history and physical examination) with or without fever (>38ºC />100.4ºF), or an oral temperature of >38ºC orally; study vaccine can be administered to persons with a minor illness; any condition that, in the opinion of the investigator, might interfere with interpretation of data supporting the primary study objectives".
Interventions	**Vaccine type:** hemagglutinin-based influenza vaccine derived from influenza A (H1N1) pdm09 virus. **Experimental:** 15, 25 or 90 mcg H1N1 with Alhydrogel. n = 30 **Control:** 15, 25 or 90 mcg H1N1. n = 30 **Aluminium concentration:** 750 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Denmark

	Vaccination schedule: two doses in the deltoid muscle of the same non-dominant arm at Day 0 and Day 21
Outcomes	**Immunogenicity:** "sera for immunogenicity assessments were collected on day 0 (pre-vaccination), day 21 (pre-vaccination) and day 56, and sent to the Influenza Division, CDC for analysis by hemagglutination inhibition assay and microneutralization assays. Immunogenicity assessments included determination of (1) geometric mean titer of hemagglutination inhibition assay and microneutralization antibody titers; (2) the proportion of subjects in each group who seroconverted on days 21 and 56; and (3) the proportion of subjects in each group who achieved an hemagglutination inhibition titer of ≥40 or an microneutralization titer of ≥80 or ≥160 on days 21 and 56". **Safety:** "proportion of subjects in each treatment group reporting one or more local or systemic solicited and unsolicited reactions within 7 days after vaccination (days 0–6 and 21–27), and the occurrence of any serious adverse events, defined as any untoward events resulting in hospitalization or death,during the study. Clinical safety laboratory assessments were performed at baseline, day of vaccination, and 7 days after each vaccination. The presence of solicited local and general signs and symptoms, including measurement of oral temperature, were assessed after each vaccination and at 1, 2, 3, 7 and 14 days post-vaccination. Also, the investigators recorded any other adverse events occurring on the first administration of the vaccine through day 56 as unsolicited adverse events".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (james.f.cummings6.mil@mail.mil, Jamesfcummings@gmail.com) were inquired about the followings: ● Method of allocation concealment ● Randomization sequence generation ● Blinding of outcome assessors ● Total number of participants with non-serious adverse event related and not related to the vaccine ● Number of participants who got influenza after the vaccine No answer received **Funding:** the HAC1 clinical trial and serologic testing were supported by the Defense Advanced Research Projects Agency. The cGMP manufacture of HAC1 was supported by DARPA under contract number # HR0011-10-C-0051.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Single blind

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	2/80 lost to follow up or discontinued
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01177202), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

De Bruyn 2016

Methods	Randomized, placebo controlled, Phase 2 study
Participants	**Sex:** male and female **Age:** 40-75 years **Ethnicity:** mixed **Diagnosis:** at risk of *Clostridium difficile* infection **Inclusion criteria:** "aged 40 to 75 years on the day of inclusion; informed consent form has been signed and dated; able to attend all scheduled visits and to comply with all trial procedures; for a woman of childbearing potential, use of an effective method of contraception or abstinence from at least 4 weeks prior to the first vaccination until at least 4 weeks after the last vaccination; at risk for developing Clostridium difficile infection during the trial because of impending elective surgery or hospitalization within 60 days of enrollment, or current or impending residence in a long-term care facility or rehabilitation facility". **Exclusion criteria:** "known pregnancy, or a positive urine pregnancy test; currently breastfeeding a child; participation in another clinical trial investigating a vaccine, drug, medical device, or medical procedure in the 4 weeks preceding the first trial vaccination; planned participation in another clinical trial during the present trial period; receipt of any vaccine in the 4 weeks preceding the first trial vaccination, except for influenza (seasonal or pandemic) and pneumococcal vaccine; planned receipt of any vaccine in the 4 weeks following any trial vaccination, except for influenza (seasonal or pandemic) and pneumococcal

	vaccines; previous vaccination against Clostridium difficile with either the trial vaccine or another vaccine; current or prior Clostridium difficile infection episode; receipt of blood or blood-derived products in the past 3 months, which might interfere with assessment of the immune response; known or suspected congenital or acquired immunodeficiency; or receipt of immunosuppressive therapy such as anti-cancer chemotherapy or radiation therapy within the preceding 6 months; or long-term systemic corticosteroid therapy (prednisone or equivalent for more than 2 consecutive weeks within the past 3 months); self-reported seropositivity for Human Immunodeficiency Virus, hepatitis B, or hepatitis C; anticipated or current receipt of kidney dialysis treatment; known systemic hypersensitivity to any of the vaccine components, or history of a life-threatening reaction to a vaccine containing any of the same substances; self-reported thrombocytopenia, contraindicating intramuscular vaccination; bleeding disorder, or receipt of anticoagulants in the 3 weeks preceding inclusion, contraindicating IM vaccination; deprived of freedom by an administrative or court order, or in an emergency setting, or hospitalized involuntarily; current alcohol abuse or drug addiction that might interfere with the ability to comply with trial procedures; chronic illness that, in the opinion of the investigator, is at a stage where it might interfere with trial conduct or completion; identified as a study site employee who is involved in the protocol and may have direct access to trial-related data; subjects who have any history of intestinal diverticular bleeding; subjects who have had surgery within the past three months for gastrointestinal malignancy".
Interventions	**Vaccine type:** *Clostridium difficile* vaccine **Experimental:** 50 or 100 mcg *Clostridium difficile* with aluminium hydroxide. n = 305 **Control:** 50 or 100 mcg *Clostridium difficile*. n = 306 **Aluminium concentration:** 400 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3-dose schedule on either days 0, 7, 180 or days 0, 30, 180 or on days 0, 7, 30
Outcomes	**Immunogenicity:** "blood samples were collected on days 0, 7, 14, 30, 60, 180, and 210 and analysed for (i) serum anti-toxin IgG to Clostridium difficile toxins A and B enzyme-linked immunosorbent assay, and (ii) anti-toxin A and B neutralizing capacity". **Safety:** "pre-defined (solicited) adverse events were automatically considered to be related to vaccination: solicited injection site (pain, erythema, swelling) and systemic(fever, headache, malaise, myalgia, arthralgia) adverse reactions were collected on the day of vaccination and for 6 days after each vaccination. Unsolicited adverse events were collected for 30 days after each vaccination.Serious adverse events were collected throughout the study.The relationship of unsolicited AEs and SAEs to vaccination was assessed by the investigator".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (guy.debruyn@sanofipasteur.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment

	● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further data or information added. The number of participants with adverse events are unclear; thus, we have not included these data in the meta-analyses. **Funding:** supported by Sanofi Pasteur

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised
Allocation concealment (selection bias)	Unclear risk	No information on allocation concealment
Blinding of participants and personnel (performance bias)	High risk	Open-label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	A safety analysis set, including all participants who received at least one trial vaccination, was used for the safety analyses. Dropouts > 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	High risk	A per protocol analysis set was defined for immunogenicity based on meeting inclusion/exclusion criteria, receipt of the correct number of doses,receipt of the correct formulation administered according to the protocol and in the proper time window, provision of pre- or post-dose 1 serology samples according to protocol-defined time windows, protocol

		deviations, lack of a valid serology result pre-or post-dose 1. Dropouts > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01230957), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Durbin 2020

Methods	Phase I randomised, placebo-controlled, double-blind study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** adult males or females in generally good health between 18 and 50 years of age, inclusive, who had previously received the TV003 or TV005 LATV dengue vaccine formulation successfully completed the NIH study of TV003 or TV005 (NCT01436422 orNCT01072786), with no safety concerns, and had seroconverted for ≥ 3 serotypes. **Exclusion criteria:** Participants were excluded if they had potential or known exposure to Flavivirus by infection or vaccination in the interim since the receipt of the last dose of the experimental dengue vaccine (TV003 or TV005) and planned travel to a DENVendemic area in the 30 days before or anytime during the current study. Other exclusion criteria included evidence of clinically significant neurologic, cardiac, pulmonary, hepatic, rheumatologic, autoimmune, or renal disease; poorly controlled diabetes mellitus; recent receipt of blood products; febrile illness £ 72 hours before the first dose; and receipt of any licensed vaccine within 14 or 30 days, respectively, before vaccination in this study.
Interventions	**Vaccine type:** tetravalent recombinant subunit dengue vaccine **Experimental:** V180 with Alhydrogel. n = 8 **Control:** V180. n = 8 **Aluminium concentration:** 230 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** one intramuscular injection
Outcomes	**Immunogenicity:** "the primary immunogenicity objective of the trial was to evaluate if at least one of the V180 formulations induced a booster-neutralizing antibody response, as measured by the PRNT50, to at least three of the four DENV serotypes at day 28 after vaccination. Exploratory immunogenicity objectives included analysis of the kinetics and durability of the virus neutralizing response using the PRNT50 performed at days 15 and 180 as well as analysis of the neutralization response using the focus reduction neutralization test (FRNT50) at days 28 and 180 after vaccination". **Safety:** "the overall safety and tolerability of the tested V180 formulations were assessed and reported as a summary of serious and non-serious solicited and unsolicited AEs".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 26/01/2021 authors (swhitehead@niaid.nih.gov) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of participants and personnel **Fundings**: this study was funded by Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Kenilworth, NJ, and in part by the Intramural Research Program of the NIAID, NIH. Serology data were available up to day 180 but were extracted in this review at day 28 due to fewer drop outs at this time-point.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only stated randomised
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. An external, independent, and blinded data safety monitoring board reviewed the trial design before initiation and reviewed data during the trial to evaluate the safety, study progress, and conduct of the study.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Double-blind. An external, independent, and blinded data safety monitoring board reviewed the trial design before initiation and reviewed data during the trial to evaluate the safety, study progress, and conduct of the study.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	An external, independent, and blinded data safety monitoring board reviewed the trial design before initiation and reviewed data during the trial to evaluate the safety, study progress, and conduct of the study.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No drop outs
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	No drop outs
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	No drop outs

Selective reporting (reporting bias)	Unclear risk	The study protocol was registered prior to randomisation (NCT02450838), but safety outcome not described in the protocol.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ehrlich 2008

Methods	Randomized, partially blinded (between groups) clinical trial
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects are 18 to 45 years of age, inclusive, on the day of screening; have an understanding of the study, agree to its provisions, and give written informed consent prior to study entry; are clinically healthy, as determined by medical history and physical examination; agree to keep a daily record of symptoms; if female and capable of bearing children, have a negative urine pregnancy test result at study entry and agree to employ adequate birth control measures for the duration of the study". **Exclusion criteria:** "subjects are at high risk of contracting H5N1 influenza infection (e.g. poultry workers); suffer from a significant neurological, cardiac, pulmonary (including asthma), hepatic, rheumatic, autoimmune, hematological or renal disorder; are unable to lead an independent life as a result of either physical or mental handicap; suffer from any kind of immunodeficiency; suffer from a disease or are undergoing a form of treatment that can be expected to influence immune response. Such treatment includes, but is not limited to, systemic or high dose inhaled (>800µg/day of beclomethasone dipropionate or equivalent) corticosteroids, radiation treatment or other immunosuppressive or cytotoxic drugs; have a history of inflammatory or degenerative neurological disease (e.g. Guillain Barré); have a history of severe allergic reactions or anaphylaxis; have a rash, dermatologic condition or tattoos which may interfere with injection site reaction rating; have received a blood transfusion or immunoglobulins within 90 days of study entry; have donated blood or plasma within 30 days of study entry; have received any live vaccine within 4 weeks or inactivated vaccine within 2 weeks prior to vaccination in this study; have undergone systemic corticoid therapy within 30 days prior to study entry; have a functional or surgical asplenia; have a known or suspected problem with alcohol or drug abuse; were administered an investigational drug within six weeks prior to study entry or are concurrently participating in a clinical study that includes the administration of an investigational product; are a member of the team conducting this study or are in a dependent relationship with the study investigator. Dependent relationships include close relatives (i.e., children, partner/spouse, siblings, parents) as well as employees of the investigator; if female, are pregnant or lactating".
Interventions	**Vaccine type:** whole-virus H5N1 influenza vaccine **Experimental:** 7.5 or 15 mcg H5N1 with aluminium. n = 91 **Control:** 7.5 or 15 mcg H5N1. n = 90 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported

	Vaccination schedule: two injections into the deltoid muscle at an interval of 21 days
Outcomes	**Immunogenicity:**" number of subjects with hemagglutination-inhibition and neutralizing antibodies to the vaccine strain (A/Vietnam/1203/2004) 21 days after the first and second doses of vaccine". **Safety:** any systemic reaction during the first 7 days after each dose of vaccine
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** 14/12/2018 authors (hartmut.ehrlich@abivax.com) were inquired about the followings ● method for random sequence generation ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further data or information added **Funding:** supported by Baxter The trial only reports on vaccine-related serious adverse events.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	High risk	Subject assignment to a particular cohort was in order of enrollment.
Blinding of participants and personnel (performance bias)	High risk	Subjects and investigators were unaware of the dose of vaccine administered within the subgroups. Investigators were partially blinded, meaning that they were blinded within the respective dosing cohorts.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Unblinded safety data from Cohorts 1a, 1b, 2 and 3 were provided to the independent Data Safety Monitoring Board (DSMB) while it was ensured that study monitors, data managers and other staff involved in the conduct of the study as well as laboratory personnel remained blinded.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Unblinded safety data from Cohorts 1a, 1b, 2 and 3 were provided to the independent Data Safety Monitoring Board (DSMB) while it was ensured that study monitors, data managers and other staff involved in the conduct of the study

		as well as laboratory personnel remained blinded.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Laboratory personnel remained blinded.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	<5 % drop-outs.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	<5 % drop-outs.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Drop outs>5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00349141), and all outcomes were reported on .
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ensoli 2009

Methods	Randomized, double blind, placebo-controlled clinical trial
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** Caucasian **Diagnosis:** healthy **Inclusion criteria:** "negative pregnancy test for women of childbearing potential within 3 days prior to baseline evaluation and use of an acceptable means of contraception (condom, hormonal or mechanical method) for one month prior to immunization and for all the duration of the study; complete blood count and differential defined as: -hematocrit >=30% for women, >= 38% for men -hemoglobin >= 9.5 g/dL -white cell counts >= 4,000 and <= 9,500 cells/mm3 -total lymphocyte count >=1000 cells/mm3 -CD4+ T cell count > 500 cells/microL based on 2 separate determinations (Hannet, 1992) -platelets (100,000-550,000/ mm3) -differential within institutional normal limits or approval of site physician; normal ALT (as defined by the range of the clinical site laboratory) and Creatinine (< 1.6 mg/dl). Normal urine dipstick with esterase and nitrite; normal thyroid function; negative ELISA for HIV-1/HIV-2 and HIV-1 viral load (plasma viremia) < 50 copies/mL within 1 month of immunization; availability for follow-up for planned duration of at least 12 months and willing to have further brief evaluations at 24 and 36 months". **Exclusion criteria:** "identifiable high-risk behavior for HIV-1 infection, including a history of injection drug use within 12 months prior to enrollment or higher-risk sexual behavior; history of neoplastic diseases, encephalopathy, neuropathy or

	unstable CNS pathology, immunodeficiency, autoimmune disease, angina or cardiac arrhythmias, or any other clinically significant medical problems; chest radiography showing evidence of active or acute cardiac or pulmonary disease; history of anaphylaxis or serious adverse reactions to vaccines as well as serum IgE levels exceeding 1000 U.I./ml; history of serious allergic reaction to any substance, requiring hospitalization or emergent medical care (e.g. Steven-Johnson syndrome, bronchospasm, or hypotension); active syphilis [NOTE. If the serology is documented to be a false positive or due to an adequately treated infection, the volunteer is eligible]; active tuberculosis [NOTE: Volunteers with a positive PPD and a normal chest X-ray showing no evidence of TB and not requiring isoniazid (INH) therapy are eligible]; medical or psychiatric condition or occupational responsibilities which preclude subject compliance with the protocol. Specifically excluded are persons with psychotic disorders, major affective disorders, suicidal ideation; current use of psychotropic drugs; participation in another experimental protocol within six months prior to pre-study screening; current or prior therapy with immunomodulators or immunosuppressive drugs and anticoagulant drugs within 30 days prior to study medication administration; any unstable cardiovascular disease (e.g. unstable hypersensitive disease needing modification or introduction of an anti-hypersensitive treatment); live attenuated vaccines within 60 days of study [NOTE: Medically indicated sub-unit or killed vaccines (e.g., influenza, pneumococcal, hepatitis A and B) are not exclusionary, but should be given at least 4 weeks away from HIV immunizations]; use of investigational agents within 90 days prior to study; receipt of blood products or immunoglobulin in the past 6 months; prior receipt of HIV-1 vaccine in a previous HIV vaccine trial; positivity for HBV antigens (HBs Ag, HBe Ag), and HCV, HTLV-I and HTLV-II antibodies; pregnant or lactating women".
Interventions	**Vaccine type:** human immunodeficiency virus (HIV)-1 Tat-based vaccine **Experimental:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine with aluminium. n = 8 **Control:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine. n = 7 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** 5 doses (weeks 0, 4, 8, 12, 16) either subcute in alum or intradermic without adjuvant
Outcomes	**Immunogenicity**: "volunteers are monitored for anti-Tat specific antibodies, anti-Tat proliferative response and in vitro gamma-IFN and IL-4 (or IL-10) production in response to Tat (Elispot)". **Safety**: "clinical observation and monitoring of haematological, biochemical, virological and immunological parameters. Safety is evaluated by monitoring of volunteers for local and systemic adverse reactions during the trial".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 12/02/2020 authors (barbara.ensoli@iss.it) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results

	• proportion of participants with any AE related and not related to the vaccine • serological response GMT for each study arm at maximum follow up • proportion of participants who developed HIV after the vaccination • random sequence generation • allocation concealment • blinding of participants and personnel • blinding of outcome assessment No answer received. **Funding:** funded by "ISS special funding for Tat vaccine trials", the Italian Concerted Action on HIV-AIDS Vaccine Development (ICAV, National AIDS Program) and the Italian Ministry of Health

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs > 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	The study protocol for one trial was registered prior to randomisation (NCT00529698), and all outcomes were reported on.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

EUCTR2009-015103-58-FI

Methods	A multicenter, randomised, double-blind, active control trial
Participants	**Sex:** male and female **Age:** 20-45 years / 12-15 months **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** afebrile healthy toddlers 12-15 months of age documented to have completed a full 3-dose infant series of Prevnar 4, and 6 months of age **Exclusion criteria:** not reported
Interventions	**Vaccine type:** pneumococcal conjugate vaccine (V114). **Experimental:** V114 aluminium adjuvated. n = 33 **Control:** V114. n = 29 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** Merck **Vaccination schedule:** one intramuscular dose
Outcomes	**Immunogenicity:** "serotype-specific IgG concentrations were measured using the Meso-Scale Discovery electrochemiluminescence assay on sera collected prior to vaccination and 30 days postvaccination. A multiplex opsonophagocytic assay assay was used to measure opsonophagocytic assay activity in the pre- and post vaccination sera. The percentage of subjects with postvaccination serotype-specific IgG concentrations corresponding to a level of ≥0.35 µg/mL in the WHO ELISA was evaluated. Other endpoints of interest were the geometric mean concentrations of IgG, the percentage of subjects with opsonophagocytic killing activity as defined by functional antibacterial opsonophagocytic assay titer ≥1:8 and the geometric mean opsonophagocytic assay titers for the serotypes contained in the V114 vaccine". **Safety:** "adverse experiences were collected using a validated vaccine report card. All adverse experiences were graded for severity. **Adult and Toddler Stages**: 1) Study subjects were observed for 30 minutes postvaccination for any immediate adverse experiences. 2) Solicited injection site adverse experiences (redness, swelling, nodules, and pain/tenderness) and solicited systemic adverse experiences (muscle pain, joint pain, and tiredness) were collected day 1 to day 14 after vaccination. 3) Any other systemic or injection site adverse experiences were collected day 1 to day 14 after vaccination. 4) Serious adverse experiences from the time the consent form was signed through completion of the subject's participation in the study at day 30 after receipt of study vaccine. 5) Body temperature was measured during day 1 to day 7 after vaccination. If fever was suspected, body temperature was also measured during day 8 to day 14. **Adults Only:** 1) CBC with differential including platelets, serum creatinine, ALK, ALT, bilirubin, and urinalysis was performed at day 1 and day 14; 2) Severity of adverse events and safety laboratory abnormalities were assessed according to a toxicity grading scale as mild (Grade I), moderate (Grade II), severe (Grade III), and potentially life threatening (Grade IV)".
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** authors not shown **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

EUCTR2010-019775-29-AT

Methods	Multicenter, randomised, double-blind, worldwide, active control trial
Participants	**Sex:** male and female **Age:** 6 to 12 weeks **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** Afebrile healthy infant subjects 6 to 12 weeks (42 to 89 days) old **Exclusion criteria:** not reported

Interventions	**Vaccine type:** pneumococcal conjugate vaccine (V114) **Experimental:** V114 aluminium adjuvated. n = 378 **Control:**V114. n = 386 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** Merck **Vaccination schedule:** 4 intramuscular doses at 2, 4, 6, and 12 to15 months of age
Outcomes	**Immunogenicity:** "serotype-specific IgG concentrations were measured using the Meso-Scale Discovery (MSD) electrochemiluminescence (ECL) assay and a multiplex opsonophagocytic assay was used to measure functional opsonophagocytic assay. The percentage of subjects with postvaccination serotype-specific IgG concentrations corresponding to a level of ≥0.35 µg/mL in the WHO ELISA was evaluated. Other endpoints of interest were the geometric mean concentrations of IgG, the percentage of subjects with opsonophagocytic killing activity as defined by functional antibacterial opsonophagocytic assay titer ≥1:8 and the geometric mean opsonophagocytic assay titers for the serotypes contained in the V114 vaccine". **Safety:** "adverse experiences were collected using a validated vaccination report card. All adverse experiences were graded for severity. (1) Study subjects were observed for 30 minutes postvaccination for any immediate adverse experiences. (2) Solicited injection-site adverse experiences (redness, swelling, hard lump, and pain/tenderness) and solicited systemic adverse experiences (irritability, drowsiness, appetite lost, and hives or welts) were collected day 1 to day 14 after vaccination. (3) Any other systemic or injection-site adverse experiences were collected day 1 to day 14 after vaccination. 4) Serious adverse experiences were collected from the time the consent form was signed through completion of the subject's adverse experiences participation in the study at day 30 after receipt of last dose of study vaccine. 5) Body temperature was measured during day 1 to day 7 after vaccination. If fever was suspected, body temperature was also measured during day 8 to day 14".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** authors not shown **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further information

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop-outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop-outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop-outs >5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Evans 2001

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-55 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "had no evidence of serious medical illness as evaluated by history, physical examination, or screening laboratory studies. The subjects were all seronegative for HIV-1 and were at low or intermediate risk of HIV infection as judged by the earlier described AIDS Vaccine Evaluation Group criteria. Women of childbearing age had negative pregnancy tests before each immunization and agreed to use birth control measures throughout the study period and follow-up". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** human immunodeficiency virus (HIV) **Experimental:** different doses of QS-21 (50 and 100 mg) with or without rsgp120 with aluminium. n =77 **Control:** different doses of QS-21 (50 and 100 mg) with or without rsgp120. n = 79 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 intramuscular injections at months 0, 1, 10 or 0, 1, 6 or 0, 1, 2

Outcomes	**Safety** and **immunogenicity**
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 25/02/2020 authors (tomevansmd@gmail.com) were inquired on the followings: ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any non-serious and serious adverse event related and not related to the vaccine for each intervention arm ● serological response GMT for each study arm at maximum follow up ● proportion of participants who developed HIV after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment Answer received about randomisation **Funding:** sponsored by the NIH Division of AIDS

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a

Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Falsey 2008

Methods	Randomized, double-blind, placebo-controlled study
Participants	**Sex:** male and female **Age:** >65 years **Ethnicity:** mixed **Diagnosis:** cardiopulmonary diseases **Inclusion criteria:** "persons with cardiopulmonary disease. Study subjects were required to be 65 years old and to have a physician-confirmed diagnosis of congestive heart failure(CHF) or active pulmonary disease that required medication to control symptoms. Active pulmonary diseases included chronic obstructive pulmonary disease (COPD), chronic bronchitis, emphysema, asthma, interstitial fibrosis, and environmental lung disease". **Exclusion criteria:** known immunodeficiency, autoimmune disease, or active malignancy, or were taking immunosuppressive medications
Interventions	**Vaccine type:** respiratory syncytial virus (RSV) vaccine **Experimental:** RSV vaccine with aluminium phosphate. n = 400 **Control:** RSV vaccine. n = 383 **Aluminium concentration:** 1500 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Sanofi Pasteur **Vaccination schedule:** one dose intramuscularly into the deltoid muscle. All participants received licensed inactivated trivalent influenza vaccine (Fluzone; sanofi pasteur) at the same visit in the opposite arm.
Outcomes	**Immunogenicity:** "assessed at 1 month and 12 months after the year 1 and year 2 vaccinations. Levels of serum antibody to RSV were assessed by a plaque reduction neutralization assay and enzyme immunoassay. Hemagglutinin-inhibition antibody assays for H1N1, H3N2, and influenza B vaccine strains were performed". **Safety:** "immediate safety was assessed on site for 30 min after receipt of vaccination. Adverse events, both local and systemic, were solicited for 7 days after vaccination by use of diary cards. Non-solicited adverse events were recorded for 4 weeks after vaccination, and serious adverse events were recorded throughout the entire study period. Surveillance for acute respiratory illness began 8 days after vaccination (starting in late September) and continued through the end of April in the following year. Weekly phone calls to subjects ensured that illnesses were evaluated. Acute respiratory illness was defined as new onset or worsening of either 2 respiratory symptoms (sore throat, nasal congestion, or discharge, hoarseness, cough, sputum, wheezing, dyspnea, or pleuritic chest pain) or 1 respiratory symptom and 1 systemic symptom (feverishness, fatigue, anorexia, headache, or myalgia). If an illness met the acute respiratory illness criteria, a nasal swab sample and blood sample were collected during the acute phase of infection. Participants filled out diary

	cards during the illness and a convalescent-phase blood sample was collected approximately 4 weeks after the illness".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (ann.falsey@viahealth.org) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** Sanofi Pasteur

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	High risk	No pre-published protocol. Serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Frasch 1988

Methods	Ransomised, double-blind study
Participants	**Sex:** male and female **Age:** 22-43 years **Ethnicity:** not reported **Diagnosis:** cardiopulmonary diseases **Inclusion criteria:** medical students and employees from Baylor College of Medicine within the Texas Medical Center in Houston **Exclusion criteria:** candidates were excluded if they had an acute or chronic illness, a strong allergic history (especially to a vaccine component), a history of unusual or severe reactions to any vaccination, or a history of meningococcal disease or if they were pregnant.
Interventions	**Vaccine type:** neisseria meningitidis serotype 2b protein-group B polysaccharide **Experimental:** aluminium hydroxide-adsorbed Neisseria meningitidis vaccine. n = 12 **Control:** neisseria meningitidis vaccine. n = 12 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 doses at day 0 and 42 intramuscularly
Outcomes	**Immunogenicity:** blood specimen obtained at vaccination, three and six weeks after the first dose, and four weeks after the second immunization. Antibody response measured by ELISA. Bactericidial assay 3 weeks after the first and four weeks after the second immunization. **Safety:** volunteers took their own temperatures 6 h after vaccination and returned at 24 and 48 h for temperature measurement, an examination of the injection site, and a general clinical exam done by study clinicians.
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** no contact found **Funding:** supported in part by contract AI-22672 from the National Institute of Allergy and Infectious Diseases

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	n.a
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Fries 2017

Methods	Phase 1, randomised, observer-blinded, placebo controlled trial
Participants	**Sex:** male and female **Age:** >60 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "adult males and females, ≥ 60 years of age, without symptomatic cardiopulmonary disease. The following are examples of subjects who may bear cardio-pulmonary diagnoses but who would remain

	eligible:subjects on stable (no change in ≥ 2 months) therapy for findings (e.g., hypertension or hyperlipidemia) that are not associated with current symptoms or disability.Subjects who receive intermittent prophylaxis for risks associated with asymptomatic findings (e.g., antibiotic prophylaxis prior to dental procedures in a subject with asymptomatic mitral valve prolapse).Other clinically insignificant findings, not deemed to be associated with increased risk due to respiratory viral infections as determined by the Investigator. Free of other illnesses which are believed to increase the risk of influenza or influenza related complications including: diabetes mellitus, congenital or acquired blood dyscrasias, renal or hepatic dysfunction, and morbid obesity. Willing and able to give informed consent prior to study enrollment. Able to comply with study requirements". **Exclusion criteria:** "subjects who have any functional limitation or symptoms related to cardiac and/or pulmonary disease (including asthma or other episodic symptoms), or who receive ongoing therapy to control symptoms or functional limitation, are not eligible. Participation in research involving investigational product (drug / biologic / device) within 45 days before planned date of first vaccination and/or planned participation at any time during the study. History of a serious reaction to any prior vaccination or known allergy to constituents of licensed TIV (e.g., egg proteins). History of Guillain-Barré Syndrome within 6 weeks following a previous influenza vaccine. Receipt of any influenza vaccine within the preceding 3 months. Receipt of any vaccine in the 4 weeks preceding the study vaccination and planned receipt of a licensed vaccine any time prior to Day 56. Receipt of an RSV vaccine at any time. Any known or suspected immunosuppressive condition, acquired or congenital, as determined by medical history and/or physical examination. Chronic administration (defined as more than 14 continuous days) of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. An immunosuppressant dose of glucocorticoid will be defined as a systemic dose ≥10 mg of prednisone per day or equivalent. The use of topical, inhaled, and nasal glucocorticoids will be permitted provided these are not administered for diagnoses inconsistent with the inclusion criteria. The use of inhaled glucocorticoids, although typically not associated with system absorption, will generally indicate the presence of a diagnosis inconsistent with inclusion criteria 1 or 2. Administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or during the study. Acute disease at the time of enrollment (defined as the presence of a moderate or severe illness with or without fever, or an oral temperature >38.0°C on the planned day of vaccine administration). Known disturbance of coagulation. Suspicion or recent history (within one year of planned vaccination) of alcohol or other substance abuse. Any condition that in the opinion of the Investigator would pose a health risk to the subject if enrolled or could interfere with evaluation of the vaccine or interpretation of study results (including neurologic or psychiatric conditions deemed likely to impair the quality of safety reporting)".
Interventions	**Vaccine type:** respiratory syncytial virus fusion protein (RSV F) nanoparticle vaccine **Experimental:** 60 or 90 mcg RSV F with aluminium phosphate. n = 80 **Control:** 60 or 90 mcg RSV F. n = 80 **Aluminium concentration:** 1200 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark

	Vaccination schedule: one intramuscular dose into the left deltoid
Outcomes	**Safety:** "time frame: day 0 to day 364. Number (and percentage) of subjects with solicited local and systemic Adverse events over the seven days post injection; all adverse events, solicited and unsolicited over 56 days post-first injection. Significant New Medical Conditions, Medically Attended Events and Serious Adverse Events will be collected for one year. Diaries were provided to all subjects to document any adverse events (both solicited and unsolicited) and to record body temperature, concomitant medications, and medical visits and procedures, starting on each vaccination day and continuing for 6 days thereafter". **Immunogenicity:** "serum IgG antibody titers for the F-Protein antigen (time frame: day 0 to day 364). Immunogenicity will be measured using derived / calculated endpoints based on: geometric mean titer, geometric mean ratio, seroconversion rate, seroresponse rate. Serum HAI titers specific for the influenza antigens contained in the seasonal vaccine, (time frame: day 0 to day 56). Subjects provided blood samples on days 0, 7, 28, 56, 119, 182, and 364. Antibody concentrations/titers for the palivizumab-competitive antibody and linear antigenic Site II peptide ELISAs; and the RSV/A and RSV/B microneutralization (MN) assays were determined. Day 0, 28, 56, and 182 sera were used for HAI titer determination for all three antigens contained in the TIV".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (vshinde@novavax.com) were inquired about the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** funded and sponsored by Novavax, Inc., 20 Firstfield Road, Gaithersburg, MD 20878, US

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Observer blind, no further details

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	The safety population was used for all safety analyses and was defined as all study subjects who provided consent, were randomised, and received any study test article.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The safety population was used for all safety analyses and was defined as all study subjects who provided consent, were randomised, and received any study test article.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	The Per protocol population was used for all immunogenicity analyses and was defined as all subjects in the safety population who received the assigned dose of the investigational vaccine according to protocol and as randomised, had RSV serology results at least at days 0, 28, and 56, and had no major protocol deviations affecting the primary immunogenicity outcomes as determined by Novavax prior to unblinding. More than 5% drop outs in the per protocol population.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01709019), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Gantt 2018

Methods	Phase 1 randomised, observer-blind, placebo-controlled
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "generally healthy adult female and male 18 to 40 years of age, inclusive; serologically confirmed to be CMV seronegative at screening, female volunteers must agree to use an adequate contraception method as deemed appropriate by the investigator". **Exclusion criteria:** "history of or current clinically significant medical illness or any other illness that in the opinion of the investigator interferes with the interpretation of the study results, clinically significant abnormal physical examination, vital signs, or clinically significant abnormal values for hematology, clinical chemistry or urinalysis at screening as determined by the investigator, previous receipt of any cytomegalovirus vaccine, history of allergic reactions or

	anaphylactic reaction to any vaccine component, pregnant or breastfeeding or plans to conceive from two weeks before the study start through six months after the last dose of study vaccine, known or suspected impairment of immunological function, including but not limited to autoimmune diseases, splenectomy, or HIV/AIDS, chronic administration (defined as more than 14 days in total) of immune-suppressive or other immune-modifying drug with six months prior to the product dose (for corticosteroids, this is defined as prednisone ≥20 mg/day or equivalent). Inhaled and topical steroids are allowed, participation in another clinical study within 30 days or plans to participate in another treatment based clinical study during the conduct of the present study, any skin abnormality or tattoo that would limit post-vaccination injection site assessment, any history of cancer requiring chemotherapy or radiation within 5 years of randomization or current disease, are family members of study center staff".
Interventions	**Vaccine type:** human cytomegalovirus **Experimental:** 1 mcg of CMV envelope glycoprotein B (gB) with aluminium. n =26 **Control:** 1 mcg of CMV envelope glycoprotein B (gB). n = 25 **Aluminium concentration:** not reported **Aluminium type:** Adju-Phos®® **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 injections on days 0, 56, and 168
Outcomes	**Safety:** "number of participants with local and systemic adverse events during seven-day follow-up period [time frame: day of vaccine administration (days 0, 56, 168) and six subsequent days]. Number of participants with any adverse event [time frame: following each of the 3 injections of study vaccine, the occurrence of adverse events was captured during a 28-day follow-up period as well as through day 336 or early withdrawal.]. Number of participants with any serious adverse event [time frame: through day 336 or early withdrawal]. Number of participants with any hematological or biochemical laboratory abnormality [time frame: through day 336 or early withdrawal]". **Immunogenicity:** "geometric mean titer of antibody binding to CMV gB [time frame: through day 336 or early withdrawal]. Geometric mean titer of antibody avidity index value against gB [time frame: through day 336 or early withdrawal]. To measure the avidity of responses against CMV gB protein, a standard ELISA assay using recombinant gB protein which did or did not include treatment with 5M urea for 30 minutes of samples after sera had been incubated with recombinant protein. The reported value, or Avidity Index, represents the percent of signal measured in ELISA after treatment with urea relative to samples not exposed to urea. Geometric mean titer of neutralizing antibody against CMV infection of fibroblast cells [time frame: through day 196 or early withdrawal]. Geometric mean titer of neutralizing antibody against CMV infection of epithelial cells [time frame: through day 336 or early withdrawal]".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 30/01/2020 authors (Joanne.Langley@Dal.Ca) were inquired about the followings: ● random sequence generation ● allocation concealment ● final aluminium concentration per dose

	● aluminium manufacturer Answer received and information added **Fundings:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A modified blocked randomization stratified by study center ensured that each study center had at least one subject randomised to every dose group. Participants were assigned to a specified treatment based on the randomization schedule.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	"This was an observer-blinded study. Both Participants and the study center staff performing outcome measurement were blinded; vaccines were administered by qualified unblinded study personnel who had no other role in the

		study. As this was a blinded study, all data presented including immunological testing and safety assessments were presented by group only to maintain the blind. The DSMB reports were provided to assist the DSMB in their safety evaluations; however, those data were blinded and presented in aggregate to maintain the blind".
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02826798), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Glenn 2013

Methods	Phase 1, placebo-controlled, observer-blinded, dose-escalating study
Participants	**Sex:** male and female **Age:** 18–49 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male or female aged 18 to 49 years inclusive; ability to provide written informed consent to participate; healthy, as determined by medical history, physical examination, vital signs, and clinical safety laboratory examinations at baseline; females are required to fulfill one of the following criteria:At least 1 year post-menopausalSurgically sterileWilling to use oral, implantable, transdermal or injectable contraceptives for 30 days prior to first vaccination and until 28 days after each vaccinationWilling to abstain from sexual intercourse or use another reliable form of contraception approved by the Investigator (eg, intrauterine device, female condom, and diaphragm with spermicide, cervical cap, use of condom by the sexual partner, or a sterile sexual partner) for study duration and until 28 days after vaccination; all female subjects must have a negative urine pregnancy test within 48 hours preceding receipt of each vaccination. Comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits and be contacted by telephone throughout the follow-up period". **Exclusion criteria:** "presence of significant uncontrolled medical or psychiatric illness (acute or chronic). This includes institution of new medical or surgical treatment, or a significant dose alteration for uncontrolled symptoms or drug toxicity within 3 months of screening and reconfirmed on Day 1 prior to vaccination; positive serology for HIV-1 or HIV-2, or HBsAg or HCV antibodies; pregnant or lactating female; females who plan to become pregnant or plan to discontinue contraceptive precautions within 30 days prior to first vaccination

	and 28 days after each vaccination; cancer, or treatment for cancer, within 3 years, excluding basal cell carcinoma or squamous cell carcinoma, which is allowed; presence of any medical condition that may be associated with impaired immune responsiveness, including diabetes mellitus; receipt (or history of receipt), during the preceding 3-month period, of any medications or other treatments that may adversely affect the immune system such as allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs or other drugs known to be frequently associated with significant major organ toxicity, or systemic corticosteroids (oral or injectable). Inhaled and topical corticosteroids will be allowed. receipt or planned administration of a non study vaccine within 30 days prior to vaccination or during the study. If a non study vaccine has been administered, administration of study vaccine injection can be delayed and given as soon as allowable within the 30-day window, provided the non study vaccine is not administered within 2 weeks prior to study enrollment. Immunization with Tetanus Toxoids Adsorbed for adult use (Td or Tdap) vaccine, on an emergency basis, up to 8 days before or at least 8 days after a dose of study vaccine will be allowed. History of illicit drug or alcohol abuse within the previous 1 year or positive drug or alcohol screen; history of anaphylactic type reaction to injected vaccines; receipt of any investigational product or non registered drug within the 30 days prior to vaccination or current enrollment in any investigational drug study or intent to enroll in such a study within the ensuing study period; receipt or donation of blood or blood products 8 weeks prior to vaccination or planned receipt or donation during the study period; acute disease, defined as the presence of a moderate or severe illness (as determined by the Investigator through medical history and physical examination) with or without fever, or an oral temperature of ≥38ºC, within 72 hours prior to vaccination. Study vaccine can be administered to persons with a minor illness, such as diarrhea or mild upper respiratory tract infection with or without low-grade fever. Vaccination can be delayed until the subject has recovered. Any condition that, in the opinion of the Investigator, would interfere with the primary study objectives".
Interventions	**Vaccine type:** recombinant respiratory syncytial virus (RSV) fusion (F) protein nanoparticle vaccine **Experimental:** 30 or 60 mcg RSV F with aluminium phosphate. n = 40 **Control:** 30 or 60 mcg RSV F. n =40 **Aluminium concentration:** 1200 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** 2 doses intramuscularly on day 0 and 30
Outcomes	**Immunogenicity:** "neutralizing antibody and total anti-F antibody of the 6 RSV-F protein particle vaccine formulations; time frame: day 60 (2 months) assessed in the following manner: neutralizing antibody against RSV measured in a PRNT assayTotal anti-F IgG measured by enzyme-linked immunosorbent assay". **Safety:** "time frame: day 60 (2 months): immediate adverse events, solicited adverse events, all serious adverse events and SNMCs, vital signs, laboratory assessments. Study subjects used diary cards to record the occurrence and intensity of solicited reactions for 7 days post each vaccination. Unsolicited events including serious adverse events were recorded at each visit through study end (day 210). A clinical chemistry panel, CBC, differential and platelet count were evaluated at day 0 and on study days 30 and 60. Tolerability of each dose level was monitored by a Safety Monitoring Committee after the first

	injection before proceeding to the next higher dose. The criteria used to evaluate safety included: immediate adverse events reported within 30 min of vaccination; solicited and unsolicited adverse events, serious adverse events; and clinical laboratory safety measurements".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (gglenn@novavax.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	High risk	Observer-blind. The site study personnel, Safety Monitoring Committee and study participants were blinded to treatment assignments. Study personnel responsible for vaccine preparation were unblinded, but had no involvement in vaccine administration or evaluation of the safety or immunogenicity results
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Blinded Safety Monitoring Committee
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Drop outs less than 5%
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Safety analysis was on total randomised population.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop-outs <5%.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Immunogenicity analysis was on the per protocol population. Drop-outs > than 5%.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01290419), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Glenn 2016

Methods	Phase 2 randomised, observer-blind, placebo-controlled study
Participants	**Sex:** female **Age:** ≥ 18 and ≤ 35 years of age **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy adult females, ≥ 18 and ≤ 35 years of age. "Healthy" shall be defined by the absence of any illness, acute or chronic, that requires ongoing systemic therapy for the control of symptoms or prevention of disability. Subjects on stable (no change in ≥ 3 months) therapy for findings (e.g., hypertension or hyperlipidemia) that are not associated with symptoms or disability are eligible, as are users of hormonal contraceptives. Subjects who receive intermittent prophylaxis for risks associated with asymptomatic findings (e.g., antibiotic prophylaxis prior to dental procedures in a subject with mitral valve prolapse) are eligible. Ongoing therapy will be defined as continuous or, if intermittent, more frequent than once every 3 months (e.g., use of an inhaled bronchodilator for exercise-induced bronchospasm more than once every 3 months). Persons being treated for illnesses or conditions that would become acutely symptomatic or disabling in the absence of treatment are not eligible. Willing and able to give informed consent prior to study enrollment. Able to comply with study requirements. Women who are not surgically sterile must have a negative urine pregnancy test prior to each vaccination; will be advised through the Informed Consent process to avoid becoming pregnant over the duration of the study, and must assert that they will employ an effective form of birth control for the duration of the study. Acceptable forms of birth control are: credible history of continuous abstinence from heterosexual activity, hormonal contraceptives (oral, injectable, implant, patch, ring), double-barrier contraceptives (condom or diaphragm, with spermicide), and IUD". **Exclusion criteria:** "participation in research involving investigational product (drug / biologic / device) within 45 days before planned date of first vaccination. History of a serious reaction to any prior vaccination. Received any vaccine in the 4 weeks preceding the study vaccination; or any RSV vaccine at any time. Any known or suspected immunosuppressive condition, acquired or congenital, as determined by history and/or physical examination. Chronic administration

	(defined as more than 14 continuous days) of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. An immunosuppressant dose of glucocorticoid will be defined as a systemic dose ≥10mg of prednisone per day or equivalent. The use of topical, inhaled, and nasal glucocorticoids will be permitted. Administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or during the study. Acute disease at the time of enrollment (defined as the presence of a moderate or severe illness with or without fever, or an oral temperature >38.0°C on the planned day of vaccine administration). Known disturbance of coagulation. Women who are pregnant or breastfeeding, or plan to become pregnant during the study. Suspicion or recent history (within one year of planned vaccination) of alcohol or other substance abuse. Any condition that in the opinion of the investigator would pose a health risk to the subject if enrolled or could interfere with evaluation of the vaccine or interpretation of study results (including neurologic or psychiatric conditions deemed likely to impair the quality of safety reporting)".
Interventions	**Vaccine type:** recombinant respiratory syncytial virus (RSV) fusion (F) protein nanoparticle vaccine **Experimental:** 60 or 90 mcg RSV F with aluminium phosphate. n =120 **Control:** 60 or 90 mcg RSV F. n =119 **Aluminium concentration:** 1200 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** one intramuscular injections into the deltoid on day 0 and a placebo dose consisted of isotonic saline on day 28 to preserve the blind or two intramuscular injections into the deltoid, on day 0 and day 28 (±2 days) in opposite arms
Outcomes	**Immunogenicity:** "serum IgG antibody titers specific for the F-Protein antigen across treatment groups, (time frame: day 0 to day 112). Immunogenicity will be measured using derived / calculated endpoints based on: geometric mean titer, geometric mean ratio, seroconversion rate, seroresponse rate". **Safety:** "time frame: day 0 to day 182: number (and percentage) of subjects with solicited local and systemic adverse events over the seven days post-injections; all adverse events, solicited and unsolicited over 56 days post-first injection. Significant new medical conditions, medically attended events and serious adverse events will be collected for six months".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (gglenn@novavax.com) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of outcome assessment ● total number of participants with non-serious adverse events ● number of patients who got respiratory syncytial virus infection after the vaccination. No answer received **Funding:** supported in part by PATH (formerly Program for Appropriate Technology in Health), Seattle, Washington

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Observer blind
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Safety analysis was on total randomised population.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Immunogenicity analysis was on the per protocol population. Drop-outs > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01704365), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Greenberg 2018

Methods	A phase 2 randomised, double-blind, multicenter study
Participants	**Sex:** male and female **Age:** 6 weeks to 12 weeks **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy infants ≥ 42 days to ≤ 89 days. Participant's parent/legal guardian understands the study procedures, alternate treatments available and risks involved with the study, and voluntarily agrees to allow the child to participate by giving written informed consent. Afebrile, with a rectal

	temperature <38.1°C (<100.5°F) or axillary temperature <37.8°C (<100.0°F) on day of vaccination. Participant's parent/legal guardian is able to read, understand, and complete study questionnaires (i.e., the Vaccination Report Card). Participant is able to attend all scheduled visits and to comply with the study procedures. Participant's parent/legal guardian has access to a telephone". **Exclusion criteria:** "prior administration of any pneumococcal vaccine. Known hypersensitivity to any component of the pneumococcal conjugate vaccine. Known or suspected impairment of immunological function. Participant has a history of congenital or acquired immunodeficiency (e.g., splenomegaly). Participant or his/her mother has documented human immunodeficiency virus infection. Functional or anatomic asplenia. History of autoimmune disease including multiple sclerosis, systemic lupus, polymyositis, inclusion body myositis, dermatomyositis, Hashimoto's thyroiditis, Sjogren's syndrome, rheumatoid arthritis, other autoimmune disorders. Known neurologic or cognitive behavioral disorders including multiple sclerosis, MS-like disease, encephalitis/myelitis, acute disseminating encephalomyelitis, pervasive developmental disorder, and related disorders. Use of any immunosuppressive therapy (Note: topical and inhaled/nebulized steroids are permitted). Participants on intramuscular, oral, or intravenous corticosteroid treatment should be excluded if they are receiving or expected to receive in the period from 30 days prior to Visit 1 through Visit 6 (30 days post-dose 4) more than 2 mg/kg per day of prednisone (or its equivalent), or more than 20 mg/d if they weigh more than 10 kg and are not otherwise immunocompromised. Excluded immunosuppressive therapies also include chemotherapeutic agents used to treat cancer or other conditions, and treatments associated with organ or bone marrow transplantation, or autoimmune disease. Participant has received other licensed non-live vaccines within the 14 days before receipt of study vaccine. Participant has received a licensed live virus vaccine within the 30 days prior of receipt of study vaccine. Prior receipt of a blood transfusion or blood products, including immunoglobulins. Investigational drugs or vaccines received within the 2 months before receipt of study vaccine. Participation in another clinical study within 42 days before the beginning or anytime during the duration of the current clinical study. History of invasive pneumococcal disease (positive blood culture, positive cerebrospinal fluid culture, or other sterile site) or known history of other culture positive pneumococcal disease. A recent febrile illness (rectal temperature ≥38.1°C [≥100.5°F]) occurring within 72 hours before receipt of study vaccine. History of failure to thrive. Participant has a coagulation disorder contraindicating IM vaccination. Participant and his/her mother have documented hepatitis B surface antigen-positive. Any infant who cannot be adequately followed for safety according to the protocol plan. Participant's parent/legal guardian is unlikely to adhere to study procedures, keep appointments, or is planning to relocate during the study. Any other reason that in the opinion of the investigator may interfere with the evaluation required by the study".
Interventions	**Vaccine type:** 15-valent pneumococcal conjugate vaccine (PCV15) **Experimental:** aluminium-adjuvanted-PCV15. n = 378 **Control:** PCV15. n = 386 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** alum phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** 4-dose regimen at 2, 4, 6, and 12–15 months of age

Outcomes	**Immunogenicity:** "percentage of participants achieving the Immunoglobulin G serotype-specific ≥0.35 µg/mL threshold value for postvaccination 3, (time frame: one month postvaccination 3). Percentage of participants meeting the serotype-specific IgG reference level (an antibody concentration measured by the pneumococcal polysaccharide electrochemiluminescence assay corresponding to the World Health Organization enzyme-linked immunosorbent assay ≥ 0.35 µg/mL) (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. IgG Geometric mean concentrations for Postvaccination 3, (time frame: one month postvaccination 3). The IgG geometric mean concentrations were evaluated as measured in the pneumococcal polysaccharide electrochemiluminescence assay, for recipients of adjuvated V114, non-adjuvanted V114, and Prevnar 13® (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. IgG geometric mean concentrations for postvaccination 4, (time frame: one month postvaccination 4). The IgG geometric mean concentrations were evaluated as measured in the pneumococcal polysaccharide electrochemiluminescence assay, for recipients of adjuvated V114, non-adjuvanted V114, and Prevnar 13® (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Percentage of participants with opsonophagocytic killing activity titer ≥ 8 as measured by fourfold multiplexed opsonization assay for postvaccination 3, (time frame: one month postvaccination 3). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity antibody responses. Percentage of participants with opsonophagocytic killing activity titer ≥ 8 as measured by fourfold multiplexed opsonization assay for postvaccination 4 (time frame: one month postvaccination 4). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity antibody responses. Opsonophagocytic killing activity geometric mean titers as measured by fourfold multiplexed opsonization assay for postvaccination 3 (time frame: one month postvaccination 3). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 3) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. The opsonophagocytic killing activity antibody responses included the percentage of participants with opsonophagocytic killing activity titer and the geometric mean titers. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity

	antibody responses.Opsonophagocytic killing activity geometric mean titers as measured by fourfold multiplexed opsonization assay for postvaccination 4 (time frame: one month postvaccination 4). The antibody responses as measured by fourfold multiplexed opsonization assay (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Functional antibody activity was assessed only in a subset of the vaccinated participants, thus the study was not powered for assessing non-inferiority with respect to the opsonophagocytic killing activity antibody responses. Percentage of participants achieving the IgG Serotype-specific Threshold Value of ≥0.35 µg/mL for Postvaccination 4, (time frame: one month postvaccination 4). Percentage of participants achieving World Health Organization predefined antibody threshold as measured by the pneumococcal polysaccharide electrochemiluminescence assay corresponding to the World Health Organization enzyme-linked immunosorbent assay value of ≥ 0.35µg/mL (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®. Percentage of participants Achieving the IgG serotype-specific threshold value of ≥1.0 µg/mL for Postvaccination 4 [(time frame: one month postvaccination 4). Percentage of participants achieving World Health Organization predefined antibody threshold as measured by the pneumococcal polysaccharide electrochemiluminescence assay corresponding to the World Health Organization enzyme-linked immunosorbent assay value of ≥ 1.0 µg/mL (post-dose 4) for the 13 serotypes in common with PREVNAR 13® (serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F) and the 2 serotypes (serotypes 22F and 33F) not in common with PREVNAR 13®". **Safety:** "number of participants with an adverse event (time frame: up to day 14 postvaccination). An adverse event is defined as any unfavorable and unintended change in the structure, function, or chemistry of the body temporally associated with the use of the Sponsor's product, whether or not considered related to the use of the product. Any worsening (i.e., any clinically significant adverse change in frequency and/or intensity) of a preexisting condition which is temporally associated with the use of the Sponsor's product, is also an adverse experience. Number of participants with an injection-site adverse event (time frame: up to day 14 postvaccination). Injection-site adverse event reported by > 0% of participants in one or more vaccination groups were assessed. Number of participants with a systemic adverse event (time frame: up to day 14 postvaccination). Systemic adverse events reported by > 0% of participants in one or more vaccination groups were assessed. Number of participants with a serious adverse event, (time frame: up to one month after last dose of study vaccine). A serious adverse event is an adverse event that results in death, is life threatening, requires or prolongs an existing hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is another important medical event deemed such by medical or scientific judgment. Number of Participants who discontinued the study due to an adverse event, (time frame: up to day 14 postvaccination). Any worsening (i.e., any clinically significant adverse change in frequency and/or intensity) of a preexisting condition which is temporally associated with the use of the Sponsor's product, is also an adverse experience".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** the email address orf the corresponding author provided in the publication was not valid. **Funding:** provided by Merck & Co., Inc., Kenilworth, NJ, USA

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop-outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	High risk	Drop-outs >5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop-outs >5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT01215188), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Harro 2001

Methods	Double-blind, randomised, placebo-controlled, phase I safety and immunogenicity trial
Participants	**Sex:** male and female **Age:** 18- to 29-years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy, human immunodeficiency virus-1-seronegative, 18- to 29-year-old volunteers. Subjects were determined by history to be at low risk for HPV16 exposure". **Exclusion criteria:** "individuals were not eligible to participate if they had a history of more than four lifetime sexual partners or more than two sexual partners within the preceding 6 months. History of abnormal cervical cytology, immunodeficiency, anaphylaxis to medicines or vaccines, receipt of blood products within 3 months of enrollment, current pregnancy or lactation, and any other condition that might interfere with the study objectives".
Interventions	**Vaccine type:** human papillomavirus 16 L1 virus-like particle vaccine (HPV16 L1 VLP) **Experimental (a):** 10 mcg HPV16 L1 VLP with aluminium potassium sulfate. n = 10 **Control (a):** 10 mcg HPV16 L1 VL. n = 10 **Experimental (b):** 50 mcg HPV16 L1 VLP with aluminium potassium sulfate. n = 10 **Control (b):** 50 mcg HPV16 L1 VL. n =10 **Aluminium concentration: (a)** 1200 mcg/dose. **(b)** 6000 mcg/dose **Aluminium type:** aluminium potassium sulfate **Aluminium manufacturer:** Merck **Vaccination schedule:** 3-doses; intramuscular injection in the deltoid region at months 0, 1, and 4
Outcomes	**Immunogenicity:** "subjects were evaluated clinically, and blood was collected for clinical and immunologic tests before each injection (months 0, 1, and 4) and 1 month after each injection. Clinical laboratory tests done at each of these visits included the following: CBC, platelet count, serum ALT, serum creatinine, urine protein, and urine hemoglobin measurements. IgG-specific HPV16 L1 VLP-based enzyme-linked immunosorbent assays were performed. In vitro HPV16 pseudovirion neutralization assays were performed". **Safety:** "after each injection, oral temperatures were recorded after 30 minutes, 6 hours, and daily for 6 days. Adverse reactions were monitored by study staff in the clinic at 30 minutes and 2 days after each injection and by telephone for the following 6 days".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (lowyd@mail.nih.gov) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel

	● number of drop outs (patients randomised but not analysed) ● funding source ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events /number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) ● serological response per patient at maximum follow up No answer received **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	To maintain blinding, the individual who prepared and administered the vaccine was not involved in data collection or clinical evaluation of volunteers.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	To maintain blinding, the individual who prepared and administered the vaccine was not involved in data collection or clinical evaluation of volunteers.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%

Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Harro 2012

Methods	Phase-1, randomised, controlled, double-blind trial
Participants	**Sex:** male and female **Age:** 18–70 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "participant was in good physical health. Participant was able to understand study procedures and agreed to participate in the study by providing written informed consent. Participant was willing and able to participate in the entire study duration planned for up to 12 months (~360 days). Female participants of reproductive potential were required to have a negative urine pregnancy test immediately prior to study vaccination. Female participants of reproductive potential must have used an acceptable method of birth control for 2 weeks prior to enrollment, and agreed to use an acceptable method of birth control for 1 month after vaccination". **Exclusion criteria:** "participant suffered from a chronic skin condition that predisposed the individual to the development of chronic skin or soft-tissue infections (e.g., psoriasis, chronic granulomatous disease, atopic dermatitis). Participant developed a serious infection (e.g., bacteremia, pneumonia, mediastinitis) attributed to S. aureus in the 12 months prior to screening. Participant had a history of anaphylaxis to aluminium-containing adjuvant or other vaccine components. Participant had a temperature of ≥100.4ºF (≥38.0ºC), oral equivalent, within 48 hours prior to receipt of V710. Participant had received a live virus vaccine within 30 days prior to receipt of V710 or was scheduled to receive vaccination with a live virus vaccine within 30 days following study entry. Participant had received any other licensed vaccine (including non-live virus vaccines) within 14 days prior to receipt of V710 or was scheduled to receive any other licensed vaccine (including non-live virus vaccines) within 30 days following study entry. Participant had received V710 in a prior clinical study (Merck V710 Protocol 001). Participant was administered immunoglobulin or blood product within 90 days prior to receipt of V710 or was scheduled to receive such products within 30 days following study entry. Participant had received treatment with systemic (intramuscular, oral, or intravenous) corticosteroids or another immunosuppressive medication (e.g., calcineurin inhibitors, mycophenolate, azathioprine) or biological agents (e.g., rituximab) in the 14 days prior to receipt of V710 or was anticipated to receive such medications for a chronic medical condition during the course of the study. Participant had a condition that required active medical intervention or monitoring to avert serious danger to the participant's health or well-being, such as diabetes mellitus, autoimmune disease, or clinically significant chronic medical conditions that were considered progressive, including but not limited to: coronary artery disease, congestive heart failure, cardiomyopathy, progressive valvular heart disease, chronic obstructive pulmonary disease, pulmonary fibrosis, active peptic ulcer disease, chronic renal disease, chronic hepatic disease, multiple sclerosis, progressive neuropathies, or seizure disorder requiring therapy in the past 3 years. Participant had known or

	suspected impairment of immunologic function including, but not limited to, the following conditions: autoimmune disease, diabetes mellitus, endstage renal disease, hepatic insufficiency/cirrhosis, splenectomy, or human immunodeficiency virus/acquired immune deficiency syndrome. Participant had a condition in which repeated venipuncture or injections posed more than minimal risk for the subject, such as hemophilia, other severe coagulation disorders, or significantly impaired venous access. Participant was pregnant or breastfeeding, or planning to conceive within the 12-month study duration period. Participant had clinically significant abnormalities based on the participant or physical examination. Participant had recent history (within the past 5 years) or current evidence of drug or alcohol abuse. Participant had a major psychiatric illness, including any history of schizophrenia or severe psychosis, bipolar disorder requiring therapy, or any subject with suicidal ideation within the previous 3 years. Participant was legally or mentally incapacitated. Participant had participated in another clinical study in the past 4 weeks, or participant planned to participate in a treatment-based study or study in which an invasive procedure was to be performed during the first 3 months of this study (through Day 84). Participant had a history of any condition which, in the opinion of the investigator, posed an additional risk for the subject or confounded the results of the study".
Interventions	**Vaccine type:** staphylococcus aureus vaccine (V710) **Experimental:** aluminium-adjuvanted V710. n = 32 **Control:** V710. n = 32 **Aluminium concentration:** not reported **Aluminium type:** amorphous aluminium hydroxyphosphate sulfate (AAHS) **Aluminium manufacturer:** Merck **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "number of participants with a positive immune response, defined as a change in antibody level greater than or equal to a 2-fold-rise at day 14 compared to baseline, (time frame: baseline and day 14 postvaccination). Immunoglobulin G antibodies were measured by a LUMINEX assay. Geometric mean antibody concentrations, (time frame: day 14 postvaccination). Immunoglobulin G antibodies were measured by a LUMINEX assay. Change in antibody concentration (titer) at day 14 compared to baseline, expressed as the geometric mean fold-rise (time frame: day 14 postvaccination). Immunoglobulin G antibodies were measured by a LUMINEX assay.The geometric mean fold-rise is the ratio of the antibody concentration at day 14 to the antibody concentration at baseline". **Safety:** "number of vaccine-related serious adverse experiences, (time frame: up to day 360 postvaccination). Investigators were instructed to determine the seriousness and causality (relatedness to test vaccine) of each adverse experiences based on criteria defined in the protocol: a serious adverse event is any adverse experiences that: results in death, is life threatening, results in a persistent or significant disability/incapacity, results in or prolongs an existing inpatient hospitalization, is a congenital anomaly/birth defect, is a cancer, is an overdose, or is another important medical event that may require medical or surgical intervention to prevent one of the outcomes listed above".
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 14/12/2018 authors (charro@jhsph.edu, nicholaskartsonis@merck.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received Results from the trial is also available here: clinicaltrials.gov/ct2/show/results/NCT01324440. There are some minor discrepancies in the number of adverse events reported in the trial report compared to the data available at clinicaltrials.gov. We used the data from clinicaltrials.gov when meta-analysing AEs as more adverse events were available here. **Funding:** funded by Merck Sharp & Dohme, a subsidiary of Merck & Co., Inc. The study sponsor was involved in the study design; data collection, analysis, and interpretation of data; in writing the report; and the decision to submit the paper for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	"Double-blind. Because of the differences in appearance between V710 with and without adjuvant, and between V710 and saline placebo, one unblinded member of the study site staff was responsible for preparing and administering the study vaccine or placebo. This person had no further involvement with any subsequent study procedure postvaccination, including safety follow-up procedures. All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained blinded to treatment group throughout the study".
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	"Double-blind. Because of the differences in appearance between V710 with and without adjuvant, and between V710 and saline placebo, one unblinded member of the study site staff was responsible for preparing and administering the study vaccine or placebo. This person had no further involvement with any subsequent study procedure postvaccination, including safety follow-up procedures. All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained

		blinded to treatment group throughout the study".
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained blinded to treatment group throughout the study
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All other individuals involved with the study conduct, including investigators, study physicians, and statisticians remained blinded to treatment group throughout the study
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	All patients in both studies were included in the primary analyses.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	All patients in both studies were included in the primary analyses.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	All patients in both studies were included in the primary analyses.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01324440), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Hung 2019

Methods	Open-label, multicenter, phase II, randomised studies
Participants	**Sex:** male and female **Age:** 2 months to 6 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "healthy children aged from 2 months to 6 years old at the time of first vaccination. Subject's guardians are able and willing to comply with study procedures and provide the signed informed consent. Subject is able and can comply with the requirements of the protocol. Subject with body temperature ≤38°C". **Exclusion criteria:** "subject with previous known exposure to Enterovirus 71 (EV71). Subject with a history of herpangina, hand-foot-mouth disease,and acute hemorrhagic conjunctivitis associated with enterovirus infection in the past 3 months. Subject with gestation < 37 weeks. Subject with birth weight <2.5 kg. Subject with a history of hypersensitivity to vaccines, or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Family history of seizures or progressive neurological disease. Family history of congenital or hereditary immunodeficiency. Severe malnutrition or dysgenopathy. Major congenital defects or serious chronic illness, including perinatal brain damage. Subject diagnosed of having autoimmune disease (e.g., celiac disease, type I diabetes, lupus, juvenile dermatomyositis, scleroderma, juvenile idiopathic arthritis, immune (or idiopathic) thrombocytopenia purpura). Bleeding disorder diagnosed by a doctor or significant bruising or hemostatic difficulties with IM injections or blood draws. Any acute infections 7 days prior to administrating the first vaccination. Use of any investigational product (including

	drug, vaccine) within 30 days prior to vaccination or planned use during the study period. Administration of any vaccines within 14 days prior to randomisation. Use of immunoglobulins or any blood products within 3 months prior to vaccination or planned use during the study period. Chronic administration (defined as > 14 days) of immunosuppressants or other immunomodulators or systemic corticosteroids within 6 months prior to vaccination or planned use during the study period. Subjects who had ever received investigational EV-71 vaccine prior to randomisation. Under anti-tuberculosis prevention or therapy. Any condition that in the opinion of the investigator may interfere with the evaluation of study objectives".
Interventions	**Vaccine type:** inactivated enterovirus A71 vaccine **Experimental:** 1.0 mcg EV-A71 with aluminium hydroxide. n = 47 **Control:** 1.0 mcg EV-A71. n =44 **Aluminium concentration:** 150 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** one intramuscular injection
Outcomes	**Immunogenicity:** "serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 56). To evaluate the immunogenicity change of serum neutralizing antibody titers induced by the EV71 vaccine from baseline on day 56. Seroconversion rate based on neutralizing antibody titers, (time frame: day 56). to evaluate the immunogenicity change of seroconversion rate from baseline on day 56. Serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 28).To evaluate the immunogenicity change of serum neutralizing antibody titers induced by the EV71 vaccine from baseline on day 28. Serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 196). To evaluate the immunogenicity change of serum neutralizing antibody titers induced by the EV71 vaccine from baseline on Day 196. Seroconversion rate based on neutralizing antibody titers, (time frame: day 28).To evaluate the immunogenicity change of seroconversion rate from baseline on Day 28. Seroconversion rate based on neutralizing antibody titers, (time frame: day 196). To evaluate the immunogenicity change of seroconversion rate from baseline on day 196. Serum neutralizing antibody titers induced by the EV71 vaccine, (time frame: day 364). T evaluate the immunogenicity of serum neutralization antibody titer induced by the EV 71 vaccine on Day 364". **Safety:** "solicited adverse events, (time frame: 7 days after each vaccination). Unsolicited adverse events, (time frame: 28 days after each vaccination). The occurrence of overall adverse events and serious adverse event, (time frame: day 0 to day 196). The occurrence of EV 71 breakthrough infection after Visit 3, (time frame: day 57 to day 364)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (mchung@vghtpe.gov.tw, kgwu@vghtpe.gov.tw) were inquired on the followings: ● aluminium manufacturer. ● allocation concealment. ● total number of participants with non-serious adverse event related and not related to the vaccine ● number of participants who got Enterovirus infection after the vaccine.

	No answer received **Funding:** supported by Enimmune Corporation

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation list was generated using SAS software (version 9.4) and contained sequential coding which was assigned to group A1, A2 and A3 in the ratio of 1:1:1.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open-label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03268083), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Jackson 2009

Methods	Randomized, controlled, single-blind, multicenter phase II study
Participants	**Sex:** male and female **Age:** 11–17 years **Ethnicity:** Asian **Diagnosis:** healthy

	Inclusion criteria: healthy adolescents between and including 11-17 years of age, who provide written informed consent. **Exclusion criteria:** subjects with a previous or suspected disease caused by N. meningitidis; or previous immunization with a meningococcal vaccine or vaccine containing meningococcal antigen(s); any serious acute, chronic or progressive disease.
Interventions	**Vaccine type:** quadrivalent meningococcal glycoconjugate vaccine (MenACWY-CRM) **Experimental:** MenACWY-CRM with aluminium phosphate. n = 164 **Control:** MenACWY-CRM. n = 151 **Aluminium concentration:** 300 mcg **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** Novartis **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "percentages of subjects with N.Meningitidis human serum bactericidal activity titers≥ 1:4, After one dose of MenACWY-CRM Vaccine, with Adjuvant or without adjuvant, (time frame:1 month after vaccination). Immune response of a single dose of MenACWY-CRM conjugate vaccine, with adjuvant or without adjuvant, one month after administration to subjects aged 11 to 17 years, as measured by the percentage of subjects with human serum bactericidal activity titers >1:4 directed against N meningitidis serogroups A, C, W and Y. Human serum bactericidal activity geometric mean titers after one dose of MenACWY-CRM Vaccine, with adjuvant or without adjuvant (time frame: 1 month after vaccination). Immune response of one dose of MenACWY-CRM conjugate vaccine, with adjuvant or without adjuvant one month after administration in subjects 11 to 17 years of age, as measured by human serum bactericidal activity geometric mean titers against N meningitidis serogroups A, C, W, and Y. Percentages of subjects with human serum bactericidal activity titers≥ 1:4, after one dose of MenACWY-CRM(Ad-) (time frame: 12 months after vaccination). Human serum bactericidal activity geometric mean titers after one dose of MenACWY-CRM(Ad-) (time frame: 12 months after vaccination)". **Safety:** "numbers of subjects 11 to 17 years of age who reported solicited local and systemic adverse events after the vaccination (time frame: day 1 to day 7). Safety was assessed as the number of subjects 11 to 17 years of age who reported solicited local and systemic adverse events from day 1 up to and including day 7 after the vaccination of MenACWY-CRM conjugate vaccine, with adjuvant or without adjuvant".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (jackson.l@ghc.org, Lisa.A.Jackson@kp.org) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events/number of patients analysed at maximum follow up (please note that we are interested in

	SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** supported by grants from Novartis Vaccines, Siena, Italy

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Single blind
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	A total of 13 serious adverse events occurred but were not assigned per arm.

Juergens 2014

Methods	Randomized, open-label, multicenter trial
Participants	**Sex:** male and female **Age:** ≥65 years **Ethnicity:** African **Diagnosis:** healthy

	Inclusion criteria: "male and female adults 65 years of age and older. Determined by medical history, physical examination, and clinical judgment to be eligible for the study. Expected to be available for the duration of the trial (up to approximately 13 months)". **Exclusion criteria:** "received previous immunization with 23vPS. Serious chronic disorders including metastatic malignancy. Known or suspected hypersensitivity to any vaccine or vaccine components".
Interventions	**Vaccine type:** 13-valent pneumococcal conjugate vaccine (PCV13) **Experimental:** PCV13 with aluminium phosphate. n = 295 **Control:** PCV13. n =299 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** one dose intramuscularly into the deltoid muscle of the arm
Outcomes	**Immunogenicity:** "PCV13 serotype-specific immune responses were measured in a central laboratory by ELISA measuring IgG binding antibodies25 or by standardized microcolony OPA assays measuring functional antibody titers". **Safety:** "participants recorded local reactions (redness, swelling, pain, and limitation of arm movement) and systemic events (oral temperature to measure fever, fatigue, headache, chills, rash, vomiting, decreased appetite, and new and aggravated generalized muscle and joint pain) daily for 14 days after vaccination in an e-diary. Other AEs were collected by the investigator at each visit (29–43 days after each vaccination) based on clinical evaluation as well as in response to nonspecific questions".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (Beate.Schmoele-Thoma@Pfizer.com) were inquired on the followings: ● aluminium manufacturer ● allocation concealment ● number of participants with mortality and non-serious adverse event related and not related to the vaccine ● number of participants who got Streptococcus pneumoniae infection after the vaccine No answer received **Funding:** funded by Pfizer Inc

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation schedule was determined through an interactive voice response telephone system.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open-label

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00269672), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kashala 2002

Methods	Randomised, double-blinded, placebo-controlled phase I trial
Participants	**Sex:** male **Age:** 18–23 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "participants were recruited from the Colombian navy personnel in Tumaco by non-coercive means under international scientific and ethical standards embodied in the Declaration of Helsinki". **Exclusion criteria:** "subjects were excluded if they had a history of allergic reactions to vaccination, vaccination with live virus within 4 weeks of the beginning of the trial; had undergone splenectomy; or had a history of malaria, malaria parasitemia, or malaria vaccination, atopy, immunosuppressive drug therapy such as steroids; abnormal haematology and/or blood chemistry values. Volunteers were also excluded if they had positive markers of active auto-immunity; had cardiovascular, renal, and hepatic dysfunction or any other medical or psychiatric condition that in the opinion of the protocol chairman would compromise the patient's ability to tolerate the treatment. Furthermore, they were excluded if they were unwilling to give informed consent, or were willing to withdraw from the study".

Interventions	**Vaccine type:** plasmodium falciparum malaria peptide vaccine SPf66 **Experimental:** SPf66 with aluminium and 50 mcg of QS-21. n = 20 **Control:** SPf66 with 50 mcg of QS-21. n =19 **Aluminium concentration:** 1250 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Superfos Biosector (Vedvaek, Denmark) **Vaccination schedule:** three doses on days 0, 30 and 180
Outcomes	**Immunogenicity:** "sera were obtained at baseline, on day 20 following each immunisation, and at 4 months after the second and third doses, respectively for measurement of vaccine-induced immune responses. The antibody response against SPf66 has been characterised in previous reports based on end-point titres, using two-fold increments, as low/no response (enzyme-linked immunosorbent assay titres <100), intermediate (enzyme-linked immunosorbent assay titre: 200–800) or high (enzyme-linked immunosorbent assay titre =1600)". **Safety:** "evaluation of the reactogenicity, tolerability and toxicity. Reactogenicity was assessed by observing the incidence and severity of local and systemic reactions within 1, 24–48, 72 h and 20 days following the injections. Assessment of local reactions included erythema, tenderness/induration, pain, local pruritus, arm motion limitation related to pain and presence of local adenopathy. Erythema, tenderness/induration were scored as none, mild (<30mm in the largest diameter), moderate (30–120mm in the largest diameter) and severe (>120mm in the largest diameter). The limitation of the range of motion was considered a surrogate measurement of local pain, and was used to determine the severity of the local pain. Pain was scored as none or mild, if the range of motion was unchanged, and the subject could raise his arm to greater than a 90○ angle, moderate if the subject could raise his arm to no more than 45–90○ angle, and severe if the subject could not raise his/her arm to a 45○ angle. Systemic reactions were also rated, including allergic reactions such as rash/hives, generalised pruritus, bronchospasm, with or without hypotension, fever (≲38 ○C), anorexia, vomiting, abdominal pain, malaise/dizziness, chills, diarrhoea, headache, hypotension, cyst/nodules, abscess, necrosis, axillary adenopathy, myalgia, arthralgia and arthritis. To monitor toxicity, the blood specimens and sera were obtained from each subject at baseline, 48 h and 20 days post vaccination for haematology and blood chemistry".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (oscar.kashala@emdpharmaceuticals.com, gerd.pluschke@swisstph.ch) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in

	SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received, but no further information or data added **Funding:** supported by the Government of Colombia, and UNDP/World Bank/WHO Special Programme for Research and Training in Tropical Diseases

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blinded, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Keitel 2008

Methods	Multicenter, randomised, double-blind, placebo-controlled clinical trial
Participants	**Sex:** male and female **Age:** 18–49 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "males or nonpregnant females between the ages of 18 and 49 years, inclusive. Agree to practice adequate contraception (i.e., barrier methods, abstinence, intrauterine devices, and licensed hormonal methods) for the entire study period if they are females of childbearing potential (not surgically sterile or postmenopausal for greater than or equal to 1 year). Are in good health as determined by vital signs, medical history to ensure stable medical condition, and targeted physical examination based on medical history. Are able to understand and comply with planned study procedures. Provide written informed consent prior to initiation of any study procedures. Stable medical condition - no change in prescription medication, dose, or frequency of medication in the last 3 months and health outcomes of the specific disease are considered to be within acceptable limits in the last 6 months. Any change that is due to change of health care provider, insurance company etc, or that is done for financial reasons, as long as in the same class of medication will not be considered a violation of this inclusion criterion. Any change in prescription medication due to improvement of a disease outcome will not be considered a violation of this inclusion criterion". **Exclusion criteria:** "have a known allergy to eggs or other components of the vaccine (including gelatin, formaldehyde, octoxinol, thimerosal, aluminium hydroxide, and chicken protein). Have a positive urine or serum pregnancy test prior to vaccination (if female of childbearing potential) or are women who are breastfeeding. Have immunosuppression as a result of an underlying illness or treatment, or use of anticancer chemotherapy or radiation therapy within the preceding 36 months. Have an active neoplastic disease or a history of any hematologic malignancy. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (>800 mcg/day of beclomethasone dipropionate or equivalent) within the preceding 6 months (Nasal and topical steroids are allowed.). Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to self or others. Are receiving psychiatric drugs (aripiprazole, clozapine, ziprasidone, haloperidol, molindone, loxapine, thioridazine, thiothixene, pimozide, fluphenazine, risperidone, mesoridazine, quetiapine, trifluoperazine, chlorprothixene, chlorpromazine, perphenazine, trifluopromazine, olanzapine, carbamazepine, divalproex sodium, lithium carbonate or lithium citrate). Subjects who are receiving a single antidepressant drug and stable for at least 3 months prior to enrollment, without de-compensating symptoms will be allowed to be enrolled in the study. Have a history of receiving immunoglobulin or other blood product within the 3 months prior to vaccination in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to vaccination in this study. Have an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses (This includes, but is not limited to, known chronic liver disease, significant renal disease, unstable or progressive neurological disorders, diabetes mellitus, and transplant

	recipients.). Have a history of severe reactions following immunization with contemporary influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week of vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 1 month prior to vaccination in this study or expect to receive an experimental agent during the 7-month study period. Have any condition that would, in the opinion of the site investigator, place them at an unacceptable risk of injury or render them unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Planned to travel outside of the USA in the time between the first vaccination and 56 days following the first vaccination. Have a history of Guillain Barre syndrome. Have any condition that the investigator believes may interfere with successful completion of the study".
Interventions	**Vaccine type:** inactivated Influenza A/H5N1 vaccine **Experimental (a):** 3.75 mcg H5N1 with aluminium hydroxide. n = 61 **Control (a):** 3.75 mcg H5N1. n = 59 **Experimental (b):** 15 or 45 mcg H5N1 with aluminium hydroxide. n = 302 **Control (b):** 15 or 45 mcg H5N1. n =295 **Aluminium concentration: (a)** 300 mcg/dose. **(b)** 600 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Sanofi **Vaccination schedule:** 2 vaccinations 1 month apart
Outcomes	**Immunogenicity:** "proportion of subjects in each group achieving a serum neutralizing antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus, (time frame: 28 days following second dose of vaccine). Proportion of subjects in each dose group achieving a serum Hemagglutination Inhibition antibody titer of greater than or equal to 1:40 against the influenza A/H5N1 virus, (time frame: 28 days following second dose of vaccine). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group, (time frame: 28 days following second dose of vaccine). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group, (time frame: 28 days following second dose of vaccine). Geometric mean titer and the frequency of 4-fold or greater increases in serum HAI antibody titers, (time frame: 1 month after receipt of each dose, and 7 months after receipt of the first dose of vaccine). Development of serum antibody responses against antigenically drifted variants of H5N1 influenza virus, (time frame: blood samples for serum assays will be collected at day 0 and at days 28, 56, and 208 after the first immunization)". **Safety:** "adverse event or serious adverse event information (solicited in the clinic and via memory aids, concomitant medications, and periodic targeted physical assessments), (time frame: adverse events will be collected through 28 days following the second dose of vaccine (approximately day 56). Serious adverse events will be collected throughout the study through day 208)".
Are there other groups in the trial not analysed in this review?	No

Notes	**Contacts with authors:** on 14/12/2018 authors (wkeitel@bcm.edu) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● Final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** financial support: Division of Microbiology and Infectious Diseases, National Institute of Allergy and Infectious Diseases, National Institutes of Health (contracts AI-25465, AI-25461, and AI-25460); University of Maryland General Clinical Research Center (grant M01 RR165001); University of Rochester Clinical and Translational Science Institute (grant UL1 RR024160 from the National Center for Research Resources, National Institutes of Health).

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not reported
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not reported

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00296634), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Keitel 2009

Methods	Multicenter, randomised, double-blind, placebo-controlled clinical trial
Participants	**Sex:** male and female **Age:** 18 and 40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "stage 1 cohort: healthy males and females aged 18 to 40 years, inclusive; females of childbearing potential must agree to practice adequate contraception for the duration of the study (not surgically sterile or Aftermenopausal for at least 1 year); good health as determined by screening medical history and physical examination; on no medications with the exception of licensed hormonal contraceptives and vitamins; willingness to comply with study procedures; ability to provide informed consent; availability for follow-up for 6 months after second vaccination; normal screening laboratory values for total WBC, platelet count, hemoglobin, creatinine, and alanine aminotransferase (Stage 1 only). Stage 2 Cohort: healthy males and non-pregnant females aged 18 to 40 years, inclusive. Females of childbearing potential must agree to practice adequate contraception for the duration of the study. Good health as determined by screening medical history and physical examination. Willingness to comply with study procedures. Ability to provide informed consent. Availability for follow-up for 6 months after second vaccination". **Exclusion criteria:** "stage 1 cohort: have a positive urine or serum pregnancy test in the 24 hours prior to vaccination (if female of childbearing potential) or are women who are breastfeeding. Have immunosuppression as a result of an underlying illness or treatment, or use of anticancer chemotherapy or radiation therapy within the preceding 36 months. Have an active neoplastic disease or a history of any hematologic malignancy. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (>800 mcg/day of beclomethasone dipropionate or equivalent) within the preceding 6 months. (Nasal and topical steroids are allowed.) Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to self or others. Have a history of receiving immunoglobulin or other blood product within the 3 months prior to vaccination in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to vaccination in this study. Have an acute or chronic medical

condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses (This includes, but is not limited to, known chronic liver disease, significant renal disease, unstable or progressive neurological disorders, diabetes mellitus, and transplant recipients.) Have a history of severe reactions following immunization with contemporary influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week of vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 1 month prior to vaccination in this study or expect to receive an experimental agent during the 7-month study period. Have any condition that would, in the opinion of the site investigator, place them at an unacceptable risk of injury or render them unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Have a history of Guillain Barre Syndrome. Have any condition that the investigator believes may interfere with successful completion of the study. Stage 2 Cohort: have a positive urine or serum pregnancy test prior to vaccination (if female of childbearing potential) or are women who are breastfeeding. Have immunosuppression as a result of an underlying illness or treatment, or use of anticancer chemotherapy or radiation therapy within the preceding 36 months. Have an active neoplastic disease or a history of any hematologic malignancy. Have long-term use of oral steroids, parenteral steroids, or high-dose inhaled steroids (>800 mcg/day of beclomethasone dipropionate or equivalent) within the preceding 6 months (nasal and topical steroids are allowed.). Have a diagnosis of schizophrenia, Bi-polar disease or other major psychiatric diagnosis. Have been hospitalized for psychiatric illness, history of suicide attempt or confinement for danger to self or others. Are receiving psychiatric drugs*. Subjects who are receiving a single antidepressant drug and stable for at least 3 months prior to enrollment, without de-compensating symptoms will be allowed to be enrolled in the study (aripiprazole, clozapine, ziprasidone, haloperidol, molindone, loxapine, thioridazine, molindone, thiothixene, pimozide, fluphenazine, fluphenazine, risperidone, mesoridazine, quetiapine, trifluoperazine, chlorprothixene, chlorpromazine, perphenazine, trifluopromazine, olanzapine, carbamazepine, divalproex sodium, lithium carbonate or lithium citrate). Have a history of receiving immunoglobulin or other blood product within the 3 months prior to vaccination in this study. Have received any other licensed vaccines within 2 weeks (for inactivated vaccines) or 4 weeks (for live vaccines) prior to vaccination in this study. Have an acute or chronic medical condition that, in the opinion of the investigator, would render vaccination unsafe or would interfere with the evaluation of responses (this includes, but is not limited to, known chronic liver disease, significant renal disease, unstable or progressive neurological disorders, diabetes mellitus, and transplant recipients.). Have a history of severe reactions following immunization with contemporary influenza virus vaccines. Have an acute illness, including an oral temperature greater than 100.4 degrees F, within 1 week of vaccination. Received an experimental agent (vaccine, drug, biologic, device, blood product, or medication) within 1 month prior to vaccination in this study or expect to receive an experimental agent during the 7-month study period. Have any condition that would, in the opinion of the site investigator, place them at an unacceptable risk of injury or render them unable to meet the requirements of the protocol. Participated in an influenza A/H5 vaccine study in the past in a group receiving vaccine (but does

	not exclude documented placebo recipients). Have a known active human immunodeficiency virus, hepatitis B, or hepatitis C infection. Have a history of alcohol or drug abuse in the last 5 years. Have a history of Guillain Barre Syndrome. Have any condition that the investigator believes may interfere with successful completion of the study".
Interventions	**Vaccine type:** inactivated whole virus Influenza A/H5N1 vaccine **Experimental:** 7.5 or 15 mcg H5N1 with aluminium hydroxide. n = 101 **Control:** 7.5 or 15 mcg H5N1. n =100 **Aluminium concentration:** 350 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses of vaccine in the deltoid muscle approximately 28 days apart
Outcomes	**Immunogenicity:** "proportion of subjects in each dose group achieving a serum neutralizing antibody titer of greater than or equal to 40 against the influenza A/H5N1 virus, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Proportion of subjects in each dose group achieving a serum hemagglutination inhibition antibody titer of greater than or equal to 40 against the influenza A/H5N1 virus, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group, (time frame: 28 days after receipt of the 2nd dose of vaccine, approximately Day 56). Geometric mean titer and frequency of 4-fold or greater increases in neutralizing antibody titers in each group, (time frame: 1 month and 7 months after receipt of the 1st dose of vaccine). Geometric mean titer and frequency of 4-fold or greater increases in serum HAI antibody titers in each group, (time frame: 1 month and 7 months after receipt of the 1st dose of vaccine)". **Safety:** "adverse event or serious adverse event information (solicited in the clinic and via memory aids, concomitant medications, periodic targeted physical assessment, and laboratory safety evaluations for a subset of subjects, (time frame: duration of study)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (wkeitel@bcm.edu) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up

	● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** National Institute of Allergy and Infectious Diseases; Division of Microbiology and Infectious Diseases (contracts NO1-AI-25465, NO1-AI-25461, NO1-AI-25462, NO1-AI-25463, NO1-AI25462); the Stanford University General Clinical Research Center (grant M01 RR-00070); the University of Maryland General Clinical Research Center (grant M01 RR165001); and the Vanderbilt University General Clinical Research Center (grant M01 RR-00095)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00382980), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kotloff 2001

Methods	Phase 1, randomised study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy adults 18 to 50 years of age were recruited from the Baltimore-Washington community and the University of Maryland College Park campus. Women were required to have a negative pregnancy test within 7 days of each vaccination". **Exclusion criteria:** "after medical history determination, physical examination, and a battery of laboratory tests, individuals meeting the following criteria were excluded: significant medical or psychiatric abnormalities, a history of gastrointestinal disease including previous antibiotic-associated diarrhea, a positive fecal *C. difficile* stool cytotoxin assay, antibiotic therapy during the previous month, and failure to achieve a passing grade (70%) on a written examination which tested the volunteer's comprehension of the trial".
Interventions	**Vaccine type:** *Clostridium difficile* toxoid vaccine **Experimental:** 6.25, 25 or 100 mcg Clostridium difficile vaccine with aluminium hydroxide. n = 15 **Control:** 6.25, 25 or 100 mcg Clostridium difficile vaccine. n = 15 **Aluminium concentration:** 723 to 757 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Reheis **Vaccination schedule:** four inoculations administered by intramuscular injection into the deltoid muscle (alternating arms) on study days 1, 8, 30, and 60
Outcomes	**Immunogenicity:** "peripheral blood mononuclear cells were collected before and 7 days after administration of each dose of vaccine to measure circulating immunoglobulin A and IgG antibody secreting cells recognizing toxin A and toxin B, by enzyme-linked immune absorbent spot assay. Sera were collected pre-vaccination and 8, 15, 30, 37, 60, 67, and 90 days after immunization and tested for antibodies to toxins A and B by enzyme-linked immunosorbent assay and by neutralization assay. Specific fecal immunoglobulin A antibody responses were measured before and 7 days after each dose of vaccine by enzyme-linked immunosorbent assay". **Safety:** "subjects were monitored at the study site for 30 min after receiving vaccine to detect immediate reactions. For 7 days after each inoculation, they kept a diary in which they recorded their evening oral temperature; the time, date, and consistency (loose or formed) of all stools passed; and the following signs and symptoms: anorexia, malaise, joint swelling, numbness or tingling in fingers or toes, rash, abdominal pain, joint pain, injection site pain, and other, unsolicited complaints. Subjects graded their pain as mild (aware of symptom but symptom easily tolerated), moderate (symptom interferes with usual activity), or severe (symptom prevents work or ability to perform usual activity). In addition, a medical history was obtained at each visit to the study site and by telephone interview 3 days after each inoculation. A panel of clinical laboratory tests (complete blood count with differential and platelet counts,

	creatinine, alanine aminotransferase, and urinalysis) was performed 1 week after each inoculation. Study personnel inspected the injection site 1 and 7 days after each inoculation for the presence of local reactions (swelling, erythema, or heat)".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** 14/12/2018 authors (kkotloff@som.umaryland.edu) were inquired on the followings: ● method for random sequence generation ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** supported by a contract from the National Institute of Allergy and Infectious Diseases to Myron Levine (NO1-AI-45251)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, no further details
Allocation concealment (selection bias)	High risk	Volunteers were sequentially assigned, in groups of 10, to receive one of three increasing doses of vaccine
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kunz 1976

Methods	Field trial
Participants	**Sex:** male and female **Age:** adults, 18-65 years old **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** volunteers, previously received detailed information about the trial, male and female, adults, 18-65 years old **Exclusion criteria:** not reported
Interventions	**Vaccine type:** H3N2 influenza **Experimental (a):** subunit - 700 IU of strain a/port chalmers/73 in 0.5 ml with aluminium. n = 15 **Control (a):** subunit - 700 IU of strain a/port chalmers/73 in 0.5 ml. n = 15 **Experimental (b):** whole virus with aluminium. n = 48 **Control (b):** whole virus. n = 48 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Sandoz **Vaccination schedule:** 1 dose
Outcomes	**Immunogenicity**: "geometric mean of hemagglutination inhibition assay (HIA) before and after vaccination; number of participants with seroconversion in HIA (4 fold increase in titer); number of participants that reach a titer over 1:64 in HIA (assumed immunity against disease level); geometric mean of antibodies in neuraminidase inhibition assay before and after vaccination". **Safety:** "questionnaire where participants record local and systemic side effects for 3 days after vaccination, participants are asked about side effects at visits on day 28 and 90".
Are there other groups in the trial not analysed in this review?	No
Notes	**Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Randomization using a continuous list
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Kutzelnigg 2009

Methods	Two phase I randomised clinical studies
Participants	**Sex:** not reported **Age:** not reported **Ethnicity:** not reported **Diagnosis:** probable Alzheimer's disease of mild to moderate severity **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** Ab Affitopes vaccines for Alzheimer's disease **Experimental:**Ab Affitopes with aluminium hydroxide. n = 12 **Control:** Ab Affitopes. n = 12 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 4 vaccinations at 4-week intervals

Outcomes	Immunogenicity and safety
Are there other groups in the trial not analysed in this review?	No
Notes	Abstract **Funding**: not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Landrum 2017

Methods	Randomized, double-blind, placebo-controlled dose-escalation study
Participants	**Sex:** male and female **Age:** 18-55 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy adult males or females, DoD beneficiaries, including active duty members, 18-55 years of age. Negative urine pregnancy test for female subjects of child bearing potential (negative test within 24 hours prior to investigational product injection) or documented surgical sterility. Female subjects of child-bearing potential must use an acceptable method of birth control, as determined by the PI. Willingness to participate in this study as evidenced by written informed consent". **Exclusion criteria:** "prior receipt of S. aureus rAT or rLukS-PV; known S. aureus infection requiring medical treatment within the 3 months prior to study injection; known active viral or bacterial infection or symptoms/signs consistent with such an infection within the 2 weeks prior to study injection (mild intercurrent viral illness with a temperature of 100.3°F (37.9°C) or less did not require exclusion, if in the judgment of the investigator or sub-investigator this illness would not interfere with the safety evaluation); seropositivity for HIV infection at the time of screening; known or suspected abuse of prescribed or illicit drugs, or alcohol in the past year; use of any new medications (except oral contraceptives, over-the-counter medications or vitamin supplements) within the 7 days prior to IP injection; use of investigational drugs, vaccines, or devices within the 30 days prior to each dose of study injection or anticipated use of such items during the study (licensed vaccines could be administered at any time with the exception of 14 days prior to and 14 days after administration of each study injection); use of systemic steroids (any dose) or high daily dose of inhaled steroids within the last month (use of low or medium daily dose inhaled, intranasal, or low potency topical steroid creams/ointments was allowed unless such medication was begun within the 7 days prior to study injection); history of a bleeding or coagulation disorder, or use of anticoagulant medications within 7 days prior to study injection; actively breastfeeding; presence of grade 1 or higher abnormality in laboratory or vital signs parameter at time of screening; or presence of any condition which, in the opinion of the investigator, placed the subject at undue risk or potentially jeopardized the quality of the data to be generated".
Interventions	**Vaccine type:** no vaccine. Only placebo arm extracted in this review **Experimental:** 0.9% saline placebo with aluminium hydroxide. n = 22 **Control:** 0.9% saline placebo. n = 22 **Aluminium concentration:** 800 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Nabi Biopharmaceuticals **Vaccination schedule:** one dose
Outcomes	**Safety:** "clinical examinations, clinical laboratory results, self-reported diary reactogenicity data and adverse event reports, (time frame: up to 6 months). Adverse events, local reactogenicity, and systemic reactogenicity were assessed through clinical examination by study providers, clinical lab results, as well as review of subject-completed diary". **Immunogenicity:** "geometric mean concentrations after first injection, complete population, (time frame: up to 3 months). Immunogenicity is the ability of a particular substance, such as an antigen or epitope, to provoke an immune

	response in the body of a human or animal. Immunogenicity was determined on the basis of anti-rAT and anti-rLukS-PV IgG concentrations assessed by enzyme-linked immunosorbent assay in sera from blood samples collected on days 0 (baseline), 14, 28 and 84 for those receiving a single dose of vaccine. For those receiving a second dose of vaccine, immunogenicity assessments were also conducted on days 98 and 112. Immunogenicity was evaluated using the following metrics: geometric mean concentrations, geometric mean fold increase and seroresponse status. Seroresponse variables are normally defined in terms of exceeding a threshold".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (tlalani@idcrp.org) were inquired on the followings: ● allocation concealment. ● blinding of outcome assessment ● number of participants who got Staphylococcus aureus infection after the vaccine. No answer received **Funding:** no funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Web-based randomisation system
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Study personnel and subjects were double-blinded to active vaccine vs. placebo within each treatment cohort, but not to volume of vaccine/placebo injected
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs more than 5%

Incomplete outcome data (attrition bias) All-cause mortality	High risk	Drop outs more than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs more than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01011335), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Langley 2009

Methods	Blinded, randomised, dose-ranging, placebo-controlled study
Participants	**Sex:** male and female **Age:** ≥65 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** participants were enrolled from the community in five urban areas of Canada (Halifax, NS; Toronto, ON; Edmonton, AB; Surrey and Port Coquitlam, BC) Eligible persons were ≥65 years of age with informed consent. **Exclusion criteria:** "immunosuppressive condition (secondary to underlying disease or to immunosuppressive therapy in the preceeding 6-months, or currently on immunosuppressive or immunomodulating medication), known impairment of neurologic function or unstable seizure disorder, receipt of blood products or immunoglobulins within the previous 3-months or planned receipt in 60 days post-immunization, known chemical dependency associated with immune suppression, previous immunization against RSV, known or suspected hypersensitivity to a previous vaccine intended for study use, or to its components, recent receipt of other vaccines (killed vaccines <14 days; live <30 days) or planned receipt of other vaccines in 32 days post-study immunization, receipt of an investigational product within a month of study enrollment, known or suspected acute infectious respiratory tract infection at time of study enrollment, acute febrile illness, other conditions which might interfere with vaccine evaluation or pose a health risk to participant according to the investigator's judgement or previous treatment with a hormone of pituitary origin".
Interventions	**Vaccine type:** respiratory syncytial virus subtype A vaccine (RSV-A) **Experimental:** 100 mcg RSV-A with aluminium. n = 114 **Control:** 100 mcg RSV-A. n = 112 **Aluminium concentration:** 1500 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** one intramuscular injection. All participants were administered influenza vaccine (Fluzone® or Vaxigrip ®) by intramuscular injection on day 32.
Outcomes	**Safety:** "study participants were observed for 30min after immunization in the clinic. Telephone contact was made by a study team member on days 1, 2 and 4 after immunization to assess solicited local and systemic adverse events. All adverse events were recorded in the participant diary. Adverse event data was

	also collected on days 120, 180 and 360 during visits. Severe/serious adverse events were assessed throughout the study. Solicited local (redness, swelling, pain, tenderness) and systemic AEs (fever, chills, sore throat, sinus/nasal congestion, cough, headache, nausea, vomiting, diarrhea, general aches, tiredness, sore joints) were categorized as mild, moderate and severe by the participants. Serious adverse events were defined as any untoward medical occurrence, which may be related to the study product, resulting in any of the following: death, is life-threatening, persistent or significant disability/incapacity, requires inpatient hospitalisation or is a congenital anomaly or birth defect. All events regardless of relationship to vaccine were recorded". **Immunogenicity:** "plaque reduction neutralization assay [RSV-A long strain and RSVB strain W/V/1467/*85 [B-1 wild type]] and enzyme immunoassay. The RSVneutralizing antibody titre was performed. The proportion of participants with RSV-A and -B NA titres ≥4-fold the baseline value were determined at day 32 (1-month post-vaccination), days 60, 180 and 360. RSV F and G glycoprotein specific antibodies and RSV M protein-specific antibodies were also assayed using the enzyme-linked immunoassay on these days. Hemagglutinin-inhibition antibody assays for H1N1 and H3N2 influenza A, and influenza B vaccine strains were performed".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 30/01/2020 authors (joanne.langley@dal.ca) were inquired on the followings: ● Method for random sequence generation ● Procedure for blinding of outcome assessment ● Funding source ● Final concentration of aluminium injected ● Aluminum manufacturer ● Number of patients who died/number of patients analysed ● Number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● Number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● Total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** no source of funding reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized. Participants were assigned by within-center block randomisation to one of five study arms.
Allocation concealment (selection bias)	Unclear risk	Not described

Blinding of participants and personnel (performance bias)	Unclear risk	Blinded. Because the un-adjuvanted preparation is a clear colorless liquid and the other alum-containing test articles were cloudy white, all study vaccines were over-labelled with a blinding label to conceal allocation.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop-outs < 5%.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	High risk	No pre-published protocol. Serious adverse events not considered to be related to the vaccine were not assigned by arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Langley 2016

Methods	Phase I randomised, observer-blind, controlled, dose escalation trial
Participants	**Sex:** male and female **Age:** 50–64 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "age 50-64 years, inclusive. Good general health status, as determined by history and physical examination no greater than 30 days prior to administration of the test article. Participants who, in the opinion of the investigator, can and will comply with the requirements of the protocol (e.g. completion of Diary Cards, return for follow-up visits). Written informed consent obtained from the participant. If female of child-bearing potential and heterosexually active, has practiced adequate contraception for 30 days prior to injection, has a negative pregnancy test on the day of injection, and has agreed to continue adequate contraception until 180 days after injection. (Please refer to the glossary for the definition of child-bearing potential and adequate contraception)".

	Exclusion criteria: "use of any investigational or non-registered product (drug or vaccine) other than the study product within 28 days preceding the dose of study product, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the participant has been or will be exposed to an investigational or a non-investigational vaccine/product (pharmaceutical product or device). Planned administration/ administration of a vaccine/product not foreseen by the study protocol within the period starting 28 days before injection of a study vaccine and ending 84 days after. Administration of immunoglobulins and/or any blood products within the 3 months preceding the dose of study product or planned administration during the study period. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. (Laboratory testing for HIV, Hepatitis C and Hepatitis B will be performed during the screening visit). Chronic administration (defined as more than 14 days in total) of immunosuppressants or other immune-modifying drug within 6 months prior to the product dose (for corticosteroids, this will mean prednisone ≥ 20 mg/day, or equivalent). Inhaled and topical steroids are allowed. Family history of congenital or hereditary immunodeficiency. History of or current autoimmune disease. History of hypersensitivity to any test article constituent or products used during the course of study procedures. Known or suspected hypersensitivity to any ingredient in the formulation or component of the container. Pregnant or lactating female. Female planning to become pregnant or planning to discontinue contraceptive precautions within 180 days of study vaccine receipt. Any hematological (hemoglobin level, white blood cell, and platelet count) and biochemical (alanine aminotransferase, aspartate aminotransferase, blood urea nitrogen and creatinine) abnormality as per local laboratory normal values considered clinically significant by the transient mild laboratory abnormalities may be rescreened and the participant will be deemed eligible if the laboratory repeat test is normal as per local laboratory normal values and investigator assessment. Any acute or chronic, clinically significant disease, as determined by physical examination or laboratory screening tests. Malignancies within previous 5 years (excluding non-melanic skin cancer) and lymphoproliferative disorders. Current alcoholism and/or drug abuse. Acute disease and/or fever at the time of Screening ≥ 38°C. Planned move to a location that will prohibit participating in the trial until study end. Any other condition that the investigator judges may interfere with study procedures (e.g. drawing blood) or findings (e.g. immune response)".
Interventions	**Vaccine type:** respiratory syncytial virus vaccine (RSV) **Experimental:** 10 or 25 mcg RSV with aluminium **Control:** 10 or 25 mcg RSV **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** not reported
Outcomes	**Immunogenicity**: blood samples for IgG Ab to SH Ag and hematologic or biochemical parameters were collected on day 0, 7, 28, 56, 63, 84 and 236 **Safety**: solicited local and general adverse events were collected on a memory aide day 0-6 post-injection and unsolicited adverse events through to day 28 post-injection. Serious adverse events, medically attended adverse events, and potentially immune mediated disease were collected to day 236.

Are there other groups in the trial not analysed in this review?	Yes
Notes	Abstract **Contacts with authors:** on 26/01/2021 authors (joanne.langley@dal.ca) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment ● number of drop outs per outcome ● aluminium concentration per dose ● aluminium manufacturer ● safety and immunogenicity results No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Participants and observers were blinded to allocation.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Participants and observers were blinded to allocation.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Participants and observers were blinded to allocation.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Participants and observers were blinded to allocation.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.

Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02472548), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Langley 2017

Methods	Randomized, controlled, observer-blinded, first in- humans, phase 1 clinical trial
Participants	**Sex:** male **Age:** 18–44 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects who, in the opinion of the investigator, can and will comply with the requirements of the protocol. Written informed consent obtained from the trial participants. A male between, and including, 18 and 44 years of age at the time of vaccination. Healthy subjects as established by medical history and clinical examination before entering into the study". **Exclusion criteria:** "use of any investigational or non-registered product other than the study vaccines within 30 days preceding the dose of study vaccine, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the subject has been or will be exposed to an investigational or a non-investigational vaccine/product. Planned administration/ administration of a vaccine not foreseen by the study protocol within the period starting 30 days before and ending 30 days after vaccination, with the exception of any licensed influenza vaccine which may be administered ≥ 15 days before or after vaccination. Previous vaccination against RSV. Chronic administration of immunosuppressants or other immune-modifying drugs within 6 months prior to the vaccine dose. Inhaled and topical steroids are allowed. Administration of long-acting immune-modifying drugs at any time during the study period. Administration of immunoglobulins and/or any blood products within the 3 months preceding the dose of study vaccine or planned administration during the study period. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. Family history of congenital or hereditary immunodeficiency. History of or current autoimmune disease. History of any reaction or hypersensitivity likely to be exacerbated by any component of the vaccines. Hypersensitivity to latex. Any clinically significant hematological (hemoglobin level, white blood cell [WBC], lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine aminotransferase, aspartate aminotransferase and creatinine) abnormality as per the opinion of the investigator, based on the local laboratory normal ranges.Subjects with hematological/ biochemical values out of normal range which are expected to be temporary may be re-screened at a later date. Any acute or chronic, clinically significant disease, as determined by physical examination or laboratory screening tests. Malignancies within previous 5 years (excluding non-melanoma skin cancer) and lymphoproliferative disorders. Current alcoholism and/or drug abuse. Acute disease and/or fever at the time of Screening.Fever is defined as temperature ≥ 37.5°C /99.5°F for oral, axillary or tympanic route, or ≥ 38.0°C /100.4°F on rectal route. The preferred route for recording temperature in this study will be oral.Subjects with a minor illness (such as mild diarrhea) without fever may be enrolled at the discretion of the investigator.Subjects with acute

	disease and/ or fever at the time of Screening may be re screened at a later date. Planned move to a location that will prohibit participating in the trial until study end. Any other condition that the investigator judges may interfere with study procedures (e.g. drawing blood) or findings (e.g. immune response)".
Interventions	**Vaccine type:** respiratory syncytial virus vaccine (RSV) **Experimental:** 10, 30 or 60 mcg RSV with aluminium. n = 48 **Control:** 10, 30 or 60 mcg RSV. n = 47 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** GSK **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "humoral immune response in terms of neutralizing antibody titers to the investigational RSV vaccines, in all subjects, in all groups, (time frame: at pre-vaccination (day 0) and post-vaccination (day 7, day 30 and day 60)). Persistence of the humoral immune response in terms of neutralizing antibody titers to the investigational RSV vaccines, in all subjects, in all groups, (time frame: at day 180 and day 360)". **Safety:** "occurrence of each solicited local and general adverse event, (time frame: during the 7 days (Days 0-6) follow-up period after vaccination). Occurrence of any hematological (hemoglobin level, white blood cells lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine amino-transferase, aspartate amino-transferase and creatinine) laboratory abnormality, (time frame: at day 0, 7 and 30). Occurrence of any hematological (hemoglobin level, whole blood cells, lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine amino-transferase, aspartate amino-transferase and creatinine) laboratory abnormality, (time frame: at day 60). Occurrence of any unsolicited adverse events, (time frame: during a 30-day (days 0-29) follow-up period after vaccination). Occurrence of any serious adverse events, (time frame: from day 0 to day 60). Occurrence of any hematological (hemoglobin level, whole blood cells, lymphocyte, neutrophil, eosinophil and platelet count) and biochemical (alanine amino-transferase, aspartate amino-transferase and creatinine) laboratory abnormality, (time frame: at day 180 and day 360). Occurrence of any serious adverse events, (time frame: from day 60 to the study conclusion (i.e. day 360))".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (joanne.langley@dal.ca) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received **Funding:** supported by GlaxoSmithKline Biologicals SA. GlaxoSmithKline Biologicals SA was involved in all stages of the study conduct and analysis.

	GlaxoSmithKline Biologicals SA paid for open access of the article.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation sequence was generated using MATerial Excellence, a software program developed for use in Statistical Analysis System (SAS®) (Cary, North Carolina) by GSK (Rixensart, Belgium). A randomisation blocking scheme was used to ensure balance between study groups in each step, and a minimization procedure accounted for age category (18–32 or 33–44 years).
Allocation concealment (selection bias)	Low risk	Study group assignment was allocated by an Internet-based central randomisation system.
Blinding of participants and personnel (performance bias)	Low risk	Participants and all study personnel, except for an unblinded nurse, whose sole role was to prepare and administer study vaccines, were not aware of treatment assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT02247726), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2015

Methods	Two phase I/II, single-blind, parallel-group studies
Participants	**Sex:** male and female **Age:** >18 **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subjects who the investigator believes will comply with the requirements of the protocol. A male or female ≥ 18 years at the time of the first vaccination. Written informed consent obtained from the subject. Free of obvious health problems as established by medical history and clinical examination before entering into the study. If the subject is female, she must be of non-childbearing potential, i.e., either surgically sterilized or one year post-menopausal; or, if of childbearing potential, she must be abstinent or have used adequate contraceptive precautions for 30 days prior to vaccination, have a negative pregnancy test and must agree to continue such precautions for two months after completion of the vaccination series". **Exclusion criteria:** "use of any investigational or non-registered product (drug or vaccine) other than the study vaccine(s) within 30 days preceding the first dose of study vaccine, or planned use during the study period or participation to another pharmaceutical/vaccine study. Chronic administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within six months prior to the first vaccine dose. Use of any anticoagulants. Planned administration/ administration of a vaccine not foreseen by the study protocol within 2 weeks of the first dose of vaccines. Previous vaccination against Streptococcus pneumoniae. Bacterial pneumonia within 3 years prior to 1st vaccination. Any confirmed or suspected immunosuppressive or immunodeficient condition, including human immunodeficiency virus (HIV) infection. History of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Current serious neurologic or mental disorders. Currently smoking > 25 cigarettes per day. Inflammatory processes such as known chronic active infections. All malignancies (excluding non-melanic skin cancer) and lymphoproliferative disorders diagnosed or treated actively during the past 5 years. History of administration of an experimental vaccine containing MPL or QS21. Acute disease at the time of enrolment. Acute or chronic, clinically significant pulmonary, cardiovascular, hepatic or renal functional abnormality, as determined by physical examination or laboratory screening tests, at the discretion of the investigator. History of chronic alcohol consumption and/or intravenous drug abuse".
Interventions	**Vaccine type:** pneumococcal protein PhtD vaccine **Experimental:** 30 mcg PhtD with aluminium. n = 50 **Control:** 30 mcg PhtD. n = 49 **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injections into the deltoid region of the upper right arm at 0 and 2 months. Influenza vaccine (FluarixTM; GlaxoSmithKline,Rixensart, Belgium) was offered
Outcomes	**Safety**: "occurrence, intensity and relationship of any solicited local and general signs and symptoms, (time frame: during a 7-day follow up period after each vaccine dose). Occurrence, intensity and relationship to vaccination of

	unsolicited local and general signs and symptoms, (time frame: during a 30-day follow up period after each vaccine dose). Occurrence of all serious adverse events, (time frame: during the entire study period). Number and percentage of subjects with normal or abnormal values for biochemical assessments and for haematological analysis, (time frame: at each scheduled time point (month 0, 1, 3, 12, 24 and 36))". **Immunogenicity**: "anti- PhtD antibody concentration, (time frame: one month after the first injection). Anti-PhtD antibody concentration, (time frame: one month after 2 injections). Anti- PhtD antibody concentration, (time frame: at 12, 24 and 36 months after the first vaccination). Anti-PhtD antibody avidity, (time frame: at month 0, 1 and 3). Evaluation of protection afforded by passive transfer of anti PhtD antibodies sera pooled from all individuals, (time frame: at month 0, 1 and 3). Frequency of PhtD specific plasma cells generated by in vitro cultivated memory B-cells, in a subset of subjects, (time frame: at month 0, 1 and 3). Frequency of CD4 and/or CD8 T cells that produce cytokines (IL-2, IL-4, IFNg, CD40L and/or GM-CSF, and TNFα), upon PhtD re-stimulation in vitro, to evaluate the T-cell response, in a subset of subjects, (time frame: at month 0, 1, 12, 24 and 36). Anti-polysaccharide total IgG concentration in Group A for all vaccine pneumococcal serotypes, (time frame: at month 0, 1, 12, 24 and 36). Anti-PS antibody avidity for 5 serotypes in Group A, (time frame: at month 0, 1). Deposition of complement components on the surface of different bacterial strains 3 strains (GSK/CDC, OPA, isogenic TIGR4) of 5 serotypes in Group A, (time frame: at month 0, 1). Opsonophagocytic activity titres in Group A to all vaccine pneumococcal serotypes, (time frame: at month 0, 1 and 12). Frequency of PS-specific plasma cells generated by in vitro cultivated memory B-cells in Group A in a subset of subjects, (time frame: at month 0, 1)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (isabel.lerouxroels@ugent.be, yves.horsmans@uclouvain.be) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further data or information added **Funding:** supported by GlaxoSmithKline (GSK) Biologicals SA Only vaccine-related serious adverse events were reported.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Centralised randomisation system on the Internet
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	The studies were single-blind, meaning that participants were unaware of the vaccination assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	The studies were single-blind, meaning that participants were unaware of the vaccination assignment.
Blinding of outcome assessment (detection bias) All-cause mortality	High risk	Single-blind
Blinding of outcome assessment (detection bias) Immunogenicity	High risk	Single-blind
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00307528), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2016a

Methods	Study 2 was a phase I, randomised,observer-blind, placebo-controlled, single-center, dose-escalation study
Participants	**Sex:** male and female **Age:** 18-70 years **Ethnicity:** Caucasian **Diagnosis:** previous or current smokers **Inclusion criteria:** "subjects who, in the opinion of the investigator, can and will comply with the requirements of the protocol. A male or female between, and including, 50 and 70 years of age at the time of the first vaccination. Written informed consent obtained from the subject. Subjects without medical history, clinical finding or laboratory finding which in the opinion of the investigator could pose a safety concern or interfere with the protocol. Current or former smokers.

	A smoking history of at least 10 pack-years. Female subjects of non-childbearing potential may be enrolled in the study. Female subjects of childbearing potential may be enrolled in the study, if the subject:has practiced adequate contraception for 30 days prior to vaccination, and has a negative pregnancy test on the day of vaccination, and has agreed to continue adequate contraception during the entire treatment period and for 2 months after completion of the vaccination series". **Exclusion criteria:** "use of any investigational or non-registered product other than the study vaccines within 30 days preceding the first dose of study vaccine, or planned use during the study period. Concurrently participating in another clinical study, at any time during the study period, in which the subject has been or will be exposed to an investigational or a non-investigational vaccine/product. Planned administration/ administration of a vaccine not foreseen by the study protocol in the period starting 30 days before the first dose and ending 30 days after the last dose of vaccine, with the exception of any influenza vaccine which may be administered ≥ 15 days preceding or following any study vaccine dose. Previous vaccination with any vaccine containing NTHi-antigens. Chronic administration of immunosuppressants or other immune-modifying drugs within six months prior to the first vaccine dose. Topical steroids are allowed. Administration of immunoglobulins and/or any blood products within the 3 months preceding the first dose of study vaccine or planned administration during the study period. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. History of or current autoimmune disease. Post-bronchodilator Forced Expiratory Volume in 1 second (FEV1) < 80% of predicted normal value. Diagnosed with a respiratory disorder. Please note that subjects with mild pulmonary obstruction can be enrolled. Laboratory evidence of clinically significant haematological abnormalities at Screening. Acute disease and/or fever at the time of enrolment. Current alcoholism and/or drug abuse. Has significant disease, in the opinion of the investigator, likely to interfere with the study and/or likely to cause death within the study duration. History of or current condition preventing intramuscular injection as bleeding or coagulation disorder. Has contraindication for spirometry testing. Malignancies within previous 5 years and lymphoproliferative disorders. Pregnant or lactating female. Female planning to become pregnant or planning to discontinue contraceptive precautions. Any other condition that the investigator judges may interfere with study findings".
Interventions	**Vaccine type:** non-typeable Haemophilus influenzae (NtHi) protein vaccine **Experimental:** 10 or 30 mcg NtHi with aluminium. n = 60 **Control:** 10 or 30 mcg NtHi. n = 60 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** alhydrogel **Aluminium manufacturer:** Brenntag Biosector, Frederikssund, Denmark **Vaccination schedule:** two doses at months 0, 2 and 6
Outcomes	**Immunogenicity:** "humoral immune response to the components of the NtHi vaccine formulations, in all subjects, in all groups, in terms of antibody concentrations, (time frame: prior to each vaccination, 30 days after each vaccination, and at study conclusion (day 420)). Cell-mediated immune responses to components of the NtHi vaccine formulations, in a sub-cohort of subjects, in all groups (time frame: prior to each vaccination, 30 days after each vaccination, and at study conclusion (day 420))". **Safety:** "occurrence of each solicited local and general adverse event, in all

	subjects, in all groups, (time frame: during a 7-day follow-up period (from day 0 to day 6) after each vaccination). Occurrence of any unsolicited adverse event, in all subjects, in all groups, (time frame: during a 30-day follow-up period (from day 0 to day 29) after each vaccination). Occurrence of haematological and biochemical laboratory abnormalities, in all subjects, in all groups, (time frame: prior to each vaccination). Occurrence of haematological and biochemical laboratory abnormalities, in all subjects, in all groups, (time frame: 7 days after each vaccination). Occurrence of haematological and biochemical laboratory abnormalities, in all subjects, in all groups, (time frame: at study conclusion (day 420)). Occurrence of any serious adverse event, in all subjects, in all groups, (time frame: from first vaccination (day 0) to study conclusion (day 420)). Occurrence of any potential immune-mediated disease in all subjects, in all groups, (time frame: from first vaccination (day 0) to study conclusion (day 420))".
Are there other groups in the trial not analysed in this review?	Yes.
Notes	**Contacts with authors:** on 14/12/2018 authors (Geert.LerouxRoels@UGent.be, Mathieu.peeters@yahoo.co.uk) were inquired on the followings: ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no further data or information added **Funding:** GlaxoSmithKline Biologicals SA was the funding source and was involved in all stages of the clinical trial conduct and analysis. GlaxoSmithKline Biologicals SA also took in charge all costs associated with the development and the publishing of the manuscript. Only vaccine-related serious adverse events were reported.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomized using a centralized randomisation system on internet (SBIR)
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Observer-blind, i.e. the vaccine recipients and those responsible for the evaluation of any study endpoint were blinded to the administered vaccines. Due to differences in the appearance of the study vaccines, vaccines were prepared and administered by authorized medical personnel who did not participate in any of the study clinical evaluations or assays.

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Data were collected in an observer-blind manner.Those responsible for the evaluation of any study endpoint were blinded to the administered vaccines.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Data were collected in an observer-blind manner.Those responsible for the evaluation of any study endpoint were blinded to the administered vaccines.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Data were collected in an observer-blind manner.Those responsible for the evaluation of any study endpoint were blinded to the administered vaccines.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	The safety analysis was performed on the total vaccinated cohort which included all vaccinated participants
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The safety analysis was performed on the total vaccinated cohort which included all vaccinated participants
Incomplete outcome data (attrition bias) Immunogenicity	High risk	. The immunogenicity analysis was based on the according-to-protocol cohort for immunogenicity Drop outs more than 5%.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01678677), and all outcomes were reported on www.gsk-clinicalstudyregister.com (GSK studies 116018 and 116647)-GlaxoSmithKline (GSK) 2838497A.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2016b

Methods	Phase Ib, randomised, observer-blind and placebo-controlled study
Participants	**Sex:** female **Age:** 18-40 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** healthy females 18-40 years of age inclusive **Exclusion criteria:** "individuals who are pregnant or nursing. Individuals who have had a previous immunization with a vaccine containing streptococcus antigens or any vaccine within 30 days of enrollment through 30 days of study completion. Individuals with a history of severe allergic reactions after previous vaccination. Individuals with designated blood tests that are not within normal range".
Interventions	**Vaccine type:** trivalent group B Streptococcus vaccine (GBS) **Experimental:** 5 or 20 mcg GBS with aluminium. n = 160 **Control:** 5 or 20 mcg GBS. n = 160 **Aluminium concentration:** 1000 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** one or two injections, administered one month apart

Outcomes	**Immunogenicity:**" percentage of subjects achieving specific thresholds of antibody concentrations for subjects in enrollment group, (time frame: day 1 and day 61). The percentage of subjects achieving a specific threshold of antibody concentrations of ≥ 0.5 µg/mL as determined by enzyme-linked immunosorbent assay at day 61 across cohorts 1 and 2 for potential use in pregnant women. Geometric mean concentrations of Serum Anti-GBS IgG antibody levels by serotype at day 61, (time frame: day 1 and day 61]. Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody levels in healthy subjects for potential use in pregnant woman was assessed in terms of geometric mean concentrations at day 61 in cohorts 1 and 2. Geometric mean ratios of serum group B streptococcus IgG antibody levels by serotype, (time frame: day 61/day 1). Geometric mean ratios relative to baseline (day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels measured at day 61 for subjects across cohorts 1 and 2. Percentage of subjects achieving specific thresholds of antibody concentrations for subjects in enrollment groups 1 & 2, (time frame: day 1 and day 361). The percentage of subjects achieving a specific threshold of antibody concentrations of ≥ 0.5 µg/mL at day 361 across all the cohorts for potential use in non-pregnant women. Geometric mean concentrations of serum anti-group B streptococcus IgG antibody levels by serotype at day 361, (time frame: day 1 and day 361). Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody levels in healthy subjects for potential use in non-pregnant woman was assessed in terms of geometric mean concentrations at day 361 across all cohorts. Geometric mean ratios of serum group B streptococcus IgG Antibody levels by serotype at day 361, (time frame: day 1 and day 361). Geometric mean ratios relative to baseline (day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels in healthy subjects for potential use in non-pregnant women, measured at day 361 across all cohorts. Persistence of antibody response in terms of geometric mean concentrations by serotype for subjects in enrollment group 1, (time frame: day 1 to day 721). The persistence of Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody response was assessed by comparing enzyme-linked immunosorbent assay responses in terms of geometric mean concentrations at day 721 after the first injection for subjects enrolled in cohorts 1 and 2. Persistence of antibody response in terms of geometric mean concentrations by serotype for subjects in enrollment group 2, (time frame: day 1 to day 721). The persistence of Serotype-specific (Ia, Ib & III) group B streptococcus IgG antibody response assessed by comparing enzyme-linked immunosorbent assay responses in terms of geometric mean concentrations at day 721 after the first injection for subjects enrolled in cohort 3 and 4. Geometric mean ratios of serum group B streptococcus IgG antibody levels by serotype for subjects in enrollment group 1, (time frame: day 1 to day 721). Geometric mean ratios was calculated at Day 721 relative to baseline (Day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels across all cohorts 1 and 2. Geometric mean ratios of serum group B streptococcus IgG antibody levels by serotype for subjects in enrollment group 2, (time frame: day 1 to day 721). Geometric mean ratio was calculated at Day 721 relative to baseline (day 1) of serotype-specific (Ia, Ib, III) group B streptococcus IgG antibody levels across cohorts 3 and 4". **Safety:** "number of participants reporting solicited adverse events after receiving the group B streptococcus trivalent vaccine in enrollment group 1, (time frame: day 1 to day 7). Safety was assessed in terms of number of participants reporting solicited local and systemic adverse events after receiving trivalent GBS vaccine from day 1 to day 7 for subjects across cohorts 1 and 2.

	Number of participants reporting solicited adverse events after receiving the group B streptococcus trivalent vaccine in enrollment group 2, (time frame: day 1 to day 7). Safety was assessed in terms of number of participants reporting solicited local and systemic adverse events after receiving GBS vaccine from day 1 to day 7 for subjects in cohorts 3 and 4. Number of participants reporting unsolicited adverse events after receiving group B streptococcus trivalent vaccine in enrollment group 1, (time frame: day 1 to day 721). Safety was assessed in terms of number of participants with unsolicited adverse events after receiving group B streptococcus trivalent vaccine, reported from day 1 to day 721 across subjects in cohorts 1 and 2. Number of participants reporting unsolicited adverse events after receiving group B streptococcus trivalent vaccine in enrollment group 2, (time frame: day 1 to day 721). Safety was assessed in terms of number of participants with unsolicited adverse events after receiving group B streptococcus trivalent vaccine, reported from day 1 to day 721 across subjects in cohorts 3 and 4".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (Geert.LerouxRoels@UGent.be, isa.x.bedell@gsk.com) were inquired on the followings: ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no further data or information added **Funding:** supported by Novartis Vaccines and DiagnosticsDivision, now part of the GSK group of companies

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A web-based randomisation system was used to randomly assign subjects into each treatment arm within the two Enrolment Groups
Allocation concealment (selection bias)	Low risk	A web-based randomisation system was used to randomly assign subjects into each treatment arm within the two Enrolment Groups
Blinding of participants and personnel (performance bias)	Unclear risk	Observer-blind, no further details

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01150123), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Leroux-Roels 2019

Methods	Phase I, observer-blind, randomised, placebo-controlled, dosage-escalation study
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** Caucasian **Diagnosis:** healthy **Inclusion criteria:** "healthy males and non-pregnant females 18 to 45 years of age at time of enrollment. Individuals who have given written consent after the nature of the study has been explained according to local regulatory requirements. Individuals in good health as determined by the outcome of the medical history, physical examination and clinical judgment of the investigator. Individuals who can comply with the study procedures and are available for follow up". **Exclusion criteria:** "individuals with any severe chronic or acute disease. Individuals with a history of illness or with an ongoing illness that may pose additional risk to the subject if he/she participates in the study, including the following: history of any chronic respiratory illness, including current diagnosis of asthma within 2 years, exercise induced wheezing, reactive airway disease, emphysema, chronic bronchitis, cystic fibrosis or chronic obstructive pulmonary disease.Any respiratory illness within 7 days prior to receiving the first study injection.Any active pulmonary infection or other inflammatory conditions, even in the absence of febrile episodes, within 14 days prior to the first study

	injection.Hepatitis B or hepatitis C infection. Individuals who have had a malignancy or lymphoproliferative disorder within the past 5 years. Individuals with known or suspected impairment of the immune system including but not limited to HIV, autoimmune disorders, immunosuppressive therapy, and diabetes mellitus. Individuals with any history of progressive or severe neurologic disorder, seizure disorder or Guillian-Barré syndrome. Individuals with a known bleeding diathesis, or any condition that may be associated with a prolonged bleeding time. Individuals with a BMI > 35 kg/m2. BMI is to be calculated by the following formula: subject weight at baseline divided by subject height in meters multiplied by the subject height in meters. The numerical result will be rounded to the nearest 0.1. Individuals who are allergic to any of the vaccine components, or with a history of anaphylaxis after vaccination. Individuals who during the 90 days prior to enrollment receive any medications or other treatments that may adversely affect the immune system such as allergy injections, immune globulin, interferon, immunomodulators, cytotoxic drugs or other drugs known to be frequently associated with significant major organ toxicity. Individuals who receive systemic immunosuppressive agents including steroids. Prior corticosteroid therapy should be discontinued 28 days prior to enrollment. Individuals using inhaled or topical corticosteroids will be permitted. Receipt or donation of blood or blood products 8 weeks prior to vaccination or planned receipt or donation during the study period. Individuals participating in any clinical trial with another investigational product 28 days prior to receiving the first study vaccination or intent to participate in another clinical study at any time during the conduct of this study. Individuals who have received any vaccine 28 days prior to enrollment in this study, or who plan to receive any non-study vaccines within 28 days of the second dose of study vaccine. Individuals with any clinically significant abnormal safety laboratory result, as judged by the investigator. If female, 'of childbearing potential', sexually active and has not used any of the 'acceptable contraceptive methods' for at least two months prior to study entry.Childbearing potential is defined as status post onset of menarche and not meeting any of the following conditions: menopausal for at least two years; sterile status after bilateral tubal ligation for at least one year, immediately after bilateral oophorectomy or after hysterectomy.Acceptable methods of birth control are defined as one or more of the following: Hormonal contraceptives.Barrier each and every time during intercourse.Intrauterine device (IUD).Monogamous relationship with vasectomized partner. Partner must have been vasectomized for at least six months prior to subject's study entry. If female subject of childbearing potential and have a positive urine pregnancy test prior to study vaccinations, or are currently lactating. If female of childbearing potential and sexually active, refusal to use an 'acceptable contraceptive method' through to three weeks after last study vaccination. Individuals with behavioral or cognitive impairment or psychiatric disease that, in the opinion of the investigator, may interfere with the subject's ability to participate in the study. Individuals with a history of drug or alcohol abuse within the past 2 years. Individuals who are acting as study personnel or immediate family members or the spouse of study personnel. Individuals with a body temperature ≥38 °C (≥100.4○F) within 3 days of intended study vaccination. Individuals with any condition that, in the opinion of the investigator, would interfere with the primary study objectives".

Interventions	**Vaccine type:** respiratory syncytial virus fusion glycoprotein F (RSV F) subunit vaccine **Experimental:** 45, 90 or 135 mcg RSV F with aluminium. n = 72 **Control:** 45, 90 or 135 mcg RSV F. n = 72 **Aluminium concentration:** 1000 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses 28 days apart
Outcomes	**Immunogenicity:** geometric mean titers of the serum anti-RSV neutralizing antibody titers, (time frame: at day 57). Immunogenicity was measured in terms of the geometric mean titers of the serum anti-RSV neutralizing antibody titers at day 57 (28 days after the second dose). Percentage of subjects with a ≥ 4-fold Increase in serum anti-RSV neutralizing antibody titers, (time frame: at day 57). Immunogenicity was measured in terms of percentage of subjects with a ≥ 4-fold increase in serum anti-RSV neutralizing antibody titers, from day 1 (baseline) to day 57 (28 days after the second dose). **Safety:** number of subjects with any solicited local symptoms, (time frame: within 30 minutes after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited local symptoms, (time frame: from day 1 (6 hour) through day 3 after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited local symptoms, (time frame: from day 4 through day 7 after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited local symptoms, (time frame: from day 1 (6 hours) to Day 7 after each vaccination). Assessed solicited local symptoms were: induration, swelling, erythema and pain. Any induration/swelling/erythema = induration/swelling/erythema spreading beyond 25 millimeters (mm) of injection site. Any pain = occurrence of the symptom regardless of intensity grade. Number of Subjects with any solicited systemic symptoms and other Indicators of reactogenicity, (time frame: within 30 minutes after each vaccination]Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited systemic symptoms and other indicators of reactogenicity, (time frame: from day 1 (6 hours) through day 3 after each vaccination). Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Number of subjects with any solicited systemic symptoms and other indicators of reactogenicity, (time frame: from day 4 through day 7 after each

vaccination). Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Subjects with any solicited systemic symptoms and other indicators of reactogenicity, (time frame: from day 1 (6 hours) to day 7 after each vaccination). Assessed solicited systemic symptoms were: nausea, fatigue, myalgia, arthralgia, headache, fever (body temperature ≥ 38.0°C), chills, coughing, diarrhea, rhinorrhea and wheezing. Other solicited data included: prevention of pain and/or fever and treatment of pain and/or fever. Any = occurrence of the symptom regardless of intensity grade. Number of subjects with unsolicited adverse events, (time frame: from day 1 to day 28 after each vaccination). An unsolicited adverse events covers any untoward medical occurrence in a clinical investigation subject temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product and reported in addition to those solicited during the clinical study and any solicited symptom with onset outside the specified period of follow-up for solicited symptoms. Any was defined as the occurrence of any unsolicited adverse events regardless of intensity grade or relation to vaccination. Number of subjects with serious adverse events and other significant adverse events, (time frame: from study start (day 1) until study completion (day 394)]. Serious adverse events assessed include medical occurrences that resulted in death, were life-threatening, required hospitalization or prolongation of hospitalization or resulted in disability/incapacity, congenital anomaly or birth defect. Any adverse events = occurrence of the serious adverse events regardless of intensity grade. Possibly or probably related serious adverse events = serious adverse events assessed by the investigator as possibly or probably related to the study vaccination.Other significant adverse events assessed include unsolicited medically attended adverse events, unsolicited adverse events leading to study withdrawal, new onset of chronic diseases and adverse events of special interest. Medically attended adverse events = an adverse event that leads to an unscheduled visit to a healthcare practitioner. New onset of chronic diseases = an adverse event that represents a new diagnosis of a chronic medical condition that was not present or suspected in a subject prior to study enrollment.

Immunogenicity: geometric mean titers of the serum anti-RSV neutralizing antibody titers, (time frame: at day 1, day 29 and day 181). Immunogenicity was measured in terms of geometric mean titers of the serum anti-RSV neutralizing antibody titers at day 1, day 29 (28 days after the first dose) and day 181 (six months after the first dose). Percentage of subjects with a ≥ 4-fold Increase in serum anti-RSV neutralizing antibody titers, (time frame: at day 1, day 29 and day 181). Immunogenicity was measured in terms of percentage of subjects with a ≥ 4-fold increase in serum anti-RSV neutralizing antibody titers from day 1 to day 29 (28 days after the first dose) and from day 1 to day 181 (six months after the first dose). Percentage of subjects with serum anti-RSV neutralizing antibody titers greater than the 3rd quartile of serum anti-RSV neutralizing antibody titers at day 1, (time frame: at day 29, day 57 and day 181). Immunogenicity was measured in terms of the percentage of subjects at day 29 (28 days after the first dose), day 57 (28 days after the second dose) and day 181 (six months after the first dose) with serum anti-RSV neutralizing antibody titers greater than the 3rd quartile of serum anti-RSV neutralizing antibody titers at day 1 (baseline). Geometric mean titers of the serum total binding antibody to each of the RSV Proteins F, G and N, (time frame: at day 1, day 29, day 57 and

	day 181). Immunogenicity was measured in terms of the geometric mean titers of the serum total binding antibody to each of the RSV proteins F, G and N at day 1 (baseline), day 29 (28 days after the first dose), day 57 (28 days after the second dose) and day 181 (six months after the first dose). Percentage of subjects with a ≥ 4-fold increase in serum total binding antibody to each of the RSV Proteins F, G and N, (time frame: at day 1, day 29, day 57 and day 181). Immunogenicity was measured in terms of the percentage of subjects with a ≥ 4-fold increase in serum total binding antibody to each of the RSV Proteins F, G and N from day 1 (baseline) to all time points (day 29 [28 days after the first dose], day 57 [28 days after the second dose], and day 181 [six months after the first dose]). Percentage of subjects with serum total binding antibody titers to each of the RSV Proteins F, G, and N greater than the 3rd quartile of serum total binding antibody titers to RSV Protein F at day 1, (time frame: at day 1, day 29, day 57 and day 181). Immunogenicity was measured in terms of the percentage of subjects at day 29 (28 days after the first dose), day 57 (28 days after the second dose) and day 181 (six months after the first dose), with serum total binding antibody titers to each of the RSV proteins F, G, and N greater than the 3rd quartile of serum total binding antibody titers to RSV protein F at day 1 (baseline). Ratio of RSV F serum neutralizing antibody titers to each of the RSV F serum total binding antibody titers to RSV Proteins F, G and N, (time frame: at day 1, day 29, day 57 and day 181). Immunogenicity was measured in terms of the ratio of RSV F serum Nab titers to each of the RSV F serum total binding antibody titers to RSV proteins F, G and N at days 1, 29, 57 and 181.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (Geert.LerouxRoels@UGent.be) were inquired on the followings: ● aluminium manufacturer. ● allocation concealment. ● blinding of outcome assessment. ● number of participants who got respiratory syncytial virus infection after the vaccine. No answer received **Funding:** sponsored by Novartis Vaccines Division. On 02 March 2015, Novartis non-influenza Vaccines Business was acquired by the GlaxoSmithKline group of companies. The Sponsor was involved in all stages of the study conduct and analysis and took charge of all costs associated with developing and publishing this manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Observer-blind

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Dropouts <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Dropouts <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Dropouts <5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02298179), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Liang 2010

Methods	Multicentre, double-blind, placebo-controlled, randomised trial
Participants	**Sex:** male and female **Age:** >3 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female aged >3 years; able to show legal identity card for the sake of recruitment; volunteers or their guardians are able to understand and sign the informed consent". **Exclusion criteria:** "cases, cured cases and close contact of influenza A (H1N1) virus. Women of pregnancy, lactation or about to be pregnant in 60 days. Subject that has a medical history of any of the following: allergic history, or allergic to any ingredient of vaccine, such as egg, egg protein, etc. Serious adverse reactions to vaccines such as anaphylaxis, hives, respiratory difficulty, angioedema, or abdominal pain. Autoimmune disease or immunodeficiency. Asthma that is unstable or required emergent care, hospitalization or intubation during the past two years or that required the use of oral or intravenous corticosteroids. Diabetes mellitus (type I or II), with the exception of gestational diabetes. History of thyroidectomy or thyroid disease that required medication within the past 12 months. Serious angioedema episodes within the previous 3 years or requiring medication in the previous two years. Bleeding disorder diagnosed by a doctor (e.g. factor deficiency, coagulopathy, or platelet disorder

	requiring special precautions) or significant bruising or bleeding difficulties with IM injections or blood draws. Active malignancy or treated malignancy for which there is not reasonable assurance of sustained cure or malignancy that is likely to recur during the period of study. Seizure disorder other than: febrile seizures under the age of two years old, seizures secondary to alcohol withdrawal more than 3 years ago, or a singular seizure not requiring treatment within the last 3 year. Asplenia, functional asplenia or any condition resulting in the absence or removal of the spleen. Guillain-Barre Syndrome. Any history of immunosuppressive medications or cytotoxic medications or inhaled corticosteroids within the past six months (with the exception of corticosteroid nasal spray for allergic rhinitis or topical corticosteroids for an acute uncomplicated dermatitis). History of any blood products or seasonal influenza vaccine administration within 3 months before the dosing. Administration of any other investigational research agents within 30 days before the dosing. Administration of any live attenuated vaccine within 30 days before the dosing. Administration of subunit or inactivated vaccines, e.g., pneumococcal vaccine, or allergy treatment with antigen injections, within 14 days before the dosing. Be receiving anti-TB prophylaxis or therapy currently. Axillary temperature > 37.0 centigrade at the time of dosing. Psychiatric condition that precludes compliance with the protocol: past or present psychoses, past or present bipolar disorder requiring therapy that has not been well controlled on medication for the past two years, disorder requiring lithium, suicidal ideation occurring within five years prior to enrollment. Any medical, psychiatric, social condition, occupational reason or other responsibility that, in the judgment of the investigator, is a contraindication to protocol participation or impairs a volunteer's ability to give informed consent".
Interventions	**Vaccine type:** 2009 pandemic influenza A H1N1 vaccine **Experimental:** 7.5, 15 or 30 mcg H5N1 with aluminium hydroxide. n = 1814 **Control:** 7.5, 15 or 30 mcg H5N1. n =8184 **Aluminium concentration:** 1000 mcg/dose or 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Sinovac Biotech **Vaccination schedule:** two doses 21 days apart
Outcomes	**Immunogenicity:** hemagglutination inhibition antibody titer, (time frame: day 0, day 21, day 35) **Safety:** local and systemic adverse reaction after vaccination, (time frame: day 0, day 42)
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/!2/2019 authors (wangyu@chinacdc.cn) were inquired on the followings: ● blinding of participants and personnel ● blinding of outcome assessment. ● number of participants who got influenza after the vaccination. No answer received **Funding:** the vaccine manufacturers funded the trials (Sinovac Biotech, Hualan Biological Bacterin, China National Biotec Group, Beijing Tiantan Biological Products, Changchun Institute of Biological Products, Changchun Changsheng Life Sciences, Jiangsu Yanshen Biological Technology Stock, Zhejiang

	Tianyuan Bio-Pharmaceutical, Lanzhou Institute of Biological Products, Shanghai Institute of Biological Products, and Dalian Aleph Biomedical).

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Vaccine and placebo used for each age-group were labelled with sequential numbers according to a random number table generated with the block method by a statistician not involved in the rest of the trial. Block size varied by age group and centre.
Allocation concealment (selection bias)	Low risk	The same sequential numbers were also assigned to participants according to their sequence of enrolment. Participants received vaccine or placebo according to their sequential numbers. Masking was achieved by covering all labels with aluminium foil.
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further info
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind, no further info
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind, no further info
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind, no further info
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00956111 and NCT00975572), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Longo 2009

Methods	Randomized, double blind, placebo-controlled phase I study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** not reported **Diagnosis:** HIV infected **Inclusion criteria:** "negative pregnancy test for women of childbearing potential within 3 days prior to baseline evaluation and use of an acceptable means of contraception (condom, hormonal or mechanical method) for one month prior to immunization and the duration of the study; clinically asymptomatic HIV-1 infected individuals (CDC Clinical category A), as determined by two positive Enzyme-linked immunosorbent assay (ELISA) and a confirmatory Western Blot; mean CD4+ T cell count >= 400 cells/microL based on 2 separate determinations, at least 2 weeks apart, within the four weeks pre-study screening period. [Note: If one of the two values is < 400 the patient will be excluded. Patients ever having had a value of CD4+ T cell number < 250 will be excluded]; plasma HIV-1 viremia levels <= 50,000 copies/mL; complete blood count and differential defined as: - Hematocrit >= 30% for women, >= 38% for men - Hemoglobin >= 9.5 g/dL - White cell counts >= 4,000 cells/mm3 - Total lymphocyte count >= 1000 cells/mm3 - platelets >= 100,000/mm3 Differential within institutional normal limits or approval of site physician. Normal ALT (as defined by the range of the clinical site laboratories) and Creatinine (<= 1.6 mg/dL). Normal urine dipstick with esterase and nitrite. Normal thyroid function. Availability for follow-up for planned duration of at least 12 months and willing to have further brief evaluations at 24 and 36 months. Signed informed consent". **Exclusion criteria:** "history of AIDS-related opportunistic or neoplastic disease; history of encephalopathy, neuropathy, or unstable CNS pathology (HIV or non-HIV related); History of non-HIV related neoplastic diseases, autoimmune diseases, angina or cardiac arthymias, or any other clinically significant medical problems; chest radiography showing evidence of active or acute cardiac or pulmonary disease; history of anaphylaxis or serious adverse reactions to vaccines as well as serum IgE levels exceeding 1000 U.I./mL; history of serious allergic reaction to any substance, requiring hospitalization or emergent medical care (e.g. Steven-Johnson syndrome, bronchospasm, or hypotension); active syphilis [NOTE. If the serology is documented to be a false positive or due to an adequately treated infection, the volunteer is eligible]; active tuberculosis [NOTE: Volunteers with a positive PPD and a normal chest X-ray showing no evidence of TB and not requiring isoniazid (INH) therapy are eligible]; medical or psychiatric condition or occupational responsibilities which preclude subject compliance with the protocol. Specifically excluded are persons with psychotic disorders, major affective disorders, suicidal ideation; current use of psychotrophic drugs; use of antiretroviral therapy within 3 months of pre-study screening. Use of any experimental HIV therapy or participation in another experimental protocol within three (3) months of pre-study screening; current or prior therapy with immunomodulators or immunosuppressive drugs and anticoagulant drugs within 30 days prior to study medication administration; any

	unstable cardio-vascular disease (e.g unstable hypertensive disease needing modification or introduction of an anti-hypertensive treatment); live attenuated vaccines within 60 days of study [NOTE: Medically indicated sub-unit or killed vaccines (e.g., influenza, pneumococcal, hepatitis A and B) are not exclusionary, but should be given at least 4 weeks away from HIV immunizations]; receipt of blood products or immunoglobulin in the past year; prior receipt of HIV-1 vaccine in a previous HIV vaccine trial; positivity for HBV antigens (HBs Ag, HBe Ag), and HCV, HTLV-I and HTLV-II antibodies; Positivity for HHV-8 antibodies and plasmaviremia (volunteers will be screened for anti-HHV-8 antibodies and positivity confirmed by PCR, only the individuals confirmed positive by PCR will be excluded); pregnant or lactating women".
Interventions	**Vaccine type:** human immunodeficiency virus (HIV)-1 Tat-based vaccine **Experimental:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine with aluminium **Control:** 7.5 mcg, 15 mcg or 30 mcg HIV-1 Tat vaccine **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** 5 doses (weeks 0, 4, 8, 12, 16) either subcute in alum or intradermic without adjuvant
Outcomes	**Immunogenicity**: "volunteers are monitored for anti-Tat specific antibodies, anti-Tat proliferative response and in vitro gamma-IFN and IL-4 (or IL-10) production in response to Tat (Elispot)". **Safety**: "clinical observation and monitoring of haematological, biochemical, virological and immunological parameters. Safety is evaluated by monitoring of volunteers for local and systemic adverse reactions during the trial".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** 12/02/2020 authors (barbara.ensoli@iss.it) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any AE related and not related to the vaccine ● serological response GMT for each study arm at maximum follow up ● proportion of participants who developed HIV after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment **Funding:** funded by “ISS special funding for Tat vaccine trials”, the Italian Concerted Action on HIV-AIDS Vaccine Development (ICAV, National AIDS Program) and the Italian Ministry of Health

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double-blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double-blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	The study protocol for one trial was registered prior to randomisation (NCT00505401), and all outcomes were reported on.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

Low 2014

Methods	Double-blinded, randomised, Phase 1 study
Participants	**Sex:** male and female **Age:** 21-64 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** participants, between 21 and 64 years of age, with satisfactory baseline medical assessment and laboratory values within the normal ranges **Exclusion criteria:** presence of acute infection during 14 days preceding the first vaccination, a temperature ≥38○C at the date of the first vaccination, and

	the receipt of immunoglobulins or blood products within 9 months prior to enrolment or during the study. Additional exclusion criteria were receipt of seasonal influenza vaccine in the past 2 years,or any licensed vaccine within 30 days prior to the first injection or HAI titers >1:40 at screening. Concomitant medications (except other vaccines) were not restricted. Women of child-bearing potential had to have a negative pregnancy test at each visit.
Interventions	**Vaccine type:** virus-like particle pandemic influenza A (H1N1) 2009 vaccine **Experimental:** 100 mcg H1N1 with Alhydrogel. n =43 **Control:** 100 mcg H1N1. n = 41 **Aluminium concentration:** 1000 mcg/dose or 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag-Biosector, Denmark **Vaccination schedule:** two intramuscular injections 21 days apart
Outcomes	**Immunogenicity:** "seroconversion against A/California/07/2009 (H1N1) by HAI on Day 21 and 42, defined as either a pre-vaccination HAI titer <10 and a post vaccinationHAI titer ≥40, or a pre-vaccination HAI titer ≥10 and minimum four-fold rise in post-vaccination HAI titer. In addition,geometric mean titers and the percentage of subjects achieving seroprotection (HAI titer ≥40), were calculated". **Safety:** "adverse events and severe adverse events were summarized by treatment group withe each subject counted once per adverse event category with the highest severity of treatment emergent adverse event (Day 0-Day 42). The following factors assessed: vital signs: (blood pressure, pulse, respiratory rates and body temperature) at all visits (Days 0, 21 and 42) and on vaccination days pre- and 60 min post-vaccination, hematology and blood biochemistry at all visits, solicited and unsolicited local/systemic events (symptom diaries) 21 days following each vaccination".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (vdiermayr@d3.a-star.edu.sg, veronica@diermayr.com) were inquired on the followings: ● procedure for blinding of outcome assessment ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received **Funding:** funded by A*Star, but the funding body was not scientifically involved in the clinical study or the decision to submit this article for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Independent statistician from the Singapore Clinical Research Institute generated the randomisation list. An unblinded research coordinator randomly assigned subjects1:1 to the adjuvated or the non-adjuvanted group. A computer-generated list (SAS®software, NC, USA) with randomly permutated block sizes of 4 and 6 was provided by SCRI.

Allocation concealment (selection bias)	Low risk	An unblinded research coordinator randomly assigned subjects1:1 to the adjuvated or the non-adjuvanted group. A computer-generated list (SAS®software, NC, USA) with randomly permutated block sizes of 4 and 6 was provided by SCRI.
Blinding of participants and personnel (performance bias)	Low risk	Double-blinded. Study personnel and participants were blinded to the treatment allocation, except for the independent statistician from the CRO, generating the randomisation list and the unblinded clinical research coordinator, mixing the vaccine with alhydrogel or diluent prior to injection. The 500 µL for injection were drawn up in a masked syringe by the unblended study coordinator prior to injection.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Immunogenicity assessments on sera of recruited volunteers from baseline (Day 0), Day 21 and Day 42 were performed on blinded samples. Unblinding was performed by the statistician at the CRO.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Manoff 2019

Methods	Randomized, placebo controlled, phase I dose-escalation, first in human, proof of principle trial
Participants	**Sex:** male and female **Age:** 18-49 years **Ethnicity:** not reported **Diagnosis:** healthy

	Inclusion criteria: "in good health. Voluntarily agrees to participate by giving written informed consent. Able to read, understand, and complete study questionnaires. Able to complete all scheduled visits and comply with study procedures. Access to a telephone. Agrees to avoid unusual, vigorous exercise from 72 hours before any dose of study vaccine/placebo through 15 days after that dose. Weighs ≥110 pounds (50 kg) and has a body mass index (BMI) of 19 to 32 kg/m^2. No fever (temperature ≥100.4°F/38.0°C) for 72 hours prior to vaccination. Females of reproductive potential agree to remain abstinent or to use 2 acceptable methods of birth control from enrollment through 6 weeks after the last dose of study vaccine/placebo". **Exclusion criteria:** "history of receiving any flavivirus vaccine (e.g. Japanese encephalitis, tick-borne encephalitis, or yellow fever) or planned receipt of any such vaccine during the study period. History of any flavivirus infection or serologic evidence of any flavivirus infection, including West Nile, dengue, yellow fever, Saint Louis encephalitis (if available), Kunjin, Murray Valley encephalitis, and Japanese encephalitis. History of residence for a cumulative period of >1 year in a country where dengue, Japanese encephalitis virus, or yellow fever virus is common. Planned travel to an area where dengue is common through 28 days after receiving the last dose of study vaccine/placebo. Known hypersensitivity to any component of the dengue vaccine. Abuse of drugs or alcohol within 12 months prior to screening. Pregnant or breastfeeding, or expecting to conceive in the time from enrollment through 6 weeks after the last dose of study vaccine/placebo. Positive serum test for human immunodeficiency virus (HIV), hepatitis B surface antigen (HBsAg), and/or hepatitis C antibody. Known, suspected, or a history of immunocompromise. History of malignancy within 5 years prior to enrollment. Poorly controlled diabetes mellitus. Use of any immunosuppressive therapy (except topical and inhaled/nebulized steroids). Receipt of any licensed non-live vaccine within 14 days prior to the first dose of study vaccine/placebo or plans to receive a licensed non-live vaccine during the time between receiving the first dose and 28 days after receiving the last dose of study vaccine/placebo. Receipt of any licensed live vaccine within 30 days prior to the first dose of study vaccine/placebo or plans to receive a licensed live vaccine during the time between receiving the first dose and 28 after receiving the last dose of study vaccine/placebo. Received investigational drugs or vaccines within 2 months prior to the first dose of study vaccine/placebo. History of receiving 1 or more doses of an investigational dengue vaccine. Participation in another clinical study within 42 days prior to enrollment, or plans to participate in another clinical study from enrollment through 1 year after the last dose of study vaccine/placebo. Planned donation of eggs or sperm from the time of enrollment through 28 days after the last dose of study vaccine/placebo. Prior receipt of a blood transfusion or blood products within 6 months prior to the first dose of study vaccine/placebo. Hospitalization for acute illness within 3 months prior to the first dose of vaccine/placebo".
Interventions	**Vaccine type:** recombinant subunit vaccine for dengue V180 **Experimental:** V180 with Alhydrogel. n = 9 **Control:** V180. n = 18 **Aluminium concentration:** 225 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** three injections at months 0, 1, and 2

Outcomes	**Immunogenicity:** seroconversion rate for each serotype, (time frame: 28 days post-dose 3 (day 84). Geometric mean titer of virus neutralizing antibodies for each serotype, (time frame: 28 days post-dose 3 (day 84) **Safety:** serious adverse events were followed for 1 year after the last injection. Adverse events were collected until 14 days from the last injection.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/12/2019 authors (beth-ann.coller@merck.com) were inquired on the followings: ● aluminium manufacturer ● allocation concealment ● random sequence generation ● blinding of participants and personnel ● blinding of outcome assessment ● number of patient analysed per outcome (i.e. number of drop-outs) ● number of participants with and non-serious adverse event related and not related to the vaccine ● number of participants who got Dengue infection after the vaccination No answer received **Funding:** funded by Merck

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	The safety was monitored and advised at pre-specified time points by an unblinded independent external Data Monitoring Committee.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01477580), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Mark 1994

Methods	Randomized, double-blind, clinical trial
Participants	**Sex:** male and female **Age:** 10 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "10 years of age due for their booster DT vaccination.". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** Diptheria Toxoid vaccine **Experimental:** DT vaccine with aluminium phosphate. n = 124 **Control:** DT vaccine. n = 111 **Aluminium concentration:** 250 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** one booster dose
Outcomes	**Safety and Immunogenicity**
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** no contact found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described.
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. No further details

Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a
Selective reporting (reporting bias)	Unclear risk	No protocol pre-published.
Other bias	Unclear risk	The trial appeared to be free of other issues that could put it at risk of bias.

Meulen 2015

Methods	Randomized, double-blind, multicenter clinical trial
Participants	**Sex:** male and female **Age:** 12-15 months **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy toddlers, 12-15 months of age who have previously completed a documented full 3 dose infant series of Prevnar™ at 2, 4, and 6 months of age. Participant's parent/legal guardian understands the study procedures, alternate treatments available and risks involved with the study, and voluntarily agrees to participate by giving written informed consent. Afebrile, with a rectal temperature <38.1°C (<100.5°F) or axillary temperature <37.8°C (<100.0°F) on day of vaccination. Participant's parent/legal guardian is able to read, understand, and complete study questionnaires (i.e., the Vaccination Report Card). Participant is able to attend all scheduled visits and to comply with the study procedures. Participant's parent/legal guardian has access to a telephone". **Exclusion criteria:** "have received less than the full 3-dose infant series of Prevnar™ or 3rd dose less than 2 months before study vaccine. Known hypersensitivity to any component of the pneumococcal conjugate vaccine. Known or suspected impairment of immunological function. Participant has a history of congenital or acquired immunodeficiency (e.g., splenomegaly). Participant or his/her mother has documented HIV infection. Functional or anatomic asplenia. History of autoimmune disease including multiple sclerosis, systemic lupus, polymyositis, inclusion body myositis, dermatomyositis, Hashimoto's thyroiditis, Sjogren's syndrome, rheumatoid arthritis, other autoimmune disorders. Known neurologic or cognitive behavioral disorders including multiple sclerosis, multiple sclerosis-like disease, encephalitis/myelitis,

	acute disseminating encephalomyelitis, pervasive developmental disorder, and related disorders. Receipt of intramuscular, oral, or intravenous corticosteroid treatment within the 2 weeks prior to vaccination. (Note: Toddlers on topical and inhaled/nebulized steroids may participate in the study.) Use of systemic steroids are only permitted when the participant is receiving less than 2 mg/kg per day of prednisone (or its equivalent), or less than 20 mg/d if they weigh more than 10 kg and are not otherwise immunocompromised. Participant has received other licensed non-live vaccines administered within the 14 days before receipt of study vaccine. Participant has received a licensed live virus vaccine within 30 days prior of receipt of study vaccine (Exception: Influenza virus vaccine given according to recommended guidelines within 7 days of receiving study vaccine). Prior receipt of a blood transfusion or blood products, including immunoglobulins. Investigational drugs or vaccines received within the 2 months before receipt of study vaccine. Participation in another clinical study within 42 days before the beginning or anytime during the duration of the current clinical study. History of invasive pneumococcal disease (positive blood culture, positive cerebrospinal fluid culture, or other sterile site) or known history of other culture positive pneumococcal disease. A recent (<72 hours) febrile illness (rectal temperature ≥38.1°C [≥100.5°F]) occurring within 48 hours before receipt of study vaccine. History of failure to thrive. Participant has a coagulation disorder contraindicating IM vaccination. Participant and his/her mother is documented hepatitis B surface antigen-positive. Any toddler who cannot be adequately followed for safety according to the protocol plan. Participant's parent/legal guardian is unlikely to adhere to study procedures, keep appointments, or is planning to relocate during the study. Any other reason that in the opinion of the investigator may interfere with the evaluation required by the study".
Interventions	**Vaccine type:** 15-valent Pneumococcal Conjugate Vaccine PCV15 **Experimental:** PCV15 with aluminium. n = 33 **Control:** PCV15. n = 29 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** Merck aluminium phosphate **Aluminium manufacturer:** Merck **Vaccination schedule:** one dose administered intramuscularly on day 1
Outcomes	**Immunogenicity:** "percentage of participants achieving the Immunoglobulin G serotype-specific Threshold value of ≥ 0.35µg/mL, (time frame: day 30 after vaccination). Immunoglobulin G for the serotypes contained in V114 was determined using an electrochemiluminescence assay. Geometric mean concentration of Pneumococcal Serotype Immunoglobulin G antibodies, (time frame: day 30 after vaccination). Pneumococcal serotype-specific IgG was measured in serum using an electrochemiluminescence assay. Percentage of participants achieving opsonophagocytic activity ≥1:8, (time frame: day 30 after vaccination). Opsonophagocytic activity for the serotypes contained in V114 was determined using a multiplex opsonophagocytic assay. Geometric mean titer of pneumococcal serotype-specific opsonophagocytic activity, (time frame: day 30 after vaccination). Opsonophagocytic activity for the serotypes contained in V114 was determined using a multiplex opsonophagocytic assay". **Safety:** "percentage of participants with any adverse event, (time frame: up to day 14 after vaccination). An adverse event is defined as any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment. An adverse even can therefore be any

	unfavourable and unintended sign, symptom, or disease temporally associated with the use of a medicinal product or protocol-specified procedure, whether or not considered related to the medicinal product or protocol-specified procedure. Any worsening of a preexisting condition that is temporally associated with the use of the Sponsor's product, is also an adverse even. Percentage of participants with any serious adverse even, (time frame: up to day 14 after vaccination). A serious adverse event is an adverse event that results in death, is life threatening, requires or prolongs an existing hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is another important medical event deemed such by medical or scientific judgment. percentage of participants with any vaccine-related adverse event, (time frame: up to day 14 after vaccination). An adverse event is defined as any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment. An adverse event can therefore be any unfavourable and unintended sign, symptom, or disease temporally associated with the use of a medicinal product or protocol-specified procedure, whether or not considered related to the medicinal product or protocol-specified procedure. Any worsening of a preexisting condition that is temporally associated with the use of the Sponsor's product, is also an adverse event. Relatedness of the adverse event to study vaccine was determined by the investigator".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/ 2018 authors (luwy_musey@merck.com, ajokestm@uw.edu) were inquired on the followings: ● method for random sequence generation ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** provided by Merck & Co., Inc

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Low risk	Subjects were allocated to treatment assignment using a randomised schedule.
Blinding of participants and personnel (performance bias)	Low risk	Double-blind. All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	All study personnel, including investigators, study site personnel, subjects, monitors and central laboratory personnel, remained blinded to treatment allocation throughout the study.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01215175), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Moustafa 2012

Methods	Randomized, blinded, placebo-controlled, phase II immunogenicity study
Participants	**Sex:** male and female **Age:** 18 to 80 years **Ethnicity:** mixed **Diagnosis:** end-stage renal disease on chronic hemodialysis **Inclusion criteria:** "participant is scheduled to undergo cardiothoracic surgery involving a full median sternotomy (not including cardiac transplantation surgery) within 14 to 60 days after vaccination. Female participants of reproductive potential are required to have a negative urine or serum pregnancy test immediately prior to study vaccination and must use an acceptable form of birth control". **Exclusion criteria:** "participants had an invasive Staphylococcus aureus infection within the past 3 months prior to study entry. A realistic (>50%) possibility that cardiothoracic surgery will be necessary sooner than 10 days after vaccination. Participant is planning to undergo cardiac transplantation surgery or sternal debridement to remedy an infection resulting from a prior cardiothoracic surgery. Participant has any type of ventricular-assist device in place at the time of study entry. Participant has a history of anaphylaxis to any of the vaccine components. Participant received V710 vaccine, any other investigational Staphylococcus

	aureus vaccine, or investigational Staphylococcus aureus antibodies. Participant has a temperature of ≥100.4ºF (≥38.0ºC), oral equivalent, within 48 hours prior to study vaccination. Participant has impairment of the immune system. Participant has a medical condition in which the expected survival is less than 90 days".
Interventions	**Vaccine type:** Staphylococcus aureus Vaccine V710 **Experimental:** "3a": V710 60 mcg with AAHSA x3, or "3b": V710 60 mcg with AAHSA + V710 60 µg with AAHSA + Placebo, or "4a": V710 60 mcg with AAHSA + Placebo + V710 60 mcg with AAHSA or "4b": V710 60 mcg with AAHSA + Placebo + Placebo. n = 75 **Control:** "1a": V710 60 mcg, or "1b": V710 60 mcg + V710 60 + Placebo, or "2a": V710 60 mcg + Placebo + V710 60 mcg, or "2b": V710 60 mcg + Placebo + Placebo. n = 73 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxyphosphate sulfate adjuvant (AAHSA) **Aluminium manufacturer:** not reported **Vaccination schedule:** all patients received their priming doses on day 1 and day 28, with a booster dose (equally divided between V710 and placebo) on day 180
Outcomes	**Immunogenicity:** "number of participants with invasive Staphylococcus aureus Infection, (time frame: up to 90 days after surgery). Diagnosis of the Staphylococcus aureus infections employed standardized definitions adapted from the Centers for Disease Control guidelines for nosocomial infections. An invasive Staphylococcus infection included bacteremia, deep sternal wound infection, deep-tissue organ/space infection at another surgical site, or any other deep-tissue infection. Number of participants with surgical-site Staphylococcus aureus infection, (time frame: up to 90 days after surgery). Diagnosis of the Staphylococcus aureus infections employed standardized definitions adapted from the Centers for Disease Control guidelines for nosocomial infections. A Staphylococcus infection surgical-site infection included any superficial incisional, deep incisional, or organ/space infection at the sternal site, the vascular harvest (donor) site, or any other site at which the surgery was performed". **Safety:** "number of participants with staphylococcus aureus bacteremia and/or deep sternal wound infection, (time frame: up to 90 days after surgery). Diagnosis of the Staphylococcus aureus infections employed standardized definitions adapted from the Centers for Disease Control guidelines for nosocomial infections (Garner JS, Jarvis WS, Emori TG, et al. CDC definitions for nosocomial infections. APIC Infect Control App Epidemiol 1996;A1-20). Bacteremia was defined as ≥1 positive blood culture for S. aureus regardless of the presence of clinical symptoms. A Staphylococcus aureus deep sternal wound infection included mediastinitis or a deep incisional surgical-site infection involving the sternal wound. Incidence rate of vaccine-related serious adverse experiences, (time frame: up to 360 days after surgery). Vaccine-related adverse experiences were those deemed by the investigator to be possibly, probably, or definitely vaccine related. A serious adverse experience was any adverse experience occurring at any dose that 1) resulted in death, 2) was life threatening, 3) resulted in a persistent or significant disability/incapacity, 4) resulted in or prolonged an existing inpatient hospitalization, 5) was a congenital anomaly/birth defect, 6) was a cancer, 7) was an overdose, or 8) jeopardized the participant and required medical or surgical intervention".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (dalya_guris@merck.com) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** funded by Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc., Whitehouse Station, NJ

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomization and allocation were assigned using an interactive voice response system. Numbered packaging was used to implement allocation.
Allocation concealment (selection bias)	Low risk	Randomization and allocation were assigned using an interactive voice response system. Numbered packaging was used to implement allocation.
Blinding of participants and personnel (performance bias)	Low risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study

Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Unblinded member of the study site staff prepared and administered vaccine or placebo but had no further involvement with any subsequent study procedure, including safety follow-up. All other individuals involved with the study, including investigators, participants, and the clinical monitors and statisticians employed by the sponsor, remained blinded to treatment group throughout the study
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00572910), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT00309647

Methods	Phase I partially-blind randomised multi-centric study
Participants	**Sex:** male and female **Age:** 18-60 years **Ethnicity:** Caucasian **Diagnosis:** healthy **Inclusion criteria:** "male or female between, and including, 18 and 60 years of age at the time of the first vaccination. Healthy subjects as established by medical history and clinical examination before entering into the study. If the subject is female, she must be of non-childbearing potential". **Exclusion criteria:** "administration of any vaccine during the period starting 15 days before the first administration of the study vaccine and ending 21 after the second one. Administration of an influenza vaccine other than the study vaccines during the entire study period. Chronic administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within six months prior to the first administration of the study vaccine. Any confirmed or suspected immunosuppressive or immunodeficient condition, based on medical history and physical examination. History of hypersensitivity to vaccines. History of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Acute clinically significant pulmonary, cardiovascular, hepatic or renal functional abnormality, as determined by physical examination or laboratory screening tests. Acute disease at the time of enrolment. Administration of immunoglobulins and/or any blood products within the three months preceding the first administration of the study vaccine or during the study. Lactating women. Use of any investigational or non-registered product (drug or vaccine) other than the study vaccine(s) within 30 days prior to the first vaccination, or planned use

	during the study period".
Interventions	**Vaccine type:** influenza H5N1 vaccine **Experimental:** 27, 15, 7.5 or 3.8 mcg H5N1 with aluminium. n = 201 **Control:** 27, 15, 7.5 or 3.8 mcg H5N1. n = 199 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular administrations at day 0 and Day 21 in the deltoid region of the non-dominant arm
Outcomes	**Immunogenicity**: "the following parameters were calculated with 95 % confidence intervals: from the anti-heamagglutinin antibody titres: geometric mean titres pre- and post-vaccinations (days 0, 21, and 42), seroconversion rates and seroconversion factors at days 21 and 42, seroprotection rates at days 0, 21 and 42. From the neutralisation antibody titres: geometric mean titres pre- and post-vaccinations (days 0, 21 and 42), seroconversion rates at days 21 and 42. In order to evaluate the cell-mediated immune response, the following parameter was calculated:frequency of influenza-specific cytokine-positive CD4/CD8 T-cells at days 0, 21 and 42". **Safety**: "recording of solicited local (pain, redness, swelling, induration, ecchymosis) and general (fatigue, headache, fever, shivering, sweating increase, myalgia and arthralgia) adverse events during a 7 days follow-up period after vaccination. Recording of unsolicited adverse events occurring during a 21-day follow-up period after the first vaccination and a 30-day follow-up period after the second vaccination. Recording of serious adverse events occurring during the entire study".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** 30/01/2020 authors (dk-info@gsk.com) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any AE related and not related to the vaccine ● proportion of participants who developed influenza after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment No answer received **Funding**: GSK

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described

Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Partially blinded
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Partially blinded
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Partially blinded
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Partially blinded
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Unclear risk	The study protocol was registered prior to randomisation (NCT00309647), and all outcomes were reported on.
Other bias	Low risk	The study was not published in a peer-reviewed journal but only on the company website.

NCT00562237

Methods	Randomized, observer-blind, placebo-controlled study
Participants	**Sex:** male and female **Age:** 18-49 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "being healthy and ≥ 18 and ≤ 49 years of age; willing and able to give informed consent". **Exclusion criteria:** "having participated in an influenza H5 vaccine trial in the past; known to be allergic to any constituent of the vaccine; serious adverse reactions to previous (influenza) vaccination; currently participating in another clinical trial or having participated in any clinical trial in the month preceding the start of the study; using medication that influences the immune system".
Interventions	**Vaccine type:** influenza H5N1 vaccine **Experimental:** S205 10 or 30 mcg HA with AlOH **Control:** S205 10 or 30 mcg HA **Aluminium concentration:** 500 mcg/dose

	Aluminium type: aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 intramuscular injections
Outcomes	**Immunogenicity**: hemagglutination inhibition titers, virus neutralization,anti-HA antibody level kinetics (time frame: one year). **Safety**: Not stated. Time frame: one year
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Solvay Pharmaceutical) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT00693615

Methods	Phase II, double-blind, randomised study
Participants	**Sex:** female **Age:** 18-30 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "females 18 through 30 years of age (must not have reached the 31st birthday). Written informed consent obtained from the volunteer. Unless previously surgically sterilized, agrees to use an effective method of birth control beginning 30 days before the first study injection and continuing through 60 days after the final study injection. Must have a negative serum pregnancy test within 21 days of study entry and must not be breast feeding. Healthy by medical history and physical examination. Seronegative for HPV-16 and HPV-18 antibody by ELISA within 21 days of study entry. Cervical specimen negative for high-risk HPV DNA using the Digene Hybrid Capture® II HPV test (high-risk types Probe B) within 21 days of study entry. Normal Pap smear, using the Cytyc ThinPrep® Pap Test, within 21 days of study entry. No evidence of anogenital HPV lesions or physical findings suggestive of other gynaecologic pathogens on pelvic examination within 21 days of study entry. Agrees to no other vaccines or experimental therapy until 30 days after the last study injection". **Exclusion criteria:** "acute illness or fever (oral temperature ≥ 99.5°F [37.5°C]) at start of the study. History or clinical manifestations of significant medical or psychiatric disorder. Use of immunosuppressive medication (inhaled and topical corticosteroids are permitted) within the previous 90 days or history of immunodeficiency. History of cancer. History of alcohol or drug abuse within the past 2 years. Abnormal laboratory blood values at screening. Other abnormal laboratory values in the screening panel which in the opinion of the principal investigator are judged to be clinically significant. Receipt of immunoglobulin or blood products within 90 days prior to study entry. History of abnormal Pap smear (other than a single prior report of ASCUS or indeterminate Pap smear with a subsequent normal report). Genital herpes disease involving the cervix or with disease characterized on examination or by history by extensive external lesions. Volunteers with a history of recurrent genital herpes disease characterized by limited external lesions are eligible to participate in the study. Positive tests for hepatitis C antibody, hepatitis B surface antigen, or HIV-1 antibody. Any prior receipt of any vaccine (experimental or otherwise) for treatment or prophylaxis of genital warts or other papillomavirus related condition. Any treatment of genital warts or other papillomavirus related condition within 6 months of randomization (local therapy for common skin and/or plantar warts is allowed) Previous administration of any components of the investigational vaccine. Receipt of any experimental vaccine within 90 days prior to entry into this study. Receipt of any experimental drug therapy within 30 days or five half-lives of the experimental drug (if the half-life is known), whichever is longer".
Interventions	**Vaccine type:** MEDI-517 HPV vaccine **Experimental:** MEDI-517 HPV vaccine with aluminium. n = 18 **Control:** MEDI-517 HPV vaccine. n = 22 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported

	Vaccination schedule: 3 intramuscular injections into the deltoid at day 0, 30, 180
Outcomes	**Immunogenicity**: "serum enzyme-linked immunosorbent assay antibody titers against HPV-16 and HPV-18 at 30 days after the third dose (day 210). Serum enzyme-linked immunosorbent assay antibody titers against HPV-16 and HPV-18 on study days 0, 7, 30, 60, 180 and 360, and at 18, 24, 36, 48 months. Neutralization antibody titers against HPV-16 and HPV-18 on study days 0, 60, 210 and 360. Inhibitory ELISAs at 18, 24, 36 and 48 months. Lymphoproliferation, interleukin-5 and interferon-gamma assays at study days 0, 60, 210 and 360, and at 18, 24 months".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 16/01/2020 authors (dk-info@gsk.com) were inquired on the followings: ● safety and immunogenicity results Answer received **Funding**: GSK Note: The serology outcome data are available up to 4 years. However, in this meta-analysis, we use the 360 days time point. At the maximum follow up there were a too high drop out rate.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Low risk	The investigators assigned a patient identification number. A randomization list provided to the study pharmacist by MedImmune contained assignment to treatment groups. The pharmacist did not reveal treatment assignment to anyone.
Blinding of participants and personnel (performance bias)	Low risk	The investigator, all clinical staff (except the study pharmacist), the volunteers and the sponsor's staff (other than the study statistician) were unaware of treatment assignment
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs < 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs < 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs more than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00693615), and all outcomes were reported on.
Other bias	Low risk	The study was not published in a peer-reviewed journal but only on the company web site.

NCT01244867

Methods	Phase 2, randomised, observer-blind, single center, dose-ranging study
Participants	**Sex:** male and female **Age:** 18-60 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male and female adults, 18 to 60 years of age; healthy as judged by the Principal Investigator (PI) and determined by medical history, physical examination, vital signs, screening laboratories and medical history conducted no more than 30 days prior to study vaccine administration; BMI of ≥18 and ≤ 32; comprehension of the study requirements, expressed availability for the required study period and ability to attend scheduled visits; accessible by telephone on a consistent basis; In the opinion of the Investigator, competence and willingness to provide written, informed consent for participation after reading the informed consent form. The subject must have adequate opportunity to discuss the study with an Investigator or qualified designee; If female and capable of child-bearing, have a negative serum pregnancy test result at study entry, has been consistently using effective birth control for the 28 days prior to study entry and agree to continue employing adequate birth control measures for the duration of the study". **Exclusion criteria:** "presence of significant acute or chronic, uncontrolled medical or neuropsychiatric illness. Any medical or neuropsychiatric condition which, in the Investigator's opinion, would render the subject incompetent to provide informed consent or unable to provide valid safety observations and reporting; confirmed or suspected immunosuppressive condition or immunodeficiency including history of human immunodeficiency virus (HIV) infection or Hepatitis B or C or presence of lymphoproliferative disease; Presence of any febrile illness, oral temperature of >38.0˚C within 24 hours of test article administration. Such subjects may be re-evaluated for enrolment after resolution of illness; History of autoimmune disease; Administration of any vaccine (including any other influenza vaccine) within a 30 day period prior to study enrolment, or planned administration within the period from the first vaccination up to blood sampling at Day 42 or within 30 days prior to blood sampling at Day 228. Immunization on an emergency basis of a tetanus and

	diphtheria toxoids adsorbed for adult use (Td) will be allowed provided the vaccine is not administered within two weeks prior to test article administration. Receipt of any other emergency immunizations (e.g. rabies) will result in a case-by-case review of continued participation; Use of any investigational or non-registered product within 30 days prior to study enrolment or planned use during the study period. Subjects may not participate in any other drug study while participating in this study; Treatment with systemic glucocorticoids at a dose exceeding ≥ 10 mg of prednisone per day, or equivalent for more than 7 consecutive days or for 10 or more days in total, within one month of first test article administration, or any other cytotoxic or immunosuppressant drug or any immune globulin preparation within three months of vaccination. Nasal or inhaled glucocorticoids are allowed; any significant disorder of coagulation or treatment with Coumadin derivatives or heparin. Persons receiving prophylactic anti-platelet medications, e.g., low-dose aspirin, and without a clinically apparent bleeding tendency are eligible; history of previous H5N1 vaccination or a history of exposure to H5N1 virus; history of allergy to any of the constituents of H5 VLP (H5N1) study vaccine, Alhydrogel® (aluminium hydroxide), or the phosphate buffer; history of severe allergic reactions or anaphylaxis; history of tobacco allergy; history of anti-histaminics used continuously for 4 weeks or more at any time in the past year; have a rash, dermatological condition or tattoos which may interfere with injection site reaction rating; have received a blood transfusion or immunoglobulins within 90 days of study entry; if female, and of childbearing potential, has not been consistently using effective birth control for the 28 days prior to study entry. An example of highly effective birth control is oral contraceptives, hormone implants, abstinence (confirmed by Investigator), or male condom plus spermicide. All female subjects, regardless of birth control history must provide a serum sample for pregnancy screening. Effective birth control must be used for the duration of the study for female of child bearing potential must have no plan to become pregnant during the study period.; among female subjects, either known pregnancy or urine beta-human chorionic gonadotropin (ß-hCG) test results consistent with pregnancy prior to test article administration on Day 0; female subjects who are lactating; Vital sign abnormalities: systolic blood pressure ≥150 mmHg, diastolic blood pressure ≥90 mmHg, resting pulse rate <40 bpm or >100 bpm or according to the Investigator's opinion; Cancer or treatment for cancer within 3 years of test article administration. Persons with a history of cancer who are disease-free without treatment for 3 years or more are eligible. Persons with treated and uncomplicated basal cell carcinoma of the skin are eligible".
Interventions	**Vaccine type:** plant-made H5 VLP influenza vaccine **Experimental:** 45 mcg VLP with aluminium **Control:** 45 mcg VLP **Aluminium concentration:** not reported **Aluminium type:** alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** two consecutive doses 21 days apart
Outcomes	**Immunogenicity**: "geometric mean titres of hemagglutination inhibition antibody on days 0, 21 and 42. Follow-up serology samples for geometric mean titres will be taken at Day 228". **Safety**: "rate and severity of solicited and unsolicited adverse events post-vaccination. A 6-month follow-up period will be performed".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Medicago) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT01447407

Methods	Partially-blind, placebo controlled Phase 1b study
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "informed of the nature of the study and have agreed to and

	are able to read, review, and sign the informed consent document prior to screening. The informed consent document will be written in English, therefore the volunteer must have the ability to read and communicate in English. Completed the screening process (as described in this protocol) within 28 days prior to dosing. Healthy male and female volunteers 18-50 years of age, inclusive, at the time of dosing. No clinically significant deviation from normal as judged by the investigator(s) in the medical history, physical examination (including but may not be limited to an evaluation of the cardiovascular, gastrointestinal, respiratory and central nervous systems), vital sign assessments, 12-lead electrocardiogram (ECG), clinical laboratory assessments, and by general observations. Fmale volunteers must be one of the following: of childbearing potential and practicing an acceptable method of birth control as judged by the Investigator(s), naturally postmenopausal (no menses) for at least 1 year and has a documented FSH level ≥ 40 mIU/mL surgically postmenopausal (bilateral oophorectomy or hysterectomy), sterile (surgically [bilateral tubal ligation] or the Essure® Procedure) Female volunteers that are surgically sterile or surgically postmenopausal must provide documentation of the bilateral tubal ligation, bilateral oophorectomy, or hysterectomy prior to dosing or the volunteer must agree to use a medically acceptable method of birth control. The Essure® Procedure must have been inserted at least 3 months prior with documentation of the Essure® confirmation test prior to Period I dosing. If the procedure was inserted less than 3 months prior to Period I dosing or proper documentation of the confirmation test is not provided, the volunteer must agree to use an additional medically acceptable method of birth control". **Exclusion criteria:** "reports receiving any investigational drug, investigational vaccine, or investigational device within 30 days prior to dosing. Reports any presence or history of a clinically significant disorder involving the cardiovascular, respiratory, renal, gastrointestinal, immunologic, hematologic, endocrine, or neurologic system(s) or psychiatric disease as determined by the Investigator(s). Clinical laboratory test values outside the accepted range. When confirmed upon additional testing, demonstrates a reactive screen for hepatitis B surface antigen, hepatitis C antibody, or HIV antibody. Demonstrates a positive drug screen for non-prescription drugs. Reports a clinically significant illness during the 28 days prior to dosing (as determined by the Investigator[s]). Reports a history of allergic response(s) to nickel or anaphylaxis (or other serious reactions) to aluminium. Reports receiving any live attenuated vaccine including FluMist® within 6 weeks prior to dosing or any licensed inactivated vaccine within 3 weeks prior to dosing. Reports the use of any immunosuppressive drugs, including systemic corticosteroids, within 4 weeks prior to dosing. Reports the use of any medications or treatments that may alter immune responses to the study vaccine within 3 weeks prior to dosing (eg, cyclosporine, tacrolimus, cytotoxic drugs, immune globulin, Bacillus Calmette-Guerin [BCG], monoclonal antibodies, radiation therapy). Reports a history of clinically significant allergies including food or drug allergies or anaphylaxis (or other serious reactions) to vaccines. Reports a history of drug or alcohol addiction or abuse within the past year. Reports receiving any blood products within 3 months prior to dosing and throughout the study. Reports donating blood within 28 days prior to dosing. All subjects will be advised not to donate blood for four weeks after completing the study. Reports donating plasma (e.g. plasmapheresis) within 14 days prior to dosing. All subjects will be advised not to donate plasma for four weeks after completing the study. Demonstrates, in the opinion of study staff, veins unsuitable for repeated

	venipuncture (e.g. veins difficult to locate, access, or puncture; veins with a tendency to rupture during or after puncture). Pregnant, lactating, breastfeeding, or intends to become pregnant over the course of the study. Demonstrates a positive pregnancy screen. Reports smoking or using tobacco products or is currently using nicotine products (patches, gums, etc). Thirty (30) days abstinence prior to dosing is required. Any other medical and/or social (e.g. uncooperative or non-compliant) reason which, in the opinion of the investigator(s), would prevent participation in the study".
Interventions	**Vaccine type:** NDV-3, directed against *Staphylococcus aureus* and Candida sp **Experimental:** NDV-3 vaccine with alum. n =41 **Control:** NDV-3 vaccine without alum. n = 40 **Aluminium concentration:** 1 mcg/dose **Aluminium type:** alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** one dose administered IM
Outcomes	**Immunogenicity**: "to compare the serum IgG immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Serum IgG will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the serum IgA1 immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Serum IgA1 will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the cervicovaginal wash IgG immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Cervicovaginal wash IgG will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the cervicovaginal wash IgA1 immune response between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. Cervicovaginal wash IgA1 will be evaluated by ELISA on serial-diluted samples, resulting in titer values of reciprocal dilution at which the ELISA readout is three times greater than the assay background value. To compare the cellular immune response for Als3-specific production of IFN-g from PBMCs between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. The IFN-g cellular immune responses will be evaluated by ELISpot using approximately 200,000 PBMCs per well. A positive response was defined as a sample with greater than 20 spot forming units per 10^6 PBMCs. To compare the cellular immune response for Als3-specific production of IL-17A from PBMCs between the 2 dose levels, routes of administration, and effects of alum adjuvant, at selected time points up to 90 days post-vaccination. The IL-17A cellular immune responses will be evaluated by ELISpot using approximately 200,000 PBMCs per well. A positive response was defined as a sample with greater than 20 spot forming units per 10^6 PBMCs. Time Frame: Baseline, Day 7, Day 14, Day 28, Day 90/Exit". **Safety**: "safety of a single dose of NDV-3 vaccine, administered either IM with or without alum adjuvant at one dose level or ID at a lower dose level, compared to placebo. Clinical evaluations will be assessed on each subject at selected time points up to 90 days post-vaccination".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/01/2020 author (john_hennessey@novadigm.net) contacted. Study results uploaded on clinicaltrial.gov after request **Funding:** NovaDigm Therapeutics

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Only states randomised. No further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Masking: Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor)
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Not described
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Not described
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Unclear risk	The study protocol for one trial was registered prior to randomisation (NCT01447407), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT01995617

Methods	Phase 1, randomised, double-blind, placebo-controlled, dose escalation study
Participants	**Sex:** male and female **Age:** 18-55 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "males and non-pregnant females, ages 18 to 55 years inclusive. Willing and able to provide written informed consent. Willing to perform and comply with all study procedures including attending clinic visits as scheduled. Willing to practice adequate contraception that may include, but is not limited to, abstinence, monogamous relationship with vasectomized partner, vasectomy, barrier methods such as condoms, diaphragms, intrauterine devices, and licensed hormonal methods for 28 days before and 90 days after receiving Study Drug". **Exclusion criteria:** "immunocompromised individuals, including those receiving immunosuppressive doses of corticosteroids (more than 20 mg of prednisone given daily or on alternative days for 2 weeks or more within 6 months prior to the first dose of Study Drug, any dose of corticosteroids within 30 days of the first dose of Study Drug, or high dose inhaled corticosteroids [> 960 µg/day of beclomethasone dipropionate or equivalent]) or other immunosuppressive agents. Presence or history of autoimmune disease, regardless of current treatment. Insulin dependent diabetes. Positive serologic test for HIV-1 or hepatitis C infection; positive hepatitis B surface antigen (HBsAg). Screening serum chemistry or hematology abnormality ≥ Grade 1. Screening urinalysis abnormality ≥ Grade 2. Positive urine drug screen or alcohol Breathalyzer test at Screening or on Study Day 1. Other active, uncontrolled co-morbidities that, in the opinion of the Investigator would make the subject unsuitable for the study or unable to comply with the study requirements.NOTE: Subjects who are taking a medication to control an underlying co-morbidity may be enrolled if there have been no changes to their medication within 60 days prior to the first dose of Study Drug. Any acute illness including, fever (>100.4 degrees F [> 38 degrees C]) within 3 days prior to the first dose of Study Drug. Pregnant or nursing women. Receipt of any investigational drug within 30 days prior to the first dose of Study Drug. Receipt of blood products within 90 days prior to the first dose of Study Drug. Donation of blood or plasma within 56 days prior to Screening. Receipt of a live vaccine within 28 days prior or a subunit vaccine within 14 days prior to the first dose of Study Drug or planned vaccination within 30 days following the last dose of Study Drug. Prior vaccination with pneumococcal vaccine. History of hypersensitivity to any component of the vaccine or history of an allergic reaction to an immunization. History of invasive pneumococcal disease (i.e., sepsis, meningitis or pneumonia with bacteremia). History of drug or alcohol abuse that, in the opinion of the Investigator, would interfere with the subject's ability to comply with the study requirements. Any other condition that, in the opinion of the Investigator would make the subject unsuitable for the study or unable to comply with the study requirements".
Interventions	**Vaccine type:** GEN-004, a *Streptococcus Pneumoniae* protein subunit vaccine **Experimental:** low, mid or high dose vaccine with aluminium **Control:** low, mid or high dose vaccine **Aluminium concentration:** 350 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported

	Vaccination schedule: 3 doses at 4 week intervals
Outcomes	**Immunogenicity** and **safety** up to 12 months after the last dose
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Genocea Biosciences) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT02022163

Methods	Phase I, randomised trial
Participants	**Sex:** male and female **Age:** 18-60 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male and female adults, 18 to 60 years of age, inclusive. Healthy as judged by the Investigator or designee and determined by medical history, complete general history/symptom-directed physical examination, vital signs, screening laboratories, and medical history conducted no more than 30 days prior to study vaccine administration. BMI of ≥18 and ≤32. Comprehension of the study requirements, expressed availability for the required study period, and ability to attend scheduled visits. Accessible by phone on a consistent basis Give his/her consent to participate in this study (by signing the ICF). In the opinion of the Investigator, competence and willingness to provide written, informed consent for participation after reading the informed consent form. The subject must have adequate opportunity to discuss the study with an Investigator or qualified designee. If female, have a negative pregnancy test result prior to first immunization. Female of childbearing potential (except subjects in a monogamous same sex relationship), must use an effective birth control for the 28 days prior to immunization and must agree to continue employing adequate birth control measures from Day 0 (first immunization) until at least 60 days post-second immunization and must have no plan to become pregnant from Day 0 (first immunization) until at least 60 days post-second immunization. Highly effective birth control includes hormonal contraceptives (e.g., injectable, topical [patch], estrogenic vaginal ring, etc.), intra-uterine device (IUD), abstinence (confirmed by Investigator), or male condom plus spermicide. Abstinent subjects should be asked what method(s) they would use, should their circumstances change, and subjects without a well-defined plan should be excluded". **Exclusion criteria:** "presence of significant acute or chronic, uncontrolled medical or neuropsychiatric illness. "Uncontrolled" is defined as: requiring a new medical or surgical treatment within one month prior to study vaccine administration, requiring a change in medication dosage in one month prior to study vaccine administration due to uncontrolled symptoms or drug toxicity (elective dosage adjustments in stable subjects are acceptable). Hospitalization or an event fulfilling the definition of a serious adverse event within one month prior to study vaccine administration. Any medical or neuropsychiatric condition or any history of excessive alcohol use or drug abuse which, in the Investigator's opinion, would render the subject incompetent to provide informed consent or unable to provide valid safety observations and reporting. Any confirmed or suspected immunosuppressive condition or immunodeficiency including history of human immunodeficiency virus (HIV) infection, Hepatitis B or C, or the presence of lymphoproliferative disease. Presence of any febrile illness, oral temperature of >38.0˚C within 24 hours prior to immunization. Such subjects may be re-evaluated for enrolment after resolution of illness. History of autoimmune disease. Administration of any vaccine (including any other influenza vaccine) within 30 days prior to study enrolment or planned administration within the period from the vaccination up to blood sampling at Day 42 or within 30 days prior to blood sampling at Day 228. Immunization on

	an emergency basis of a tetanus and diphtheria toxoids adsorbed for adult use (Td) will be allowed provided the vaccine is not administered within two weeks prior to study vaccine administration. Receipt of any other emergency immunizations (e.g., rabies) will result in a case-by-case review by the medical monitor of continued participation. Administration of any adjuvated or investigational influenza vaccine other than a 'simple' seasonal Trivalent Inactivated Vaccine (TIV) or Quadrivalent Inactivated Vaccine (QIV) within 1 year prior to study enrolment or planned administration prior to the end of this trial (Day 228). Use of any investigational or non-registered product within 30 days prior to study enrolment or planned use during the study period. Subjects may not participate in any other investigational or marketed drug study while participating in this study. Treatment with systemic glucocorticoids at a dose exceeding 10 mg of prednisone per day, or equivalent for more than 7 consecutive days or for 10 or more days in total, within one month of study vaccine administration, any other cytotoxic or immunosuppressant drug, or any globulin preparation within 3 months of vaccination. Low doses of nasal or inhaled glucocorticoids are generally allowed. Use of high dose inhaled steroids or oral and parenteral high dose steroid medications. Nasal steroids are allowed. Any significant disorder of coagulation or treatment with warfarin derivatives or heparin. Persons receiving prophylactic anti-platelet medications, e.g., low-dose aspirin [≤ 325 mg/day (1 regular adult aspirin) or ≤ 81 mg/day (1 baby aspirin)], and without a clinically apparent bleeding tendency are eligible. History of previous H7N9 vaccination or a history of exposure to H7N9 virus. Any subject that was enrolled on previous H7N9 studies (except the ones that received placebo) would not be eligible. Are at high risk of contracting H7N9 influenza infection (e.g. poultry workers). History of allergy to any of the constituents of the H7 VLP (H7N9) study vaccine, Alhydrogel® (aluminium hydroxide), or to the phosphate-buffered saline (PBS) (used as placebo). History of severe allergic reactions (including anaphylaxis) to any food, medication or bee sting or previous severe asthma. History of tobacco allergy. Continuous use of anti-histamines in the last 4 weeks prior to first immunization or use of anti-histamines 48 hours prior to each study immunization. Have a rash, dermatological condition, tattoos, or muscle mass at injection site which may interfere with injection site reaction rating. Have received a blood transfusion within 90 days prior to study vaccination. If female, positive pregnancy test results at screening, and prior to immunizations on Day 0 and Day 21. Female subjects who are lactating. Any vital sign abnormalities: systolic blood pressure, diastolic blood pressure, resting heart rate not well controlled or according to the Investigator's opinion. Cancer or treatment for cancer within 3 years of study vaccine administration. Persons with a history of cancer who are disease-free without treatment for 3 years or more are eligible. Persons with treated and uncomplicated basal cell carcinoma of the skin are eligible".
Interventions	**Vaccine type:** plant-made H7 virus-like particle (VLP) influenza vaccine **Experimental:** high dose of H7 VLP vaccine + Alhydrogel **Control:** high dose of H7 VLP vaccine **Aluminium concentration:** not reported **Aluminium type:** Alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 intramuscular injections

Outcomes	**Immunogenicity**: "geometric mean titers of hemagglutination inhibition (HI antibody on days 0, 21 and 42 to measure geometric mean fold rise and seroconversion rates). Follow-up serology samples for geometric mean titers will be taken at day 228". **Safety**: "rate, severity and relationship to vaccination of solicited and unsolicited adverse events post-vaccinations. A 6-month follow-up period will be performed".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Medicago) contacted. No answer **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT02304185

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "participant must be healthy on the basis of physical examination, medical history, electrocardiogram, laboratory criteria, and vital signs measurement performed at Screening. Participants are negative for human immunodeficiency virus (HIV) infection at Screening (negative United States Food and Drug Administration-approved HIV diagnostic blood test). All female participants of childbearing potential must have a negative serum (beta human chorionic gonadotropin) at Screening, and a negative urine pregnancy test pre-dose on Day 1 and Day 29. A woman must agree not to donate eggs (ova, oocytes) for the purposes of assisted reproduction until 3 months after receiving the last dose of study vaccine. A man must agree not to donate sperm until 3 months after receiving the last dose of study vaccine. Participants are assessed by the clinic staff as being at low risk for HIV infection". **Exclusion criteria:** "participant has chronic active hepatitis B or active hepatitis C, active syphilis infection, chlamydia, gonorrhea, or trichomonas. Active syphilis documented by exam or serology unless positive serology is due to past treated infection. In the 12 months prior to enrollment, participant has a history of newly acquired syphilis, gonorrhea, non-gonococcal urethritis, herpes simplex virus type 2 (HSV2), chlamydia, pelvic inflammatory disease, trichomonas, mucopurulent cervicitis, epididymitis, proctitis, lymphogranulomavenereum, chancroid, or hepatitis B. Participant has any clinically significant acute or chronic medical condition that in the opinion of the Investigator would preclude participation (for example, history of seizure disorders, bleeding/clotting disorder, autoimmune disease, active malignancy, poorly controlled asthma, active tuberculosis or other systemic infections). Participant has had major surgery within the 4 weeks prior to study entry or planned major surgery through the course of the study. Participant has had a thyroidectomy, or thyroid disease requiring medication during the last 12 months".
Interventions	**Vaccine type:** trimeric glycoprotein140 (gp140) **Experimental:** gp140, 50 mcg with aluminium phosphate **Control:**gp140, 50 mcg **Aluminium concentration:** 425 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 intramuscular injections on day 1 and day 29
Outcomes	**Immunogenicity:** "envelop binding antibody titer at week 8" **Safety**: "number of participants with local and systemic reactogenicity events over 8 days after first and second vaccination. Number of participants with adverse events or serious adverse events (time frame: week 52)"
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** 17/01/2020 authors (RA-RNDUS-JANSSENRND@its.jnj.com) were inquired on the safety and immunogenicity results. Answer received. **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT02777411

Methods	Randomised, open-label, dose-finding, phase II study
Participants	**Sex:** male and female **Age:** 6 months to 6 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy children aged from 3 to 6 years old (i.e. ≥ 3 years old and < 7 years old) for Part A; from 6 to 35 months old (i.e. ≥ 6 months old

	and < 36 months old) for Part B at the time of first vaccination. Subject's guardians were able and willing to comply with study procedures and give written informed consent. Subject was able and could comply with the requirements of the protocol. Subject with body temperature ≤38°C" **Exclusion criteria:** "subject with previous known exposure to Enterovirus 71 (EV71). Subject with a history of herpangina, hand-foot-mouth disease, acute hemorrhagic conjunctivitis associated with enterovirus infection in the past 3 months. Subject with gestation < 37 weeks. Subject with birth weight <2.5 kg. Subject with a history of hypersensitivity to vaccines, or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Family history of seizures or progressive neurological disease. Family history of congenital or hereditary immunodeficiency. Severe malnutrition or dysgenopathy. Major congenital defects or serious chronic illness, including perinatal brain damage. Subject diagnosed of having autoimmune disease (e.g., celiac disease, type I diabetes, lupus (SLE), juvenile dermatomyositis, scleroderma, juvenile idiopathic arthritis (JIA), immune (or idiopathic) thrombocytopenia purpura). Bleeding disorder diagnosed by a doctor or significant bruising or bleeding difficulties with IM injections or blood draws. Any acute infections 7 days prior to administrate the first vaccination. Use of any investigational product (including drug, vaccine) within 30 days prior to vaccination or planned use during the study period.Administration of any attenuated live vaccine within 7 days prior to vaccination. Use of immunoglobulins or any blood products within 3 months prior to vaccination or planned use during the study period. Chronic administration (defined as > 14 days) of immunosuppressants or other immunomodulators or systemic corticosteroids within 6 months prior to vaccination. Under anti-tuberculosis prevention or therapy. Any condition that in the opinion of the investigator may interfere with the evaluation of study objectives".
Interventions	**Vaccine type:** enterovirus EV71 **Experimental:** 1 or 2 mcg of EV71 with aluminium phosphate **Control:** 1 or 2 mcg of EV71 **Aluminium concentration:** 150 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two vaccinations at 28 days apart
Outcomes	**Immunogenicity**: "immunoglobulin G titers induced by the EV71 vaccine (determined by ELISA), time frame: day 28, day 56, day 196; serum neutralizing antibody titers induced by the EV71 vaccine; seroconversion rate based on neutralizing antibody titers; change in the laboratory results based on hematology tests in each visit; change in the laboratory results based on biochemistry tests in each visit (time frame: day 28, day 56, day 196)" **Safety**: "solicited adverse events, time frame: 7 days after each vaccination. Unsolicited adverse events, time frame: 28 days after each vaccination. The occurrence of overall adverse events and serious adverse event, time frame: day 0 to day 196. Change in heart rates at each visit; change in blood pressures at each visit (for the 3 to 6 years old group only)(time frame: day 28, day 56, day 196)"
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** company (Enimmune Corporation) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	Open label
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Open label
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Open label
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03010228

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-39 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy adults ≥18 to <40 years of age (for US healthy adults ≥ 19 years to <40 years) at the time of screening. Health status is assessed by investigator at time of screening based on medical history, physical

examination, and laboratory parameters.
Written informed consent obtained from the subject prior to any study related procedures.
BMI ≥18.5 and <30 at Visit 0 (Screening Visit). Men or women; women require a negative pregnancy test at screening. Women with childbearing potential must agree to use an adequate contraception during the entire study".
Exclusion criteria: "pathological findings in any of the investigations (i.e. medical history, physical examination) as deemed clinically relevant by the investigator or any abnormal laboratory parameter of hematology, clinical chemistry, coagulation, RF (Rheumatoid Factor) or ACPA (Anti-citrullinated protein antibodies) at the Screening Visit. Medical history of severe cardiovascular, respiratory (including asthma), metabolic, neurological, hepatic, rheumatic, hematological, gastrointestinal, renal disorders. Medical history of or current musculoskeletal disorders as deemed clinically relevant by the investigator, arthritis or chronic musculoskeletal pain. Previous vaccination against Lyme borreliosis with any (investigational) vaccine. Use of any other investigational or non-registered medicinal product within 30 days prior to VLA15 vaccination at Visit 1 (Day 0) and throughout the entire study period. Chronic illness related to Lyme borreliosis (LB), a history of or active symptomatic LB as suspected or diagnosed by a physician. Subjects with a positive serology test result for Borrelia burgdorferi sensu lato (s.l.) antibodies at screening are excluded. Tick bite within 3 weeks prior to screening, or tick bite during vaccination period (i.e. Day 0 to Day 56). Known active infection with Babesia microti (babesiosis) or Anaplasma phagocytophilum (ehrlichiosis). Active or passive immunization four weeks before first vaccination at Visit 1 and up to Day 84 (i.e. 4 weeks after the last VLA15 immunization). Afterwards, vaccinations should be avoided, except for influenza (seasonal or pandemic) vaccines which may be administered after Day 84 (i.e. 4 weeks after the last VLA15 immunization). Subjects susceptible to require a vaccine during the study period (e.g. due to planned travel) should be excluded at screening. Known congenital, hereditary or acquired immunodeficiency, including infection with human immunodeficiency virus (HIV), status post organ transplantation or immuno-suppressive therapy within 30 days prior to Day 0 and up to Day 84. Immuno-suppressive therapy is defined as administration of chronic (longer than 14 days) prednisone or equivalent ≥ 0.05 mg/kg/day. Topical and inhaled steroids are allowed.
Planned intake of NSAID (Nonsteroidal anti-inflammatory drug) within three days prior and within seven days after any VLA15 vaccination. History of severe hypersensitivity reactions and anaphylaxis. History of allergic bronchial asthma and severe allergic rhinoconjunctivitis. Known hypersensitivity or allergic reactions to one of the components of the vaccine. History of autoimmune disease, including Type I Diabetes mellitus. Subjects with vitiligo or thyroid disease taking thyroid hormone replacement are not excluded. Any malignancy in the past 5 years. If treatment for cancer was successfully completed more than 5 years ago and the malignancy is considered to be cured, the subject may be enrolled. Acute febrile infections within 4 weeks prior to first vaccination and body temperature >37.8 C (oral) prior to each vaccination. Known or suspected alcohol abuse, alcohol dependence, i.e. an average of more than approximately 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day] or illicit drug use within the last year. Inability or unwillingness to avoid more than the usual intake of alcohol (i.e. not more than 60 g alcohol (approximately 1 liter of beer or 0.5 liter of wine) per day) during the 48 hours after vaccination. Pregnancy (positive pregnancy test), lactation or inadequate contraception in

	women with childbearing potential. Inability or unwillingness to provide informed consent or not able to understand and comply with protocol requirements, instructions and protocol-stated restrictions. Donation of blood or blood-derived products (e.g. plasma) within 4 weeks prior to Visit 0 (Screening Visit) and during the entire study. Receipt of blood or blood-derived products in the past 3 months prior to Visit 0 (Screening Visit) or anticipation of such products during the entire study. Mental disorder as deemed clinically relevant by the investigator. History of Guillain-Barré-Syndrome (GBS). Any condition which might interfere with study objectives or that would limit the subject's ability to complete the study in the opinion of the investigator. Persons who are committed to an institution (by virtue of an order issued either by the judicial or the administrative authorities). Persons who are in a dependent relationship with the sponsor, an investigator or other study team members, or the study center. Dependent relationships include close relatives and household members (i.e. children, partner/spouse, siblings, parents) as well as employees of the investigator or study center personnel".
Interventions	**Vaccine type:** VLA15 vaccine against Lyme borreliosis **Experimental:** VLA15 with aluminium **Control:** VLA15 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 3 intramuscular vaccinations into the deltoid region on days 0, 28 and 56
Outcomes	**Immunogencity:** "GMTs (Geometric Mean Titre) for IgG against each OspA serotype ST1 to ST6, determined by ELISA, (time frame: day 0, 28, 56, 84, 180, 236 and 365). SCRs (Seroconversion Rate, defined based on fold increase of each OspA serotype specific IgG (ST1 to ST6) as compared to baseline), (time frame: day 28, 56, 84, 180, 236 and 365). GMFR (Geometric Mean of the fold rise as compared to baseline) for IgG against each OspA serotype ST1 to ST6, determined by ELISA,(time frame: day 28, 56, 84, 180, 236 and 365)" **Safety**: "rate of SAEs, rate of any solicited or unsolicited Grade 3 or Grade 4 events to day 84. Rate of solicited, unsolicited local and systemic AEs within 7 days after each and after any vaccination up to day 84. Rate of SAEs during the entire study period, up to day 365. Rate of any solicited or unsolicited Grade 3 or Grade 4 AEs during the entire study period".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** company (Valneva Austria GmbH) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described

Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	n.a.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03026348

Methods	Randomized, observer-blind, trial
Participants	**Sex:** male and female **Age:** 60-80 years **Ethnicity:** not reported **Diagnosis:** clinically stable for underling conditions **Inclusion criteria:** "males and females 60 through 80 years of age who are ambulatory and live in the community or in an assisted-living facility that provides minimal assistance, such that the subject is primarily responsible for self-care and activities of daily living. Subjects may have one or more chronic medical diagnoses, but should be clinically stable as assessed by: absence of changes in medical therapy within one month due to treatment failure or toxicity (dose adjustments of ongoing therapies for optimal effect, or replacements within a class of drugs due to convenience or cost, will be deemed acceptable),Absence of medical events qualifying as SAEs within one month of the planned vaccination on Day 0, andAbsence of known, current, and life-limiting diagnoses which, in the opinion of the investigator, render survival to completion of the protocol unlikely. Willing and able (on both a physical and cognitive basis) to give informed consent prior to study enrollment. To complete the consent process, all qualifying subjects will correctly answer at least 4 out of 5 questions of the informed consent form (ICF) comprehension assessment in no more than 2 attempts.Able to comply with study requirements. As the

	protocol procedures involve telephone contacts for safety ascertainment, eligible subjects must have a reliable access to a telephone". **Exclusion criteria:** "received any prior RSV vaccine. Participation in research involving any additional investigational product (drug / biologic / device) within 45 days before planned date of first vaccination. History of a serious reaction to any prior vaccination or a history of Guillain-Barré syndrome (GBS) within 6 weeks of any prior influenza immunization. Receipt of inactivated influenza vaccine within 14 days prior to the Day 0 dose of test article or any other vaccine within the 4 weeks prior to the Day 0 dose of test article. Any known or suspected immunosuppressive condition, acquired or congenital, as determined by history and/or physical examination. Chronic administration (defined as more than 14 continuous days) of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. An immunosuppressant dose of glucocorticoid will be defined as a systemic dose ≥ 10 mg of prednisone per day or equivalent. The use of topical, inhaled, and nasal glucocorticoids will be permitted. Administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or during the study. Acute disease at the time of enrollment (defined as the presence of a moderate or severe illness with or without fever, or an oral temperature ≥ 38.0°C on the planned day of vaccine administration). Known disturbance of coagulation. Potential subjects receiving aspirin, clopidogrel, prasugrel, dipyridamole, dabigatran, apixaban, rivaroxaban, or warfarin under good control for cardiovascular prophylaxis or prophylaxis of thromboembolic disease or stroke in the setting of atrial fibrillation will NOT be excluded. Suspicion or recent history (within one year of planned vaccination) of alcohol or other substance abuse. Any condition that in the opinion of the investigator would pose a health risk to the subject if enrolled or could interfere with evaluation of the vaccine or interpretation of study results (including neurologic, cognitive, or psychiatric conditions deemed likely to impair the quality of study compliance or safety reporting)".
Interventions	**Vaccine type:** RSV F vaccine **Experimental:** RSV F vaccine with aluminium **Control:** RSV F vaccine **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two injections at day 1 and 21
Outcomes	**Immunogenicity**: "neutralizing antibody titers to at least one RSV/A strain, (time frame: day 0, 21, 28). Serum concentrations of antibodies competitive with palivizumab for binding to the RSV F protein, (time Frame: day 0, 21, 28, 56, 119, 385). Serum IgG antibody concentrations as ELISA units specific for the F protein antigen, (time frame: day 0, 21, 28, 56, 119, 385). Counts of IFN-γ spot forming units following in vitro stimulation of day 0, day 7, and day 28 PBMC isolates with RSV F peptides, (time frame: day 0, 7, 28). Counts and proportions of Day 0, Day 7, and Day 28 peripheral blood T cells positive by intracellular staining for IL-2, IFN-γ, or TNF-α production (alone or any combination thereof) following in vitro stimulation with RSV F peptides, (time frame: day 0, 7, 28)" **Safety**: "subjects with solicited local and systemic AEs occurring within the 7-day period following dosings on Day 0 and Day 21 and all adverse events, solicited and unsolicited, occurring within the 56-day period of Day 0, (time frame: Day 0 - Day 6, Day 21 - Day 27; Day 0 - Day 56)"

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Novavax) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03284710

Methods	Phase 1/2a Partially Double-blinded, Randomized Clinical Trial
Participants	**Sex:** male and female **Age:** 18-40 years **Ethnicity:** black **Diagnosis:** healthy

Inclusion criteria: "Age of 18 to 40 years. Access to a participating HVTN CRS and willingness to be followed for the planned duration of the study
Ability and willingness to provide informed consent. Assessment of understanding: volunteer demonstrates understanding of this study; completes a questionnaire prior to first vaccination with verbal demonstration of understanding of all questionnaire items answered incorrectly.Agrees not to enroll in another study of an investigational research agent. Good general health as shown by medical history, physical exam, and screening laboratory. testsHIV-Related Criteria: Willingness to receive HIV test results. Willingness to discuss HIV infection risks and amenable to HIV risk reduction counseling. Assessed by the clinic staff as being at "low risk" for HIV infection and committed to maintaining behavior consistent with low risk of HIV exposure through the last required protocol clinic visit.Laboratory Inclusion ValuesHemogram/Complete blood count (CBC). Hemoglobin ≥ 11.0 g/dL for volunteers who were born female, ≥ 13.0 g/dL for volunteers who were born male. White blood cell count = 3,300 to 12,000 cells/mm^3. Total lymphocyte count ≥ 800 cells/mm^3
Remaining differential either within institutional normal range or with site physician approval. Platelets = 125,000 to 550,000/mm^3. Chemistry panel: ALT, AST, and ALP < 1.25 times the institutional upper limit of normal; creatinine ≤ institutional upper limit of normal.Virology. Negative HIV-1 and -2 blood test: Sites may use locally available assays that have been approved by HVTN Laboratory Operations. Negative Hepatitis B surface antigen (HBsAg). Negative anti-Hepatitis C virus antibodies (anti-HCV), or negative HCV polymerase chain reaction (PCR) if the anti-HCV is positive Urine. Normal urine: negative urine glucose, and negative or trace urine protein, andNegative or trace urine hemoglobin (if trace hemoglobin is present on dipstick, a microscopic urinalysis with red blood cells levels within institutional normal range). Volunteers who were born female: negative serum or urine beta human chorionic gonadotropin (β-HCG) pregnancy test performed prior to vaccination on the day of initial vaccination. Persons who are NOT of reproductive potential due to having undergone total hysterectomy or bilateral oophorectomy (verified by medical records), are not required to undergo pregnancy testing. A volunteer who was born female must: Agree to consistently use effective contraception for sexual activity that could lead to pregnancy from at least 21 days prior to enrollment through the last required protocol clinic visit. Effective contraception is defined as using condoms (male or female), or diaphragm or cervical cap, PLUS 1 of the following methods: Intrauterine device (IUD), Hormonal contraception (in accordance with Republic of South Africa: National Contraception Clinical Guidelines), successful vasectomy in the male partner (considered successful if a volunteer reports that a male partner has [1] documentation of azoospermia by microscopy, or [2] a vasectomy more than 2 years ago with no resultant pregnancy despite sexual activity postvasectomy); or any other contraceptive method approved by the HVTN 107 Protocol Safety Review TeamOr not be of reproductive potential, such as having reached menopause (no menses for 1 year) or having undergone hysterectomy, bilateral oophorectomy, or tubal ligation;Or be sexually abstinent. Volunteers who were born female must also agree not to seek pregnancy through alternative methods, such as artificial insemination or in vitro fertilization until after the last required protocol clinic visit. Volunteers who were born female consenting to provide cervical samples: pap smear within the 3 years prior to enrollment, with the latest result reported as normal or ASCUS (atypical squamous cells of undetermined significance); for those 21 years and older that have not had a

pap smear within the last 3 years prior to enrollment, must be willing to undergo a pap smear with the result reported as normal or ASCUS prior to sample collection."

Exclusion criteria: "Blood products received within 120 days before first vaccination. Investigational research agents received within 30 days before first vaccination. Body mass index (BMI) ≥ 40; or BMI ≥ 35 with 2 or more of the following: systolic blood pressure > 140 mm Hg, diastolic blood pressure > 90 mm Hg, current smoker, known hyperlipidemia. Intent to participate in another study of an investigational research agent or any other study that requires non-HVTN HIV antibody testing during the planned duration of the HVTN 107 study. Pregnant or breastfeedingVaccines and other Injections
HIV vaccine(s) received in a prior HIV vaccine trial. For volunteers who have received control/placebo in an HIV vaccine trial, the HVTN 107 PSRT will determine eligibility on a case-by-case basis. Non-HIV experimental vaccine(s) received within the last 5 years in a prior vaccine trial. Exceptions may be made for vaccines that have subsequently undergone licensure in a volunteer's country of residence. For volunteers who have received control/placebo in an experimental vaccine trial, the HVTN 107 PSRT will determine eligibility on a case-by-case basis. For volunteers who have received an experimental vaccine(s) greater than 5 years ago, eligibility for enrollment will be determined by the HVTN 107 PSRT on a case-by-case basis.Live attenuated vaccines other than influenza vaccine received within 30 days before first vaccination or scheduled within 14 days after injection (eg, measles, mumps, and rubella [MMR]; oral polio vaccine [OPV]; varicella; yellow fever). Influenza vaccine or any vaccines that are not live attenuated vaccines and were received within 14 days prior to first vaccination (eg, tetanus, pneumococcal, Hepatitis A or B). Allergy treatment with antigen injections within 30 days before first vaccination or that are scheduled within 14 days after first vaccinationImmune System. Immunosuppressive medications received within 168 days before first vaccination. (Not exclusionary: [1] corticosteroid nasal spray; [2] inhaled corticosteroids; [3] topical corticosteroids for mild, uncomplicated dermatitis; or [4] a single course of oral/parenteral corticosteroids at doses < 2 mg/kg/day and length of therapy < 11 days with completion at least 30 days prior to enrollment.) Serious adverse reactions to vaccines or to vaccine components such as eggs, egg products, or neomycin, including history of anaphylaxis and related symptoms such as hives, respiratory difficulty, angioedema, and/or abdominal pain. (Not excluded from participation: a volunteer who had a nonanaphylactic adverse reaction to pertussis vaccine as a child.). Immunoglobulin received within 60 days before first vaccination. Autoimmune disease. ImmunodeficiencyClinically significant medical conditions. Untreated or incompletely treated syphilis infection. Clinically significant medical condition, physical examination findings, clinically significant abnormal laboratory results, or past medical history with clinically significant implications for current health. A clinically significant condition or process includes but is not limited to: Any medical, psychiatric, occupational, or other condition that, in the judgment of the investigator, would interfere with, or serve as a contraindication to, protocol adherence, assessment of safety or reactogenicity, or a volunteer's ability to give informed consent. Psychiatric condition that precludes compliance with the protocol. Specifically excluded are persons with psychoses within the past 3 years, ongoing risk for suicide, or history of suicide attempt or gesture within the past 3 years. Current anti-tuberculosis (TB) prophylaxis or therapy
Asthma other than mild, well-controlled asthma. (Symptoms of asthma severity as defined in the most recent National Asthma Education and Prevention

		Program (NAEPP) Expert Panel report).Exclude a volunteer who uses a short-acting rescue inhaler (typically a beta 2 agonist) daily, orUses moderate/high dose inhaled corticosteroids, orIn the past year has either of the following: greater than 1 exacerbation of symptoms treated with oral/parenteral corticosteroids;Needed emergency care, urgent care, hospitalization, or intubation for asthma. Diabetes mellitus type 1 or type 2, including cases controlled with diet alone. (Not excluded: history of isolated gestational diabetes.). Thyroidectomy, or thyroid disease requiring medication during the last 12 months Hypertension: if a person has been found to have elevated blood pressure or hypertension during screening or previously, exclude for blood pressure that is not well controlled. Well-controlled blood pressure is defined as consistently ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic, with or without medication, with only isolated, brief instances of higher readings, which must be ≤ 150 mm Hg systolic and ≤ 100 mm Hg diastolic. For these volunteers, blood pressure must be ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic at enrollment.If a person has NOT been found to have elevated blood pressure or hypertension during screening or previously, exclude for systolic blood pressure ≥ 150 mm Hg at enrollment or diastolic blood pressure ≥ 100 mm Hg at enrollment. Bleeding disorder diagnosed by a doctor (eg, factor deficiency, coagulopathy, or platelet disorder requiring special precautions) Malignancy (Not excluded from participation: Volunteer who has had malignancy excised surgically and who, in the investigator's estimation, has a reasonable assurance of sustained cure. or who is unlikely to experience recurrence of malignancy during the period of the study) Seizure disorder: History of seizure(s) within past three years. Also exclude if volunteer has used medications in order to prevent or treat seizure(s) at any time within the past 3 years. Asplenia: any condition resulting in the absence of a functional spleen. Current anti-tuberculosis (TB) prophylaxis or therapy Asthma other than mild, well-controlled asthma. (Symptoms of asthma severity as defined in the most recent National Asthma Education and Prevention Program (NAEPP) Expert Panel report).Exclude a volunteer who: uses a short-acting rescue inhaler (typically a beta 2 agonist) daily, orUses moderate/high dose inhaled corticosteroids, orIn the past year has either of the following: Greater than 1 exacerbation of symptoms treated with oral/parenteral corticosteroids;Needed emergency care, urgent care, hospitalization, or intubation for asthma. Diabetes mellitus type 1 or type 2, including cases controlled with diet alone. (Not excluded: history of isolated gestational diabetes.). Thyroidectomy, or thyroid disease requiring medication during the last 12 months Hypertension: If a person has been found to have elevated blood pressure or hypertension during screening or previously, exclude for blood pressure that is not well controlled. Well-controlled blood pressure is defined as consistently ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic, with or without medication, with only isolated, brief instances of higher readings, which must be ≤ 150 mm Hg systolic and ≤ 100 mm Hg diastolic. For these volunteers, blood pressure must be ≤ 140 mm Hg systolic and ≤ 90 mm Hg diastolic at enrollment.If a person has NOT been found to have elevated blood pressure or hypertension during screening or previously, exclude for systolic blood pressure ≥ 150 mm Hg at enrollment or diastolic blood pressure ≥ 100 mm Hg at enrollment. Bleeding disorder diagnosed by a doctor (eg, factor deficiency, coagulopathy, or platelet disorder requiring special precautions). Malignancy (Not excluded from participation: Volunteer who has had malignancy excised surgically and who, in

	the investigator's estimation, has a reasonable assurance of sustained cure. or who is unlikely to experience recurrence of malignancy during the period of the study). Seizure disorder: History of seizure(s) within past three years. Also exclude if volunteer has used medications in order to prevent or treat seizure(s) at any time within the past 3 years. Asplenia: any condition resulting in the absence of a functional spleen. History of hereditary angioedema, acquired angioedema, or idiopathic angioedema.
Interventions	**Vaccine type:** Clade C ALVAC and gp120 HIV Vaccine **Experimental:** ALVAC-HIV + gp120/Al(OH)3 **Control:** ALVAC-HIV + gp120/ **Aluminium concentration:** 625mcg/dose **Aluminium type:** aluminum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** participants will receive the ALVAC-HIV vaccine in the left deltoid at months 0, 1, 3, 6, and 12, and Bivalent Subtype C gp120 in the right deltoid at months 3, 6, and 12.
Outcomes	**Immunogenicity**: "occurrence of Vaccine-induced Systemic IgG Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 2 [Time Frame: Measured at Month 6.5]Serum IgG responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:50 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Samples from post-enrollment visits have positive responses if they meet three criteria: (1) net MFI greater than or equal to an antigen-specific response threshold (defined as the maximum of 100 and the 95th percentile of baseline net MFI), (2) net MFI values are greater than 3 times baseline net MFI, and (3) experimental antigen MFI values are greater than 3 times baseline MFI. Level of Vaccine-induced Systemic IgG Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 2 [Time Frame: Measured at Month 6.5]Serum IgG responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:50 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Comparisons were performed among positive responders only (positivity criteria are described in Outcome 1). Occurrence of Vaccine-induced Serum IgA Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 3 [Time Frame: Measured at Month 6.5]Serum IgA responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:10 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus

reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Samples from post-enrollment visits have positive responses if they meet three criteria: (1) net MFI greater than or equal to an antigen-specific response threshold (defined as the maximum of 100 and the 95th percentile of baseline net MFI), (2) net MFI values are greater than 3 times baseline net MFI, and (3) experimental antigen MFI values are greater than 3 times baseline MFI.
Level of Vaccine-induced Serum IgA Ab Binding to the 3 gp120 Env Proteins Contained in the Vaccine Regimen (ZM96, TV1.C, and 1086.C) in Group 1 and Group 3 [Time Frame: Measured at Month 6.5]Serum IgA responses were measured on a Bio-Plex instrument using a standardized custom Luminex assay, run at 1:10 dilution. The readout is background-subtracted mean fluorescent intensity (MFI), with background adjustment for an antigen-specific plate level control. For each sample, response magnitude is net MFI, defined as experimental antigen MFI minus reference antigen MFI. Net MFI < 1 is set to 1, and net MFI > 22,000 is set to 22,000. Data are excluded if blood draw date was outside the allowable window, a participant was HIV-infected, reference antigen > 5,000 MFI, or baseline net MFI > 6,500. Comparisons were performed among positive responders only.
Safety: number of Participants Reporting Local Reactogenicity Signs and Symptoms: Pain and/or Tenderness [Time Frame: Measured through Month Measured through 3 days after each vaccine dose at Months 0, 1, 3, 6, and 12]Graded according to the Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events, Version 2.1 [July 2017]. The maximum grade observed for each symptom over the time frame is presented.
Number of Participants Reporting Local Reactogenicity Signs and Symptoms: Erythema and/or Induration [Time Frame: Measured through Month Measured through 3 days after each vaccine dose at Months 0, 1, 3, 6, and 12]Graded according to the Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events, Version 2.1 [July 2017]. The maximum grade observed for each symptom over the time frame is presented.
Number of Participants Reporting Systemic Reactogenicity Signs and Symptoms [Time Frame: Measured through Month Measured through 3 days after each vaccine dose at Months 0, 1, 3, 6, and 12]Graded according to the Division of AIDS (DAIDS) Table for Grading the Severity of Adult and Pediatric Adverse Events, Version 2.1 [July 2017]. The following symptoms are considered as systemic reactogenicity if the onset date was within the periods of assessment specified in the protocol: malaise and/or fatigue, myalgia, headache, nausea, vomiting, chills, arthralgia, and body temperature. The item Max. Systemic Symptoms is the maximum of the individual systemic reactogenicities excluding body temperature for a participant. Number of Participants Reporting Adverse Events (AEs), by Relationship to Study Product [Time Frame: Measured through 30 days after each vaccine dose at Months 0, 1, 3, 6, and 12]For participants reporting multiple AEs over the time frame, the maximum relationship is counted. Number of Participants Reporting Adverse Events (AEs), by Severity Grade [Time Frame: Measured through 30 days after each vaccine dose at Months 0, 1, 3, 6, and 12]For participants reporting multiple AEs over the time frame, the maximum severity grade is counted.
Number of Participants Reporting Serious Adverse Events (SAEs) [Time Frame: Measured through Month 18]Measured as outlined in Version 2.0 (January 2010) of the Manual for Expedited Reporting of Adverse Events to DAIDS (DAIDS EAE Manual). Number of Participants Reporting Adverse

	Events of Special Interest (AESIs) [Time Frame: Measured through Month 18]There were no adverse events of special interest reported by any participant. Number of Participants Reporting New Chronic Conditions (Requiring Medical Intervention for >= 30 Days) [Time Frame: Measured through Month 18]. Chemistry and Hematology Laboratory Measures - ALT (SGPT), AST, Alkaline Phosphatase [Time Frame: Measured during screening, Days 7, 42, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Creatinine [Time Frame: Measured during screening, Days 7, 42, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Hemoglobin [Time Frame: Measured during screening, Days 7, 42, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Lymphocytes, Neutrophils [Time Frame: Measured during screening, Days 0, 1, 3, 7, 42, 84, 85, 87, 91, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Chemistry and Hematology Laboratory Measures - Platelets, WBC [Time Frame: Measured during screening, Days 0, 1, 3, 7, 42, 84, 85, 87, 91, 98, 182, 378, and 455]For each local laboratory measure, summary statistics were presented by treatment group and timepoint for the overall population. Numbers of Participants With Grade 1 or Higher Local Laboratory Results [Time Frame: Measured during screening, Days 0, 1, 3, 7, 42, 84, 85, 87, 91, 98, 182, 378, and 455]The number (percentage) of participants with local laboratory values recorded as meeting Grade 1 AE criteria or above as specified in the DAIDS AE Grading Table were tabulated by treatment arm for each post-vaccination time point. Number of Participants With Early Study Termination Associated With an AE or Reactogenicity [Time Frame: Measured through Month 18]There were no early study terminations associated with an AE or reactogenicity reported by any participant. Number of Participants With Study Product Discontinuation Associated With an AE or Reactogenicity [Time Frame: Measured through Month 18]From the study product discontinuation form, study product discontinuation reasons are tabulated by treatment arm.".
Are there other groups in the trial not analysed in this review?	Yes.
Notes	**Contacts with authors:** on 23/08/2021 authors (paulg@uab.edu; kmngadi@auruminstitute.org) were inquired on the followings: ● random sequence generation ● allocation concealment ● blinding of participants, personnel and outcome assessors ● aluminium manufacturer ● proportion of participants per group who developed HIV ● geometric mean titres after vaccination per group No answer received **Funding:** funded by National Institute of Allergy and Infectious Diseases (NIAID)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized. No further information
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind. No further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. No further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. No further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind. No further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03284710), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

NCT03295318

Methods	Phase 2, observer-blind, randomised, controlled study
Participants	**Sex:** male and female **Age:** 12-16 months **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "male and female children between 12 months and 16 months old inclusive (minimum 365 days of age and maximum 16 months plus 29 days of age); for whom parent(s)/legal guardian(s) have given written informed consent after the nature of the study has been explained according to local regulatory requirements; who the investigator believes that their

	parent(s)/guardian(s) will be available for all the subject visits and would comply with the requirements of the protocol (e.g. timely reporting of adverse events, availability for study site visits and home visits); individuals in good health as determined by the outcome of medical history, physical examination and clinical judgment of the investigator. Individuals who completed their local infant EPI schedule through 9 months of age (except MenAfriVac dose). A birth dose of OPV is not required)". **Exclusion criteria:** "history of any meningococcal vaccine administration. Current or previous, confirmed or suspected disease caused by N. meningitidis. Household contact with and/or intimate exposure to an individual with any laboratory confirmed N. meningitidis infection within 60 days of enrolment. History of severe allergic reactions after previous vaccinations or hypersensitivity to any study vaccine component including tetanus, diphtheria and diphtheria toxoid (CRM197). Acute or chronic, clinically significant pulmonary, cardiovascular, metabolic, neurological, hepatic, or renal functional abnormality, as determined by medical history or physical examination. Any confirmed or suspected condition with impaired/altered function of immune system (immunodeficient or autoimmune conditions). Have any bleeding disorder which is considered as a contraindication to intramuscular injection or blood draw. Severe acute malnutrition. A family history of congenital or hereditary immunodeficiency. History of either hepatitis B or hepatitis C virus infection or human immunodeficiency virus infection. Major congenital defects. Chronic administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within three months prior to the study vaccination or planned use throughout the study period. (For corticosteroids, this means prednisone, or equivalent, ≥ 0.5 mg/kg per day. Inhaled, intranasal and topical steroids are allowed). Administration of blood, blood products and/or plasma derivatives or any parenteral immunoglobulin preparation in the past 3 months or planned use throughout the study period. Administration of any vaccine within 28 days prior to enrolment in the study or planned administration of any vaccine within 14 days before or after any study vaccination. Use of any investigational or non-registered drug or vaccine within 30 days prior to the administration of study vaccines or planned during the study. Malaria infection as confirmed by a Rapid Diagnostic Test. Individuals who are close family members of individuals conducting this study. Have experienced a moderate or severe acute infection and/or fever (defined as temperature ≥ 37.5°C) within 3 days prior to enrolment. Have received systemic antibiotic treatment within 3 days prior to enrolment. Non-residence in the study area or intent to move out within six months. Any condition which, in the opinion of the investigator, might pose additional risk to the subject due to participation in the study".
Interventions	**Vaccine type:** meningococcal ACYWX conjugate vaccine NmCV-5 **Experimental:** NmCV-5 with aluminium **Control:** NmCV-5 **Aluminium concentration:** not reported **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** two dose series separated by at least 84 days
Outcomes	**Immunogenicity**: "seroprotective rSBA titres, (time frame: 112 days). Percentage of subjects with rSBA titer ≥ 8 against serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Long term protective rSBA titres, (time frame: 112 days). Percentage of subjects with rSBA titer ≥ 128 against

	serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Rise in rSBA titres, (time frame: 112 days). Percentage of subjects with fourfold rise in rSBA titers against serogroups A, C, W, Y and X at Visits Day 28 and Day 112. Geometric mean of rSBA titres, (time frame: 112 Days). rSBA GMT for serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112" **Safety**: percentage of subjects with at least one severe solicited AE within 7 days after any study vaccination (Days 0-6 and Days 84-90). Solicited local and systemic AEs reported during the 7 days after each vaccination (Days 0-6 and Days 84-90); unsolicited AEs reported during 28 days after each vaccination (Days 0-27 and Days 84-111); AEs leading to premature withdrawal during the entire study period; SAEs reported during the entire study period
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** company (Serum Institute of India Pvt. Ltd) were inquired on the safety and immunogenicity results. No answer received **Funding:** not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further information
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further information
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.

Selective reporting (reporting bias)	Unclear risk	No results or publication
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Nicholson 2009

Methods	Randomized, phase I trial
Participants	**Sex:** male and female **Age:** >18 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "mentally competent adults, who have signed an informed consent form after having received a detailed explanation of the study protocol. Male or female subjects over 18 years who are either healthy or have a stable medical condition. Able to understand and comply with all study procedures and to complete study diaries. Individuals who can be contacted throughout the study and are available for all study visits. Females should either be using secure contraceptive precautions including a) the oral contraceptive pill, b) condom/barrier contraception c) partner has had a vasectomy, d) be surgically sterilised, or e) post- (defined as at least two years since the last menstrual period)" **Exclusion criteria:** "any clinically significant concurrent illness or unstable medical condition including: malignant tumours, autoimmune illnesses (including rheumatoid arthritis), acute or progressive renal or hepatic pathology, chronic obstructive pulmonary disease requiring oxygen therapy, and any active neurological disorder. Individuals with a history of anaphylaxis or serious reactions to vaccines; hypersensitivity to eggs, chicken protein, chicken feathers, influenzal viral protein, neomycin or polymixin, or products containing mercury. Persons with known immunosuppressive disease or who use systemic immunosuppressive drugs or other drugs listed in section 8 of the British National Formulary (BNF) or chloroquine, gold or penicillamine or other drugs listed in section 10.1.3 of the BNF to suppress a chronic disease process, or have received in the last 6 months radiotherapy or chemotherapy. Subjects who are at high risk of developing illnesses of the immune system. Individuals who are taking immunostimulant therapy or interferon. Individuals who have received blood products or immunoglobulins parenterally during the preceding three months. Women should not be pregnant or lactating. Women who refuse to use a reliable contraceptive method throughout the study. Known or suspected drug abuse. Individuals who have received another vaccine or experimental (investigational) drug in the preceding 4 weeks. Individuals who have previously received H9N2 vaccine. Unable to lead an independent life either physically or mentally. Regularly drink more than 40 units of alcohol weekly. Individuals who have had acute respiratory pathology or infections requiring systemic antibiotic or antiviral therapy during the preceding 7 days (chronic antibiotic therapy for prevention of urinary tract infections is acceptable). Individuals who had a temperature over 38 degrees C in the preceding 3 days. Individuals who, in the opinion of the investigator, have conditions that might complicate interpretation of the study results".
Interventions	**Vaccine type:** whole-virus influenza A/H9N2 vaccine **Experimental:** 1.7, 5, 15 or 45 mcg H9N2 with aluminium. n = 101 **Control:** 1.7, 5, 15 or 45 mcg H9N2. n = 101

	Aluminium concentration: 250 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses of vaccine, 21 days apart, either by intramuscular injection into the deltoid muscle of the non-dominant arm, or by intradermal injection into skin overlying the deltoid
Outcomes	**Immunogenicity:** "geometric mean antibody titres of antibody to influenza H9 by HI and neutralising antibody assays, (time frame: pre-vaccination, 21 days and 42 days). Seroprotective antibody titres to influenza H9 by HI and neutralising antibody, (time frame: pre-vaccination, 21 days and 42 days)" **Safety:** "local and systemic reactogenicity of influenza H9 vaccine, (time frame: within 7 days of vaccination)"
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (karlgnicholson@doctors.org.uk) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no further information or data added **Funding:** the funder, the UK Department of Health approved the study design, but took no part in data collection, data analysis, data interpretation, or writing of the report Only vaccine-related serious adverse events were reported.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	A vaccine randomisation list with a randomisation block size of 14 was used to maintain group balance.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Vaccine injections were given by a nurse who was not involved in further assessment. The contents of the syringe were shielded from the volunteers' view.

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00814229), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Nolan 2008

Methods	Randomised, dose comparison, parallel assignment, multicentre trial
Participants	**Sex:** male and female **Age:** 18-45 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy adults. Men or women aged 18–45 years, and for women of childbearing potential, a negative urine pregnancy test and willingness to use adequate methods of contraception". **Exclusion criteria:** "hypersensitivity to vaccine components (including eggs, chicken protein, neomycin, polymixin, aluminium or thiomersal); immunosuppression; administration of immunoglobulins and/or any blood products (within 90 days); participation in a clinical study or use of an investigational compound (within 90 days); recent vaccination with a registered vaccine (within 30 days); known history of Guillain–Barre´ Syndrome; active neurological disease or current treatment with an anticoagulant".
Interventions	**Vaccine type:** inactivated split-virus influenza A (H5N1) vaccine **Experimental:** 7.5 or 15 mcg H5N1 with aluminium + 50 mcg thiomerosal. n = 200 **Control:** 7.5 or 15 mcg H5N1 + 50 mcg thiomerosal. n = 200 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium phosphate

	Aluminium manufacturer: not reported **Vaccination schedule:** two single doses by intramuscular or deep subcutaneous injection into the deltoid muscle, 21 days apart
Outcomes	**Immunogenicity:** "haemagglutinin inhibition with horse erythrocytes and virus microneutralisation assays were conducted. Blood samples (10mL) were collected immediately before each vaccine dose, on Day 0, Day 21, Day 42, Month 6". **Safety:** "participants were observed for 30min post-vaccination and were given diary cards to record solicited local and systemic adverse events on the evening of vaccination and 6 days thereafter; 21-day diary cards were issued for any unsolicited adverse events. Local reaction measurement cards were used to measure redness, swelling and ecchymosis at the injection site and a digital thermometer was provided for measuring oral temperature. Serious adverse events were to be reported within 12 months of the last dose of study vaccine".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (terry.nolan@mcri.edu.au, t.nolan@unimelb.edu.au) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received, but no further information or data added **Funding:** CSL Limited, Australia

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation code was prepared before each trial commenced using a SAS database (Version 9.1.3, SAS Institute,Inc., Cary, NC).
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Participants and investigators were blinded to treatment allocation in both trials. In the phase I trial there was a visible difference between vaccines with and without adjuvant. Therefore, personnel who prepared and administered the

		study vaccine had no further involvement in the study.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT00136331), and all outcomes were reported on. 23 SAEs reported in the phase I trial but data not shown.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Pan 2013

Methods	Prospective, randomised, open label, observer-blind, single-center study
Participants	**Sex:** male and female **Age:** 20-60 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "healthy male and female individuals (age 20 and 60 years at the time of enrolment). The female participants were: (1) of non childbearing potential, i.e. surgically sterilized (defined as having undergone hysterectomy and/or bilateral oophorectomy and/or bilateral salpingectomy); (2) 1 year postmenopausal; or (3) had agreed to practice abstinence or use adequate contraceptive precautions for 30 days prior to the study and for 2 months after study completion. Female participants were also required to have a negative pregnancy test and to not be breastfeeding for the duration of the study". **Exclusion criteria:** "previous known or potential exposure to avian influenza virus or any adjuvated or unadjuvanted H5N1 HA antigen vaccine; vaccination with any influenza vaccine within 6 months prior to study enrolment or at any time during the study period; presence of confirmed or suspected abnormal immune function, immunosuppressive, or immunodeficient condition, including human immunodeficiency virus infection; history of hypersensitivity to vaccines

	or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine; chronic administration (defined as >14 days) of immunosuppressants or other".
Interventions	**Vaccine type:** influenza A/H5N1 vaccine **Experimental:** 3 or 6 mcg H5N1 with aluminium. n = 30 **Control:** 3 or 6 mcg H5N1. n = 30 **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular doses 21 days apart
Outcomes	**Immunogenicity:** serum samples were obtained prior to vaccination, as well as on days 21 and 42 after vaccination. Antibody titers were measured by hemagglutination-inhibition assays and virus neutralization assays. **Safety:** all study participants were monitored for self-reported as well as investigator-assessed local and general adverse events during the 6 weeks following the first vaccination. Complete blood count, serum biochemistry, adverse events, and serious adverse events were evaluated.
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (hsmaids@hotmail.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** supported by National Health Research Institution (NSC grant: 98A1-VCSP01-014) and National Taiwan University Hospital (DOH-101-TD-B-111-001)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open label

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer blind, no further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer blind, no further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Paoletti 2001

Methods	Phase I, randomised clinical trial
Participants	**Sex:** male and female **Age:** 18-50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** group B Streptococcal type III conjugate vaccine (GBS-III) **Experimental:** 12.5 mcg GBS-III with aluminium. n = 15 **Control:** 12.5 mcg GBS-III. n = 15 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Superfos Biosector a/s, Vedbæk, Denmark **Vaccination schedule:** one dose
Outcomes	**Immunogenicity:** "GBS type III CPS-specific IgG in sera from all subjects was measured by an enzyme-linked immunosorbent assay". **Safety:** "subjects in the first trial were instructed regarding the completion of a vaccine diary form, how to record oral temperature, and regarding the occurrence of systemic symptoms or injection site symptoms or signs daily for 8 days. For the second trial, the vaccine diary was completed for 3 days unless symptoms existed, which were monitored until resolution. Any volunteer with a temperature of 100.0°F or with grade 3 symptoms within 48 h of vaccination came to the clinic for examination by a study physician".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 13/12/2018 authors (lawrance_paoletti@hms.harvard.edu, lpaoletti@channing.harvard.edu) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** supported by NIH-NIAID contracts AI-25152 and AI-75326

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	For assessment of reactogenicity and safety in the alum trial, subjects and study personnel, except for the nurse administering the vaccine injections, were blinded to vaccine group assignment. The second dose study of GBS III-TT conjugate vaccine was of open-label design.

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	Dropouts <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	Dropouts <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Phanuphak 1989

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 19-52 years **Ethnicity:** Asian **Diagnosis:** healthy **Inclusion criteria:** "none of the subjects had a history of previous rabies vaccination. None were taking antimalarials, corticosteroids or immunosuppressive drugs" **Exclusion criteria:** not reported
Interventions	**Vaccine type:** combined rabies and tetanus vaccine (PVRV) **Experimental:** PVRV with aluminium. n = 15 **Control:** PVRV. n = 15 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses intramuscularly on both deltoid areas on day 0, and one dose on days 7 and 21
Outcomes	**Immunogenicity:** "neutralizing antirabies antibody was determined by rapid fluorescent focus inhibition test. Antitetanus antibody was determined by enzyme-linked immunosorbent assay according, cell-mediated immune response to rabies antigen was determined by antigen-stimulated lymphocyte transformation test". **Safety:** "during each follow-up visit, the subject was asked about any possible side-effects from the preceding vaccination session according to a set of checklist questions".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** contact not found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Pillet 2019

Methods	Randomized, phase II, clinical trial
Participants	**Sex:** male and female **Age:** > 50 years **Ethnicity:** white **Diagnosis:** healthy **Inclusion criteria:** "male and female subjects, 50 years of age or older. BMI of ≥18 and ≤32. Give his/her consent to participate in this study (by signing the informed consent form). In the opinion of the Investigator, competence and willingness to provide written, informed consent for participation after reading the informed consent form. The subject must have adequate opportunity to discuss the study with an Investigator or qualified designee. Healthy as judged by the Investigator or designee and determined by general physical examination, vital signs, clinical laboratory tests, and medical history conducted

	no more than 90 days prior to study vaccine administration. Subjects with a pre-existing chronic disease will be allowed to participate if the disease is stable and, according to the Investigator's judgment, the condition is unlikely confound the results of the study or pose additional risk to the subject by participating in the study. Stable disease is defined as no new onset of exacerbation of pre-existing chronic disease 6 months prior to immunization. Based on Investigator's judgment, a subject with more recent stabilisation of a disease may still be eligible. Comprehension of the study requirements, expressed availability for the required study period, ability to attend scheduled visits, accessible by phone on a consistent basis. If female, have a negative serum pregnancy test result at screening and negative urine pregnancy test on Day 0 prior to immunization. Female of childbearing potential must use an effective method of contraception for 1 month prior to immunization and agrees to continue employing adequate birth control measures for at least 60 days post-immunization. Moreover, she must have no plan to become pregnant for at least 2 months post-immunization. Abstinent subjects should be asked what method(s) they would use, should their circumstances change, and subjects without a well-defined plan should be excluded.The following relationship or methods of contraception are considered to be effective: hormonal contraceptives (e.g., injectable, topical [patch], estrogenic vaginal ring, etc.);Intra-uterine device (IUD) with or without hormonal release;Male partner using a condom plus spermicide or sterilized partner (at least 1 year prior to immunization);History of credible abstinence (self-reported);Heterosexual abstinence at least 60 days post-immunization;Female partner. Non-childbearing females defines as: surgically-sterile (defined as bilateral tubal ligation or hysterectomy performed more than 1 month prior to immunization);Post-menopausal (absence of menses for 24 consecutive months and age consistent with natural cessation of ovulation)". **Exclusion criteria:** "according to Investigator's opinion, presence of significant acute or chronic, uncontrolled medical or neuropsychiatric illness. "Uncontrolled" is defined as:Requiring a new medical or surgical treatment within one month prior to study vaccine administration which would interfere with vaccine evaluation or study completion (subject requiring medical or surgical treatment that would not interfere with evaluation [e.g., minor knee surgery, cataracts, etc.] would be eligible);Requiring a change in medication dosage in one month prior to study vaccine administration due to uncontrolled symptoms or drug toxicity (elective dosage adjustments in stable subjects are acceptable). Any medical or neuropsychiatric condition or any history of excessive alcohol use or drug abuse which, in the Investigator's opinion, would render the subject unable to provide informed consent or unable to provide valid safety observations and reporting. Any autoimmune disease or any confirmed or suspected immunosuppressive condition or immunodeficiency including history of human immunodeficiency virus (HIV) infection, Hepatitis B or C, or the presence of lymphoproliferative disease. Administration of any vaccine (including any other influenza vaccine) within 30 days prior to study enrolment or planned administration within the period from the vaccination up to blood sampling at Day 21 or within 30 days prior to blood sampling at Day 201. Immunization on an emergency basis will be evaluated in a case-by-case by the Investigator. Administration of any adjuvated or investigational influenza vaccine within 1 year prior to study enrolment or planned administration prior to the end of this trial (Day 201). Administration of any 'standard', not adjuvated influenza vaccine (e.g., live attenuated Trivalent Inactivated Influenza Vaccine (TIV) or Quadrivalent Inactivated Vaccine (QIV) vaccine IN or split TIV or QIV vaccine by

	either intra-dermal or intramuscular route) prior to the 30-day exclusion period mentioned above would be acceptable. Use of any investigational or non-registered product within 30 days prior to study enrolment or planned use during the study period. Subjects may not participate in any other investigational or marketed drug study while participating in this study (approximately 7-12 months [depending when subjects undergo screening session and Day 201 visit]). Treatment with systemic glucocorticoids at a dose exceeding 10 mg of prednisone per day, or equivalent for more than 7 consecutive days or for 10 or more days in total, within one month of study vaccine administration, any other cytotoxic or immunosuppressant drug, or any immunoglobulin preparation within 3 months of vaccination. Routine use of standard doses of nasal or inhaled glucocorticoids is allowed. Any significant disorder of coagulation or treatment with warfarin derivatives or heparin. Persons receiving prophylactic anti-platelet medications, e.g., low-dose aspirin (no more than 325 mg/day), and without a clinically apparent bleeding tendency are eligible. Subjects treated with new generation drugs that will not increase risk of intramuscular bleeding (such as clopidogrel) are also eligible. History of allergy to any of the constituents of the VLP quadrivalent study vaccine, Alhydrogel® (aluminium hydroxide) or to the phosphate buffered saline (PBS; used as placebo), or tobacco allergy. History of anaphylaxis reaction to any food, medication or bee sting. Any history of severe asthma (e.g., status asthmatic, hospitalization for asthma control) or recurrent asthma episodes requiring medical attention in the last 3 years (≥ 1 episode/year). Continuous use of anti-histamines in the last 4 weeks prior to immunization or use of anti-histamines 48 hours prior to study immunization. Have a rash, dermatological condition, tattoos, muscle mass, or any other abnormalities at injection site which may interfere with injection site reaction rating. Have received a blood transfusion within 90 days prior to study vaccination. If female, have a positive or doubtful pregnancy test prior to immunization or lactating females. Vital sign abnormalities (systolic blood pressure and/or diastolic blood pressure, heart rate). Although a vital signs measurement is out of the acceptable ranges, a subject may be included in the study based on Investigator's judgment. Presence of any febrile illness (including oral temperature (OT) ≥38.0˚C within 24 hours prior to immunization). Such subjects may be re-evaluated for enrolment after resolution of illness. Cancer or treatment for cancer within 3 years of study vaccine administration. Persons with a history of cancer who are disease-free without treatment for 3 years or more are eligible. Persons with treated and uncomplicated basal cell carcinoma of the skin are eligible. Person with non-treated, non-disseminated local prostate cancer are eligible. Identified as an Investigator or employee of the Investigator or study center with direct involvement in the proposed study, or identified as an immediate family member (i.e., parent, spouse, natural or adopted child) of the Investigator or employee with direct involvement in the proposed study".
Interventions	**Vaccine type:** quadrivalent plant-derived virus like particle influenza vaccine candidate **Experimental:** 15 mcg QVLP with aluminium. n = 75 **Control:** 15 mcg QVLP. n = 75 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag Canada Inc, Toronto **Vaccination schedule:** 1 intramuscular injection

Outcomes	**Immunogenicity**: "(time frame: 21 days after injection) geometric mean titers of hemagglutination inhibition antibodies against the vaccine strains on days 0 and 21 and assessed by measuring geometric mean fold rise, seroconversion rate and seroprotection rate. Follow-up serology samples for geometric mean titers will be collected at day 201. Immunogenicity will be evaluated by geometric mean titers of microneutralization antibodies (days 0 and 21) against the vaccine strains and assessed by measuring geometric mean fold rise ans seroconversion rate. Cross-reactivity of antibodies induced by a single dose of the quadrivalent VLP vaccine will also be assessed and measured by HI and microneutralization antibody titers for heterologous influenza strains. This will be assessed using the measure of geometric mean fold rise, seroconversion rate and seroprotection rate (for hemagglutination inhibition only)". **Safety**: "solicited systemic and local reactions (time frame: 21 days after injection). Safety and tolerability will be assessed by the rate, severity and relationship to vaccination of solicited and unsolicited adverse events post-vaccination. A 6-month follow-up period will be performed".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 29/01/2020 authors (landryn@medicago.com) were inquired on the followings: ● blinding of safety assessors ● total number of patients with any adverse events/number of patients analysed at 6 months follow up (please note that we are interested in AEs related and not related to the vaccine). ● total number of patients with any serious adverse events/number of patients analysed at 6 months follow up (please note that we are interested in SAEs related and not related to the vaccine). ● proportion of participants who developed influenza after the vaccination ● GMT values for 15 mcg VLP with and without aluminium No answer received **Funding:** this work was supported in part by awards to BJW and SP from Canadian Institutes of Health Research (http://www.cihr-irsc.gc.ca/e/193. html) and the Ministère de l'Economie, de la Science et de l'Innovation du Quebec (https://www. economie.gouv.qc.ca/accueil/), and by Medicago Inc. Additionally, Medicago Inc. provided support for this study in the form of salaries for NL, JC, and SP. Caprion Biosciences Inc. also provided support for this study in the form of salaries for JFP and BYD. BG has received honoraria from Medicago Inc. The specific roles of these authors are articulated in the 'author contributions' section. As well, Medicago Inc. designed the protocols of the two clinical trials reported in this paper, decided to publish the results, and was involved in the preparation of the manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation is performed using random block permutations obtained with the PLAN procedure of SAS.

Allocation concealment (selection bias)	Low risk	The site will be provided a pre-determined sequence of randomisation number according to the randomisation code.
Blinding of participants and personnel (performance bias)	High risk	Vaccine or placebo is administer by unblinded staff.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Observer-blind. Immunogenicity assessors were blinded. No further description for safety assessors.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Observer-blind. Immunogenicity assessors were blinded. No further description for safety assessors.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Observer-blind. Immunogenicity assessors were blinded. No further description for safety assessors.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02236052), and all outcomes were reported on. However 6-month follow-up period for safety not reported.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Pressler 1982

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 18–50 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** H3N2 influenza vaccine **Experimental:** Influenza vaccine with aluminium. n = 94 **Control:** Influenza vaccine. n = 95 **Aluminium concentration:** 2400 mcg/dose **Aluminium type:** aluminium oxide **Aluminium manufacturer:** not reported **Vaccination schedule:** one intramuscular injection

Outcomes	**Immunogenicity**: haemagglutiantion inhibition assay and neuraminidase inhibition test at day 0, 30, 90 and 180 **Safety**: only most severe reactions assessed
Are there other groups in the trial not analysed in this review?	No
Notes	**Funding**: not stated

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Rello 2017

Methods	Phase II, randomised, placebo-controlled, partially blinded, parallel-group, multicenter study
Participants	**Sex:** male and female **Age:** 18–80 years **Ethnicity:** mixed **Diagnosis:** intensive care unit patients on mechanical ventilation **Inclusion criteria:** "male or female patients aged between 18 and 80 years. Patients admitted to an ICU with a need for mechanical ventilation for more than 48 hours at visit 0. At high risk for acquiring infection against P. aeruginosa at visit 0. Patients who, as determined by the investigator, have a high probability of survival for at least 48 hours. In females, either childbearing potential terminated by surgery or 1 year post menopausal, or a negative pregnancy test and the willingness of practicing a reliable methods of contraception. Written informed consent or waiver according to the national regulations". **Exclusion criteria:** "known use of any other investigational or non-registered drug within 30 days prior to IC43 vaccination at Visit 0. Low severity of illness defined by an acute physiology score < 8 at visit 0. Patients < 6 months post organ transplantation. Severe thrombocytopenia or other coagulopathy which in the opinion of the investigator makes the patient unsuitable for intramuscular injection. Pregnancy, lactation. Persons who have been committed involuntarily to an institution, e.g. mental health facility or prison, will not participate in the study".
Interventions	**Vaccine type:** *Pseudomonas aeruginosa* vaccine IC43 **Experimental:** 100 mcg IC43 with aluminium. n = 104 **Control:** 100 mcg IC43. n = 98 **Aluminium concentration:** 1400 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** the authors do not want to disclose this information (email conversation with the author). **Vaccination schedule:** two doses in a 7-day interval
Outcomes	**Immunogenicity:** OprF/I specific IgG antibody titer measured by enzyme-linked immunosorbent assay, (time frame: day 7 and 14) **Safety:** rate of serious adverse events, safety laboratory parameters, systemic tolerability, local tolerability, (time frame: day 7)
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (jrello@crips.es) were inquired on the followings: ● final concentration of aluminium injected ● aluminium manufacturer ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received **Funding:** financed by Valneva Austria GmbH

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation lists were generated by a statistician at a contract research organization (Assign Data Management and Biostatistics GmbH, Innsbruck, Austria), using nQuery Advisor.
Allocation concealment (selection bias)	Low risk	Each subject was assigned the next free medication kit (randomisation) number for the corresponding study center in ascending order. Sealed envelopes containing details of treatment allocation (one per patient) were provided to study sites in case emergency un-blinding was required.
Blinding of participants and personnel (performance bias)	High risk	Partially blinded. IC43 without adjuvant (i.e. purified OprF/I fusion protein) was administered un-blinded, and IC43 with adjuvant and placebo were administered blinded because of different appearance and storage conditions.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	A blinded independent Clinical Endpoint Committee reviewed and confirmed diagnosis of P. aeruginosa infections. A blinded independent DSMB evaluated safety data and provided recommendations to the sponsor.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	A blinded independent Clinical Endpoint Committee reviewed and confirmed diagnosis of P. aeruginosa infections. A blinded independent DSMB evaluated safety data and provided recommendations to the sponsor.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Primary immunogenicity and safety analyses were based on the intention to treat population.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Primary immunogenicity and safety analyses were based on the intention to treat population.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Primary immunogenicity and safety analyses were based on the intention to treat population.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00876252), and all outcomes were reported on.
Other bias	Low risk	"After 137 patients had been enrolled, data became available from a phase-I trial, in which both 100 µg IC43 without adjuvant and 100 µg IC43 with adjuvant had a favorable safety profile and a similar immunogenicity profile. Based on the newly available data and following the recommendation of a Data Safety Monitoring Board (DSMB), a protocol amendment was introduced to add a fourth treatment group

		in this phase II study, to assess IC43 100 µg without aluminium hydroxide adjuvant. From this point in the study, randomisation was performed in a 2:1:1:1 ratio (double the number of patients randomised to 100 µg IC43 without adjuvant to achieve equal numbers per group)".

Riddle 2016

Methods	Randomised, single-blind study
Participants	**Sex:** male and female **Age:** 18–50 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female volunteers, age of 18 to 50 years (inclusive) at the time of enrollment. Signed informed consent form. Completion and review of comprehension test (achieved >70% accuracy). Available for the required follow-up period and scheduled clinic visits. Women: negative pregnancy test with understanding (through informed consent process) to not become pregnant or or breast feed during the study or within twelve (12) weeks after the last vaccine dose". **Exclusion criteria:** "health condition that, in the opinion of the investigator, may interfere with optimal participation in the study or place the volunteer at increased risk of adverse Events (AEs). Study clinicians, in consultation with the PI, will use clinical judgment on a case by-case basis to assess safety risks under this criterion. The PI will consult with the Research Monitor as appropriate. Clinically significant abnormalities on physical examination. Clinically significant abnormalities on basic laboratory screening. Presence of significant unexplained laboratory abnormalities that, in the opinion of the PI, may potentially confound the analysis of the study results. Regular use of constipation, antacid or anti-diarrheal medications or treatments. Abnormal stool pattern (fewer than 3 stools per week or more than 3 per day) or loose/liquid stools more than occasionally. Use of immunosuppressive drugs such as corticosteroids or chemotherapeutics that may influence antibody development. Women currently nursing. Participation in research involving another investigational product (defined as receipt of investigational product or exposure to invasive investigational device) within 30 days of planned date of first vaccination or anytime throughout the duration of the study. Positive blood test for HBsAg, hepatitis C Virus (HCV), HIV-1. Positive blood test for HLA-B27. Immunosuppressive illness or immunoglobulin deficiency (serum immunoglobulin A level < 7 mg/dL or limit of detection of assay). Family history of congenital or hereditary immunodeficiency. Treatment with immunoglobulins or blood products within 3 months from first candidate vaccine injection. History of microbiologically confirmed Shigella infection. Personal or family history of inflammatory arthritis. Personal or family history of irritable bowel syndrome. Received previous experimental Shigella vaccine or live Shigella challenge. Have had diarrhea while travelling outside the United States or lived for 2 or more months during the past 3 years in a country with potentially higher Shigella infection rates, including Africa, South America, Central America, and Asia (except Japan). Occupation involving handling of Shigella bacteria currently, or in the past 3 years. History of allergy to any vaccine. History of allergy to aluminium hydroxide. Serum immunoglobulin G endpoint titer ≥ 2500 to Shigella Lipopolysaccharide".

Interventions	**Vaccine type:** bioconjugate vaccine against Shigella flexneri 2a (Flexyn2a) **Experimental:** 10 mcg Flexyn2a with alhydrogel. n = 12 **Control:** 10 mcg Flexyn2a. n =12 **Aluminium concentration:** not reported **Aluminium type:** alhydrogel **Aluminium manufacturer:** not reported **Vaccination schedule:** two injections 4 weeks apart
Outcomes	**Immunogenicity:** "evaluation of antigen-specific antibodies between baseline (day 0) and after injection for all groups, (time frame: until day 56). Immunogenicity will be evaluated after the first and second injection and compared to pre-immune levels. Evaluation of antigen-specific antibodies between subjects receiving the candidate vaccine with and without adjuvant, (time frame: until day 56). Immunogenicity will be compared between subjects receiving candidate vaccine with and without adjuvant after the first and second injection". **Safety:** "occurrence and severity of adverse events, (time frame: until day 56). Number and severity of local site injection and general adverse events will be collected and compared between the different arms of the study".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (mark.s.riddle10.mil@mail.mil, veronica.gambillara@lmtbio.com) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** funded by Wellcome Trust (100527/B/12/Z)

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described

Blinding of participants and personnel (performance bias)	High risk	Single-blind. Only volunteers were blinded to their group assignment.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	Single-blind
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Single-blind
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Single-blind
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT02388009), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Rosenqvist 1998

Methods	Randomized, phase II trial
Participants	**Sex:** male and female **Age:** not reported **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy adult volunteers. Subjects were laboratory personnel or medical students from Oslo and Bergen". **Exclusion criteria:** not reported
Interventions	**Vaccine type:** group B neisseria meningitidis outer membrane vesicle vaccine (PC) **Experimental:** PC with aluminium **Control:** PC **Aluminium concentration:** 3300 mcg/ml **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Superfos Biosector, Vedbiek, Denmark **Vaccination schedule:** intramuscularly in the upper arm twice, six weeks apart

Outcomes	**Immunogenicity:** antibody response was evaluated by enzyme-linked immunosorbent assay. Bactericidal assay was performed **Safety:** local and systemic side effects were reported on a standard form for the first two days after vaccination
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** contact not found **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

RPCEC00000101

Methods	Randomized controlled, double Blind, parallel phase 2 clinical trial
Participants	**Sex:** male and female **Age:** 15-18 months **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "1) Voluntary participation (signature of informed consent by parents). 2) Healthy children of both sexes, aged 15-18 months of age, with history and normal clinical examination. 3) Nutritional evaluation over 10 percentile. 4) History of full primary immunization series against Hib (2-4-6 months)" **Exclusion criteria:** "1) Having received the booster dose against Haemophilus influenzae type b. 2) Submit acute infectious disease at the time of vaccination or within three days prior to the administration of it. 3) Higher temperature at 37oC at the time of the visit to administer the vaccine or from the previous 24 hours. 4) History of chronic diseases eg. Bronchial Asthma, Congenital or acquired Cardiopathies, neuropathies, haematological diseases, etc. 5) Thiomersal allergy or allergy to any components of the vaccine 6) History of convulsive or non-convulsive encephalopathy 7) Underlying immunosuppressive disease, intake of immunosuppressor drugs (including steroids) in the six month period prior to study entry"
Interventions	**Vaccine type:** haemophilus influenzae type b infection **Experimental:** alum phosphate adjuvated Quimi-Hib, 10 mcg **Control:** Quimi-Hib, 10 mcg **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** intramuscular route
Outcomes	**Immunogenicity**: "IgG antiPRP geometric mean titre, Short-term (=0,15 mcg/mL) and long-term (=1 mcg/mL) seroprotection percentages, seroconversion rate (increase in =2 folds in baseline IgG antiPRP antibody titer measured at day 30 after administration of the last vaccine dose)" **Safety**: percentage of local and systemic adverse events within 72 hours, and at 7 and 30 days after vaccination
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 94/04/2020 authors (biomed@cigb.edu.cu) were inquired on the followings: • type of aluminium used • concentration of aluminium injected per dose • aluminium manufacturer • all-cause mortality results • proportion of participants with any AE related and not related to the vaccine • proportion of participants with any SAE related and not related to the vaccine • serological response GMT for each study arm at maximum follow up • proportion of participants who developed influenza after the vaccination • random sequence generation • allocation concealment

	● blinding of participants and personnel ● blinding of outcome assessment No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind. No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind. No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-punished protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

RPCEC00000135

Methods	Multicenter, controlled, randomized, doubled blind clinical trial
Participants	**Sex:** male and female **Age:** 6-7 weeks **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "1.Signing of the written inform consent Act by parents/legal guardians. 2.6 weeks old healthy infants, both sexes, with normal clinical examination and medical history. 3.Nutritional evaluation over 10 percentile

	4.History of no primary immunization series against Hib or DTP. 5. Administration of the scheduled Hepatitis B vaccine dose at birth" **Exclusion criteria:** "1. Administration of other Investigation Medicinal Product (IMP) different to IMP under study in the 30 days previous to administration of the 1st vaccine dose or its intention of use during the period planned for the clinical trial. 2. Acute infectious illness or body temperature over 37?C at the moment of vaccination visit. 3. History of preterm delivery (birth before 36th weeks of pregnancy). 4. Major congenital defects or serious chronic disease. 5. Infants with history of convulsive or non-convulsive encephalopathy. 6. Chronic medication (defined as more than 21 days of treatment) with immune-suppressor or immuno-modulatory drugs prior to administration of the 1st vaccine dose. 7. History of invasive Hib disease. 8. Any suspected or diagnosed immune-suppression or immunodeficiency (congenital or secondary), including HIV infection, based on the medical history and physical examination. 9. History of allergic reactions or allergic disease with probability to be exacerbated by any component of the vaccine under study (including thiomersal allergy). 10. Gammaglobulin administration or other blood –derived components during the period prior to administration of the 1st vaccine dose or the intention of administration during the planned trial period. 11. Maternal history for HIV or HBV infection, based on the medical records"
Interventions	**Vaccine type:** haemophilus influenzae type b infection **Experimental:** alum phosphate adjuvated Quimi-Hib, 10 mcg **Control:** Quimi-Hib, 10 mcg **Aluminium concentration:** not reported **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** intramuscular route, 6-10-14 weeks schedule
Outcomes	**Immunogenicity**: IgG antiPRP geometric mean titre, Short-term (=0,15 mcg/mL) and long-term (=1 mcg/mL) seroprotection percentages, seroconversion rate (increase in =2 folds in baseline IgG antiPRP antibody titer measured at day 30 after administration of the last vaccine dose) **Safety**: percentage of local and systemic adverse events at 1h, 24h, 48h, 72h, Days 7 and 30 post-vaccination in each dose
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 04/02/2020 authors (biomed@cigb.edu.cu) were inquired on the followings: ● type of aluminium used ● concentration of aluminium injected per dose ● aluminium manufacturer ● all-cause mortality results ● proportion of participants with any AE related and not related to the vaccine ● proportion of participants with any SAE related and not related to the vaccine ● serological response GMT for each study arm at maximum follow up ● proportion of participants who developed influenza after the vaccination ● random sequence generation ● allocation concealment ● blinding of participants and personnel ● blinding of outcome assessment No answer received

	Funding: not reported No results

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind.No further details
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Double blind.No further details
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Double blind.No further details
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Double blind.No further details
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Ruckwardt 2021

Methods	Randomised, open-label, phase 1 clinical trial.
Participants	**Sex:** male and female **Age:** adults **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "18 to 50 years of age. Willing and able to complete the informed consent process. Available for clinic visits through 44 weeks after enrollment. Able to provide proof of identity to the satisfaction of the study clinician completing the enrollment process. Willing to donate blood and mucosal samples to be stored and used for future research. In good general

	health without clinically significant medical history. Physical examination and laboratory results without clinically significant findings and a Body Mass Index (BMI) less than or equal to 40 within the 56 days prior to enrollment. Laboratory criteria within 56 days prior to enrollment: White Blood Cell (WBC) and differential either within institutional normal range or accompanied by Principal Investigator (PI) or designee approval. Platelets = 125,000-500,000/mm^3. Hemoglobin within institutional normal range. Creatinine less than or equal to 1.1 x upper limit of normal (ULN). Alanine aminotransferase (ALT) less than or equal to 1.25 x ULN. Negative for HIV infection by an FDA approved method of detection.Criteria applicable to women of childbearing potential: Negative result on a human chorionic gonadotropin pregnancy test on day of enrollment before receiving study product. Agree to use effective means of birth control from at least 21 days before enrollment through 4 weeks after the last injection." **Exclusion criteria:** breast-feeding or planning to become pregnant through 4 weeks after the last injection.Subject has received any of the following: More than 10 days of systemic immunosuppressive medications or cytotoxic medications within the 4 weeks prior to enrollment or any within the 14 days prior to enrollment.Blood products within 16 weeks prior to enrollment. Live attenuated vaccines within 4 weeks prior to enrollment. Inactivated vaccines within 2 weeks prior to enrollment. Investigational research agents within 4 weeks prior to enrollment or planning to receive investigational products while on the study. Current allergen immunotherapy with antigen injections, unless on maintenance schedule. Current anti-tuberculosis(TB) prophylaxis or therapy.Subject has any of the following: Serious reactions to vaccines that preclude receipt of study injections as determined by the investigator. Hereditary angioedema, acquired angioedema, or idiopathic forms of angioedema. Asthma that is not well controlled. Diabetes mellitus (type I or II), with the exception of gestational diabetes. Thyroid disease that is not well controlled. Hypertension that is not well controlled. Evidence of autoimmune disease or immunodeficiency. Idiopathic urticaria within the past year. Bleeding disorder diagnosed by a doctor (e.g. factor deficiency, coagulopathy, or platelet disorder requiring special precautions) or significant bruising or bleeding difficulties with IM injections or blood draws. Malignancy that is active or history of malignancy that is likely to recur during the study. Seizure disorder other than: 1) febrile seizures, 2) seizures secondary to alcohol withdrawal more than 3 years ago, or 3) seizures that have not required treatment within the last 3 years. Asplenia, functional asplenia or any condition resulting in the absence or removal of the spleen. Psychiatric condition that precludes compliance with the protocol; past or present psychoses; or within 5 years prior to enrollment, a history of suicide plan or attempt. Any medical, psychiatric, or social condition that, in the judgment of the investigator, is a contraindication to protocol participation or impairs a subject s ability to give informed consent.
Interventions	**Vaccine type:** respiratory syncytial virus prefusion F subunit vaccine DS-Cav1 **Experimental:** 50 µg, 150 µg, or 500 µg DS-Cav1 with aluminium hydroxide **Control:** 50 µg, 150 µg, or 500 µg DS-Cav1 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** Brenntag-Biosector **Vaccination schedule:** administered in the deltoid muscle by needle and syringe at weeks 0 and 12.

Outcomes	**Safety**: number of subjects reporting local and systemic reactogenicity signs and symptoms within 7 days of the first, second or any dose of the product administration of DS-Cav1 alone or with alum adjuvant. Number of subjects with abnormal laboratory measures of safety up tp day 308. Number of subjects with one or more unsolicited non-serious adverse events (time frame: day 0 through day 28 after product administration). Number of subjects with serious adverse events (time frame: day 0 through day 308). Number of subjects who had respiratory syncytial virus infection following product sdministration (time frame: day 0 through day 308). **Immunogenicity**: respiratory syncytial virus subtype A (RSV A) antigen-specific neutralizing antibody geometric mean titers at 4 weeks after the first or second product administration of DS-Cav1 alone or with alum adjuvant.
Are there other groups in the trial not analysed in this review?	No
Notes	**Funding:** this work was supported with intramural funding from the NIAID. The funding source had no involvement with the study design, execution, or preparation of the manuscript.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomized using permuted block randomisation with block sizes of 4 or 6, chosen at random.
Allocation concealment (selection bias)	Low risk	The protocol statistician generated the randomisation sequence which was programmed into the database. Participants were assigned to a group upon electronic enrolment by a study clinician.
Blinding of participants and personnel (performance bias)	High risk	Open-label
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	High risk	Open-label
Blinding of outcome assessment (detection bias) All-cause mortality	High risk	Open-label
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Open-label
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%

Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs <5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03049488), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Sheldon 2016

Methods	Phase 1, first-in-human, dose-escalating, placebo-controlled, randomised, observer-blinded study
Participants	**Sex:** male and female **Age:** 50-85 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "evidence of a personally signed and dated informed consent document. Healthy male and female adults aged 50 to 85 years at enrollment as determined by medical history, physical examination, and the clinical judgment of the investigator to be eligible for the study. Subjects with preexisting chronic medical conditions determined to be stable may be included. Male subjects who, in the opinion of the investigator, are biologically capable of fathering children, and who are sexually active with women of childbearing potential must agree to use a highly effective method of contraception from the time of informed consent through at least 28 days after the last dose of investigational product (through Visit 9 at Month 7). Female subjects who are not of childbearing potential (ie, meet at least one of the following criteria): have undergone hysterectomy or bilateral oophorectomy;Have medically confirmed ovarian failure orAre medically confirmed to be postmenopausal. Availability for the entire duration of the study, and able to comply with scheduled visits, treatment plan, laboratory tests, and other study procedures including completion of the electronic diary (eDiary) from day 1 to day 7 following each vaccination. Ability to be contacted by telephone during study participation". **Exclusion criteria:** "previous administration of an investigational C. difficile vaccine or C. difficile monoclonal antibody therapy. Proven or suspected prior episode of CDAD. Unstable chronic medical condition or disease requiring significant change in therapy or hospitalization for worsening disease within 12 weeks before receipt of study vaccine. Serious chronic medical disorders including metastatic malignancy, severe chronic obstructive pulmonary disease (COPD) , end stage renal disease with or without dialysis, clinically unstable cardiac disease, or any other disorder that in the investigator's opinion precludes the subject from participating in the study. Donation of blood volume of 250 mL or greater, or donation of plasma within 3 months prior to enrollment or during the conduct of the study. Bleeding diathesis or condition associated with prolonged bleeding time that may contraindicate intramuscular injection or blood draw including subjects taking anticoagulant, antiplatelet and/or antithrombotic agents except for low dose daily aspirin (≤325mg per day) within 30 days before enrollment through completion of Visit 9 (Month 7). Any

	contraindication to vaccination or vaccine components. "Immunocompromised persons or subjects currently on immunosuppressive therapy or with a history of immunosuppressive therapy, including chemotherapy agents for treatment of diseases including, but not limited to cancer, inflammatory bowel disease or autoimmune disease. Recent history (within the past 6 months) of long term (7 days or longer) systemic corticosteroid use." Subjects who received oral or parenteral antibiotics within 1 month before enrollment. Topical antibiotics are allowed. Receipt of blood products or immunoglobulins (including monoclonal antibodies) within 6 months before enrollment through conclusion of the study. Participation in other investigational or interventional studies within 30 days before the current study begins and/or during study participation. Participation in purely observational studies is acceptable. Subjects who are investigational site staff members or relatives of those site staff members, or subjects who are Pfizer employees directly involved in the conduct of the trial. Females of childbearing potential; males of childbearing potential not using highly effective contraception or not agreeing to continue highly effective contraception for at least 28 days after last dose of investigational product (through Visit 9 at Month 7). Females receiving exogenous estrogen therapy. Residence in a nursing home, long-term care facility, requirement for semiskilled nursing care or assisted living. An ambulatory subject who lives in an autonomous manner in a retirement home or village is eligible for the trial. Any abnormality in screening hematology and/or blood chemistry laboratory values according to the toxicity grading scale. Subjects with stable Grade 1 abnormalities for hemoglobin, leukocyte count, and platelets may be considered eligible at the discretion of the investigator. A positive screening test or known infection with human immunodeficiency virus (HIV), hepatitis B virus (HBV) and/or hepatitis C virus (HCV). Other severe acute or chronic medical or psychiatric condition or laboratory abnormality that may increase the risk associated with study participation or investigational product administration or may interfere with the interpretation of study results and, in the judgment of the investigator, would make the subject inappropriate for entry into this study".
Interventions	**Vaccine type:** *Clostridium difficile* vaccine **Experimental:** 50, 100 or 200 mcg *Clostridium difficile* vaccine with aluminium. n = 73 **Control:** 50, 100 or 200 mcg *Clostridium difficile* vaccine. n = 72 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag **Vaccination schedule:** 3-dose vaccine regimen administered on day 1, Month 1,and Month 6 by intramuscular injection into the upper deltoid muscle
Outcomes	**Immunogenicity:** "neutralizing antibody levels at Month 2, (time frame: month 2). Neutralizing antibody levels after each vaccination dose, (time frame: baseline, day 14, month 1, day 37, month 6, day 187, month 7). Number of subjects in each treatment group with 4 and higher fold-rises in neutralizing antibody levels after each vaccination dose, (time frame: baseline, day 14, month 1, day 37, month 6, day 187, month 7). Number of subjects in each treatment group with neutralizing antibody levels ≥ a specified threshold after each vaccination dose., (time frame: baseline, day 14, month 1, day 37, month 6, day 187, month 7)". **Safety:** "number subjects reporting local reactions (pain, erythema, and induration) and their severity, as self reported on eDiaries for 7 days following each vaccination, (time frame: 7 days). Number of subjects reporting systemic

	reactions (fever, vomiting, diarrhea, headache, fatigue, new or worsening muscle pain, and new or worsening joint pain) and their severity, as self reported on eDiaries for 7 days following each vaccination, (time frame: 7 days). Number of subjects reporting adverse events from day 1 (at the time of vaccination) up to 28 days post dose 3 (visit 9, month 7) and serious adverse events throughout the study period, (time frame: 7 months (adverse events) 12 months (serious adverse events)). Number of subjects with abnormal hematology and blood chemistry laboratory assessments after each vaccination dose, (time frame: day 3, day 14, month 1, day 37, month 6, day 187)"
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2028 authors (nicholas.kitchin@pfizer.com) were inquired on the followings: ● method for allocation concealment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received **Funding:** sponsored by Pfizer Inc. The sponsor was responsible for the design of the study, monitoring, analysis, and interpretation of the data, as well as the decision to submit the data for publication.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Subjects were randomised via an interactive response technology system.
Allocation concealment (selection bias)	Low risk	Allocation to investigational product was via an interactive response technology system
Blinding of participants and personnel (performance bias)	Low risk	Observer-blinded study. The study staff dispensing and administering the vaccine were unblinded. The subject, all other study site members, including the investigator and study site coordinator,remained blinded.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Those performing the clinical assessments including the investigator and study site coordinator, and the laboratory staff performing the immunogenicity testing, remained blinded.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Those performing the clinical assessments including the investigator and study site coordinator, and the laboratory staff performing the immunogenicity testing, remained blinded.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Those performing the clinical assessments including the investigator and study site coordinator, and the laboratory staff performing the immunogenicity testing, remained blinded.

Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01706367), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Tapia 2021

Methods	Phase 2, observer-blind, randomized, controlled Study
Participants	**Sex:** male and female **Age:** 12 months to 16 months (child) **Ethnicity:** African **Diagnosis:** healthy **Inclusion criteria:** "male and female children between 12 months and 16 months old inclusive (minimum 365 days of age and maximum 16 months plus 29 days of age); for whom parent(s)/legal guardian(s) have given written informed consent after the nature of the study has been explained according to local regulatory requirements; who the investigator believes that their parent(s)/ guardian(s) will be available for all the subject visits and would comply with the requirements of the protocol (e.g., timely reporting of adverse events, availability for study site visits and home visits); Individuals in good health as determined by the outcome of medical history, physical examination and clinical judgment of the investigator.Individuals who completed their local infant EPI schedule through 9 months of age (except MenAfriVac dose). A birth dose of OPV is not required)". **Exclusion criteria:** "history of any meningococcal vaccine administration. Current or previous, confirmed or suspected disease caused by N. meningitidis. Household contact with and/or intimate exposure to an individual with any laboratory confirmed N. meningitidis infection within 60 days of enrolment. History of severe allergic reactions after previous vaccinations or hypersensitivity to any study vaccine component including tetanus, diphtheria and diphtheria toxoid (CRM197). Acute or chronic, clinically significant pulmonary, cardiovascular, metabolic, neurological, hepatic, or renal functional abnormality, as determined by medical history or physical examination. Any confirmed or suspected condition with impaired/altered function of immune system (immunodeficient or autoimmune conditions). Have any bleeding disorder which is considered as a contraindication to intramuscular injection or blood draw. Severe acute malnutrition. A family history of congenital or hereditary immunodeficiency. History of either hepatitis B or hepatitis C virus infection or human immunodeficiency virus infection. Major congenital defects. Chronic

	administration (defined as more than 14 days) of immunosuppressants or other immune-modifying drugs within three months prior to the study vaccination or planned use throughout the study period. (For corticosteroids, this means prednisone, or equivalent, ≥ 0.5 mg/kg per day. Inhaled, intranasal and topical steroids are allowed). Administration of blood, blood products and/or plasma derivatives or any parenteral immunoglobulin preparation in the past 3 months or planned use throughout the study period. Administration of any vaccine within 28 days prior to enrolment in the study or planned administration of any vaccine within 14 days before or after any study vaccination. Use of any investigational or non-registered drug or vaccine within 30 days prior to the administration of study vaccines or planned during the study. Malaria infection as confirmed by a Rapid Diagnostic Test. Individuals who are close family members of individuals conducting this study. Have experienced a moderate or severe acute infection and/or fever (defined as temperature ≥ 37.5°C) within 3 days prior to enrolment. Have received systemic antibiotic treatment within 3 days prior to enrolment. Non-residence in the study area or intent to move out within six months. Any condition which, in the opinion of the investigator, might pose additional risk to the subject due to participation in the study.
Interventions	**Vaccine type:** meningococcal ACYWX conjugate vaccine **Experimental:** NmCV-5 with aluminium phosphate **Control:** NmCV-5 **Aluminium concentration:** 125 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** intramuscularly in two doses 12 weeks apart
Outcomes	**Safety**: percentage of subjects with at least one severe solicited AE within 7 days after any study vaccination. Solicited local and systemic AEs reported during the 7 days after each vaccination. Unsolicited AEs reported during 28 days after each vaccination. AEs leading to premature withdrawal during the entire study period. SAEs reported during the entire study period. **Immunogenicity**: percentage of subjects with rSBA titer ≥ 8 against serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Percentage of subjects with rSBA titer ≥ 128 against serogroups A, C, W, Y and X at Visits Day 0, Day 28, Day 84 and Day 112. Percentage of subjects with fourfold rise in rSBA titers against serogroups A, C, W, Y and X at Visits Day 28 and Day 112. Geometric mean of rSBA titres.
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 23/08/2021 authors (drpsk@seruminstitute.com') were inquired on the followings: ● aluminium manufacturer ● proportion of participants per group who developed meningitis ● total number of participants per group who experienced any adverse events Answer received but no data provided. **Funding:** U.K. Foreign, Commonwealth, and Development Office.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Quadruple (participant, care provider, investigator, outcomes assessor)
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	Drop outs less than 5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT03295318), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Vandenberghe 2016

Methods	Phase 2, 90-week, randomised, double-blind, placebo-controlled, multicenter study
Participants	**Sex:** male and female **Age:** <85 years **Ethnicity:** mixed **Diagnosis:** mild Alzheimer's disease **Inclusion criteria:** "male and/or female patients below 85 years of age (inclusive) Diagnosis of mild Alzheimer's Disease Mini-Mental State Examination (MMSE) 20 to 26 (inclusive) at screening, untreated or on stable dose of cholinesterase inhibitor or memantine over the last 4 weeks prior to

	clinical assessments" **Exclusion criteria:** "previously participated in an AD vaccine study and received active treatment. History or presence of an active autoimmune disease. History or presence of seizure disorder. Presence of significant coronary heart disease and/or cerebrovascular disease. Presence of other neurodegenerative disease and/or psychiatric disorders (with the exception of successfully treated depression). Advanced, severe, progressive or unstable disease that might interfere with the safety, tolerability and pharmacodynamic assessments of the patient Other protocol-defined inclusion/exclusion criteria may apply".
Interventions	**Vaccine type:** CAD106 Alzheimer's disease vaccine **Experimental:** 450 mcg CAD106 with aluminium. n = 21 **Control:** 450 mcg CAD106. n = 16 **Aluminium concentration:** 450 mcg/dose **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** seven intramuscular injections at weeks 0, 6, 12, 24, 36, 48, and 60
Outcomes	**Immunogenicity:** "amyloid beta-specific and Qβ carrier-specific antibody response to CAD106 in serum and cerebrospinal fluid, (time frame: screening and through the end of the study to week 90). Amyloid beta-specific and Qβ carrier-specific T-cell response to CAD106 using peripheral blood mononuclear cells, (time frame: screening and at week 8). Changes over time of the concentrations of disease related markers (Aβ1-40 and Aβ1-42 in plasma; Aβ1-40, Aβ1-42, total-tau, phospho-tau in CSF, or other markers) in patients with mild Alzheimer's disease receiving CAD106 compared to placebo, (time frame: screening and through the end of the study to week 90)". **Safety:** "physical/neurological examinations, electrocardiogram, vital signs, standard and special laboratory evaluations, magnetic resonance imaging, adverse events / serious adverse events monitoring, (time frame: screening and through the end of the study to week 90)".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (rik.vandenberghe@uz.kuleuven.ac.be) were inquired on the followings: ● aluminium type ● aluminium manufacturer ● allocation concealment ● random sequence generation ● blinding of immunogenicity outcome assessment ● number of participants with mortality, serious and non-serious adverse event related and not related to the vaccine per study arm No answer received **Funding:** sponsored by Novartis Pharma AG, Basel, Switzerland

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Patients were centrally randomised.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Low risk	Double-blind. CAD106 and adjuvant doses were prepared by an unblinded pharmacist and administered by an unblinded study nurse. Patients, investigators, and the site personnel were masked to the treatment assigned.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	An independent unblinded data monitoring committee monitored safety data including adverse events, vital signs, weight, electrocardiograms, and safety laboratory measurements regularly throughout the study. Image analysis blinded to treatment allocation was performed by Avid Radiopharmaceuticals Inc.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	An independent unblinded data monitoring committee monitored safety data including adverse events, vital signs, weight, electrocardiograms, and safety laboratory measurements regularly throughout the study. Image analysis blinded to treatment allocation was performed by Avid Radiopharmaceuticals Inc.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	No drop outs
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT01097096), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Verdijk 2013

Methods	Randomized, controlled, double blind, phase I 'proof-of-concept' trial
Participants	**Sex:** male **Age:** 18-49 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "age 18 to 49, inclusive at the time of enrolment; in good health as determined by the outcome of medical history, physical examination screening/baseline labs and clinical judgment of the investigator. Male must have received poliovirus vaccinations with OPV according to the Polish National Immunization Program as a child based on vaccination cards or information given by subjects during interview with investigator. Preferred: number (and date) of poliovirus vaccinations known. Willingness and ability to adhere to the study regimen. Having a signed informed consent form" **Exclusion criteria:** "IPV or OPV booster dose after the age of 12 years. Positive for HIV, Hepatitis B or Hepatitis C. Known or suspected allergy against any of the vaccine components History of unusual or severe reactions to any previous vaccination. Known or suspected disease or use of medication that may influence the immune system. Known or suspected immune deficiency. Systemic treatment with corticosteroids within one month before screening. Administration of plasma (including immunoglobulins) or blood products three months prior to the study. Blood donation within one month before screening. Any vaccination within three months before screening and during the study until the last visit. History of any neurological disorder including epilepsy or febrile seizures. Evidence of excessive alcohol use or drug use. Any infectious disease at the time of screening and/or inclusion. Participation in another clinical trial within three months before screening. Abnormal pre-treatment laboratory parameters which are clinically relevant according to the investigator. Bleeding disorders or the usage of anticoagulants".
Interventions	**Vaccine type:** inactivated poliovirusvirus vaccine Sabin-IPV **Experimental:** Sabin-IPV with aluminium. n = 15 **Control:** Sabin-IPV. n = 15 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag, Denmark **Vaccination schedule:** a single intramuscular injection
Outcomes	**Immunogencity:** "immunogenicity of Sabin-IPV and adjuvated Sabin-IPV was evaluated by virus neutralizing titers against type , type 2 and type 3 poliovirusviruses before and after vaccination" **Safety:** "subjects were observed for 30 min after vaccination. Subjects were required to complete a diary on the vaccination day and during the first three days following the day of vaccination in which solicited and unsolicited adverse events were to be recorded. When adverse reactions were still present after these three days, subjects were to continue to monitor these reactions until they had disappeared. Blood samples were taken before vaccination and on day 28 (range 25–31) after vaccination. A follow-up phone call was made 6 months after vaccination to assess whether any serious adverse event had occurred in the period in between".

Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (pauline.verdijk@intravacc.nl, Wilfried.bakker@intravacc.nl) were inquired on the followings: ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received **Funding:** supported by the World Health Organization using funds provided by a Grant from the Bill and Melinda Gates Foundation. The World Health Organization was involved in the design of the clinical trial.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The statistician prepared a randomisation list with a block size of three using SAS version 9.1.3. Randomization numbers were printed on the label of each vial with IMP.
Allocation concealment (selection bias)	Low risk	Randomization was performed using SAS version 9.1.3. SAS procedure plan was used to randomly assign treatment to subject numbers with an appropriate block size. The output of this procedure was used to prepare the randomisation list, the emergency list and emergency envelopes. Subject numbers were printed on the label of each vial with IMP. Each subject receives a vial with his own subject number.
Blinding of participants and personnel (performance bias)	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.

Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded and coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	No drop outs. Six months after vaccination, a follow-up call was performed to verify whether any serious adverse events had occurred. Two subjects could not be contacted - not stated which arm.
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01708720), and all outcomes were reported on. However, serious adverse events occurred but were not assigned per arm.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Verdijk 2014

Methods	Double-blind, randomised,controlled, phase I/IIa dose-escalation trial in the target population
Participants	**Sex:** male and female **Age:** infants aged 56–63 days **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "age 8 weeks (56-63 days) at the time of first vaccination. Infants in good general health, eligible to be vaccinated according to the Polish national vaccination program. The same health criteria apply as used in well-baby clinics when a child receives a vaccination, e.g. also children with a small increase in temperature (< 38.5° C) or with a common cold (runny nose etc) are seen as children with normal health. The parent(s)/legally representative(s) have to be willing and able to allow their child to participate in the trial according to the described procedures. Presence of a signed informed consent in which the parent(s)/legally representative(s) have given written informed consent after receiving oral and written information (signature from

	one parent in case of single-parent family)". **Exclusion criteria:** "any IPV or OPV dose. Known or suspected allergy against any of the vaccine components. History of unusual or severe reactions to any previous vaccination administration of plasma (including immunoglobulins) or blood products prior and during the study. Any vaccination less than 14 days before or after each vaccination with the IMP History of any neurological disorder including epilepsy or febrile seizures. Any infectious disease at the time of screening and/or inclusion that might interfere with the results of the study. Present evidence of serious disease(s) demanding immunosuppressive medical treatment, like cytostatics and prednisolones, that might interfere with the results of the study Any known or suspected primary or secondary immunodeficiency. Communication problems interfering in the study realization according to the judgment of the investigator Bleeding disorders and use of anticoagulants. Premature birth (<37 weeks). Participation in another clinical trial".
Interventions	**Vaccine type:** inactivated poliovirusvirus vaccine Sabin-IPV **Experimental:** low, middle or high dose sIPV with aluminium. n = 60 **Control:** low, middle or high dose sIPV. n = 60 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** Alhydrogel **Aluminium manufacturer:** Brenntag, Denmark **Vaccination schedule:** one dose of 0.5 mL intramuscularly in the right thigh. Infants participating in the trial also received the regular booster dose at 15–18 months with wIPV
Outcomes	**Immunogenicity:** level of virus neutralizing titers in serum, (time frame: 28 days after last vaccination) **Safety:** number of adverse reactions following vaccination, (time frame: 5 days or until adverse reactions have resolved). A follow-up telephone call was made 6 months after the last vaccination with the IMP to assess whether any serious adverse event had occurred during that period.
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (pauline.verdijk@intravacc.nl, Wilfried.bakker@intravacc.nl) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of participants and personnel ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received

	Funding: supported by the World Health Organization using funds provided by a grant from the Bill and Melinda Gates Foundation. The World Health Organization was involved in the design of the clinical trial.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Inclusion and randomisation was performed in three steps according to a randomisation list prepared by the statistician.
Allocation concealment (selection bias)	Low risk	At the sites vaccines with lowest tracking/identification number was given to the patients in chronological order of recruitment date. The investigator and other site staff were blinded during the conduct of the study
Blinding of participants and personnel (performance bias)	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) All-cause mortality	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Blinding of outcome assessment (detection bias) Immunogenicity	Low risk	Study unblinding was performed after database lock, including immunogenicity analysis and assessment of adverse events. All samples for immunogenicity were blinded were coded only with subject number and visit(date). The investigator and other site staff, the sponsor clinical study manager and the laboratory staff were blinded throughout the study until formal study unblinding.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%

Incomplete outcome data (attrition bias) Immunogenicity	Low risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01709071), and all outcomes were reported on.
Other bias	Low risk	Serious adverse events occurred but were not assigned per arm in the publication. Data added after contact with authors.

Vernacchio 2002

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** infants aged 15-36 months **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** healthy toddlers aged 15–36 months **Exclusion criteria:** not reported
Interventions	**Vaccine type:** pneumococcal–CRM197 conjugate vaccine (PCV9) **Experimental:** 10 or 25 mcg PCV9 with aluminium. n = 38 **Control:** 10 or 25 mcg PCV9. n = 40 **Aluminium concentration:** 500 mcg/dose **Aluminium type:** aluminium phosphate **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses 60 days apart
Outcomes	**Immunogenicity:** "serum IgG antibodies to pneumococcal capsular polysaccharides were measured by enzyme-linked immunosorbent assay. Anti-CRM197 IgG concentrations were measured by standard enzyme-linked immunosorbent assay. Microneutralization assay was performed". **Safety:** "following each immunization, parents were asked to complete a 7-day diary card and record daily a single rectal temperature, local site reactions and systemic side effects. On day 7 after each immunization (range 7–10 days), study subjects were examined by trained research nurses and the recorded safety data were reviewed with the parents. Blood samples were obtained before the first immunization and on days 7 and 30 following the second immunization for safety and immunogenicity analyses. The safety evaluation consisted of monitoring a complete blood count, aspartate aminotransferase, alanine aminotransferase and creatinine concentrations".
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (louis.vernacchio@childrens.harvard.edu) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed

	● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further details
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	n.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	n.a.
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs more than 5%
Selective reporting (reporting bias)	High risk	No pre-published protocol. Four serious adverse events occurred but were not assigned by arm because considered as not related to the vaccine.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Warrel 1984

Methods	Randomised clinical trial
Participants	**Sex:** male and female **Age:** 9–69 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** at very low risk of rabies infection **Exclusion criteria:** not reported
Interventions	**Vaccine type:** HDCSV rabies vaccine **Experimental:** HDCSV with aluminium. n = 20 **Control:** HDCSV without aluminium. n = 20 **Aluminium concentration:** 980 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** unclear
Outcomes	**Immunogenicity** and **safety**
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Funding**: the Wellcome Trust of Great Britain

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.

Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Westritschnig 2014

Methods	Phase I, randomised, placebo-controlled, observer blinded, multicenter trial
Participants	**Sex:** male and female **Age:** 18–65 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "written informed consent obtained prior to study entry; healthy adults aged between 18 and 65 years; no clinically relevant pathological findings in any of the investigations at the Screening visit. Minor deviations of laboratory values from the normal range may be accepted, if judged by the investigator to have no clinical relevance. In female subjects either childbearing potential terminated by surgery or a negative serum pregnancy test during screening and the willingness not to become pregnant during the entire study period by practicing reliable methods of contraception". **Exclusion criteria:** "history of autoimmune diseases and malignancies. Active or passive vaccination 4 weeks before and during the entire study protocol. Use of any other investigational or non-registered drug or vaccine in addition to the study vaccine during the study period or within 30 days preceding the first dose of study vaccine. History of severe hypersensitivity reactions and anaphylaxis. Known hypersensitivity or allergic reactions to one of the components of the vaccine. Clinically significant diseases as judged by the investigator. Immunodeficiency due to immunosuppressive therapy. A family history of congenital or hereditary immunodeficiency".
Interventions	**Vaccine type:** pseudomonas aeruginosa hybrid outer membrane protein OprF/I vaccine (IC43) **Experimental:** 100 mcg IC43 with aluminium. n = 33 **Control:** 100 mcg IC43. n = 32 **Aluminium concentration:** not reported **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two intramuscular injections 7 days apart
Outcomes	**Immunogenicity:** "measurement of fOprF/I-specific IgG antibodies by enzyme-linked immunosorbent assay at day 14 (7 days after second vaccination), measurement of antibody avidity on days 7 and 14, measurement of anti-histidine antibodies on days 7, 14, 90, and 180. Functional antibodies; avidity of OprF/I-specific IgG antibodies; anti-histidine IgG antibodies; peripheral blood mononuclear cell analysis were performed". **Safety:** rate of serious adverse events during vaccination period until 6 months after first vaccination, safety laboratory parameters at intervals up to day 180,

	systemic and local tolerability at intervals up to day 180
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 14/12/2018 authors (bernd.jilma@meduniwien.ac.at) were inquired on the followings: ● Final concentration of aluminium injected ● Aluminum manufacturer ● Number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up Answer received but no further information or data added **Funding:** not reported

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomised list was created using nQuery Advisor by a statistician at a contract research organization (Assign Data Management and Biostatistics Gmb).
Allocation concealment (selection bias)	Low risk	Subjects were randomly allocated. Each subject was assigned the next free medication kit (randomisation) number for the corresponding study center in ascending order.
Blinding of participants and personnel (performance bias)	High risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	The trial was blinded for the subjects and not blinded for the investigator applying the vaccination, but blinded for a second independent investigator assessing local and systemic tolerability.

Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00778388), and all outcomes were reported on.
Other bias	Low risk	"Major protocol deviations included: incorrect study treatment dosage (21 subjects [65.6%] in the IC43 50 µg group, and 2 subjects [6.1%] in the IC43 100 µg with adjuvant group); forbidden concomitant medication (1 subject [3.0%] in the IC43 100 µg with adjuvant group); treatment kit number issue (1 subject [3.1%] in the IC43 100 µg without adjuvant group); and second vaccination not given (2 subjects [6.1%] in the IC43 200 µg group, 1 subject [3.1%] in the IC43 50 µg group, and 1 subject [3.0%] in the placebo group)".

Wressnigg 2013

Methods	Double blind, randomised, dose-escalation, phase 1/2 study
Participants	**Sex:** male and female **Age:** 18–70 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subject is 18 to 70 years old at the time of screening. Subject has an understanding of the study, agrees to its provisions, and gives written informed consent prior to study entry. Subject is generally healthy, as determined by the investigator's clinical judgment through collection of medical history and the performance of a physical examination. If female of childbearing potential, presents with a negative urine pregnancy test, and agrees to employ adequate birth control measures for the duration of the study. Additional inclusion criterion for seropositive subjects in Section 2 only: subject is seropositive for Borrelia burgdorferi sensu lato antibodies at study entry". **Exclusion criteria:** "subject has a physician-diagnosed chronic illness related to Lyme borreliosis (LB) or active LB. Subject has been treated for LB with antibiotics within 3 months of study entry. Subject had a tick bite within 3 weeks prior to screening or first vaccination. Subject has a history or active infection with Babesia microti (babesiosis) or Anaplasma. phagocytophilum (ehrlichiosis). Subject currently has or has a history of significant cardiovascular, respiratory (including asthma), metabolic, neurological, hepatic, rheumatic, autoimmune, hematological, gastrointestinal or renal disorder. Subject has clinically significant abnormal laboratory values at screening. Subject currently has or has a history of immunodeficiency.Subject tests positive for human immunodeficiency virus, hepatitis B surface antigen or hepatitis C virus. Subject has a disease or is currently undergoing a form of treatment or was undergoing a form of treatment within 30 days prior to study entry that could be expected to influence immune response. Subject has a history of anaphylaxis or severe allergic reactions. Subject has received any live vaccine within 4 weeks or inactivated vaccine within 2 weeks prior to vaccination in this study. Subject is pregnant or lactating at the time of study enrollment. Additional exclusion criterion for subjects in Section 1 and seronegative subjects in Section 2: subject is seropositive for Borrelia burgdorferi sensu lato antibodies at study entry".

Interventions	**Vaccine type:** multivalent OspA vaccine against Lyme borreliosis **Experimental:** 30, 60 or 90 mcg OspA with aluminium. n = 151 **Control:** 30, 60 or 90 mcg OspA. n = 149 **Aluminium concentration:** 1000 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** three intramuscular immunisations with intervals of 28 days, and a booster 9–12 months after the first dose
Outcomes	**Immunogenicity:** "antibody response to the vaccine, (time frame: 28 days after the third vaccination (= day 85)). Antibody response, (time frame: at baseline, 28 days after each vaccination (i.e. days 29, 57 and 85), 180 and 270 days after the first vaccination (day 181, day 271) and 180 days after the booster vaccination (day 361 or day 451 - 546)). Fold increase in antibody titer compared to baseline, (time frame: 28 days after each vaccination, 180 and 270 days after the first vaccination and 180 days after the booster vaccination). Seroconversion rate (at least 4-fold increase of each rOspA type-specific Immunoglobulin G titer) as compared to baseline, (time frame: 28 days after each vaccination, 180 and 270 days after the first vaccination and 180 days after the booster vaccination)". **Safety:** "frequency and severity of injection site and systemic reactions; (time frame: within 7 days after each vaccination (i.e. days 8, 36 and 64))"
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on 14/12/2018 authors (noel_barrett@baxter.com) were inquired on the followings: ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up No answer received **Funding:** funded by Baxter. Baxter employees were responsible for study design, data collection, data analysis, data interpretation, writing of the report, and the decision to submit for publication

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was done centrally via an electronic data capture system.
Allocation concealment (selection bias)	Unclear risk	Participants were recruited sequentially and randomly assigned to one of six study groups in equal ratios.
Blinding of participants and personnel (performance bias)	Unclear risk	Double-blind, no further information

Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	n.a.
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01504347), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Wu 2017

Methods	Phase I/II randomised trial
Participants	**Sex:** male and female **Age:** 20–60 years **Ethnicity:** mixed **Diagnosis:** healthy **Inclusion criteria:** "subject within the age limit range, is free of obvious health problems as judged by the investigator before entering the study. Female subjects must be either of non-childbearing potential, or one year post-menopausal; or, if of childbearing potential, must be abstinent or have used adequate contraceptive precautions for 30 days prior to receiving the study vaccination, have a negative pregnancy test and must agree to continue such precautions for two months after completion of the vaccination series. Subject is willing and able to comply with all required study visits and follow-up required by this protocol. Subject must provide written informed consent or the Subjects' legal representative must understand and consent to the procedure". **Exclusion criteria:** "subjects with previous known or potential exposure to avian influenza virus H7N9 HA antigen and any other avian influenza including H5N1. Subjects who have had seasonal influenza vaccine within 6 months prior to enrolment or who will have seasonal influenza vaccine at any time during the study period. Subjects administered with any other vaccine during the period within 6 weeks before the first administration of study vaccine. Subjects with

	confirmed or suspected abnormal immune function, immunosuppressive or immunodeficient condition, including human immunodeficiency virus (HIV) infection, based on medical history and physical examination. Subjects with a history of hypersensitivity to vaccines or a history of allergic disease or reactions likely to be exacerbated by any component of the vaccine. Subjects with a history of inflammatory or degenerative neurological disease. Subjects receiving chronic administration of immunosuppressants or other immune-modifying drugs within 6 months prior to the administration of the study vaccine. Inhaled and topical steroids are allowed. Known HIV, hepatitis B (HBsAg) or hepatitis C seropositivity. Subjects, judged by the investigator, with any clinically significant medical illness including acute pulmonary, cardiovascular, hepatic or renal functional abnormality, as determined by medical history, physical examination and/or laboratory screening tests. The subjects would not be suitable for the vaccination if the screening laboratory tests show any of the follows: ALT or AST > upper limit of normal range; serum creatinine > upper limit of normal range; any significant laboratory abnormality as judged by the investigator. Subjects receiving administration of immunoglobulins and/or any blood products within the 3 months preceding the administration of the study vaccine or at any time during the study. Subjects with acute disease at the time of enrolment. Use of any investigational or non-registered product other than the study vaccine within 30 days prior to the first vaccination, or planned use of the abovementioned product during the entire study period. Psychiatric, addictive, or any disorder, which may compromise the ability to give a truly informed consent for participation of this study or adequate compliance. Any clinically relevant disease and/or abnormal laboratory findings (, which, in the opinion of the Investigator, may either put the patient at risk because of participation in the study, or influence the results of the study, or the patient's ability to participate in the study. Breast feeding or pregnant women or refusal to submit to a urine test to rule out pregnancy prior to enrolment and during the study".
Interventions	**Vaccine type:** inactivated cell culture-derived H7N9 influenza vaccine **Experimental:** 15 or 30 mcg H7N9 vaccine with aluminium. n = 100 **Control:** 15 or 30 mcg H7N9. n = 100 **Aluminium concentration:** 300 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 2 doses at 21-day intervals
Outcomes	**Immunogenicity:** "HI and neutralizing antibody titers, (time frame: 1 year from the start of study enrolment). Change of antibody titers from baseline to day 22 and day 43. Geometric mean titer of both serum HI titers and neutralizing antibody titers, pre-vaccination and day 22 and day 43 post-vaccination" **Safety:** "adverse event, (time frame: 1 year from the start of study enrolment). Percentage, intensity and relationship to vaccination of solicited local and general signs and symptoms during a 7-day follow-up period after the first and second dose of administered vaccine. Percentage, intensity and relationship to vaccination of unsolicited local and general signs and symptoms during a 21-day follow-up period after each administered vaccine. Occurrence of overall adverse events and serious adverse events during the entire period of study"
Are there other groups in the trial not analysed in this review?	No

Notes	**Contacts with authors:** on17/12/2019 authors (changsc@ntu.edu.tw) were inquired on the followings: ● aluminium manufacturer ● allocation concealment ● random sequence generation ● blinding of participants and personnel ● blinding of outcome assessment ● number of participants with any serious and non-serious adverse event related and not related to the vaccine. Please kindly provide this number out of total number of participants analysed per arm No answer received **Funding:** funded by Medigen Vaccine Biologics Co., Taiwan

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized using the blocking factor to ensure that the balance between the 4 study groups was maintained.
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) All-cause mortality	Low risk	Drop outs less than 5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs > 5%
Selective reporting (reporting bias)	High risk	The study protocol was registered prior to randomisation (NCT02436928), and all outcomes were reported on. One serious adverse event was not assigned to any arm.

Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Wuorimaa 2001

Methods	Randomised clinical study
Participants	**Sex:** male and female **Age:** infant **Ethnicity:** Finnish and Israeli **Diagnosis:** healthy **Inclusion criteria:** not reported **Exclusion criteria:** not reported
Interventions	**Vaccine type:** 11-valent pneumococcal conjugate vaccine **Experimental:** 11-valent pneumococcal conjugate vaccine with aluminium **Control:** 11-valent pneumococcal conjugate vaccine **Aluminium concentration:** 300 mcg/dose **Aluminium type:** aluminium hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** 4 doses at the age of 2, 4, 6, and 12 months
Outcomes	**Immunogenicity**: "functional activity of antibodies to five capsular serotypes (4, 6B, 14, 19F, and 23F). OPA of antibodies against serotypes 4, 6B, 14, 19F, and 23F were analysed. Serum samples obtained 1 month after the third dose (at 7 months), and before (at 12 months) and 1 month after (at 13 months) the fourth dose was used. Enzyme immunoassay to measure IgG antibody concentrations to capsular polysaccharides was used. The OPA of sera was determined with the viable cell killing assay".
Are there other groups in the trial not analysed in this review?	No
Notes	**Contacts with authors:** on14/12/2018 authors (helena.kayhty@thl.fi) were inquired on the followings: ● method for random sequence generation ● method for allocation concealment ● procedure for blinding of participants and personnel ● procedure for blinding of outcome assessment ● number of drop outs (patients randomised but not analysed) ● Final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) Answer received but no information added. **Funding:** Sanofi Pasteur provided the financial support

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized, no further information
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Unclear risk	N.a.
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	N.a.
Incomplete outcome data (attrition bias) Immunogenicity	Unclear risk	N.a.
Selective reporting (reporting bias)	Unclear risk	No pre-published protocol
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Zhu 2009

Methods	Randomized, double-blind, placebo-controlled, clinical trial
Participants	**Sex:** male and female **Age:** 3 years to 90 years **Ethnicity:** not reported **Diagnosis:** healthy **Inclusion criteria:** "healthy male or female aged 3 and older. Be able to show legal identity card for the sake of recruitment. Volunteers or their guardians are able to understand and sign the informed consent". **Exclusion criteria:** "cases, cured cases and close contact of influenza A (H1N1) virus. Women of pregnancy, lactation or about to be pregnant in 60 days. Subject that has a medical history of any of the following: allergic history, or allergic to any ingredient of vaccine, such as egg, egg protein, etc. Serious

	adverse reactions to vaccines such as anaphylaxis, hives, respiratory difficulty, angioedema, or abdominal pain. Autoimmune disease or immunodeficiency. Asthma that is unstable or required emergent care, hospitalization or intubation during the past two years or that required the use of oral or intravenous corticosteroids. Diabetes mellitus (type I or II), with the exception of gestational diabetes. History of thyroidectomy or thyroid disease that required medication within the past 12 months. Serious angioedema episodes within the previous 3 years or requiring medication in the previous two years. Bleeding disorder diagnosed by a doctor (e.g. factor deficiency, coagulopathy, or platelet disorder requiring special precautions) or significant bruising or bleeding difficulties with IM injections or blood draws. Active malignancy or treated malignancy for which there is not reasonable assurance of sustained cure or malignancy that is likely to recur during the period of study. Seizure disorder other than: febrile seizures under the age of two years oldSeizures secondary to alcohol withdrawal more than 3 years ago, or a singular seizure not requiring treatment within the last 3 years. Asplenia, functional asplenia or any condition resulting in the absence or removal of the spleen Guillain-Barre Syndrome. Any history of immunosuppressive medications or cytotoxic medications or inhaled corticosteroids within the past six months (with the exception of corticosteroid nasal spray for allergic rhinitis or topical corticosteroids for an acute uncomplicated dermatitis). History of any blood products or seasonal influenza vaccine administration within 3 months before the dosing. Administration of any other investigational research agents within 30 days before the dosing. Administration of any live attenuated vaccine within 30 days before the dosing. Administration of subunit or inactivated vaccines, e.g., pneumococcal vaccine, or allergy treatment with antigen injections, within 14 days before the dosing. Be receiving anti-TB prophylaxis or therapy currently. Axillary temperature > 37.0 centigrade at the time of dosing. Psychiatric condition that precludes compliance with the protocol: past or present psychoses; past or present bipolar disorder requiring therapy that has not been well controlled on medication for the past two years; disorder requiring lithium; suicidal ideation occurring within five years prior to enrollment. Any medical, psychiatric, social condition, occupational reason or other responsibility that, in the judgment of the investigator, is a contraindication to protocol participation or impairs a volunteer's ability to give informed consent".
Interventions	**Vaccine type:** Influenza A (H1N1) vaccine **Experimental:** 15 or 30 mcg H1N1 with aluminium. n = 770 **Control:** 15 or 30 mcg H1N1. n = 770 **Aluminium concentration:** 1200 mcg/ml **Aluminium type:** not reported **Aluminium manufacturer:** not reported **Vaccination schedule:** two vaccinations intramuscularly, in the deltoid muscle, 21 days apart
Outcomes	**Immunogenicity:** proportions of subjects with an increase in the hemagglutination- inhibition titer by a factor of 4 or more on day 21 after the first dose and on day 35 (14 days after the second dose) **Safety:** presence of any systemic reaction or injection-site reaction 21 days after the first dose and 14 days after the second dose
Are there other groups in the trial not analysed in this review?	Yes

Notes	**Contacts with authors:** on 14/12/2018 authors (jszfc@vip.sina.com) were inquired on the followings: ● procedure for blinding of participants and personnel ● final concentration of aluminium injected ● aluminium manufacturer ● number of patients who died/number of patients analysed ● number of patients with serious adverse events (SAEs)/number of patients analysed at maximum follow up (please note that we are interested in SAEs related and not related to the vaccine) ● number of patients with the disease against which they were vaccinated/number of patients analysed at maximum follow up ● total number of patients with any adverse events/number of patients analysed at maximum follow up (please note that we are interested in AEs related and not related to the vaccine) No answer received **Funding:** sponsored by Hualan Biological Bacterin Company Serious adverse events are reported only as vaccine-related.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation schedule was created with the use of SAS software (version 9.1). The block size was defined by an independent statistician at Southeast University (Nanjing, China).Randomization-code numbers were assigned to subjects in chronological order by the investigator.
Allocation concealment (selection bias)	Low risk	Randomization-code numbers were assigned to subjects in chronological order by the investigator.
Blinding of participants and personnel (performance bias)	Unclear risk	Double blind, no further information
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	Not described
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	Not described
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	All samples were assayed in a blinded manner, in duplicate, and were double-checked by two investigators at the Chinese National Institute for the Control of Pharmaceuticals and Biological Products.
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs >5%

Incomplete outcome data (attrition bias) All-cause mortality	High risk	Drop outs >5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT00975572), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Zhu 2013

Methods	Randomised, double-blind, placebo-controlled, phase 2 trial
Participants	**Sex:** male and female **Age:** 6-36 months old **Ethnicity:** Chinese **Diagnosis:** healthy **Inclusion criteria:** "healthy children aged from 6 to 36 months old as established by medical history and clinical examination. The subjects' guardians are able to understand and sign the informed consent. Had never received the vaccine against EV71. Subjects who can and will comply with the requirements of the protocol. Subjects with temperature <=37.0°C on axillary setting". **Exclusion criteria:** "subject who has a medical history of HFMD. <= 37 weeks gestation. Subjects with a birth weight <2.5 kg. Subject that has a medical history of any of the following: allergic history, or allergic to any ingredient of vaccine. Family history of seizures or progressive neurological disease. Family history of congenital or hereditary immunodeficiency. Severe malnutrition or dysgenopathy. Major congenital defects or serious chronic illness, including perinatal brain damage. Autoimmune disease. Bleeding disorder diagnosed by a doctor or significant bruising or bleeding difficulties with IM injections or blood draws. Any acute infections in last 7 days. Any prior administration of immunodepressant or corticosteroids in last month. Any prior administration of blood products in last 3 month. Any prior administration of other research medicines in last 1 month. Any prior administration of attenuated live vaccine in last 28 days. Any prior administration of subunit or inactivated vaccines in last 14 days, such as pneumococcal vaccine. Under the anti-TB prevention or therapy. Any condition that in the opinion of the investigator, may interfere with the evaluation of study objectives. Exclusion Criteria for the second dose: had any Grade 3 or Grade 4 adverse reaction within 7 days after first dose. Any situation meet the exclusion criteria stated in the exclusion criteria for first dose. Had any SAE related to first dose during the following-up of first dose. Any condition that in the opinion of the investigator, or IRB".
Interventions	**Vaccine type:** enterovirus 71 vaccine (EV71) **Experimental:** 640 U EV71 with aluminium. n = 240 **Control:** 640 U EV71. n = 240 **Aluminium concentration:** 180 mcg/dose **Aluminium type:** alum hydroxide **Aluminium manufacturer:** not reported **Vaccination schedule:** two doses given on day 0 and day 28

Outcomes	**Immunogenicity:** "geometric mean titre of anti-EV71 antibodies in serum after first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination). Seroconversion rate of anti-EV71 antibodies in serum after first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination). The clinical abnormality of hematological examination, blood biochemical test and urinalysis after first vaccination in children, (time frame: 3 days after first vaccination) and the second vaccination, (time frame: 3 days after second vaccination). The persistence of immunogenicity of the EV71vaccine after two doses in children and infants,(time frame: 6 months after blood collection at day 56)". **Safety:** "frequency of systemic and local adverse reactions after the first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination). Frequency of adverse events and any SAE after the first vaccination, (time frame: 28 days after first vaccination) and the second vaccination, (time frame: 28 days after second vaccination)"
Are there other groups in the trial not analysed in this review?	Yes
Notes	**Contacts with authors:** on 17/12/2019 authors (hen54@x263.net, wangjz@nicpbp.org.cn) were inquired on the followings: ● aluminium manufacturer. ● blinding of immunogenicity outcome assessment. ● number of participants who got enterovirus infection after the vaccination. Email address wangjz@nicpbp.org.cn not valid **Funding:** The National Science and Technology Major Project (2011ZX10004-902) of the Chinese Ministry of Science and Technology, China's 12–5 National Major Infectious Disease Program (2012ZX10002-001), and Beijing Vigoo Biological. One of the sponsors of the study (Beijing Vigoo Biological) was involved in trial design, but none of the sponsors were involved in data collection, statistical analysis, interpretation, or writing of the report.

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	"We stratified eligible participants by age (6–11 months [infants] and 12–36 months [children]) and assigned each a sequential number according to their sequence of enrolment. Participants were then randomly allocated (1:1:1:1:1) to receive either placebo, EV71 vaccine of 160 U with adjuvant, 320 U with adjuvant, 640 U with adjuvant, or 640 U without adjuvant, according to a blocked (block size of 20) randomisation list generated by the statistical analysis program SAS".
Allocation concealment (selection bias)	Low risk	The four vaccine formulations and placebo had identical packaging and were blindly labelled with the sequential numbers, which were the only identifiers.

Blinding of participants and personnel (performance bias)	Low risk	Double-blind. Individuals involved in randomisation and masking did not participate in any other process of the trial. All participants, their guardians, and investigators were masked to the treatment allocation
Blinding of outcome assessment (detection bias) Serious and non serious adverse events	Unclear risk	An independent data and safety monitoring committee was set up to oversee the recording of serious adverse events and provide final decisions of causality between SAEs and vaccination before unbinding. The committee also supervised the updating of the safety data within 28 days after each injection.
Blinding of outcome assessment (detection bias) All-cause mortality	Unclear risk	An independent data and safety monitoring committee was set up to oversee the recording of serious adverse events and provide final decisions of causality between SAEs and vaccination before unbinding. The committee also supervised the updating of the safety data within 28 days after each injection.
Blinding of outcome assessment (detection bias) Immunogenicity	Unclear risk	Not described
Incomplete outcome data (attrition bias) Serious and non-serious adverse events	Low risk	Drop outs <5%
Incomplete outcome data (attrition bias) All-cause mortality	Unclear risk	Drop outs <5%
Incomplete outcome data (attrition bias) Immunogenicity	High risk	Drop outs >5%
Selective reporting (reporting bias)	Low risk	The study protocol was registered prior to randomisation (NCT01399853), and all outcomes were reported on.
Other bias	Low risk	The trial appeared to be free of other issues that could put it at risk of bias.

Footnotes

Characteristics of excluded studies

August 2017

Reason for exclusion	Saline placebo without a vaccine

Bravo 2020

Reason for exclusion	Wrong comparator

Brown 2014

Reason for exclusion	Placebo contained aluminium

De Carvalho 2009

Reason for exclusion	Placebo contained aluminium

De Carvalho 2010

Reason for exclusion	Placebo contained aluminium

Garland 2009

Reason for exclusion	Wrong comparator

Garland 2013

Reason for exclusion	Wrong comparator

Giannini 2006

Reason for exclusion	Placebo contained aluminium

GlaxoSmithKline Vaccine HPV-007 Study 2009

Reason for exclusion	Placebo contained aluminium

Harper 2004

Reason for exclusion	Placebo contained aluminium

Lindgren 2017

Reason for exclusion	Wrong comparator

Naud 2014

Reason for exclusion	Placebo contained aluminium

NCT00851643

Reason for exclusion	Wrong comparator

Paavonen 2007

Reason for exclusion	Wrong comparator

Rivera 2018

Reason for exclusion	Wrong comparator

Roteli-Martins 2012

Reason for exclusion	Placebo contained aluminium

Saez-Llorens 2020

Reason for exclusion	Wrong comparator

Skinner 2014

Reason for exclusion	Placebo contained aluminium

Sow 2013

Reason for exclusion	Placebo contained aluminium

Sullivan 2018

Reason for exclusion	Placebo alone without vaccine

Szarewski 2009

Reason for exclusion	Wrong comparator

Teixeira 2017

Reason for exclusion	Placebo contained aluminium

Van Damme 2014

Reason for exclusion	Placebo contained aluminium

Villa 2006

Reason for exclusion	Placebo contained aluminium

Wheeler 2016

Reason for exclusion	Placebo contained aluminium

Zhu 2014

Reason for exclusion	Placebo contained aluminium

Footnotes

Characteristics of studies awaiting classification

Chua 2019

Methods	Phase 1A randomised controlled trial
Participants	HIV-1 uninfected healthy volunteers aged 18 to 45 years
Interventions	GP120-CD4 chimeric subunit vaccine
Outcomes	Safety and immunogenicity
Notes	Palcebo not described. Author (jchua@ihv.umaryland.edu) contacted

EUCTR2004-001896-21-FI

Methods	A Phase II, Randomized, Observer Blind, Multi-Center, Active Controlled Study
Participants	Healthy Children Aged 12-59 Months
Interventions	MenACWY with or without adjuvant
Outcomes	Safety and immunogenicity
Notes	Adjuvant not described. Company (Novartis) contacted

Lin 2006

Methods	Phase I randomised controlled trial
Participants	Healthy individuals aged 18–60 years in and around Beijing
Interventions	Two doses of placebo or an inactivated whole-virion influenza A (H5N1) vaccine with 1·25 μg, 2·5 μg, 5 μg, or 10 μg haemagglutinin per dose with aluminium hydroxide adjuvant on day 0 and 28
Outcomes	Safey and immunogenicity
Notes	Placebo content not described. Authors (yinwd@sinovac.com) contacted 3 times but no answer received

NCT00001053

Methods	A Phase I Safety and Immunogenicity Study
Participants	Healthy Volunteers
Interventions	HIV p17/p24:Ty-VLP (virus-like particles) vaccine with or without aluminium
Outcomes	No information
Notes	Company (protocol@tech-res.com) contacted about more information on study arms

NCT00100724

Methods	A Phase 2 Multi-Center, Randomized Dose-Finding Trial
Participants	Healthy Volunteers
Interventions	rPA102 vaccine given intramuscularly at 0 and 4 weeks over 2 dose ranges and 4 adjuvant levels.
Outcomes	Safety and immunogenicity.
Notes	Adjuvants not described.

NCT01594320

Methods	Phase 1 Randomized, Observer-Blinded, Dose-Ranging Study.
Participants	Healthy Young Adults.
Interventions	Monovalent A/Indonesia/05/05 (H5N1) Virus-Like Particle Avian Influenza Antigen (Recombinant) With and Without Adjuvant 1
Outcomes	Immunogenicity and Safety
Notes	Adjuvants and arms not described. Company (ir@novavax.com) contacted

NCT01596725

Methods	Phase 1 randomised, observer-blinded, dose-ranging study
Participants	Healthy young adults
Interventions	Monovalent A/Indonesia/05/05 (H5N1) Virus-Like Particle Avian Influenza Antigen (Recombinant) With and Without Adjuvant 2
Outcomes	Immunogenicity and Safety
Notes	Adjuvants and arms not described. Company (ir@novavax.com) contacted

NCT03572062

Methods	A phase1/2, placebo-controlled, randomised, observer-blind, dose-fining, first-in-human study
Participants	65 years to 85 years (Older Adult)
Interventions	62 subjects will be randomised 1:1 to receive a dose of high dose adjuvated RSV vaccine or placebo followed by a second dose 2 months later
Outcomes	Immunogenicity and Safety
Notes	Adjuvant not described. Company (ClinicalTrials.gov_Inquiries@pfizer.com) contacted

Footnotes

Characteristics of ongoing studies

NCT03049488

Study name	Dose, Safety, Tolerability and Immunogenicity of a Stabilized Prefusion RSV F Subunit Protein Vaccine, VRC-RSVRGP084-00-VP (DS-Cav1), Alone or With Alum Adjuvant, in Healthy Adults
Methods	Phase I Randomized, Open-Label Clinical Trial
Participants	Healthy Adults
Interventions	VRC-RSVRGP084-00-VP (DS-Cav1) was developed by VRC, NIAID and is composed of the respiratory syncytial virus (RSV) fusion (F) glycoprotein ectodomain. Adjuvant is an aluminium hydroxide suspension (alum) provided in a sterile, pyrogenfree suspension at a concentration of 5 mg/mL in 3 mL glass vials filled to 0.7 +/- 0.10 mL. The alum dose is 500 mcg ● Group: 1; Subjects: 15; Dose: 50mcg; Day 0: DS-Cav1; Week 12 [1]: DS-Cav1 ● Group: 2; Subjects: 15; Dose: 50mcg; Day 0: DS-Cav1 + alum; Week 12 [1]: DS-Cav1 + alum ● Group: 3; Subjects: 15; Dose: 150mcg; Day 0: DS-Cav1; Week 12 [1]: DS-Cav1 ● Group: 4; Subjects: 15; Dose: 150mcg; Day 0: DS-Cav1 + alum; Week 12 [1]: DS-Cav1 + alum ● Group: 5; Subjects: 15; Dose: 500mcg; Day 0: DS-Cav;1 Week 12 [1]: DS-Cav1 ● Group: 6; Subjects: 15; Dose: 500mcg; Day 0: DS-Cav1 + alum; Week 12 [1]: DS-Cav1 + alum
Outcomes	Safet and immunogenicity
Starting date	February 9, 2017
Contact information	Grace L Chen, M.D. National Institute of Allergy and Infectious Diseases (NIAID)
Notes	

NCT03776994

Study name	Venezuelan Equine Encephalitis Monovalent Virus-Like Particle Vaccine (VEEV)
Methods	Phase 1 dose escalation radnomised study
Participants	Healthy volunteers between 18 and 50 years
Interventions	Non-adjuvanted and adjuvated monovalent VEE VLP Vaccine administered via intramuscular injection at escalating doses of 2 µg, 10 µg, and 20 µg as a 2-dose primary series (Day 0, Day 28) with a Day 140 booster dose
Outcomes	Safety and immunogenicity
Starting date	December 17, 2018
Contact information	SRI Biosciences Clinical Trials Unit, Plymouth, Michigan, United States, 48170. Gloria Jen, MS 734-527-4201 gloria.jen@sri.com
Notes	

NCT03879603

Study name	VRC 313: A Trivalent Virus-like Particle (VLP) Encephalitis Vaccine (WEVEE) in Healthy Adults
Methods	Phase 1, randomised, open-label, dose-escalation study
Participants	Healthy Adults
Interventions	WEVEE vaccine alone (Group 1, Group 3, and Group 5) or, WEVEE vaccine mixed with alum (Group 2, Group 4, and Group 6, respectively)
Outcomes	Safety and immunogenicity
Starting date	March 19, 2019
Contact information	Julie E Ledgerwood, DO, National Institute of Allergy and Infectious Diseases (NIAID)
Notes	

NCT04009824

Study name	Evaluating the Safety and Immunogenicity of AGS-v PLUS, a Universal Mosquito-Borne Disease and Mosquito Control Vaccine
Methods	Randomized, Double-Blind, Placebo-Controlled, Single-Center, Phase 1 Study
Participants	Healthy Volunteers
Interventions	Participants in Group 1 will receive placebo on Days 1 and 22. Participants in Group 2 will receive unadjuvanted AGS-v PLUS vaccine on Days 1 and 22. Participants in Group 3 will receive Montanide ISA-51 adjuvated AGS-v PLUS vaccine on Day 1 and placebo on Day 22. Participants in Group 4 will receive Montanide ISA-51 adjuvated AGS-v PLUS vaccine on Days 1 and 22. Participants in Group 5 will receive Alhydrogel® adjuvated AGS-v PLUS vaccine on Days 1 and 22.
Outcomes	Safety and immunogenicity
Starting date	July 5, 2019
Contact information	Matthew B. Laurens, MD, MPH, Matthew J. Memoli, MD, MS, National Institute of Allergy and Infectious Diseases (NIAID)
Notes	

NCT04056117

Study name	Safety and Immunogenicity of a Shigella-Tetravalent Bioconjugate Vaccine: A Phase 1/2 Randomized Controlled and Age Descending Study Including Dose Finding in 9 Month Old Infants
Methods	Phase 1/2 trial is a multicentre, randomised, controlled, double blind, with two steps; a step 1 with a safety cohort and a step 2 with a dose-finding cohort
Participants	8 Months to 50 Years (Child, Adult)
Interventions	Tetravalent bioconjugate candidate vaccine Shigella4V. Adults will receive a 2 dose schedule, children and infants will receive a 3 dose schedule. For each vaccine dose, formulation with and without Aluminium adjuvant will be tested.

Outcomes	Safety and immunogenicity
Starting date	August 14, 2019
Contact information	Mainga Hamaluba, MD Mainga Hamaluba <MHamaluba@kemri-wellcome.org>
Notes	

NCT04563533

Study name	Clinical Trial of Quadrivalent Recombinant Norovirus Vaccine (Pichia Pastoris)
Methods	A single-center, randomised, blinded, multi-cohort, placebo-controlled trial
Participants	6 Weeks and older (Child, Adult, Older Adult)
Interventions	Tetravalent recombinant Norovirus vaccine There are 10 cohorts, randomly divided into experimental group (low-dose or high-dose), placebo group (containing aluminium), and the random ratio of each entry stage is 2:1. Among them, infants and the elderly will be added a placebo group without aluminium adjuvant, test group (low dose or high dose), placebo group (aluminium), placebo (aluminum-free), the random ratio is 2:1:1; Dose expansion stage: It is planned to enroll 240 subjects in Rong'an County. The elderly, infants, and infants will be divided into 3 cohorts, and each age group will be randomly divided into low-dose or high-dose groups, with a random ratio of 1:1.
Outcomes	Safety and immunogenicity
Starting date	September 24, 2020
Contact information	huangt24@126.com Liuzhou Center for Disease Control and Prevention
Notes	

Footnotes

Supplementary material

Aluminium adjuvants versus placebo or no intervention in vaccine development randomised clinical trials. A systematic review with meta-analysis and Trial Sequential Analysis

Sara Russo Krauss (0000-0002-5761-2118), research associate[1], Marija Barbateskovic, clinical research consultant [1], Sarah Louise Klingenberg, information specialist[1], Snezana Djurisic, clinical research consultant[1], Sesilje B Petersen, scientist,[2], Mette Kenfelt, chief executive officer,[3] De Zhao Kong, deputy chief physician and head of department[4,5], Janus C Jakobsen, professor and chief physician[1,6], Christian Gluud (0000-0002-8861-0799), professor and head of department[1,6]

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[3] Birkerød, Denmark.

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[5] Liaoning University of Traditional Chinese Medicine, Shen Yang, China

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Database	Time span	Search strategy
Cochrane Central Register of Controlled Trials in the Cochrane Library	2021, Issue 6	#1 MeSH descriptor: [Adjuvants, Immunologic] explode all trees #2 MeSH descriptor: [Vaccine Excipients] explode all trees #3 ((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*)) #4 #1 or #2 or #3 #5 MeSH descriptor: [Vaccines] explode all trees #6 vaccin* #7 (biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #8 #5 or #6 or #7 #9 #4 and #8
MEDLINE Ovid	1946 to June 2021	1. exp Adjuvants, Immunologic/ 2. exp Vaccine Excipients/ 3. ((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 4. 1 or 2 or 3 5. exp Vaccines/ 6. vaccin*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 7. (biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 8. 5 or 6 or 7 9. 4 and 8 10. randomized controlled trial.pt. 11. controlled clinical trial.pt. 12. clinical trials as topic.sh. 13. (random* or placebo*).ab. or trial.ti. 14. 10 or 11 or 12 or 13 15. exp animals/ not humans.sh.

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		16. 14 not 15 17. 9 and 16
Embase Ovid	1974 to June 2021	1. exp immunological adjuvant/ 2. exp vaccine excipient/ 3. ((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 4. 1 or 2 or 3 5. exp vaccine/ 6. vaccin*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 7. (biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 8. 5 or 6 or 7 9. 4 and 8 10. Randomized controlled trial/ or Controlled clinical trial/ or trial.ti. 11. (random* or blind* or placebo*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 12. 10 or 11 13. Animal experiment/ not (human experiment/ or human/) 14. 12 not 13 15. 9 and 14
LILACS (Bireme)	1982 to June 2021	((immunologic$ or alum$ or vaccine$) and (adjuvan$ or excipient$)) [Words] and (vaccin$ or biothrax or DT$ or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) [Words]

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BIOSIS (Web of Science)	1969 to June 2021	#5 #4 AND #3 #4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*) #3 #2 AND #1 #2 TS=(vaccin* or biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #1 TS=((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*))
Science Citation Index Expanded (Web of Science)	1900 to June 2021	#5 #4 AND #3 #4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*) #3 #2 AND #1 #2 TS=(vaccin* or biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #1 TS=((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*))
Conference Proceedings Citation Index – Science (Web of Science)	1990 to June 2021	#5 #4 AND #3 #4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*) #3 #2 AND #1 #2 TS=(vaccin* or biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #1 TS=((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*))
CNKI	September, 2021	#1 **主**题检索：铝佐剂、硫酸铝、氢氧化铝、磷酸铝、羟基磷酸铝、铵明矾、钾明矾、硫酸铝铵、硫酸铝钾 #2 **主**题检索：疫苗 #3 **全文**检索：随机 #4 #1 AND #2 AND #3

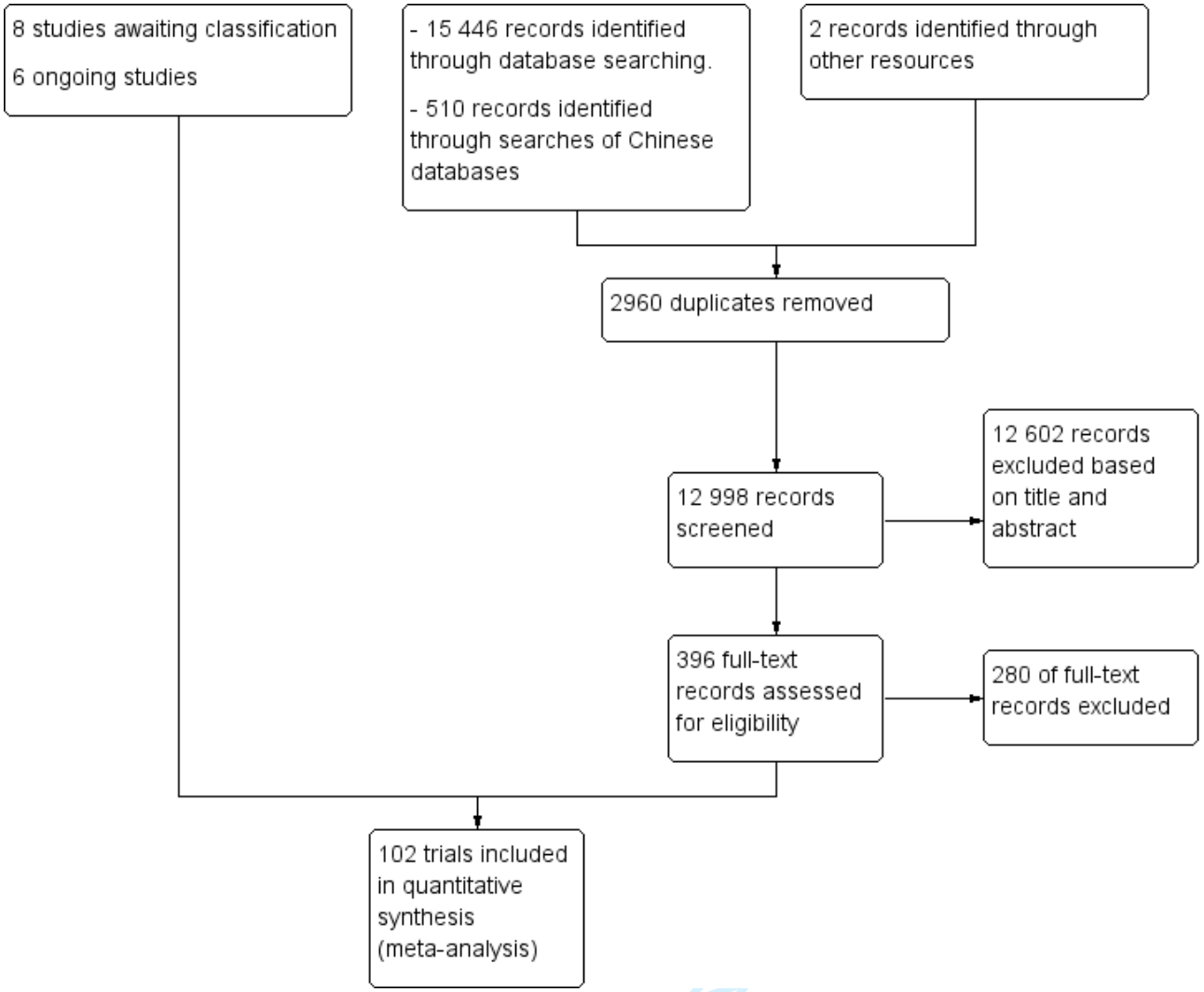
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Wanfang	September, 2021	主题:(铝佐剂 + 硫酸铝 + 氢氧化铝 + 磷酸铝 + 羟基磷酸铝 + 铵明矾 + 钾明矾 + 硫酸铝铵 + 硫酸铝钾)*主题:(疫苗)*全部:(随机)
VIP	September, 2021	任意字段：铝佐剂 OR 硫酸铝 OR 氢氧化铝 OR 磷酸铝 OR 羟基磷酸铝 OR 铵明矾 OR 钾明矾 OR 硫酸铝铵 OR 硫酸铝钾 任意字段：疫苗 注：加上随机之后变成0
Sinomed	September, 2021	("随机"[全部字段:智能]) AND ("疫苗"[全部字段:智能]) AND ("铝佐剂"[全部字段:智能] OR "硫酸铝"[全部字段:智能] OR "氢氧化铝"[全部字段:智能] OR "磷酸铝"[全部字段:智能] OR "羟基磷酸铝"[全部字段:智能] OR "铵明矾"[全部字段:智能] OR "钾明矾"[全部字段:智能] OR "硫酸铝铵"[全部字段:智能] OR "硫酸铝钾"[全部字段:智能])

Supplementary Table S1. Search strategies

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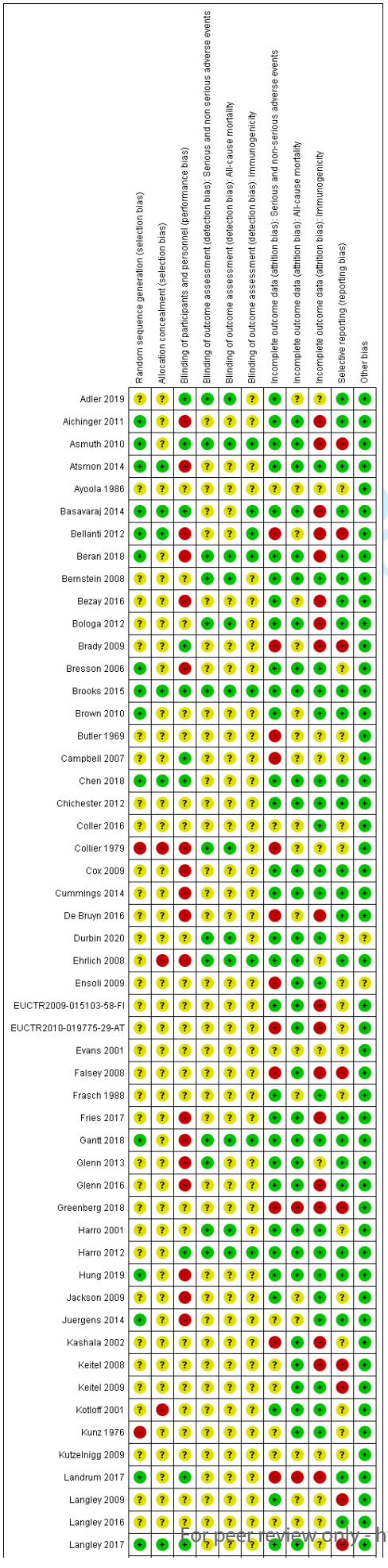
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Supplementary Figure S1. Study flow diagram. Date of search June 30, 2021.

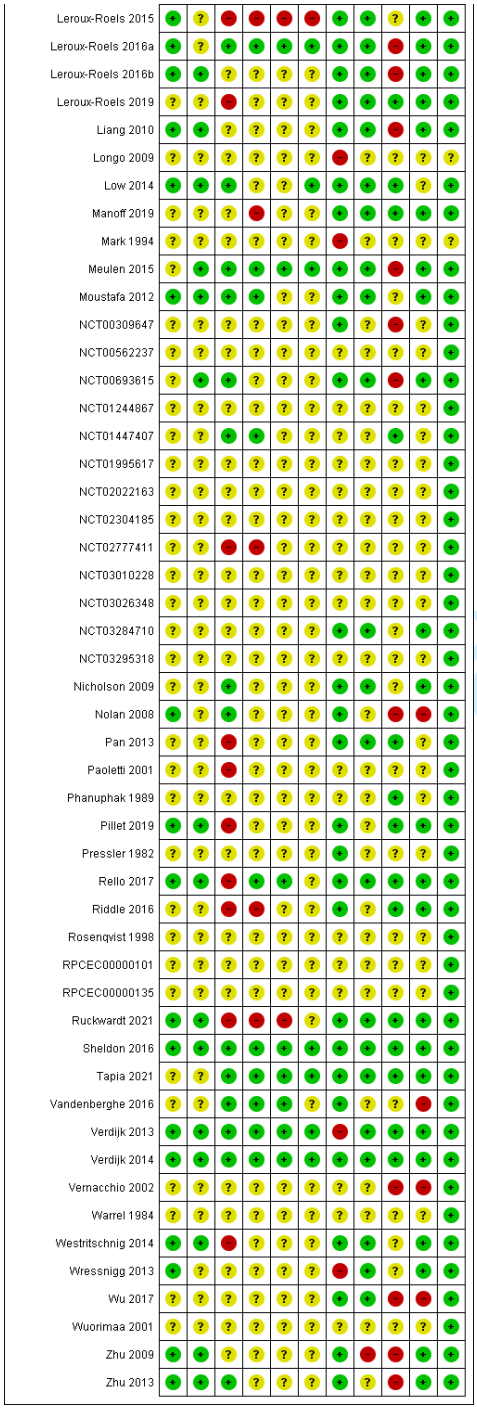
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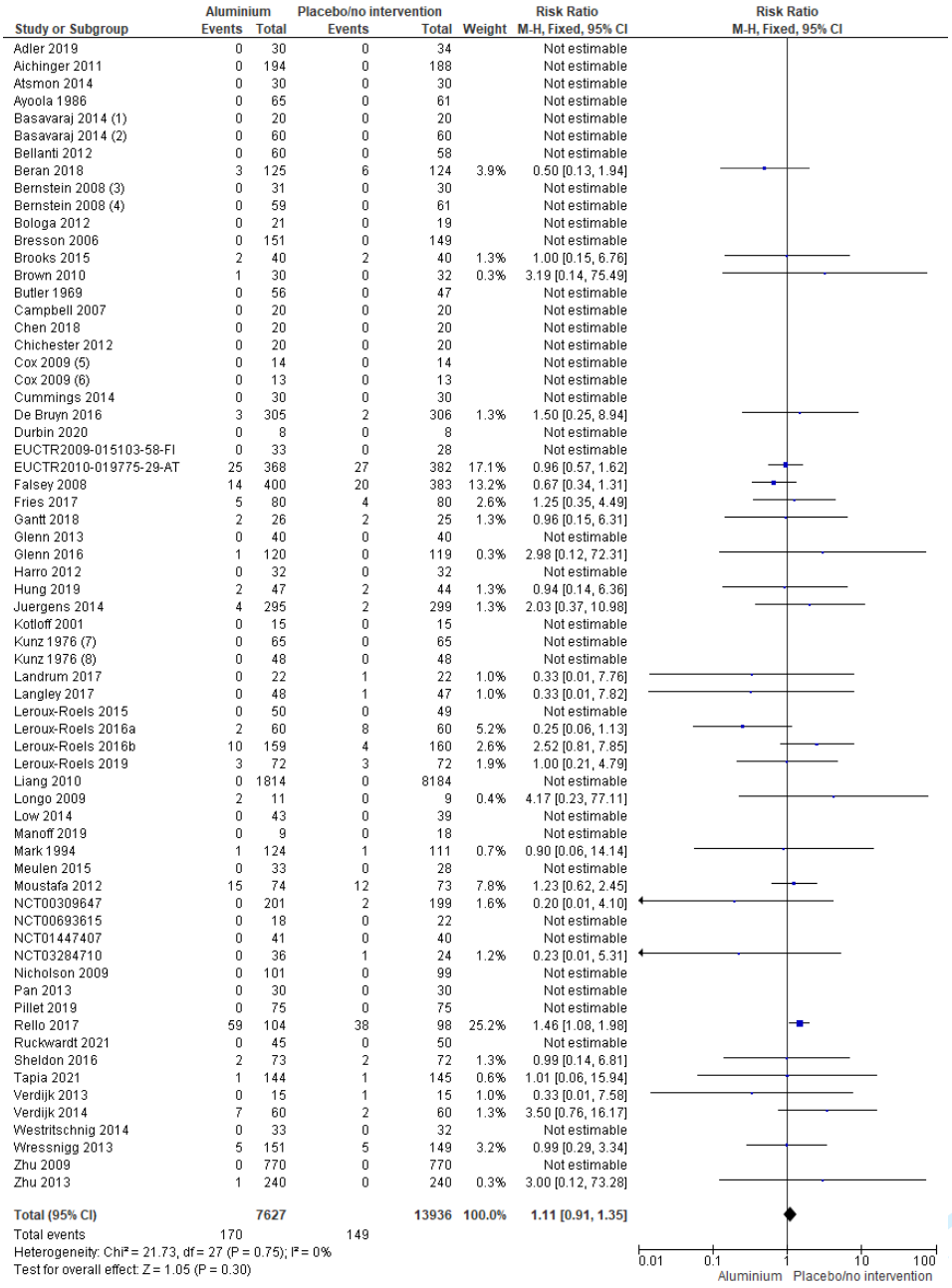
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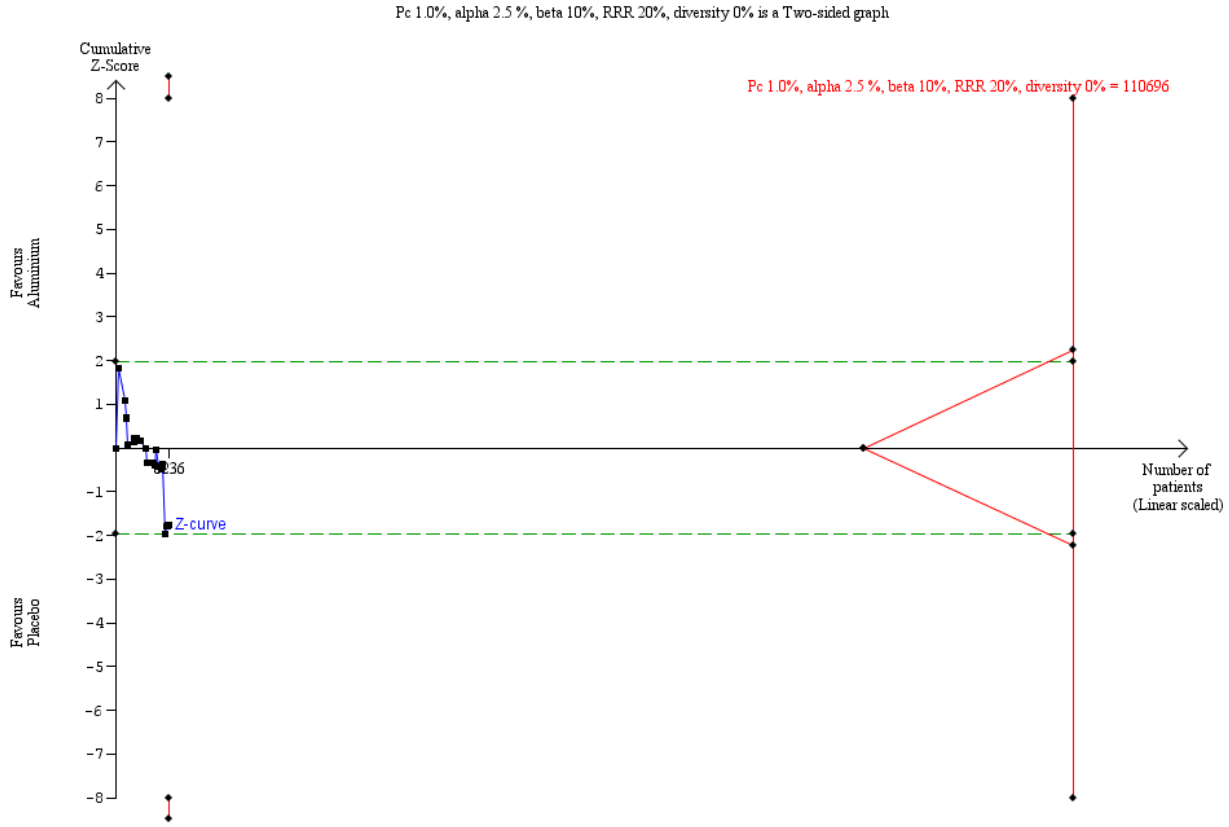
Supplementary Figure S2. Risk of bias summary: review authors' judgements about each risk of bias item for each included trial.



Supplementary Figure S3

Fixed-effect meta-analysis of the effect of aluminium adjuvant compared with placebo or no intervention on the proportion of participants with one or more serious adverse events.

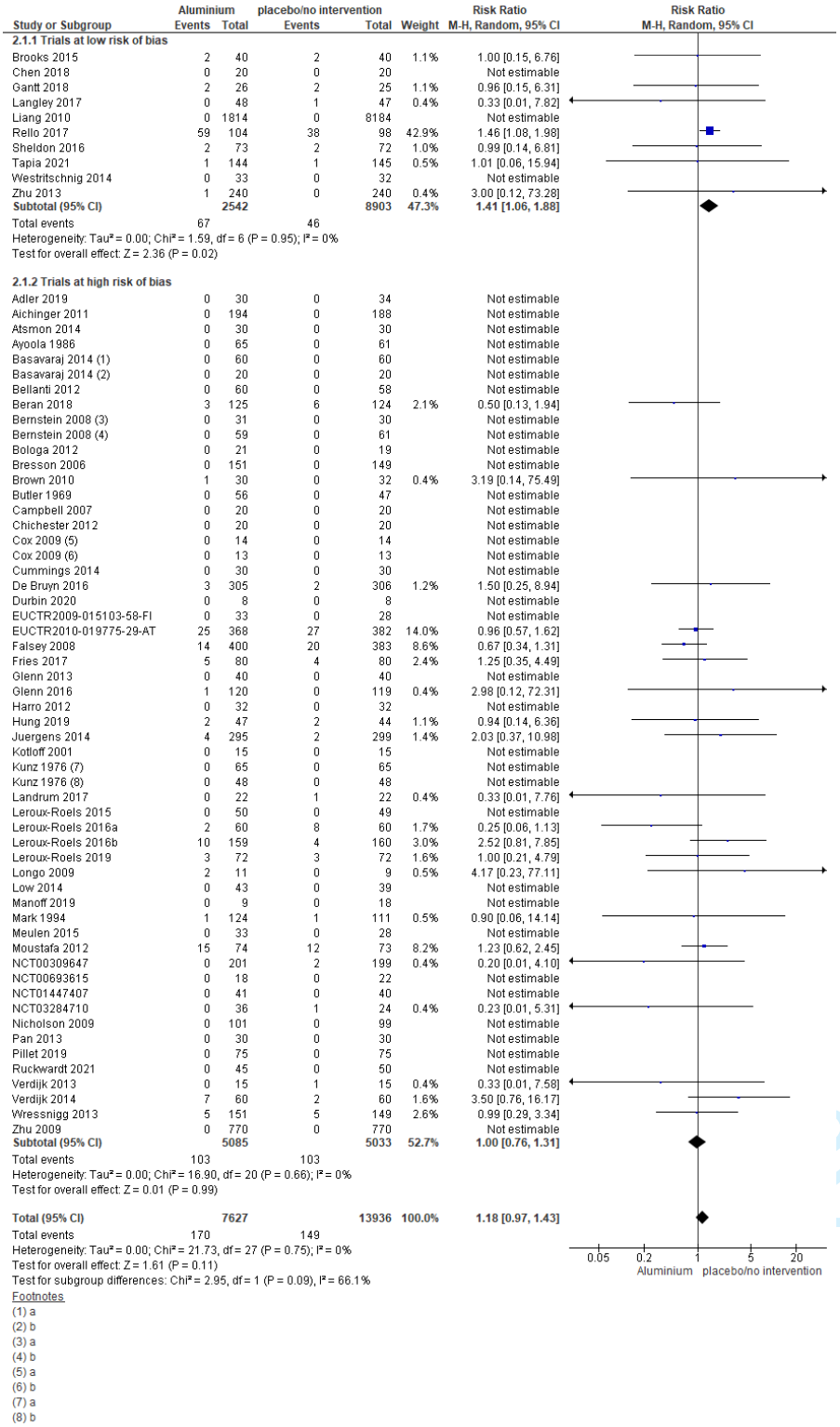
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Supplementary Figure S4. Trial Sequential Analysis using random-effects meta-analysis of the effects of aluminium adjuvants versus placebo or no intervention on the proportion of participants with serious adverse events. The analysis was based on a proportion in the control group (Pc) of 1.0 %, a relative risk reduction (RRR) of 20%, an alpha of 2.5%, a beta of 10%, and diversity of 0%. None of the trial sequential monitoring boundaries have been reached or surpassed.

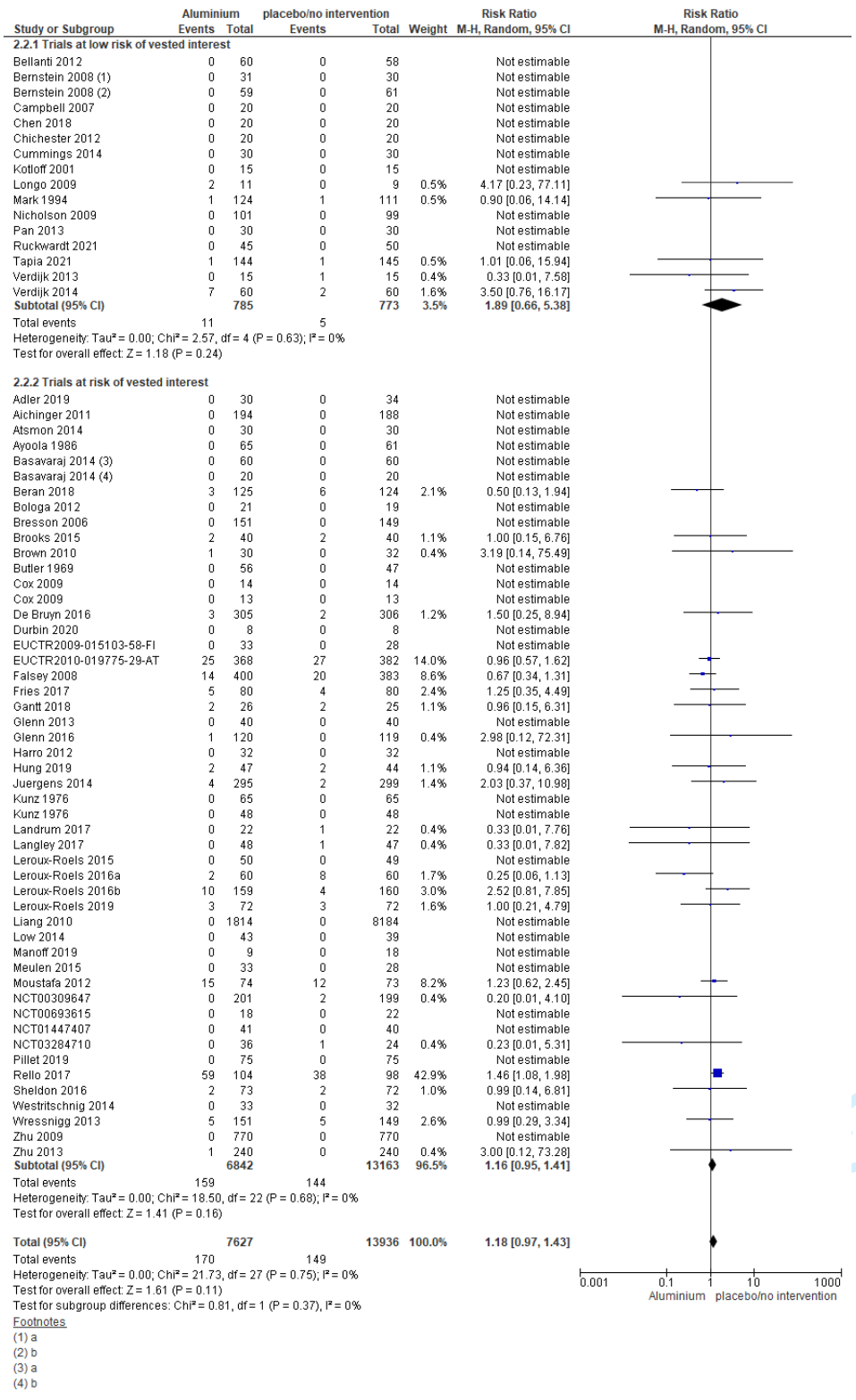
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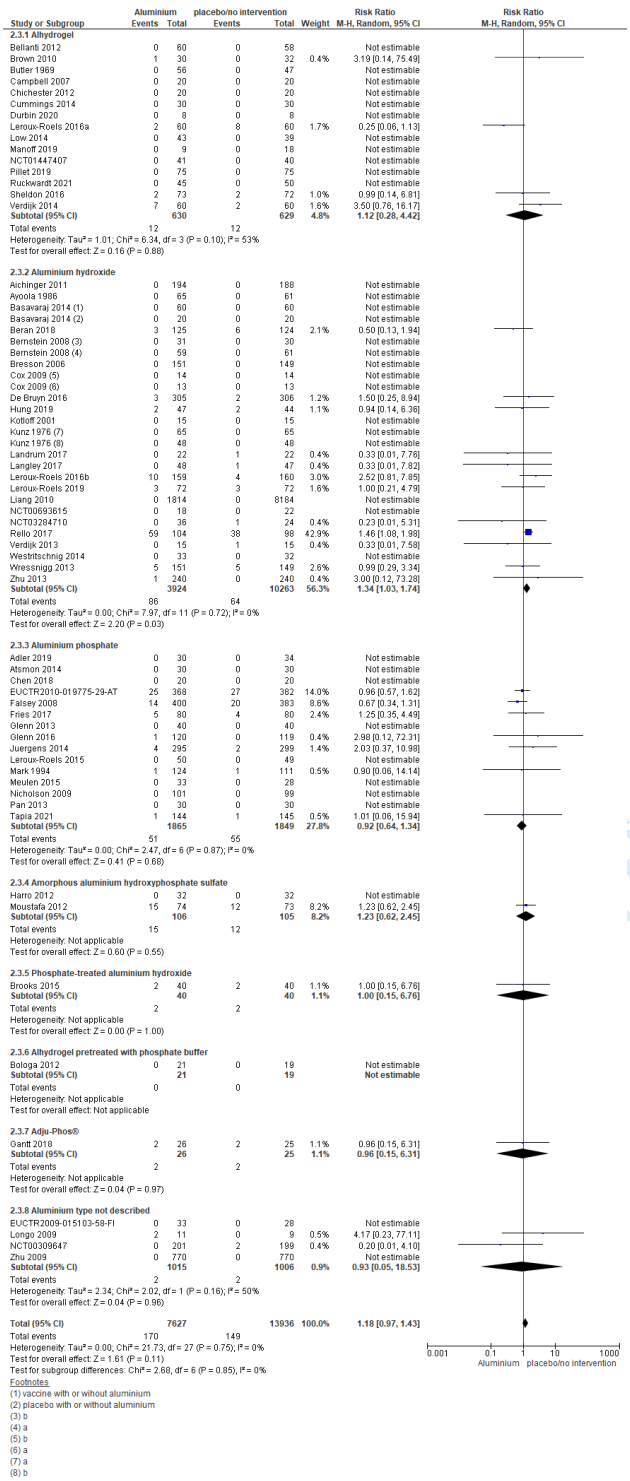
Supplementary Figure S5. Subgroup analysis for serious adverse events according to risk of bias.

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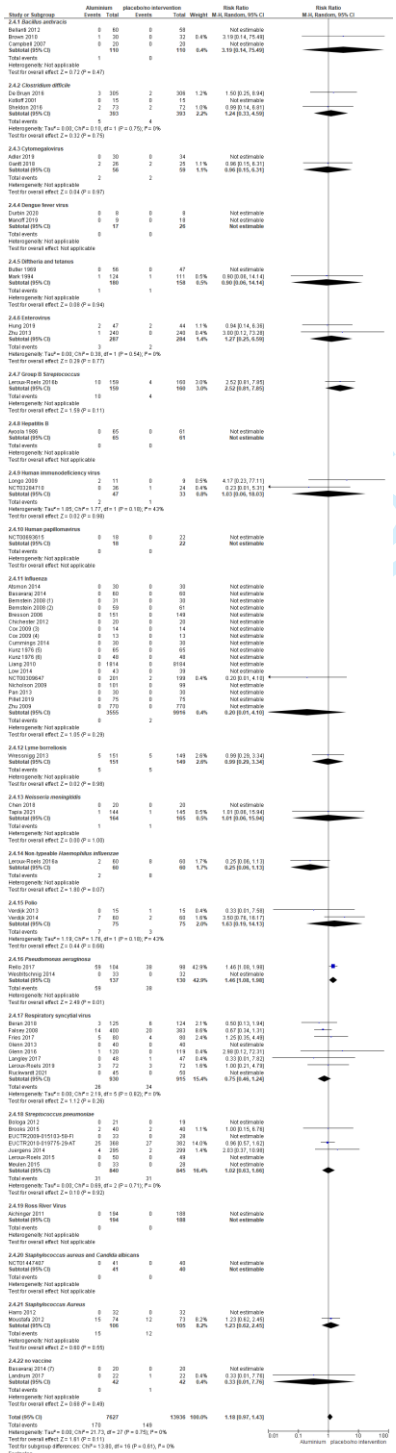
Supplementary Figure S6. Subgroup analysis of serious adverse events according to risk of vested interested.

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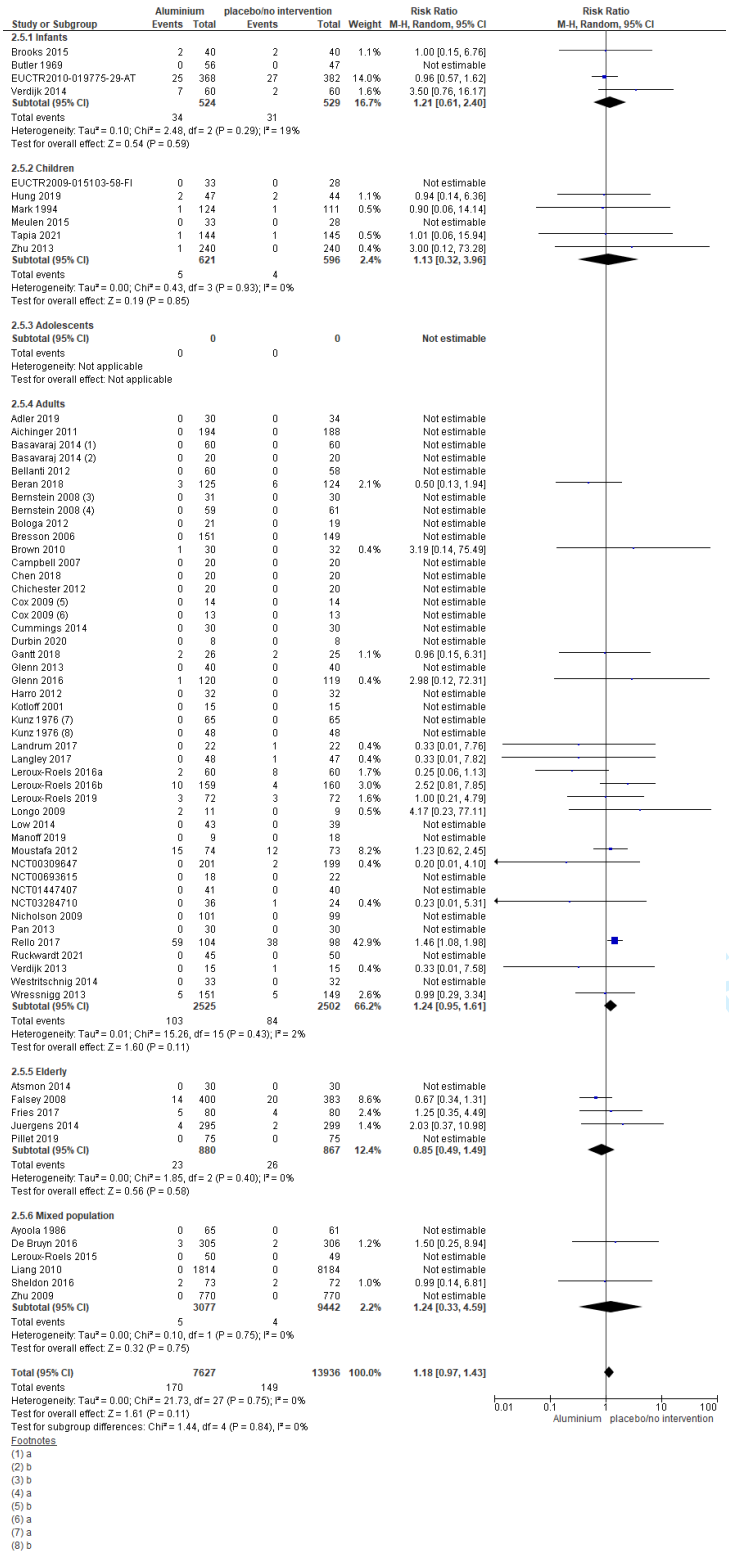
Supplementary Figure S7. Subgroup analysis of serious adverse events according to adjuvant type.

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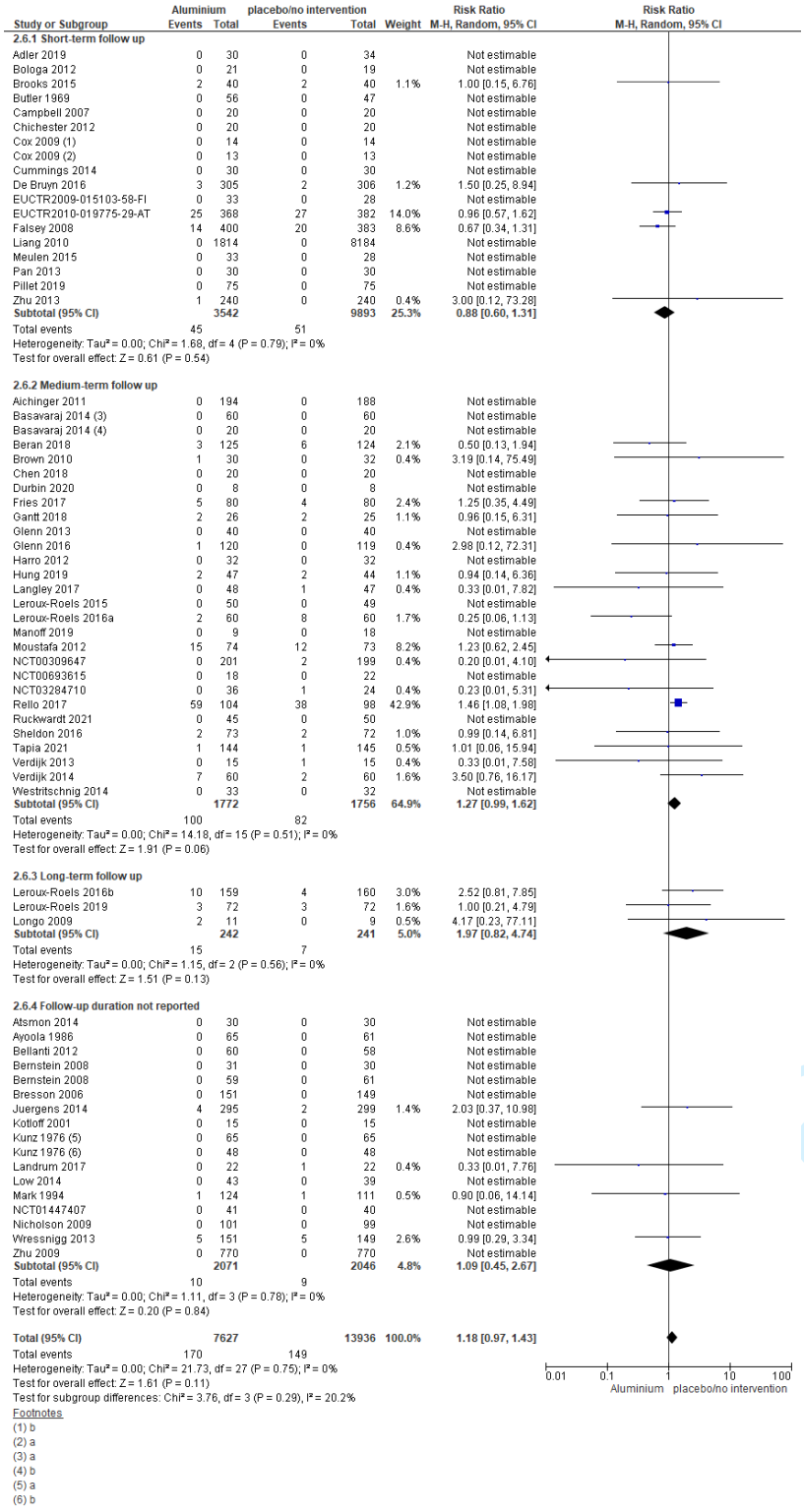
Supplementary Figure S8. Subgroup analysis of serious adverse events according to vaccine type.

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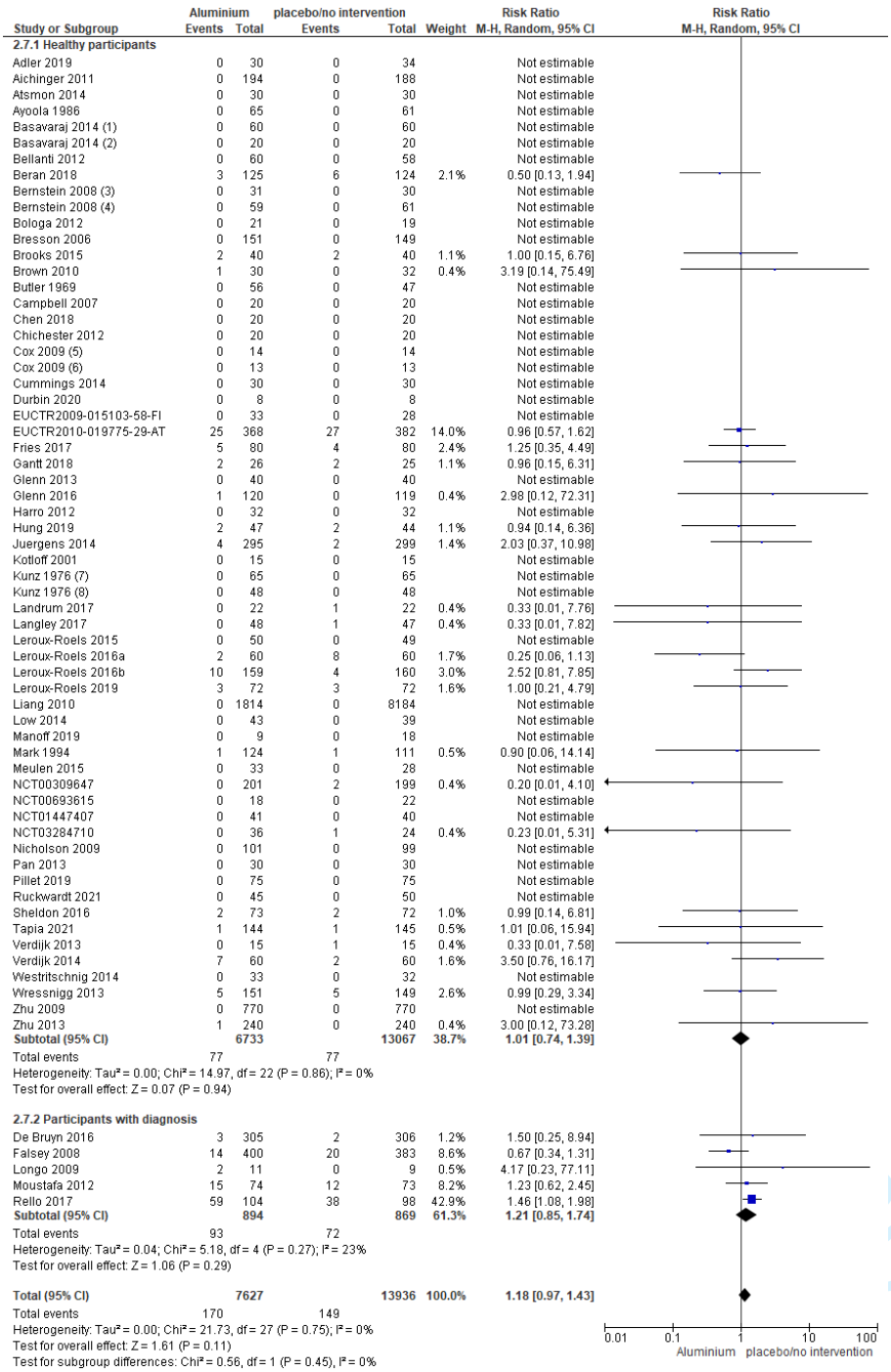
Supplementary Figure S9. Subgroup analysis of serious adverse events according to age of the participants.

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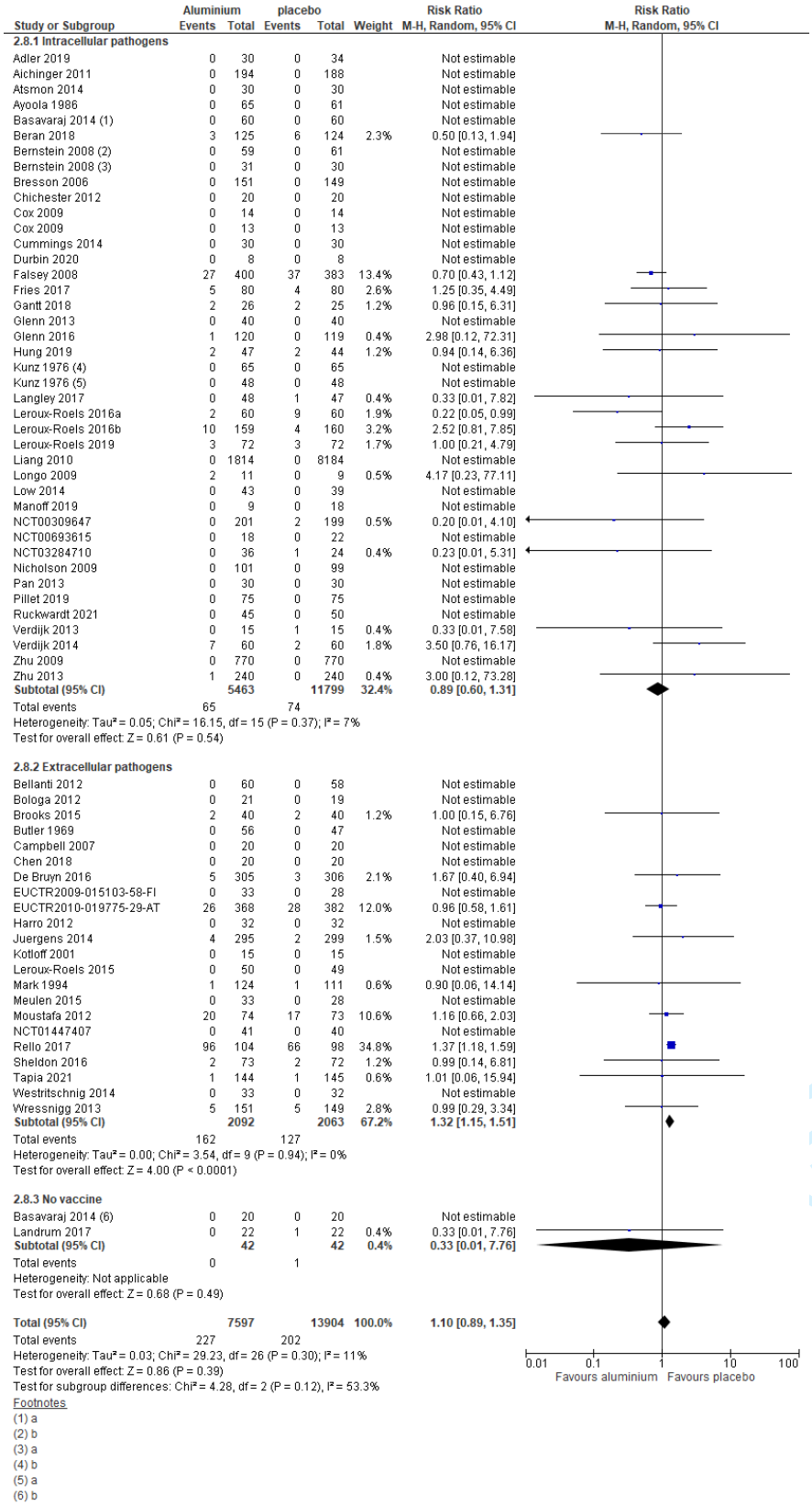
Supplementary Figure S10. Subgroup analysis of serious adverse events according to follow up duration.

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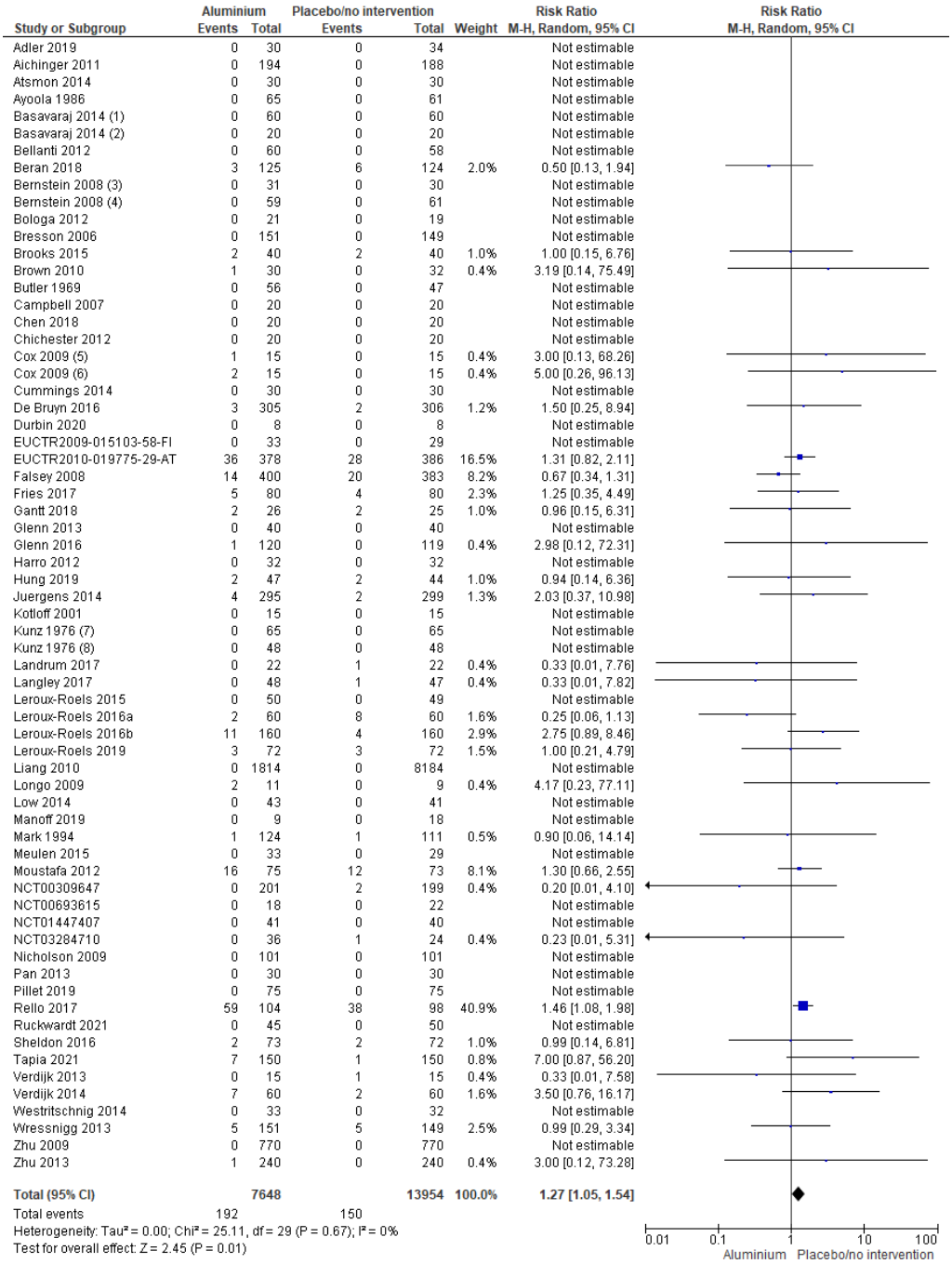
Supplementary Figure S11. Subgroup analysis of serious adverse events according to heath status of the participants.

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Supplementary Figure S12. Subgroup analysis of serious adverse events according to pathogen type the vaccine is directed against.

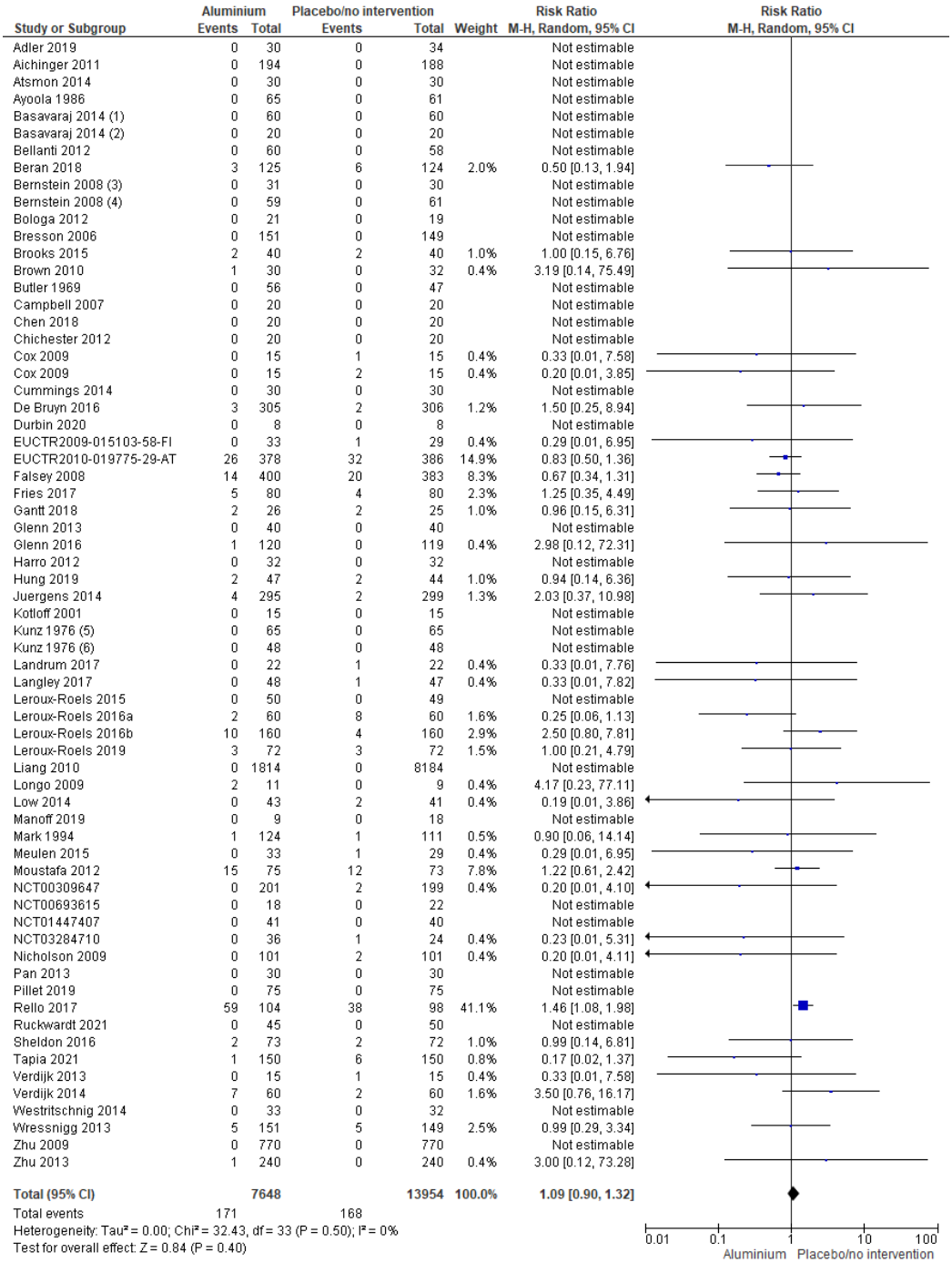
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Supplementary Figure S13. Forest plot of 'worst best' sensitivity analysis of serious adverse events.

We assessed the potential impact of missing data in the 'worst-best-case' scenario: it was assumed that all participants lost to follow-up in the experimental group had a serious adverse event; and all those with missing outcomes in the control group had no serious adverse events.

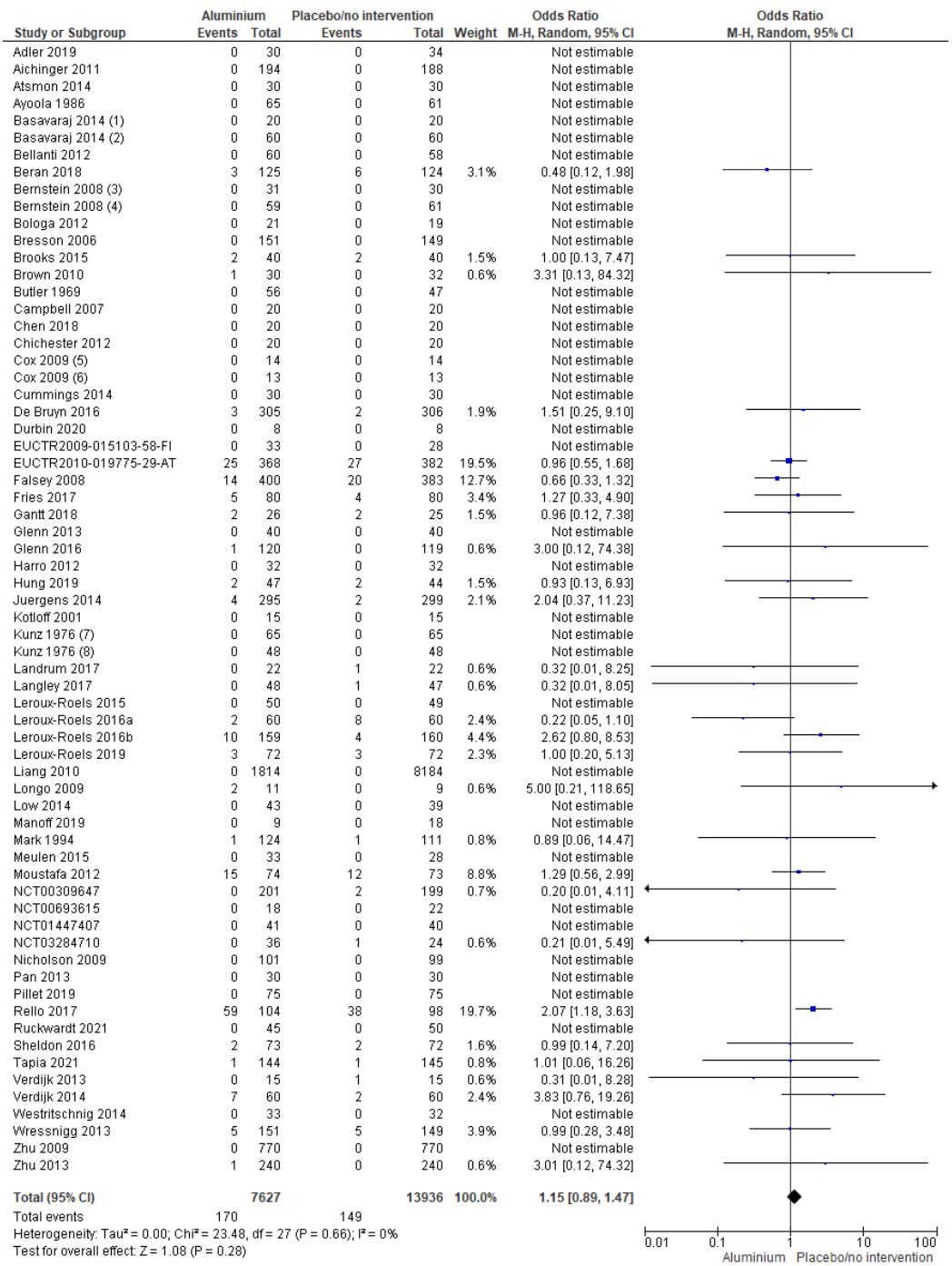
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Supplementary Figure S14. Forest plot of 'best worst' sensitivity analysis of serious adverse events.

We assessed the potential impact of missing data in the 'best-worst-case' scenario: it was assumed that all participants lost to follow-up in the control group had a serious adverse event; and all those with missing outcomes in the experimental group had no serious adverse events.

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Study or Subgroup	Aluminium Events	Aluminium Total	Placebo/no intervention Events	Placebo/no intervention Total	Weight	Odds Ratio M-H, Random, 95% CI
Adler 2019	0	30	0	34		Not estimable
Aichinger 2011	0	194	0	188		Not estimable
Atsmon 2014	0	30	0	30		Not estimable
Ayoola 1986	0	65	0	61		Not estimable
Basavaraj 2014 (1)	0	20	0	20		Not estimable
Basavaraj 2014 (2)	0	60	0	60		Not estimable
Bellanti 2012	0	60	0	58		Not estimable
Beran 2018	3	125	6	124	3.1%	0.48 [0.12, 1.98]
Bernstein 2008 (3)	0	31	0	30		Not estimable
Bernstein 2008 (4)	0	59	0	61		Not estimable
Bologa 2012	0	21	0	19		Not estimable
Bresson 2006	0	151	0	149		Not estimable
Brooks 2015	2	40	2	40	1.5%	1.00 [0.13, 7.47]
Brown 2010	1	30	0	32	0.6%	3.31 [0.13, 84.32]
Butler 1969	0	56	0	47		Not estimable
Campbell 2007	0	20	0	20		Not estimable
Chen 2018	0	20	0	20		Not estimable
Chichester 2012	0	20	0	20		Not estimable
Cox 2009 (5)	0	14	0	14		Not estimable
Cox 2009 (6)	0	13	0	13		Not estimable
Cummings 2014	0	30	0	30		Not estimable
De Bruyn 2016	3	305	2	306	1.9%	1.51 [0.25, 9.10]
Durbin 2020	0	8	0	8		Not estimable
EUCTR2009-015103-58-FI	0	33	0	28		Not estimable
EUCTR2010-019775-29-AT	25	368	27	382	19.5%	0.96 [0.55, 1.68]
Falsey 2008	14	400	20	383	12.7%	0.66 [0.33, 1.32]
Fries 2017	5	80	4	80	3.4%	1.27 [0.33, 4.90]
Gantt 2018	2	26	2	25	1.5%	0.96 [0.12, 7.38]
Glenn 2013	0	40	0	40		Not estimable
Glenn 2016	1	120	0	119	0.6%	3.00 [0.12, 74.38]
Harro 2012	0	32	0	32		Not estimable
Hung 2019	2	47	2	44	1.5%	0.93 [0.13, 6.93]
Juergens 2014	4	295	2	299	2.1%	2.04 [0.37, 11.23]
Kotloff 2001	0	15	0	15		Not estimable
Kunz 1976 (7)	0	65	0	65		Not estimable
Kunz 1976 (8)	0	48	0	48		Not estimable
Landrum 2017	0	22	1	22	0.6%	0.32 [0.01, 8.25]
Langley 2017	0	48	1	47	0.6%	0.32 [0.01, 8.05]
Leroux-Roels 2015	0	50	0	49		Not estimable
Leroux-Roels 2016a	2	60	8	60	2.4%	0.22 [0.05, 1.10]
Leroux-Roels 2016b	10	159	4	160	4.4%	2.62 [0.80, 8.53]
Leroux-Roels 2019	3	72	3	72	2.3%	1.00 [0.20, 5.13]
Liang 2010	0	1814	0	8184		Not estimable
Longo 2009	2	11	0	9	0.6%	5.00 [0.21, 118.65]
Low 2014	0	43	0	39		Not estimable
Manoff 2019	0	9	0	18		Not estimable
Mark 1994	1	124	1	111	0.8%	0.89 [0.06, 14.47]
Meulen 2015	0	33	0	28		Not estimable
Moustafa 2012	15	74	12	73	8.8%	1.29 [0.56, 2.99]
NCT00309647	0	201	2	199	0.7%	0.20 [0.01, 4.11]
NCT00693615	0	18	0	22		Not estimable
NCT01447407	0	41	0	40		Not estimable
NCT03284710	0	36	1	24	0.6%	0.21 [0.01, 5.49]
Nicholson 2009	0	101	0	99		Not estimable
Pan 2013	0	30	0	30		Not estimable
Pillet 2019	0	75	0	75		Not estimable
Rello 2017	59	104	38	98	19.7%	2.07 [1.18, 3.63]
Ruckwardt 2021	0	45	0	50		Not estimable
Sheldon 2016	2	73	2	72	1.6%	0.99 [0.14, 7.20]
Tapia 2021	1	144	1	145	0.8%	1.01 [0.06, 16.26]
Verdijk 2013	0	15	1	15	0.6%	0.31 [0.01, 8.28]
Verdijk 2014	7	60	2	60	2.4%	3.83 [0.76, 19.26]
Westritschnig 2014	0	33	0	32		Not estimable
Wressnigg 2013	5	151	5	149	3.9%	0.99 [0.28, 3.48]
Zhu 2009	0	770	0	770		Not estimable
Zhu 2013	1	240	0	240	0.6%	3.01 [0.12, 74.32]
Total (95% CI)		**7627**		**13936**	**100.0%**	**1.15 [0.89, 1.47]**
Total events	170		149			

Heterogeneity: Tau² = 0.00; Chi² = 23.48, df = 27 (P = 0.66); I² = 0%
Test for overall effect: Z = 1.08 (P = 0.28)

Footnotes
(1) placebo with or without aluminium
(2) vaccine with or without aluminium
(3) 15 mcg of antigen
(4) 30 mcg of antigen
(5) 12 mcg antigen
(6) 24 mcg antigen
(7) subunit vaccine
(8) whole virus vaccine

Supplementary Figure S15. Meta-analysis of serious adverse events with odds ratio.

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Supplementary Analysis S1.

Individual serious adverse events analysis

Meta-analysis showed no evidence of a difference between aluminium adjuvants versus control when assessing individual serious adverse events:

pneumonia (RR 0.59, 95% CI 0.16 to 2.12; trials = 6);

severe pulmonary embolism (RR 0.33, 95% CI 0.03 to 3.14; trials = 2);

gastroenteritis (RR 1.23, 95% CI 0.75 to 2.02; trials = 9);

pancreatitis (RR 0.99, 95% CI 0.10 to 9.44; trials = 2);

diarrhea (RR 0.82, 95% CI 0.17 to 3.86; trials = 4);

breast cancer (RR 0.33, 95% CI 0.05 to 2.08; trials = 3);

ligament rupture (RR 0.33, 95% CI 0.03 to 3.18; trials = 2);

depression (RR 0.33, 95% CI 0.03 to 3.08; trials = 2);

otitis (RR 0.22, 95% CI 0.03 to 1.92; trials = 2);

coronary artery disease (RR 2.98, 95% CI 0.31 to 28.28; trials = 2);

cardiac arrest (RR 1.61, 95% CI 0.08 to 31.86; trials = 2);

sepsis (RR 0.94, 95% CI 0.14 to 6.34; trials = 2).

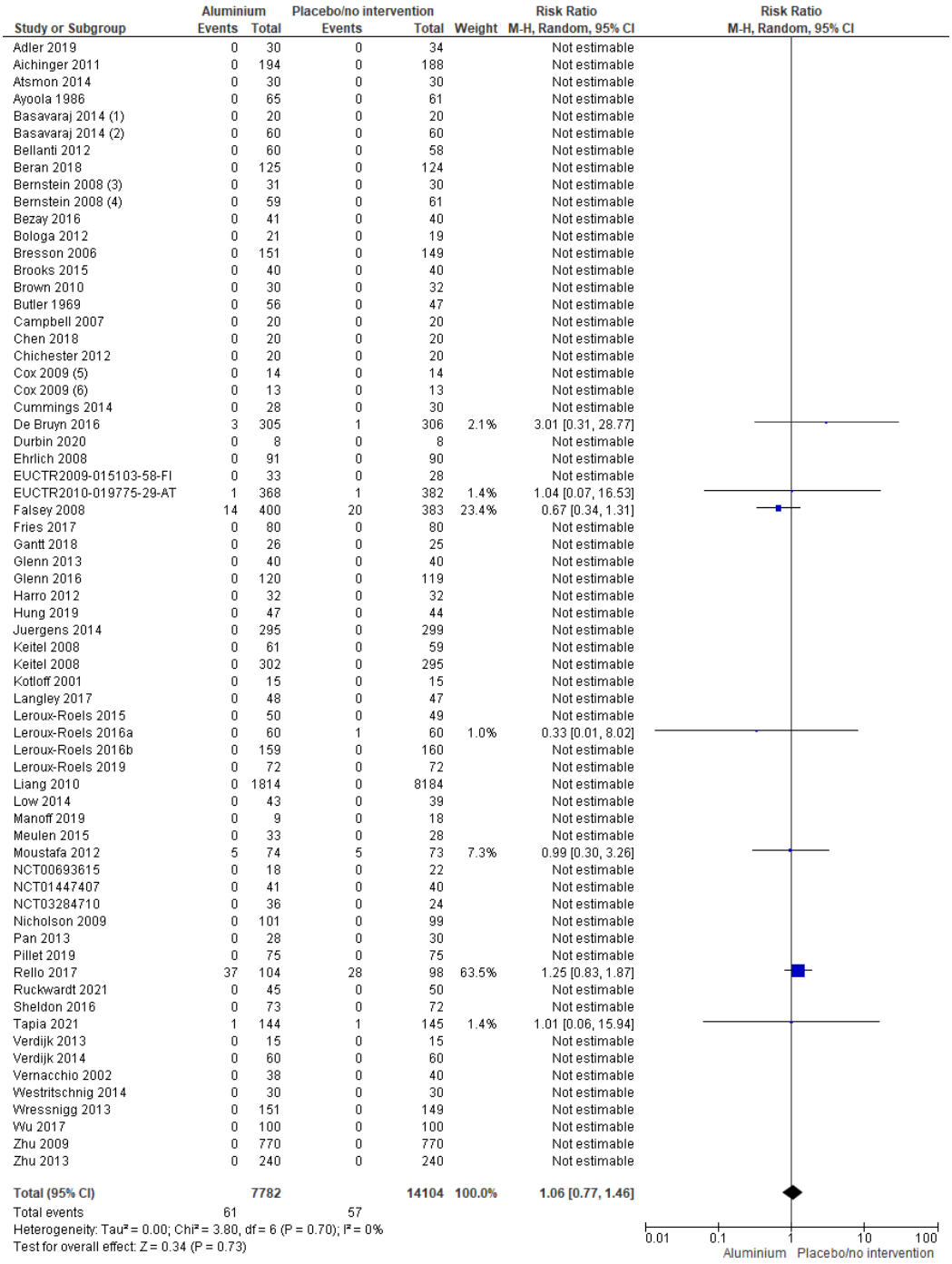
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Serious adverse events reported in the aluminum-adjuvanted group versus the placebo group		
Serious adverse event	**Aluminium event/total (study ID)**	**Placebo event/total (study ID)**
Astrocytoma	0/22 (Landrum2017)	1/22 (Landrum2017)
Congenital laryngomalacia	1/40 (Brooks 2015)	0/40 (Brooks 2015)
Acute appendicitis	1/30 (Glenn 2016)	0/29 (Glenn 2016)
Traumatic left knee dislocation	0/16 (Langley 2017)	1/16 (Langley 2017)
Tonsillitis	1/26 (Hung 2019)	1/24 (Hung 2019)
Cardiac failure	1/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Pulmonary hemorrhage	1/104 (Rello 2017)	0/98 (Rello 2017)
Shock	1/104 (Rello 2017)	0/98 (Rello 2017)
Prostate cancer	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Myocardial infarction	0/15 (Verdrijk 2013	1/15 (Verdrijk 2013
Moderate infective bursitis	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Severe cerebral metastasis and bleed	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Lung cancer	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Cubital tunnel syndrome	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Intravertebral disc protusion	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Pulmonary embolism and deep vein	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Bronchopneumonia	0/26 (Hung 2019)	1/24 (Hung 2019)
Viral encephalitis	1/120 (Zhu 2013)	0/120 (Zhu 2013)
Benign prostatic hyperplasia	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Elective abortion	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Musculoskeletal injuries	0/125 (Beran 2018)	1/124 (Beran 2018)
Spontaneous abortion	1/125 (Beran 2018)	0/124 (Beran 2018)
Transient ischemic attack	0/125 (Beran 2018)	1/124 (Beran 2018)
Umbilical hernia	1/125 (Beran 2018)	0/124 (Beran 2018)
Virally induced asthma	1/125 (Beran 2018)	0/124 (Beran 2018)
Colitis	1/125 (Beran 2018)	0/124 (Beran 2018)
Incisional hernia	1/125 (Beran 2018)	0/124 (Beran 2018)
Weber b fracture right ankle	1/125 (Beran 2018)	0/124 (Beran 2018)
Tibia fracture	1/125 (Beran 2018)	0/124 (Beran 2018)
Erysipelas cheek	1/125 (Beran 2018)	0/124 (Beran 2018)
Arrhythmia	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Cardiogenic Shock	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Gastric ulcer haemorrhage	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Gangrene	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Osteomyelitis bacteria	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Staphylococcal infection	1/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Ankle fracture	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Arteriovenous fistula occlusion	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Joint injury	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Diabetic ketoacidosis	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Azotaemia	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
COPD	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Pulmonary oedema	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Thrombosis	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Roseola infantum	0/26 (Hung 2019)	1/24 (Hung 2019)
Dehydration	0/26 (Hung 2019)	2/24 (Hung 2019)
Hand, foot, mouth disease	1/26 (Hung 2019)	0/24 (Hung 2019)
Clostridium difficile infection	1/26 (Hung 2019)	0/24 (Hung 2019)
Acute pyelonephritis	0/72 (Lerouz-Roels 2019)	1/72 (Lerouz-Roels 2019)
Cartilage injury	1/72 (Lerouz-Roels 2019)	0/72 (Lerouz-Roels 2019)
Head injury	0/72 (Lerouz-Roels 2019)	1/72 (Lerouz-Roels 2019)
Anaphylactic shock	1/72 (Lerouz-Roels 2019)	0/72 (Lerouz-Roels 2019)
Transient paralysis	1/25 (Bellino 2009)	0/22 (Bellino 2009)
Lipothymia-nausea	1/25 (Bellino 2009)	0/22 (Bellino 2009)
Laceration	0/26 (Gantt 2018)	1/25 (Gantt 2018)
Infections	1/124 (Mark 1994)	1/111 (Mark 1994)
Deep vein thrombosis	1/26 (Gantt 2018)	0/25 (Gantt 2018)

Supplementary Table S2. List of individual serious adverse events reported in only one trial. It was our aim to examine if the serious adverse events in the aluminium-adjuvanted groups of the included trials resembled the syndromes described in the literature related to aluminium toxicity. All included trials reporting serious adverse events were screened for detailed information about the events. However, the reporting was too sketchy to use the information for comparison with the syndromes. In general, the types of serious adverse events were not specified in the trials, and it was impossible to assess the total number of serious adverse events for each trial participant. We can therefore not evaluate whether the serious adverse event may be or maybe be not related to what is described as 'aluminium toxicity' in the literature.

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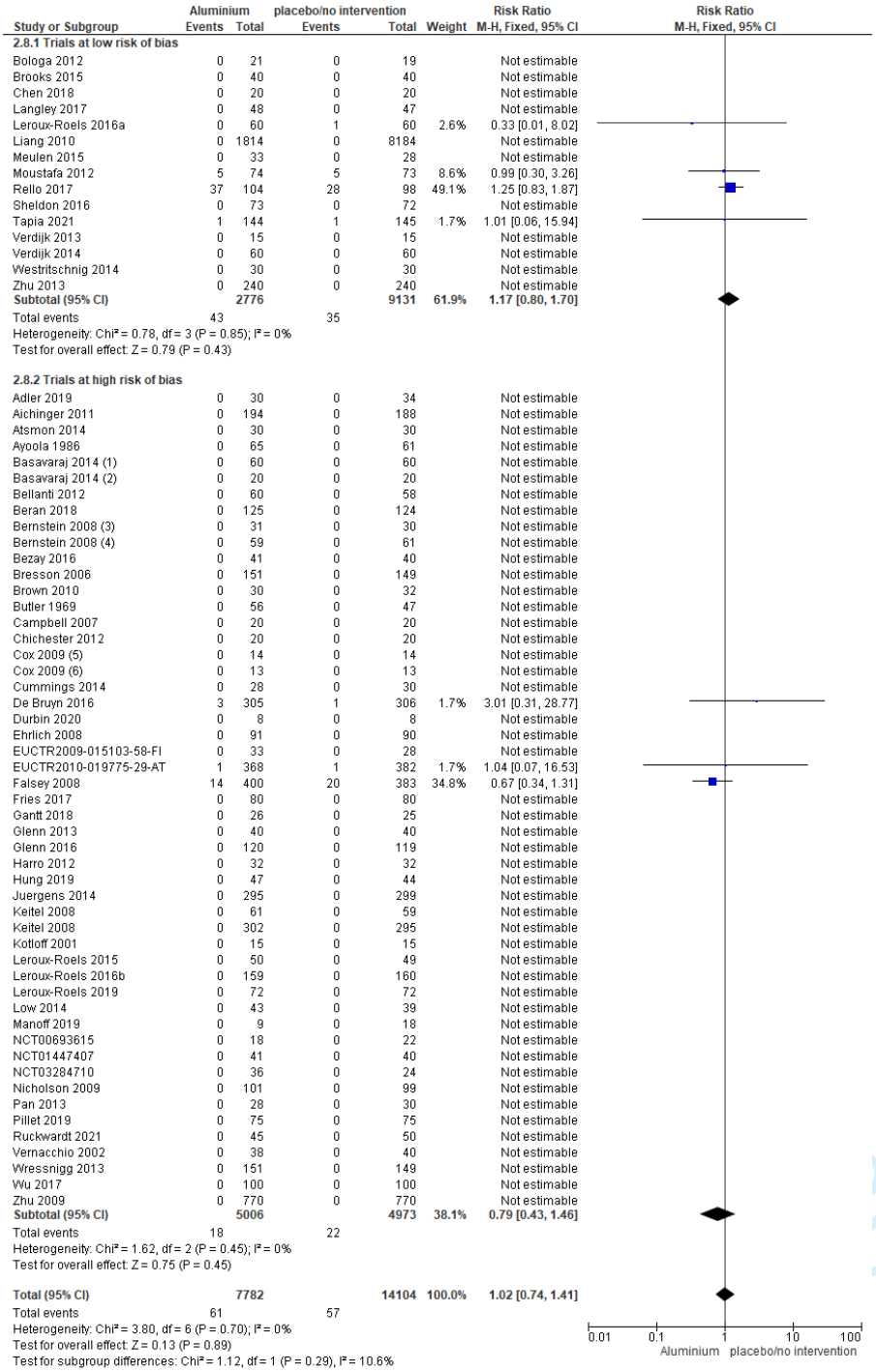


Footnotes
(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 15 mcg of antigen
(4) 30 mcg of antigen
(5) 12 mcg antigen
(6) 24 mcg antigen

Supplementary Figure S16.

Random-effects meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on all-cause mortality.

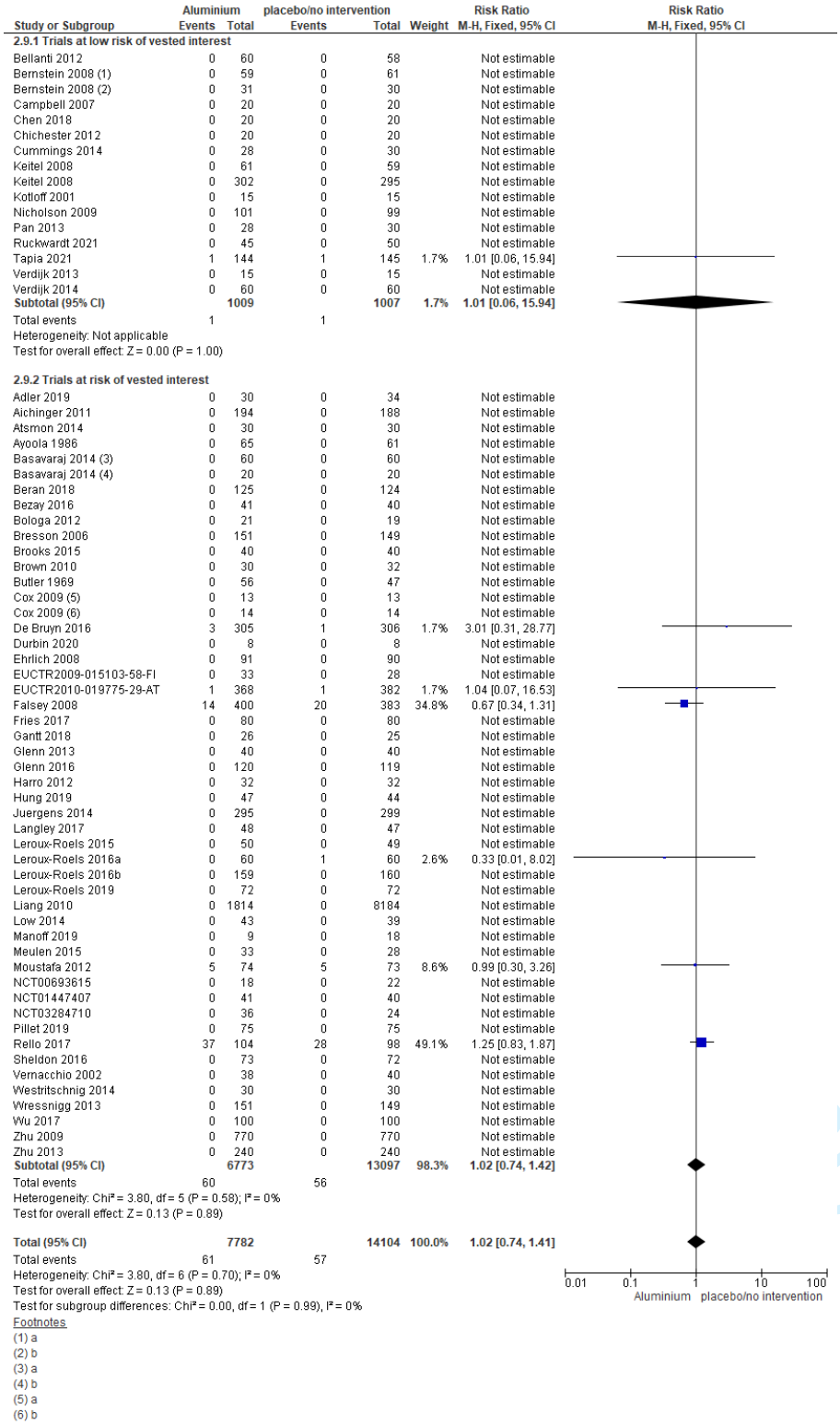
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Supplementary Figure S17.

Subgroup analysis of all-cause mortality according to risk of bias.

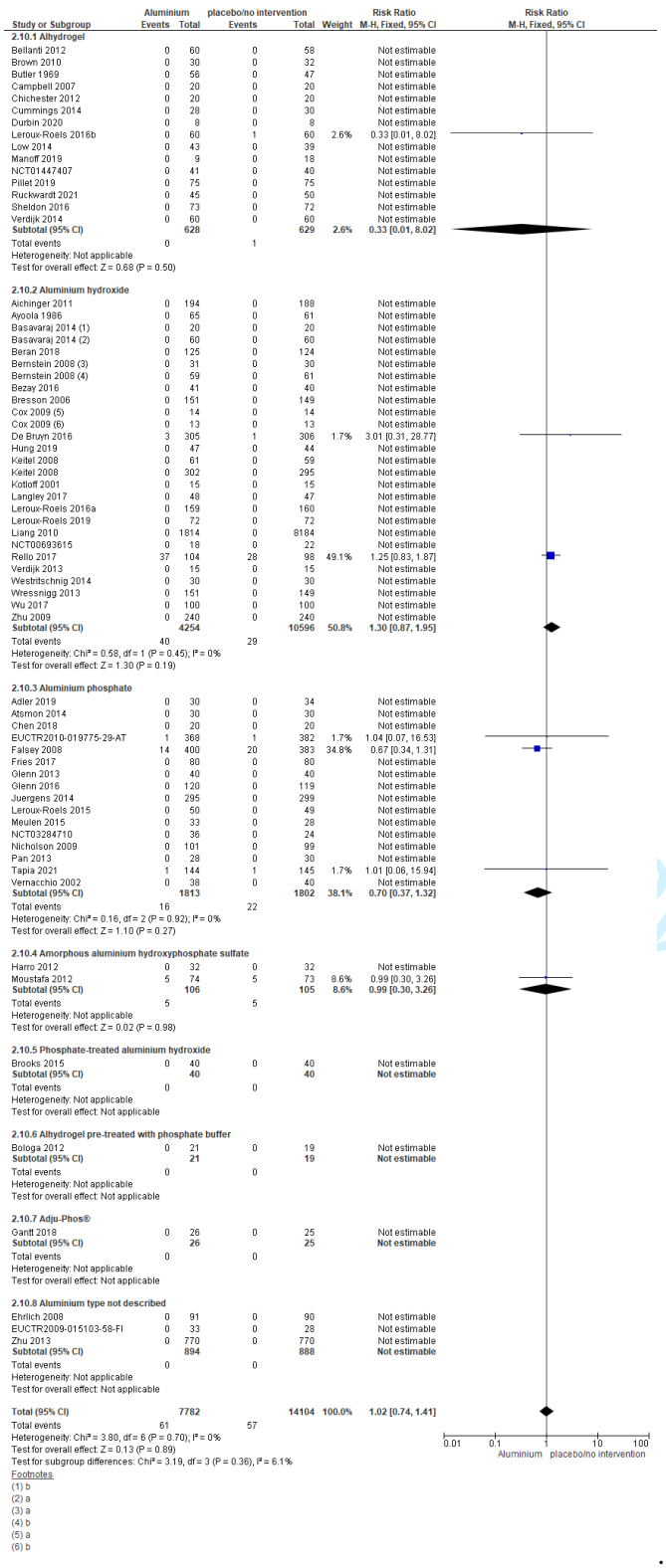
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Supplementary Figure S18.

Subgroup analysis of all-cause mortality according to risk of vested interest.

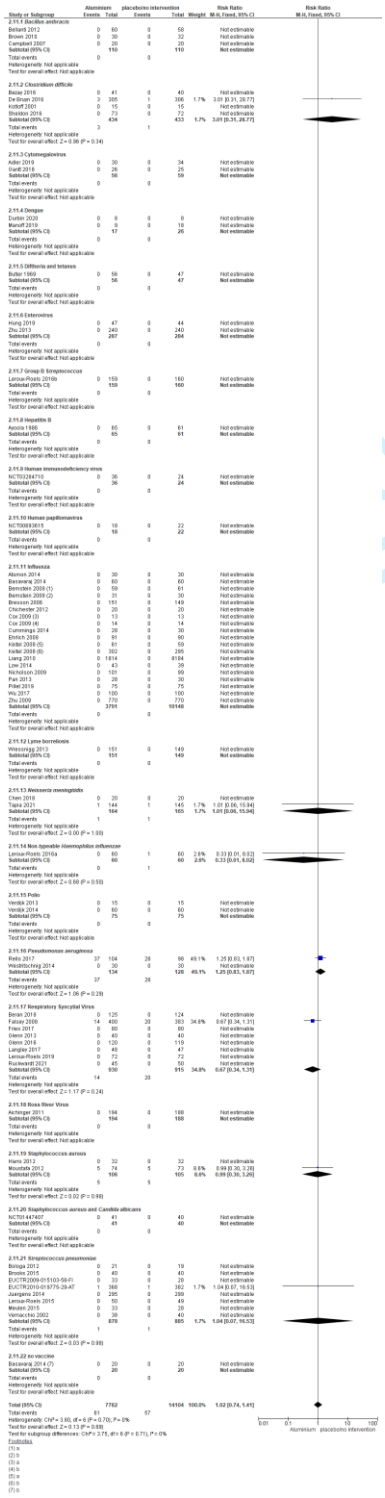
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Supplementary Figure S19.

Subgroup analysis of all-cause mortality according to aluminium adjuvant type.

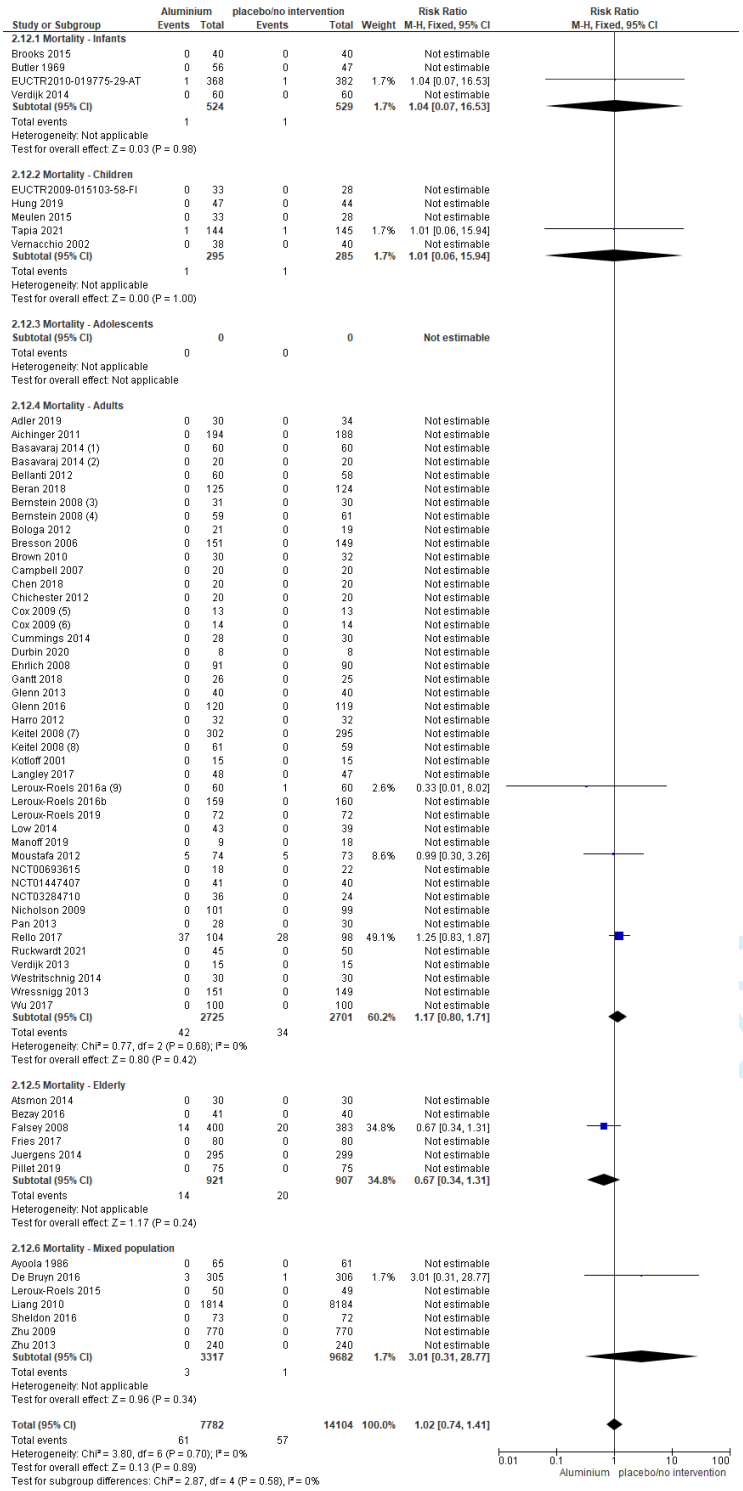
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Supplementary Figure S20.

Subgroup analysis of all-cause mortality according to vaccine type.

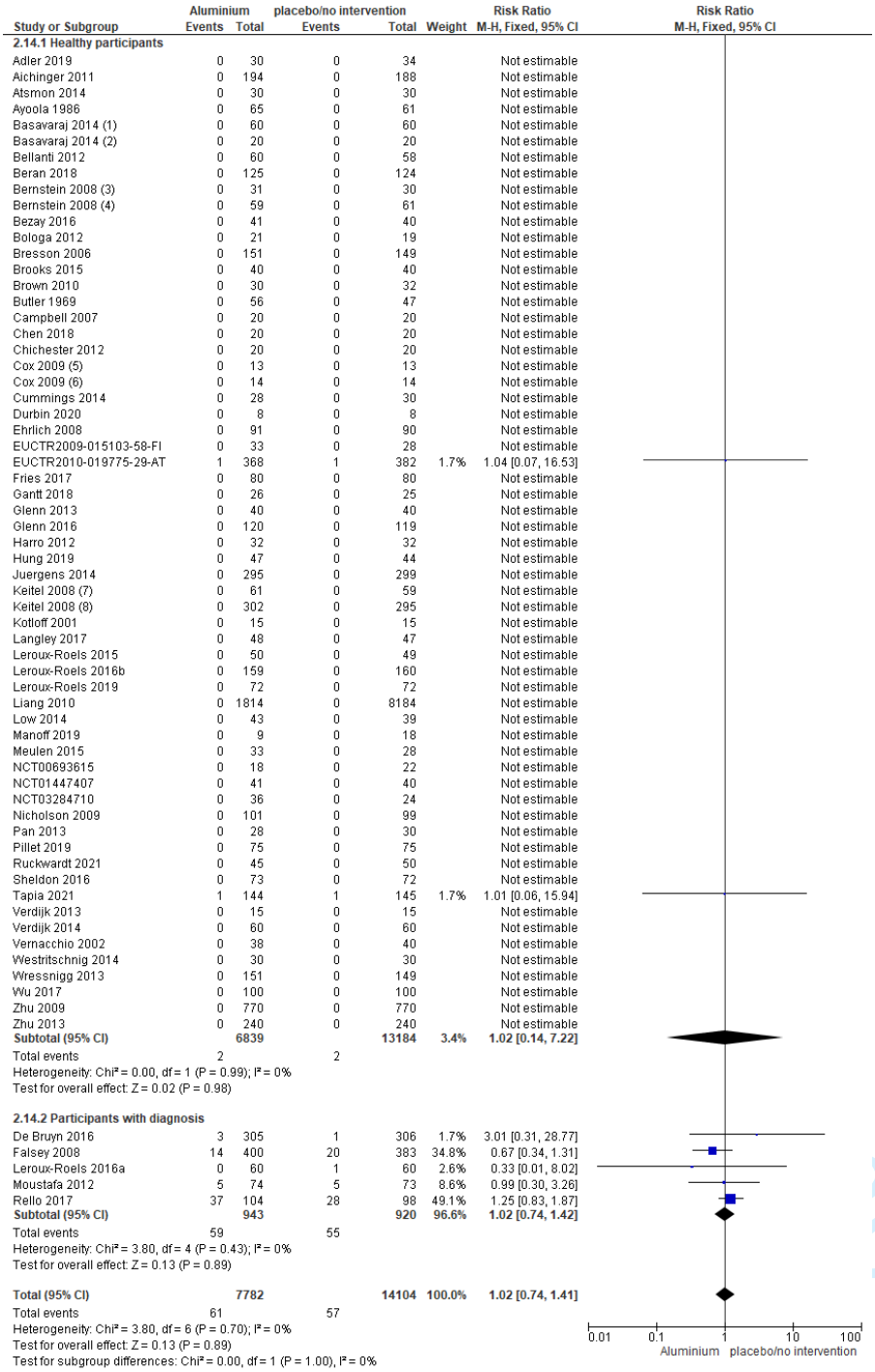
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Supplementary Figure S21.

Subgroup analysis of all-cause mortality according to age of the participants.

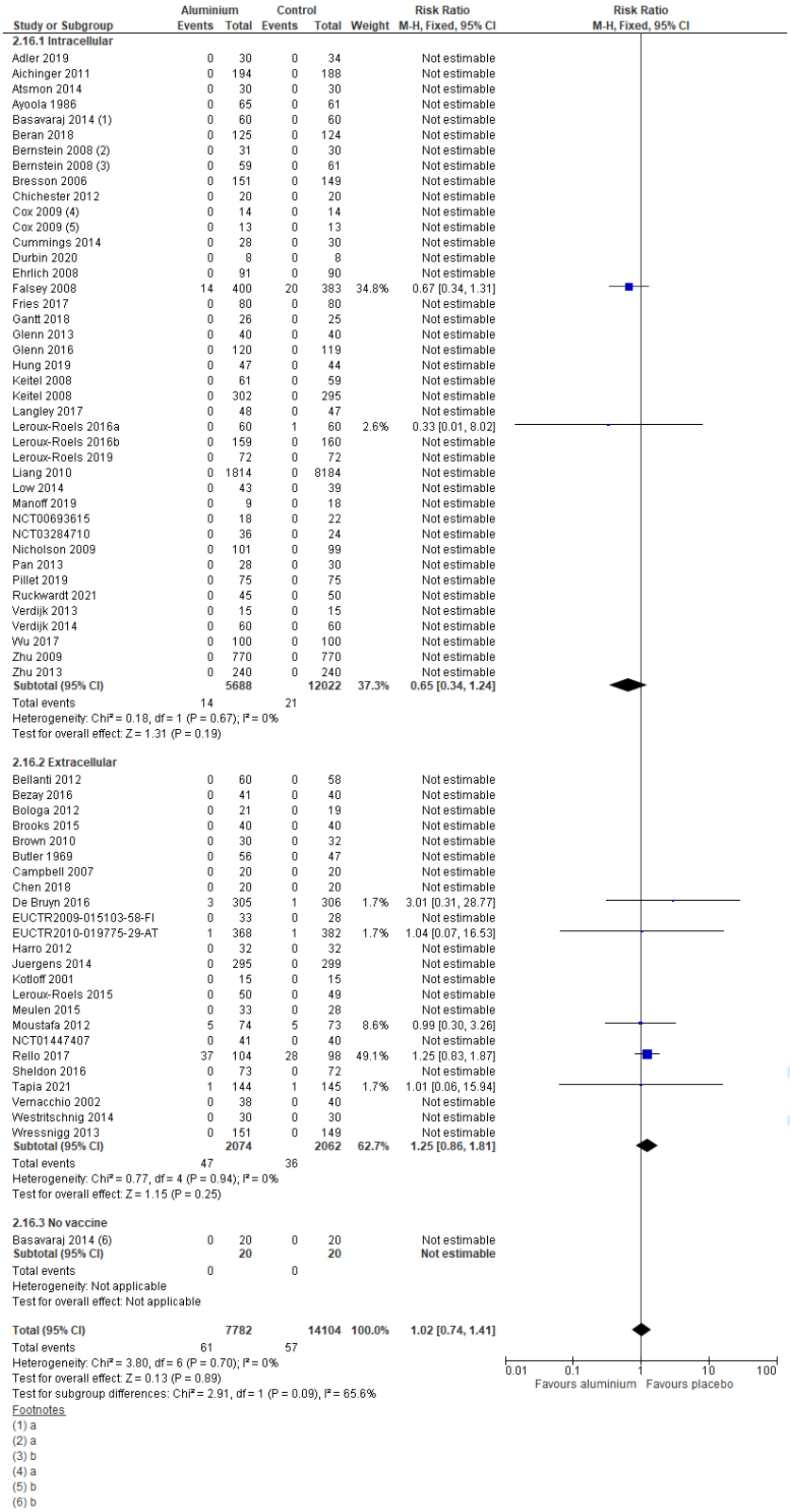
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Supplementary Figure S22.

Subgroup analysis of all-cause mortality according to health status of the participants.

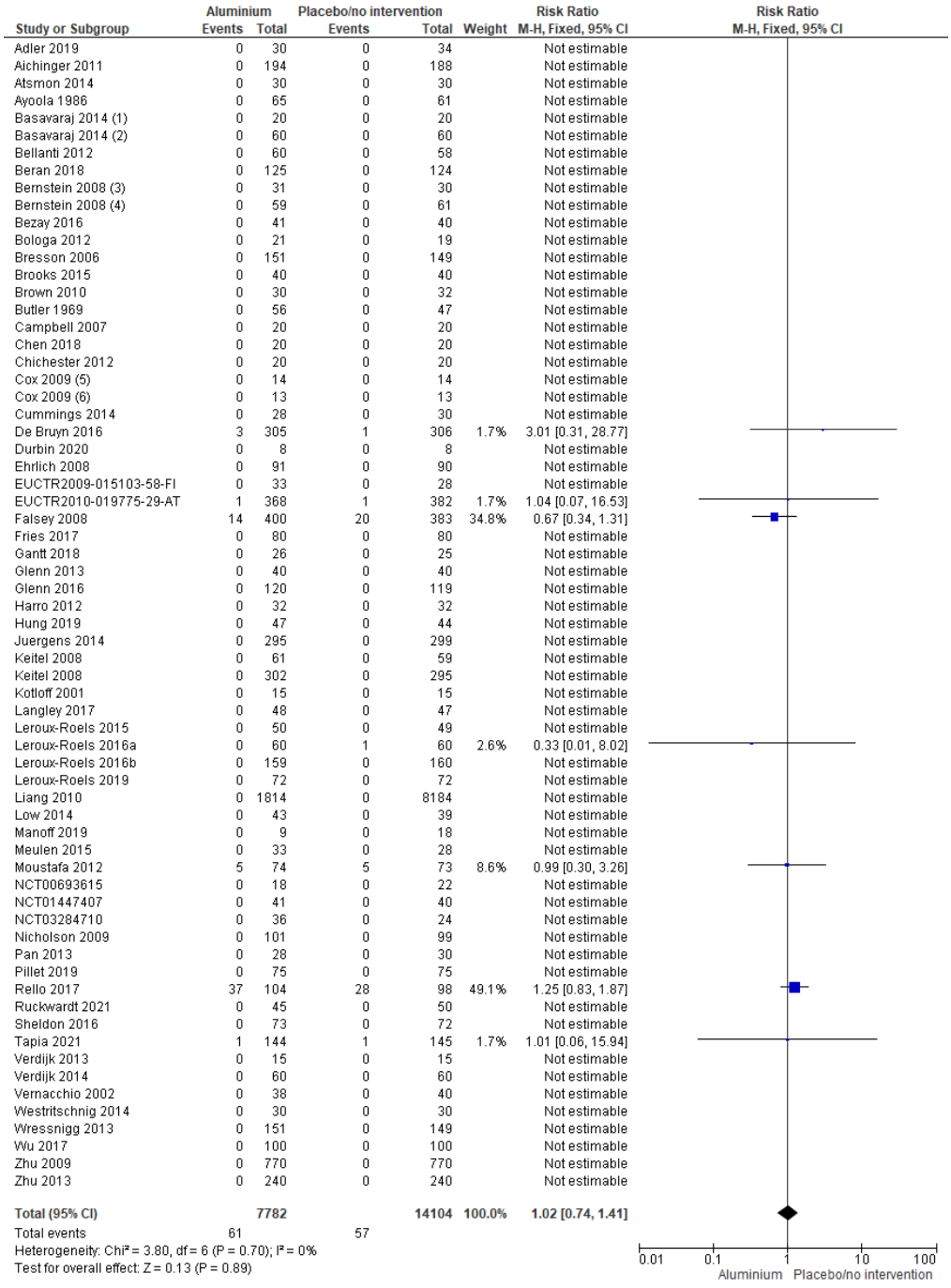
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Supplementary Figure S23.

Subgroup analysis of all-cause mortality according to pathogen type the vaccine is directed against.

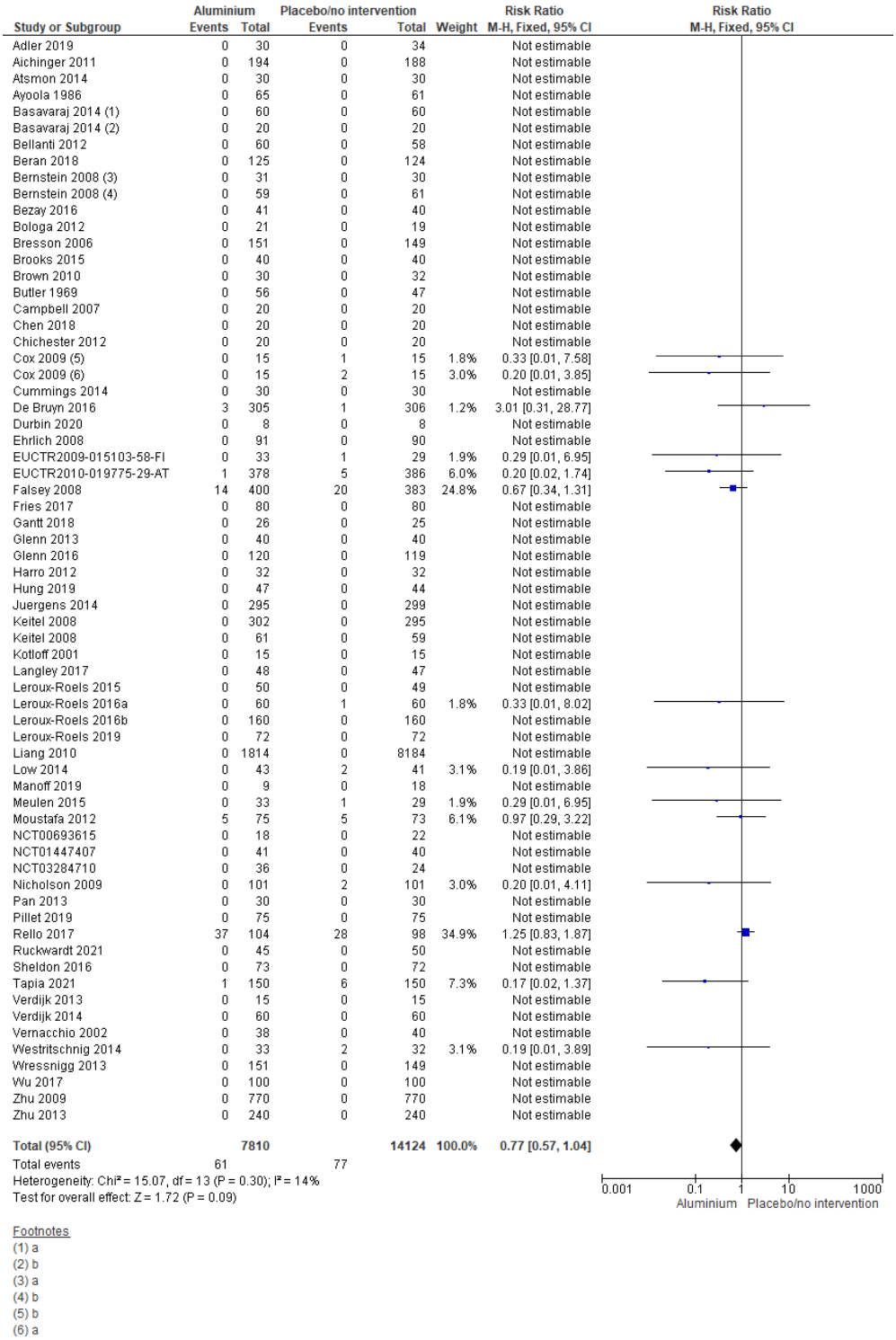
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Supplementary Figure S24.

Forest plot of 'worst best' sensitivity analysis of all-cause mortality. We assessed the potential impact of missing data in the 'Worst-best-case' scenario: it was assumed that all participants lost to follow-up in the experimental group did not survive; and all those with missing outcomes in the control group did survive.

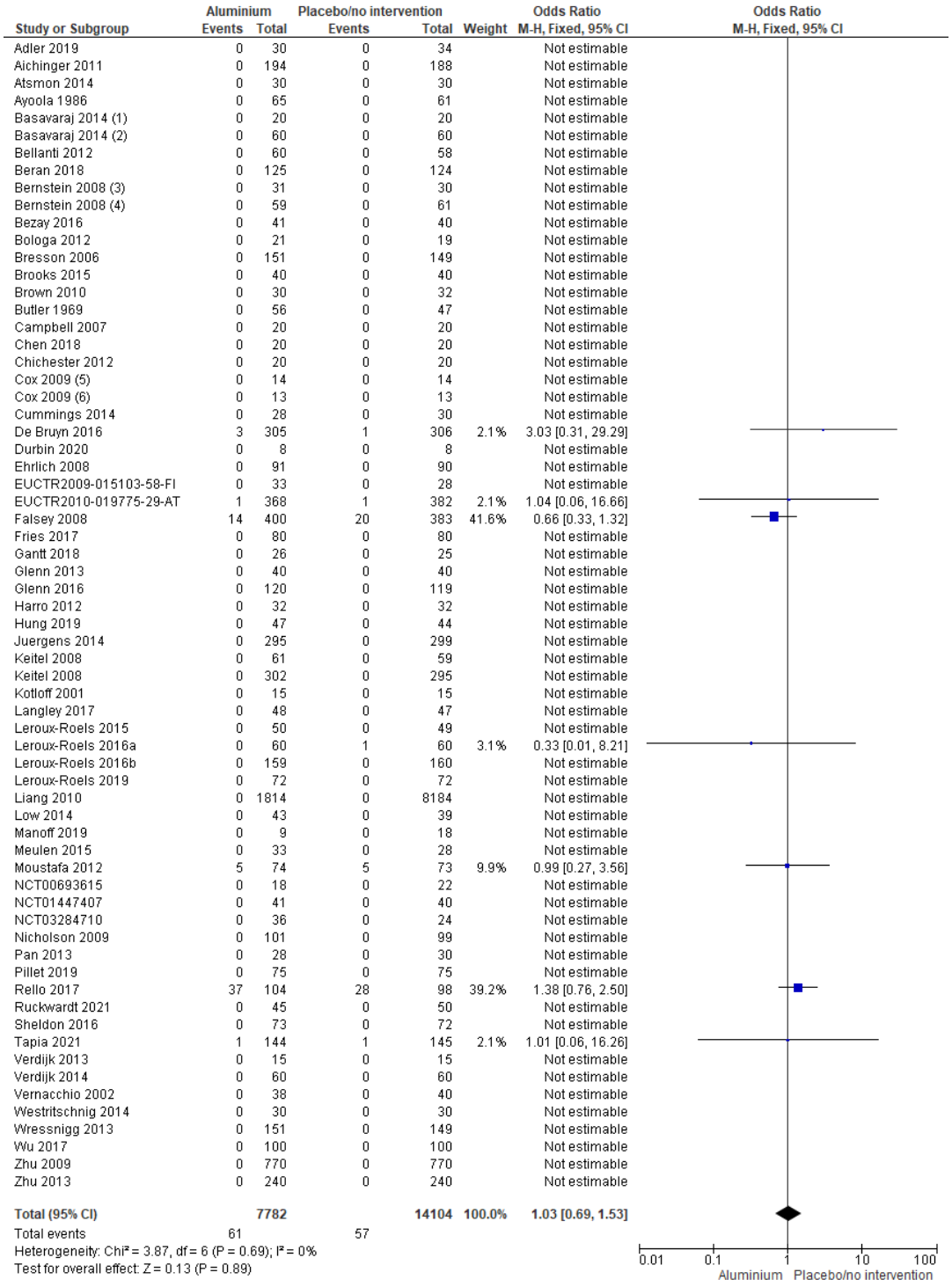
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Supplementary Figure S25.

Forest plot of 'best worst' sensitivity analysis of all-cause mortality. We assessed the potential impact of missing data in the 'best-worst-case' scenario: it was assumed that all participants lost to follow-up in the control group did not survive; and all those with missing outcomes in the experimental group did survive.

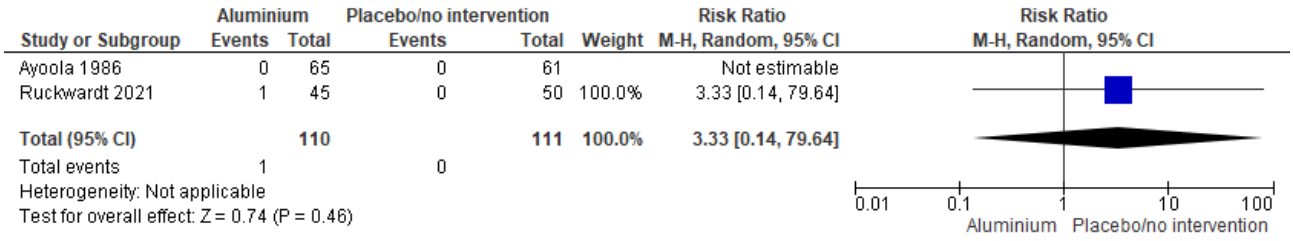
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Supplementary Figure S26.

Meta analysis of all-cause mortality with odds ratio.

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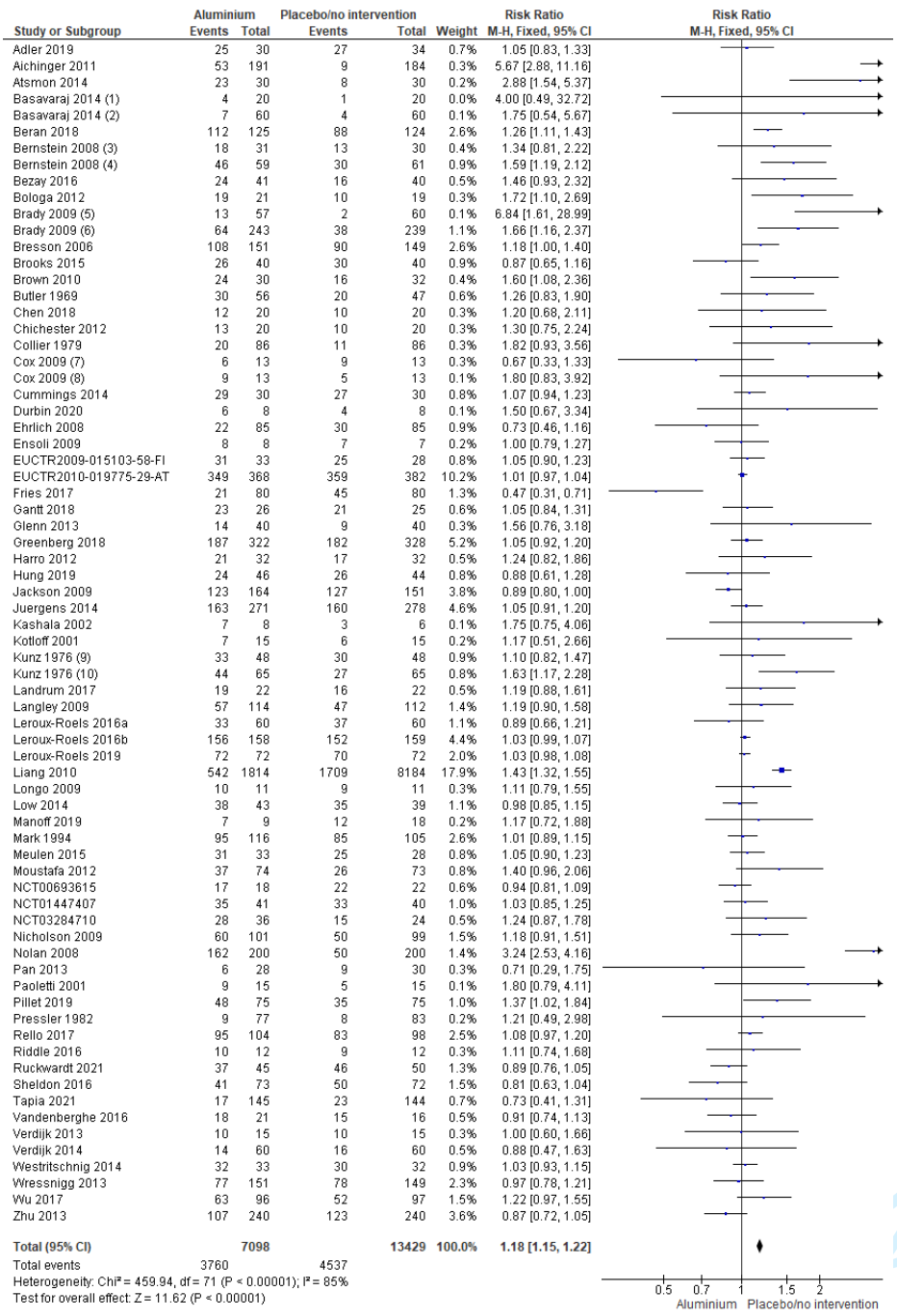


Supplementary Figure S27.

Meta-analysis of the effect of aluminium adjuvants compared with placebo on the proportion of participants with disease.

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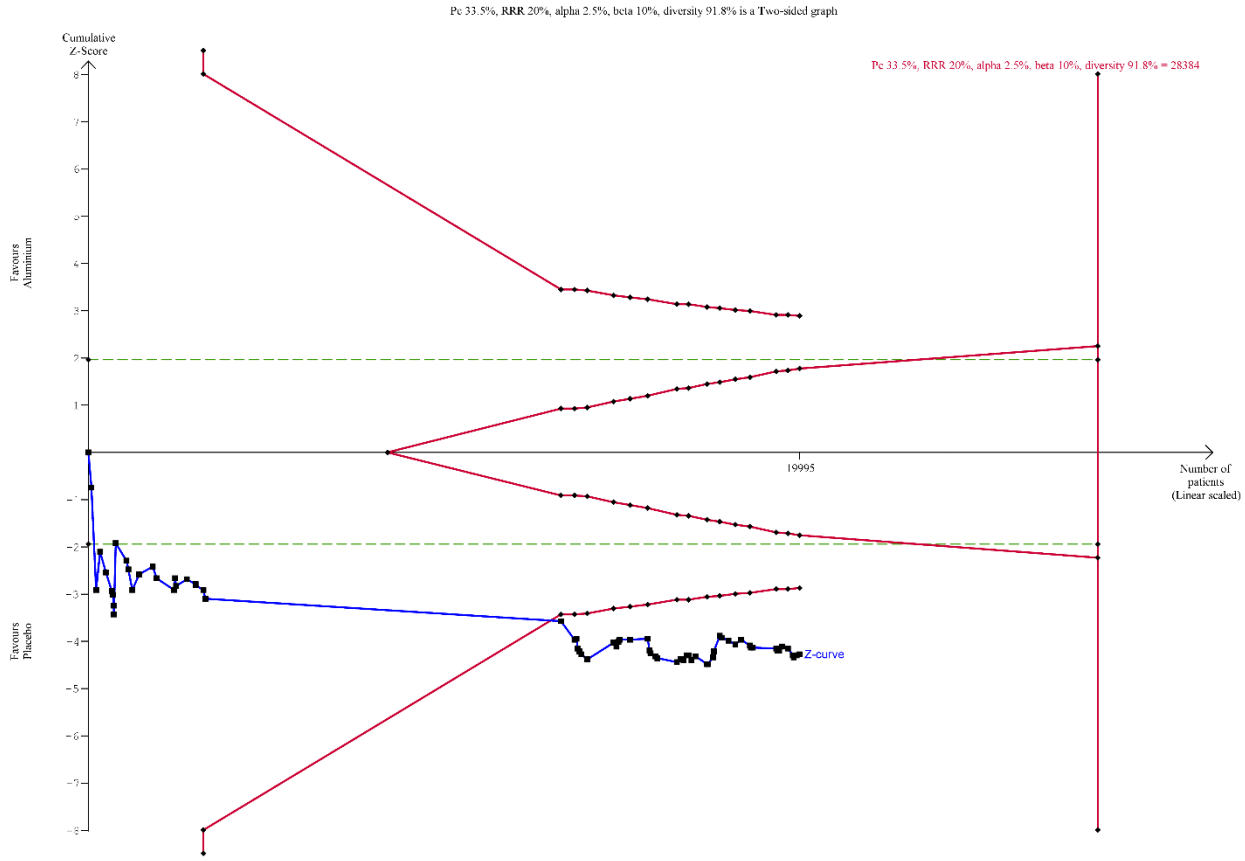


Footnotes
(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 15 mcg of antigen
(4) 30 mcg of antigen
(5) 300 mcg/dose aluminium
(6) 600 mcg/dose aluminium
(7) 12 mcg antigen
(8) 24 mcg antigen
(9) subunit vaccine
(10) whole virus vaccine

Supplementary Figure S28.

Fixed-effect meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the proportion of participants with one or more non-serious adverse events.

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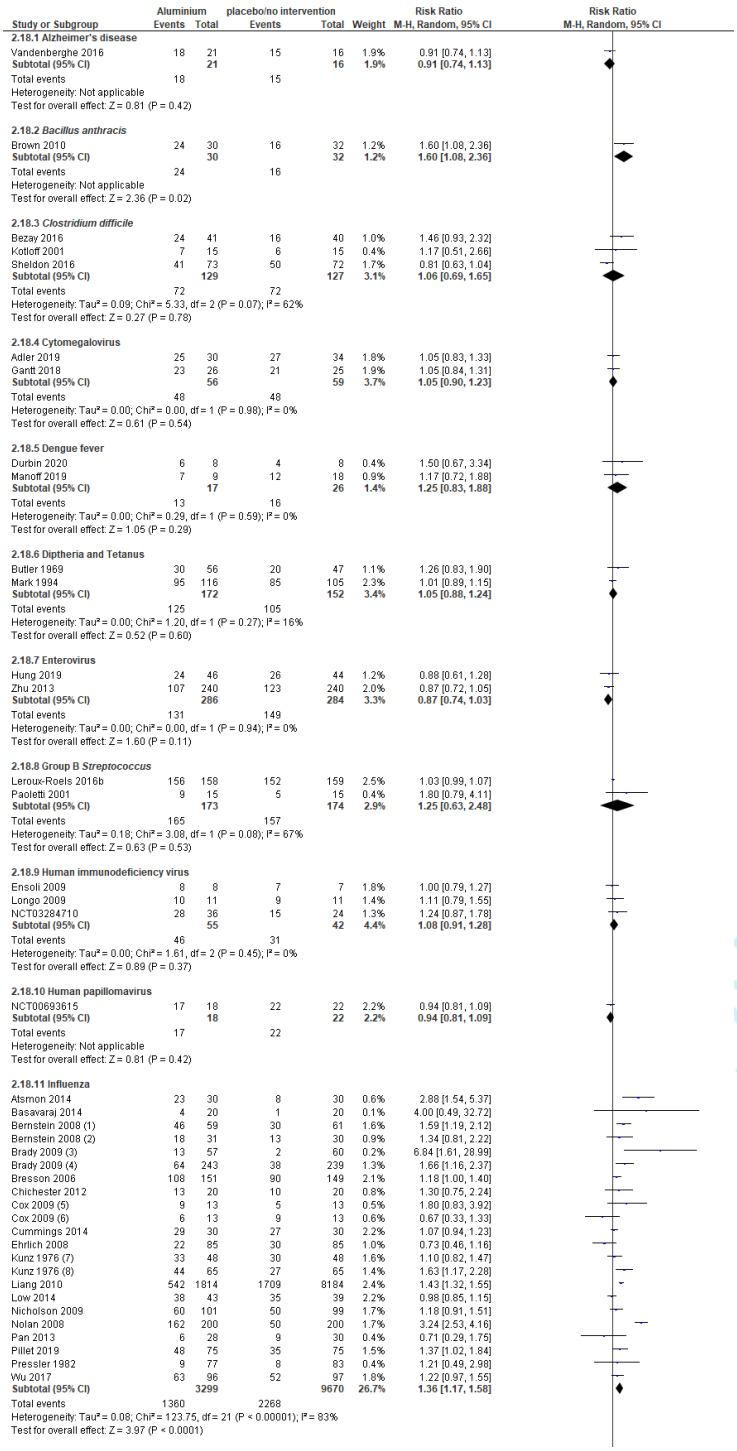


Supplementary Figure S29.

Trial Sequential Analysis using random-effects meta-analysis of the effects of aluminium adjuvants versus placebo or no intervention on the proportion of participants with adverse events considered not-serious. The analysis was based on a proportion in the control group (Pc) of 33.8%, a relative risk reduction (RRR) of 20%, an alpha of 2.5%, a beta of 10% and diversity of 92%. The graph shows that there was enough information to confirm that aluminium adjuvants compared with placebo increases the risk of one or more non-serious adverse events.

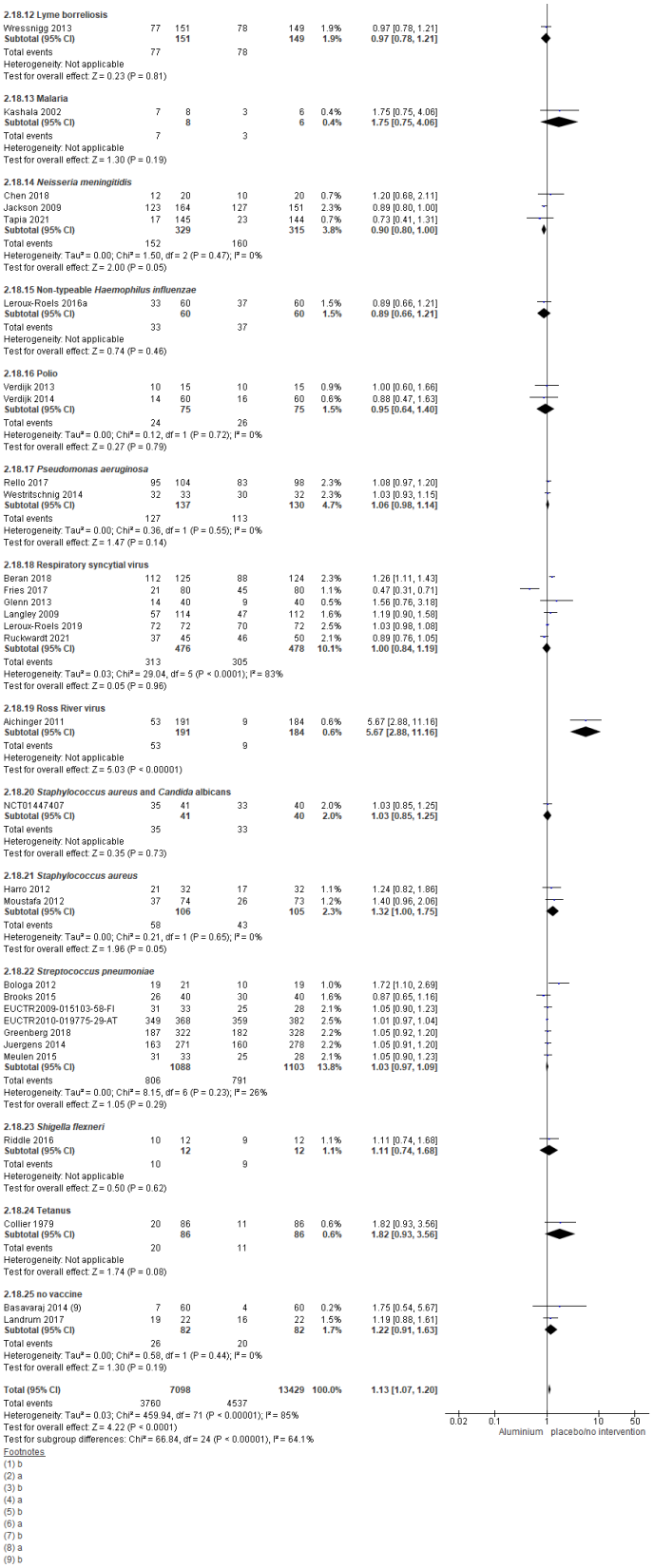
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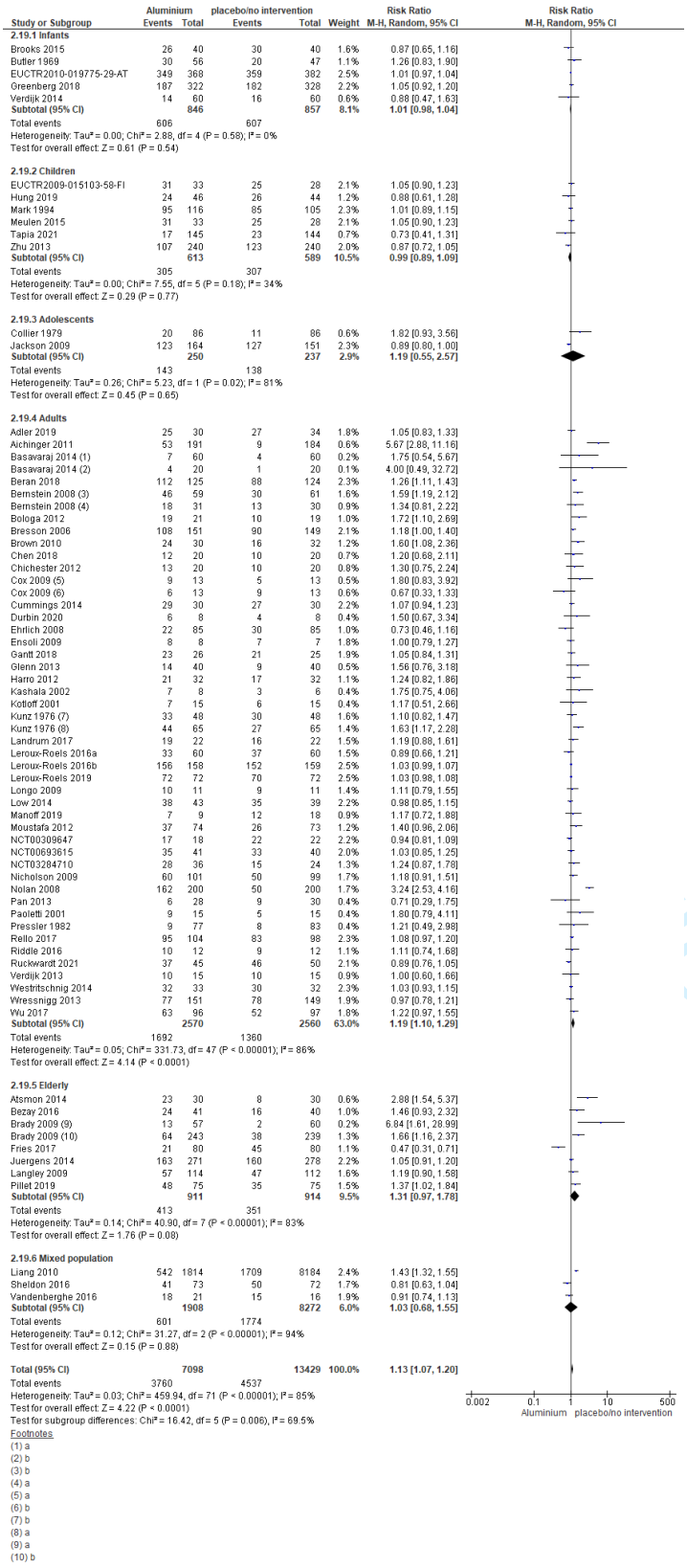
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Supplementary Figure S30.

Subgroup analyses of non-serious adverse events according to vaccine type.

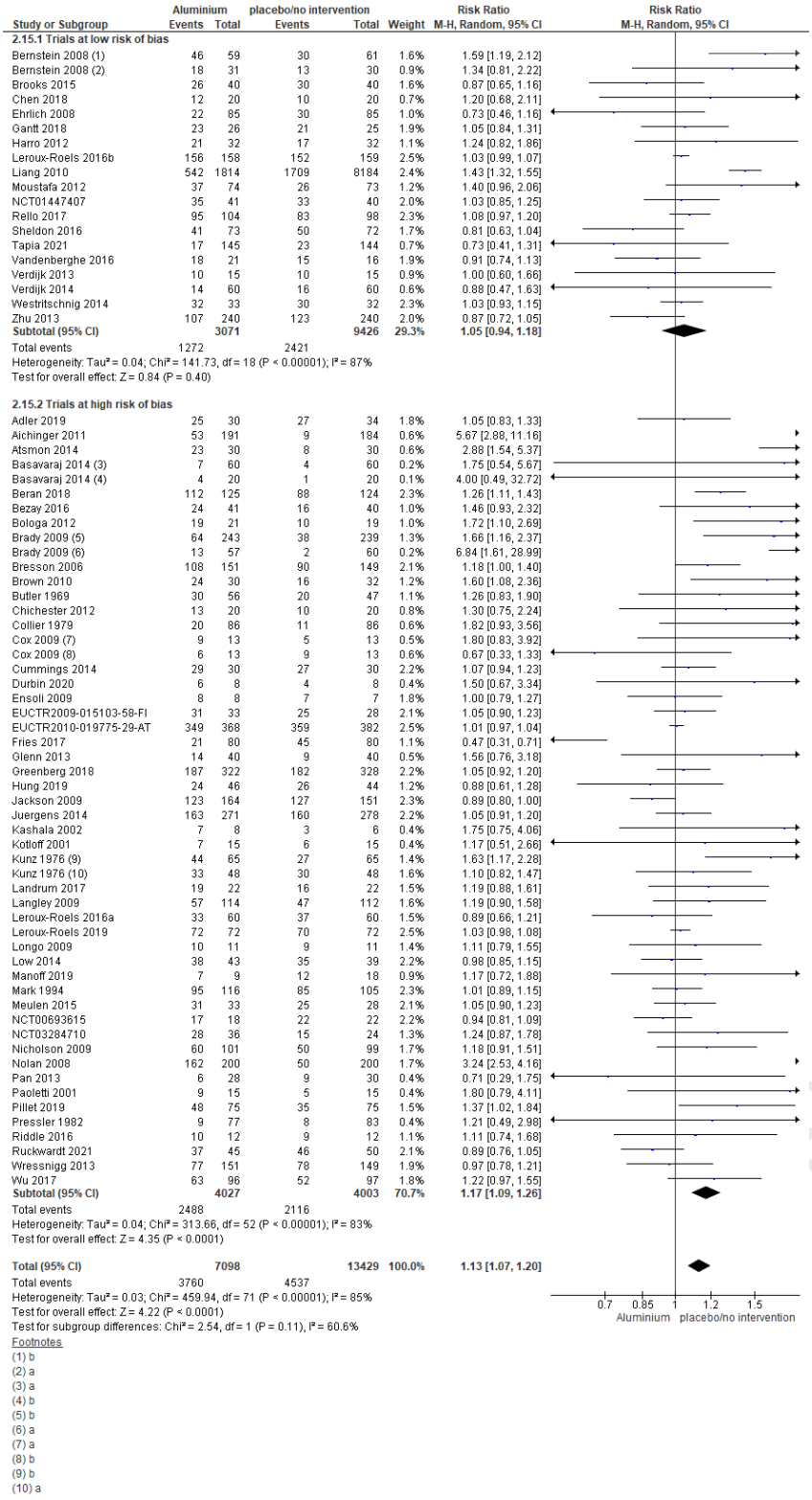
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Supplementary Figure S31.

Subgroup analyses of non-serious adverse events according to age of the participants.

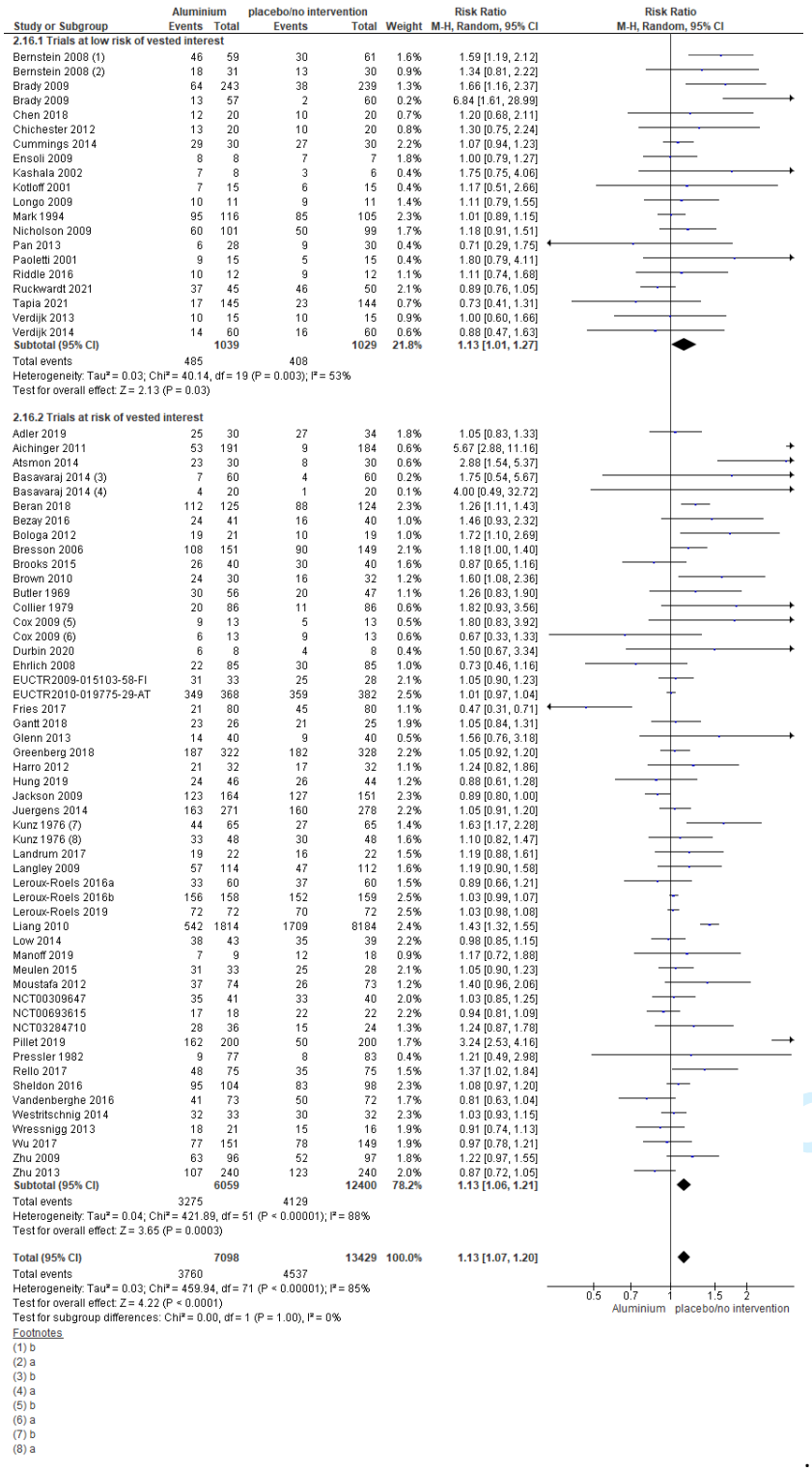
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Supplementary Figure S32.

Subgroup analyses of adverse events considered non-serious according to risk of bias.

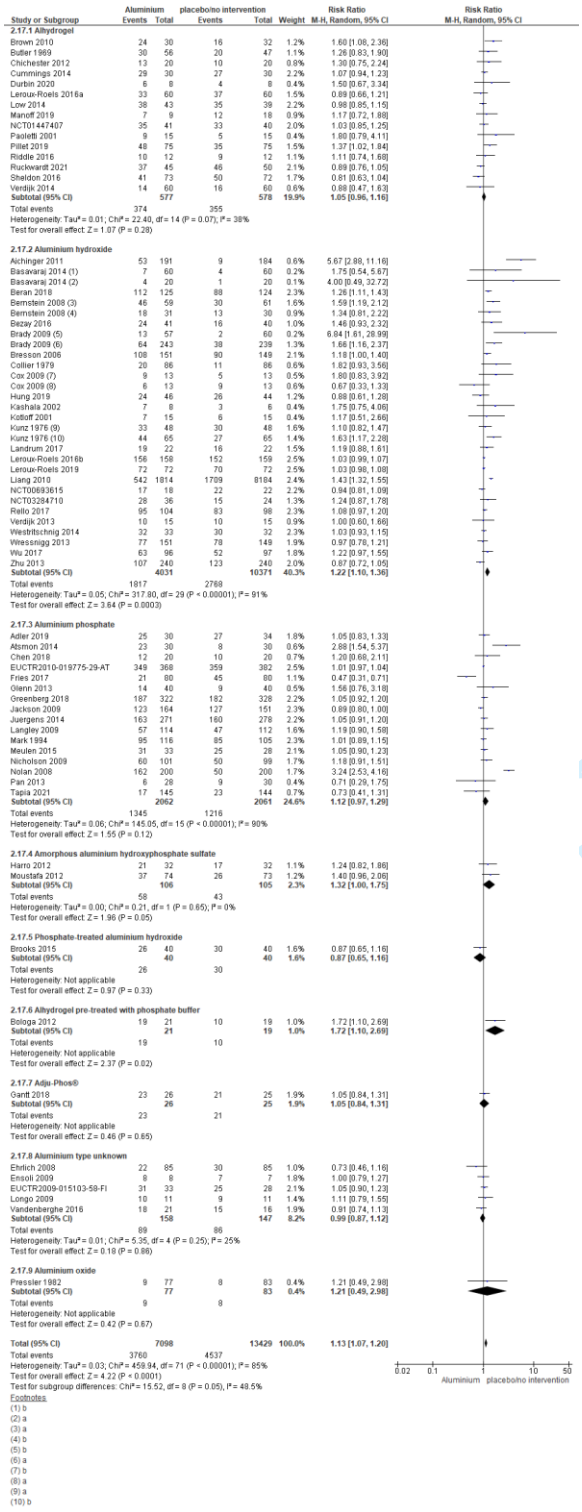
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Supplementary Figure S33.

Subgroup analyses of adverse events considered non-serious according to vested interest.

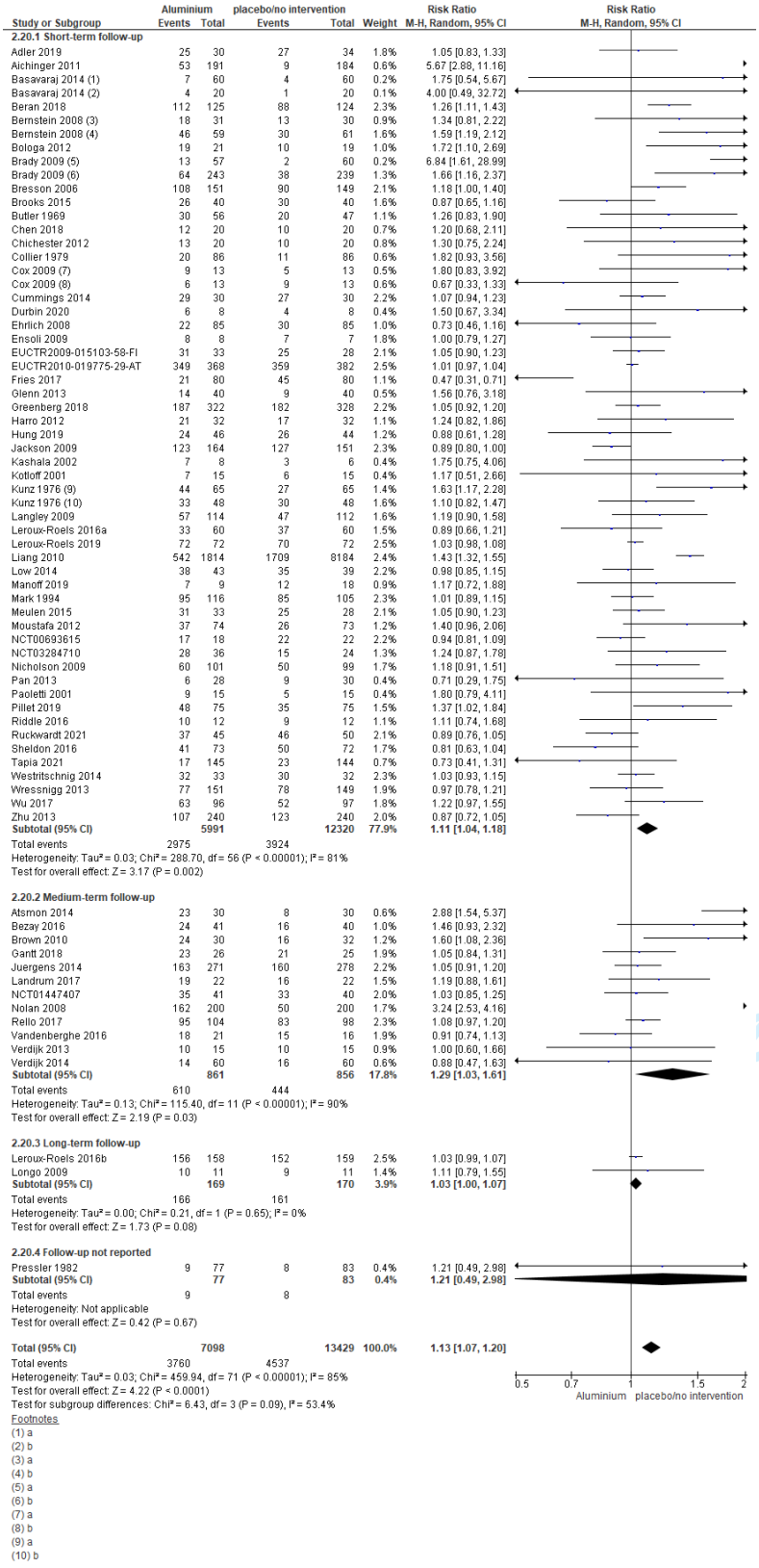
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Supplementary Figure S34.

Subgroup analyses of adverse events considered non-serious according to aluminium type.

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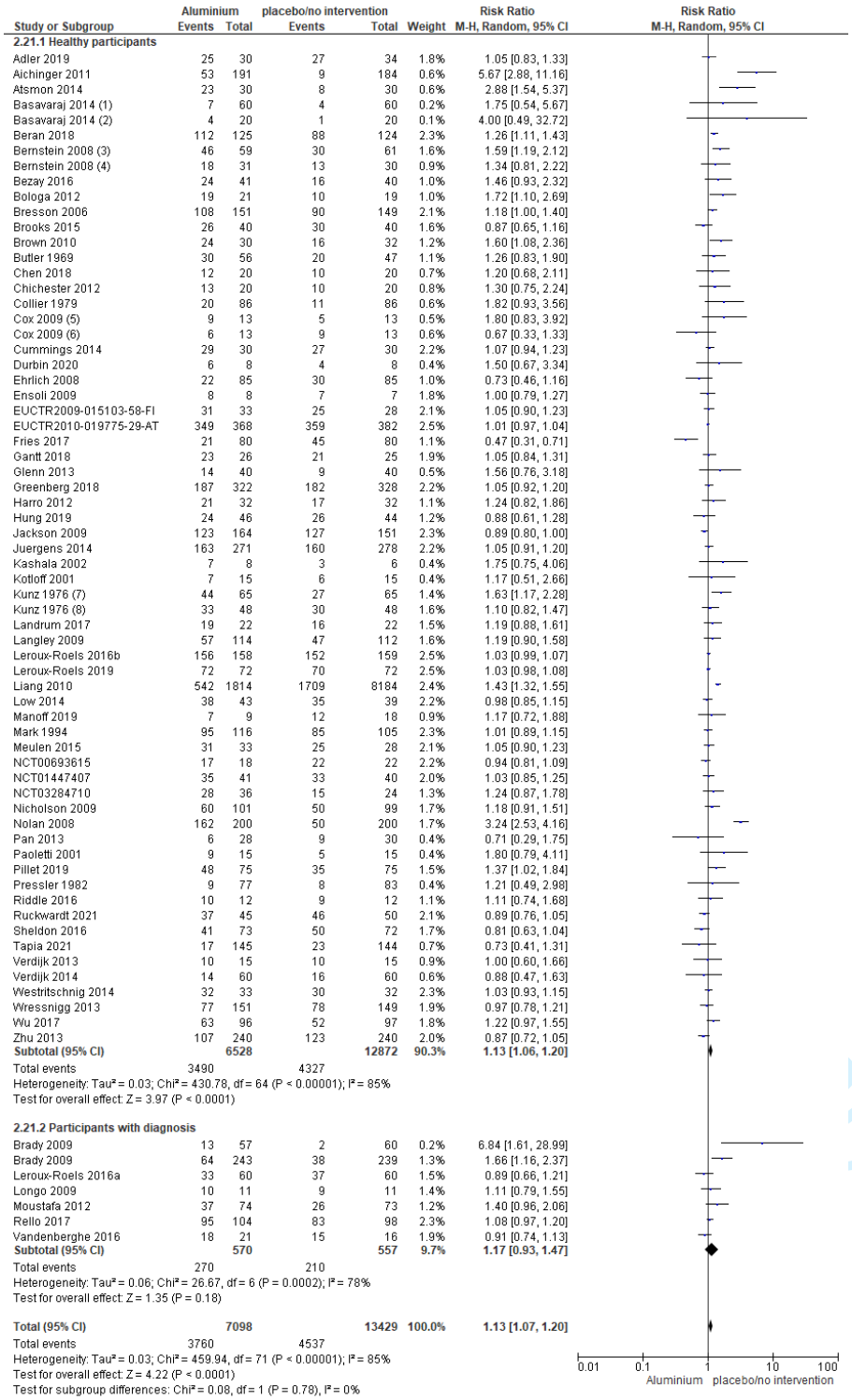
Supplementary Figure S35.

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Subgroup analyses of adverse events considered non-serious according to follow-up duration of the participants.

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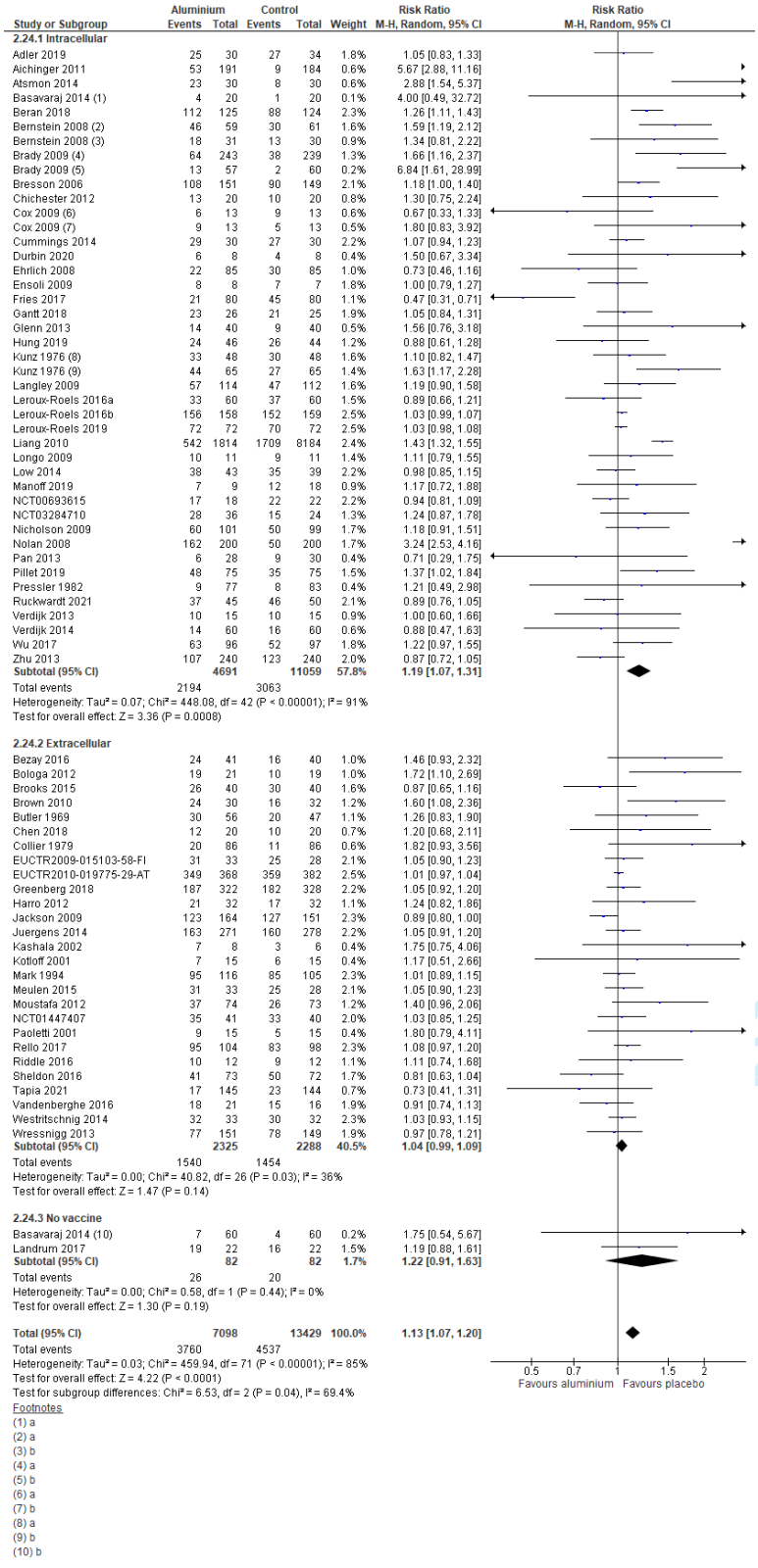
Supplementary Figure S36.

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Subgroup analyses of adverse events considered non-serious according to health status of the participants.

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Supplementary Figure S37.

Subgroup analyses of adverse events considered non-serious according to type of pathogen the vaccine is directed against.

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Supplementary Analysis S2.

Individual adverse events considered non-serious analyses

Meta-analysis showed evidence of a harmful effect of aluminium adjuvants compared with placebo or no intervention when assessing the following adverse events considered non-serious:

swelling (RR 1.38, 95% CI 1.17 to 1.62);

injection site pain (RR 1.38, 95% CI 1.19 to 1.61);

tenderness (RR 1.67, 95% CI 1.28 to 2.17);

muscle pain (RR 1.30, 95% CI 1.05 to 1.61);

erythema (RR 1.23, 95% CI 1.02 to 1.50);

musculoskeletal and connective tissue disorders (RR 1.90, 95% CI 1.16 to 3.14);

pain (RR 1.32, 95% CI 1.19 to 1.46);

injection site reaction (RR 2.02, 95% CI 1.23 to 3.32);

myalgia proximal (RR 2.12, 95% CI 1.66 to 2.72); and

rash (RR 1.92, 95% CI 0.99 to 3.72).

Meta-analysis showed no evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing the following non-serious adverse events: induration, redness, fever onset within 7 days, fever, malaise, fatigue, headache, nausea, vomiting, joint pain, lymph node swelling, itching, sore throat, feeling hot, feverishness, myalgia, ecchymosis, shivering, loss of appetite, arthralgia, stayed home, upper respiratory tract infection, back pain, oropharyngeal pain, hypertension, bruising, cough, hypotension, anorexia, numbness, muscle stiffness, sweating, exanthema, rhinorrhoea, dizziness, nasopharyngitis, urinary tract infection, change in feeding pattern, more crying, infection, bronchitis, injury, poisoning and procedural complications, flu-like symptoms, joint sprain, warmth, insomnia, constipation, toothache, alanine aminotransferase increased, arthropod bite, aspartate aminotransferase increased, contusion, dysphonia, flushing, vertigo positional, vulvovaginal candidiasis, pruritus, pharyngitis, influenza-like illness, rhinitis, chills, gastrointestinal symptoms, blood and lymphatic disorders, gastrointestinal disorders, flatulence, injection site haemorrhage, acute sinusitis, conjunctivitis, herpes zoster, ligament sprain, blood creatinine increased, blood pressure increased, hemoglobin decreased, bursitis, neck pain, tendonitis, nervous system disorders, sinus congestion, skin and subcutaneous disorders, ear and labyrinth disorders, eye disorders, dental caries, food poisoning, gastritis, gingival disorder, edema peripheral, hepato-biliary disorders, renal and urinary disorders, blood bilirubin increased, blood urea increased, epistaxis/nose bleeding, nasal congestion, hardening, less appetite, local administration site disorders, cardio-vascular disorders, general disorders, kidney and urinary disorders, psychiatric disorders, asthma, dyspepsia, cystitis, influenza, oral herpes, hyperglycaemia, ear infection, reproductive system and breast disorders, swelling in the neck, asthenia, diarrhea, and gastroenteritis.

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Individual non-serious adverse events reported by only one trial that could not be meta-analysed are shown in the table below.

Non-serious adverse events reported in the aluminum-adjuvanted groups versus the placebo group or no intervention		
Non-serious adverse events	**Aluminium event/total (study ID)**	**Placebo event/total (study ID)**
Sinus headache	1/32 (Harro 2012)	0/32 (Harro 2012)
Body pain	2/60 (Basavaraj 2014)	1/60 (Basavaraj 2014)
Common cold	2/60 (Basavaraj 2014)	1/60 (Basavaraj 2014)
Allergy on face	1/60 (Basavaraj 2014)	0/60 (Basavaraj 2014)
Burning sensation	1/60 (Basavaraj 2014)	0/60 (Basavaraj 2014)
Lower limb pain	0/60 1/20 (Basavaraj 2014)	1/60 0/20 (Basavaraj 2014)
Systolic hpertension	1/80 (Fries 2017)	4/80 (Fries 2017)
Abnormal crying	9/40 (Brooks 2015)	9/40 (Brooks 2015)
Respiratory symptoms	5/14 0/13 (Cox 2009)	0/14 3/13 (Cox 2009)
Contralateral inflammation	2/8 (Kashala 2002)	1/6 (Kashala 2002)
Joint swelling	1/15 (Kotloff 2001)	0/15 (Kotloff 2001)
Limitation of movement	13/43 (Low 2014)	13/39 (Low 2014)
Nodule	10/33 (Meulen 2015)	3/28 (Meulen 2015)
Muscle rigidity	1/60 (Verdrijk 2014)	1/60 (Verdrijk 2014)
Change in stool	12/60 (Verdrijk 2014)	9/60 (Verdrijk 2014)
Change in sleeping pattern	14/60 (Verdrijk 2014)	16/60 (Verdrijk 2014)
Change in behavior	10/60 (Verdrijk 2014)	11/60 (Verdrijk 2014)
Pain in extremity	7/33 (Westritschnig 2014)	0/32 (Westritschnig 2014)
Eclampsia	1/120 (Zhu 2013)	2/120 (Zhu 2013)
Exanthema	2/46 (Hung 2019)	1/44 (Hung 2019)
Blood fibrinogen decreased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Blood lactate dehydrogenase increased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Bone contusion	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Excoriation	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Infected dermal cyst	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Platelet count increased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Rosacea	1/60 (Leroux-Roels 2016b)	0/60 (Leroux-Roels 2016b)
Injury	1/18 (NCT00693615)	6/22 (NCT00693615)
Infection viral	1/20 (Chen 2018)	0/20 (Chen 2018)
Stress	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Myalgia - distal	39 /200 (Nolan 2008)	35/200 (Nolan 2008)
Cardiac disorders	1/73 (Sheldon 2016)	2/72 (Sheldon 2016)
Palpitations	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Abdominal distension	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Infected bites	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Fall	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Periorbital hematoma	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Post-traumatic neck syndrome	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Aspartate aminotransferase increased	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Blood cholesterol increased	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Dyslipidemia	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Musculoskeletal and connective tissue disorders	5/73 (Sheldon 2016)	5/72 (Sheldon 2016)
Neoplasms benign malignant and unspecified (including cysts and polyps)	1/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Limb discomfort	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Heart valve stenosis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Vertigo postural	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Endocrine disorders	2/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Hypothyroidism	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Testicular failure	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Vitreous detachment	0/73 (Sheldon 2016)	2/72 (Sheldon 2016)
Cellulitis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Cholelithiasis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Folliculitis	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Hand fracture	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Osteoporosis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Urethral stenosis	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Prostatism	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Blood cortisol increased	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Local reactions	118/770 (Zhu 2009 all)	55/770 (Zhu 2009 all)
Neurologic	5/770 (Zhu 2009 all)	4/770 (Zhu 2009 all)
Systemic	65/770 (Zhu 2009 all)	60/770 (Zhu 2009 all)
Bacterial vaginosis	1/20 (Chen 2018)	0/20 (Chen 2018)
Blood pressure decreased	1/30 (Atsmon 2014)	0/30 (Atsmon 2014)
Herpangina	1/46 (Hung 2019)	0/44 (Hung 2019)
Papules	0/46 (Hung 2019)	1/44 (Hung 2019)
Lymph node pain	1/32 (Harro 2012)	0/32 (Harro 2012)

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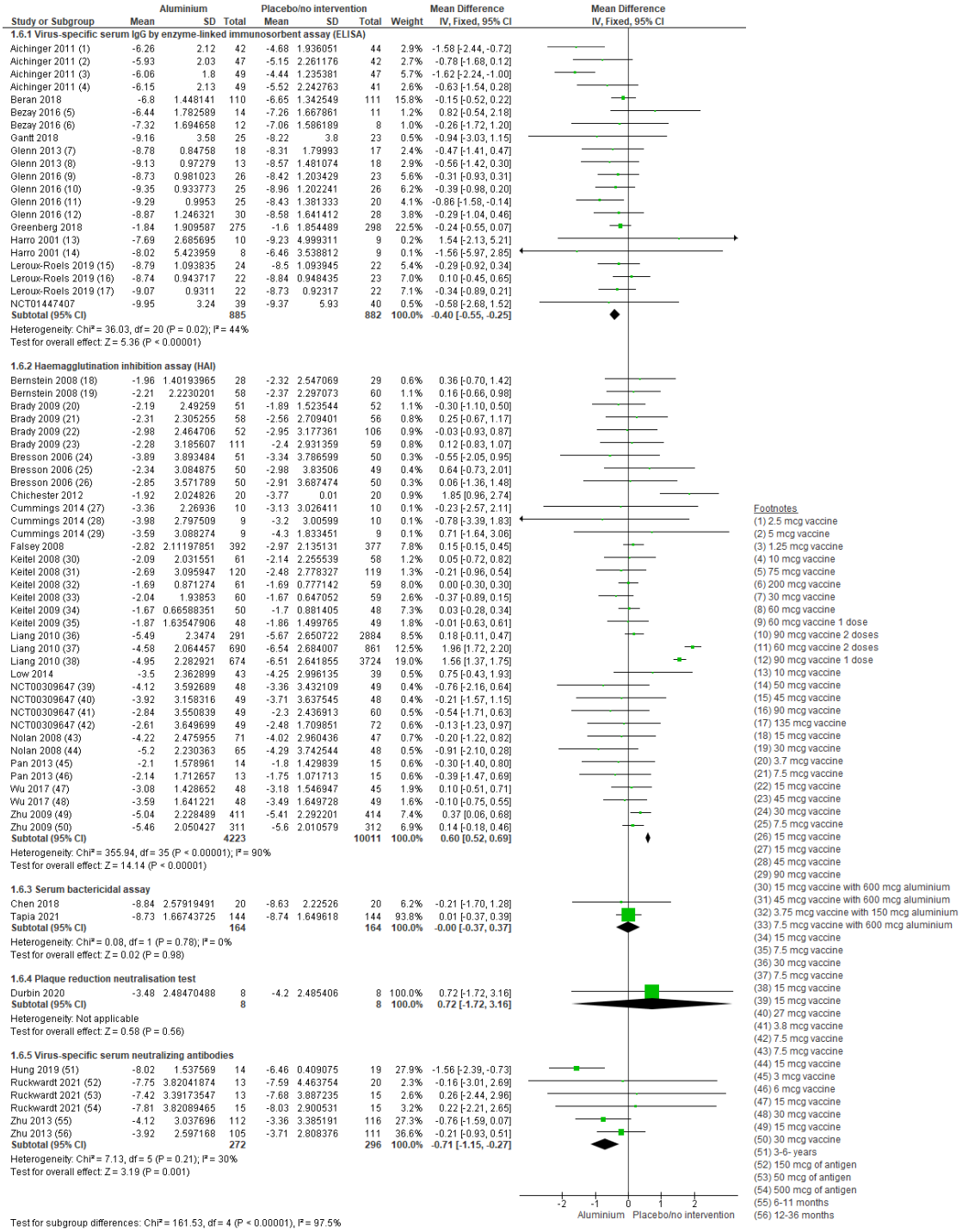
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Tooth disorder	2/72 (Leroux-Roels 2019)	0/72 (Leroux-Roels 2019)
Tinea Pedis	2/72 (Leroux-Roels 2019)	0/72 (Leroux-Roels 2019)
Wheezing	5/72 (Leroux-Roels 2019)	5/72 (Leroux-Roels 2019)
Hemoglobin abnormal	2/22 (Landrum 2017)	1/22 (Landrum 2017)
Blood LDH increase	1/22 (Landrum 2017)	1/22 (Landrum 2017)
Whole blood count increased	2/22 (Landrum 2017)	3/22 (Landrum 2017)
Neutrophil count increased	1/22 (Landrum 2017)	2/22 (Landrum 2017)
Hypoglycaemia	0/22 (Landrum 2017)	1/22 (Landrum 2017)
Blood sodium increased	2/22 (Landrum 2017)	1/22 (Landrum 2017)
Teething	0/33 (Meulen 2015)	1/28 (Meulen 2015)
Laryngitis	0/33 (Meulen 2015)	1/28 (Meulen 2015)
Respiratory syncytial virus bronchiolitis	1/33 (Meulen 2015)	0/28 (Meulen 2015)
Dysmenorrhea	0/18 (NCT00693615)	2/22 (NCT00693615)
Flu syndrome	1/18 (NCT00693615)	2/22 (NCT00693615)
Depression	0/18 (NCT00693615)	1/22 (NCT00693615)
Urogenital neoplasia	0/18 (NCT00693615)	1/22 (NCT00693615)
Vaginal moniliasis	0/18 (NCT00693615)	1/22 (NCT00693615)
Injection site vesicles	0/8 (Durbin 2020)	1/8 (Durbin 2020)
Armpit swelling	0/25 (Gantt 2018)	1/23 (Gantt 2018)
Gastroesophageal reflux	1/25 (Gantt 2018)	0/23 (Gantt 2018)
Herpes simplex	0/25 (Gantt 2018)	2/23 (Gantt 2018)
Laceration	0/25 (Gantt 2018)	1/23 (Gantt 2018)
Otitis	3/33 (Meulen 2015)	1/28 (Meulen 2015)
Head injury	1/33 (Meulen 2015)	0/28 (Meulen 2015)
Bacterial vaginosis	0/36 (NCT03284710)	2/24 (NCT03284710)
Orchitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Muscle twitching	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Dysuria	1/36 (NCT03284710)	0/24 (NCT03284710)
Otitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Discomfort	6/119 (Mark 1994)	4/105 (Mark 1994)
Neutropenia	3/45 (Ruckwardt 2021)	3/50 (Ruckwardt 2021)
Eosinophilia	1/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Leukopenia	1/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Lymphopenia	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Bradycardia	2/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Tachycardia	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Burns first degree	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Hepatic enzyme increased	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Abnormal dreams	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Laryngeal pain	0/45 (Ruckwardt 2021)	2/50 (Ruckwardt 2021)
Body tinea	0/36 (NCT03284710)	1/24 (NCT03284710)
Fungal skin infection	0/36 (NCT03284710)	1/24 (NCT03284710)
Furuncle	0/36 (NCT03284710)	1/24 (NCT03284710)
Genitourinary chlamydia infection	5/36 (NCT03284710)	0/24 (NCT03284710)
Genitourinary tract gonococcal infection	1/36 (NCT03284710)	1/24 (NCT03284710)
Pelvic inflammatory disease	1/36 (NCT03284710)	0/24 (NCT03284710)
Subcutaneous abscess	1/36 (NCT03284710)	0/24 (NCT03284710)
Tooth infection	0/36 (NCT03284710)	1/24 (NCT03284710)
Vulvovaginitis trichomonal	1/36 (NCT03284710)	0/24 (NCT03284710)
Arthropod sting	1/36 (NCT03284710)	0/24 (NCT03284710)
Limb injury	0/36 (NCT03284710)	1/24 (NCT03284710)
Skin abrasion	0/36 (NCT03284710)	1/24 (NCT03284710)
Sunburn	1/36 (NCT03284710)	0/24 (NCT03284710)
neutrophil count decreased	1/36 (NCT03284710)	0/24 (NCT03284710)
Abnormal loss of weight	0/36 (NCT03284710)	1/24 (NCT03284710)
Musculoskeletal pain	0/36 (NCT03284710)	1/24 (NCT03284710)
Glycosuria	1/36 (NCT03284710)	1/24 (NCT03284710)
Dysfunctional uterine bleeding	1/36 (NCT03284710)	0/24 (NCT03284710)
Prostatitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Vaginal discharge	1/36 (NCT03284710)	0/24 (NCT03284710)
Hand dermatitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash maculo-papular	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash papular	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash pruritic	1/36 (NCT03284710)	0/24 (NCT03284710)
Infection	4/36 (NCT03284710)	11/24 (NCT03284710)
Impetigo	1/145 (Tapia 2021)	1/144 (Tapia 2021)
Pharyngitis	5/145 (Tapia 2021)	4/144 (Tapia 2021)
Varicella	0/145 (Tapia 2021)	1/144 (Tapia 2021)
Thermal burn	1/145 (Tapia 2021)	0/144 (Tapia 2021)
erythema/induration	9/36 (NCT03284710)	6/24 (NCT03284710)
malaise and or fatigue	8/36 (NCT03284710)	3/24 (NCT03284710)
Max. Systemic Symptoms	13/36 (NCT03284710)	7/24 (NCT03284710)
Ear pain	0/36 (NCT03284710)	0/24 (NCT03284710)
Abdominal pain lower	0/36 (NCT03284710)	0/24 (NCT03284710)
Peripheral swelling	0/36 (NCT03284710)	0/24 (NCT03284710)
Vessel puncture site phlebitis	0/36 (NCT03284710)	0/24 (NCT03284710)
Lower respiratory tract infection	0/36 (NCT03284710)	0/24 (NCT03284710)
Respiratory tract infection	0/36 (NCT03284710)	0/24 (NCT03284710)
Urinary tract infection bacterial	0/36 (NCT03284710)	0/24 (NCT03284710)
Vulvovaginitis gonococcal	0/36 (NCT03284710)	0/24 (NCT03284710)
Eye injury	0/36 (NCT03284710)	0/24 (NCT03284710)
Human bite	0/36 (NCT03284710)	0/24 (NCT03284710)
Skin laceration	0/36 (NCT03284710)	0/24 (NCT03284710)
Investigation	14/36 (NCT03284710)	6/24 (NCT03284710)
Cardiac murmur	14/36 (NCT03284710)	6/24 (NCT03284710)
Blood bicarbonate decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
Platelet count decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
White blood cell count decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
Vulvovaginal rash	14/36 (NCT03284710)	6/24 (NCT03284710)
Rhinalgia	14/36 (NCT03284710)	6/24 (NCT03284710)
Dermatitis	14/36 (NCT03284710)	6/24 (NCT03284710)
Ingrowing nail	14/36 (NCT03284710)	6/24 (NCT03284710)
Musculoskeletal chest pain	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)

Supplementary Table S3.

List of individual adverse events considered non-serious reported in only one trial.

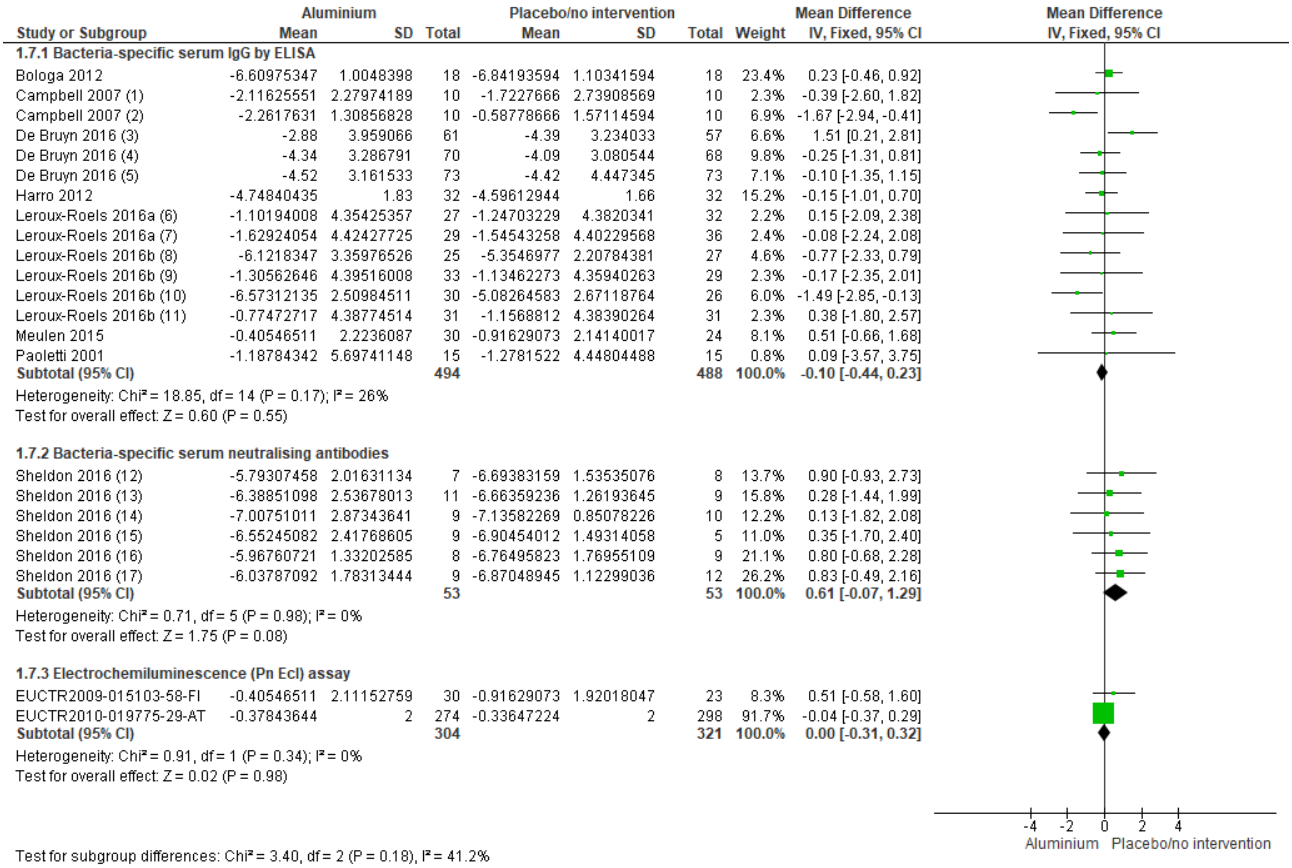
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Supplementary Figure S38.

Fixed-effect meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean titres (GMT) grouped by analytical assay.

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Supplementary Figure S39.

Fixed-effect meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean concentrations (GMC) grouped by analytical assay.

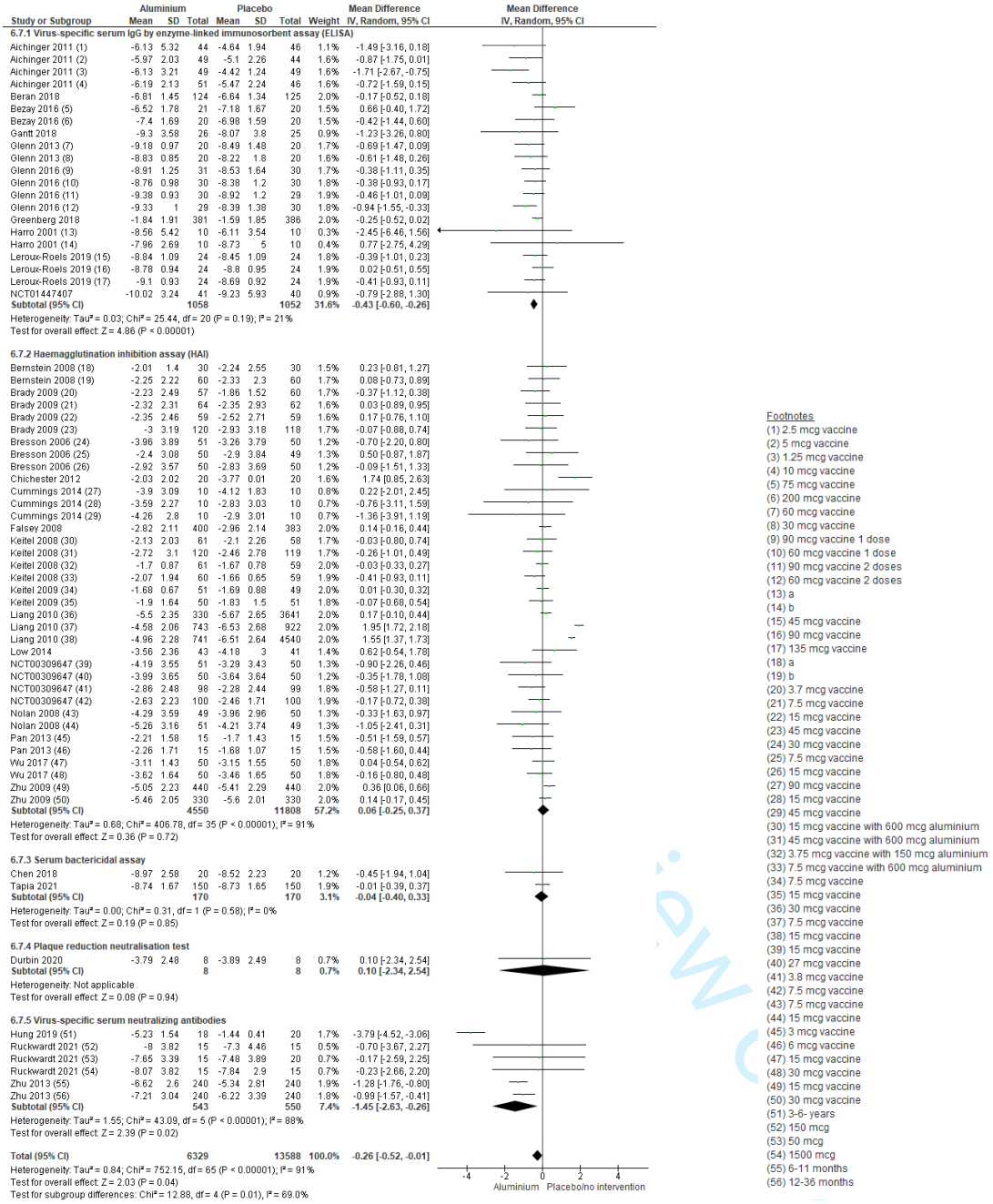
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Supplementary Figure S40.

Overview of the serology analysis with subgroup analysis. We performed subgroup analyses on the most frequently reported analytical assays: enzyme-linked immunosorbent assay (ELISA) and haemagglutination inhibition assay (HAI). We observed a beneficial effect of aluminium adjuvants on serological response in sporadic subgroups (based on vaccine types and on pathogen types).

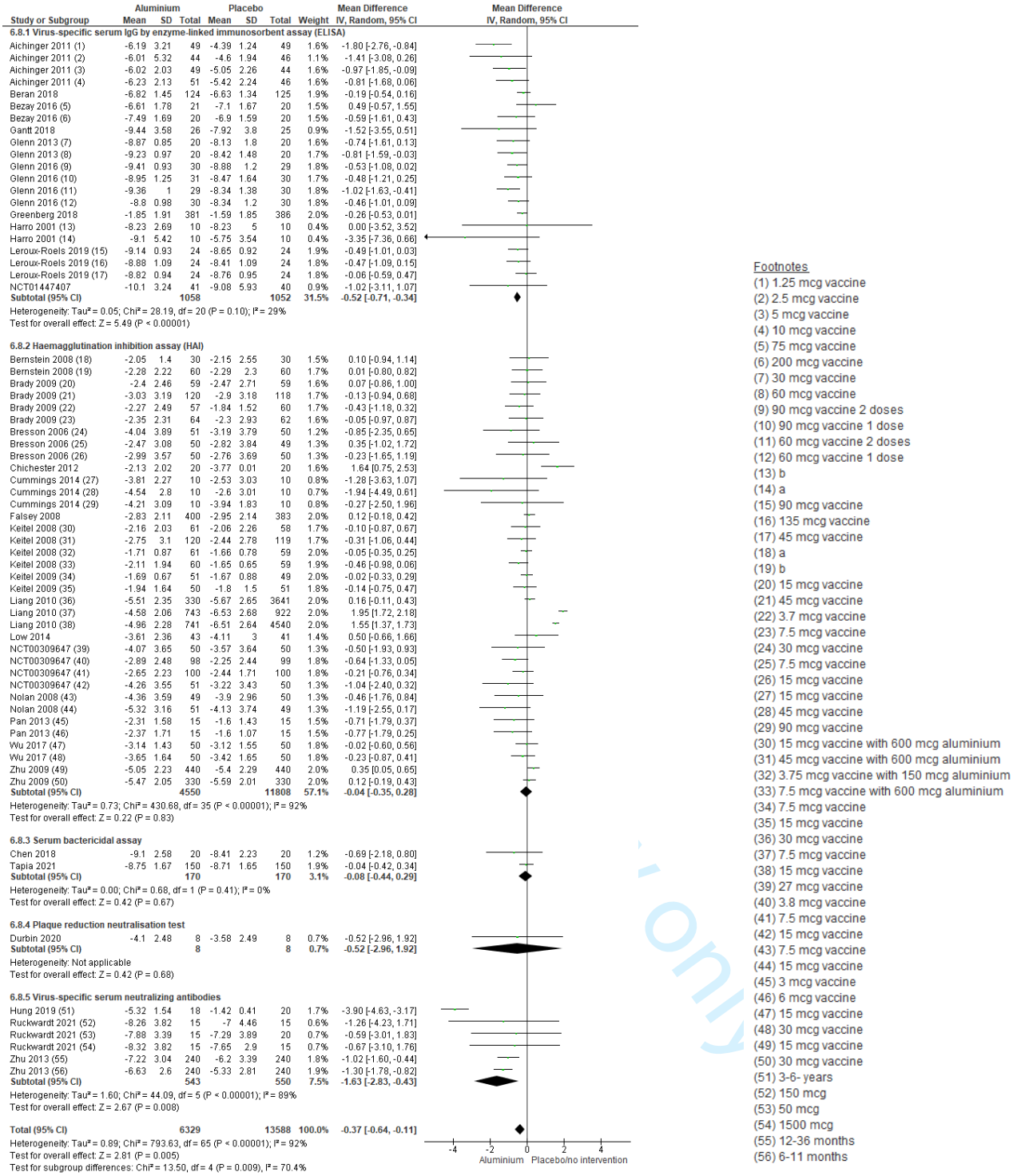
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Supplementary Figure S41.

Sensitivity analysis. Best-worst case scenario with 1 standard deviation for GMT. For serology analysed with geometric mean titre (GMT), 777/3950 (19.7%) participants in the intervention group and 2231/10,217 (21.8%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'best-worst' case scenario, so that there was a beneficial effect of aluminium adjuvants in random-effects meta-analysis (best worst with 1 standard deviation: MD -0.26, 95% CI -0.52 to -0.01; participants = 19 917; studies = 66; I^2 = 91%. Best worst with 2 standard deviations: MD -0.37, 95% CI -0.64 to -0.11; participants = 19 917; studies = 66; I^2 = 92%) Incomplete outcome data did not have the potential to influence the result in the 'worst-best' analysis with 1 or 2 standard deviations (data not shown).

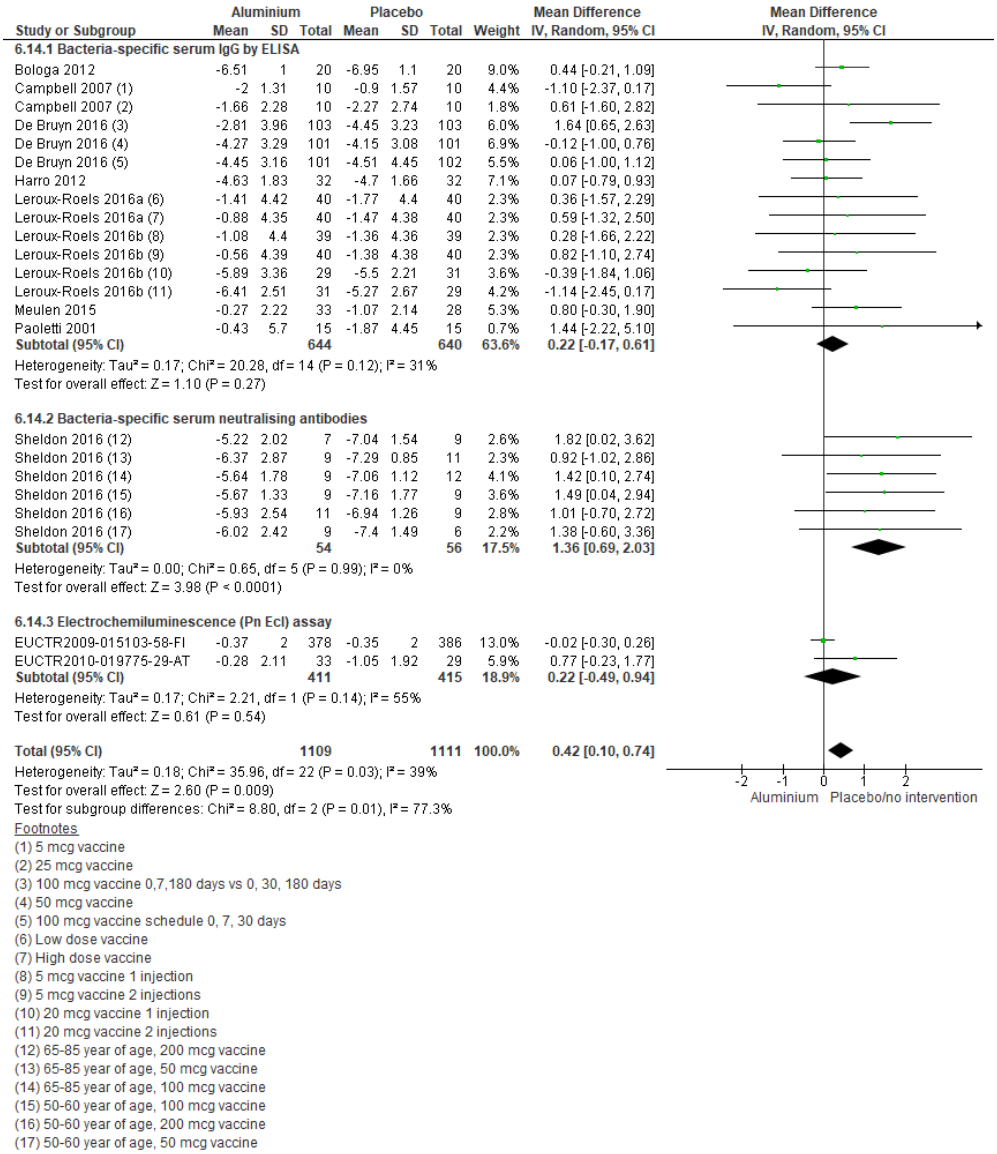
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Supplementary Figure S42.

Sensitivity analysis. Best-worst case scenario with 2 standard deviations for serology analysed with geometric mean titre (GMT).

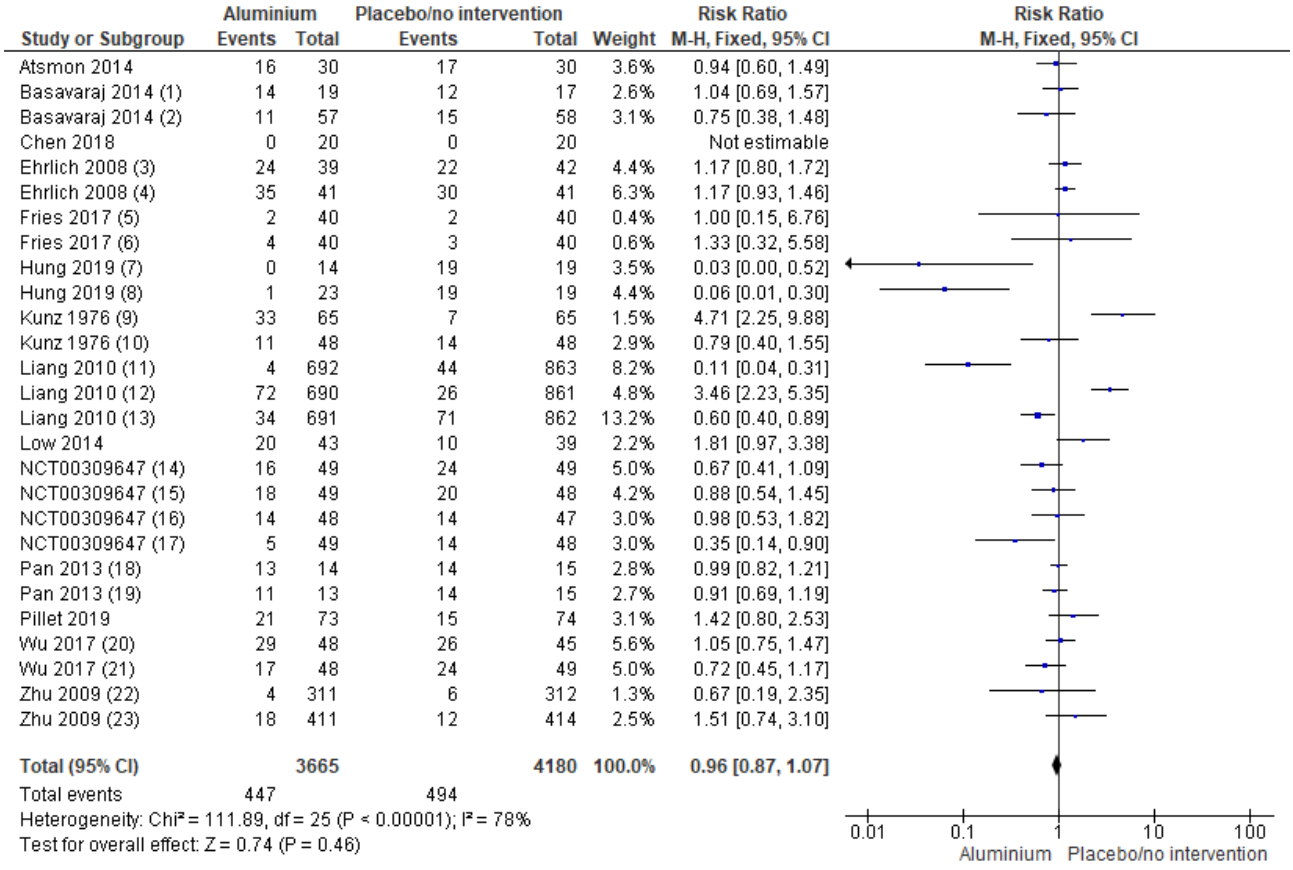
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Supplementary Figure S43.

Sensitivity analysis. Worst-best case scenario with 2 standard deviations for serology analysed with geometric mean concentration (GMC). For serology analysed with GMC, 258/784 (32.9%) participants in the intervention group and 249/478 (52.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'worst-best' case scenario, so that there was a harmful effect of aluminium adjuvants in random-effects meta-analysis (2 standard deviations: MD 0.42, 95% CI 0.10 to 0.74; participants = 2220; studies = 23; I^2 = 39%). Incomplete outcome data did not have the potential to influence the result in the 'worst-best' analysis with 1 standard deviation. Incomplete outcome data did not have the potential to influence the result in the 'best-worst' analysis. Meta-analysis of trials with imputed SD showed no evidence of a difference between aluminium adjuvants versus placebo (GMT: MD -0.11, 95% CI -0.33 to 0.12; GMC: MD 0.04, 95% CI -0.17 to 0.26).

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Footnotes
(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 7.5 mcg vaccine
(4) 15 mcg vaccine
(5) 60 mcg vaccine
(6) 90 mcg vaccine
(7) 3-6 years of age
(8) 2-35 months of age
(9) subunit vaccine
(10) whole virus vaccine
(11) 30 mcg vaccine
(12) 7.5 mcg vaccine
(13) 15 mcg vaccine
(14) 15 mcg vaccine
(15) 27 mcg vaccine
(16) 3.8 mcg vaccine
(17) 7.5 mcg vaccine
(18) 3 mcg vaccine
(19) 6 mcg vaccine
(20) 15 mcg vaccine
(21) 30 mcg vaccine
(22) 30 mcg vaccine
(23) 15 mcg vaccine

Supplementary Figure S44.

Fixed-effect meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on seroprotection.

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Supplementary Analysis S3.

Subgroup and sensitivity analyses for Seroprotection

Test for subgroup differences showed no difference when comparing trials at high risk of bias to trials at low risk of bias; trials at risk of vested interest to trials at low risk of vested interest; trials with different aluminium types and trials with vaccines against different pathogen types.

Test for subgroup differences was statistically significant (P = 0.0005) in the subgroup analysis according to vaccine type.

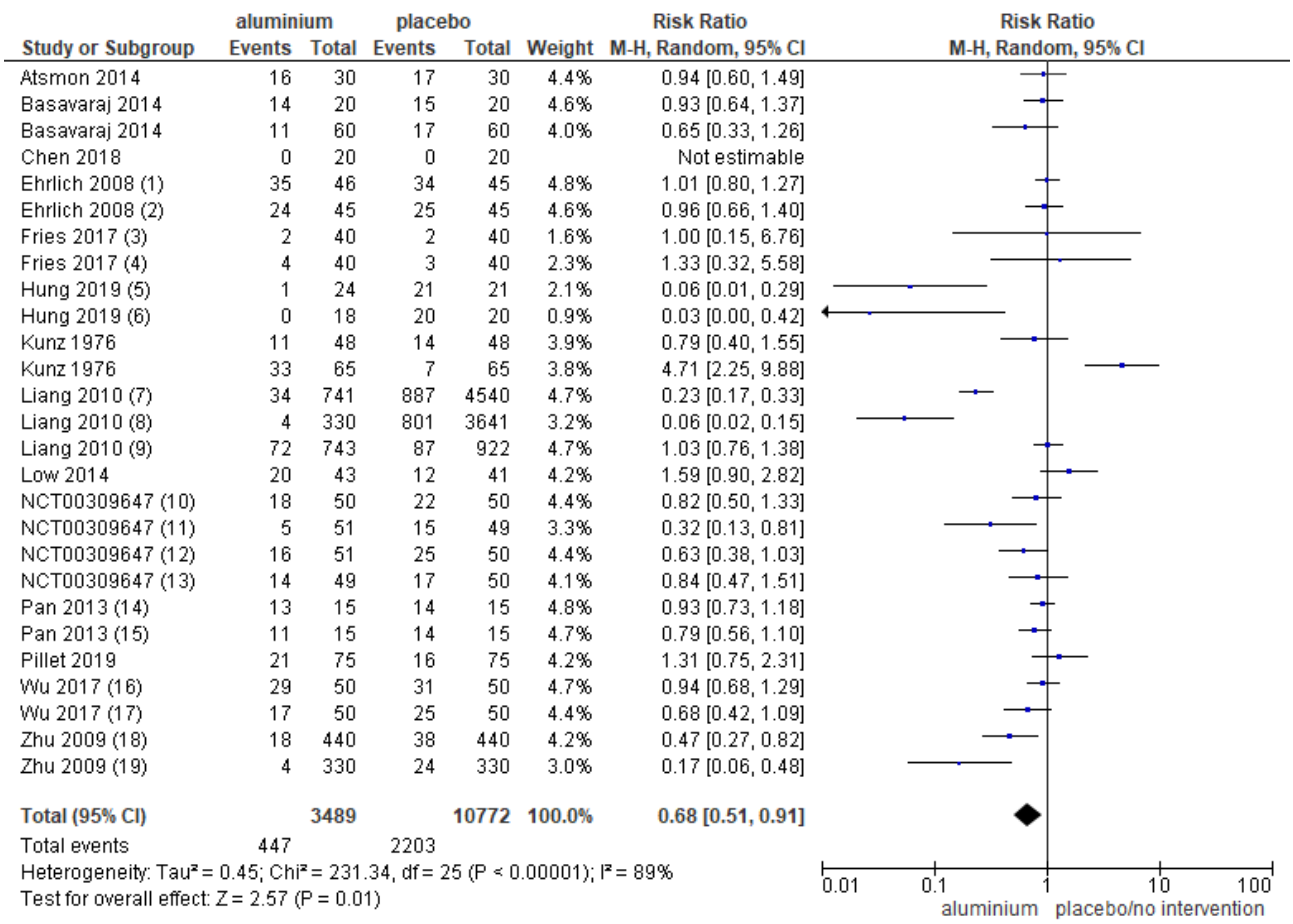
Test for subgroup differences was statistically significant (P = 0.0004) in the subgroup analysis according to age.

Trial Sequential Analysis showed that the accrued information for seroprotection was below 5% of the diversity-adjusted required information size (n = 52 586; Pc 11%, RRR 20%, alpha 2.5%, beta 10%, diversity 82.6%).

242/3165 (7.6%) participants in the intervention group and 1709/9986 (17.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'best-worst' case scenario. The 'best-worst' case analysis showed that incomplete outcome data did not have the potential to influence the result.

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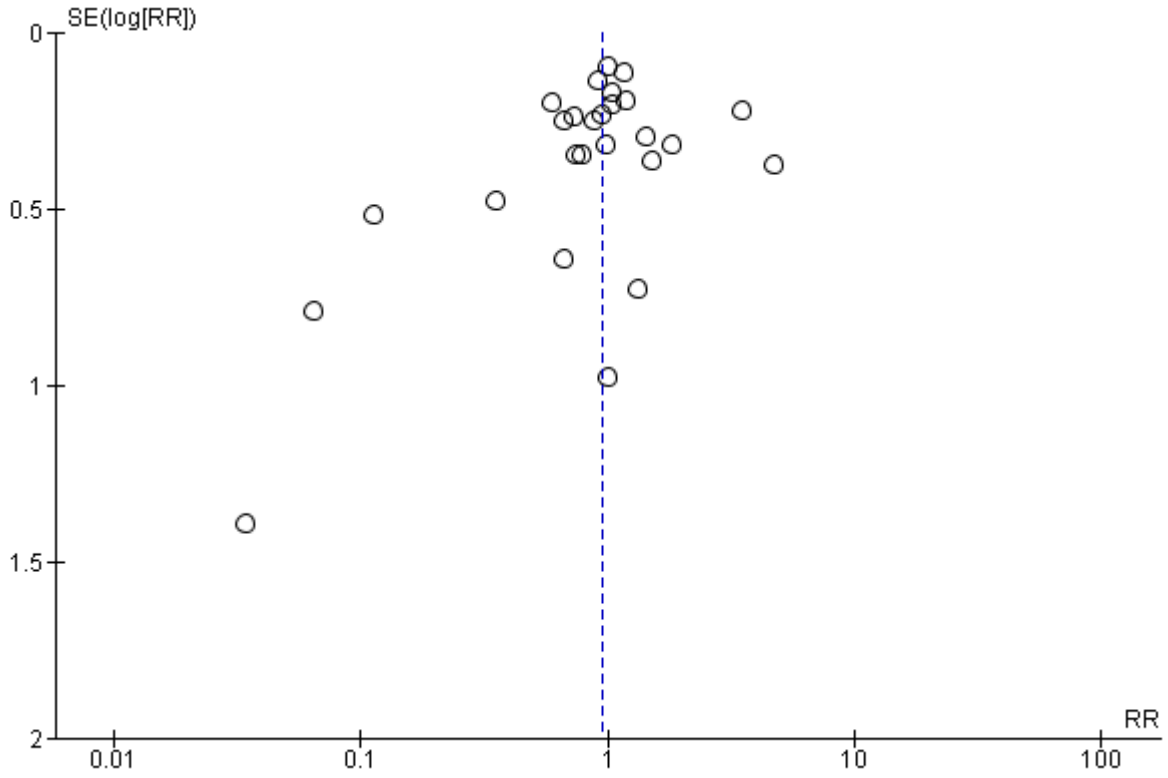


Footnotes
(1) 7.5 mcg vaccine
(2) 15 mcg vaccine
(3) 90 mcg vaccine
(4) 60 mcg vaccine
(5) 3-6 years of age
(6) 2-35 months of age
(7) 15 mcg vaccine, adults
(8) 7.5 mcg vaccine, adults
(9) 15 mcg vaccine, elderly
(10) 7.5 mcg vaccine
(11) 27 mcg vaccine
(12) 15 mcg vaccine
(13) 3.8 mcg vaccine
(14) 3 mcg vaccine
(15) 6 mcg vaccine
(16) 15 mcg vaccine
(17) 30 mcg vaccine
(18) 15 mcg vaccine
(19) 30 mcg vaccine

Supplementary Figure S45.

Sensitivity analysis of seroprotection. Best-worst case scenario for seroprotection. Incomplete outcome data did not have the potential to influence the result in the 'worst-best' (data not shown).

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Supplementary Figure S46.

Funnel plot for seroprotection outcome. Visual inspection of funnel plot may suggest potential publication bias for smaller trials reporting a harmful effect of aluminium adjuvants in the intervention group. Regression-based Harbord test showed no small-study effects (beta = 0.99).

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Supplementary Analysis S4.

Recency of the different trials

We performed our analysis using Review Manager 5.4. In this software, we used the inbuilt function of sorting the forest plots entries by year and we visually inspected the forest plots to account for the recency of different trials. Together with the inspection of the I^2, we did not observe an influence on the analysis stemming from the date of when the studies were performed.

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Supplementary methods

For the serological response, we used the geometric mean concentration or titre (GMC or GMT) extracted from the first assay reported in the journal publication by the trialists. If a trial tested multiple antigens, we included data from the first antigen of the first assay as presented in the publication. For the meta-analysis, we transformed the GMT in the following way: we calculated the natural Log for both the GMT (or GMC) and for the CI.

If SD were not reported, we calculated them using data from the trial, if possible. When standard deviations (SDs) were not reported, we calculated the SD from the CI, we first calculated a t value from the number of participants (n) analysed for that outcome (t value=TINV(1-0.95;n-1), and then we calculated the SD as follows: (SQRT(n))*(LogCIhigh-LogCIlow) / t value). In the meta-analysis, we then plotted the GMT (or GMC) multiplied by -1 and the SD calculated as described above. When imputing the SD due to lack of reporting of CI by the trialists, an average SD was calculated from the other trials and used in the meta-analysis and presented in sensitivity analyses.

We did not impute missing values for any outcomes in our primary analysis. In our sensitivity analysis for dichotomous and continuous outcomes, we did impute data

Unit of analysis issues

We included data from trials where participants were individually randomised to two or more intervention groups. When trials had more than two intervention groups, we considered the following scenarios.

Experimental intervention	Control
Vaccine with different antigen concentrations and same aluminium concentration	Same vaccine with different antigen concentrations without aluminium
Vaccine with different antigen concentration and different aluminium concentration	Same vaccine with different antigen concentration without aluminium
Different aluminium concentration	Placebo

We analysed the scenarios above as follows:

- when the intervention groups had different antigen concentration but same aluminium concentration, we pooled the data from all the intervention groups for the safety analyses, and we analysed each intervention group separately for serology.

- when the intervention groups had different antigen concentration and different aluminium concentration, we analysed each intervention group separately both for safety and for serology.

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- when we presented analyses from multiple groups from the same trial, the ID of the same trial appears multiple times in the forest plot together with a footnote describing the difference in the vaccine composition (i.e. antigen concentration) in that trial group.

Differences between protocol and review

We did not specifically search for observational studies (quasi-randomised studies; cohort studies; and patient series), and we abandoned our potential plan to provide a narrative account of such data. If and when valid benefits of adding aluminium adjuvants one day may be identified on patient-relevant outcomes, observational studies should be scrutinized for rare and late occurring harms.

We originally planned to base our results, conclusions and 'Summary of findings' (SoF) table on trials at low risk of bias. However, due to lack of relevant data, we based our results, conclusions and SoF table on all trials regardless of the bias assessment.

We originally planned to perform the subgroup analysis on the effects of nanoparticle aluminium adjuvant compared to microparticle aluminium adjuvant regardless of type of aluminium salt. However, due to lack of data, we could not perform such analysis.

We originally planned to perform the subgroup analysis on the effect of total administration of aluminium between trials. However, this was replaced by a meta-regression analysis.

We originally planned to perform the subgroup analysis on the effects of aluminium adjuvant administration between trials with different maximal follow-up periods, by defining the follow up period 'from randomisation until the end of follow up'. However, we decided to change the definition of the follow up period to 'from end of the vaccination until the end of follow up'.

We added a subgroup analysis for trials assessing vaccines against different pathogen types in response to a peer reviewer. We did not describe this analysis in our protocol.

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Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Manuscript page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Manuscript page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Manuscript page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Manuscript page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Manuscript page 4; Supplementary page 50; Protocol
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Manuscript page 5; Supplementary Table 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary Table 1;
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 5,6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Manuscript page 5,6; Protocol
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Manuscript page 5,6; Protocol; Supplementary Page 51
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Protocol; Supplementary Page 50
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 6; Protocol
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Manuscript page 6; Supplementary Page 50; Protocol
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Manuscript page 4; Supplementary page 50; Protocol
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing	Supplementary page 50

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Section and Topic	Item #	Checklist item	Location where item is reported
		summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Supplementary Table2,3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Manuscript page 4,5; Supplementary page 50; Protocol
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Manuscript page 5,6; 7 Protocol
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Manuscript page 4,5; Supplementary page 50; Protocol
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Manuscript page 5; Supplementary page 50; Protocol
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Manuscript page 7; Protocol
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Manuscript Page 7; Supplementary Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Appendix 1
Study characteristics	17	Cite each included study and present its characteristics.	Appendix 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Manuscript Page 11 to 17
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	n.a.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Manuscript Page 11 to 17 Supplementary material for subgroup analysis
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Manuscript Page 11 to 17; Supplementary material for subgroup analysis
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Manuscript Page 11 to 17; Supplementary material for sensitivity analysis
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Manuscript Page 11 to 17;

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Section and Topic	Item #	Checklist item	Location where item is reported
			Supplementary material for sensitivity analysis
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Manuscript page 18 to 22
	23b	Discuss any limitations of the evidence included in the review.	Manuscript page 18 to 22
	23c	Discuss any limitations of the review processes used.	Manuscript page 18 to 22
	23d	Discuss implications of the results for practice, policy, and future research.	Manuscript page 18 to 22
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	abstract
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	abstract
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Supplementary page 51
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Manuscript page 23,24
Competing interests	26	Declare any competing interests of review authors.	Manuscript page 25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	n.a.

Supplementary Table S4. PRISMA checklist.

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Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Manuscript page 1
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Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Manuscript page 5,6; Protocol; Supplementary Page 51
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Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Manuscript page 6; Supplementary Page 50; Protocol
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Manuscript page 4; Supplementary page 50; Protocol
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing	Supplementary page 50

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Section and Topic	Item #	Checklist item	Location where item is reported
		summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Supplementary Table2,3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Manuscript page 4,5; Supplementary page 50; Protocol
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Manuscript page 5,6; 7 Protocol
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Manuscript page 4,5; Supplementary page 50; Protocol
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Manuscript page 5; Supplementary page 50; Protocol
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	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Appendix 1
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Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	n.a.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Manuscript Page 11 to 17 Supplementary material for subgroup analysis
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Manuscript Page 11 to 17; Supplementary material for subgroup analysis
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Manuscript Page 11 to 17; Supplementary material for sensitivity analysis
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Manuscript Page 11 to 17; Supplementary material

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Section and Topic	Item #	Checklist item	Location where item is reported
			for sensitivity analysis
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Manuscript page 18 to 22
	23b	Discuss any limitations of the evidence included in the review.	Manuscript page 18 to 22
	23c	Discuss any limitations of the review processes used.	Manuscript page 18 to 22
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Competing interests	26	Declare any competing interests of review authors.	Manuscript page 25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	n.a.

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